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NCERT based · Rationalised syllabus

CBSE Class 12

BIOLOGY

KHATAM IN 199 PAGES

Session 2026-27

Reproduction | Genetics & Evolution | Human Welfare | Biotechnology | Ecology

By Nitesh Bhaiya

One Shot handwritten notes · NCERT diagrams · Most Important Questions in every chapter

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Puri Biology, Ek Notebook Mein

Most asked: 13 chapters | notes + Most Important Questions with answers | last-night revision

1. Unit VI: Reproduction

Ch	Chapter	Page
1	Sexual Reproduction in Flowering Plants	3
2	Human Reproduction	15
3	Reproductive Health	30

2. Unit VII: Genetics and Evolution

Ch	Chapter	Page
4	Principles of Inheritance and Variation	39
5	Molecular Basis of Inheritance	59
6	Evolution	83

3. Unit VIII: Biology and Human Welfare

Ch	Chapter	Page
7	Human Health and Disease	96
8	Microbes in Human Welfare	111

4. Unit IX: Biotechnology

Ch	Chapter	Page
9	Biotechnology: Principles and Processes	120
10	Biotechnology and its Applications	135

5. Unit X: Ecology

Ch	Chapter	Page
11	Organisms and Populations	146
12	Ecosystem	157
13	Biodiversity and Conservation	168

6. Revision

Section	Page
Last-night revision: one-liners, differences, scientists, numbers, full forms, most expected questions, must-draw diagrams	179
All the best from Paheli, Bujho and Nitesh Bhaiya	199

Nitesh Bhaiya ki baat

Is notebook mein poori NCERT Biology hai, board exam wali language mein. Har chapter ke end mein Most Important Questions answers ke saath. Roz ek chapter + uske questions, aur board exam bas ek aur test lagega!

Nitesh Bhaiya | Eduspark

7. Board Paper Pattern (Theory 70 + Practical 30)

Section	Type of questions	Marks
A	16 MCQ / Assertion-Reason x 1	16
B	5 very short answer x 2	10
C	7 short answer x 3	21
D	2 case-based x 4	8
E	3 long answer x 5	15
Total	**33 questions, 3 hours**	**70**

8. Unit-wise Marks

Unit	Marks
VI. Reproduction (Ch 1-3)	16
VII. Genetics and Evolution (Ch 4-6)	20
VIII. Biology and Human Welfare (Ch 7-8)	12
IX. Biotechnology and its Applications (Ch 9-10)	12
X. Ecology and Environment (Ch 11-13)	10
Total	**70**

Is book ko kaise padhein

1. Pehle notes padho: yellow = must learn, red = keyword.
2. Har box ka kaam: **Yaad karo** = trick, **Exam Alert** = marks katne wali galti, **Quick fact** = 1-mark wali line.
3. Phir chapter ke **Most Important Questions** khud likh ke solve karo, answer green mein match karo.
4. Diagrams pencil se khud banao; labels yaad karo.
5. Exam se pehle wali raat: page 179 se revision.

Exam hall tips

→ 15 min reading time mein choice wale questions pehle tick kar lo.

→ Diagram pencil se, labels seedhi lines se, heading ke saath.

→ Long answers mein points + diagram + keywords underline: full marks ka formula.

Last 7 days plan

Day	Kya karna hai
1	Ch 1-3 (Reproduction) + diagrams
2-3	Ch 4-6 (Genetics, Molecular basis, Evolution)
4	Ch 7-8 (Health, Microbes)
5	Ch 9-10 (Biotechnology)
6	Ch 11-13 (Ecology)
7	Revision p179 + one sample paper, timed

Sexual Reproduction in Flowering Plants

Most asked: double fertilisation, embryo sac (7-celled, 8-nucleate), microsporangium wall layers, outbreeding devices, apomixis, false fruit

The **flower** is the site of sexual reproduction in angiosperms. This chapter follows the full story: **pollen and ovule formation, pollination, double fertilisation, seed and fruit.**

1. Flower: A Fascinating Organ of Angiosperms

Flowers have aesthetic, ornamental, social, religious and cultural value, but for a biologist they are morphological and embryological marvels and the **sites of sexual reproduction.**

Much before a flower is seen, the decision to flower is taken: hormonal and structural changes lead to differentiation and development of the **floral primordium.** Inflorescences bear floral buds and then flowers.

→ **Androecium** = whorl of **stamens**: the male reproductive organ.
→ **Gynoecium** = **pistil(s)**: the female reproductive organ.

Quick fact

Sepals (calyx) and petals (corolla) are the **accessory whorls**; stamens and pistil are the essential whorls, because they take part directly in sexual reproduction.

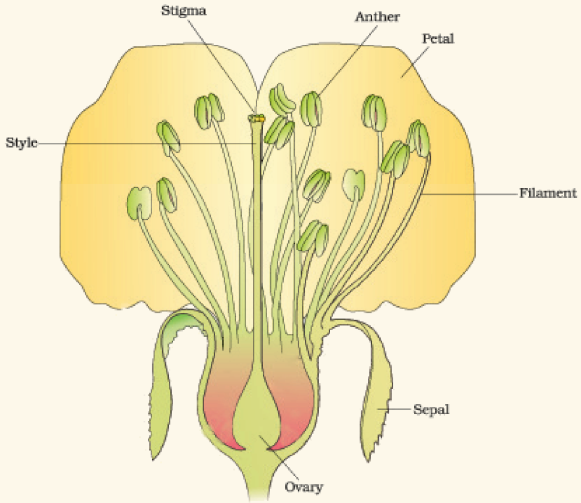


Fig. 1.1 Diagrammatic L.S. of a flower

2. Pre-fertilisation: Structures and Events

A. Stamen

A typical stamen has two parts: a long slender stalk, the **filament**, and a terminal, generally bilobed structure, the **anther**. The proximal end of the filament is attached to the **thalamus or petal.** The number and length of stamens vary with species.

A typical angiosperm anther is bilobed; each lobe has **two theca** (dithecous), separated by a longitudinal groove. The anther is a **tetragonal** structure with four microsporangia, two in each lobe. Microsporangia develop further and become **pollen sacs**, packed with pollen grains.

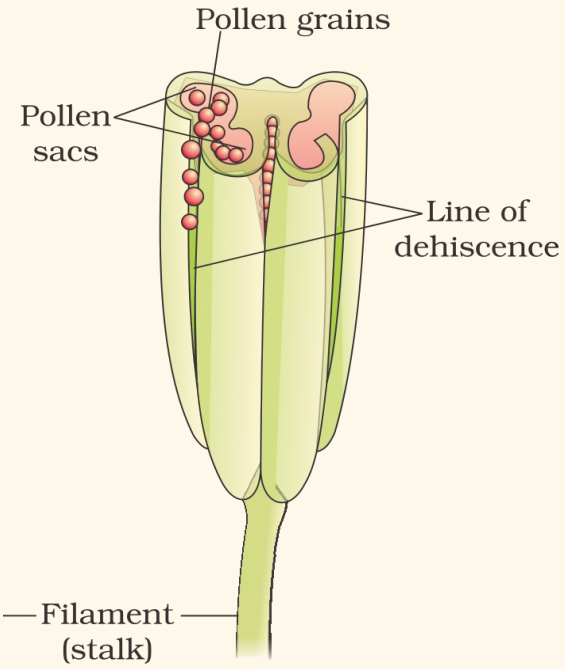


Fig. 1.2 (b) Three-dimensional cut section of an anther

B. Structure of microsporangium

In transverse section a microsporangium is **nearly circular.** It is surrounded by four wall layers:

Wall layer (outside to inside)	Function
1. Epidermis	protection
2. Endothecium	protection; helps in **dehiscence** of anther to release pollen
3. Middle layers (2-3)	protection; help in dehiscence
4. Tapetum (innermost)	**nourishes** the developing pollen grains

Tapetum: its cells have **dense cytoplasm** and generally more than one nucleus. When the anther is young, a group of compactly arranged homogenous cells called **sporogenous tissue** occupies the centre of each microsporangium. PYQ

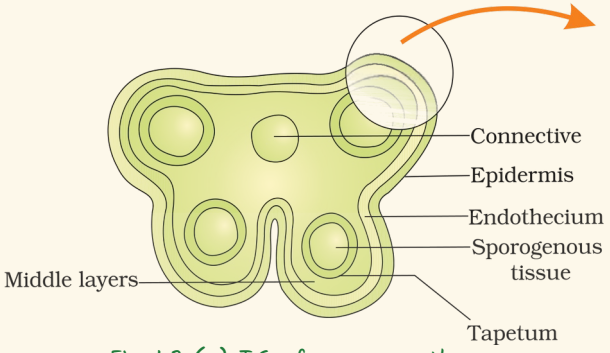


Fig. 1.3 (a) T.S. of a young anther

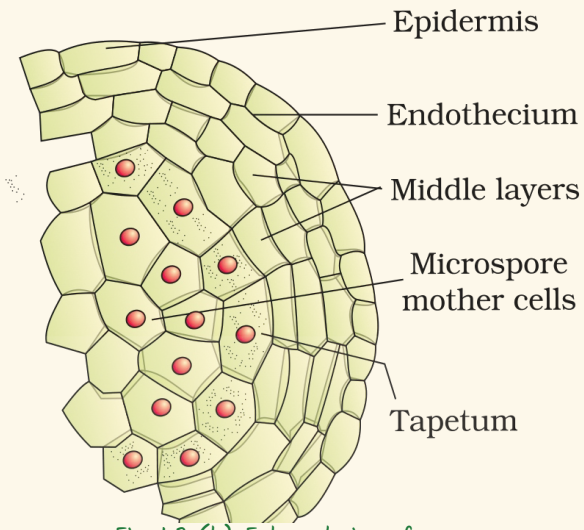


Fig. 1.3 (b) Enlarged view of one microsporangium showing wall layers

Yaad karo: wall layers (bahar se andar)

Every **E**ngineer **M**akes **T**ea = Epidermis, Endothecium, Middle layers, Tapetum. Tapetum = andar wali **mummy**: pollen ko khana deti hai!

C. Microsporogenesis

Each cell of the sporogenous tissue can act as a **pollen mother cell (PMC)** or microspore mother cell, which undergoes **meiosis** to form a cluster of four cells, the **microspore tetrad**. The process of formation of microspores from a PMC through meiosis is called **microsporogenesis**.

Sporogenous tissue (2n) → PMC (2n) → Meiosis → Microspore tetrad (n) → Pollen grains (n)

As the anthers mature and dehydrate, the microspores **dissociate** from each other and develop into pollen grains. Each microsporangium forms **thousands of pollen grains**, released by dehiscence of the anther.

D. Pollen grain (male gametophyte)

Pollen grains are generally **spherical**, about **25–50 micrometres** in diameter, with a prominent **two-layered wall**:

- → **Exine** (outer): hard, made of **sporopollenin**, one of the **most resistant organic materials** known. It withstands high temperature, strong acids and alkali; **no enzyme** that degrades it is known. This is why pollen is well preserved as **fossils**.
- → **Germ pores**: apertures on the exine where sporopollenin is **absent**; the pollen tube comes out through them.
- → **Intine** (inner): thin, continuous layer of **cellulose and pectin**.

The cytoplasm is surrounded by a plasma membrane. A mature pollen grain contains **two cells**:

Vegetative cell	Generative cell
bigger	small
abundant food reserve	spindle-shaped, dense cytoplasm
large, irregular nucleus	floats in the cytoplasm of vegetative cell
forms the pollen tube	divides mitotically into **two male gametes**

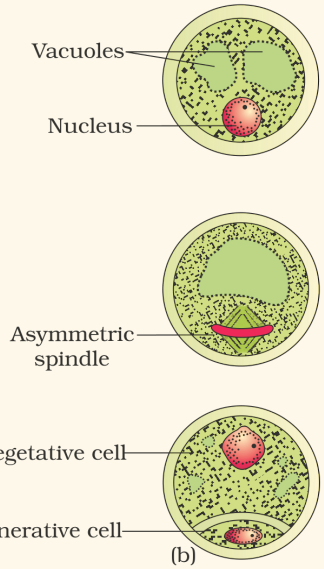


Fig. 1.5 (b) Stages of a microspore maturing into a pollen grain

In over **60% of angiosperms**, pollen grains are shed at the **2-celled stage**. In the rest, the generative cell divides mitotically to give two male gametes before shedding (**3-celled stage**). PYQ

Quick fact: pollen in daily life

- → Pollen of some plants (e.g. **Parthenium** or carrot grass, which came to India with imported wheat) causes **allergies** and respiratory disorders like asthma and bronchitis.
- → Pollen is rich in nutrients; **pollen tablets** are used as food supplements, and pollen extract is used by athletes and race horses.
- → Pollen viability: **rice and wheat** lose it within **30 minutes**; some members of Rosaceae, Leguminosae and Solanaceae stay viable for **months**.
- → Pollen can be stored for years in **liquid nitrogen (−196 °C)** in **pollen banks**, like seed banks, for crop breeding.

Paheli-Bujho

Bujho: Paheli, pollen fossil kaise ban jaata hai? Itna chhota toh hai!

Paheli: Iski exine sporopollenin ki bani hai. Acid, garmi, enzyme: koi isko tod nahi paata.

Bujho: Matlab pollen ka bulletproof jacket!

Paheli: Bilkul! Bas germ pores pe jacket nahi hota.

E. The pistil, megasporangium (ovule) and embryo sac

The gynoecium may be **monocarpellary** (single pistil) or **multicarpellary** (more than one pistil). Pistils may be fused (**syncarpous**, e.g. *Papaver*) or free (**apocarpous**, e.g. *Michelia*).

- → **Stigma**: landing platform for pollen grains.
- → **Style**: elongated slender part beneath the stigma.
- → **Ovary**: basal bulged part; inside its cavity (**ovarian cavity, locule**) is the **placenta**, from which **megasporangia (ovules)** arise.

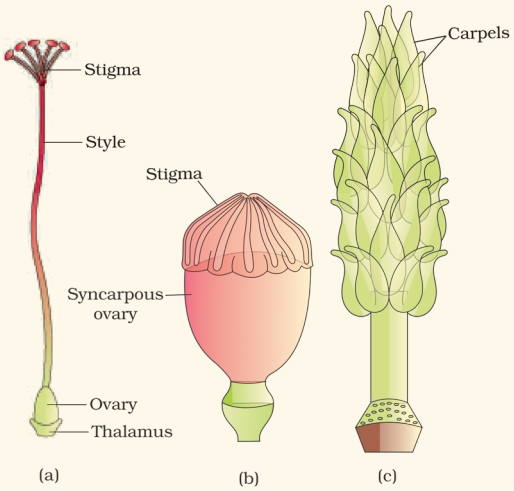


Fig. 1.7 (a) Pistil of Hibiscus; (b) multicarpellary syncarpous pistil of Papaver; (c) apocarpous gynoecium of Michelia

The number of ovules in an ovary may be **one** (wheat, paddy, mango) or **many** (papaya, watermelon, orchids).

F. Structure of a typical ovule (megasporangium)

- → **Funicle**: stalk attaching the ovule to the placenta.
- → **Hilum**: junction between ovule and funicle.
- → **Integuments**: one or two protective envelopes around the ovule.
- → **Micropyle**: small opening at the tip, not covered by integuments.
- → **Chalaza**: basal part, opposite the micropylar end.
- → **Nucellus**: mass of cells inside the integuments with abundant **reserve food**.
- → **Embryo sac** (female gametophyte): located in the nucellus. An ovule generally has a single embryo sac formed from a megaspore.

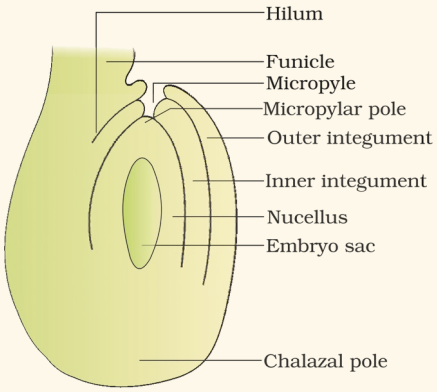


Fig. 1.7 (d) Diagrammatic view of a typical anatropous ovule

G. Megasporogenesis

The process of formation of **megaspores** from the **megaspore mother cell (MMC)** is called megasporogenesis. Ovules generally differentiate a single MMC in the **micropylar region of the nucellus**. It is a large cell with dense cytoplasm and a prominent nucleus.

MMC (2n) → Meiosis → 4 megaspores (n) → 3 degenerate → 1 functional megaspore

Exam Alert: microsporogenesis vs megasporogenesis

Microsporogenesis	Megasporogenesis
in the **microsporangium** (anther)	in the **nucellus** of the ovule
starts from pollen mother cell (PMC)	starts from megaspore mother cell (MMC)
many PMCs in each microsporangium	generally a **single MMC** per ovule
all 4 microspores form pollen grains	only **1 of 4** megaspores is functional
product: microspore tetrad, then pollen	product: functional megaspore, then embryo sac

H. Female gametophyte (embryo sac)

In a majority of flowering plants, one megaspore is **functional** while the other three degenerate. The functional megaspore develops into the embryo sac. This method of embryo sac formation from a single megaspore is called **monosporic development**. PYQ

1. The nucleus of the functional megaspore divides mitotically to form **two nuclei**, which move to opposite poles (**2-nucleate** embryo sac).
2. Two more sequential mitotic divisions give the **4-nucleate** and then the **8-nucleate** stages.
3. These divisions are free nuclear: nuclear divisions are **not followed immediately** by cell wall formation.
4. After the 8-nucleate stage, cell walls are laid down, organising the typical female gametophyte.

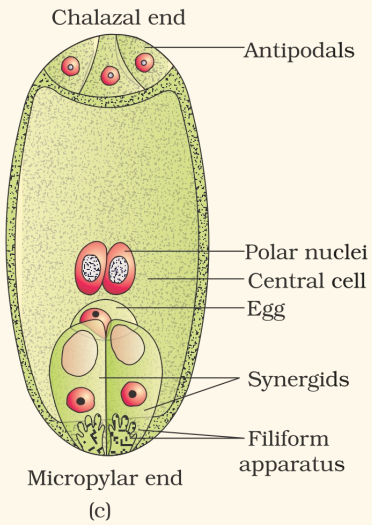


Fig. 1.8 (c) A diagrammatic representation of the mature embryo sac

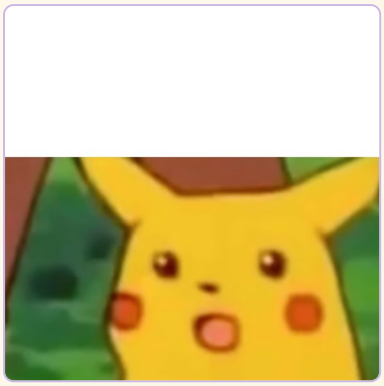
Position	Cells	Number
Micropylar end	**egg apparatus**: 2 synergids + 1 egg cell	3 cells
Chalazal end	antipodals	3 cells
Centre	large **central cell** with 2 polar nuclei	1 cell, 2 nuclei

So a typical angiosperm embryo sac at maturity is 8-nucleate but 7-celled. The synergids have special cellular thickenings at

the micropylar tip called the **filiform apparatus**, which guides the pollen tube into the synergid. Board fav

Yaad karo: 7 cells, 8 nuclei

3 + 3 + 1 = 7 cells (egg apparatus + antipodals + central cell). Nuclei: **3 + 3 + 2 = 8**, kyunki central cell ke paas **2 polar nuclei** hain. Ek cell ka extra nucleus = 7 vs 8 ka raaz!



Mera reaction jab suna: embryo sac **7-celled** hai par **8-nucleate**.

Asli baat: 3 micropylar (egg apparatus) + 3 antipodals + 1 binucleate central cell.

3. Pollination

Both male and female gametes are **non-motile**, so pollen must be brought to the stigma. Pollination is the transfer of pollen grains (shed from the anther) to the **stigma** of a pistil.

A. Kinds of pollination

Type	Pollen goes from	Note
Autogamy	anther to stigma of the **same flower**	needs synchrony in pollen release and stigma receptivity
Geitonogamy	one flower to **another flower of the same plant**	functionally cross-pollination (needs an agent) but **genetically similar** to autogamy
Xenogamy	anther to stigma of a **different plant**	the only type that brings genetically different pollen

Chasmogamous flowers: open flowers with exposed anthers and stigma (most plants). **Cleistogamous flowers:** do **not open at all**; anthers and stigma lie close together, so they are invariably autogamous and produce **assured seed-set** even without pollinators. Disadvantage: no cross-pollination. Examples: *Viola* (common pansy), *Oxalis*, *Commelina* (which bear both types). PYQ

Exam Alert

→ Cross-pollination **cannot** occur in cleistogamous flowers: they never open.

→ Geitonogamy = **functionally cross**, **genetically self**. Ye line 1 mark pakka.

B. Agents of pollination

Abiotic (wind, water) and **biotic** (animals). A majority of plants use biotic agents; only a small proportion use abiotic agents. Pollination by abiotic agents is more a matter of chance, so such plants produce **enormous amounts of pollen**.

Wind-pollinated (anemophily)	Water-pollinated (hydrophily)
light, **non-sticky** pollen	pollen protected from wetting by a **mucilaginous covering**
well-exposed stamens	*Vallisneria*: female flower reaches the surface on a long stalk; male flowers released on the surface
large, often **feathery stigma**	sea grasses like *Zostera*: female flowers stay submerged, long ribbon-like pollen released inside water
often a single ovule per ovary; numerous flowers packed in an inflorescence	fairly limited, about **30 genera**, mostly monocots: *Vallisneria*, *Hydrilla*, *Zostera*
common in grasses; corn cob tassels are stigma and style	not all aquatic plants: **water hyacinth, water lily** are pollinated by insects or wind

Wind and water pollinated flowers are not very colourful and **do not produce nectar**.

C. Insect (animal) pollination

Bees, butterflies, flies, beetles, wasps, ants, moths, birds (**sunbirds, humming birds**) and bats are common pollinators; among animals, insects, particularly bees, are dominant. Some primates (lemurs), arboreal rodents and reptiles (gecko lizard, garden lizard) also pollinate.

→ Flowers are large, **colourful, fragrant** and rich in **nectar**; small flowers are clustered in an inflorescence.

→ Flies and beetles are attracted by **foul odours**; pollen grains are generally sticky.

→ **Floral rewards:** nectar and pollen grains. Some plants give **safe places to lay eggs**: the tallest flower of *Amorphophallus* (about 6 feet); a species of **moth and *Yucca*** cannot complete their life cycles without each other.

→ **Pollen/nectar robbers** consume pollen or nectar without bringing about pollination.

Wind-pollinated flower	Insect-pollinated flower
small, not colourful, no nectar, no fragrance	large, colourful, fragrant, nectar-rich
pollen light, dry, non-sticky, produced in huge amounts	pollen sticky, fewer in number
stigma large, feathery, exposed	stigma sticky, often inside the flower
e.g. grasses, maize	e.g. *Yucca*, *Amorphophallus*, sunflower

D. Outbreeding devices

Continued self-pollination leads to **inbreeding depression**. Flowering plants have developed many devices to discourage self-pollination and encourage cross-pollination: PYQ

1. Pollen release and stigma receptivity are **not synchronised**: pollen is released before the stigma becomes receptive, or vice versa.
2. Anther and stigma are placed at **different positions**, so pollen cannot reach the stigma of the same flower.

3. **Self-incompatibility:** a genetic mechanism that prevents self-pollen (same flower or other flowers of the same plant) from fertilising the ovules, by inhibiting **pollen germination or pollen tube growth** in the pistil.
4. Production of **unisexual flowers**: if male and female flowers are on the same plant (**monoecious**, e.g. castor, maize), autogamy is prevented but not geitonogamy; in **dioecious** plants (papaya) both are prevented.

E. Pollen-pistil interaction

The pistil can recognise the pollen: compatible (right type) or incompatible. If compatible, the pistil accepts it and promotes post-pollination events; if incompatible, it rejects it by preventing germination or pollen tube growth. This dialogue is mediated by **chemical components** of the pollen interacting with those of the pistil.

After compatible pollination, the pollen grain germinates on the stigma to produce a **pollen tube** through one of the germ pores. The contents of the pollen grain move into it; the tube grows through the stigma and style and reaches the ovary. In plants shed at the 2-celled stage, the generative cell divides to form two male gametes **during the growth of the pollen tube**.

Pollen on stigma → Germination → Pollen tube in style → Enters ovule via micropyle → Enters a synergid through filiform apparatus

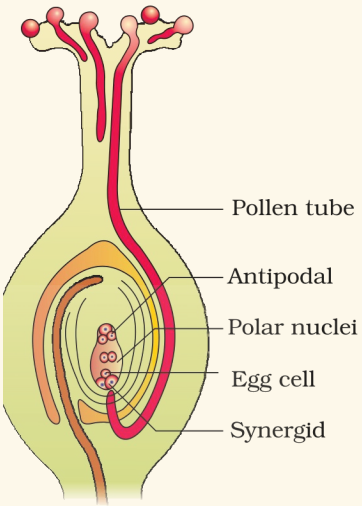


Fig. 1.12 (c) L.S. of pistil showing the path of the pollen tube

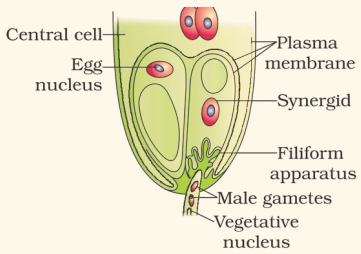


Fig. 1.12 (d) Enlarged view of the egg apparatus

All these events, from the deposition of pollen on the stigma until the pollen tube enters the ovule, are together called pollen-pistil interaction. It is a dynamic process involving pollen recognition followed by promotion or inhibition of the pollen.

F. Artificial hybridisation

A major approach of **crop improvement**: only the desired pollen grains are used for pollination and the stigma is protected from contamination by unwanted pollen. PYQ

- → **Emasculation**: removal of anthers from the flower bud **before the anther dehisces**, using forceps (needed when the female parent bears bisexual flowers).
- → **Bagging**: the emasculated flower is covered with a bag of suitable size, generally of **butter paper**, to prevent contamination of its stigma with unwanted pollen.
- → When the stigma becomes receptive, mature pollen collected from the male parent is **dusted on the stigma** and the flower is **rebagged** for the fruits to develop.
- → If the female parent produces **unisexual flowers**, emasculation is not needed: the female flower buds are bagged before the flowers open.

Plant breeder ka look jab wo **emasculation** aur **bagging** karne nikalta hai.

Asli baat: Artificial hybridisation: anthers hatao (emasculation), bag lagao, desired pollen dust karo, rebag karo.

Paheli-Bujho

Bujho: Paheli, plant breeder phoolon ko thaili kyun pehnata hai?
Paheli: Usse bagging kehte hain. Galat pollen stigma tak na pahunche.
Bujho: Aur anther kyun nikaal diye?
Paheli: Emasculation! Taaki self-pollination na ho, sirf chosen pollen lage.

4. Double Fertilisation

After entering a synergid, the pollen tube releases **two male gametes** into its cytoplasm. Two fusions now occur inside one embryo sac:

Syngamy	*Triple fusion*
1st male gamete + **egg cell**	2nd male gamete + **two polar nuclei** in the central cell
forms **diploid zygote** (2n)	forms **triploid primary endosperm nucleus**, PEN (3n)
zygote develops into the **embryo**	central cell becomes the **primary endosperm cell (PEC)** and forms the **endosperm**

Since two types of fusions, syngamy and triple fusion, take place in an embryo sac, the phenomenon is called **double fertilisation**, an event **unique to flowering plants.** Board fav

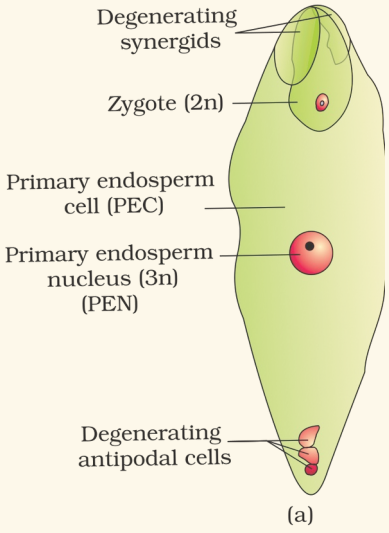


Fig. 1.13 (a) Fertilised embryo sac showing zygote and PEN

Exam Alert: ploidy table

Structure	Ploidy
pollen grain, egg, synergid, antipodal, male gamete, megaspore	n
MMC, PMC, nucellus, integument, zygote, embryo	2n
primary endosperm nucleus, endosperm	3n

Paheli-Bujho

Bujho: Ek pollen tube mein do male gametes? Ek extra kyun?

Paheli: Ek egg se milke zygote banata hai, doosra do polar nuclei se milke endosperm.

Bujho: Matlab ek baby banata hai, doosra uska tiffin!

Paheli: Haha, exactly! Isliye isse double fertilisation kehte hain.

5. Post-fertilisation: Structures and Events

Following double fertilisation, events of **endosperm and embryo development**, maturation of ovule(s) into seed(s) and ovary into fruit are collectively termed post-fertilisation events.

A. Endosperm

Endosperm development precedes embryo development. The PEC divides repeatedly to form a **triploid endosperm tissue**, filled with reserve food, which **nourishes the developing embryo.** PYQ

In the most common type, the PEN undergoes successive nuclear divisions to give **free nuclei** (free-nuclear endosperm); cell wall formation follows and the endosperm becomes **cellular**. The number of free nuclei before cellularisation varies greatly.

Quick fact: coconut

The **coconut water** of tender coconut is **free-nuclear endosperm** (thousands of nuclei), and the surrounding **white kernel** is the **cellular endosperm.**

The endosperm may be **completely consumed** by the developing embryo before seed maturation (pea, groundnut, beans) or may **persist** in the mature seed (castor, coconut) and be used up during germination.

B. Embryo

The embryo develops at the **micropylar end** of the embryo sac where the zygote is situated. Most zygotes divide only after a certain amount of endosperm is formed: an adaptation to provide assured nutrition to the developing embryo. PYQ

Zygote → Proembryo → Globular → Heart-shaped → Mature embryo

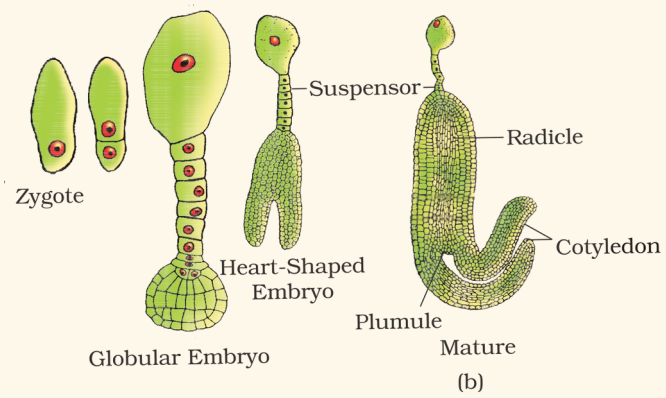


Fig. 1.13 (b) Stages in embryo development in a dicot

The early stages of embryo development (**embryogeny**) are similar in both monocots and dicots.

Dicot embryo	Monocot (grass) embryo
an embryonal axis and **two cotyledons**	only **one cotyledon**, called scutellum, situated lateral to the axis
epicotyl: part of the axis above the cotyledons; ends in **plumule** (stem tip)	**coleoptile**: hollow foliar structure enclosing the shoot apex and leaf primordia
hypocotyl: part below the cotyledons; ends in **radicle** (root tip)	**coleorrhiza**: undifferentiated sheath covering the radicle and root cap
root tip covered with a root cap	epiblast present in some grasses

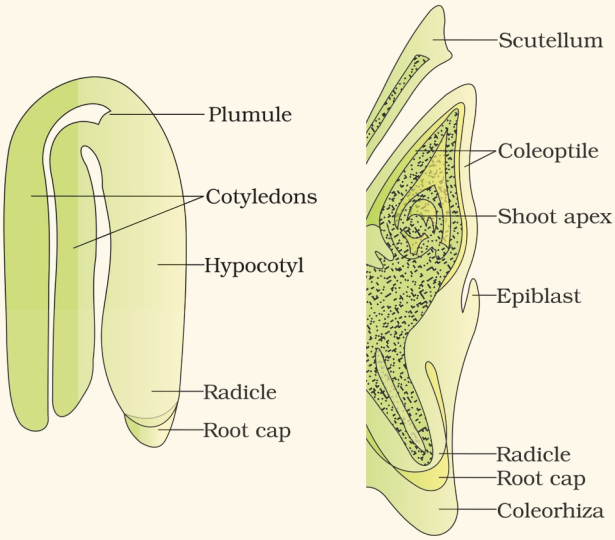


Fig. 1.14 (a) A typical dicot embryo; (b) L.S. of an embryo of grass

Yaad karo: epi upar, hypo neeche

Epicotyl = cotyledon ke **upar** (plumule tak); **Hypo**cotyl = **neeche** (radicle tak). Monocot mein: **coleo**ptile = plumule ka cover, **coleo**rrhiza = radicle ka cover.

C. Seed

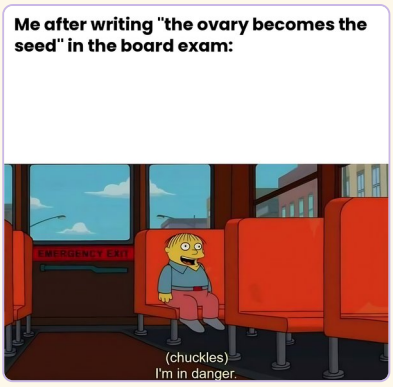
In angiosperms the seed is the final product of sexual reproduction, often described as a **fertilised ovule**. Seeds form inside fruits. A seed typically consists of **seed coat(s), cotyledon(s) and an embryo axis.**

Term	Meaning
Non-albuminous (ex-albuminous)	no residual endosperm, fully consumed: pea, groundnut
Albuminous	part of the endosperm retained: wheat, maize, barley, castor
Perisperm	residual, persistent **nucellus**: black pepper, beet
Testa and tegmen	integuments of the ovule harden into tough protective **seed coats**
Micropyle	remains as a small pore in the seed coat; lets in **oxygen and water** during germination

As the seed matures, its **water content is reduced**: seeds become relatively dry (10–15% moisture by mass). General metabolic activity slows, and the embryo may enter a state of inactivity called **dormancy**. Given favourable conditions (adequate moisture, oxygen and suitable temperature), they **germinate**.

D. Fruit

As ovules mature into seeds, the **ovary develops into a fruit**: the transformation of ovules into seeds and ovary into fruit proceeds simultaneously. The **wall of the ovary** develops into the wall of the fruit, called pericarp, which may be fleshy (guava, orange, mango) or dry (groundnut, mustard).



Board exam mein ek galti aur pura answer gaya.
Asli baat: ***Ovule*** becomes the ***seed***; ***ovary*** becomes the ***fruit***. Ulta mat likhna!

True fruit	False fruit
develops only from the **ovary**	the **thalamus** also contributes to fruit formation
other floral parts degenerate and fall off	e.g. **apple, strawberry, cashew**
most plants, e.g. mango, tomato	in apple, the fleshy edible part is the thalamus

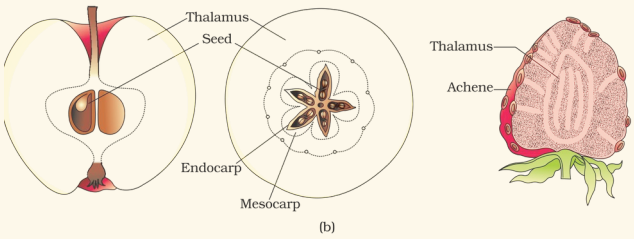


Fig. 1.15 (b) False fruits of apple and strawberry

Parthenocarpic fruits: fruits developed without fertilisation; they are **seedless**, e.g. **banana**. Parthenocarpy can be induced by application of **growth hormones**. PYQ

E. Why seeds matter

- → Seeds have better **adaptive strategies** for dispersal to new habitats and help the species colonise other areas.
- → They have food reserves, so young seedlings are nourished until they photosynthesise; the hard seed coat protects the young embryo.
- → Being products of sexual reproduction, they generate **new genetic combinations** leading to variations.
- → Old seeds: *Lupinus arcticus* seed excavated from Arctic Tundra germinated after an estimated record of **10,000 years** of dormancy; a **2000 years** old viable seed of date palm, *Phoenix dactylifera*, was found near the Dead Sea.
- → Seeds are the basis of **agriculture**: dehydration and dormancy of mature seeds are crucial for storage, so seeds can be used as food throughout the year and to raise crops in the next season.
- → Some orchid fruits contain thousands of tiny seeds; similar in parasitic species like *Orobanche* and *Striga*.

Not sure if apple ek fruit hai... ya **thalamus** ne dhokha diya hai.

Asli baat: Apple, strawberry, cashew = false fruits: fleshy part thalamus se banta hai, sirf ovary se nahi.

Key terms: what becomes what after fertilisation

Before	After
ovule	seed
ovary	fruit
ovary wall	pericarp
outer / inner integument	testa / tegmen (seed coats)
zygote (2n)	embryo
PEN (3n)	endosperm
micropyle	micropyle of seed (pore)
nucellus	consumed, or persists as perisperm
synergids, antipodals	degenerate

6. Apomixis and Polyembryony

Apomixis: some species, such as certain members of **Asteraceae and grasses**, have evolved a special mechanism to produce seeds without fertilisation. Apomixis is a form of **asexual reproduction that mimics sexual reproduction.** Board fav

→ In some species, the **diploid egg cell** is formed without reduction division and develops into the embryo without fertilisation.

→ More often, as in many **Citrus and mango** varieties, some **nucellar cells** surrounding the embryo sac start dividing, protrude into the embryo sac and develop into embryos.

Polyembryony: occurrence of more than one embryo in a seed, as in **Citrus** (orange) and mango, due to nucellar embryos.

Key terms

→ **Hybrid seeds** must be produced every year: if seeds of hybrids are collected and sown, the progeny **segregate** and lose hybrid characters, which makes hybrid seed costly for farmers.

→ If hybrids are made into **apomicts**, there is no segregation of characters in the hybrid progeny; farmers can keep using the hybrid seeds year after year.

→ Hence the active research the world over to understand the **genetics of apomixis** and to transfer apomictic genes into hybrid varieties.

Flower to seed: the full journey

PMC (2n) → meiosis → pollen (n) → pollination → pollen tube → 2 male gametes → double fertilisation → zygote + PEN → embryo + endosperm → seed → fruit

1 minute recap

1. Anther: 4 microsporangia; wall = epidermis, endothecium, middle layers, tapetum (nourishes).
2. PMC (2n) undergoes meiosis to form microspore tetrad; pollen: exine (sporopollenin) + intine; 2-celled (60%) or 3-celled.
3. Ovule: funicle, hilum, integuments, micropyle, chalaza, nucellus, embryo sac; MMC gives 1 functional megaspore (monosporic).
4. Embryo sac: 7-celled, 8-nucleate; filiform apparatus guides the pollen tube.
5. Autogamy, geitonogamy, xenogamy; cleistogamous flowers give assured seed-set; outbreeding devices prevent inbreeding depression.
6. Double fertilisation = syngamy (2n zygote) + triple fusion (3n PEN).
7. Endosperm before embryo; ovule becomes seed, ovary becomes fruit; apple is a false fruit; banana parthenocarpic.
8. Apomixis = seeds without fertilisation; polyembryony in Citrus and mango.

MOST IMPORTANT QUESTIONS

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM Board fav The innermost wall layer of the microsporangium that nourishes the developing pollen grains is:
(a) Endothecium
(b) Middle layer
(c) Tapetum
(d) Epidermis

Ans: (c) Tapetum

Q2. IM The exine of a pollen grain is made of:
(a) Cellulose
(b) Sporopollenin
(c) Pectin
(d) Lignin

Ans: (b) Sporopollenin

Q3. IM Board fav A mature embryo sac of a typical angiosperm is:
(a) 8-celled, 8-nucleate
(b) 8-celled, 7-nucleate
(c) 7-celled, 7-nucleate
(d) 7-celled, 8-nucleate

Ans: (d) 7-celled and 8-nucleate

Q4. 1M In which of the following is the endosperm completely consumed before the seed matures?
(a) Pea
(b) Castor
(c) Wheat
(d) Coconut
Ans: (a) Pea

Q5. 1M Board fav The ploidy of nucellus, primary endosperm nucleus and egg cell respectively is:
(a) n, 3n, 2n
(b) 2n, 3n, n
(c) 2n, 2n, n
(d) 3n, 2n, n
Ans: (b) 2n, 3n, n

Q6. 1M Geitonogamy is genetically similar to:
(a) Autogamy
(b) Xenogamy
(c) Apomixis
(d) Cleistogamy
Ans: (a) Autogamy

Q7. 1M The coconut water of tender coconut and the white kernel are respectively:
(a) both cellular endosperm
(b) cellular and free-nuclear endosperm
(c) free-nuclear and cellular endosperm
(d) perisperm and endosperm
Ans: (c) free-nuclear endosperm and cellular endosperm

Q8. 1M Residual persistent nucellus in the seed of black pepper and beet is called:
(a) Pericarp
(b) Endosperm
(c) Testa
(d) Perisperm
Ans: (d) Perisperm

Q9. 1M Board fav Name the structure in the synergids that guides the pollen tube.
Ans: Filiform apparatus.

Q10. 1M Name the part of the flower that contributes to fruit formation in apple and strawberry along with the ovary.
Ans: Thalamus (so they are false fruits).

Q11. 1M Which plant produces both chasmogamous and cleistogamous flowers?
Ans: *Viola* (common pansy), *Oxalis* and *Commelina*.

Assertion–Reason (1 mark each)

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q12. 1M Board fav **A:** Pollen grains are well preserved as fossils. **R:** Exine is made of sporopollenin, which is not degraded by any known enzyme.
Ans: (a) Both true and R explains A.

Q13. 1M **A:** Cleistogamous flowers produce assured seed-set. **R:** Cleistogamous flowers are invariably autogamous.
Ans: (a) Both true and R explains A.

Q14. 1M **A:** Apple is called a false fruit. **R:** In apple, the fleshy edible part develops from the ovary wall.
Ans: (c) A is true but R is false: the fleshy part develops from the thalamus.

Q15. 1M **A:** Emasculation is needed when the female parent bears unisexual flowers. **R:** Emasculation removes anthers before dehiscence.
Ans: (d) A is false (emasculation is needed for bisexual flowers), R is true.

Short answer (2–3 marks)

Q16. 3M Board fav Differentiate between microsporogenesis and megasporogenesis. Which type of cell division occurs in both? Name the structures formed at the end.
Ans: Microsporogenesis: formation of microspores from a pollen mother cell in the anther; ends in a **microspore tetrad**. Megasporogenesis: formation of megaspores from the megaspore mother cell in the nucellus of the ovule; ends in **four megaspores** (one functional). Both involve **meiosis**.

Q17. 2M Arrange in correct developmental sequence: pollen grain, sporogenous tissue, microspore tetrad, pollen mother cell, male gametes.
Ans: Sporogenous tissue, pollen mother cell, microspore tetrad, pollen grain, male gametes.

Q18. 2M What is meant by monosporic development of the female gametophyte?
Ans: Development of the embryo sac (female gametophyte) from a **single functional megaspore**, the other three megaspores of the tetrad degenerating.

Q19. 2M Why is the tapetum important? Mention two features of its cells.
Ans: The tapetum **nourishes** the developing pollen grains. Its cells have dense cytoplasm and generally more than one nucleus.

Q20. 2M Board fav Mention two strategies evolved to prevent self-pollination in flowers.
Ans: (i) Pollen release and stigma receptivity are not synchronised. (ii) Anther and stigma are at different positions. (Also: self-incompatibility; unisexual flowers.)

Q21. 3M What is self-incompatibility? Why does self-pollination not lead to seed formation in self-incompatible species?
Ans: A genetic mechanism that prevents self-pollen from fertilising the ovules. The pistil inhibits **pollen germination or pollen tube growth**, so the male gametes never reach the egg and no seed forms.

Q22. 2M Board fav What is bagging technique? How is it useful in a plant breeding programme?
Ans: Covering the emasculated flower with a bag (usually butter paper) to prevent contamination of its stigma with unwanted pollen. It ensures that only the **desired pollen** is used, so the intended cross is obtained.

Q23. 3M Board fav What is triple fusion? Where and how does it take place? Name the nuclei involved.
Ans: Fusion of the second male gamete with the **two polar nuclei** in the central cell of the embryo sac, forming the triploid primary endosperm nucleus. Nuclei involved: one male gamete nucleus and two polar nuclei.

Q24. 2M Why do you think the zygote is dormant for some time in a fertilised ovule?
Ans: The zygote divides only after some **endosperm** is formed. This adaptation provides assured nutrition to the developing embryo.

Q25. 3M Board fav Differentiate between (a) hypocotyl and epicotyl, (b) coleoptile and coleorrhiza.
Ans: (a) Epicotyl: part of the embryonal axis above the cotyledons, ends in the plumule; hypocotyl: part below the cotyledons, ends in the radicle. (b) Coleoptile: hollow foliar sheath enclosing the shoot apex and leaf primordia; coleorrhiza: sheath enclosing the radicle and root cap.

Q26. 3M Differentiate between (a) integument and testa, (b) perisperm and pericarp.
Ans: (a) Integument: protective envelope of the ovule; testa: outer seed coat formed from the integument after fertilisation. (b) Perisperm: residual persistent nucellus in some seeds (black pepper, beet); pericarp: wall of the fruit formed from the ovary wall.

Q27. 2M Why is apple called a false fruit? Which part(s) of the flower form the fruit?
Ans: In apple the **thalamus** also contributes to the fruit; the fleshy edible part is the thalamus. The fruit forms from the ovary along with the thalamus.

Q28. 3M What are chasmogamous flowers? Can cross-pollination occur in cleistogamous flowers? Give reasons.
Ans: Chasmogamous flowers are open flowers with exposed anthers and stigma. No, cross-pollination cannot occur in cleistogamous flowers because they **never open**; anthers and stigma lie close together, so they are autogamous.

Q29. 3M Board fav What is apomixis? What is its importance for farmers?
Ans: Formation of seeds **without fertilisation**. If hybrids are made apomictic, the hybrid characters do not segregate, so farmers can reuse hybrid seed every year instead of buying new hybrid seed.

Q30. 2M Name two features of wind-pollinated flowers and two of water-pollinated flowers.
Ans: Wind: light, non-sticky pollen; well-exposed stamens; large feathery stigma. Water: pollen protected by a mucilaginous covering; long ribbon-like pollen in sea grasses (*Zostera*).

Q31. 2M What is polyembryony? How do nucellar cells form embryos?
Ans: Occurrence of more than one embryo in a seed. In Citrus and mango, nucellar cells around the embryo sac divide, protrude into the embryo sac and develop into embryos.

Q32. 3M Why is the seed considered the final product of sexual reproduction? Give two advantages of seeds.
Ans: It is the fertilised ovule containing the embryo. Advantages: dormancy and hard seed coat help dispersal and survival; food reserves nourish the seedling; new genetic combinations give variation.

Q33. 2M Mention the fate of the following after fertilisation: (a) synergids, (b) integuments, (c) ovary wall, (d) nucellus.
Ans: (a) Degenerate. (b) Harden into seed coats (testa and tegmen). (c) Pericarp (fruit wall). (d) Mostly consumed; persists as perisperm in black pepper and beet.

Q34. 2M Board fav What are parthenocarpic fruits? Give an example and one way to induce parthenocarpy.
Ans: Fruits that develop without fertilisation and are seedless, e.g. banana. Induced by applying growth hormones.

Q35. 2M List two floral rewards offered to insects. Give one example where the plant and pollinator depend on each other completely.
Ans: Nectar and pollen grains (also safe places to lay eggs). Example: a species of moth and the plant *Yucca*.

Q36. 2M Why do wind-pollinated plants produce a large number of pollen grains, and why do their flowers lack nectar and colour?
Ans: Wind pollination depends on chance, so enormous pollen improves the chance of reaching a stigma. They do not need to attract animals, so colour, fragrance and nectar are absent.

Q37. 2M Explain how pollen grains can be stored for long periods. Why is this useful?
Ans: In liquid nitrogen at −196 °C in pollen banks. Stored pollen can be used later in crop breeding programmes.

Q38. 2M Pollen grains of rice lose viability within 30 minutes of release. What does this mean for pollination in rice?
Ans: Pollen must land on a receptive stigma within about 30 minutes; after that it cannot germinate, so timing of pollen release and stigma receptivity matters.

Q39. 2M Name the cell of the pollen grain that forms (a) the pollen tube and (b) the male gametes. Which division forms the male gametes?
Ans: (a) Vegetative cell. (b) Generative cell, which divides by mitosis to form two male gametes.

Q40. 2M Board fav A flower of a dioecious plant cannot undergo autogamy or geitonogamy. Explain. Give an example.
Ans: Male and female flowers are on different plants, so pollen of the same flower or the same plant can never reach the stigma; only xenogamy is possible. Example: papaya.

Q41. 2M Board fav Name the parts of an angiosperm flower in which the male and the female gametophytes develop.
Ans: Male gametophyte (pollen grain): in the microsporangium (pollen sac) of the anther. Female gametophyte (embryo sac): in the nucellus of the ovule, inside the ovary.

Q42. 2M An anther with a malfunctioning tapetum often fails to produce viable male gametophytes. Give one reason.
Ans: The tapetum nourishes the developing pollen grains; without this nourishment the microspores cannot develop into viable pollen.

Q43. 2M If you could induce parthenocarpy with hormones, which kind of fruit would you choose and why?
Ans: Fruits eaten for their fleshy part whose seeds are a nuisance, e.g. orange, lemon, watermelon or guava: seedless fruits are preferred by consumers.

Q44. 2M Why does the embryo develop at the micropylar end of the embryo sac?
Ans: The zygote is formed at the micropylar end (the egg is part of the egg apparatus there, where the pollen tube enters), so the embryo develops at that end.

Q45. 2M Give the ploidy of a synergid and of an endosperm cell. Why is the endosperm triploid?
Ans: Synergid: n. Endosperm: 3n, because the PEN forms by fusion of one male gamete (n) with two polar nuclei (n + n).

Q46. 2M Which wall layer of the anther helps in dehiscence? How many microsporangia are present in a typical anther?
Ans: Endothecium (with the middle layers and epidermis) helps in dehiscence. A typical anther has four microsporangia, two in each lobe.

Q47. 2M *Board fav* What is the function of the filiform apparatus? Where is it present?
Ans: It is a group of special cellular thickenings at the micropylar tip of the synergids; it guides the entry of the pollen tube into the synergid.

Q48. 3M Define pollen-pistil interaction. Why is it called a dynamic process?
Ans: All events from pollen deposition on the stigma until the pollen tube enters the ovule. It is dynamic because the pistil recognises the pollen and then either promotes it (compatible) or inhibits it (incompatible), through chemical dialogue between pollen and pistil.

Q49. 2M Seeds are dry at maturity. How does this help? What conditions are needed to end dormancy and start germination?
Ans: Water content falls to 10–15%, metabolism slows and the embryo becomes dormant, so the seed can survive storage and unfavourable periods. Germination needs adequate moisture, oxygen and a suitable temperature.

Q50. 2M In most angiosperms pollen is shed at the 2-celled stage. How and where are the two male gametes formed in such plants?
Ans: This is true for over 60% of angiosperms. The generative cell divides mitotically into two male gametes; this happens inside the pollen tube as it grows through the style.

Q51. 2M Why is a monoecious plant like maize able to prevent autogamy but not geitonogamy?
Ans: Its flowers are unisexual, so pollen cannot reach the stigma of the same flower (no autogamy). But male and female flowers are on the same plant, so pollen of one flower can reach another flower of the same plant (geitonogamy).

Q52. 2M State two differences between an albuminous and a non-albuminous seed with one example each.
Ans: Albuminous: part of the endosperm persists in the mature seed (wheat, maize, castor). Non-albuminous: endosperm is completely used up by the developing embryo (pea, groundnut).

Q53. 3M *Board fav* Draw a labelled diagram of a mature dicot embryo. Name the parts that develop into the shoot and the root.
Ans: Diagram as Fig. 1.14(a): cotyledons, plumule, epicotyl, hypocotyl, radicle, root cap. Plumule (at the end of the epicotyl) gives the shoot; radicle (at the end of the hypocotyl) gives the root.

Q54. 2M Some flowers are large, colourful and fragrant while others are small and dull. Relate each type to its pollinating agent with one reason.
Ans: Large, colourful, fragrant, nectar-rich flowers are animal (insect) pollinated: these features attract pollinators. Small, dull flowers without nectar are wind or water pollinated: they need not attract animals and instead produce large amounts of pollen.

Q55. 2M *Board fav* Differentiate between geitonogamy and xenogamy. Which of the two brings genetically different pollen to the stigma?
Ans: Geitonogamy: transfer of pollen from the anther to the stigma of another flower of the **same plant**; functionally cross-pollination (needs an agent) but genetically similar to autogamy. Xenogamy: transfer of pollen to the stigma of a flower on a **different plant**. Only xenogamy brings genetically different pollen.

Q56. 3M Describe the embryo of a grass. What is the scutellum?
Ans: A grass (monocot) embryo has **one cotyledon**, the **scutellum**, situated towards one side (lateral) of the embryonal axis. At its lower end the axis has the radicle and root cap enclosed in a sheath, the **coleorrhiza**. The part above the scutellum is the epicotyl, with the shoot apex and leaf primordia enclosed in a hollow foliar structure, the **coleoptile**.

Q57. 2M What is the micropyle of a seed? State its function.
Ans: In a mature seed the micropyle remains as a **small pore in the seed coat**. It facilitates the entry of **oxygen and water** into the seed during germination.

Q58. 3M Board fav How does endosperm development in coconut show both types of endosperm? Why does endosperm development precede embryo development?

Ans: The PEN undergoes successive nuclear divisions without wall formation, giving **free-nuclear endosperm** (the tender coconut water); cell walls then form, giving **cellular endosperm** (the white kernel). Endosperm develops first because the developing embryo needs its stored food for nourishment.

Long answer (5 marks)

Q59. 5M Board fav (a) With a neat labelled diagram, describe the parts of a typical anatropous ovule. (b) Describe the development of a 7-celled, 8-nucleate embryo sac.

Ans: (a) Diagram as Fig. 1.7(d). Funicle attaches the ovule to the placenta; hilum is the junction; one or two integuments; micropyle at the tip; chalaza at the base; nucellus with reserve food; embryo sac inside the nucellus. (b) Functional megaspore nucleus divides mitotically (free nuclear) into 2, then 4, then 8 nuclei, four at each pole. Walls are laid down: 3 cells at the micropylar end form the egg apparatus (2 synergids + egg), 3 at the chalazal end form antipodals; one nucleus from each pole moves to the centre as polar nuclei in the large central cell. Hence 7 cells and 8 nuclei.

Q60. 5M (a) Draw and label the T.S. of a mature microsporangium. Write the functions of its wall layers. (b) Describe the structure of a mature pollen grain.

Ans: (a) Diagram as Fig. 1.3(b). Epidermis, endothecium and middle layers protect and help in dehiscence; tapetum nourishes developing pollen. (b) Spherical, 25–50 micrometres; exine of sporopollenin with germ pores; intine of cellulose and pectin. Contains a large vegetative cell (food reserve, irregular nucleus) and a small spindle-shaped generative cell that forms two male gametes.

Q61. 5M Board fav (a) What is double fertilisation? Describe the events from the entry of the pollen tube into the embryo sac to the formation of the zygote and PEN. (b) Write the fate of each.

Ans: (a) Two fusions in one embryo sac. The pollen tube enters a synergid through the filiform apparatus and releases two male gametes. One fuses with the egg (syngamy) forming the diploid zygote; the other fuses with the two polar nuclei (triple fusion) forming the triploid PEN; the central cell becomes the primary endosperm cell. (b) Zygote develops into the embryo (proembryo, globular, heart-shaped, mature). PEC forms the endosperm that nourishes the embryo.

Q62. 5M (a) Explain the different devices that encourage cross-pollination. (b) Describe the steps of artificial hybridisation.

Ans: (a) Non-synchronised pollen release and stigma receptivity; different positions of anther and stigma; self-incompatibility; unisexual flowers (monoecious prevents autogamy, dioecious prevents autogamy and geitonogamy). (b) Emasculation of the bisexual female parent before anther dehiscence; bagging with butter paper; when stigma is receptive, dusting desired pollen; rebagging till fruit develops. Unisexual female flowers are only bagged.

Case-based (4 marks)

Q63. 4M Board fav A plant breeder wants to cross a tall wheat variety (female) with a disease-resistant variety (male). Wheat has bisexual flowers. (a) What must be done to the female flower first, and when? (b) Why is the flower covered with butter paper? (c) When is pollen dusted on it? (d) Why is this needed every year for hybrid seed?

Ans: (a) Emasculation: anthers removed before they dehisce. (b) Bagging prevents contamination of the stigma by unwanted pollen. (c) When the stigma becomes receptive. (d) Seeds of hybrids segregate in the next generation and lose hybrid characters.

Q64. 4M A student observed a tender coconut: watery fluid inside, soft white kernel around it. Her teacher said both are the same tissue at two stages. (a) Name the tissue and its ploidy. (b) Which is free-nuclear and which cellular? (c) Name a seed in which this tissue is fully used up before maturity. (d) Which develops first: this tissue or the embryo?

Ans: (a) Endosperm, triploid (3n). (b) Coconut water is free-nuclear; the white kernel is cellular. (c) Pea / groundnut / beans. (d) Endosperm develops first, to nourish the embryo.

Q65. 4M Board fav In some Citrus varieties, a single seed gives rise to several seedlings, and some grasses form seeds without fertilisation. (a) Name both phenomena. (b) Give the origin of the extra embryos in Citrus. (c) Are such embryos genetically identical to the parent? (d) One benefit of transferring this trait to hybrids.

Ans: (a) Polyembryony and apomixis. (b) Nucellar cells surrounding the embryo sac. (c) Yes, nucellar embryos are diploid copies of the mother plant (no fertilisation). (d) Hybrid characters do not segregate, so farmers reuse hybrid seeds.

Q66. 4M In *Vallisneria*, the female flower reaches the water surface on a long stalk, while in sea grasses like *Zostera* the female flowers stay submerged. (a) Name this type of pollination. (b) How do male flowers or pollen reach the female flower in each? (c) How is the pollen protected? (d) Are water hyacinth and water lily pollinated this way?

Ans: (a) Pollination by water (hydrophily). (b) *Vallisneria*: male flowers or pollen are released on the surface and carried passively by water currents to the female flowers. *Zostera*: long ribbon-like pollen grains are released inside the water. (c) By a mucilaginous covering that prevents wetting. (d) No. Their flowers emerge above the water and are pollinated by insects or by wind.

Exam Alert: diagrams to practise

- → Fig. 1.3(b) T.S. of microsporangium (4 wall layers), Fig. 1.7(d) anatropous ovule, Fig. 1.8(c) mature embryo sac: ye teen sabse zyada aate hain.
- → Labels seedhi line se, right side mein; embryo sac mein micropylar end neeche/upar clearly likho.
- → Double fertilisation ke 2 fusions + products (2n zygote, 3n PEN): 3 marks pakke.
- → Differentiate wale pairs: epicotyl/hypocotyl, coleoptile/coleorrhiza, perisperm/pericarp, integument/testa.

HUMAN REPRODUCTION

Most asked: spermatogenesis and oogenesis, menstrual cycle and its hormones, fertilisation and implantation, placenta and its hormones, parturition

Humans are **sexually reproducing and viviparous**. This chapter covers the **reproductive systems, gametogenesis, menstrual cycle, fertilisation, pregnancy, parturition and lactation**.

The reproductive events in humans occur in a fixed sequence:

Gametogenesis → Insemination → Fertilisation → Cleavage and blastocyst → Implantation → Gestation → Parturition

- → **Gametogenesis**: formation of gametes, **sperms** in males and **ovum** in females.
- → **Insemination**: transfer of sperms into the female genital tract.
- → **Fertilisation**: fusion of male and female gametes to form a **zygote**.
- → **Implantation**: attachment of the **blastocyst** to the uterine wall.
- → **Gestation** = embryonic development; **parturition** = delivery of the baby.

The reproductive events occur after puberty. There are remarkable differences between males and females: sperm formation continues even in **old men**, but formation of ova ceases in women around the age of **fifty years**.

1. The Male Reproductive System

It is located in the **pelvis region** and includes a pair of testes, along with **accessory ducts, glands** and the **external genitalia**.

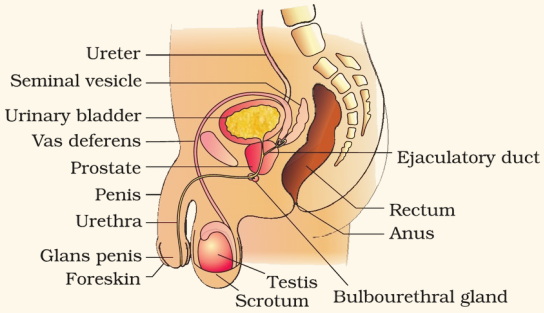


Fig. 2.1 (a) Diagrammatic sectional view of male pelvis showing reproductive system

A. Testes

- → Situated **outside the abdominal cavity** within a pouch called the scrotum.
- → The scrotum keeps the testes at a temperature **2-2.5 °C lower** than normal internal body temperature, which is necessary for spermatogenesis. PYQ
- → In adults, each testis is **oval**, about **4-5 cm long** and **2-3 cm wide**, covered by a dense covering.
- → Each testis has about 250 compartments called **testicular lobules**; each lobule has **one to three** highly coiled seminiferous tubules in which sperms are produced.

B. Seminiferous tubule

Each seminiferous tubule is lined on its inside by two types of cells:

Cell	Nature and function
Spermatogonia (male germ cells)	undergo **meiotic divisions** finally leading to sperm formation
Sertoli cells	provide **nutrition** to the germ cells

The regions outside the seminiferous tubules are called **interstitial spaces**. They contain small blood vessels and interstitial cells or Leydig cells, which **synthesise and secrete androgens** (testicular hormones). Other **immunologically competent cells** are also present. PYQ

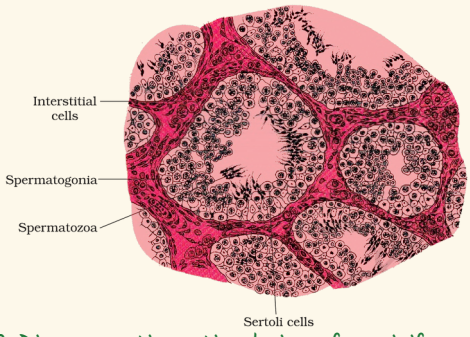


Fig. 2.2 Diagrammatic sectional view of seminiferous tubule

Exam Alert

- → **Sertoli** = nutrition (nurse cells); **Leydig** = androgens (testosterone). Ye dono swap karna sabse common galti hai.
- → Leydig cells seminiferous tubule ke **bahar** (interstitial space) hote hain, andar nahi.

C. Male accessory ducts

The male sex accessory ducts are the **rete testis, vasa efferentia, epididymis and vas deferens**. The seminiferous tubules open into the vasa efferentia through the rete testis.

- → Vasa efferentia leave the testis and open into the **epididymis**, located along the posterior surface of each testis.
- → The epididymis leads to the **vas deferens**, which ascends to the abdomen and loops over the urinary bladder.
- → It receives a duct from the **seminal vesicle** and opens into the urethra as the ejaculatory duct.
- → These ducts **store and transport** the sperms from the testis to the outside through the urethra.

→ The **urethra** originates from the urinary bladder, extends through the penis and opens outside at the urethral meatus.

Yaad karo: sperm ka raasta = SEVEN UP

Seminiferous tubules, (rete testis, vasa efferentia), **E**pididymis, **V**as deferens, **E**jaculatory duct, **N**othing, **U**rethra, **P**enis. Seven Up yaad, toh path yaad!

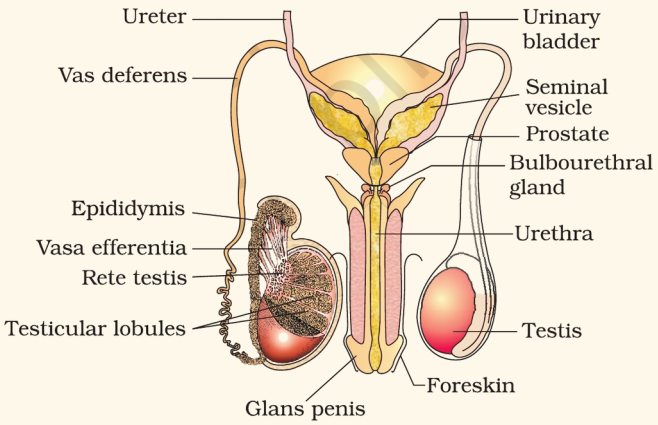


Fig. 2.1 (b) Diagrammatic view of male reproductive system (part of testis is open to show inner details)

D. External genitalia: penis

The **penis** is the male external genitalia. It is made up of special tissue that helps in **erection** of the penis to facilitate insemination. Its enlarged end is the **glans penis**, covered by a loose fold of skin called the **foreskin**.

Key terms

- → **Rete testis**: network of tubules that receives sperms from seminiferous tubules.
- → **Vasa efferentia**: ducts that carry sperms from rete testis to the epididymis.
- → **Epididymis**: coiled duct on the posterior surface of the testis; stores sperms and helps them mature.
- → **Urethral meatus**: external opening of the urethra.

E. Male accessory glands

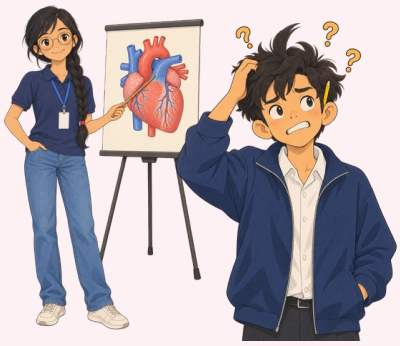
Gland	Number	Secretion / role
Seminal vesicles	paired	part of **seminal plasma**
Prostate	single	part of seminal plasma
Bulbourethral glands	paired	secretion helps in **lubrication of the penis**

Seminal plasma is rich in fructose, calcium and certain enzymes. PYQ Fructose gives energy to the sperms.

Quick fact

The scrotum hangs outside the body only because the testes need a cooler temperature. If the testes fail to descend into the scrotum, sperm formation does not take place normally, which causes **sterility**.

Paheli-Bujho



Bujho: Testes bahar scrotum mein kyun latakte hain? Ajeeb design hai!

Paheli: Sperms ko body se thoda thanda ghar chahiye, 2-2.5 degree kam.

Bujho: Matlab scrotum sperms ka AC room hai!

Paheli: Bilkul! Isliye testes andar reh jaayein toh sperm nahi bante.

2. The Female Reproductive System

It consists of a pair of ovaries along with a pair of **oviducts, uterus, cervix, vagina** and the **external genitalia**, located in the pelvic region. These parts, along with a pair of **mammary glands**, are integrated structurally and functionally to support ovulation, fertilisation, pregnancy, birth and child care.

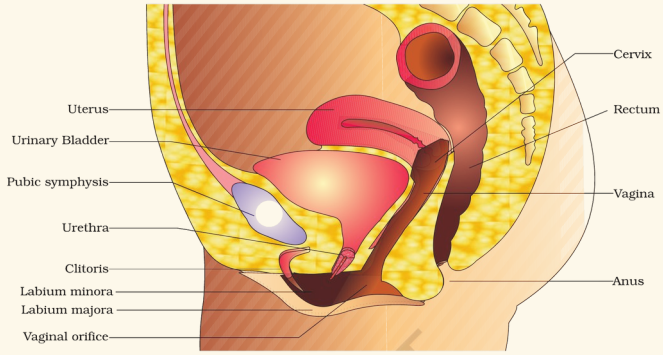


Fig. 2.3 (a) Diagrammatic sectional view of female pelvis showing reproductive system

A. Ovaries

- → The **primary female sex organs**: they produce the female gamete (**ovum**) and several steroid hormones (**ovarian hormones**).
- → Located one on each side of the **lower abdomen**; each ovary is about 2 to 4 cm long.
- → Connected to the pelvic wall and uterus by **ligaments**.
- → Each ovary is covered by a thin **epithelium** enclosing the **ovarian stroma**, which has two zones: a peripheral cortex and an inner medulla.

B. Oviducts (fallopian tubes)

Each fallopian tube is about 10-12 cm long and extends from the periphery of each ovary to the uterus. It has three parts:

Part	Features
Infundibulum	funnel-shaped part closer to the ovary; edges have finger-like projections called fimbriae, which help collect the ovum after ovulation
Ampulla	wider part; site of **fertilisation** PYQ
Isthmus	last part, with a **narrow lumen**; joins the uterus

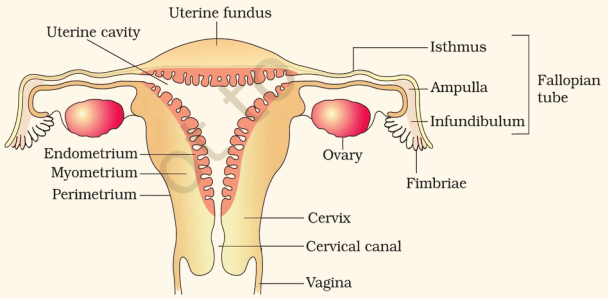


Fig. 2.3 (b) Diagrammatic sectional view of the female reproductive system

C. Uterus

→ Single; also called the **womb**; shape like an inverted pear; supported by ligaments attached to the pelvic wall.

→ The uterus opens into the vagina through a narrow **cervix**. The cavity of the cervix is the **cervical canal**, which along with the vagina forms the birth canal.

The wall of the uterus has **three layers** of tissue:

Layer	Features
Perimetrium	external thin membranous layer
Myometrium	middle thick layer of **smooth muscle**; shows strong contractions during **delivery** of the baby
Endometrium	inner glandular layer lining the uterine cavity; undergoes cyclical changes during the menstrual cycle

Yaad karo: uterus ki deewar (bahar se andar)

Papa Maange Endless chai = Perimetrium, Myometrium (muscle = delivery push), Endometrium (har mahine badalti hai).

D. External genitalia

→ **Mons pubis**: cushion of fatty tissue covered by skin and pubic hair.

→ **Labia majora**: fleshy folds of tissue that extend down from the mons pubis and surround the vaginal opening.

→ **Labia minora**: paired folds of tissue under the labia majora.

→ **Hymen**: a membrane that often partially covers the opening of the vagina.

→ **Clitoris**: a tiny finger-like structure at the upper junction of the two labia minora, above the urethral opening.

Quick fact: hymen

The hymen is often torn during the first coitus, but it can also break by a sudden fall or jolt, insertion of a vaginal tampon, or active sports like horseback riding and cycling. In some women it persists even after coitus. So the presence or absence of hymen is **not a reliable indicator** of virginity or sexual experience.

E. Mammary glands

A functional mammary gland is characteristic of all **female mammals**. The glands are paired (breasts) and contain **glandular tissue** and a variable amount of **fat**. The glandular tissue of each breast is divided into 15–20 mammary lobes containing clusters of cells called **alveoli**. The cells of alveoli secrete milk, which is stored in the cavities (lumens) of the alveoli.

Alveoli → Mammary tubules → Mammary duct → Mammary ampulla → Lactiferous duct → Milk sucked out

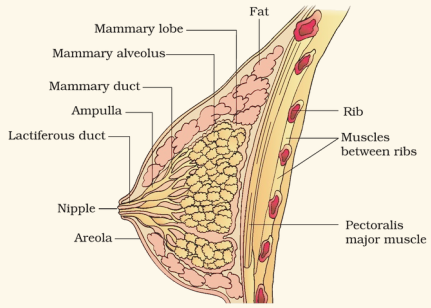


Fig. 2.4 A diagrammatic sectional view of mammary gland

3. Gametogenesis

The primary sex organs, the **testis** in males and the **ovaries** in females, produce gametes (sperms and ovum) by the process called gametogenesis.

A. Spermatogenesis

In the testis, the immature male germ cells (**spermatogonia**) produce sperms by spermatogenesis, which **begins at puberty**.

→ Spermatogonia (sing. spermatogonium) on the inside wall of seminiferous tubules multiply by **mitotic division**. Each is **diploid** with **46 chromosomes**.

→ Some spermatogonia, called primary spermatocytes, periodically undergo **meiosis**.

→ A primary spermatocyte completes the first meiotic division (**reduction division**), forming two equal, haploid **secondary spermatocytes** with 23 chromosomes each.

→ The secondary spermatocytes undergo the second meiotic division to produce four equal, haploid spermatids.

Spermatogonium (2n) → mitosis → Primary spermatocyte (2n) → meiosis I → 2 secondary spermatocytes (n) → meiosis II → 4 spermatids (n) → spermiogenesis → 4 sperms (n)

Spermiogenesis: spermatids are transformed into spermatozoa (sperms). **Spermiation**: after spermiogenesis, sperm heads are embedded in the **Sertoli cells** and are finally released from the seminiferous tubules. PYQ

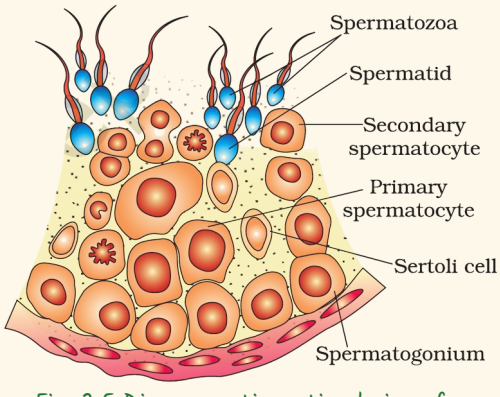


Fig. 2.5 Diagrammatic sectional view of a seminiferous tubule (enlarged)

B. Hormonal control of spermatogenesis

Spermatogenesis starts at puberty due to a significant increase in the secretion of gonadotropin releasing hormone (GnRH), a hypothalamic hormone.

Hypothalamus → GnRH → Anterior pituitary → LH + FSH

Hormone	Acts on	Effect
LH	Leydig cells	stimulates synthesis and secretion of **androgens**, which in turn stimulate spermatogenesis
FSH	Sertoli cells	stimulates secretion of some **factors** that help in spermiogenesis

C. Structure of a sperm

A sperm is a **microscopic** structure with a **head, neck, middle piece and tail.** A **plasma membrane** envelops the whole body of the sperm.

- → **Head**: an elongated **haploid nucleus**; its anterior portion is covered by a cap-like structure, the acrosome, filled with **enzymes** that help fertilisation of the ovum.
- → **Middle piece**: possesses numerous **mitochondria**, which produce energy for the movement of the tail.
- → **Tail**: facilitates **sperm motility**, essential for fertilisation.

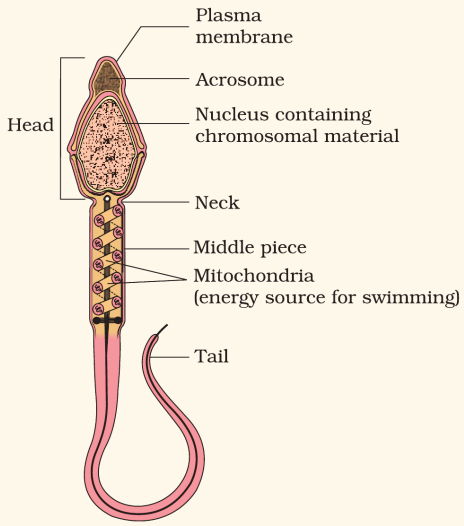


Fig. 2.6 Structure of a sperm

The human male ejaculates about 200 to 300 million sperms during a coitus. For normal fertility, at least **60%** of sperms must have **normal shape and size** and at least **40%** of them must show **vigorous motility.** PYQ

Sperms released from the seminiferous tubules are transported by the accessory ducts. Secretions of the epididymis, vas deferens, seminal vesicle and prostate are essential for **maturation and motility** of sperms. The seminal plasma along with the sperms constitutes the semen. The functions of male sex accessory ducts and glands are maintained by the testicular hormones (**androgens**).

D. Oogenesis

The process of formation of a mature female gamete is called oogenesis. It is markedly different from spermatogenesis.

- → Oogenesis is **initiated during embryonic development**, when a couple of million gamete mother cells (**oogonia**) are formed within each foetal ovary. No more oogonia are formed and added after birth.
- → Oogonia start division, enter **prophase I** of meiosis and get temporarily arrested at that stage. These are the **primary oocytes.**
- → Each primary oocyte gets surrounded by a layer of **granulosa cells**: this is the primary follicle.
- → A large number of these follicles **degenerate** from birth to puberty, so at puberty only 60,000-80,000 primary follicles are left in each ovary. PYQ
- → Primary follicles get surrounded by more layers of granulosa cells and a new **theca**: these are **secondary follicles.**
- → The secondary follicle soon becomes a **tertiary follicle**, with a fluid-filled cavity called the antrum. The theca layer is organised into an inner **theca interna** and an outer **theca externa.**

At this stage the primary oocyte within the tertiary follicle grows in size and **completes its first meiotic division.** It is an unequal division, forming a large haploid **secondary oocyte** and a tiny **first polar body.** The secondary oocyte retains the bulk of the **nutrient-rich cytoplasm** of the primary oocyte.

The tertiary follicle further changes into the mature follicle or Graafian follicle. The secondary oocyte forms a new membrane called the **zona pellucida** around it. The Graafian follicle now ruptures to release the secondary oocyte (ovum) from the ovary by the process called ovulation.

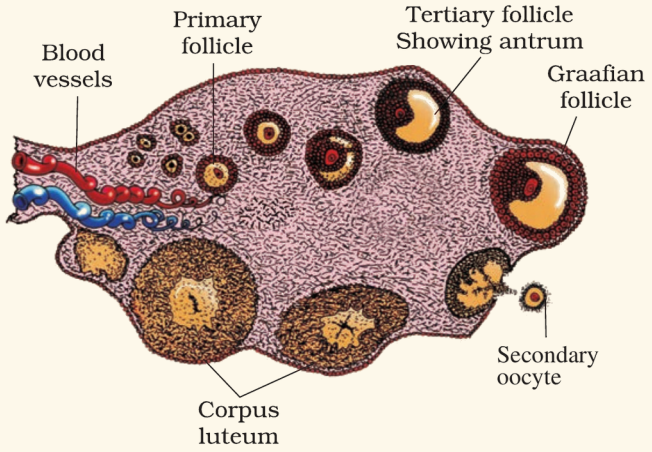


Fig. 2.7 Diagrammatic section view of ovary

Oogonium (2n) → mitosis (foetal life) → Primary oocyte (2n), arrested in prophase I → meiosis I (at puberty) → Secondary oocyte (n) + 1st polar body → ovulation → meiosis II only after sperm entry → Ovum (n) + 2nd polar body

Exam Alert

- → Ovulation mein **secondary oocyte** release hota hai, mature ovum nahi. Meiosis II sperm ke entry ke baad hi complete hota hai.
- → One oogonium = **1 ovum** (+ polar bodies); one spermatogonium (primary spermatocyte) = **4 sperms**.
- → Primary follicle: oocyte + 1 layer granulosa; tertiary follicle: **antrum** + theca interna/externa.

Spermatogenesis vs oogenesis Board fav

Spermatogenesis	Oogenesis
occurs in **testes** (seminiferous tubules)	occurs in **ovaries**
begins at **puberty**, continues till old age	begins in the **foetal ovary**; stops around 50 years (menopause)
meiotic divisions are **equal**	meiotic divisions are **unequal** (polar bodies)
1 primary spermatocyte gives **4 sperms**	1 primary oocyte gives **1 ovum**
continuous process, no arrest	arrested in prophase I; meiosis II completes only after sperm entry
spermiogenesis needed to form motile sperm	no such step; ovum is non-motile

Ploidy of cells in gametogenesis PYQ

Male line	Ploidy	Female line	Ploidy
Spermatogonium	2n (46)	Oogonium	2n (46)
Primary spermatocyte	2n	Primary oocyte	2n
Secondary spermatocyte	n (23)	Secondary oocyte, 1st polar body	n
Spermatid	n	Ootid (ovum), 2nd polar body	n
Sperm	n	Zygote (after fertilisation)	2n

Sperm vs ovum

Sperm	Ovum
small, microscopic, **motile** (tail)	large, **non-motile**
little cytoplasm, no food reserve	**nutrient-rich** cytoplasm
acrosome with enzymes	surrounded by **zona pellucida** and corona radiata cells
carries X or Y	always carries X

4. Menstrual Cycle

The reproductive cycle in female primates (monkeys, apes and human beings) is called the **menstrual cycle**. The first menstruation begins at puberty and is called menarche. In human females, menstruation is repeated at an average interval of about **28/29 days**; the cycle of events from one menstruation till the next is the menstrual cycle. **One ovum is released (ovulation) during the middle of each cycle.**

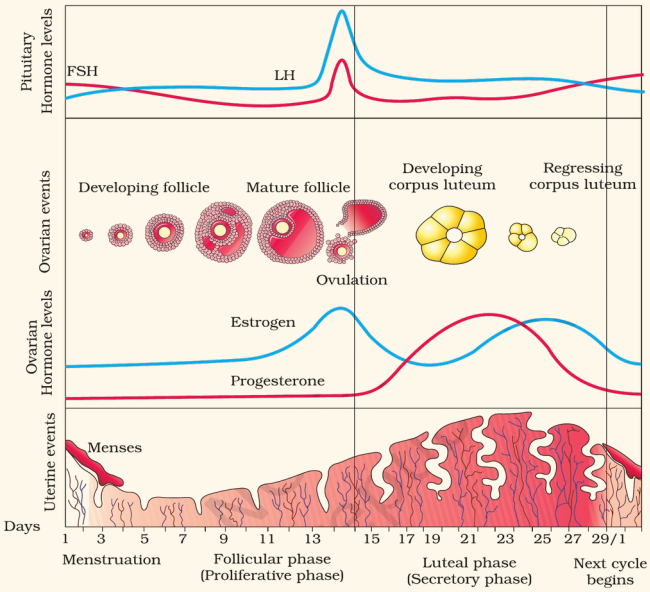


Fig. 2.9 Diagrammatic presentation of various events during a menstrual cycle

Phase (days)	Ovary	Uterus (endometrium)	Hormones
Menstrual (1-5)	no ovulation of the previous ovum; corpus luteum has degenerated	breakdown of endometrial lining and its blood vessels; flows out as menstrual fluid	low estrogen and progesterone
Follicular / proliferative (6-13)	primary follicles grow into a mature **Graafian follicle**	endometrium **regenerates** through proliferation	LH, FSH rise gradually; growing follicles secrete **estrogens**
Ovulatory (about day 14)	Graafian follicle ruptures, **ovulation**	endometrium thick	LH surge; LH and FSH peak
Luteal / secretory (15-28)	remaining Graafian follicle forms **corpus luteum**	endometrium maintained, ready for implantation	high **progesterone** from corpus luteum

The menstrual flow results from the breakdown of the endometrial lining of the uterus and its blood vessels, and lasts for **3-5 days**. Menstruation only occurs if the released ovum is not fertilised. Lack of menstruation may indicate pregnancy, but it may also be caused by other factors such as **stress and poor health**.

The rapid secretion of LH, reaching a maximum during the mid-cycle, is called the LH surge; it induces rupture of the Graafian follicle and thereby the release of the ovum (**ovulation**). PYQ

The corpus luteum secretes large amounts of **progesterone**, which is essential for the maintenance of the endometrium.

Such an endometrium is necessary for implantation of the fertilised ovum and other events of pregnancy. **During pregnancy all events of the menstrual cycle stop** and there is no menstruation.

In the absence of fertilisation, the corpus luteum **degenerates**. This causes disintegration of the endometrium, leading to menstruation and a new cycle. In human beings, menstrual cycles cease around 50 years of age; this is called **menopause**. Cyclic menstruation is an indicator of the **normal reproductive phase** and extends between menarche and menopause.

Yaad karo: cycle ke 4 phases

Mummy Fir Order Layi = Menstrual (1–5), Follicular (6–13), Ovulation (14), Luteal (15–28). Corpus luteum = temporary manager jo progesterone se endometrium sambhalta hai.

Quick fact: menstrual hygiene

- → Change sanitary napkins or clean homemade pads every **4–5 hours** as per requirement.
- → Dispose of used napkins properly, wrapped in a used paper; do not throw them in drainpipes of toilets or in the open.
- → Take a bath regularly and wash hands with soap after handling the napkin.

5. Fertilisation and Implantation

During copulation (coitus), semen is released by the penis into the vagina; this is called insemination. The motile sperms swim rapidly, pass through the cervix, enter the uterus and finally reach the **junction of the isthmus and ampulla** (ampullary–isthmic junction) of the fallopian tube. The ovum released by the ovary is also transported to the ampullary region, where fertilisation takes place. PYQ

Fertilisation can occur only if the ovum and sperms are transported **simultaneously** to the ampullary region. This is why **not all copulations lead to fertilisation and pregnancy**.

A. Events of fertilisation

1. A sperm comes in contact with the **zona pellucida** layer of the ovum and induces changes in the membrane that block the entry of additional sperms. This ensures that only one sperm fertilises an ovum.
2. Secretions of the **acrosome** help the sperm enter the cytoplasm of the ovum through the zona pellucida and the plasma membrane.
3. This induces **completion of the meiotic division** of the secondary oocyte. The second meiotic division is also unequal and forms a **second polar body** and a haploid ovum (**ootid**).
4. The haploid nucleus of the sperm and that of the ovum fuse together to form a diploid zygote.

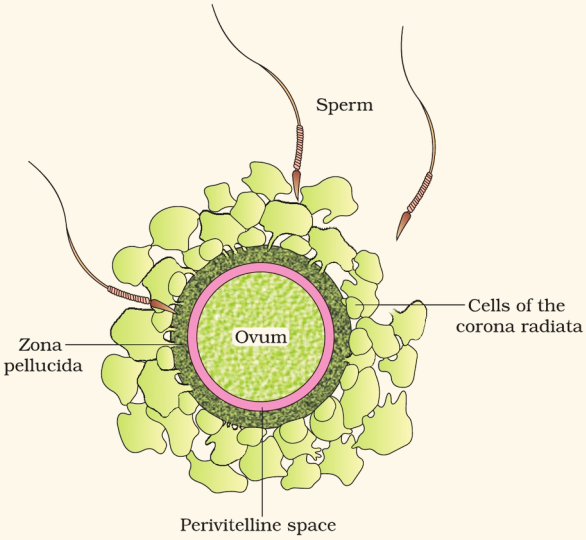


Fig. 2.10 Ovum surrounded by few sperms

Pehla sperm zona pellucida tak pahuncha: khushi. Baaki crores ke liye: entry band.

Asli baat: pehle sperm ke contact se membrane badalti hai, extra sperms block (polyspermy nahi).

B. Sex determination: who decides?

The chromosome pattern in the human female is **XX** and in the male is **XY**. So all the haploid gametes produced by the female (ova) have the sex chromosome **X**, whereas in the male gametes (sperms) the sex chromosome can be **either X or Y**: 50% of sperms carry X and 50% carry Y.

Ovum	Sperm	Zygote	Baby
X	X	XX	Female
X	Y	XY	Male

Hence, the sex of the baby is decided by the father and not by the mother, so blaming women for giving birth to daughters is scientifically wrong. Board fav

Paheli–Bujho

Bujho: Mohalle wali aunty bol rahi thi, beti hui toh maa ki galti.

Paheli: Galat! Maa ke saare ova sirf X chromosome laate hain.

Bujho: Toh Y kaun laata hai?

Paheli: Papa ka sperm!

X mila toh beti, Y mila toh beta.

Quick fact: twins

Identical (monozygotic) twins come from one ovum fertilised by one sperm; the early embryo splits into two, so they are genetically identical and of the same sex.
Fraternal (dizygotic) twins come from two ova released together and fertilised by two different sperms, so they may differ in sex and looks.

C. Cleavage, blastocyst and implantation

The mitotic division (cleavage) starts as the zygote moves through the isthmus of the oviduct towards the uterus and forms **2, 4, 8, 16 daughter cells** called **blastomeres**. The embryo with 8 to 16 blastomeres is called a morula. The morula continues to divide and transforms into a blastocyst as it moves further into the uterus.

- → Blastomeres in the blastocyst are arranged into an outer layer called the trophoblast and an inner group of cells attached to it called the inner cell mass.
- → The **trophoblast** layer gets attached to the endometrium; the **inner cell mass** gets differentiated as the **embryo**.
- → After attachment, uterine cells divide rapidly and cover the blastocyst. As a result, the blastocyst becomes embedded in the endometrium of the uterus. This is called implantation, and it leads to pregnancy. PYQ

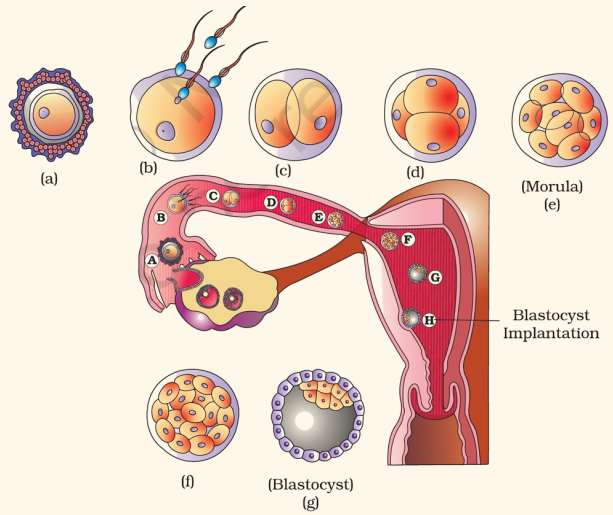


Fig. 2.11 Transport of ovum, fertilisation and passage of growing embryo through fallopian tube

Zygote (ampulla) → Cleavage → Morula (8–16 cells) → Blastocyst → Implantation in endometrium → Pregnancy

Key terms

- → **Ootid**: haploid ovum formed after meiosis II at fertilisation.
- → **Blastomeres**: daughter cells formed by cleavage.
- → **Morula**: embryo with 8–16 blastomeres.
- → **Trophoblast**: outer layer of blastocyst; attaches to endometrium.
- → **Inner cell mass**: forms the embryo; contains stem cells.

6. Pregnancy and Embryonic Development

After implantation, finger-like projections appear on the trophoblast, called chorionic villi, which are surrounded by the uterine tissue and maternal blood. The chorionic villi and uterine tissue become **interdigitated** with each other and jointly form a structural and functional unit between the developing embryo (foetus) and the maternal body, called the placenta. PYQ

Functions of the placenta:

- → Facilitates the supply of **oxygen and nutrients** to the embryo.
- → Removes **carbon dioxide and excretory/waste materials** produced by the embryo.
- → The placenta is connected to the embryo through an umbilical cord, which helps in the transport of substances to and from the embryo.
- → Acts as an **endocrine tissue** and produces several hormones.

Hormone	Source	Note
hCG (human chorionic gonadotropin)	placenta	detected in pregnancy tests
hPL (human placental lactogen)	placenta	pregnancy only
Estrogens, progestogens	placenta (also ovary)	maintain pregnancy
Relaxin	**ovary**, later phase of pregnancy	pregnancy only

hCG, hPL and relaxin are produced in women only during pregnancy. In addition, the levels of other hormones like estrogens, progestogens, cortisol, prolactin and thyroxine increase several-fold in maternal blood. These are essential for supporting foetal growth, metabolic changes in the mother and maintenance of pregnancy.

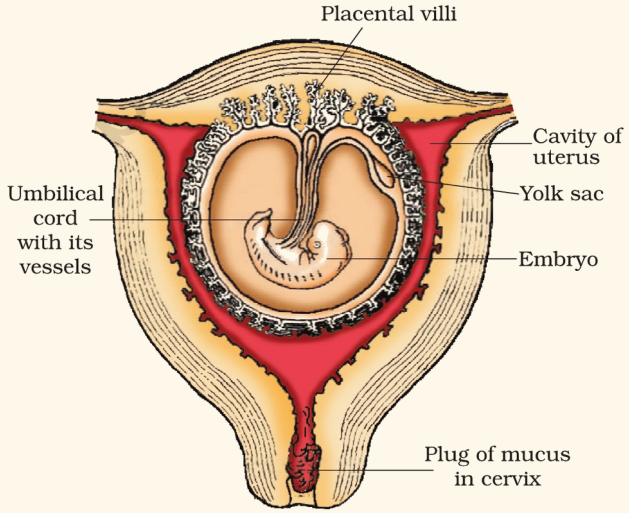


Fig. 2.12 The human foetus within the uterus

Germ layers and stem cells

Immediately after implantation, the inner cell mass (embryo) differentiates into an outer layer called **ectoderm** and an inner layer called **endoderm**. A **mesoderm** soon appears between them. These three layers give rise to all tissues (organs) in adults. The inner cell mass contains certain cells called stem cells, which have the potency to give rise to all the tissues and organs.

Exam Alert

→ Germ layers **inner cell mass** se bante hain; trophoblast se chorionic villi aur placenta ka part.

→ Pehla sign of growing foetus = **heart sound** (about 1 month).

Development of the human foetus PYQ

Time	Development
After **1 month**	**heart** is formed; heart sound is the first sign of a growing foetus
End of **2nd month**	**limbs and digits** develop
End of **12 weeks** (1st trimester)	most **major organ systems** formed; limbs and external genital organs well developed
5th month	first **movements** of foetus; hair appear on the head
End of **24 weeks** (2nd trimester)	body covered with fine hair; **eyelids separate**; eyelashes formed
End of **9 months**	fully developed; ready for delivery

Yaad karo: foetus ka calendar

1 = dil (heart), **2** = haath-pair ki ungliyan, **3** = saare organs, **5** = hilna + sir pe baal, **6** = palkein, **9** = baby ready! (months)

7. Parturition and Lactation

The average duration of human pregnancy is about **9 months**, called the gestation period. Vigorous contraction of the uterus at the end of pregnancy causes expulsion (delivery) of the foetus. This process of delivery is called parturition. It is induced by a complex **neuroendocrine mechanism**.

Signals from fully developed foetus and placenta →
Mild uterine contractions (**foetal ejection reflex**) →
Oxytocin from maternal pituitary →
Stronger uterine contractions → More oxytocin →
Baby expelled through birth canal

→ This **stimulatory reflex** between uterine contraction and oxytocin secretion continues, resulting in stronger and stronger contractions. PYQ

→ Soon after the infant is delivered, the **placenta** is also expelled out of the uterus.

→ Doctors inject **oxytocin** to induce delivery when needed.

The mammary glands of the female undergo differentiation during pregnancy and start producing milk towards the end of pregnancy by the process called lactation. This helps the mother feed the new-born. The milk produced during the initial few days of lactation is called colostrum; it contains several **antibodies** absolutely essential to develop resistance in new-born babies. **Breast-feeding** during the initial period of infant growth is recommended by doctors for bringing up a healthy baby. PYQ

Paheli-Bujho

Bujho: Ovulation mein ovum release hota hai na, poora ready?

Paheli: Nahi, wo secondary oocyte hai. Meiosis II aadha pending hai.

Bujho: Toh poora kab hota hai?

Paheli: Jab sperm andar aata hai, tab second polar body nikalti hai.

Differences boards love Board fav

Term 1	Term 2
Insemination: transfer of semen into the vagina	**Fertilisation**: fusion of sperm and ovum in the ampulla
Ovulation: release of secondary oocyte from Graafian follicle (day 14)	**Implantation**: embedding of blastocyst in endometrium
Morula: solid ball of 8–16 blastomeres	**Blastocyst**: trophoblast + inner cell mass
Menarche: first menstruation at puberty	**Menopause**: cessation of cycles (about 50 years)
Follicular phase: estrogen; endometrium proliferates	**Luteal phase**: progesterone; endometrium maintained
Parturition: delivery of the foetus	**Lactation**: milk production by mammary glands

Hormones of gametogenesis and the cycle Board fav

Hormone	Source	Main role
GnRH	hypothalamus	stimulates release of FSH and LH
FSH	anterior pituitary	male: Sertoli cells, spermiogenesis; female: follicle growth, estrogen secretion
LH	anterior pituitary	male: Leydig cells, androgens; female: LH surge causes ovulation, corpus luteum formation
Androgens	Leydig cells	spermatogenesis, male accessory organs, male characters
Estrogen	growing follicles	proliferation of endometrium
Progesterone	corpus luteum	maintains endometrium, supports pregnancy

Hormones of pregnancy, birth and lactation

Hormone	Source	Main role
hCG, hPL	placenta	maintain pregnancy, foetal growth
Relaxin	ovary	later phase of pregnancy
Oxytocin	posterior pituitary	uterine contractions in parturition; milk ejection
Prolactin	anterior pituitary	milk production (lactation)

Exam Alert

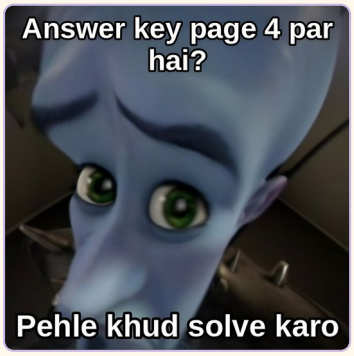
→ Relaxin ka source **ovary** hai, placenta nahi: MCQ mein trap.
→ hCG, hPL, relaxin: sirf pregnancy mein banate hain.
→ Umbilical cord embryo ko placenta se jodti hai; placenta = chorionic villi + uterine tissue.

1 minute recap

1. Testes in scrotum (2–2.5 °C lower); 250 lobules; seminiferous tubules: spermatogonia + Sertoli; Leydig cells make androgens.
2. Sperm path: seminiferous tubules, rete testis, vasa efferentia, epididymis, vas deferens, ejaculatory duct, urethra; glands: seminal vesicles, prostate, bulbourethral.
3. Ovary (cortex, medulla); oviduct: infundibulum (fimbriae), ampulla, isthmus; uterus wall: perimetrium, myometrium, endometrium.
4. Spermatogenesis: 1 primary spermatocyte gives 4 sperms; GnRH, LH (Leydig), FSH (Sertoli).
5. Oogenesis starts in foetus; primary oocyte arrested in prophase I; ovulation releases secondary oocyte.
6. Menstrual cycle 28 days: menstrual, follicular, ovulation (LH surge, day 14), luteal (progesterone).
7. Fertilisation in ampulla; sperm decides sex; morula, blastocyst, implantation.
8. Placenta: hCG, hPL, estrogens, progestogens; relaxin from ovary; parturition by oxytocin; colostrum has antibodies.

MOST IMPORTANT QUESTIONS

Objective, assertion-reason, short, long and case-based, with answers



Answers green mein neeche hi hain, chhupaye nahi.
Bas pehle khud try!

Objective (1 mark each)

Q1. IM Board fav Leydig cells are found in the:
(a) seminiferous tubules
(b) ovary
(c) interstitial spaces of the testis
(d) epididymis
Ans: (c) interstitial spaces of the testis

Q2. IM The temperature of the scrotum necessary for spermatogenesis is lower than the body temperature by:
(a) 0.5 °C
(b) 2–2.5 °C
(c) 5 °C
(d) 7 °C
Ans: (b) 2–2.5 °C

Q3. IM Board fav Which of the following secretes the hormone relaxin during the later phase of pregnancy?
(a) Placenta
(b) Pituitary
(c) Uterus
(d) Ovary
Ans: (d) Ovary

Q4. IM At puberty, the number of primary follicles left in each ovary is:
(a) 60,000–80,000
(b) 2 million
(c) 2,00,000
(d) 6,000–8,000
Ans: (a) 60,000–80,000

Q5. IM Board fav The site of fertilisation in humans is the:
(a) uterus
(b) ampullary region of the fallopian tube
(c) infundibulum
(d) cervix
Ans: (b) ampullary region of the fallopian tube

Q6. IM Which hormone level shows a surge on about the 14th day of the menstrual cycle?
(a) Progesterone
(b) FSH only
(c) LH
(d) Relaxin
Ans: (c) LH

Q7. IM The embryo with 8–16 blastomeres is called:
(a) morula
(b) blastocyst
(c) gastrula
(d) zygote
Ans: (a) morula

Q8. 1M Board fav The number of sperms formed from 25 primary spermatocytes and the number of ova formed from 25 primary oocytes are:
(a) 25 and 25
(b) 100 and 25
(c) 100 and 100
(d) 50 and 25
Ans: (b) 100 and 25

Q9. 1M Name the hormone detected in pregnancy test kits.
Ans: hCG (human chorionic gonadotropin).

Q10. 1M Name the cells that provide nutrition to the germ cells in seminiferous tubules.
Ans: Sertoli cells.

Q11. 1M What is the first milk after childbirth called? Why is it important?
Ans: Colostrum; it contains antibodies that give the new-born passive immunity.

Q12. 1M Board fav NCERT: Fill in the blanks. (a) Humans reproduce ____ (b) Humans are ____ (c) Fertilisation is ____ in humans (d) Gametes are ____ (e) Zygote is ____ (f) Release of ovum from a mature follicle is ____ (g) Ovulation is induced by ____ (h) Fusion of gametes is ____ (i) Fertilisation takes place in ____ (j) Zygote divides to form ____ which is implanted (k) Vascular connection between foetus and uterus is ____
Ans: (a) sexually (b) viviparous (c) internal (d) haploid (e) diploid (f) ovulation (g) luteinising hormone (LH) (h) fertilisation (i) ampullary region of the fallopian tube (j) blastocyst (k) placenta.

Q13. 1M The part of the fallopian tube closest to the ovary is the:
(a) isthmus
(b) ampulla
(c) infundibulum
(d) cervix
Ans: (c) infundibulum

Q14. 1M Board fav The secondary oocyte is released from the ovary at the stage of:
(a) prophase I
(b) metaphase II of meiosis
(c) completed meiosis
(d) interphase
Ans: (b) metaphase II of meiosis

Q15. 1M Which layer of the blastocyst gets attached to the endometrium?
(a) Trophoblast
(b) Inner cell mass
(c) Endoderm
(d) Mesoderm
Ans: (a) Trophoblast

Q16. 1M The milk ejection and uterine contractions during parturition are both caused by:
(a) prolactin
(b) relaxin
(c) progesterone
(d) oxytocin
Ans: (d) oxytocin

Q17. 1M The cavity of the cervix together with the vagina forms the:
(a) uterine cavity
(b) birth canal
(c) antrum
(d) ampulla
Ans: (b) birth canal

Q18. 1M Board fav Which of these is haploid?
(a) Spermatogonium
(b) Primary oocyte
(c) Secondary spermatocyte
(d) Zygote
Ans: (c) Secondary spermatocyte

Assertion-Reason (1 mark each)

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q19. 1M Board fav **A:** The sex of the baby is decided by the father. **R:** The male produces two kinds of sperms, X-bearing and Y-bearing.
Ans: (a) Both true and R explains A.

Q20. 1M **A:** Only one sperm fertilises an ovum. **R:** The contact of a sperm with the zona pellucida induces changes that block the entry of additional sperms.
Ans: (a) Both true and R explains A.

Q21. 1M **A:** Menstruation does not occur during pregnancy. **R:** The corpus luteum keeps secreting progesterone, which maintains the endometrium.
Ans: (a) Both true and R explains A.

Q22. 1M **A:** The ovum released at ovulation is haploid and has completed meiosis. **R:** The first meiotic division of the primary oocyte forms a secondary oocyte and the first polar body.
Ans: (d) A is false (the secondary oocyte completes meiosis II only after sperm entry), R is true.

Short answer (2-3 marks)

Q23. 2M Board fav NCERT: Write two major functions each of testis and ovary.
Ans: Testis: (i) produces sperms (spermatogenesis); (ii) secretes androgens (testosterone) from Leydig cells. Ovary: (i) produces ova (oogenesis); (ii) secretes ovarian hormones, estrogen and progesterone.

Q24. 3M Board fav NCERT: Describe the structure of a seminiferous tubule.
Ans: Each testicular lobule has 1–3 highly coiled seminiferous tubules. Each tubule is lined on its inside by spermatogonia (male germ cells), which undergo meiosis to form sperms, and Sertoli cells, which nourish the germ cells. The interstitial spaces outside the tubules have blood vessels, Leydig cells (secrete androgens) and immunologically competent cells.

Q25. 3M Board fav NCERT: What is spermatogenesis? Briefly describe the process.
Ans: Formation of sperms from spermatogonia in the seminiferous tubules, beginning at puberty. Spermatogonia (2n) multiply by mitosis; some become primary spermatocytes, which complete meiosis I to form two haploid secondary spermatocytes; these undergo meiosis II to form four haploid spermatids. Spermatids change into sperms by spermiogenesis, and the sperms are released by spermiation.

Q26. 2M NCERT: Name the hormones involved in regulation of spermatogenesis.
Ans: GnRH from the hypothalamus; LH (acts on Leydig cells to secrete androgens) and FSH (acts on Sertoli cells to secrete factors for spermiogenesis) from the anterior pituitary; androgens (testosterone) from the testis.

Q27. 2M Board fav NCERT: Define spermiogenesis and spermiation.
Ans: Spermiogenesis: transformation of spermatids into spermatozoa (sperms). Spermiation: release of sperms from the seminiferous tubules after their heads were embedded in Sertoli cells.

Q28. 2M Board fav NCERT: Draw a labelled diagram of a sperm.
Ans: Diagram as Fig. 2.6. Labels: plasma membrane, acrosome, nucleus (head), neck, middle piece with mitochondria, tail.

Q29. 2M NCERT: What are the major components of seminal plasma?
Ans: Secretions of seminal vesicles, prostate and bulbourethral glands, rich in fructose, calcium and certain enzymes. Semen = seminal plasma + sperms.

Q30. 3M NCERT: What are the major functions of male accessory ducts and glands?
Ans: Ducts (rete testis, vasa efferentia, epididymis, vas deferens) store and transport sperms; epididymis also helps sperm maturation. Glands (seminal vesicles, prostate) form the seminal plasma that gives fructose (energy), calcium and enzymes for motility and maturation; bulbourethral glands lubricate the penis.

Q31. 3M Board fav NCERT: What is oogenesis? Give a brief account of oogenesis.
Ans: Formation of a mature female gamete. Oogonia form in the foetal ovary, enter prophase I and stay arrested as primary oocytes. Each with a layer of granulosa cells forms a primary follicle; at puberty 60,000–80,000 remain per ovary. They become secondary follicles (more granulosa layers + theca), then tertiary follicles (antrum). The primary oocyte completes meiosis I unequally: secondary oocyte + first polar body. The Graafian follicle releases the secondary oocyte at ovulation; meiosis II completes only after sperm entry.

Q32. 3M Board fav NCERT: Draw a labelled diagram of a section through the ovary.
Ans: Diagram as Fig. 2.7. Labels: primary follicle, secondary follicle, tertiary follicle with antrum, Graafian follicle, ovum (secondary oocyte), corpus luteum, blood vessels.

Q33. 2M NCERT: Draw a labelled diagram of a Graafian follicle.
Ans: A large mature follicle with a big fluid-filled antrum; the secondary oocyte surrounded by the zona pellucida sits on one side among granulosa cells (corona radiata); the follicle wall has granulosa cells, theca interna and theca externa.

Q34. 3M Board fav NCERT: Name the functions of (a) corpus luteum (b) endometrium (c) acrosome (d) sperm tail (e) fimbriae.
Ans: (a) Secretes progesterone to maintain the endometrium. (b) Undergoes cyclic changes and supports implantation and placenta formation. (c) Its enzymes help the sperm penetrate the ovum. (d) Motility of the sperm. (e) Collect the ovum after ovulation.

Q35. 3M NCERT: True/False, correct the false ones: (a) Androgens are produced by Sertoli cells. (b) Spermatozoa get nutrition from Sertoli cells. (c) Leydig cells are found in ovary. (d) Leydig cells synthesise androgens. (e) Oogenesis takes place in corpus luteum. (f) Menstrual cycle ceases during pregnancy. (g) Presence or absence of hymen is not a reliable indicator of virginity.
Ans: (a) False: androgens are produced by Leydig cells. (b) True. (c) False: Leydig cells are found in the testis. (d) True. (e) False: oogenesis takes place in the ovary (follicles). (f) True. (g) True.

Q36. 2M Board fav NCERT: What is menstrual cycle? Which hormones regulate it?
Ans: The cycle of events in female primates from one menstruation to the next, about 28 days. Regulated by GnRH (hypothalamus), FSH and LH (anterior pituitary), and estrogen and progesterone (ovary).

Q37. 2M Board fav NCERT: What is parturition? Which hormones are involved in its induction?
Ans: Expulsion of the fully developed foetus from the uterus at the end of pregnancy. Signals from the foetus and placenta cause mild contractions (foetal ejection reflex), which trigger oxytocin release from the maternal pituitary; oxytocin causes stronger contractions, which release still more oxytocin until the baby is delivered.

Q38. 3M Board fav NCERT: In our society women are often blamed for giving birth to daughters. Explain why this is not correct.
Ans: Women have XX chromosomes and all ova carry X. Men have XY, so 50% sperms carry X and 50% carry Y. If an X-sperm fertilises the ovum the baby is a girl (XX); a Y-sperm gives a boy (XY). So the sex of the baby is decided by the father, not the mother.

Q39. 2M NCERT: How many eggs are released by a human ovary in a month? How many if the mother gave birth to identical twins? Would the answer change for fraternal twins?
Ans: One egg per month. Identical twins: one egg (one zygote splits into two embryos). Fraternal twins: two eggs, each fertilised by a different sperm, so yes, the answer changes.

Q40. 1M NCERT: How many eggs were released by the ovary of a female dog that gave birth to 6 puppies?
Ans: At least 6 eggs (assuming each puppy came from a separate fertilised egg; dogs are multiparous).

Q41. 2M NCERT: Describe the parts of the oviduct (fallopian tube).
Ans: About 10–12 cm long: infundibulum (funnel near the ovary, with fimbriae that collect the ovum), ampulla (wider middle part, site of fertilisation) and isthmus (narrow lumen, joins the uterus).

Q42. 2M Board fav Differentiate between Sertoli cells and Leydig cells.
Ans: Sertoli cells are inside seminiferous tubules and nourish germ cells (FSH acts on them). Leydig cells are in interstitial spaces outside the tubules and secrete androgens (LH acts on them).

Q43. 2M Why is the second meiotic division of the secondary oocyte important? When does it occur?
Ans: It forms the haploid ovum (ootid) and the second polar body, so that only a haploid nucleus fuses with the sperm nucleus. It occurs only when a sperm enters the secondary oocyte during fertilisation.

Q44. 2M Board fav What is the LH surge? What is its role?
Ans: The rapid rise of LH to its maximum in the middle of the menstrual cycle (about day 14). It induces rupture of the Graafian follicle and release of the ovum (ovulation).

Q45. 2M Name the layers of the uterine wall. Which layer undergoes cyclic changes and which helps in delivery?
Ans: Perimetrium, myometrium and endometrium. The endometrium undergoes cyclic changes in the menstrual cycle; the smooth muscle myometrium contracts strongly during delivery.

Q46. 2M Trace the path of milk from the alveoli to the outside in a mammary gland.
Ans: Alveoli, mammary tubules, mammary duct, mammary ampulla, lactiferous duct, out through the nipple.

Q47. 2M Board fav What is implantation? Which layer of the blastocyst helps in it?
Ans: Embedding of the blastocyst in the endometrium of the uterus; uterine cells divide and cover it. The trophoblast attaches to the endometrium; the inner cell mass forms the embryo.

Q48. 2M Board fav Name the hormones produced in women only during pregnancy. Name their sources.
Ans: hCG and hPL from the placenta; relaxin from the ovary (later phase of pregnancy).

Q49. 2M Why are stem cells of the inner cell mass important?
Ans: They have the potency to give rise to all tissues and organs of the body (through ectoderm, mesoderm and endoderm).

Q50. 3M Board fav Give the role of the placenta as an endocrine gland and as a connecting organ.
Ans: Connection: supplies oxygen and nutrients and removes CO_2 and wastes of the foetus through the umbilical cord. Endocrine: secretes hCG, hPL, estrogens and progestogens to maintain pregnancy and support foetal growth.

Q51. 3M Why does menstruation stop during pregnancy but start again if the ovum is not fertilised?
Ans: After fertilisation the corpus luteum keeps secreting progesterone (supported by hCG), so the endometrium is maintained. Without fertilisation the corpus luteum degenerates, progesterone falls and the endometrium breaks down as menstruation.

Q52. 2M Differentiate between spermiogenesis and spermatogenesis.
Ans: Spermatogenesis is the whole process of forming sperms from spermatogonia (mitosis + meiosis + spermiogenesis). Spermiogenesis is only its last step: transformation of non-motile spermatids into motile sperms (no division).

Q53. 2M What are chorionic villi? What is their role?
Ans: Finger-like projections of the trophoblast after implantation, surrounded by uterine tissue and maternal blood. They interdigitate with uterine tissue to form the placenta, the site of exchange between mother and foetus.

Q54. 2M Board fav Mention the changes in the foetus by the end of the first and second trimesters.
Ans: End of 12 weeks (1st trimester): most major organ systems formed, limbs and external genital organs well developed. End of 24 weeks (2nd trimester): body covered with fine hair, eyelids separate and eyelashes form.

Q55. 2M Why is breast-feeding recommended in the initial days after birth?
Ans: The first milk, colostrum, contains several antibodies essential to develop resistance in the new-born; breast milk also gives complete nutrition for healthy growth.

Q56. 2M What is menarche and menopause? What indicates the normal reproductive phase?
Ans: Menarche is the first menstruation at puberty; menopause is the stopping of menstrual cycles at about 50 years. Cyclic menstruation between menarche and menopause indicates the normal reproductive phase.

Q57. 3M List any three differences between a primary follicle and a Graafian follicle.
Ans: Primary follicle: primary oocyte with one layer of granulosa cells, no theca, no antrum. Graafian follicle: secondary oocyte with zona pellucida, many granulosa layers, theca interna and externa, large fluid-filled antrum; ready to rupture at ovulation.

Q58. 2M Where are the following located? (a) Leydig cells (b) fimbriae (c) acrosome (d) antrum
Ans: (a) Interstitial spaces of the testis. (b) Edges of the infundibulum of the fallopian tube. (c) Anterior part of the sperm head. (d) Tertiary/Graafian follicle of the ovary.

Q59. 2M What is the role of the epididymis and the seminal vesicle?
Ans: Epididymis: stores sperms and helps in their maturation, and passes them to the vas deferens. Seminal vesicle: its secretion (rich in fructose) forms a major part of seminal plasma; its duct joins the vas deferens to form the ejaculatory duct.

Q60. 2M Board fav Explain why not all copulations lead to fertilisation.
Ans: Fertilisation happens only if the ovum and sperms reach the ampullary region at the same time. The ovum is released only once in the middle of the cycle, so copulation at other times does not lead to fertilisation.

Q61. 2M How does the second polar body form? Is it formed in males?
Ans: When a sperm enters the secondary oocyte, the oocyte completes the unequal second meiotic division, giving a large haploid ootid (ovum) and a tiny second polar body. No; spermatogenesis has equal divisions and forms no polar bodies.

Q62. 2M Name the three germ layers and the cells that form them. Why are they important?
Ans: Ectoderm (outer), endoderm (inner) and mesoderm (between them), formed from the inner cell mass after implantation. All tissues and organs of the adult develop from these three layers.

Q63. 2M Board fav What is the foetal ejection reflex?
Ans: Mild uterine contractions induced by signals from the fully developed foetus and placenta at the end of pregnancy; they trigger oxytocin release from the maternal pituitary, starting parturition.

Q64. 2M Mention the role of FSH and LH in the female.
Ans: FSH stimulates growth of follicles and secretion of estrogen by them. LH rises with FSH in the follicular phase; its mid-cycle surge causes ovulation and helps form the corpus luteum.

Q65. 2M Board fav What is the role of the corpus luteum, and what happens to it if fertilisation does not occur?
Ans: It forms from the ruptured Graafian follicle and secretes large amounts of progesterone to maintain the endometrium for implantation. Without fertilisation it degenerates, progesterone falls and menstruation follows.

Q66. 2M Why is the ovum released from the ovary called a secondary oocyte?
Ans: Because at ovulation the oocyte has completed only meiosis I (forming the first polar body); it completes meiosis II only after a sperm enters it.

Q67. 1M Name the parts of the female external genitalia.
Ans: Mons pubis, labia majora, labia minora, hymen and clitoris.

Q68. 2M What is the function of the myometrium and the endometrium during pregnancy and delivery?
Ans: Endometrium: receives the blastocyst for implantation and, with chorionic villi, forms the placenta. Myometrium: its smooth muscle shows strong contractions during parturition to expel the baby.

Q69. 2M Why are the testes of man called both exocrine and endocrine glands?
Ans: Exocrine: they produce sperms, which are carried out through ducts. Endocrine: their Leydig cells secrete androgens directly into the blood.

Q70. 2M Board fav What is the role of the acrosome and of the zona pellucida in fertilisation?
Ans: Acrosome enzymes help the sperm pass through the zona pellucida and plasma membrane into the ovum. The zona pellucida, after the first sperm touches it, changes to block other sperms (prevents polyspermy).

Q71. 2M What are the functions of the scrotum and the cervix?
Ans: Scrotum: keeps the testes 2–2.5 °C below body temperature for spermatogenesis. Cervix: connects the uterus to the vagina; its canal with the vagina forms the birth canal.

Q72. 2M Why does the ovum have a large amount of cytoplasm while the polar bodies have very little?
Ans: Both meiotic divisions of the oocyte are unequal, so the ovum retains the bulk of the nutrient-rich cytoplasm needed for early development, while the polar bodies get very little and degenerate.

Long answer (5 marks)

Q73. 5M Board fav (a) Draw a labelled diagram of the male reproductive system. (b) Describe the accessory ducts and glands and their functions.
Ans: (a) Diagram as Fig. 2.1(b): testis in scrotum, seminiferous tubules, rete testis, vasa efferentia, epididymis, vas deferens, seminal vesicle, ejaculatory duct, prostate, bulbourethral gland, urethra, penis, glans penis, foreskin, urinary bladder, ureter. (b) Ducts store and transport sperms: seminiferous tubules to rete testis, vasa efferentia, epididymis (along posterior surface of testis), vas deferens (loops over the bladder), joined by the seminal vesicle duct to form the ejaculatory duct, which opens into the urethra; urethra opens at the urethral meatus. Glands: paired seminal vesicles and single prostate form seminal plasma rich in fructose, calcium and enzymes; paired bulbourethral glands lubricate the penis.

Q74. 5M Board fav (a) Draw a labelled diagram of the female reproductive system. (b) Describe the structure of the ovary, fallopian tube and uterus.
Ans: (a) Diagram as Fig. 2.3(b): ovaries, fimbriae, infundibulum, ampulla, isthmus, uterine cavity, perimetrium, myometrium, endometrium, cervix, cervical canal, vagina. (b) Ovary: 2–4 cm, attached by ligaments; epithelium encloses stroma with cortex and medulla; produces ova and hormones. Fallopian tube: 10–12 cm; infundibulum with fimbriae, ampulla (fertilisation), isthmus. Uterus: single, inverted pear, supported by ligaments, opens through the cervix into the vagina; wall of perimetrium, myometrium (contracts at delivery) and endometrium (cyclic changes).

Q75. 5M Board fav Describe the menstrual cycle with the help of a diagram showing the changes in the ovary, uterus and hormone levels.
Ans: Diagram as Fig. 2.9. Menstrual phase (days 1–5): endometrium and its blood vessels break down. Follicular/proliferative phase (6–13): FSH and LH rise; primary follicles grow into a Graafian follicle; follicles secrete estrogen; endometrium regenerates. Ovulation (about day 14): LH surge ruptures the Graafian follicle, ovum released. Luteal/secretory phase (15–28): remaining follicle forms corpus luteum, which secretes progesterone to maintain the endometrium. If no fertilisation, the corpus luteum degenerates and menstruation follows; in pregnancy the cycle stops. Menarche starts and menopause (about 50 years) ends the cycles.

Q76. 5M Board fav Describe the events from fertilisation to implantation in humans, with a diagram.
Ans: Diagram as Fig. 2.11. Sperm and ovum reach the ampulla. The sperm contacts the zona pellucida; changes block other sperms; acrosomal enzymes help entry; the secondary oocyte completes meiosis II forming the ovum (ootid) and second polar body; nuclei fuse to form a diploid zygote. Cleavage in the isthmus gives 2, 4, 8, 16 blastomeres; 8–16 cells form the morula, which becomes the blastocyst in the uterus, with outer trophoblast and inner cell mass. The trophoblast attaches to the endometrium; uterine cells cover it and the blastocyst is embedded (implantation), leading to pregnancy.

Q77. 5M Board fav (a) What is a placenta? Describe its structure and functions. (b) Describe the development of the human foetus month by month.
Ans: (a) After implantation the trophoblast forms chorionic villi, which interdigitate with uterine tissue to form the placenta, connected to the embryo by the umbilical cord. Functions: supplies oxygen and nutrients, removes CO_2 and wastes, and acts as an endocrine tissue secreting hCG, hPL, estrogens and progestogens. (b) 1 month: heart formed; 2 months: limbs and digits; 12 weeks: major organ systems, limbs and external genitalia; 5th month: first movements and hair on the head; 24 weeks: fine body hair, eyelids separate, eyelashes; 9 months: fully developed and ready for delivery.

Q78. 5M Board fav (a) Describe spermatogenesis with a labelled flow chart, showing the ploidy at each stage. (b) Explain the hormonal control of spermatogenesis.
Ans: (a) Spermatogonium (2n) by mitosis; primary spermatocyte (2n), meiosis I, 2 secondary spermatocytes (n); meiosis II, 4 spermatids (n); spermiogenesis, 4 sperms (n), released by spermiation. Starts at puberty in seminiferous tubules. (b) Hypothalamus releases more GnRH at puberty; it makes the anterior pituitary release LH and FSH. LH acts on Leydig cells to secrete androgens that stimulate spermatogenesis; FSH acts on Sertoli cells to secrete factors that help spermiogenesis.

Case-based (4 marks)

Q79. 4M Board fav A couple has been trying for a baby. The husband's semen report shows 120 million sperms per ejaculation; 35% have normal shape and 25% are actively motile. (a) Is he likely to be fertile? Give the normal values. (b) Which part of the sperm gives energy for motility? (c) Name the glands that add fructose to semen. (d) Why must the testes stay in the scrotum?

Ans: (a) Probably not: normal is 200–300 million sperms, with at least 60% normal shape and 40% vigorous motility. (b) Mitochondria of the middle piece. (c) Seminal vesicles (with prostate) form seminal plasma rich in fructose. (d) The scrotum keeps testes 2–2.5 °C below body temperature, needed for spermatogenesis.

Q80. 4M Board fav A woman with a regular 28-day cycle took a pregnancy test on day 35 and it was positive. (a) Which hormone did the kit detect and where is it produced? (b) Why did menstruation not occur? (c) Name two hormones produced only during pregnancy besides this one. (d) Where did fertilisation take place?

Ans: (a) hCG, produced by the placenta (trophoblast). (b) The corpus luteum continued to secrete progesterone, maintaining the endometrium. (c) hPL (placenta) and relaxin (ovary). (d) Ampullary region of the fallopian tube.

Q81. 4M A doctor injected a hormone to a woman at full term whose labour was not progressing. (a) Name the hormone and its source in the body. (b) What starts the natural process? (c) Why do contractions keep getting stronger? (d) Name the first milk and give its importance.

Ans: (a) Oxytocin, from the posterior pituitary. (b) Signals from the fully developed foetus and placenta cause mild uterine contractions (foetal ejection reflex). (c) Contractions cause more oxytocin release, which causes stronger contractions (positive feedback). (d) Colostrum; rich in antibodies, gives the new-born resistance.

Q82. 4M Board fav Study the hormone pattern of a 28-day cycle: hormone X peaks sharply on day 14; hormone Y is low in the first half and highest around days 18–24; hormone Ƶ rises in the follicular phase. (a) Identify X, Y and Ƶ with their sources. (b) Which event does the peak of X cause? (c) What happens to Y if fertilisation does not occur, and what follows? (d) Name the phase in which the endometrium regenerates.

Ans: (a) X = LH (anterior pituitary); Y = progesterone (corpus luteum); Ƶ = estrogen (growing follicles). (b) Rupture of the Graafian follicle, ovulation. (c) The corpus luteum degenerates, progesterone falls and the endometrium breaks down: menstruation and a new cycle. (d) Follicular (proliferative) phase.

Q83. 4M A student drew a sequence: zygote, 2 cells, 4 cells, 8–16 cells, a hollow ball with an outer layer and an inner cluster, embedding in the uterus. (a) Name the 8–16 celled stage and the hollow ball. (b) Name the outer layer and the inner cluster, and give one role of each. (c) Where does cleavage begin? (d) What is the embedding called?

Ans: (a) Morula; blastocyst. (b) Trophoblast: attaches to the endometrium and later forms chorionic villi; inner cell mass: forms the embryo and contains stem cells. (c) In the isthmus of the oviduct, as the zygote moves towards the uterus. (d) Implantation.

Q84. 4M Board fav A 13-year-old girl had her first period. Her mother told her that she was born with all the egg cells she would ever have. (a) Name the first menstruation. (b) Is the mother right? Explain. (c) At which stage are these cells arrested till puberty? (d) About how many primary follicles are left in each ovary at puberty?

Ans: (a) Menarche. (b) Yes: oogonia form only in the foetal ovary; no new oogonia are formed after birth. (c) Primary oocytes arrested in prophase I of meiosis. (d) 60,000–80,000.

Q85. 4M A sample from a seminiferous tubule shows large cells near the basal wall, cells with sperm heads embedded in them, and cells outside the tubules near blood vessels. (a) Name the cells at the basal wall and their ploidy. (b) Name the cells holding sperm heads; give their role. (c) Name the cells outside the tubule and the hormone acting on them. (d) Name the process of release of sperms from these tubules.

Ans: (a) Spermatogonia, diploid (2n = 46). (b) Sertoli cells; they nourish the germ cells and, under FSH, secrete factors for spermiogenesis. (c) Leydig (interstitial) cells; LH acts on them to secrete androgens. (d) Spermiation.

Q86. 4M Riya and Siya are twins of the same sex who look exactly alike; their cousins Aman and Anya are twins of different sex. (a) Which pair is monozygotic? (b) How many ova and sperms formed each pair? (c) Why can Aman and Anya differ in sex? (d) Who decided the sex of each child?

Ans: (a) Riya and Siya. (b) Riya–Siya: one ovum and one sperm (the embryo split into two). Aman–Anya: two ova and two sperms. (c) The two ova were fertilised by different sperms, one carrying X and one carrying Y. (d) The father, through the X- or Y-bearing sperm.

Exam Alert: diagrams to practise

- → Fig. 2.1(b) male system, Fig. 2.3(b) female system, Fig. 2.5 seminiferous tubule, Fig. 2.6 sperm, Fig. 2.7 ovary: ye paanch har saal rotate hote hain.
- → Menstrual cycle graph (Fig. 2.9): LH ka peak day 14 aur progesterone ka peak luteal phase mein dikhana.
- → Numbers yaad rakho: 250 lobules, 60,000–80,000 follicles, 200–300 million sperms, 60% / 40%.

REPRODUCTIVE HEALTH

Most asked: contraceptive methods and their mode of action, ART (IVF, ZIFT, GIFT, ICSI, AI), MTP Act, STIs and their prevention, amniocentesis ban

Reproductive health means total well-being in all aspects of reproduction. This chapter covers **RCH programmes, population, contraception, MTP, STIs and infertility.**

In simple words, reproductive health means **healthy reproductive organs with normal functions.** However, it has a broader perspective and includes the emotional and social aspects of reproduction also.

WHO definition PYQ

According to the **World Health Organisation (WHO)**, reproductive health means a total well-being in all aspects of reproduction, i.e., **physical, emotional, behavioural and social.** A society whose people have physically and functionally normal reproductive organs and normal emotional and behavioural interactions in all sex-related aspects is called **reproductively healthy.**

1. Reproductive Health: Problems and Strategies

India was amongst the first countries in the world to initiate action plans and programmes at a national level to attain total reproductive health as a social goal.

Programme	Year / features
Family planning	initiated in **1951**; periodically assessed over the decades
Reproductive and Child Health Care (RCH)	improved programmes covering wider reproduction-related areas; currently in operation

Paheli-Bujho

Bujho: Paheli, school mein sex education? Sab sharma jaayenge!

Paheli: Sharam se myths badhte hain. Sahi jaankari se hi safety aati hai.

Bujho: Matlab WhatsApp wale gyaan se behtar teacher ki class.

Paheli: Bilkul! Isliye NCERT ise encourage karta hai.

Major tasks under these programmes: creating awareness and providing facilities and support for a reproductively healthy society.

A. Creating awareness

- → **Audio-visual and print media**: governmental and non-governmental agencies spread awareness about reproduction-related aspects.
- → **Parents, close relatives, teachers and friends** also have a major role in spreading this information.
- → Sex education in schools should be encouraged to give the right information to the young and to discourage myths and misconceptions about sex. PYQ
- → Proper information about reproductive organs, **adolescence** and related changes, safe and hygienic sexual practices, **STDs, AIDS**, etc., helps adolescents lead a reproductively healthy life.
- → Educating **fertile couples** and those of marriageable age about birth control options, care of pregnant mothers, post-natal care of mother and child, importance of **breast feeding** and equal opportunities for the male and the female child.
- → Awareness about problems of **uncontrolled population growth** and social evils like sex-abuse and sex-related crimes.

B. Medical facilities and new strategies

Successful implementation needs strong **infrastructural facilities, professional expertise and material support** to give medical assistance in problems like pregnancy, delivery, STDs, abortions, contraception, menstrual problems and infertility.

- → Statutory ban on amniocentesis for sex determination, to legally check the increasing menace of **female foeticide.** Board fav
- → **Massive child immunisation** programmes.
- → **Research** on reproduction-related areas is supported by governmental and non-governmental agencies. Example: Saheli, a new oral contraceptive for females, was developed by scientists at the **Central Drug Research Institute (CDRI), Lucknow.**

Quick fact: amniocentesis

In **amniocentesis**, some of the **amniotic fluid** of the developing foetus is taken to analyse the foetal cells and dissolved substances. It is used to test for genetic disorders such as **Down syndrome, haemophilia and sickle-cell anaemia**, and to determine the survivability of the foetus. Its misuse for sex determination is banned.

Indicators of improved reproductive health in society PYQ

- ✓ Better awareness about sex-related matters.
- ✓ More medically assisted deliveries and better post-natal care, so **decreased maternal and infant mortality rates**.
- ✓ More couples with **small families**.
- ✓ Better detection and cure of STDs; more medical facilities for sex-related problems.

2. Population Stabilisation and Birth Control

Increased health facilities along with better living conditions had an **explosive impact** on the growth of population.

Population	Figure
World, 1900	about **2 billion**
World, 2000	about **6 billion**
World, 2011	**7.2 billion**
India, at independence	about **350 million**
India, 2000	close to **1 billion**
India, May 2011	crossed **1.2 billion**

Reasons for population explosion PYQ

- → Rapid decline in **death rate**.
- → Decline in **maternal mortality rate (MMR)** and **infant mortality rate (IMR)**.
- → Increase in the number of people in the **reproducible age**.

According to the **2011 census**, the population growth rate was less than **2 per cent (20/1000/year)**, a rate at which the population could still increase rapidly and cause scarcity of basic needs: **food, shelter and clothing**.

Steps taken to check population growth:

- → Motivating smaller families through contraceptive methods; slogan **Hum Do Hamare Do**; many young urban couples adopt a **one child norm**.
- → Statutory raising of the **marriageable age**: females 18 years and males 21 years. PYQ
- → **Incentives** to couples with small families.

Ideal contraceptive Board fav

- ✓ User-friendly and easily available.
- ✓ Effective and **reversible**.
- ✓ No or least side-effects.
- ✓ Does not interfere with the sexual drive, desire and/or the sexual act.

Natural → Barrier → IUDs → Oral pills → Injectables → Implants → Surgical

Yaad karo: methods ka order

New **B**oys **I**n **O**ur **I**gloo **S**leep = Natural, Barrier, IUDs, Oral pills, Injectables / Implants, Surgical.

Natural vs artificial methods

Natural methods	Artificial methods
no medicines or devices	use devices, drugs or surgery
side effects almost nil	possible side effects
high chances of failure	more effective
e.g. periodic abstinence	e.g. condoms, IUDs, pills, sterilisation

A. Natural / traditional methods

They work on the principle of avoiding the chances of ovum and sperms meeting.

Method	How it works
Periodic abstinence	couple avoids coitus from **day 10 to 17** of the cycle, when ovulation is expected; this is the fertile period
Withdrawal (coitus interruptus)	male withdraws the penis from the vagina just before ejaculation, to avoid insemination
Lactational amenorrhea	absence of menstruation: ovulation and the cycle do not occur during intense lactation after parturition; effective only up to six months PYQ

As no medicines or devices are used, **side effects are almost nil**, but the **chances of failure are high**.

B. Barrier methods

Ovum and sperms are prevented from physically meeting with the help of **barriers**, available for both males and females.

- → **Condoms**: thin **rubber/latex sheath** covering the penis (male) or the vagina and cervix (female) just before coitus, so that ejaculated semen does not enter the female tract. **Nirodh** is a popular brand of male condom.
- → Condom use has increased because they also protect from STIs and AIDS. Both male and female condoms are **disposable**, can be **self-inserted** and give privacy. PYQ
- → **Diaphragms, cervical caps and vaults**: rubber barriers inserted into the female tract to **cover the cervix** during coitus; they block entry of sperms and are reusable.
- → **Spermicidal creams, jellies and foams** are used along with these barriers to increase their efficiency.

C. Intra Uterine Devices (IUDs)

IUDs are inserted by **doctors or expert nurses** in the uterus through the vagina. They are one of the most widely accepted methods of contraception in India.

Type	Examples	Action
Non-medicated	**Lippes loop**	increase **phagocytosis** of sperms in the uterus
Copper releasing	**CuT, Cu7, Multiload 375**	**Cu ions** suppress sperm motility and fertilising capacity
Hormone releasing	**Progestasert, LNG-20**	also make the uterus **unsuitable for implantation** and the cervix hostile to sperms

IUDs are ideal for females who want to delay pregnancy and/or space children. Board fav

D. Oral contraceptives (pills)

→ Small doses of either **progestogens** or **progestogen-estrogen** combinations, taken as tablets.

→ Taken daily for 21 days, starting preferably within the **first five days** of the menstrual cycle; after a gap of **7 days** (when menstruation occurs) repeated in the same pattern.

→ They inhibit ovulation and implantation and alter the quality of **cervical mucus** to prevent or retard entry of sperms. PYQ

→ Very effective, with lesser side effects; well accepted by females.

Quick fact: Saheli

Saheli, the new oral contraceptive for females, contains a non-steroidal preparation. It is a **once a week** pill with very few side effects and high contraceptive value, developed at CDRI, Lucknow.

Exam Alert

→ Pills **ovulation aur implantation dono** rokti hain + cervical mucus badalti hain: 3 actions likhna.

→ Saheli = **non-steroidal**, once a week; baaki pills steroidal (progestogen/estrogen).

E. Injections, implants and emergency contraceptives

→ Progestogens alone or with estrogen can also be used as **injections** or **implants under the skin**. Their mode of action is like pills, but their effective periods are much longer.

→ **Emergency contraceptives**: progestogens, progestogen-estrogen combinations or IUDs given within 72 hours of coitus are very effective to avoid possible pregnancy due to rape or casual unprotected intercourse. Board fav

F. Surgical methods (sterilisation)

Advised as a **terminal method** to prevent any more pregnancies. Surgical intervention blocks gamete transport and thereby prevents conception.

Vasectomy (male)	Tubectomy (female)
small part of the **vas deferens** removed or tied up	small part of the **fallopian tube** removed or tied up
through a small incision on the **scrotum**	through a small incision in the **abdomen** or through the vagina

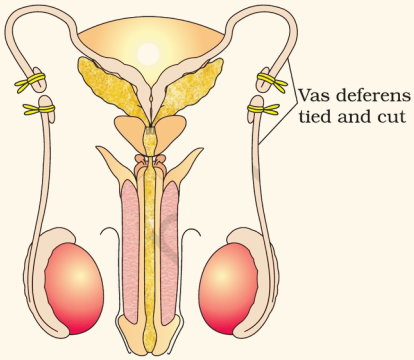


Fig. 3.4 (a) Vasectomy

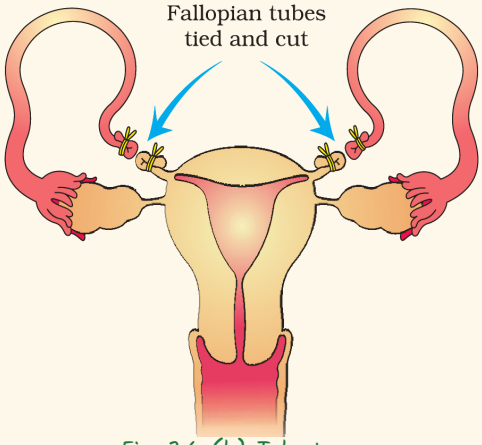


Fig. 3.4 (b) Tubectomy

These techniques are **highly effective** but their reversibility is very poor.

Exam Alert

→ Surgical methods gamete **formation** nahi rokte, sirf gamete **transport** block karte hain.

→ Removal of gonads contraceptive nahi hai: hormones bhi band ho jaayenge aur ye irreversible hai.

→ IUD = delay/spacing; sterilisation = terminal method.

Selection of a contraceptive should always be done in consultation with **qualified medical professionals**. Contraceptives are not regular requirements for reproductive health; they are practised against a natural event (conception). Possible ill-effects: **nausea, abdominal pain, breakthrough bleeding, irregular menstrual bleeding or even breast cancer**, though not very significant, should not be ignored.

Contraceptives at a glance Board fav

Method	Example	Main action
Natural	periodic abstinence, withdrawal, lactational amenorrhea	avoid meeting of ovum and sperm
Barrier	condom, diaphragm, cervical cap, vault	physical block to sperms
IUD	Lippes loop, CuT, LNG-20	phagocytosis of sperms; Cu ions; hostile uterus
Pills	progestogen or progestogen-estrogen; Saheli	inhibit ovulation and implantation
Injectables / implants	progestogens under skin	like pills, longer effect
Surgical	vasectomy, tubectomy	block gamete transport

3. Medical Termination of Pregnancy (MTP)

Intentional or voluntary termination of pregnancy before full term is called medical termination of pregnancy (MTP) or **induced abortion**. Nearly **45 to 50 million** MTPs are performed in a year all over the world, about **1/5th** of all conceived pregnancies.

→ Government of India legalised MTP in 1971 with strict conditions to avoid its misuse, especially to check illegal female foeticide. PYQ

→ Why MTP: to get rid of unwanted pregnancies due to casual unprotected intercourse, failure of the contraceptive, or rapes; also when continuing the pregnancy could be harmful or fatal to the mother, the foetus or both.

→ MTPs are relatively safe during the first trimester (up to 12 weeks); second trimester abortions are much riskier.

MTP (Amendment) Act, 2017	Opinion needed
within the first **12 weeks**	**one** registered medical practitioner
more than 12 but fewer than **24 weeks**	**two** registered medical practitioners

Grounds: (i) continuing the pregnancy would risk the woman's life or cause grave injury to her physical or mental health; or (ii) substantial risk that the child would be born with serious physical or mental abnormalities.

Exam Alert: disturbing trends

→ Zyaadatar MTPs **unqualified quacks** illegally karte hain: unsafe aur fatal ho sakta hai.

→ **Amniocentesis** se sex pata karke female foetus ka MTP: totally illegal.

→ Solution: counselling on unprotected coitus, risks of illegal abortion, and more health care facilities.

4. Sexually Transmitted Infections (STIs)

Infections or diseases transmitted through sexual intercourse are collectively called sexually transmitted infections (STI), **venereal diseases (VD)** or **reproductive tract infections (RTI)**.

STI	Pathogen
Gonorrhoea	*Neisseria gonorrhoeae* (bacterium)
Syphilis	*Treponema pallidum* (bacterium)
Chlamydiasis	*Chlamydia trachomatis* (bacterium)
Trichomoniasis	*Trichomonas vaginalis* (protozoan)
Genital herpes	Herpes simplex virus
Genital warts	Human papilloma virus (HPV)
Hepatitis-B	Hepatitis B virus
AIDS	HIV (Chapter 7)

→ **Hepatitis-B and HIV** can also spread by sharing injection needles and surgical instruments, blood transfusion, or from an infected mother to the foetus.

→ Except hepatitis-B, genital herpes and HIV, other STIs are completely curable if detected early and treated properly. Board fav

→ Early symptoms are minor: **itching, fluid discharge, slight pain, swellings** in the genital region. Infected females may often be **asymptomatic**.

→ Late or no treatment leads to **pelvic inflammatory diseases (PID)**, abortions, still births, **ectopic pregnancies**, infertility or even cancer of the reproductive tract.

→ Incidence is very high in the age group of 15–24 years.

Yaad karo: 3H jo poori tarah cure nahi hote

Hepatitis-B, Herpes (genital), HIV. Baaki STIs early detection aur poore treatment se **completely curable** hain.

Prevention of STIs (NCERT) PYQ

1. Avoid sex with unknown partners or multiple partners.
2. Always try to use condoms during coitus.
3. In case of doubt, go to a qualified doctor for early detection and get complete treatment.

Paheli-Bujho

Bujho: STI ho gaya toh chupke se medical store se dawai le lunga.

Paheli: Galat! Qualified doctor ke paas jao, poora treatment lo.

Bujho: Par sab toh cure ho jaate hain na?

Paheli: Nahi, hepatitis-B, herpes aur HIV completely curable nahi hain.

5. Infertility

Couples who are unable to produce children in spite of unprotected sexual co-habitation are infertile. Reasons can be **physical, congenital, diseases, drugs, immunological or even psychological.**

In India the female is often blamed for being childless, but **more often than not the problem lies in the male partner.** Infertility clinics help in diagnosis and treatment; where correction is not possible, couples are helped by assisted reproductive technologies (ART).

Assisted reproductive technologies Board fav

Technique	What is done
IVF + ET (test tube baby)	ova from wife/donor and sperms from husband/donor fuse to form a zygote **in the laboratory** under simulated conditions; then embryo transfer
ZIFT (zygote intra fallopian transfer)	zygote or early embryo up to 8 blastomeres transferred into the **fallopian tube**
IUT (intra uterine transfer)	embryo with more than 8 blastomeres transferred into the **uterus**

Technique	What is done
GIFT (gamete intra fallopian transfer)	ovum from a donor transferred into the fallopian tube of a female who cannot produce one but can provide a suitable environment for fertilisation
ICSI (intra cytoplasmic sperm injection)	a sperm is **directly injected into the ovum** to form an embryo in the laboratory
AI (artificial insemination)	semen from husband or a healthy donor introduced into the vagina or into the uterus (**IUI**, intra-uterine insemination); used when the male cannot inseminate or has very low sperm count

Embryos formed by **in-vivo fertilisation** (fusion of gametes within the female) can also be used for such transfer.

Yaad karo: ZIFT vs IUT

8 tak = tube (ZIFT), **8 se zyada = uterus** (IUT). GIFT mein **G** = Gamete (ovum) jaata hai, zygote nahi.

Limitations: need high precision handling by specialised professionals and expensive instruments, so available in very few centres and affordable to few. Emotional, religious and social factors are also deterrents. Since India has many orphaned and destitute children, legal adoption is one of the best methods for couples looking for parenthood.

1 minute recap

1. WHO: total well-being, physical, emotional, behavioural and social; India started family planning in 1951; now RCH.
2. Ban on amniocentesis for sex determination; Saheli (CDRI Lucknow), non-steroidal, once a week.
3. Population: rapid fall in death rate, MMR, IMR; marriage age 18 (F) / 21 (M).
4. Natural (periodic abstinence day 10–17, withdrawal, lactational amenorrhea up to 6 months), barrier, IUDs (Cu ions, phagocytosis), pills (21 days + 7 gap), injections/implants, emergency within 72 h, vasectomy/tubectomy.
5. MTP legal since 1971; Act 2017: up to 12 weeks one doctor, 12–24 weeks two doctors.
6. STIs: hepatitis-B, herpes, HIV not completely curable; 15–24 years most affected.
7. ART: IVF-ET, ZIFT (up to 8 cells, tube), IUT (more than 8, uterus), GIFT, ICSI, AI/IUI; adoption.

Notes khatam! Ab choice tumhari: scroll ya MCQ?

MOST IMPORTANT QUESTIONS

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM Board fav Saheli, the oral contraceptive pill, was developed at:
(a) AIIMS, Delhi
(b) CDRI, Lucknow
(c) NBRI, Lucknow
(d) IARI, New Delhi
Ans: (b) CDRI, Lucknow

Q2. IM Board fav Which of the following is a hormone releasing IUD?
(a) Lippes loop
(b) Multiload 375
(c) LNG-20
(d) Cu7
Ans: (c) LNG-20

Q3. IM Lactational amenorrhea is effective as a contraceptive up to a maximum of:
(a) six months after parturition
(b) one year
(c) two months
(d) three years
Ans: (a) six months after parturition

Q4. IM Board fav In ZIFT, the embryo transferred into the fallopian tube has:
(a) 16 blastomeres
(b) more than 8 blastomeres
(c) 32 cells
(d) up to 8 blastomeres
Ans: (d) up to 8 blastomeres

Q5. IM Which STI is NOT completely curable?
(a) Gonorrhoea
(b) Genital herpes
(c) Syphilis
(d) Trichomoniasis
Ans: (b) Genital herpes

Q6. IM The statutory marriageable age for females and males in India is respectively:
(a) 21 and 18
(b) 16 and 18
(c) 18 and 21 years
(d) 20 and 25
Ans: (c) 18 and 21 years

Q7. IM Which of the following is NOT an ART?
(a) GIFT
(b) ICSI
(c) Amniocentesis
(d) ZIFT
Ans: (c) Amniocentesis

Q8. 1M Emergency contraceptives are effective if used within ____ of coitus.
Ans: 72 hours.

Q9. 1M Name the surgical method of sterilisation in males.
Ans: Vasectomy.

Q10. 1M What is the fertile period in a 28-day cycle?
Ans: Day 10 to 17 of the menstrual cycle.

Assertion-Reason (1 mark each)

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q11. 1M Board fav **A:** Use of condoms has increased in recent years. **R:** Condoms also protect the user from STIs and AIDS.
Ans: (a) Both true and R explains A.

Q12. 1M **A:** Copper releasing IUDs are effective contraceptives. **R:** Cu ions suppress sperm motility and the fertilising capacity of sperms.
Ans: (a) Both true and R explains A.

Q13. 1M Board fav **A:** Surgical methods of contraception prevent gamete formation. **R:** In vasectomy, a part of the vas deferens is removed or tied.
Ans: (d) A is false (they block gamete transport, not formation), R is true.

Q14. 1M **A:** MTP is relatively safe up to 12 weeks of pregnancy. **R:** Second trimester abortions are much riskier.
Ans: (b) Both true, but R does not explain A.

Short answer (2-3 marks)

Q15. 2M Board fav NCERT: What is the significance of reproductive health in a society?
Ans: A reproductively healthy society has physically and functionally normal reproductive organs and normal emotional and behavioural interactions. It means fewer STDs, safe pregnancies, lower maternal and infant mortality, planned small families, healthy children and control of population growth.

Q16. 3M NCERT: Suggest the aspects of reproductive health that need special attention in the present scenario.
Ans: Awareness and sex education for adolescents; prevention and early cure of STIs including AIDS; safe contraception and family planning; care of pregnant mothers and post-natal care; stopping female foeticide (amniocentesis misuse) and illegal MTPs; help for infertile couples.

Q17. 2M Board fav NCERT: Is sex education necessary in schools? Why?
Ans: Yes. It gives correct information about reproductive organs, adolescence, safe and hygienic sexual practices, STDs and AIDS, and removes myths and misconceptions, so the young can lead a reproductively healthy life.

Q18. 3M NCERT: Has reproductive health in our country improved in the past 50 years? Mention some areas of improvement.
Ans: Yes: better awareness of sex-related matters, more medically assisted deliveries, better post-natal care, lower maternal and infant mortality rates, more couples with small families, better detection and cure of STDs, and more medical facilities for reproductive problems.

Q19. 2M Board fav NCERT: What are the suggested reasons for population explosion?
Ans: Rapid decline in death rate, maternal mortality rate (MMR) and infant mortality rate (IMR), and an increase in the number of people in the reproducible age, due to better health facilities and living conditions.

Q20. 2M NCERT: Is the use of contraceptives justified? Give reasons.
Ans: Yes. They help check population growth, prevent unwanted pregnancies, allow delaying and spacing of children, and condoms also prevent STIs. They should be chosen with medical advice, as some have side effects.

Q21. 2M Board fav NCERT: Removal of gonads cannot be considered a contraceptive option. Why?
Ans: Gonads also produce hormones (androgens, estrogen, progesterone) needed for secondary sexual characters and body functions. Their removal is irreversible and harmful, whereas contraceptives should be reversible with minimal side effects.

Q22. 2M Board fav NCERT: Amniocentesis for sex determination is banned in our country. Is this ban necessary? Comment.
Ans: Yes. Amniocentesis is meant to detect genetic disorders, but it was misused to find the sex of the foetus followed by abortion of female foetuses. The ban checks female foeticide and protects the sex ratio.

Q23. 2M NCERT: Suggest some methods to assist infertile couples to have children.
Ans: Diagnosis and treatment in infertility clinics; ART: IVF with embryo transfer (ZIFT, IUT), GIFT, ICSI, artificial insemination (AI/IUI); legal adoption.

Q24. 2M Board fav NCERT: What measures should one take to prevent contracting STDs?
Ans: Avoid sex with unknown or multiple partners; always use condoms during coitus; consult a qualified doctor early in case of doubt and complete the treatment; do not share needles; use screened blood.

Q25. 3M NCERT: True/False with explanation: (a) Abortions could happen spontaneously too. (b) Infertility is always due to defects in the female partner. (c) Complete lactation could help as a natural method of contraception. (d) Creating awareness about sex-related aspects improves reproductive health.

Ans: (a) True: spontaneous abortion (miscarriage) can occur naturally. (b) False: it may be due to either partner, and more often the male. (c) True: during intense lactation there is no ovulation (up to 6 months). (d) True: awareness prevents STDs, unwanted pregnancies and myths.

Q26. 3M *Board fav* NCERT: Correct the statements: (a) Surgical methods prevent gamete formation. (b) All STDs are completely curable. (c) Oral pills are very popular among rural women. (d) In ET techniques, embryos are always transferred into the uterus.

Ans: (a) Surgical methods block gamete transport, not formation. (b) Hepatitis-B, genital herpes and HIV are not completely curable. (c) Oral pills are popular among urban women (IUDs are widely accepted). (d) Embryos up to 8 blastomeres go into the fallopian tube (ZIFT); more than 8 into the uterus (IUT).

Q27. 3M *Board fav* How do oral contraceptive pills prevent pregnancy? How are they taken?

Ans: They contain progestogens or progestogen-estrogen combinations; they inhibit ovulation and implantation and change cervical mucus to retard sperm entry. Taken daily for 21 days from the first five days of the cycle, then a 7-day gap, repeated.

Q28. 3M *Board fav* Give the mode of action of the three types of IUDs with one example each.

Ans: Non-medicated (Lippes loop): increase phagocytosis of sperms. Copper releasing (CuT, Multiload 375): Cu ions reduce sperm motility and fertilising capacity. Hormone releasing (Progestasert, LNG-20): also make the uterus unsuitable for implantation and the cervix hostile to sperms.

Q29. 2M Differentiate between vasectomy and tubectomy.

Ans: Vasectomy: male sterilisation; a small part of the vas deferens is removed or tied through a small incision on the scrotum. Tubectomy: female sterilisation; a small part of the fallopian tube is removed or tied through a small incision in the abdomen or through the vagina.

Q30. 3M *Board fav* Why is MTP legalised in India, and under what conditions does the MTP (Amendment) Act, 2017 allow it?

Ans: To reduce illegal abortions and maternal deaths. Up to 12 weeks on the opinion of one registered medical practitioner; 12-24 weeks on the opinion of two, when continuing the pregnancy risks the woman's life or health, or the child would have serious abnormalities.

Q31. 2M Name any two STIs caused by bacteria and two caused by viruses.

Ans: Bacteria: gonorrhoea, syphilis (also chlamydiasis). Viruses: genital herpes, genital warts (also hepatitis-B, AIDS).

Q32. 2M *Board fav* Differentiate between ZIFT and GIFT.

Ans: ZIFT: zygote or early embryo (up to 8 blastomeres) formed by IVF is transferred into the fallopian tube. GIFT: an ovum (gamete) from a donor is transferred into the fallopian tube of a female who cannot produce ova; fertilisation happens inside her body.

Q33. 2M What is ICSI? When is artificial insemination used?

Ans: ICSI: a sperm is directly injected into the ovum in the laboratory to form an embryo. AI is used when the male cannot inseminate or has very low sperm counts; semen from the husband or a donor is introduced into the vagina or uterus (IUI).

Q34. 3M Why are STIs often detected late, and what complications can follow?

Ans: Early symptoms are minor or absent (especially in females), and social stigma stops people from getting tested. Complications: PID, abortions, still births, ectopic pregnancy, infertility, cancer of the reproductive tract.

Q35. 2M What is lactational amenorrhea? Why is it called a natural method?

Ans: Absence of menstruation during intense lactation after parturition, because ovulation does not occur; while the mother fully breast-feeds, chances of conception are almost nil (up to 6 months). No medicines or devices are used, so it is natural.

Q36. 2M Why are diaphragms and cervical caps used with spermicides?

Ans: They cover the cervix and block sperms; spermicidal creams, jellies or foams kill sperms that escape, increasing contraceptive efficiency.

Q37. 2M Give two possible ill-effects of contraceptives. Why should they be used only with medical advice?

Ans: Nausea, abdominal pain, breakthrough bleeding, irregular menstrual bleeding, even breast cancer (any two). A qualified professional selects the method that suits the person and minimises side effects.

Q38. 2M *Board fav* What is amniocentesis? Give its legitimate use.

Ans: Taking some amniotic fluid of the developing foetus to analyse foetal cells and dissolved substances. It is used to detect genetic disorders like Down syndrome, haemophilia and sickle-cell anaemia and to judge foetal survivability.

Q39. 2M Why do hormone releasing IUDs work better than copper IUDs in some respects?

Ans: Besides the sperm effects, they make the uterus unsuitable for implantation and the cervix hostile to sperms, giving a double barrier to conception.

Q40. 2M Board fav Name the measures taken by the government to check population growth.
Ans: Promoting small families (Hum Do Hamare Do, one child norm) through contraceptives; raising the marriageable age to 18 (females) and 21 (males); incentives to couples with small families.

Q41. 3M Why are STIs said to be a major threat to a healthy society? Which age group is most affected?
Ans: They are often silent at first, carry stigma and lead to PID, infertility, ectopic pregnancy, still births and cancer; hepatitis-B and HIV also spread through blood and needles and are not fully curable. The 15–24 years age group is most affected.

Q42. 2M Define the fertile period. How does periodic abstinence prevent conception?
Ans: Day 10 to 17 of the menstrual cycle, when ovulation is expected and chances of fertilisation are very high. Avoiding coitus during these days prevents the ovum and sperms from meeting.

Q43. 1M Why are condoms and female condoms said to give privacy to the user?
Ans: They are disposable and can be self-inserted by the user, so no doctor or nurse is needed.

Q44. 2M Why is legal adoption suggested as one of the best options for childless couples in India?
Ans: ART is costly, needs specialised professionals and is available in very few centres; emotional, religious and social factors also deter its use. India has many orphaned and destitute children, and adoption gives them a family.

Long answer (5 marks)

Q45. 5M Board fav (a) What is MTP? Why is it needed? (b) Give the provisions of the MTP (Amendment) Act, 2017. (c) Mention two disturbing trends linked with MTP in India and how to reverse them.
Ans: (a) Intentional termination of pregnancy before full term (induced abortion); needed to end unwanted pregnancies from casual unprotected intercourse, contraceptive failure or rape, and when continuing the pregnancy is harmful to the mother or foetus. (b) Up to 12 weeks: opinion of one registered medical practitioner; 12–24 weeks: two practitioners; grounds: risk to the woman's life or health, or substantial risk of serious abnormality in the child. (c) Illegal MTPs by unqualified quacks; misuse of amniocentesis for sex determination followed by abortion of female foetuses. Reverse by counselling against unprotected coitus, awareness of risks and more health care facilities.

Q46. 5M Board fav Describe the different contraceptive methods with their mode of action and one example of each.
Ans: Natural: avoid meeting of gametes: periodic abstinence (day 10–17), withdrawal, lactational amenorrhea (up to 6 months). Barrier: condoms (male, female), diaphragm, cervical cap, vault, with spermicides: stop sperms reaching the ovum. IUDs: Lippes loop (phagocytosis), CuT (Cu ions), LNG-20 (also blocks implantation). Oral pills: progestogens or combinations, inhibit ovulation and implantation, alter cervical mucus; Saheli weekly. Injections and implants: like pills, longer action. Emergency: within 72 h. Surgical: vasectomy and tubectomy block gamete transport; terminal, poorly reversible.

Q47. 5M Board fav What is infertility? Describe the assisted reproductive technologies that help infertile couples.
Ans: Inability to produce children despite unprotected co-habitation; causes may be physical, congenital, diseases, drugs, immunological or psychological, in either partner. ART: IVF-ET (test tube baby): gametes fuse in the laboratory; ZIFT: zygote/embryo up to 8 blastomeres into fallopian tube; IUT: embryo with more than 8 blastomeres into uterus; GIFT: donor ovum into fallopian tube; ICSI: sperm injected into ovum; AI/IUI: semen into vagina or uterus. Costly and few centres; adoption is a good option.

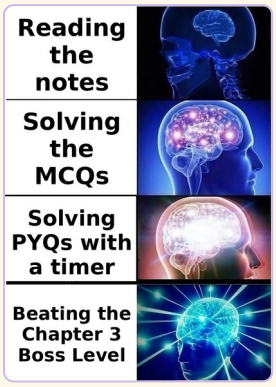
Q48. 5M Explain the strategies adopted in India to achieve reproductive health and to check population growth.
Ans: Family planning (1951), now RCH programmes; awareness through media, parents, teachers; sex education in schools; education of couples on birth control, maternal and child care, breast feeding and equal opportunity for girls; medical facilities for pregnancy, delivery, STDs, contraception, infertility; ban on amniocentesis for sex determination; child immunisation; research (Saheli). For population: Hum Do Hamare Do, one child norm, marriage age 18/21, incentives to small families.

Case-based (4 marks)

Q49. 4M Board fav A young mother who fully breast-feeds her 3-month-old baby has not menstruated since delivery. Her friend, a working woman, wants to space her next child by 3 years. (a) Why has the mother not menstruated? (b) How long can she rely on this as contraception? (c) Suggest the best method for the friend and give its action. (d) Name a weekly pill option.
Ans: (a) Lactational amenorrhea: no ovulation during intense lactation. (b) Up to a maximum of six months after parturition. (c) An IUD such as CuT: Cu ions reduce sperm motility and fertilising capacity; ideal for spacing. (d) Saheli.

Q50. 4M A couple could not conceive after years of trying. Tests showed the wife's fallopian tubes are healthy but she produces no ova; the husband has a very low sperm count. (a) Which ART suits the wife's condition? (b) Which technique can use the husband's few sperms most directly? (c) Where would an embryo of 16 cells be transferred and what is the method called? (d) Give one limitation of ART.

Ans: (a) GIFT, with a donor ovum transferred into her fallopian tube. (b) ICSI (or AI/IUI). (c) Into the uterus: IUT. (d) Expensive and available only in few centres; needs highly skilled professionals.



MCQ se case-based tak pahunche? Boss level clear!

Q51. 4M Board fav A 20-year-old college student noticed itching and discharge but felt shy to see a doctor. (a) What kind of infection could this be? (b) Why is his age group important? (c) Give two complications of delay. (d) List the NCERT principles to prevent it.

Ans: (a) A sexually transmitted infection (STI/RTI). (b) Incidence of STIs is highest in the 15–24 years age group. (c) PID, infertility, ectopic pregnancy, still birth, cancer of reproductive tract (any two). (d) Avoid unknown or multiple partners; always use condoms; consult a qualified doctor early and complete treatment.

Q52. 4M Board fav A pregnant woman in her 4th month was offered a test in which a little fluid around the foetus is drawn. Her relatives wanted to use it to know if the baby is a girl. (a) Name the test. (b) What is its proper use? (c) Is the relatives' demand legal? Why is it banned? (d) Name the national programme that covers such issues.

Ans: (a) Amniocentesis. (b) To detect genetic disorders like Down syndrome, haemophilia, sickle-cell anaemia, and to check survivability of the foetus. (c) No; sex determination by amniocentesis is statutorily banned to stop female foeticide. (d) Reproductive and Child Health Care (RCH) programme.

Q53. 4M Board fav Match each couple to a suitable method with its action: (a) A couple who never want more children. (b) A woman who wants a once-a-week pill. (c) A woman who had unprotected intercourse last night. (d) A couple who also want protection from STIs.

Ans: (a) Sterilisation (vasectomy or tubectomy): blocks gamete transport; terminal. (b) Saheli: non-steroidal weekly pill. (c) Emergency contraceptive (progestogen or progestogen-estrogen pills, or IUD) within 72 hours. (d) Condoms: barrier that stops semen entry and protects from STIs and AIDS.

Q54. 4M Data: world population was about 2 billion in 1900, 6 billion in 2000 and 7.2 billion in 2011; India crossed 1.2 billion in May 2011 with a growth rate below 2 per cent. (a) Give two reasons for this rise. (b) Why is even a 2 per cent rate alarming? (c) Name two government measures to check it. (d) Mention one feature of an ideal contraceptive.

Ans: (a) Rapid decline in death rate, MMR and IMR; more people in the reproducible age. (b) At 20/1000/year the population still grows fast and can cause scarcity of food, shelter and clothing. (c) Raising the marriageable age to 18/21; incentives to small families; Hum Do Hamare Do campaign (any two). (d) User-friendly, easily available, effective, reversible, least side effects, no interference with sexual drive (any one).

Q55. 4M Read: "In a village health camp, a doctor found that many women relied on withdrawal and periodic abstinence and had frequent unplanned pregnancies. She suggested IUDs and explained the RCH programme." (a) Why did the natural methods fail often? (b) Why are IUDs a good choice here? (c) Who should insert an IUD? (d) State one task of the RCH programme.

Ans: (a) Natural methods have high chances of failure (wrong timing, sperms may be released before withdrawal). (b) IUDs are effective, reversible, long-acting and ideal for spacing children; widely accepted in India. (c) A doctor or expert nurse, through the vagina into the uterus. (d) Creating awareness about reproduction-related aspects / providing facilities and support for a reproductively healthy society.

Exam Alert: last minute

- → Numbers: 1951, 1971, 2017, 12/24 weeks, 18/21 years, day 10–17, 21+7 days, 72 hours, 6 months, 15–24 years.
- → Examples ko type ke saath likho: Lippes loop (non-medicated), CuT (copper), LNG-20 (hormonal).
- → ZIFT/IUT/GIFT ka difference 2 marks ka pakka sawaal hai.
- → NCERT ke 12 exercise questions upar Q15–Q26 mein hain: inhe zaroor likh ke practise karo.

Principles of Inheritance and Variation

Most asked: monohybrid and dihybrid crosses, test cross, ABO blood groups (co-dominance, multiple alleles), sex determination, sickle-cell anaemia, pedigree analysis, Down, Klinefelter and Turner syndromes

Genetics deals with **inheritance** (passing of characters from parents to progeny) and **variation** (differences among individuals). This chapter covers **Mendel's laws, their exceptions, chromosomal basis, sex determination, mutation and genetic disorders.**

Humans knew from as early as **8000–1000 B.C.** that one of the causes of variation was hidden in sexual reproduction. They exploited the variations in wild populations through artificial selection and domestication. Example: the **Sahiwal cows** of Punjab.

→ **Inheritance**: the process by which characters are passed on from parent to progeny; it is the basis of **heredity**.

→ **Variation**: the degree by which progeny differ from their parents.

1. Mendel's Laws of Inheritance

Gregor Mendel conducted hybridisation experiments on garden peas for **seven years (1856–1863)** and proposed the laws of inheritance. He was the first to apply statistical analysis and mathematical logic to problems in biology, and his experiments had a **large sampling size**, which gave credibility to his data.

He used several true-breeding pea lines. A true-breeding line has undergone continuous self-pollination and shows stable trait inheritance and expression for several generations. Mendel selected **14 true-breeding varieties**, as pairs similar except for one character with **contrasting traits.** PYQ

Character	Dominant	Recessive
Stem height	tall	dwarf
Flower colour	violet	white
Flower position	axial	terminal
Pod shape	inflated (full)	constricted
Pod colour	green	yellow
Seed shape	round	wrinkled
Seed colour	yellow	green

For each character, Mendel first crossed two true-breeding plants with contrasting traits, then let the hybrids self-pollinate, and counted the progeny of every generation.

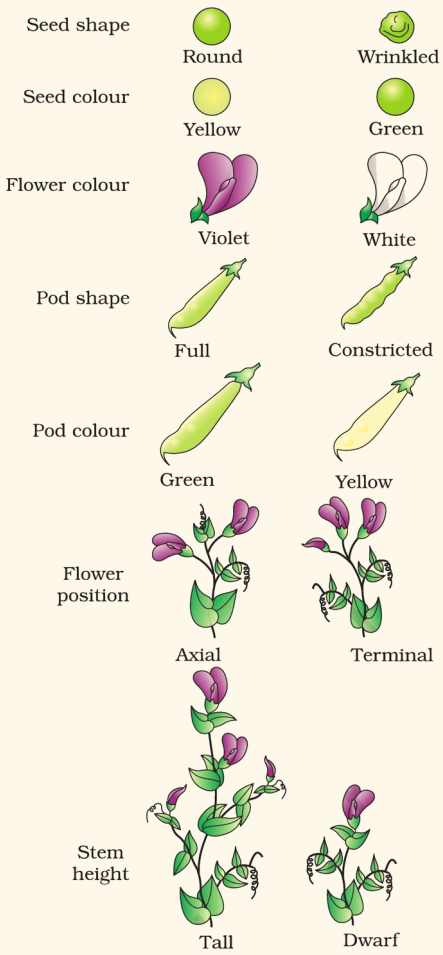


Fig. 4.1 Seven pairs of contrasting traits in pea plant studied by Mendel

Why pea? (advantages) Board fav

→ Many clear **contrasting traits** are available.
→ Flowers are bisexual and naturally **self-pollinating**, so pure lines are easy; cross-pollination can also be done artificially.
→ Short life cycle; each cross gives a large number of seeds.
→ Easy to grow; F1 hybrids are fertile.

Mendel's method of crossing

1. **Emasculation**: anthers of the flower chosen as female parent are removed before they mature, to prevent self-pollination.
2. **Bagging**: the emasculated flower is covered to protect it from unwanted pollen.
3. **Artificial pollination**: pollen from the chosen male parent is dusted on the stigma; the flower is bagged again.
4. Seeds formed are collected and grown to obtain the next generation.

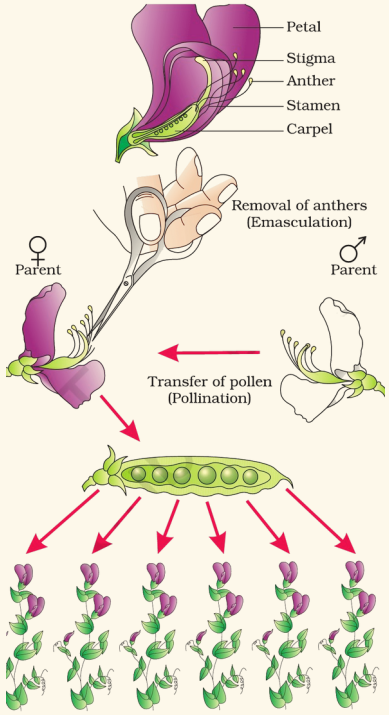


Fig. 4.2 Steps in making a cross in pea

Yaad karo: 7 traits

Stem, Flower colour, Flower position, Pod shape, Pod colour, Seed shape, Seed colour = **SFFPPSS**: "Sir Fir Fail, Papa Pitenge Sach Se!"

2. Inheritance of One Gene

Mendel crossed **tall and dwarf** pea plants. The first hybrid generation is the Filial1 progeny (F1). All F1 plants were **tall**, like one parent; none was dwarf. He self-pollinated the tall F1 plants: in the **F2 generation**, some plants were dwarf. The proportion of dwarf plants was 1/4th and tall plants **3/4th**. No intermediate height was seen: the dwarf trait reappeared unchanged.

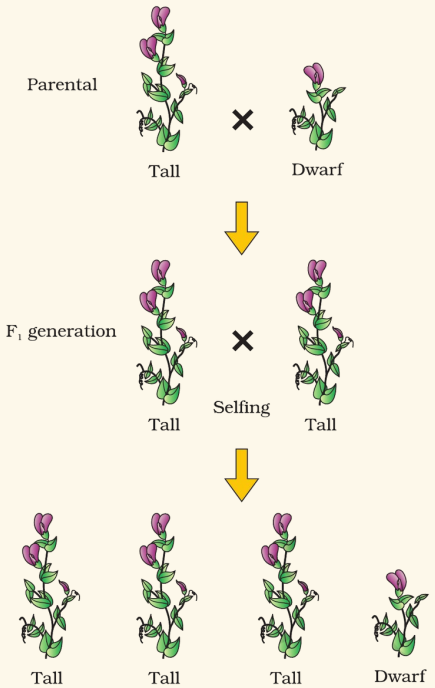


Fig. 4.3 Diagrammatic representation of monohybrid cross

Mendel's conclusions:

- → Something is stably passed down, unchanged, from parent to offspring through the gametes: Mendel called these things **factors**; now they are called genes, the units of inheritance.
- → Genes coding for a pair of contrasting traits are alleles: slightly different forms of the same gene.
- → Alleles are written with letters: **T** for tall (the trait seen in F1) and **t** for dwarf. A plant may be **TT, Tt or tt**.
- → In a true-breeding tall or dwarf plant the allelic pair is identical (**homozygous**: TT or tt); in Tt the alleles are dissimilar (**heterozygous**). Tt is a **hybrid**; a cross for one character is a monohybrid cross.

Term	Meaning
Genotype	genetic make-up, e.g. TT, Tt, tt
Phenotype	observable trait, e.g. tall or dwarf
Dominant	allele expressed in the heterozygote (T)
Recessive	allele not expressed in the heterozygote (t)

Key terms

- → **Gene**: unit of inheritance; a segment of DNA controlling a character.
- → **Locus**: fixed position of a gene on a chromosome.
- → **Pure line / true-breeding**: line giving the same trait for generations on selfing.
- → **F1, F2**: first and second filial generations.
- → **Hybrid**: offspring of parents differing in one or more traits.
- → **Back cross**: cross of F1 with either parent; with the recessive parent it is a test cross.

A. Punnett square

Developed by the British geneticist Reginald C. Punnett, it is a graphical representation to calculate the probability of all possible genotypes of offspring. Gametes of one parent are written on the top row and of the other on the left column; the boxes show the zygotes.

In F2: phenotypic ratio 3 tall : 1 dwarf; genotypic ratio 1 TT : 2 Tt : 1 tt. Mendel thus showed that in TT and Tt the plants are tall, though their genotypes differ. PYQ

Quick fact: the binomial way

Gametes of a Tt plant are T and t, each with probability 1/2. The F2 can be found by the binomial expression $(1/2\ T + 1/2\ t)^2 = 1/4\ TT + 1/2\ Tt + 1/4\ tt$, which is the same 1:2:1 genotypic ratio seen in the Punnett square.

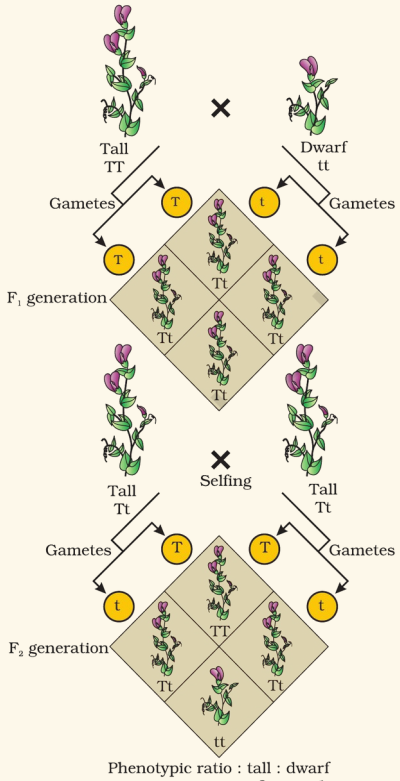


Fig. 4.4 A Punnett square used to understand a typical monohybrid cross

B. Test cross

To find the genotype of a tall plant (TT or Tt), it is crossed with a recessive (dwarf) parent. This cross of an organism showing a dominant phenotype with the recessive parent is called a test cross. Board fav

Tall plant tested	Cross	Progeny	Conclusion
TT	TT x tt	all tall (Tt)	homozygous
Tt	Tt x tt	1 tall : 1 dwarf	heterozygous

How to read a test cross

- → The recessive parent (tt) can give only t gametes, so it never hides anything.
- → So the progeny phenotypes directly show the **gametes of the tested plant.**
- → All dominant: the plant makes only T gametes (TT). Half recessive: it makes T and t equally (Tt).

C. Mendel's first two laws

Law	Statement
Law of Dominance	(i) characters are controlled by discrete units (factors); (ii) factors occur in pairs; (iii) in a dissimilar pair, one member dominates (dominant) and the other does not (recessive). It explains why only one parental trait appears in F1 and why both appear in F2 in 3:1
Law of Segregation	alleles do not show any blending and both characters are recovered as such in F2, though one is not seen in F1. During gamete formation the factors (alleles) of a pair segregate, so each gamete gets **only one** of the two (gametes are pure)

Exam Alert

- → Homozygous parent sirf **ek type** ke gametes banata hai; heterozygote **do type** ke equal frequency mein.
- → Test cross mein 1:1 aaye = heterozygous; sab dominant = homozygous.
- → Ratio yaad: monohybrid F2 phenotype 3:1, genotype 1:2:1.

Reciprocal cross

A pair of crosses in which the sexes of the parents are interchanged (tall female x dwarf male, and dwarf female x tall male). Mendel found the same F1 and F2 results in both, showing that the trait is **not influenced by the sex of the parent** (autosomal inheritance).

D. Incomplete dominance

When the F1 has a phenotype that does not resemble either parent and is **in between the two**, it is incomplete dominance. Example: flower colour in **dog flower (snapdragon, *Antirrhinum* sp.).**

- → Red-flowered (RR) x white-flowered (rr) gives **pink (Rr)** F1.
- → F1 selfed gives F2 1 red (RR) : 2 pink (Rr) : 1 white (rr).
- → Here the **genotypic and phenotypic ratios are the same (1:2:1)**; R is not completely dominant over r. PYQ

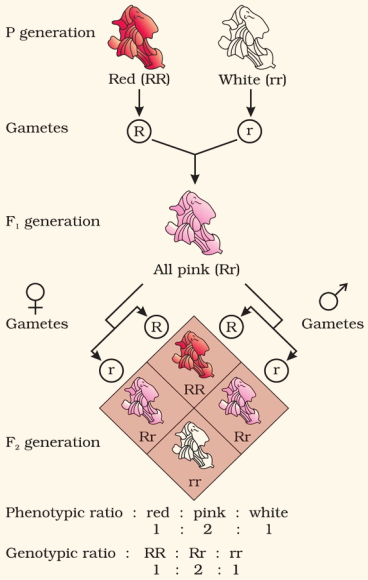


Fig. 4.6 Results of monohybrid cross in the plant snapdragon, where one allele is incompletely dominant over the other

E. Explaining the concept of dominance

Every gene has information to express a trait. In a diploid organism there are two copies of each gene (two alleles). If one allele is modified (say it codes for an enzyme), it may produce: (i) the **normal or less efficient enzyme**, (ii) a **non-functional enzyme**, or (iii) **no enzyme** at all.

In case (i) the modified allele gives the same phenotype, so it is **equivalent** to the unmodified allele. In cases (ii) and (iii) the phenotype depends only on the unmodified allele, which becomes the dominant allele; the modified one is **recessive.**

So the recessive trait is seen due to a non-functional enzyme or no enzyme.

F. Co-dominance and multiple alleles

In co-dominance, the F1 generation resembles both parents. Example: **ABO blood groups** in humans. ABO blood groups are controlled by the gene **I**; the plasma membrane of RBCs has sugar polymers that protrude from the surface, and the kind of sugar is controlled by this gene. Board fav

- → Gene I has three alleles: I^A, I^B and i. I^A and I^B produce slightly different forms of the sugar; i does not produce any sugar.
- → Each person has any **two** of the three alleles. I^A and I^B are completely dominant over i; when I^A and I^B are present together, **both express** their own types of sugars: this is co-dominance. Hence RBCs of AB have both A and B sugars.
- → Since there are three alleles, there are **six genotypes** and **four phenotypes**.
- → ABO is also an example of multiple alleles: more than two alleles govern the same character. Multiple alleles can be found only in a **population**, as an individual has only two alleles.

Allele from parent 1	Allele from parent 2	Genotype	Blood group
I^A	I^A	I^AI^A	A
I^A	I^B	I^AI^B	AB
I^A	i	I^Ai	A
I^B	I^A	I^AI^B	AB
I^B	I^B	I^BI^B	B
I^B	i	I^Bi	B
i	i	ii	O

Punnett square: I^Ai x I^Bi

Gametes	I^B	i
I^A	I^AI^B (AB)	I^Ai (A)
i	I^Bi (B)	ii (O)

Paheli-Bujho

Bujho: Mera blood group O hai, mummy A, papa B. Kya main adopted hoon?

Paheli: Nahi! Mummy I^Ai aur papa I^Bi ho sakte hain.

Bujho: Toh dono ne mujhe chupke se i de diya?

Paheli: Exactly, ii = O. Tum bilkul apne hi ho!

Complete vs incomplete vs co-dominance Board fav

Feature	Complete	Incomplete	Co-dominance
F1 phenotype	like dominant parent	intermediate	both parental traits
Example	tall pea (Tt)	pink snapdragon (Rr)	AB blood group
F2 phenotypic ratio	3:1	1:2:1	1:2:1
F2 genotypic ratio	1:2:1	1:2:1	1:2:1

Quick fact: dominance depends on the trait studied

Starch synthesis in pea seeds is controlled by one gene with alleles **B** and **b**. BB makes large starch grains, bb has lesser efficiency and makes smaller grains (seeds become wrinkled). For **seed shape**, B is dominant (Bb round). For **starch grain size**, Bb has intermediate grains: incomplete dominance. So dominance is not an autonomous feature of a gene; it depends on the gene product and the phenotype examined.

3. Inheritance of Two Genes

Mendel crossed pea plants differing in two characters: **yellow, round** seeds (RRYY) and **green, wrinkled** seeds (rryy). All F1 plants had **yellow and round** seeds (RrYy): yellow is dominant over green and round over wrinkled. Such a cross is a dihybrid cross.

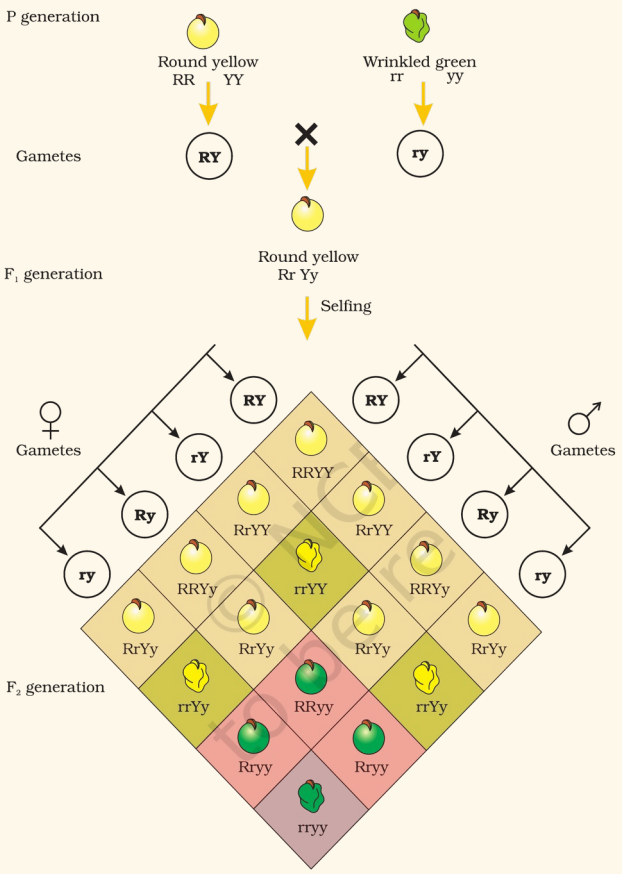
- → On selfing F1, each character still segregated in **3:1** in F2 (yellow:green and round:wrinkled), as in a monohybrid cross.
- → F1 (RrYy) produces **four kinds of gametes** in equal frequency: **RY, Ry, rY, ry** (each 25%).
- → Random fusion of these gametes gives 16 combinations in F2.

Dihybrid cross F2: Punnett square Board fav

Gametes	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

Phenotype	Number (of 16)
Yellow round	9
Green round	3
Yellow wrinkled	3
Green wrinkled	1

Phenotypic ratio 9:3:3:1. Genotypic ratio is **1:2:2:4:1:2:1:2:1** (9 genotypes).



Phenotypic ratio : round yellow : round green : wrinkled yellow : wrinkled green
9 3 3 1

Fig. 4.7 Results of a dihybrid cross where the two parents differed in two pairs of contrasting traits: seed colour and seed shape

9:3:3:1 dekh ke dost ne ye gift bheja.

Asli baat: Punnett square box by box bharo: 16 boxes, 9 genotypes, 4 phenotypes.

Law of Independent Assortment PYQ

When two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters. Example: R/r segregates independently of Y/y, which is why the dihybrid F2 ratio (9:3:3:1) is the product of two 3:1 ratios.

Yaad karo: ratio wala jadoo

Monohybrid: **3:1** (phenotype), **1:2:1** (genotype). Dihybrid: **9:3:3:1**. Incomplete dominance: **1:2:1 = 1:2:1**. Test cross: **1:1** (mono), **1:1:1:1** (di).

A. Chromosomal theory of inheritance

Mendel published his work in **1865**, but it remained unrecognised till **1900**. Reasons:

→ Communication was not easy, so his work could not be widely publicised.

→ His concept of genes (factors) as stable, discrete units that did not blend was not accepted by his contemporaries as an explanation for the apparently continuous variation in nature.

→ His **mathematical approach** to explain biological phenomena was totally new and unacceptable to many biologists then.

→ He could not provide any **physical proof** for the existence of factors.

In **1900**, three scientists (**de Vries, Correns and von Tschermak**) independently rediscovered Mendel's results. By then, chromosomes (seen by staining) and their behaviour in mitosis and meiosis were known. Walter Sutton and Theodor Boveri (1902) noted that the behaviour of chromosomes was **parallel to the behaviour of genes**, and proposed the chromosomal theory of inheritance: genes are located on chromosomes, and the pairing and separation of a pair of chromosomes leads to segregation of the pair of factors they carry. PYQ

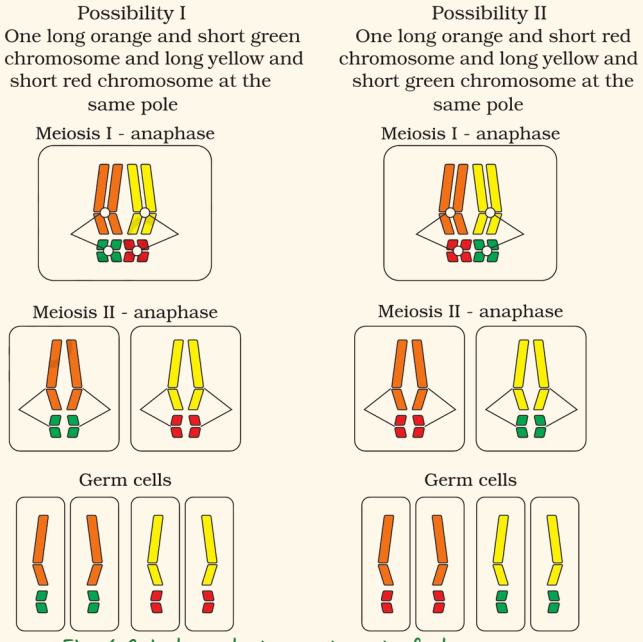


Fig. 4.9 Independent assortment of chromosomes

Chromosomes	Genes
occur in pairs	occur in pairs
segregate at gamete formation so that only one of each pair goes to a gamete	segregate at gamete formation, only one of each pair is transmitted
independent pairs segregate independently of each other	one pair segregates independently of another pair

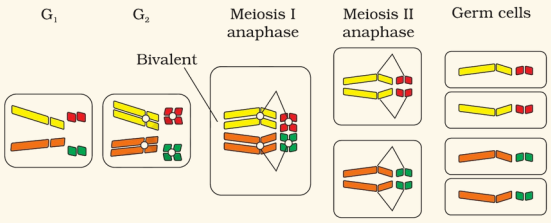


Fig. 4.8 Meiosis and germ cell formation in a cell with four chromosomes

The chromosomal theory was verified experimentally by Thomas Hunt Morgan and his colleagues, which led to the discovery of the basis of variation produced by sexual reproduction.

B. Linkage and recombination

Morgan worked with tiny fruit flies, *Drosophila melanogaster*, which are very suitable for genetic studies:

- → Can be grown on a **simple synthetic medium** in the laboratory.
- → Complete their life cycle in about **two weeks**.
- → A single mating produces a **large number of progeny** flies.
- → Clear differentiation of sexes: male and female flies are easily distinguishable.
- → Many types of hereditary variations can be seen with **low power microscopes**.

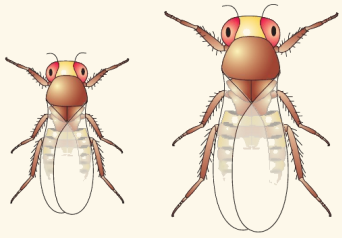


Fig. 4.10 Drosophila melanogaster (a) Male (b) Female

Morgan carried out dihybrid crosses in *Drosophila* to study sex-linked genes, e.g. crossing **yellow-bodied, white-eyed females** with **brown-bodied, red-eyed (wild type) males** and intercrossing the F1. The F2 ratios deviated significantly from 9:3:3:1.

- → Morgan found that when two genes in a dihybrid cross were on the **same chromosome**, the proportion of **parental gene combinations** was much higher than the non-parental type.
- → He attributed this to the physical association or linkage of the two genes; he coined the term linkage to describe this physical association of genes on a chromosome.
- → The term recombination describes the generation of non-parental gene combinations.
- → Genes white (w) and yellow (y) were very tightly linked and showed only **1.3%** recombination, while white and miniature wing (m) showed **37.2%** recombination. PYQ

Morgan's student **Alfred Sturtevant** used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and mapped their positions on the chromosome. Genetic maps are now used extensively as a starting point in the sequencing of whole genomes, as in the Human Genome Project.

Exam Alert

- → Tightly linked genes = **low** recombination (y-w 1.3%); loosely linked = **high** (w-m 37.2%).
- → Recombination frequency = distance ka measure (Sturtevant, gene mapping).
- → Linkage ke kaaran F2 mein **parental types zyada**, 9:3:3:1 nahi aata.

Independent assortment	Linkage
genes on **different** chromosomes	genes on the **same** chromosome
F2 dihybrid ratio 9:3:3:1	parental types far more than recombinants
test cross 1:1:1:1	test cross: more parental types

C. Polygenic inheritance

Some traits are generally controlled by **three or more genes**; these are polygenic traits. Besides multiple genes, polygenic inheritance also takes into account the influence of **environment**. Example: **human skin colour**; also human height. PYQ

- → In a polygenic trait the phenotype reflects the contribution of **each allele**: the effect of each allele is additive.
- → Suppose three genes A, B, C control skin colour, with dominant forms A, B, C for dark colour and recessive a, b, c for light colour. **AABBCC** = darkest; **aabbcc** = lightest; **AaBbCc** (three dominant alleles) = intermediate colour.
- → The number of each type of allele in the genotype determines the darkness or lightness of the skin.

D. Pleiotropy

A single gene can exhibit multiple phenotypic expressions; such a gene is a **pleiotropic gene**. The underlying mechanism in most cases is the effect of a gene on **metabolic pathways**, which contribute towards different phenotypes.

Example: phenylketonuria, a disease in humans caused by a **single gene mutation**, shows itself through multiple phenotypic expressions: **mental retardation** and a **reduction in hair and skin pigmentation**.

Polygenic inheritance	Pleiotropy
many genes control one trait	one gene controls many traits
effects of alleles are additive	one gene affects several metabolic pathways
e.g. human skin colour, height	e.g. phenylketonuria

Paheli-Bujho

Bujho: Ek hi gene se baal ka colour aur dimaag dono? Ye kaisa combo offer hai?

Paheli: Pleiotropy! Ek gene metabolic pathway badal deta hai, effects kai jagah.

Bujho: Aur skin colour mein ulta?

Paheli: Haan, wahan kai genes milke ek trait banate hain: polygenic.

4. Sex Determination

The first clue about the genetic basis of sex determination came from insects. Henking (1891) traced a specific nuclear structure all through spermatogenesis in a few insects and observed that **50% of sperms** received it after spermatogenesis, while the other 50% did not. He named it the **X body**; later it was named the X-chromosome.

→ **Autosomes**: chromosomes other than sex chromosomes.

→ **Sex chromosomes**: X and Y (or Z and W) that decide sex.

Type	Female	Male	Example	Heterogametic sex
XO	XX	XO (only one X)	grasshopper, many insects	male
XY	XX	XY	humans, *Drosophila*	male
ZW	ZW	ZZ	birds	female

→ In XO type all eggs bear an X besides the autosomes; some sperms bear X and some do not. Eggs fertilised by X-bearing sperms become **females**, others become **males**.

→ In XO and XY types, males produce two different types of gametes: this is male heterogamety.

→ In birds, the female has one Z and one W chromosome (ZW), and males have a pair of Z chromosomes (ZZ): female heterogamety. Board fav

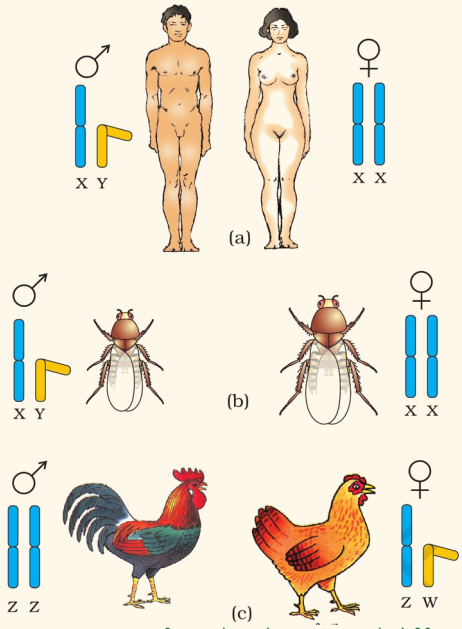
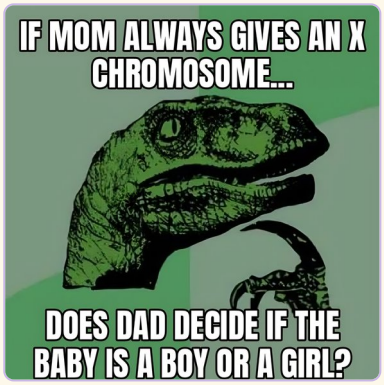


Fig. 4.12 Determination of sex by chromosomal differences: (a, b) both in humans and in Drosophila, the female has a pair of XX chromosomes and the male has XY (c) in many birds, female has ZW and male has ZZ



Haan bhai, mummy hamesha X deti hai; X ya Y wala sperm papa ka.

Asli baat: Humans mein male heterogametic hai: sex father decide karta hai.

Sex determination in humans

Humans have **23 pairs** of chromosomes: **22 pairs of autosomes** (same in both sexes) and **one pair of sex chromosomes**: XX in females and XY in males. During spermatogenesis males produce two types of gametes: 50% carry X and 50% carry Y. Females produce only X-bearing ova. An ovum fertilised by an X-sperm develops into a **female (XX)**, by a Y-sperm into a **male (XY)**. So the genetic make-up of the sperm determines the sex, and in each pregnancy the probability of a male or female child is always **50%**. PYQ

Punnett square: sex of the child

Sperm / Ovum	X (ovum)	X (ovum)
X (sperm)	XX girl	XX girl
Y (sperm)	XY boy	XY boy

Sex determination in honey bee

It is based on the number of **sets of chromosomes** an individual receives. This is the haplodiploid sex-determination system.

Feature	Female (queen or worker)	Male (drone)
Develops from	**fertilised** egg (union of sperm and egg)	**unfertilised** egg, by parthenogenesis
Ploidy	diploid, **32** chromosomes	haploid, **16** chromosomes
Gametes	eggs by meiosis	sperms by **mitosis**

Queen (2n = 32) → meiosis: eggs (n = 16) → fertilised: female (2n = 32)

Unfertilised egg (n = 16) → parthenogenesis → drone (n = 16) → mitosis: sperms (n = 16)

→ Males **do not have a father** and thus cannot have sons, but they have a **grandfather** and can have **grandsons**.

Exam Alert

→ Birds mein **female** heterogametic (ZW): ye MCQ trap hai.

→ Honey bee drone haploid (16) hai, isliye sperms **mitosis** se banta hai, meiosis se nahi.

→ Grasshopper male XO: total chromosome number female se ek kam.

5. Mutation

Mutation is a phenomenon that results in **alteration of DNA sequences** and consequently results in changes in the genotype and the phenotype of an organism. Besides recombination, mutation is another phenomenon that leads to **variation in DNA**.

→ One DNA helix runs continuously from one end to the other of each chromatid in a highly supercoiled form; so **loss (deletion) or gain (insertion/duplication) of a segment** of DNA results in alteration in chromosomes. These are chromosomal aberrations, commonly observed in **cancer cells**.

→ Mutation can also be due to change in a **single base pair** of DNA: this is point mutation. A classical example is **sickle-cell anaemia**. PYQ

→ Deletions and insertions of base pairs of DNA cause frame-shift mutations.

→ There are many chemical and physical factors that induce mutations: these are **mutagens**. **UV radiation** can cause mutations in organisms.

Type of mutation	Change	Example
Point mutation	change in a single base pair	sickle-cell anaemia
Frame-shift mutation	insertion or deletion of base pairs; reading frame shifts	(see Chapter 5)
Chromosomal aberration	loss or gain of a segment of DNA / chromosome	common in cancer cells

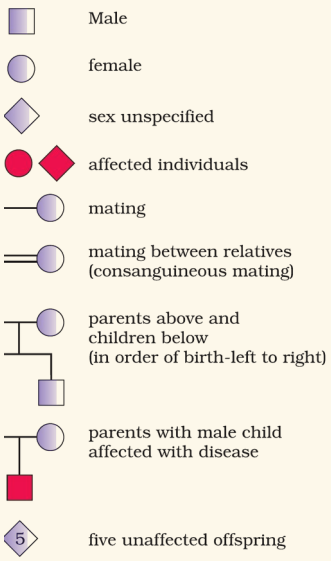
6. Genetic Disorders

A. Pedigree analysis

Control crosses that can be done in pea plants are not possible in human beings. So the study of family history about the inheritance of a particular trait over several generations is done; it is called pedigree analysis. A trait may be **dominant or recessive**, and **autosomal or sex-linked**. Board fav

→ Each and every feature in an organism is controlled by one or the other gene on the DNA of the chromosome; DNA carries genetic information and transmits it without change, but changes occasionally happen (mutations).

→ Pedigree analysis is a strong tool used in **human genetics** to trace the inheritance of a trait, abnormality or disease, and in genetic counselling to avoid disorders like haemophilia, sickle-cell anaemia and others.



Symbols used in the human pedigree analysis

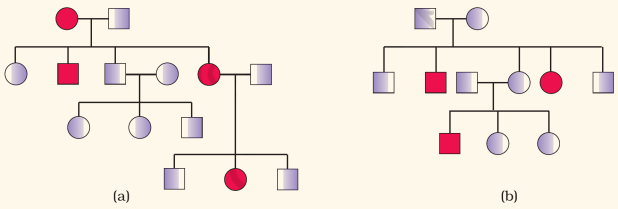


Fig. 4.14 Representative pedigree analysis of (a) autosomal dominant trait (for example: myotonic dystrophy) (b) autosomal recessive trait (for example: sickle-cell anaemia)

Reading a pedigree: quick clues PYQ

Pattern	Clues
Autosomal dominant	appears in every generation; affected child has at least one affected parent; both sexes equally
Autosomal recessive	skips generations; unaffected (carrier) parents can have affected children; both sexes equally
X-linked recessive	mostly males affected; passed from carrier mothers to sons; no father-to-son transmission
X-linked dominant	affected father passes it to all daughters and no sons

Quick fact: genetic counselling

Families with a history of a genetic disorder can consult a genetic counsellor, who uses **pedigree analysis** to identify carriers and work out the risk for future children. This helps couples take informed decisions, e.g. about sickle-cell anaemia or thalassemia.

B. Mendelian disorders

These are mainly determined by **alteration or mutation in a single gene**, are transmitted to the offspring on the same lines as the principles of inheritance, and can be traced by pedigree analysis. Examples: haemophilia, cystic fibrosis, sickle-cell anaemia, colour blindness, phenylketonuria, thalassemia. Such disorders may be **dominant or recessive**.

Disorder	Type	Cause / key point
Colour blindness	sex-linked recessive	defect in red or green cone; about **8%** males, **0.4%** females
Haemophilia	sex-linked recessive	a clotting protein affected; non-stop bleeding from a simple cut; Queen Victoria was a carrier
Sickle-cell anaemia	autosomal recessive	Glu replaced by Val at 6th position of beta globin (GAG to GUG)
Phenylketonuria	autosomal recessive	enzyme converting phenylalanine to tyrosine is absent
Thalassemia	autosomal recessive	reduced synthesis of alpha or beta globin chains

Colour blindness: failure to discriminate between **red and green** colour, due to mutation in certain genes on the **X chromosome**. The son of a woman who carries the gene has a 50% chance of being colour blind. The mother is not colour blind because the gene is recessive. A daughter will not normally be colour blind unless her mother is a carrier and her father is colour blind. PYQ

Haemophilia: shows transmission from an **unaffected carrier female** to some of the male progeny. The heterozygous carrier female may transmit the disease to sons. A haemophilic female is extremely rare, because her mother must be at least a carrier and her father haemophilic (unviable in the later stage of life). The family pedigree of **Queen Victoria** shows a number of haemophilic descendants, as she was a carrier of the disease.

Paheli-Bujho

Bujho: Colour blindness ladkon mein itni zyada kyun hoti hai?
Paheli: Gene X pe hai, aur ladkon ke paas sirf ek X hota hai.
Bujho: Matlab ladkiyon ke paas backup X hai!
Paheli: Haan, isliye wo zyaadatar sirf carrier banti hain.

X-linked crosses (colour blindness / haemophilia) Board fav

Parents	Daughters	Sons
carrier mother (X^HX^h) x normal father (X^HY)	50% normal, 50% carrier	50% normal, 50% affected
normal mother (X^HX^H) x affected father (X^hY)	all carriers	all normal
carrier mother x affected father	50% carrier, 50% affected	50% normal, 50% affected

Sickle-cell anaemia: an autosome-linked recessive trait transmitted when **both partners are carriers**. Controlled by a single pair of alleles, Hb^A and Hb^S. Only Hb^SHb^S show the disease; Hb^AHb^S appear unaffected but are **carriers** (sickle-cell trait), with 50% probability of passing the mutant gene.

→ Cause: substitution of glutamic acid (Glu) by valine (Val) at the **sixth position of the beta globin chain** of haemoglobin.

→ This is due to a single base substitution at the sixth codon of the beta globin gene from **GAG to GUG** (point mutation). Board fav

→ The mutant haemoglobin undergoes polymerisation under **low oxygen tension**, changing RBCs from biconcave discs to elongated sickle-like shape.

Position in beta chain	1	2	3	4	5	6	7
Normal Hb (Hb^A)	Val	His	Leu	Thr	Pro	**Glu**	Glu
Sickle Hb (Hb^S)	Val	His	Leu	Thr	Pro	**Val**	Glu

Phenylketonuria: an inborn error of metabolism, autosomal recessive. The affected person lacks the enzyme that converts **phenylalanine into tyrosine**. Phenylalanine accumulates and is converted into **phenylpyruvic acid** and other derivatives. Their accumulation in the brain causes mental retardation; they are also excreted in urine due to poor absorption by the kidney.

Thalassemia: an autosome-linked recessive blood disease transmitted when both parents are unaffected carriers (heterozygous). Mutation or deletion reduces the rate of synthesis of one of the globin chains (alpha or beta) of haemoglobin, forming abnormal haemoglobin and causing **anaemia**.

Alpha thalassemia	Beta thalassemia
production of alpha globin chain affected	production of beta globin chain affected
controlled by two closely linked genes **HBA1 and HBA2** on **chromosome 16** of each parent	controlled by a single gene **HBB** on **chromosome 11** of each parent
mutation or deletion of one or more of the **four** genes; more genes affected, less alpha globin	mutation of one or both the genes

Thalassemia vs sickle-cell anaemia PYQ

Thalassemia is a quantitative problem: too few globin molecules are synthesised. Sickle-cell anaemia is a qualitative problem: an incorrectly functioning globin is synthesised.

C. Chromosomal disorders

These are caused by **absence or excess or abnormal arrangement** of one or more chromosomes.

→ Failure of segregation of chromatids during cell division results in the gain or loss of a chromosome(s): aneuploidy. Example: Down's syndrome (gain of an extra chromosome 21), Turner's syndrome (loss of an X in females).

→ Failure of **cytokinesis** after telophase stage of cell division results in an increase in a whole set of chromosomes: polyploidy, often seen in **plants**.

→ A normal human cell has **46** chromosomes (23 pairs): 22 pairs autosomes and 1 pair sex chromosomes. An additional copy of a chromosome is **trisomy**; lack of one chromosome of a pair is **monosomy**. Such situations lead to very serious consequences.

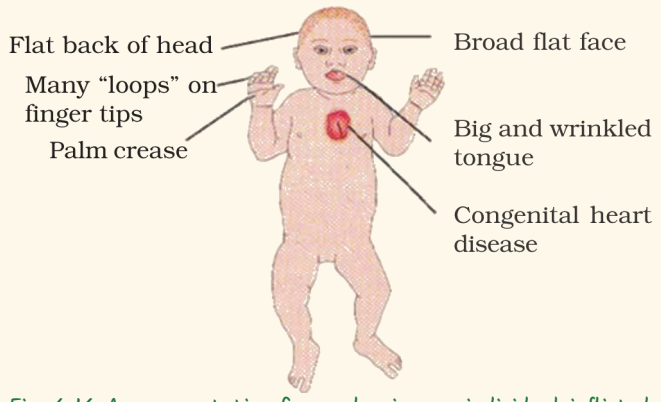


Fig. 4.16 A representative figure showing an individual inflicted with Down's syndrome

Syndrome	Karyotype	Features
Down's	trisomy of chromosome **21** (47)	first described by **Langdon Down (1866)**; short statured, small round head, furrowed tongue, partially open mouth; broad palm with characteristic palm crease; physical, psychomotor and mental development retarded
Klinefelter's	**47, XXY**	overall masculine development with feminine development of breast (**gynaecomastia**); tall; **sterile**
Turner's	**45, X0**	females; **sterile** as ovaries are rudimentary; lack of other secondary sexual characters; short stature

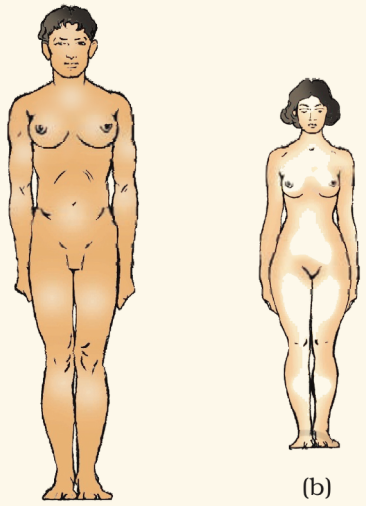


Fig. 4.17 Genetic disorders due to sex chromosome composition in humans: (a) Klinefelter's syndrome (b) Turner's syndrome

Mendelian disorders	Chromosomal disorders
mutation in a **single gene**	absence, excess or abnormal arrangement of **chromosomes**
follow Mendelian inheritance; traced by pedigree	usually not inherited in a Mendelian way
e.g. haemophilia, sickle-cell anaemia, PKU	e.g. Down, Klinefelter, Turner syndromes

Yaad karo: syndromes

Down = **21** extra (2+1 = 3 copies). Klinefelter = King size **XXY** (47, tall male). Turner = Tota ek X (**45, X0**, female).

Deviations from Mendelian ratios at a glance Board fav

Phenomenon	What happens	Example
Incomplete dominance	heterozygote intermediate	snapdragon flower colour
Co-dominance	both alleles expressed	AB blood group
Multiple alleles	more than two alleles in population	I^A, I^B, i
Pleiotropy	one gene, many effects	phenylketonuria
Polygenic inheritance	many genes, one trait	human skin colour
Linkage	genes on same chromosome inherited together	*Drosophila* y and w genes

Who did what

Scientist	Contribution
Gregor Mendel	laws of inheritance (pea, 1856–1863; published 1865)
de Vries, Correns, von Tschermak	rediscovered Mendel's work (1900)
Sutton and Boveri	chromosomal theory of inheritance (1902)
T. H. Morgan	linkage and recombination in *Drosophila*
Alfred Sturtevant	genetic (gene) mapping
Henking	X body in insects (1891)
Reginald C. Punnett	Punnett square
Langdon Down	described Down's syndrome (1866)

7. Genetics Numericals: Solved Examples

Example 1: monohybrid

Q. A heterozygous tall plant is selfed and produces 800 plants. How many are dwarf? **Solution:** Tt x Tt gives 1 TT : 2 Tt : 1 tt. Dwarf = 1/4 x 800 = 200; homozygous tall = 200; heterozygous tall = 400.

Example 2: number of gametes

Q. How many kinds of gametes does AaBbCC form? **Solution:** 2^n, where n = number of heterozygous pairs = 2. So 4 kinds: ABC, AbC, aBC, abC.

Example 3: probability in a dihybrid

Q. In RrYy x RrYy, what fraction of progeny is round green, and what fraction is RRYY? **Solution:** Round = 3/4, green = 1/4, so round green = 3/16. RR = 1/4, YY = 1/4, so RRYY = 1/16.

Example 4: blood groups

Q. Can parents with groups A and B have a child with group O? **Solution:** Yes, if both are heterozygous: I^Ai x I^Bi gives AB, A, B and O each with probability 1/4. Parents with AB and O can never have an O or AB child (only A or B).

Example 5: sickle-cell carriers

Q. Two carriers (Hb^AHb^S) marry. What is the chance of an affected child? **Solution:** Hb^AHb^S x Hb^AHb^S gives 1 Hb^AHb^A : 2 Hb^AHb^S : 1 Hb^SHb^S. Affected = 25%; carriers 50%; normal 25%.

Example 6: haemophilia

Q. A haemophilic man marries a normal (non-carrier) woman. What are the chances for their children? **Solution:** X^hY x X^HX^H: all daughters X^HX^h (carriers), all sons X^HY (normal). No son gets the disease from the father, because a son gets Y from him.

Example 7: test cross

Q. A violet-flowered pea plant on test cross gives 52 violet and 48 white. Give its genotype. **Solution:** Ratio is about **1:1**, so the plant is heterozygous Ww (Ww x ww).

Example 8: incomplete dominance

Q. A pink snapdragon is crossed with a white one. Give the progeny. **Solution:** Rr x rr gives 1 pink (Rr) : 1 white (rr); no red flowers appear.

Example 9: trihybrid gametes and ratio

Q. How many phenotypes appear in F2 of a trihybrid cross (AaBbCc selfed, complete dominance)? **Solution:** 2^3 = 8 phenotypes, in the ratio 27:9:9:9:3:3:3:1; gametes per parent = 8; genotypes = 3^3 = 27.

Example 10: colour blindness

Q. A colour-blind woman marries a normal man. Predict the children. **Solution:** X^cX^c x X^CY: all sons X^cY (colour blind), all daughters X^CX^c (carriers). This mother-to-son pattern is called criss-cross inheritance.

Example 11: polygenic shades

Q. If skin colour is controlled by 3 genes, how many phenotypic classes appear in F2 of AaBbCc x AaBbCc? **Solution:** Number of dominant alleles can be 0 to 6, so 7 classes (ratio 1:6:15:20:15:6:1).

Example 12: probability of carriers

Q. A normal child of two carrier (Aa) parents: what is the chance that it is a carrier? **Solution:** Normal children are AA (1) and Aa (2), so carrier = 2/3.

Example 13: reading a pedigree

Q. In a family, an affected father and a normal mother have all daughters affected and all sons normal. Which pattern is this? **Solution:** An affected father passes his X to every daughter and Y to every son, so the trait is X-linked dominant.

Example 14: chance of two events

Q. Two carriers of an autosomal recessive disease plan two children. What is the chance that both are affected? **Solution:** Each child 1/4, independent events: 1/4 x 1/4 = 1/16.

Chapter map

Mendel: 3 laws → Deviations: incomplete, co-dominance, multiple alleles → Chromosomal theory → Linkage, recombination, mapping → Polygenic, pleiotropy → Sex determination → Mutation → Pedigree, Mendelian and chromosomal disorders

Exam Alert: NCERT lines jo seedhe MCQ bante hain

- → Mendel: 7 saal (1856–1863), 14 true-breeding varieties, 1865 publish, 1900 rediscovery.
- → y–w 1.3%, w–m 37.2%; Henking 1891; Langdon Down 1866.
- → Honey bee: 32 (female) / 16 (male); grasshopper male XO; birds ZW female.
- → HBA1, HBA2 on chromosome 16; HBB on chromosome 11; colour blindness 8% males, 0.4% females.

1 minute recap

1. Mendel: pea, 7 years, 14 true-breeding lines, 7 traits; statistics + large samples.
2. Monohybrid F2: 3:1 phenotype, 1:2:1 genotype; test cross with recessive parent (1:1 = heterozygous).
3. Laws: dominance, segregation, independent assortment (dihybrid 9:3:3:1).
4. Incomplete dominance (snapdragon 1:2:1); co-dominance and multiple alleles (ABO: I^A, I^B, i; 6 genotypes, 4 phenotypes).
5. Sutton and Boveri: chromosomal theory; Morgan: linkage and recombination in *Drosophila* (y–w 1.3%, w–m 37.2%); Sturtevant: maps.
6. Polygenic (skin colour) vs pleiotropy (phenylketonuria).
7. Sex: XO (grasshopper), XY (humans), ZW (birds, female heterogametic); honey bee haplodiploid (32/16).
8. Point mutation: sickle-cell (GAG to GUG, Glu to Val); thalassemia quantitative; Down 21 trisomy, Klinefelter 47 XXY, Turner 45 XO.

Exam Alert: common galtiyan

- → Gametes likhte waqt ek gamete mein **har gene ka sirf ek allele**: RY sahi, RRYY galat.
- → Ratio ko "phenotypic" ya "genotypic" label karna mat bhoolo.
- → Blood group mein "I^AI^B" likho, "AB gene" nahi; O = ii.
- → X-linked cross mein Y par koi allele nahi likhte.
- → Dono events saath = probabilities **multiply**; koi ek event = **add**.

Most Important Questions

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM The number of true-breeding pea varieties selected by Mendel was:
(a) 7
(b) 21
(c) 14
(d) 12
Ans: (c) 14

Q2. IM Board fav The F2 genotypic ratio of a monohybrid cross is:
(a) 3:1
(b) 1:2:1
(c) 9:3:3:1
(d) 1:1
Ans: (b) 1:2:1

Q3. IM Board fav In snapdragon, a cross between red (RR) and white (rr) flowered plants gives F2 phenotypic ratio:
(a) 1 red : 2 pink : 1 white
(b) 3 red : 1 white
(c) all pink
(d) 9:3:3:1
Ans: (a) 1 red : 2 pink : 1 white

Q4. IM Board fav How many genotypes and phenotypes are possible for ABO blood groups?
(a) 3 and 4
(b) 4 and 6
(c) 6 and 3
(d) 6 genotypes, 4 phenotypes
Ans: (d) 6 genotypes, 4 phenotypes

Q5. IM Board fav Female heterogamety is found in:
(a) humans
(b) grasshopper
(c) birds
(d) *Drosophila*
Ans: (c) birds

Q6. IM Sickle-cell anaemia is caused by substitution of:
(a) glutamic acid by valine at the 6th position of beta globin
(b) valine by glutamic acid in alpha globin
(c) lysine by valine
(d) valine by glycine
Ans: (a) glutamic acid by valine at the 6th position of beta globin

Q7. IM Karyotype 47, XXY is found in:
(a) Turner's syndrome
(b) Klinefelter's syndrome
(c) Down's syndrome
(d) Haemophilia
Ans: (b) Klinefelter's syndrome

Q8. IM A drone (male honey bee) has:
(a) 32 chromosomes
(b) 16 chromosomes from a fertilised egg
(c) 16 chromosomes and develops by parthenogenesis
(d) 48 chromosomes
Ans: (c) 16 chromosomes and develops by parthenogenesis

Q9. IM Which pair of genes in *Drosophila* showed 1.3% recombination?
(a) white and yellow
(b) white and miniature wing
(c) yellow and miniature wing
(d) red and brown
Ans: (a) white and yellow

Q10. IM Board fav The chromosomal theory of inheritance was proposed by:
(a) Mendel
(b) Sutton and Boveri
(c) Morgan
(d) Sturtevant
Ans: (b) Sutton and Boveri

Q11. IM Phenylketonuria is an example of:
(a) polygenic inheritance
(b) co-dominance
(c) incomplete dominance
(d) pleiotropy
Ans: (d) pleiotropy

Q12. IM The proportion of tt plants in the F2 of a monohybrid cross is:
(a) 1/4
(b) 1/2
(c) 3/4
(d) 1/16
Ans: (a) 1/4

Q13. IM Board fav Mendel's law that is NOT universally applicable is the law of:
(a) segregation
(b) purity of gametes
(c) independent assortment
(d) none
Ans: (c) independent assortment (it fails for linked genes)

Q14. IM The genotypic ratio of a dihybrid F2 is:
(a) 9:3:3:1
(b) 1:2:2:4:1:2:1:2:1
(c) 1:1:1:1
(d) 3:1
Ans: (b) 1:2:2:4:1:2:1:2:1

Q15. IM Turner's syndrome is:
(a) 45, X0
(b) 47, XXY
(c) 47, trisomy 21
(d) 46, XY
Ans: (a) 45, X0

Q16. IM Board fav Which is a sex-linked recessive disorder?
(a) Sickle-cell anaemia
(b) Haemophilia
(c) Phenylketonuria
(d) Thalassemia
Ans: (b) Haemophilia

Q17. IM Polyploidy results from failure of:
(a) DNA replication
(b) crossing over
(c) segregation of chromatids
(d) cytokinesis after telophase
Ans: (d) cytokinesis after telophase

Q18. IM In grasshopper, the male has:
(a) XO, one chromosome less than the female
(b) XY
(c) ZW
(d) ZZ
Ans: (a) XO, one chromosome less than the female

Q19. IM Beta thalassemia is controlled by the gene:
(a) HBA1 on chromosome 16
(b) HBA2 on chromosome 11
(c) HBB on chromosome 11
(d) HBB on chromosome 16
Ans: (c) HBB on chromosome 11

Q20. IM Name the scientist who coined the term linkage.
Ans: T. H. Morgan.

Q21. IM What is the genotype of a person with blood group AB?
Ans: I^AI^B.

Q22. IM Name the mutation caused by insertion or deletion of base pairs.
Ans: Frame-shift mutation.

Assertion-Reason (1 mark each)

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q23. 1M Board fav **A:** In a dihybrid cross, the F2 phenotypic ratio is 9:3:3:1. **R:** The two pairs of alleles segregate independently of each other.
Ans: (a) Both true and R explains A.

Q24. 1M Board fav **A:** Haemophilia is more common in males. **R:** The gene for haemophilia is on the Y chromosome.
Ans: (c) A is true, R is false: the gene is on the X chromosome; males have only one X.

Q25. 1M **A:** In birds, the sex of the offspring is decided by the female. **R:** Females in birds are heterogametic (ZW).
Ans: (a) Both true and R explains A.

Q26. 1M **A:** Multiple alleles can be studied only in a population. **R:** An individual can have only two alleles of a gene.
Ans: (a) Both true and R explains A.

Short answer (2-3 marks)

Q27. 2M Board fav NCERT: Mention the advantages of selecting pea plant for experiment by Mendel.
Ans: Many clearly contrasting traits; naturally self-pollinating bisexual flowers, so pure lines are easy and artificial cross-pollination is possible; short life cycle; large number of seeds per cross; easy to grow.

Q28. 3M Board fav NCERT: Differentiate between (a) dominance and recessive (b) homozygous and heterozygous (c) monohybrid and dihybrid.
Ans: (a) Dominant allele expresses itself in the heterozygote; recessive expresses only when homozygous. (b) Homozygous has identical alleles (TT, tt) and forms one type of gamete; heterozygous has different alleles (Tt) and forms two types. (c) Monohybrid cross studies one character (3:1 in F2); dihybrid studies two characters (9:3:3:1).

Q29. 2M Board fav NCERT: A diploid organism is heterozygous for 4 loci. How many types of gametes can be produced?
Ans: $2^n = 2^4 =$ **16** types of gametes (if the loci assort independently).

Q30. 3M Board fav NCERT: Explain the Law of Dominance using a monohybrid cross.
Ans: Tall (TT) x dwarf (tt) gives all tall F1 (Tt): the dominant T masks t. Selfing F1 gives 3 tall : 1 dwarf in F2, showing that the recessive factor was present but not expressed. Characters are controlled by paired factors, and in a dissimilar pair one factor (dominant) expresses over the other (recessive).

Q31. 2M Board fav NCERT: Define and design a test cross.
Ans: Crossing an organism showing the dominant phenotype (genotype unknown) with the homozygous recessive parent. TT x tt gives all tall (homozygous); Tt x tt gives 1 tall : 1 dwarf (heterozygous).

Q32. 3M NCERT: Using a Punnett square, work out the F1 phenotypes after a cross between a homozygous female and a heterozygous male for a single locus.
Ans: Taking TT female x Tt male: gametes T (female); T and t (male). Offspring: TT and Tt in 1:1, all tall (100% dominant phenotype). (If the female is tt: tt x Tt gives 1 tall : 1 dwarf.)

Q33. 3M Board fav NCERT: TtYy (tall, yellow) is crossed with Ttyy (tall, green). What proportion will be (a) tall and green (b) dwarf and green?
Ans: Tt x Tt gives 3/4 tall, 1/4 dwarf; Yy x yy gives 1/2 yellow, 1/2 green. (a) Tall green = 3/4 x 1/2 = **3/8**. (b) Dwarf green = 1/4 x 1/2 = **1/8**.

Q34. 3M NCERT: Two heterozygous parents are crossed. If the two loci are linked, what would be the distribution of phenotypes in F1 of a dihybrid cross?
Ans: With complete linkage the alleles travel together, so the progeny show mostly parental combinations, about 3:1 (like a monohybrid cross) instead of 9:3:3:1; with incomplete linkage a few recombinants also appear, but parental types far exceed them.

Q35. 3M Board fav NCERT: Briefly mention the contribution of T. H. Morgan in genetics.
Ans: Gave experimental proof of the chromosomal theory using *Drosophila*; discovered linkage and recombination (coined the term linkage); showed tightly linked genes have low recombination (y-w 1.3%) and loosely linked high (w-m 37.2%); studied sex-linked inheritance; his work led to gene mapping by Sturtevant.

Q36. 2M Board fav NCERT: What is pedigree analysis? How is it useful?
Ans: Study of a trait in a family over several generations using standard symbols. It shows whether a trait is dominant or recessive, autosomal or sex-linked, helps predict carriers and the chance of disease in children, and is used in genetic counselling.

Q37. 2M Board fav NCERT: How is sex determined in human beings?
Ans: XY type: females XX produce only X ova; males XY produce 50% X and 50% Y sperms. X-sperm + ovum gives XX (girl); Y-sperm + ovum gives XY (boy). So the father's sperm decides the sex; chance of each is 50%.

Q38. 3M Board fav NCERT: A child has blood group O. The father has A and the mother B. Work out the genotypes of the parents and possible genotypes of other offspring.
Ans: Child ii gets i from each parent, so father I^Ai and mother I^Bi. Cross gives I^AI^B (AB), I^Ai (A), I^Bi (B), ii (O) in 1:1:1:1.

Q39. 3M Board fav NCERT: Explain with examples (a) co-dominance (b) incomplete dominance.
Ans: (a) Both alleles express fully in F1: I^AI^B gives AB blood group with both A and B sugars on RBCs. (b) F1 is intermediate: red (RR) x white (rr) snapdragon gives pink (Rr); F2 1 red : 2 pink : 1 white.

Q40. 2M Board fav NCERT: What is point mutation? Give one example.
Ans: Change in a single base pair of DNA. Example: sickle-cell anaemia, where GAG changes to GUG in the beta globin gene, replacing glutamic acid with valine.

Q41. 1M NCERT: Who proposed the chromosomal theory of inheritance?
Ans: Walter Sutton and Theodor Boveri (1902); experimentally verified by T. H. Morgan.

Q42. 2M Board fav NCERT: Mention any two autosomal genetic disorders with their symptoms.
Ans: Sickle-cell anaemia: RBCs become sickle-shaped under low oxygen, anaemia. Phenylketonuria: phenylalanine accumulates, mental retardation, reduced hair and skin pigmentation. (Also thalassemia: anaemia; Down's syndrome: retarded development, palm crease.)

Q43. 2M Give two reasons why Mendel's work remained unrecognised till 1900.
Ans: Poor communication; factors as stable non-blending units were not accepted; his mathematical approach was new; no physical proof of factors (any two).

Q44. 2M Why are males of honey bee said to have no father but a grandfather?
Ans: A drone develops from an unfertilised egg (parthenogenesis), so it has only a mother. The mother queen came from a fertilised egg, so the drone has a maternal grandfather; it can have grandsons through its daughters but no sons.

Q45. 2M Board fav Differentiate between male heterogamety and female heterogamety with examples.
Ans: Male heterogamety: males produce two types of gametes (XO in grasshopper, XY in humans and *Drosophila*). Female heterogamety: females produce two types of eggs (ZW in birds), males are ZZ.

Q46. 3M Explain how dominance can be understood at the level of the enzyme a gene codes for.
Ans: A modified allele may make a normal/less efficient enzyme (same phenotype), a non-functional enzyme or no enzyme. In the last two cases the phenotype depends on the unmodified allele, which is dominant; the modified allele is recessive.

Q47. 2M Board fav Differentiate between polygenic inheritance and pleiotropy with one example each.
Ans: Polygenic: many genes control one trait additively, influenced by environment (skin colour). Pleiotropy: one gene shows multiple phenotypic effects through metabolic pathways (phenylketonuria: mental retardation, less pigmentation).

Q48. 2M Board fav Why is the frequency of colour blindness higher in males than in females?
Ans: The gene is recessive and X-linked. Males (XY) have one X, so one recessive allele makes them colour blind. Females need two recessive alleles (XcXc), which is rarer; heterozygous females are carriers.

Q49. 3M Board fav Differentiate between thalassemia and sickle-cell anaemia.
Ans: Thalassemia: quantitative defect, too few alpha or beta globin chains; alpha on chromosome 16 (HBA1, HBA2), beta on chromosome 11 (HBB). Sickle-cell: qualitative defect, abnormal beta globin (Glu to Val), RBCs sickle under low oxygen.

Q50. 2M Differentiate between aneuploidy and polyploidy.
Ans: Aneuploidy: gain or loss of one or a few chromosomes due to failure of segregation of chromatids (e.g. trisomy 21). Polyploidy: increase in whole sets of chromosomes due to failure of cytokinesis after telophase; common in plants.

Q51. 3M Board fav Give the karyotype and two features each of Klinefelter's and Turner's syndromes.
Ans: Klinefelter: 47, XXY; masculine with gynaecomastia, tall, sterile. Turner: 45, X0; female with rudimentary ovaries, sterile, lack of secondary sexual characters, short stature.

Q52. 2M What is a test cross for a dihybrid? Give its ratio.
Ans: Crossing the F1 dihybrid (RrYy) with the double recessive (rryy). Progeny: yellow round, yellow wrinkled, green round, green wrinkled in 1:1:1:1, showing independent assortment.

Q53. 2M Why did Morgan get a ratio different from 9:3:3:1 in dihybrid crosses of *Drosophila*?
Ans: The two genes were on the same X chromosome (linked), so parental combinations appeared far more often than recombinants; recombination depended on how close the genes were.

Q54. 1M What are mutagens? Give one example.
Ans: Physical or chemical agents that induce mutations; e.g. UV radiation.

Q55. 2M What is the role of Sturtevant's work in genetics?
Ans: He used recombination frequency between gene pairs on the same chromosome as a measure of the distance between them and mapped gene positions; genetic maps are the starting point for genome sequencing.

Q56. 2M Why is the human skin colour said to show a range of shades in a population?
Ans: It is polygenic: three or more genes contribute additively, and each dominant allele adds some darkness; the many combinations (aabbcc to AABBCC) plus environmental effects give a continuous range of shades.

Q57. 2M In a cross between a pure tall and a pure dwarf plant, what proportion of F2 tall plants are true-breeding? Why?
Ans: 1/3 of the tall plants (TT among TT and Tt: 1 of 3), since F2 tall plants are 1 TT : 2 Tt.

Q58. 2M What are the characteristic features of an autosomal recessive pedigree?
Ans: Trait skips generations; unaffected carrier parents can have affected children; both sexes are affected equally; affected individuals are homozygous recessive.

Q59. 2M Board fav A man with blood group AB marries a woman with blood group O. Give the possible blood groups of their children.
Ans: I^AI^B x ii gives I^Ai (A) and I^Bi (B) in 1:1; no AB or O child is possible.

Q60. 2M Why are haemophilic females very rare?
Ans: A female must get the recessive allele from both parents: her mother must be at least a carrier and her father haemophilic, and haemophilic males often do not survive to later life.

Q61. 2M Explain why the law of segregation is also called the law of purity of gametes.
Ans: Alleles of a pair separate during gamete formation and do not blend, so each gamete carries only one allele of each gene and is pure for that character.

Q62. 3M Explain the significance of a test cross with a dihybrid example.
Ans: RrYy x rryy gives four classes in 1:1:1:1, showing the F1 makes four gametes in equal numbers and confirming independent assortment; any deviation (more parental types) indicates linkage. It also reveals unknown genotypes.

Q63. 2M What is criss-cross inheritance? Give an example.
Ans: Transfer of an X-linked trait from father to grandson through a carrier daughter, or from mother to son. Example: colour blindness or haemophilia (X^cX^c mother gives all sons colour blind).

Q64. 2M Why did Mendel get only parental phenotypes in F1 but new combinations in F2 of a dihybrid cross?
Ans: F1 is RrYy and shows only dominant traits. In F2, independent assortment of the two allele pairs gives new combinations (round green, wrinkled yellow) besides the parental ones.

Q65. 3M Board fav How is pedigree analysis used in genetic counselling? Give an example.
Ans: By tracing a trait over generations, counsellors find carriers and calculate the risk of a disorder in future children, so couples can decide accordingly; e.g. two sickle-cell carriers have a 25% risk per child.

Q66. 2M Board fav Explain why a man cannot pass haemophilia to his son.
Ans: The son receives the Y chromosome from his father and the X from his mother; since the haemophilia gene is on X, the father passes it only to daughters (carriers).

Q67. 2M What are autosomes and sex chromosomes? How many of each are in a human sperm?
Ans: Autosomes are chromosomes other than sex chromosomes (same in both sexes); sex chromosomes (X, Y) decide sex. A human sperm has 22 autosomes and one sex chromosome (X or Y).

Q68. 3M Explain the starch grain example to show that dominance depends on the phenotype studied.
Ans: Gene B controls starch synthesis. For seed shape, Bb is round like BB (B dominant). For starch grain size, BB large, bb small and Bb intermediate (incomplete dominance). So the same alleles show different dominance relations.

Q69. 2M Mention two differences between Klinefelter's and Turner's syndromes.
Ans: Klinefelter: male, 47, XXY (extra X), tall, gynaecomastia. Turner: female, 45, X0 (one X missing), short, rudimentary ovaries. Both are sterile.

Q70. 2M What is the probability of a dwarf plant in Tt x Tt and of a green wrinkled seed in RrYy x RrYy?
Ans: Dwarf (tt) = 1/4. Green wrinkled (rryy) = 1/4 x 1/4 = 1/16.

Q71. 2M Board fav Write the amino acid change in sickle-cell anaemia and explain why RBCs become sickle-shaped.
Ans: Glutamic acid at the 6th position of the beta globin chain is replaced by valine (GAG to GUG). The mutant haemoglobin polymerises under low oxygen tension, changing the biconcave RBC into a sickle shape.

Q72. 3M Why is phenylketonuria called an inborn error of metabolism? Mention its effects.
Ans: It is an inherited (autosomal recessive) lack of the enzyme that converts phenylalanine into tyrosine. Phenylalanine and phenylpyruvic acid accumulate, causing mental retardation and reduced hair and skin pigmentation; they are also excreted in urine.

Q73. 2M Differentiate between alpha and beta thalassemia.
Ans: Alpha: alpha globin chain affected; two closely linked genes HBA1 and HBA2 on chromosome 16; mutation or deletion of one or more of four genes. Beta: beta globin chain affected; single gene HBB on chromosome 11; mutation of one or both genes.

Q74. 2M Give reasons: (a) Male honey bees have 16 chromosomes and females 32. (b) Drones produce sperms by mitosis.
Ans: (a) Males develop from unfertilised eggs (haploid) by parthenogenesis; females from fertilised eggs (diploid). (b) A haploid cell cannot undergo meiosis normally, so sperms are made by mitosis.

Q75. 3M What are the reasons for the success of Mendel's experiments?
Ans: He studied one character at a time; used true-breeding lines with clear contrasting traits; used large samples and statistical analysis; kept careful records over several generations; chose pea, which allows controlled crosses.

Q76. 2M In a monohybrid cross, how will you prove that the F1 tall plant is heterozygous?
Ans: By a test cross with the dwarf (tt) parent: Tt x tt gives 1 tall : 1 dwarf; or by selfing, which gives 3 tall : 1 dwarf.

Q77. 2M Why is Turner's syndrome female sterile?
Ans: Due to absence of one X chromosome (45, X0) the ovaries remain rudimentary, so no ova are formed; secondary sexual characters also do not develop.

Q78. 2M Explain with the binomial expression how a 1:2:1 genotypic ratio arises in a monohybrid F2.
Ans: F1 Tt forms gametes T and t each with probability 1/2. $(1/2\ T + 1/2\ t)^2 = 1/4\ TT + 1/2\ Tt + 1/4\ tt$, i.e. 1:2:1.

Q79. 2M What is a reciprocal cross? What did it show Mendel?
Ans: A cross repeated with the sexes of the parents interchanged. Mendel got identical results, showing the traits were not linked to the sex of the parent.

Q80. 2M Name the three alleles of the ABO gene and their products.
Ans: I^A and I^B make slightly different sugar polymers on the RBC surface (A and B); i makes no sugar.

Q81. 2M Explain why the frequency of recombination is used to map genes.
Ans: The farther apart two genes are on a chromosome, the more often crossing over separates them, so recombination frequency is roughly proportional to the distance between them; Sturtevant used it to order genes.

Q82. 2M Explain why a son of a colour-blind father and a normal (non-carrier) mother has normal vision.
Ans: The son gets the Y chromosome from his father and a normal X from his mother; the colour blindness gene on the father's X goes only to daughters.

Q83. 3M Give two features each of autosomal dominant and autosomal recessive inheritance in a pedigree.
Ans: Dominant: appears in every generation; affected children have an affected parent. Recessive: skips generations; normal carrier parents can have affected children. Both sexes affected equally in both.

Q84. 2M What do you mean by true-breeding? Why did Mendel use true-breeding lines?
Ans: A line that, on continuous self-pollination, shows stable inheritance and expression of a trait for several generations. It ensured that the parents were homozygous, so the results of crosses could be interpreted clearly.

Q85. 3M Board fav In pea, tall (T) is dominant over dwarf and round (R) over wrinkled. Give the gametes of TtRr and of ttrr, and the progeny ratio of their cross.
Ans: TtRr: TR, Tr, tR, tr; ttrr: tr only. Progeny: tall round, tall wrinkled, dwarf round, dwarf wrinkled = 1:1:1:1.

Q86. 2M Explain how the behaviour of chromosomes during meiosis explains Mendel's law of segregation.
Ans: Homologous chromosomes carrying the two alleles pair and then separate in anaphase I, so each gamete receives one homologue and hence one allele of each gene.

Q87. 2M What is the difference between genotype and phenotype? Can two plants with the same phenotype have different genotypes?
Ans: Genotype is the genetic make-up (TT, Tt); phenotype is the observable character (tall). Yes, TT and Tt are both tall.

Q88. 2M Why are recessive autosomal disorders more common in children of consanguineous (close-relative) marriages?
Ans: Close relatives are more likely to carry the same recessive allele inherited from a common ancestor, so the chance that both parents are carriers, and the child homozygous recessive, increases.

Q89. 1M Give one example each of an autosomal dominant and an X-linked recessive trait in humans.
Ans: Autosomal dominant: myotonic dystrophy. X-linked recessive: haemophilia or colour blindness.

Q90. 2M State two features of the sex chromosomes in a human female and a human male.
Ans: Female: two X chromosomes (homogametic), all ova carry X. Male: one X and one Y (heterogametic), sperms carry X or Y in equal numbers.

Q91. 3M Why is *Drosophila* called the "darling of geneticists"? Give three reasons.
Ans: Grows on a simple synthetic medium; completes its life cycle in about two weeks; a single mating gives many progeny; sexes are easily distinguished; many hereditary variations are visible under a low power microscope (any three).

Q92. 2M A plant with genotype AaBb is selfed. Assuming independent assortment, what fraction of progeny will be aabb, and what fraction will show both dominant traits?
Ans: aabb = 1/4 x 1/4 = 1/16; both dominant (A_B_) = 3/4 x 3/4 = 9/16.

Q93. 2M What is the chance that a couple's third child is a girl if the first two are girls? Why?
Ans: 50%. Each fertilisation is an independent event; the chance of an X- or Y-bearing sperm fertilising the ovum is always 1/2.

Long answer (5 marks)

Q94. 5M Board fav Explain a dihybrid cross with a Punnett square. State the law that it establishes.
Ans: Parents: yellow round (RRYY) x green wrinkled (rryy); F1 RrYy all yellow round. F1 gametes RY, Ry, rY, ry (1/4 each). 16-box Punnett square gives F2: 9 yellow round, 3 yellow wrinkled, 3 green round, 1 green wrinkled (9:3:3:1); each trait 3:1. Law of Independent Assortment: when two pairs of traits are combined in a hybrid, segregation of one pair is independent of the other.

Q95. 5M Board fav Describe the mechanism of sex determination in (a) humans (b) birds (c) grasshopper (d) honey bee.
Ans: (a) XY: female XX, male XY; sperms 50% X, 50% Y; father decides. (b) ZW: female ZW (heterogametic), male ZZ; the egg decides. (c) XO: female XX, male XO (one chromosome less); sperms with or without X; eggs fertilised by X-sperm become female. (d) Haplodiploid: fertilised egg gives diploid female (32), unfertilised egg gives haploid male (16) by parthenogenesis; males make sperms by mitosis, have no father and no sons.

Q96. 5M Board fav What are Mendelian disorders? Explain sickle-cell anaemia, haemophilia and phenylketonuria.
Ans: Disorders due to mutation in a single gene, inherited by Mendelian principles, traceable by pedigree. Sickle-cell: autosomal recessive (Hb^SHb^S); Glu replaced by Val at 6th position of beta globin due to GAG to GUG; RBCs sickle at low oxygen; carriers Hb^AHb^S. Haemophilia: X-linked recessive; a clotting protein is affected, non-stop bleeding; carrier mothers pass it to sons; Queen Victoria's pedigree. Phenylketonuria: autosomal recessive; enzyme converting phenylalanine to tyrosine absent; phenylpyruvic acid accumulates in brain, mental retardation; excreted in urine.

Q97. 5M Explain incomplete dominance and co-dominance with suitable crosses. How do they differ from complete dominance?
Ans: Incomplete dominance: RR (red) x rr (white) snapdragon gives Rr pink; F2 1:2:1 phenotype = genotype. Co-dominance: I^A and I^B both expressed in AB; I^Ai x I^Bi gives AB, A, B, O (1:1:1:1). Complete dominance: heterozygote looks like the dominant parent (Tt tall), F2 3:1. In incomplete dominance the heterozygote is intermediate; in co-dominance both traits appear fully.

Q98. 5M Board fav (a) What is linkage? How does it differ from recombination? (b) Describe Morgan's experiments and why *Drosophila* was suitable.
Ans: (a) Linkage: physical association of genes on the same chromosome, so they are inherited together; recombination: generation of non-parental gene combinations. (b) Morgan crossed yellow-bodied white-eyed females with brown-bodied red-eyed males and intercrossed F1; F2 deviated from 9:3:3:1 with more parental types. y-w showed 1.3% and w-m 37.2% recombination, so tightly linked genes recombine less. *Drosophila*: simple synthetic medium, two-week life cycle, many progeny per mating, clear sex difference, variations visible under low power microscope.

Q99. SM Board fav (a) What are chromosomal disorders? Distinguish aneuploidy and polyploidy. (b) Give the cause and symptoms of Down's, Klinefelter's and Turner's syndromes.
Ans: (a) Disorders due to absence, excess or abnormal arrangement of chromosomes. Aneuploidy: gain or loss of chromosome(s) due to failure of segregation of chromatids (trisomy, monosomy). Polyploidy: increase in whole sets due to failure of cytokinesis after telophase, common in plants. (b) Down's: trisomy 21; short stature, small round head, furrowed tongue, partially open mouth, broad palm with crease, retarded development (Langdon Down, 1866). Klinefelter's: 47, XXY; masculine with gynaecomastia, sterile. Turner's: 45, X0; sterile female with rudimentary ovaries, no secondary sexual characters.

Q100. SM State and explain Mendel's laws of dominance and segregation with a monohybrid cross, and show how a test cross identifies the genotype of a tall plant.
Ans: TT x tt: F1 all Tt tall (dominance: of two dissimilar factors, one expresses). F1 selfed: gametes T and t separate equally (segregation), F2 TT : Tt : tt = 1:2:1, tall : dwarf = 3:1; dwarf reappears unchanged, so factors do not blend. Test cross: tall x tt; all tall means TT, 1 tall : 1 dwarf means Tt.

Q101. SM (a) What is polygenic inheritance? Explain with human skin colour. (b) How does it differ from pleiotropy? Give an example of pleiotropy and explain its basis.
Ans: (a) Traits controlled by three or more genes, with additive allele effects and environmental influence. With genes A, B, C: AABBCC darkest, aabbcc lightest, AaBbCc intermediate; the number of dominant alleles decides the shade, giving a continuous range. (b) Pleiotropy: one gene with multiple phenotypic effects, due to its effect on metabolic pathways. Phenylketonuria: one mutant gene causes mental retardation and reduced hair and skin pigmentation.

Case-based (4 marks)

Genetics ke numericals ke baad answer key se match karte waqt:

Q102. 4M Board fav Study the pedigree: unaffected parents have an affected daughter and an unaffected son; the trait appears in both sexes and skips generations. (a) Is the trait dominant or recessive? (b) Autosomal or sex-linked? Why? (c) Give the genotypes of the parents. (d) Name one disorder inherited this way.
Ans: (a) Recessive (unaffected parents have an affected child). (b) Autosomal: an affected daughter has an unaffected father, which is not possible for X-linked recessive. (c) Both heterozygous carriers, Aa. (d) Sickle-cell anaemia (also phenylketonuria, thalassemia).

Q103. 4M Board fav A woman with normal vision whose father was colour blind marries a man with normal vision. (a) Give her genotype. (b) What proportion of their sons can be colour blind? (c) What proportion of daughters will be carriers? (d) Can any daughter be colour blind?
Ans: (a) X^CX^c (carrier). (b) 50% of sons (X^cY). (c) 50% of daughters (X^CX^c). (d) No, because the father gives a normal X^C.

Q104. 4M In a hospital, a mix-up of babies was suspected. Baby 1 has blood group O, baby 2 has AB. Couple P: both group O. Couple Q: father A, mother B. (a) Which baby belongs to couple P? Why? (b) Can couple Q have both babies' groups? (c) Name the inheritance shown by group AB. (d) How many alleles control ABO and why is it called multiple allelism?
Ans: (a) Baby 1 (O): O x O (ii x ii) can only give ii. (b) Yes, if they are I^Ai and I^Bi they can have AB and O; so baby 2 (AB) is theirs. (c) Co-dominance. (d) Three alleles (I^A, I^B, i); more than two alleles of one gene exist in the population.

Q105. 4M Board fav A boy has short stature, small round head, furrowed tongue, partially open mouth and a broad palm with a characteristic crease. His karyotype shows 47 chromosomes. (a) Name the disorder and its cause. (b) Who first described it? (c) What kind of chromosomal change is this called? (d) Name a disorder with 45 chromosomes.
Ans: (a) Down's syndrome: trisomy of chromosome 21. (b) Langdon Down (1866). (c) Aneuploidy (trisomy) due to failure of segregation of chromatids. (d) Turner's syndrome (45, X0).

Q106. 4M Board fav In *Drosophila*, a cross between flies differing in body colour and eye colour gave in F2: parental types 98.7% and recombinants 1.3%; another cross gave parental 62.8% and recombinants 37.2%. (a) Which gene pair is more tightly linked? (b) What causes recombinants? (c) Who used such data to map genes? (d) Why do linked genes not show 9:3:3:1?
Ans: (a) The first pair (yellow-white, 1.3%). (b) Crossing over between homologous chromosomes during meiosis. (c) Alfred Sturtevant. (d) They are on the same chromosome and tend to be inherited together, so parental combinations dominate.

Q107. 4M A teacher showed Mendel's dihybrid result: 315 round yellow, 108 round green, 101 wrinkled yellow, 32 wrinkled green. (a) Give the ratio. (b) What were the F1 genotype and phenotype? (c) Which law is proved? (d) What fraction of F2 is homozygous for both genes?

Ans: (a) About 9:3:3:1. (b) RrYy, round yellow. (c) Law of Independent Assortment. (d) 4/16 = 1/4 (RRYY, RRyy, rrYY, rryy).

Q108. 4M Board fav A couple, both of normal phenotype, have a son with haemophilia. (a) Give the genotypes of the parents. (b) What is the chance that their next son is haemophilic? (c) What is the chance that a daughter is a carrier? (d) Why is haemophilia called a "bleeder's disease"?

Ans: (a) Mother X^HX^h (carrier), father X^HY. (b) 50% (1/2) of sons. (c) 50% of daughters. (d) A clotting protein is affected, so a simple cut causes non-stop bleeding.

Q109. 4M A geneticist crossed a red-flowered snapdragon with a white-flowered one; F1 was pink. On selfing, F2 had 61 red, 119 pink and 58 white. (a) Name the phenomenon. (b) Give the F2 ratio and genotypes. (c) How does it differ from Mendel's tall-dwarf result? (d) Is the law of segregation violated?

Ans: (a) Incomplete dominance. (b) About 1:2:1: RR red, Rr pink, rr white. (c) In tall-dwarf, F1 resembles the dominant parent and F2 is 3:1; here F1 is intermediate and phenotype = genotype ratio. (d) No; alleles still segregate and reappear unchanged (red and white in F2).

Q110. 4M Board fav Mendel crossed a pure tall plant with violet flowers (TTVV) with a pure dwarf plant with white flowers (ttvv). (a) Give the F1 phenotype and genotype. (b) Name the gametes of F1. (c) What fraction of F2 will be dwarf violet? (d) What fraction of F2 will breed true for both traits?

Ans: (a) Tall violet, TtVv. (b) TV, Tv, tV, tv in equal proportion. (c) 3/16 (ttV_). (d) 4/16 = 1/4 (TTVV, TTvv, ttVV, ttvv).

Q111. 4M Board fav A man of blood group A whose mother was group O marries a woman of group B whose father was group O. (a) Give the genotypes of the man and woman. (b) List the possible genotypes and blood groups of their children with probability. (c) Which concept explains the AB child? (d) Why can an O child appear?

Ans: (a) Man I^Ai, woman I^Bi. (b) I^AI^B (AB), I^Ai (A), I^Bi (B), ii (O), each 1/4. (c) Co-dominance of I^A and I^B. (d) Both parents carry the recessive i, and a child may receive i from each.

Q112. 4M In a family, a normal woman whose brother is haemophilic marries a normal man. (a) What is the probability that she is a carrier? (b) If she is a carrier, what fraction of their sons will be haemophilic? (c) Give the overall chance that their first son is haemophilic. (d) Name the protein-related defect in haemophilia.

Ans: (a) 1/2 (her mother must be a carrier, so she has a 50% chance). (b) 1/2 of sons. (c) 1/2 x 1/2 = 1/4. (d) One of the proteins of the blood-clotting cascade is affected.

Q113. 4M A farmer wants pea plants that are always tall with round seeds. He has a tall round plant of unknown genotype. (a) How can he find its genotype? (b) What result shows it is TTRR? (c) What result shows TtRr? (d) Name the law shown by the result in (c).

Ans: (a) Test cross it with a dwarf wrinkled (ttrr) plant. (b) All progeny tall round. (c) Tall round, tall wrinkled, dwarf round and dwarf wrinkled in 1:1:1:1. (d) Law of Independent Assortment.

Q114. 4M A couple with normal skin pigmentation has a child with phenylketonuria. (a) Give the inheritance pattern. (b) Give the genotypes of the parents. (c) What is the risk for their next child? (d) Why does the child show both mental retardation and light hair and skin?

Ans: (a) Autosomal recessive. (b) Both heterozygous carriers (Pp). (c) 1/4 (25%). (d) Pleiotropy: lack of the enzyme converting phenylalanine to tyrosine affects several metabolic pathways; phenylpyruvic acid harms the brain and less tyrosine means less pigment.

Q115. 4M Read: "In a population of honey bees, the queen lays both fertilised and unfertilised eggs." (a) What will the unfertilised eggs develop into? (b) Give the chromosome number of a worker and a drone. (c) Name this sex-determination system. (d) How are sperms formed in drones?

Ans: (a) Drones (males), by parthenogenesis. (b) Worker 32, drone 16. (c) Haplodiploid sex-determination system. (d) By mitosis.

Exam Alert: numericals ka tarika

- → Pehle parents ke genotypes likho, phir gametes, phir Punnett square: steps likhne par poore marks.
- → Probability: alag-alag traits ki probability multiply karo (3/4 x 1/2 = 3/8).
- → Pedigree: unaffected parents ka affected beta/beti = recessive; affected father ki saari betiyan affected = X-linked dominant ka hint.
- → Beta ka X hamesha **maa** se aata hai, isliye X-linked trait father se son ko nahi jaata.

Molecular Basis of Inheritance

Most asked: DNA double helix features, Hershey-Chase experiment, Meselson-Stahl experiment, transcription unit, lac operon, genetic code, DNA fingerprinting

DNA stores the genetic information and RNA helps express it. This chapter covers the full flow: **DNA structure and packaging, replication, transcription, genetic code, translation, gene regulation, HGP and DNA fingerprinting.**

1. The DNA

DNA (deoxyribonucleic acid) and **RNA (ribonucleic acid)** are the two types of nucleic acids found in living systems. DNA is the genetic material in most organisms; RNA is the genetic material in some viruses but mostly works as a **messenger**, an **adapter**, a **structural** molecule and sometimes a **catalyst**.

DNA is a long **polymer of deoxyribonucleotides**. Its length is usually given as the number of nucleotides or base pairs (bp), and this is characteristic of an organism:

Organism	Length of DNA
Bacteriophage φX174	**5386 nucleotides** (single-stranded)
Bacteriophage lambda (λ)	**48502 bp**
Escherichia coli	**4.6×10^6 bp**
Human (haploid content)	**3.3×10^9 bp**

A. Structure of a polynucleotide chain

A **nucleotide** has three components: a nitrogenous base, a pentose sugar (ribose in RNA, deoxyribose in DNA) and a phosphate group.

Purines (2 rings)	Pyrimidines (1 ring)
Adenine (A), Guanine (G)	Cytosine (C), Thymine (T), Uracil (U)
present in both DNA and RNA	C in both; **T only in DNA**; **U only in RNA**

- → **Nucleoside** = nitrogenous base + pentose sugar, joined at the 1′ carbon by an **N-glycosidic linkage**. e.g. adenosine, guanosine, cytidine, uridine, deoxythymidine.
- → **Nucleotide** = nucleoside + phosphate, attached to the 5′-OH of the sugar by a **phosphoester linkage**.
- → Two nucleotides join by a **3′–5′ phosphodiester linkage** to form a dinucleotide; many join to form a polynucleotide.
- → A polynucleotide has a free phosphate at the **5′-end** of the sugar and a free 3′-OH at the **3′-end**: this gives the chain its polarity.
- → The **backbone** is formed by sugar and phosphates; the nitrogenous bases project from it.
- → In RNA, every nucleotide has an extra **–OH at the 2′ position** of ribose, and **uracil** replaces thymine (5-methyl uracil).

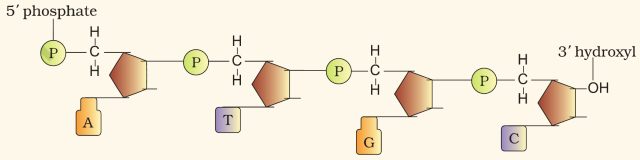


Fig. 5.1 A polynucleotide chain

Yaad karo: purines vs pyrimidines

Pure As Gold = **Pu**rines: Adenine, Guanine (2 rings, bade). **CUT the Py** = **Py**rimidines: Cytosine, Uracil, Thymine (1 ring, chhote).

B. History of DNA structure

- → **Friedrich Meischer (1869)** first identified DNA as an acidic substance in the nucleus and named it Nuclein.
- → **Maurice Wilkins and Rosalind Franklin** produced **X-ray diffraction** data of DNA.
- → **Erwin Chargaff**: in double-stranded DNA, A = T and G = C, so purines = pyrimidines and (A+G)/(T+C) = 1.
- → Using these data, **James Watson and Francis Crick (1953)** proposed the double helix model of DNA. PYQ

C. Salient features of the double helix PYQ

1. DNA is made of **two polynucleotide chains**; the backbone is sugar-phosphate and the **bases project inside**.
2. The two chains have anti-parallel polarity: if one is 5′ → 3′, the other is 3′ → 5′.
3. Bases of the two strands pair through **hydrogen bonds**: A = T (2 H-bonds) and G ≡ C (3 H-bonds). A purine always comes opposite a pyrimidine, giving a **uniform distance** between the strands.
4. The two chains are coiled in a **right-handed** fashion. **Pitch = 3.4 nm**, with roughly **10 bp per turn**, so the distance between two bp is about 0.34 nm.
5. The plane of one base pair stacks over the other. This **stacking**, together with H-bonds, gives stability to the helix.

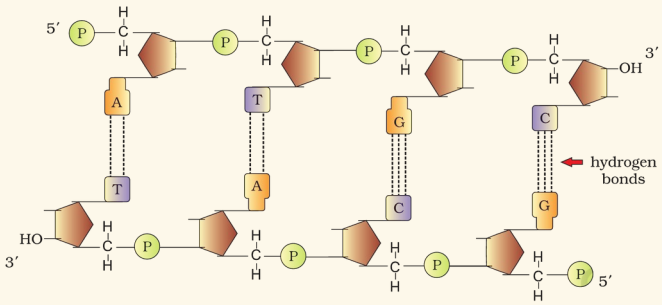


Fig. 5.2 Double-stranded polynucleotide chain

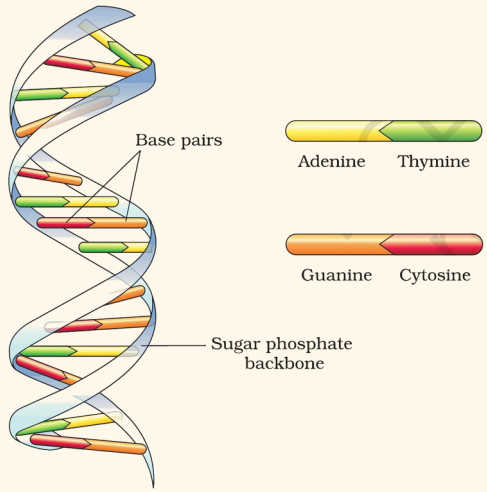


Fig. 5.3 DNA double helix

Quick fact: base-pairing logic

Because of **complementarity**, the sequence of one strand fixes the sequence of the other. If one strand is 5′-ATGC-3′, the other must be 3′-TACG-5′. This idea is used in replication, transcription and probes.

Exam Alert: Chargaff numericals

- → A = T aur G = C. Agar C = 20%, toh G = 20%, (A + T) = 60%, isliye **A = 30%** (NCERT exercise).
- → Ratio A+T / G+C species ke hisaab se badalta hai, par **A+G / T+C = 1** hamesha (double-stranded DNA).
- → Single-stranded DNA (φX174) mein Chargaff rule apply nahi hota.

Exam Alert: DNA numericals

- → **H-bonds** = 2 × (A-T pairs) + 3 × (G-C pairs). 100 bp DNA mein 30 A ho toh A-T = 30, G-C = 70, H-bonds = 60 + 210 = **270**.
- → **Length** = bp × 0.34 nm. 1000 bp = **340 nm**.
- → **Turns** = bp / 10. 1000 bp = **100 turns**.
- → Phosphodiester bonds in a linear dsDNA of n bp = 2(n − 1).

D. Central dogma

Francis Crick proposed the **central dogma** of molecular biology: genetic information flows DNA → RNA → Protein.

DNA → replication → DNA → transcription → mRNA → translation → Protein

In some viruses (**retroviruses**, e.g. HIV), information flows in the **reverse direction**, from RNA to DNA, by the enzyme reverse transcriptase (reverse transcription).

E. Packaging of the DNA helix

Length of DNA in a typical mammalian cell = 6.6×10^{9} bp × 0.34×10^{-9} m/bp = about 2.2 metres, far greater than a typical nucleus (**about 10^{-6} m**). So DNA must be tightly packed. PYQ

- → **Prokaryotes** (e.g. *E. coli*): no defined nucleus. The DNA is **negatively charged** and is held with some **positively charged proteins** in a region called the nucleoid, organised in large loops.
- → **Eukaryotes**: a set of positively charged, basic proteins called histones. They are rich in the basic amino acids **lysine and arginine**, whose side chains carry positive charges.
- → Eight histone molecules form a **histone octamer**. Negatively charged DNA wraps around it to form a nucleosome; a typical nucleosome contains **200 bp** of DNA helix.
- → Nucleosomes are the repeating unit of **chromatin**. Under the electron microscope chromatin looks like beads-on-string.
- → The beads-on-string structure is packed into **chromatin fibres**, which coil and condense at metaphase to form **chromosomes**.
- → Higher-level packaging needs an additional set of proteins: Non-histone Chromosomal (NHC) proteins.

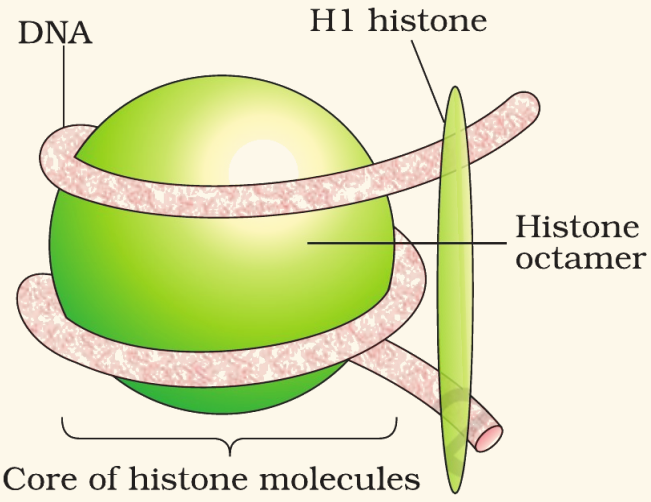


Fig. 5.4 (a) Nucleosome

DNA (2 nm) → nucleosome (beads-on-string) → chromatin fibre → looped domains → metaphase chromosome

Euchromatin	Heterochromatin
loosely packed	**densely** packed
stains **light**	stains **dark**
transcriptionally active	transcriptionally **inactive**

Feature	Prokaryotic DNA	Eukaryotic DNA
location	nucleoid (no membrane)	nucleus (inside nuclear envelope)
proteins	some positively charged proteins	histones (octamer) + NHC proteins
form	usually circular	linear, in many chromosomes
packaging unit	large loops	nucleosome (200 bp)

Paheli-Bujho

Bujho: Paheli, 2.2 metre DNA itne chhote nucleus mein kaise ghus gaya?
Paheli: Histone octamer pe lipat ke! Har nucleosome mein 200 bp DNA.
Bujho: Matlab DNA ne khud ko earphone ki tarah lapet liya?
Paheli: Haan, par bina uljhe. Beads-on-string, phir fibre, phir chromosome.

Key terms

- → **Histone octamer**: 8 basic histone proteins forming the core of a nucleosome.
- → **Nucleosome**: histone octamer + about 200 bp of DNA wrapped around it.
- → **NHC proteins**: non-histone proteins needed for higher-level packaging.
- → **Nucleoid**: DNA region of a prokaryotic cell, not bound by a membrane.

2. The Search for Genetic Material

Even though nuclein (Meischer) and Mendel's principles were discovered at almost the same time, proof that **DNA is the genetic material** came much later, through a series of experiments.

A. Transforming principle: Griffith (1928) PYQ

Frederick Griffith worked with *Streptococcus pneumoniae* (bacterium causing pneumonia) and mice.

Strain	Colony	Coat	Effect on mice
S strain	smooth, shiny	**polysaccharide** (mucous) coat	**virulent**: mice die of pneumonia
R strain	rough	no coat	**non-virulent**: mice live

1. S strain injected → **mice die.**
2. R strain injected → **mice live.**
3. Heat-killed S strain → **mice live.**
4. Heat-killed S + live R strain → mice die, and **living S bacteria** are recovered from the dead mice.

Conclusion: the R strain was **transformed** by some "transforming principle" transferred from the heat-killed S strain, which enabled R to make a smooth coat and become virulent. This must be due to transfer of genetic material. The **biochemical nature** of the genetic material was not defined by Griffith.

B. Biochemical characterisation: Avery, MacLeod and McCarty (1933-44)

Before their work the genetic material was thought to be a **protein**. They purified biochemicals (proteins, DNA, RNA etc.) from heat-killed S cells to see which one could transform live R cells into S cells.

- → **Protease** (protein-digesting) and **RNase** (RNA-digesting) enzymes did **not** affect transformation.
- → **DNase** inhibited transformation.
- → Conclusion: DNA caused the transformation of R cells. However, not all biologists were convinced. PYQ

Yaad karo: kis enzyme ne kya kiya

Protease = protein gaya, transformation phir bhi hua. **RNase** = RNA gaya, phir bhi hua. **DNase** = DNA gaya, transformation **band**. Toh asli culprit = **DNA**!

C. The genetic material is DNA: Hershey and Chase (1952) PYQ

Alfred Hershey and Martha Chase worked with viruses that infect bacteria, called **bacteriophages**. A bacteriophage attaches to the bacterium and its genetic material enters the bacterial cell, which then treats it as its own and makes more virus particles.

- → Some viruses were grown on a medium with radioactive phosphorus (^{32}P): DNA contains phosphorus, protein does not, so their **DNA** became labelled.
- → Others were grown on radioactive sulfur (^{35}S): protein contains sulfur, DNA does not, so their **protein coat** became labelled.
- → The radioactive phages were allowed to attach to *E. coli*. After infection, the viral coats were removed from the bacteria by agitating in a **blender**, and the virus particles were separated from bacteria by spinning in a **centrifuge**.
- → ^{32}P (radioactive DNA) phages: bacteria were radioactive, so DNA entered the cells. ^{35}S (radioactive protein) phages: bacteria were not radioactive; radioactivity stayed in the **supernatant** with the coats.
- → Conclusion: DNA, not protein, is the genetic material passed from virus to bacteria.

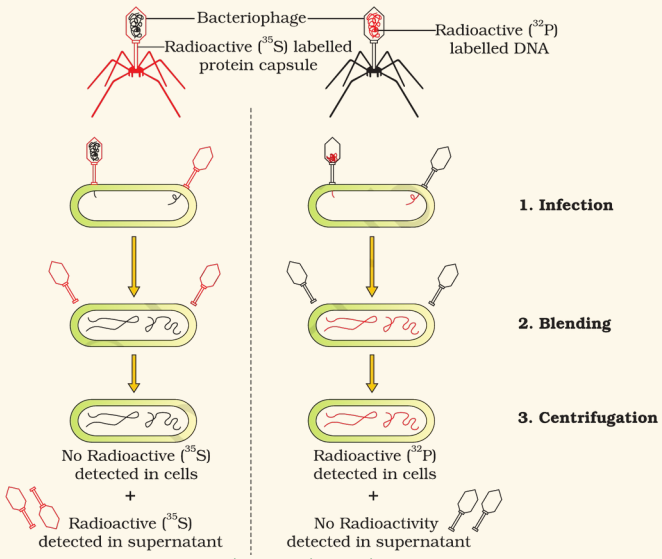


Fig. S.5 The Hershey-Chase experiment

Step	Blending	Centrifugation
What it does	separates the **empty viral coats** from the bacterial surface	separates **heavy bacteria** (pellet) from **light viral particles** (supernatant)

Hershey-Chase ka **blender + centrifuge** wala step: board mein pakka aayega.

Asli baat: ^{32}P = DNA (andar gaya), ^{35}S = protein (bahar reh gaya).

D. Properties of genetic material (DNA versus RNA)

A molecule that can act as genetic material must fulfil these criteria: PYQ

1. It should be able to **generate its replica** (replication).
2. It should be **chemically and structurally stable**.
3. It should provide scope for **slow changes (mutation)** required for evolution.
4. It should be able to **express itself** in the form of "Mendelian characters".

→ **Replication:** because of base pairing and complementarity, both DNA and RNA can direct their duplication. Proteins fail the first criterion.

→ **Stability:** genetic material should not change with stage of life, age or physiology. Griffith's heat-killed bacteria showed that DNA survives heat. Both strands of DNA are complementary and re-join when separated.

→ The 2'-OH group on every nucleotide of RNA is **reactive** and makes RNA labile and easily degradable. RNA is also catalytic, hence reactive. So DNA is chemically less reactive and more stable.

→ The presence of **thymine** in place of uracil also gives DNA **extra stability**.

→ **Mutation:** both can mutate, but RNA, being unstable, mutates **faster**. So viruses with RNA genomes (shorter life span) mutate and evolve faster.

→ **Expression:** RNA can directly code for protein synthesis and is easily expressed. DNA depends on RNA for protein synthesis. The protein-synthesising machinery has evolved around RNA.

Conclusion: DNA is better for storage of genetic information, while RNA is better for transmission (expression) of information. That is why cellular organisms keep DNA as their genetic material, while only some viruses (with short life spans) use RNA. PYQ

DNA	RNA
deoxyribose sugar (no 2'-OH)	ribose sugar (2'-OH present)
bases A, T, G, C	bases A, U, G, C
usually double-stranded	usually single-stranded
chemically stable, mutates slowly	reactive, labile, mutates fast
genetic material in most organisms	genetic material in some viruses (e.g. TMV, Qβ bacteriophage); messenger, adapter, catalyst

3. RNA World

→ RNA was the first genetic material. Essential life processes such as metabolism, translation and splicing evolved around RNA.

→ RNA acted both as **genetic material** and as a **catalyst** (some important biochemical reactions in living systems are catalysed by RNA catalysts, not protein enzymes).

→ Being a catalyst, RNA was **reactive and hence unstable**. So DNA evolved from RNA with **chemical modifications** that made it more stable.

→ DNA, being double-stranded with a **complementary strand**, resists change by evolving a process of repair.

Quick fact

A catalytic RNA is called a **ribozyme**. The **23S rRNA** in bacteria catalyses peptide bond formation during translation: living proof of the RNA world.

4. Replication

While proposing the double helix, **Watson and Crick** also proposed the scheme for replication: the two strands **separate** and each acts as a **template** for a new complementary strand. After replication, each DNA molecule has one parental and one newly synthesised strand. This is semi-conservative replication. PYQ

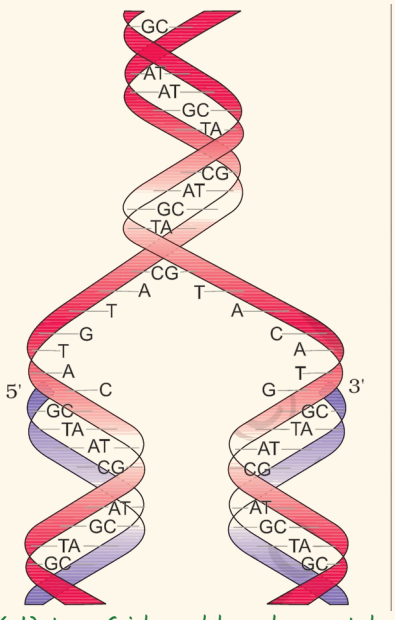


Fig. 5.6 Watson-Crick model: each parental strand acts as a template

Exam Alert: NCERT exercise

Kaunsi property se semi-conservative idea aaya? Dono strands ka **complementary base pairing**. Ek strand ka sequence doosre ko fix karta hai, isliye har strand template ban sakta hai.

A. The experimental proof: Meselson and Stahl (1958) PYQ

1. *E. coli* was grown for many generations in a medium with $^{15}NH_4Cl$ (^{15}N = heavy isotope) as the only nitrogen source. ^{15}N got incorporated into the new DNA (and other N-compounds), making heavy DNA.
2. Heavy DNA can be separated from normal DNA by **centrifugation in a caesium chloride (CsCl) density gradient**. (^{15}N is not radioactive.)
3. The cells were then transferred to a medium with normal $^{14}NH_4Cl$, and samples were taken at intervals as cells multiplied.
4. The DNA extracted in each generation was checked in the CsCl gradient for its density.

Generation (*E. coli* divides in 20 min)	DNA band observed
0 (in ^{15}N)	only heavy ($^{15}N^{15}N$)
1st (after 20 min)	**100% hybrid** or intermediate ($^{15}N^{14}N$)
2nd (after 40 min)	**50% hybrid + 50% light** ($^{14}N^{14}N$)
3rd (after 60 min)	25% hybrid + 75% light

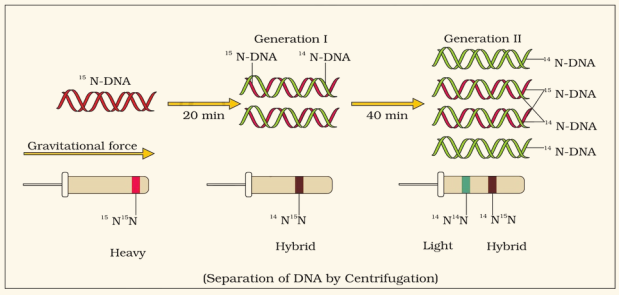


Fig. 5.7 Meselson and Stahl's experiment

Similar experiments with radioactive thymidine on *Vicia faba* (faba beans) by **Taylor and colleagues (1958)** proved that DNA in chromosomes also replicates semi-conservatively.

Paheli-Bujho

Bujho: Paheli, 2nd generation mein heavy DNA kitna bachega?
Paheli: Zero! Sirf hybrid aur light. Heavy strand akela kabhi nahi rehta.
Bujho: Kyunki har nayi DNA mein ek purana strand hota hai?
Paheli: Bilkul. Yahi semi-conservative ka matlab hai.

B. The machinery and the enzymes

- In living cells such as *E. coli*, replication needs a set of catalysts (enzymes). The main enzyme is DNA-dependent DNA polymerase: it uses a DNA template to catalyse polymerisation of deoxynucleotides.
- These enzymes are **highly efficient**: *E. coli* (4.6×10^6 bp) completes replication in **about 18 minutes**, i.e. an average rate of about 2000 bp per second.
- They must also be **highly accurate**; any mistake would result in mutations.
- **Deoxyribonucleoside triphosphates** (dATP, dGTP, dCTP, dTTP) act as substrates and also provide **energy** for polymerisation (two terminal phosphates are released).
- Replicating the long DNA in one go is not possible because of the very high energy needed, so a small opening of the helix, the replication fork, is made, where replication occurs.
- DNA-dependent DNA polymerases catalyse polymerisation **only in the 5′ → 3′ direction**. So on the template with polarity **3′ → 5′**, replication is continuous (leading strand); on the template with polarity **5′ → 3′** it is discontinuous (lagging strand).
- The discontinuously synthesised fragments (**Okazaki fragments**) are later joined by the enzyme DNA ligase.
- DNA polymerases cannot initiate replication on their own, and replication does not start randomly. It begins at a definite region called the origin of replication (ori). Because of this, a piece of DNA must have an origin to be propagated during recombinant DNA work (vectors).
- In eukaryotes, replication occurs in the **S-phase** of the cell cycle; replication and cell division must be **highly coordinated**. A failure in cell division after DNA replication results in **polyploidy**.

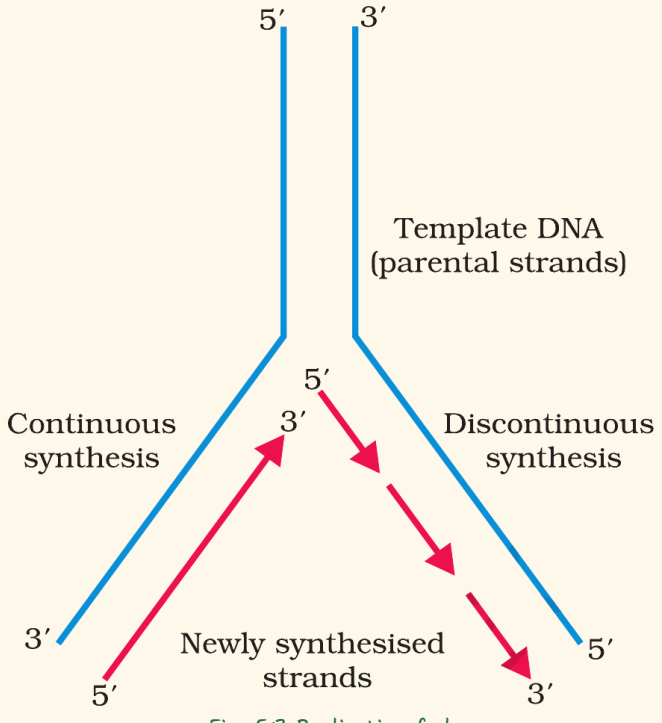


Fig. 5.8 Replicating fork

Leading strand	Lagging strand
template polarity 3′ → 5′	template polarity 5′ → 3′
synthesised **continuously**	synthesised **discontinuously** as fragments
moves towards the fork	moves away from the fork
one continuous strand, no fragments to join	fragments joined by **DNA ligase**

DNA polymerase	RNA polymerase
makes DNA (replication)	makes RNA (transcription)
substrates: dNTPs (dATP, dGTP, dCTP, dTTP)	substrates: NTPs (ATP, GTP, CTP, UTP)
copies **both** strands, whole genome	copies **one** strand, only a segment (gene)
cannot initiate on its own; starts at ori	initiates at the promoter (with σ in bacteria)

Key terms

- → **Template**: the parental strand on which a new complementary strand is built.
- → **Replication fork**: the Y-shaped region where the helix opens and replication occurs.
- → **Ori**: origin of replication, the definite site where replication begins.
- → **DNA ligase**: enzyme that joins DNA fragments (the "molecular glue").

Who did what: scientists of this chapter

Scientist	Contribution
Friedrich Meischer (1869)	identified DNA, named it nuclein
Erwin Chargaff	A = T, G = C (base equivalence)
Wilkins and Franklin	X-ray diffraction of DNA
Watson and Crick (1953)	double helix; semi-conservative replication scheme
Francis Crick	central dogma; adapter hypothesis
Griffith (1928)	transformation
Avery, MacLeod, McCarty (1933-44)	DNA is the transforming principle
Hershey and Chase (1952)	DNA is the genetic material (phage)
Meselson and Stahl (1958)	semi-conservative replication (*E. coli*)
Taylor et al. (1958)	semi-conservative replication (*Vicia faba*)
George Gamow	triplet code
H. G. Khorana	chemical synthesis of RNA with defined bases
Marshall Nirenberg	cell-free system, decoded the code
Severo Ochoa	polynucleotide phosphorylase
Jacob and Monod	lac operon
Frederick Sanger	DNA sequencing method
Alec Jeffreys	DNA fingerprinting (VNTR)

5. Transcription

The process of copying genetic information from **one strand of DNA into RNA** is transcription. The principle of complementarity governs it, except that **adenine pairs with uracil** instead of thymine.

Unlike replication (where the whole DNA is copied), only a **segment of DNA and only one of the strands** is copied into RNA. Why both strands are not copied: PYQ

- → If both strands acted as templates, they would code for RNA with **different sequences**, so one segment would code for **two different proteins**, complicating the genetic information transfer.
- → The two RNA molecules, being complementary, would **form a double-stranded RNA** and prevent translation. So transcription would be futile.

A. Transcription unit PYQ

A transcription unit in DNA is defined by three regions:

1. **Promoter**: the DNA sequence that provides the **binding site for RNA polymerase**; it determines which strand is template and which is coding. It is located towards the **5′-end (upstream)** of the structural gene (with reference to the coding strand).
2. **Structural gene**: the region that is transcribed.
3. **Terminator**: located towards the **3′-end (downstream)** of the coding strand; it defines the **end** of transcription.

- → **Template strand**: the strand with polarity **3′ → 5′**; RNA polymerase uses it, since RNA is made 5′ → 3′.
- → **Coding strand**: the other strand, polarity **5′ → 3′**. It has the same sequence as the RNA (with T in place of U) but does not code for anything. All reference points in a transcription unit are written with respect to the coding strand.
- → Additional **regulatory sequences** may be present further upstream or downstream of the promoter.

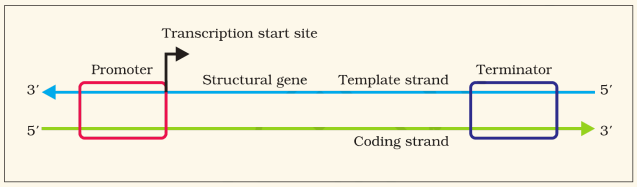


Fig. 5.9 Schematic structure of a transcription unit

Exam Alert: coding strand se mRNA

Coding strand 5′-ATGCATGC-3′ ho toh mRNA = **5′-AUGCAUGC-3′** (bas T ko U kar do). Template strand diya ho toh pehle complementary likho, phir T → U. Polarity hamesha likho, warna marks katenge.

B. Transcription unit and the gene

- → A gene is the functional unit of inheritance. Though genes are located on DNA, it is difficult to define a gene only by DNA sequence: DNA coding for tRNA or rRNA also defines a gene.

→ A **cistron** is a segment of DNA coding for a polypeptide. The structural gene in a transcription unit may be monocistronic (mostly in **eukaryotes**) or polycistronic (mostly in **bacteria or prokaryotes**).

→ In eukaryotes, the monocistronic structural genes have **interrupted coding sequences**, i.e. the genes are split genes.

→ The coding or expressed sequences are **exons**: they appear in mature or processed RNA. Exons are interrupted by **introns** (intervening sequences), which do not appear in mature RNA.

→ The **inheritance of a character** is also affected by promoter and regulatory sequences of a structural gene; these are sometimes loosely called regulatory genes, though they do not code for any RNA or protein.

Exons	*Introns*
coding (expressed) sequences	non-coding (intervening) sequences
present in mature mRNA	removed by splicing; absent in mature mRNA
carry the information for the polypeptide	occur in eukaryotic split genes; reminiscent of antiquity

C. Types of RNA PYQ

RNA	*Role in protein synthesis*
mRNA (messenger)	provides the **template** (codons)
tRNA (transfer)	brings amino acids and reads the genetic code (**adapter**)
rRNA (ribosomal)	**structural** role and **catalytic** role during translation

All three RNAs are needed to synthesise a protein in a cell.

D. Process of transcription in bacteria PYQ

In bacteria, a single DNA-dependent RNA polymerase catalyses transcription of **all types of RNA**. It works in three steps:

1. **Initiation**: RNA polymerase binds the **promoter**; it associates transiently with the initiation factor (σ, sigma) to initiate transcription.
2. **Elongation**: it uses **nucleoside triphosphates** as substrate and polymerises in a template-dependent manner following complementarity. It also helps open the helix and continues elongation. Only a short stretch of RNA remains bound to the enzyme.
3. **Termination**: once the polymerase reaches the **terminator**, the nascent RNA and the polymerase fall off; termination is facilitated by the termination factor (ρ, rho).

The RNA polymerase is only capable of catalysing elongation; it associates with σ and ρ to initiate and terminate respectively. In bacteria, mRNA needs **no processing**, and since there is no nuclear membrane, transcription and translation can be coupled (occur together, even before the mRNA is fully transcribed).

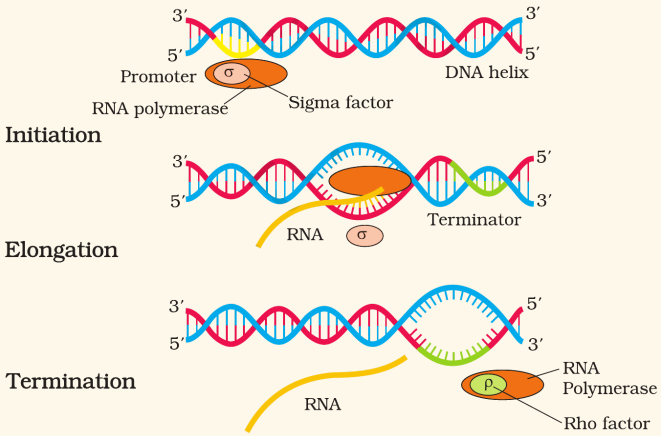


Fig. 5.10 Process of transcription in bacteria

E. Transcription in eukaryotes: two extra complexities PYQ

1. There are at least **three RNA polymerases** in the nucleus (besides the one in organelles), with a clear division of labour:

Enzyme	*Transcribes*
RNA polymerase I	rRNAs: **28S, 18S and 5.8S**
RNA polymerase II	precursor of mRNA: heterogeneous nuclear RNA (hnRNA)
RNA polymerase III	**tRNA, 5S rRNA and snRNAs** (small nuclear RNAs)

2. The primary transcript contains **both exons and introns** and is non-functional. It undergoes processing:

1. **Splicing**: introns are removed and exons are joined in a defined order.
2. **Capping**: an unusual nucleotide, methyl guanosine triphosphate, is added to the **5′-end** of hnRNA.
3. **Tailing**: adenylate residues (200-300) are added at the **3′-end** in a template-independent manner (poly-A tail).

The fully processed hnRNA, now called **mRNA**, is transported out of the nucleus for translation.

Feature	*Bacteria (prokaryotes)*	*Eukaryotes*
RNA polymerase	**one** (with σ and ρ)	**three** in nucleus (I, II, III)
structural gene	mostly polycistronic, no introns	mostly monocistronic, split (exons + introns)
primary transcript	functional mRNA directly	hnRNA, needs splicing, capping, tailing
site of translation	cytoplasm, coupled with transcription	cytoplasm, after mRNA leaves the nucleus

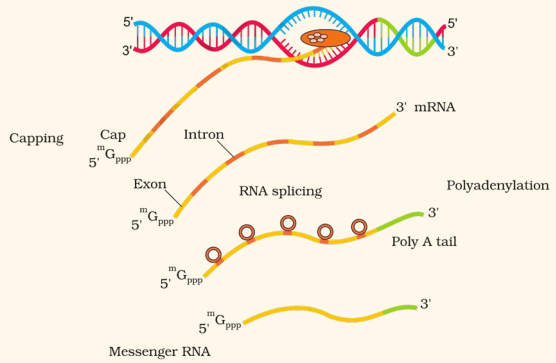


Fig. 5.11 Process of transcription in eukaryotes

Quick fact: why split genes matter

Split genes are an **ancient feature** of the genome. The presence of introns is reminiscent of **antiquity**, and splicing represents the dominance of the RNA world.

Yaad karo: RNA polymerases

1-2-3 = R-M-T: Pol I = RRNA (28S, 18S, 5.8S), Pol II = MRNA ka precursor (hnRNA), Pol III = TRNA (+ 5S rRNA, snRNA). "Ram Mohan Tiwari" bol do, yaad ho jayega!

6. Genetic Code

During replication and transcription a nucleic acid is copied into another nucleic acid, so complementarity is easy to follow. **Translation** transfers information from a polymer of nucleotides to a **polymer of amino acids**, with no complementarity between them. So a genetic code must direct the sequence of amino acids.

→ Physicist **George Gamow** argued: there are only 4 bases to code for 20 amino acids, so the code must be a combination of bases. A triplet code gives 4^3 = **64 codons**, more than enough for 20 amino acids. PYQ

→ **Har Gobind Khorana** developed the **chemical method** of synthesising RNA molecules with defined combinations of bases (homopolymers and copolymers).

→ **Marshall Nirenberg**'s cell-free system for protein synthesis finally helped decipher the code.

→ **Severo Ochoa**'s enzyme polynucleotide phosphorylase helped polymerise RNA with defined sequences in a template-independent manner (enzymatic RNA synthesis).

→ Finally, a **checker-board** for the genetic code was prepared (Table 5.1).

Salient features of the genetic code PYQ

1. The codon is a triplet. **61 codons** code for amino acids and **3 codons** do not code for any amino acid; they are **stop codons**.
2. Some amino acids are coded by more than one codon, hence the code is degenerate.
3. The codon is read in mRNA in a **contiguous** fashion: there are no punctuations (comma-less).
4. The code is nearly universal: e.g. UUU codes for phenylalanine (Phe) from bacteria to human. Some exceptions exist in **mitochondrial codons** and in some protozoans.
5. One codon codes for only one amino acid, hence it is **unambiguous and specific**.
6. AUG has dual functions: it codes for **methionine (Met)** and also acts as the **initiator codon**.
7. **UAA, UAG and UGA** are stop (terminator) codons.

First position	Second position U	Second position C	Second position A	Second position G	Third position
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA Stop	UGA Stop	A
	UUG Leu	UCG Ser	UAG Stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

Table 5.1 The codons for the various amino acids

Yaad karo: stop codons

U Are Away (UAA), **U Are Gone** (UAG), **U Go Away** (UGA). Teeno "U" se shuru, aur teeno pe protein ka **full stop**. Start sirf ek: **AUG** = "Aao Ullu Ghar" (Met).

Exam Alert: NCERT in-text

mRNA: 5'-AUG UUU UUC UUC UUU UUU UUC-3' ka protein: **Met-Phe-Phe-Phe-Phe-Phe-Phe** (UUU aur UUC dono Phe). Ulta bhi poocha jaata hai: Met-Phe-Phe ke liye possible mRNA likho: AUG UUU UUC (degenerate code ki wajah se kai answers sahi).

Exam Alert: common galtiyan

→ **Codon** mRNA pe hota hai, **anticodon** tRNA pe. Codon AUG ka anticodon = UAC.

→ Coding strand ka sequence = mRNA (T ki jagah U). Template strand se mRNA likhoge toh complementary banana padega.

→ Stop codons ke liye **koi tRNA nahi**, aur ye kisi amino acid ko code nahi karte.

→ "Universal" nahi, **nearly universal** likho (mitochondria aur kuch protozoa exceptions).

A. Mutations and genetic code

→ The relationship between genes and DNA is best understood by mutation studies. Effects of large **deletions and rearrangements** in a segment of DNA are easy to understand: they may cause loss or gain of a gene and so a function.

→ The effect of a **point mutation** is explained by the classic example of the gene for **β-globin chain**: change of a single base pair changes the amino acid residue glutamate to valine (GAG → GUG in mRNA). This results in the disease sickle cell anaemia. PYQ

Effect of insertion or deletion is shown by a sentence of three-letter words, read like codons:

Change	Reads as	Effect
original	RAM HAS RED CAP	meaningful
insert one letter (B)	RAM HAB SRE DCA P	reading frame shifts after insertion
insert three letters (BIG)	RAM HAS BIG RED CAP	**one extra word** (amino acid); rest unchanged
delete one letter (R)	AMH ASR EDC AP	reading frame shifts
delete three letters	RAM RED CAP	**one word** (amino acid) lost; rest unchanged

Insertion or deletion of **one or two bases** changes the reading frame from the point of change: these are frameshift insertion or deletion mutations. Insertion or deletion of **three or multiples of three bases** inserts or deletes one or more codons, i.e. one or more amino acids, and the reading frame stays unaltered after that point. This proves that the codon is a triplet and is read contiguously.

B. tRNA: the adapter molecule PYQ

→ From the beginning, **Francis Crick** postulated an adapter molecule that would read the code on one hand and bind specific amino acids on the other. tRNA was known before the genetic code was postulated and was then called sRNA (soluble RNA).

→ tRNA has an anticodon loop with bases complementary to the code, and an amino acid acceptor end to which it binds the amino acid.

→ tRNAs are **specific** for each amino acid. For initiation there is another specific tRNA, the initiator tRNA.

→ There are **no tRNAs for stop codons**.

→ Its secondary structure (drawn in 2D) looks like a **clover-leaf**; in reality it is a compact molecule that looks like an inverted L (3D).

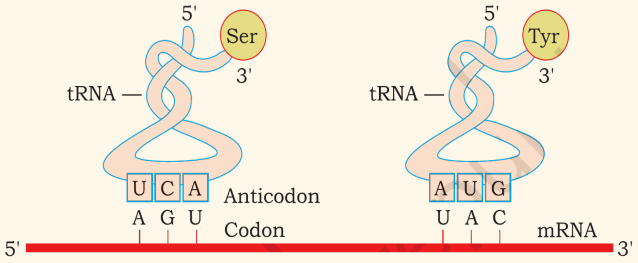


Fig. S.12 tRNA: the adapter molecule

7. Translation

Translation is the process of **polymerisation of amino acids** to form a polypeptide. The order and sequence of amino acids are defined by the sequence of bases in the mRNA. Amino acids are joined by a bond called the peptide bond, whose formation needs energy. PYQ

1. **Charging (aminoacylation) of tRNA**: amino acids are **activated in the presence of ATP** and linked to their cognate tRNA. If two such charged tRNAs come close enough, peptide bond formation between them is energetically favoured. A catalyst enhances the rate of peptide bond formation.
2. **Ribosome**: the cellular factory for protein synthesis. It consists of **structural RNAs and about 80 different proteins**. In its inactive state it exists as two subunits, a **large** and a **small** subunit.
3. **Initiation**: when the small subunit encounters an mRNA, translation begins. The ribosome binds the mRNA at the start codon (AUG), recognised only by the **initiator tRNA**.
4. **Elongation**: the large subunit has **two sites** for amino acids to bind, close enough for a peptide bond. The ribosome also acts as a catalyst (**23S rRNA in bacteria is the enzyme, a ribozyme**). Complexes of amino acid-tRNA sequentially bind to the appropriate codon by complementary base pairing with the anticodon. The ribosome moves **codon to codon** along the mRNA, adding amino acids one by one.
5. **Termination**: at the end, a release factor binds to the stop codon, terminating translation and releasing the complete polypeptide from the ribosome.

An mRNA also has some additional sequences that are not translated, the untranslated regions (UTRs). UTRs are present at both the **5'-end (before the start codon)** and the **3'-end (after the stop codon)**; they are required for an efficient translation process.

Exam Alert: NCERT exercise (ribosome ke 2 roles)

→ Ribosome = **platform**: mRNA aur charged tRNAs ko sahi jagah pakad ke rakhta hai (codon-anticodon pairing).

→ Ribosome = **catalyst**: 23S rRNA (ribozyme) peptide bond banata hai.

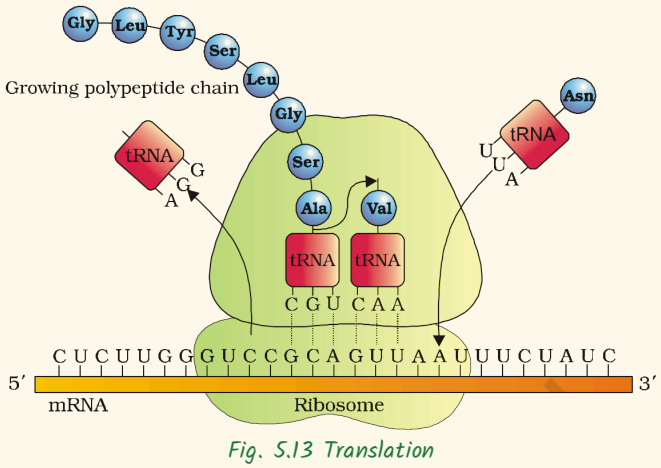
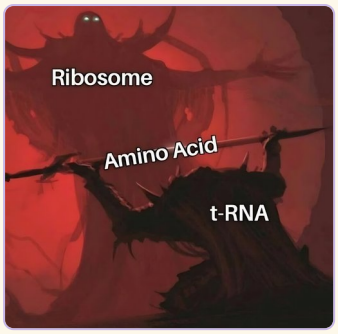


Fig. S.13 Translation



tRNA jab ribosome ko **sahi amino acid** deliver karta hai: "order aapka, sir".

Asli baat: Anticodon codon ko padhta hai, ribosome peptide bond banata hai.

Feature	*Replication*	*Transcription*	*Translation*
template	both DNA strands	one DNA strand (3′ → 5′)	mRNA
product	DNA	RNA	polypeptide
enzyme	DNA polymerase, ligase	RNA polymerase	ribosome (peptidyl transferase, 23S rRNA)
substrate	dNTPs	NTPs	amino acids (charged tRNAs)
site (eukaryote)	nucleus (S-phase)	nucleus	cytoplasm

DNA to protein: the full journey

DNA (genes) → replication (semi-conservative) → transcription (RNA pol) → hnRNA → splicing, capping, tailing → mRNA → translation (ribosome + tRNA) → polypeptide

Paheli-Bujho

Bujho: Paheli, AUG pe protein shuru, toh UAA pe kaun rokta hai?

Paheli: Release factor! Stop codon ke liye koi tRNA hi nahi hota.

Bujho: Toh ribosome bina tRNA ke ruk jaata hai, kitna emotional!

Paheli: Haha, aur polypeptide free ho jaata hai. Yaad rakhna: UTR translate nahi hote.

8. Regulation of Gene Expression

Regulation of gene expression is a very broad term that may occur at various levels. Since gene expression results in a polypeptide, it can be regulated at several levels in eukaryotes: PYQ

1. **Transcriptional level**: formation of the primary transcript.
2. **Processing level**: regulation of splicing.
3. **Transport** of mRNA from the nucleus to the cytoplasm.
4. **Translational level.**

→ Genes in a cell are expressed to perform a particular function or set of functions. Example: if **β-galactosidase** is synthesised by *E. coli*, it hydrolyses the disaccharide **lactose into galactose and glucose**; the bacteria use these as a source of energy.

→ If the bacteria have no lactose around them, there is no need to make β-galactosidase. So metabolic, physiological or environmental conditions regulate gene expression.

→ The development and differentiation of the embryo into adult organisms are also a result of the coordinated regulation of expression of several sets of genes.

→ In prokaryotes, control of the **rate of transcriptional initiation** is the predominant site of control. In a transcription unit, the activity of RNA polymerase at a promoter is regulated by interaction with **accessory proteins**, which affect its ability to recognise start sites.

→ These regulatory proteins can act both **positively (activators)** and **negatively (repressors)**. The accessibility of promoter regions of prokaryotic DNA is often regulated by interaction of proteins with sequences called operators.

→ In most operons, the operator region is **adjacent to the promoter**, and in most cases the operator sequences bind a **repressor protein**. Each operon has its specific operator and specific repressor; e.g. the **lac** operator is present only in the lac operon and interacts only with the lac repressor.

A. The lac operon PYQ

The elucidation of the lac operon was a result of a close association between a geneticist, **Francois Jacob**, and a biochemist, **Jacques Monod**. They were the first to explain a transcriptionally regulated system.

In bacteria, the arrangement where a polycistronic structural gene is regulated by a **common promoter and regulatory genes** is called an operon. Examples: *lac* operon, *trp* operon, *ara* operon, *his* operon, *val* operon.

The lac operon (lac = lactose) has **one regulatory gene (i)** and **three structural genes (z, y, a)**:

Gene	*Codes for*	*Role*
i (inhibitor; not "inducer")	**repressor** of the lac operon	made all the time (constitutively)
z	**β-galactosidase**	hydrolyses lactose into galactose and glucose
y	**permease**	increases permeability of the cell to β-galactosides (lactose)
a	**transacetylase**	transfers an acetyl group (acetylation)

All three gene products are required for metabolism of lactose. In most cases, genes in an operon function together in the same or related metabolic pathway.

B. How the lac operon is switched on and off

→ Lactose is the substrate for β-galactosidase and it regulates switching on and off of the operon, hence it is termed the **inducer**.

→ **Absence of inducer**: the repressor (made constitutively from the i gene) **binds to the operator** and prevents RNA polymerase from transcribing the operon. The operon is **switched off**.

→ **Presence of inducer**: in the absence of a preferred carbon source such as glucose, if lactose is provided, it is transported into the cells by permease (a very low level of expression of the lac operon must be present all the time, otherwise lactose cannot enter). Lactose (or **allolactose**) binds and inactivates the repressor. RNA polymerase gets access to the promoter and transcription proceeds.

→ So regulation of the lac operon can be seen as regulation of enzyme synthesis by its substrate.

→ **Glucose or galactose cannot act as inducers** for the lac operon.

→ Regulation of the lac operon by the repressor is called negative regulation. The lac operon is under positive regulation as well, but that is beyond the scope at this level.

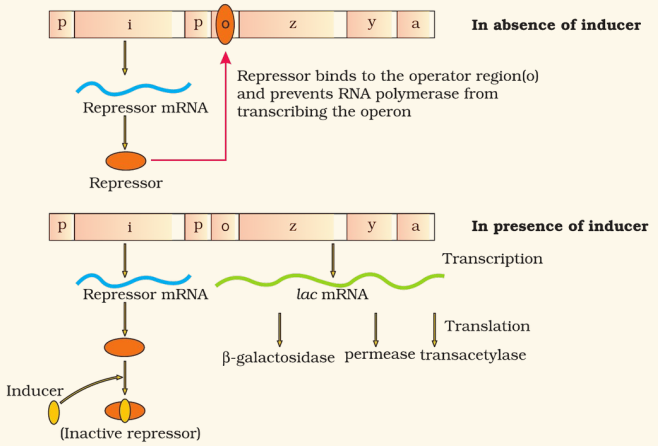


Fig. 5.14 The lac operon

Exam Alert: NCERT exercise (lac operon band kyun?)

Lactose dene ke kuch der baad operon **shut down** ho jaata hai kyunki β-galactosidase saara lactose tod deta hai. Inducer khatam = repressor phir free = operator pe baith jaata hai = transcription band. Isliye operon tab tak hi ON rehta hai jab tak lactose hai.

Paheli-Bujho

Bujho: Paheli, lac operon ka repressor ghar ka bouncer hai kya?

Paheli: Haan! Operator pe khada rehta hai, RNA polymerase ko andar nahi jaane deta.

Bujho: Aur lactose aaye toh?

Paheli: Lactose bouncer ko pakad leta hai, darwaza khul jaata hai: z, y, a ON!

9. Human Genome Project (HGP)

Genetic make-up of an organism lies in its DNA sequences: if two individuals differ, their DNA sequences must differ, at least at some places. Finding the complete DNA sequence of the human genome was set as a goal in **1990**, using genetic engineering techniques to isolate and clone DNA and new rapid sequencing methods.

HGP is called a mega project because: PYQ

→ The human genome has about 3×10^9 **bp**; at the cost of **US $3 per bp** (the estimated cost in the beginning), the total cost would be about 9 billion US dollars.

→ If the obtained sequences were stored in typed form in books with 1000 letters on each of 1000 pages, information from the DNA of a single human cell would need about 3300 books.

→ The enormous amount of data needed **high-speed computational devices** for storage, retrieval and analysis. HGP was closely associated with the rapid development of a new area of biology called bioinformatics.

A. Goals of HGP

1. Identify all the approximately **20,000–25,000 genes** in human DNA.
2. Determine the sequences of the **3 billion chemical base pairs** that make up human DNA.
3. **Store** this information in databases.
4. Improve tools for data **analysis**.
5. **Transfer** related technologies to other sectors, such as industries.
6. Address the ethical, legal and social issues (ELSI) that may arise from the project.

HGP was a **13-year** project coordinated by the **U.S. Department of Energy and the National Institute of Health**. During the early years, the **Wellcome Trust (U.K.)** became a major partner; additional contributions came from Japan, France, Germany, China and others. The project was completed in 2003.

Knowledge about the effects of DNA variations among individuals can lead to revolutionary new ways to **diagnose, treat and someday prevent** the thousands of disorders that affect human beings. Learning about non-human organisms' DNA sequences can help understand their natural capabilities, useful in solving challenges in **health care, agriculture, energy production and environmental remediation**. Many non-human model organisms have also been sequenced: bacteria, yeast, *Caenorhabditis elegans* (a free-living non-pathogenic nematode), *Drosophila* (fruit fly), plants (rice and *Arabidopsis*), etc.

B. Methodologies PYQ

Expressed Sequence Tags (ESTs)	Sequence Annotation
identifying all the genes that are **expressed as RNA**	blind approach: sequencing the **whole set of genome** (coding and non-coding), then assigning different regions with functions

1. Total DNA from a cell is isolated and converted into **random fragments** of relatively smaller sizes (DNA is a very long polymer and there are technical limits to sequencing very long pieces).
2. The fragments are **cloned in suitable hosts** using specialised vectors. Cloning amplifies each piece so it can be sequenced easily. The commonly used hosts were **bacteria and yeast**, and the vectors were BAC (bacterial artificial chromosomes) and YAC (yeast artificial chromosomes).
3. The fragments were sequenced using **automated DNA sequencers** that worked on the principle of a method developed by Frederick Sanger (Sanger is also credited with developing methods to determine amino acid sequences in proteins).
4. The sequences were arranged based on **overlapping regions** present in them. This needed generation of overlapping fragments for sequencing; aligning them was impossible by hand, so **specialised computer-based programs** were developed.
5. These sequences were then **annotated** and assigned to each chromosome. The sequence of **chromosome 1 was completed only in May 2006** (the last of the 24 human chromosomes: 22 autosomes and X and Y).
6. Another challenging task was assigning the genetic and physical maps on the genome, using information on polymorphism of **restriction endonuclease recognition sites** and some repetitive DNA sequences called **microsatellites**.

C. Salient features of the human genome PYQ

1. The human genome contains 3164.7 million nucleotide bases.
2. The average gene consists of **3000 bases**, but sizes vary greatly; the largest known human gene, dystrophin, is at **2.4 million bases**.
3. The total number of genes is estimated at 30,000, much lower than previous estimates of 80,000 to 1,40,000 genes. Almost all (**99.9 per cent**) nucleotide bases are exactly the same in all people.
4. The functions are unknown for **over 50 per cent** of the discovered genes.
5. Less than 2 per cent of the genome codes for proteins.
6. Repeated sequences make up a very large portion of the human genome. **Repetitive sequences** are stretches of DNA repeated many times, sometimes a hundred to a thousand times. They are thought to have no direct coding functions, but they shed light on chromosome structure, dynamics and evolution.
7. **Chromosome 1** has the most genes (2968), and the **Y chromosome** has the fewest (231).
8. Scientists have identified about 1.4 million locations where single-base DNA differences, **SNPs (single nucleotide polymorphisms, "snips")**, occur in humans. This helps find chromosomal locations for disease-associated sequences and trace human history.

Yaad karo: HGP ke numbers

1990 se 2003 (13 saal), **3164.7 million** bases, **30,000** genes, **99.9%** same, **<2%** protein coding, **dystrophin 2.4 million** (sabse bada), **Chr 1 = 2968** (sabse zyada genes), **Y = 231** (sabse kam), **1.4 million SNPs**.

Quick fact: bioinformatics

HGP produced data far too large to read by hand. **Bioinformatics** uses computers to store, retrieve, align and annotate sequences; the overlapping-fragment assembly of HGP itself was done by specialised computer programs.

D. Applications and future challenges

→ Mapping the human genome has opened new avenues in **biotechnology** and in understanding the **biology of diseases**.

→ In the future, rather than studying one gene at a time, we will be able to study all the genes in a genome, all the transcripts in a particular tissue or organ, or tumour, and how tens of thousands of genes and proteins work together in interconnected networks.

10. DNA Fingerprinting

As stated, **99.9 per cent** of base sequence among humans is the same. The remaining differences make every individual unique. To find genetic differences between two individuals or in a population, sequencing the whole DNA every time would be very expensive. DNA fingerprinting is a very quick way to compare the DNA sequences of any two individuals. PYQ

→ It involves identifying differences in some specific regions of DNA called repetitive DNA, where a small stretch of DNA is repeated many times.

→ These repetitive DNA are separated from bulk genomic DNA as different peaks during **density gradient centrifugation**. The bulk DNA forms a major peak and the other small peaks are called satellite DNA.

→ Satellite DNA is classified into many categories (**micro-satellites, mini-satellites**, etc.) based on base composition (A:T rich or G:C rich), length of segment and number of repetitive units. These sequences normally do not code for any proteins, but form a large portion of the human genome.

→ These sequences show a high degree of polymorphism and form the basis of DNA fingerprinting. Since DNA from every tissue (blood, hair follicle, skin, bone, saliva, sperm etc.) of an individual shows the same degree of polymorphism, they are very useful in **forensic** applications.

→ The polymorphisms are **inheritable** from parents to children, so DNA fingerprinting is the basis of **paternity testing** in case of disputes.

Quick fact: polymorphism

Polymorphism (variation at the genetic level) arises due to **mutations**. A new mutation in an individual is passed on to the offspring only if it occurs in **germ cells**. If an inheritable mutation is observed in a population at high frequency (more than 0.01), it is called a DNA polymorphism. Such variations are more likely in non-coding DNA, since mutations there have no immediate effect on reproductive ability.

A. Technique PYQ

The technique was initially developed by **Alec Jeffreys**. He used a satellite DNA as a probe that shows a very high degree of polymorphism: Variable Number of Tandem Repeats (VNTR). The technique involved **Southern blot hybridisation** using radiolabelled VNTR as a probe. Steps:

1. **Isolation** of DNA.
2. **Digestion** of DNA by restriction endonucleases.
3. **Separation** of DNA fragments by **electrophoresis**.
4. **Transferring (blotting)** of separated DNA fragments to synthetic membranes such as **nitrocellulose or nylon**.
5. **Hybridisation** using a labelled VNTR probe.
6. **Detection** of hybridised DNA fragments by autoradiography.

DNA isolation → restriction digestion → gel electrophoresis → blotting → VNTR probe hybridisation → autoradiography

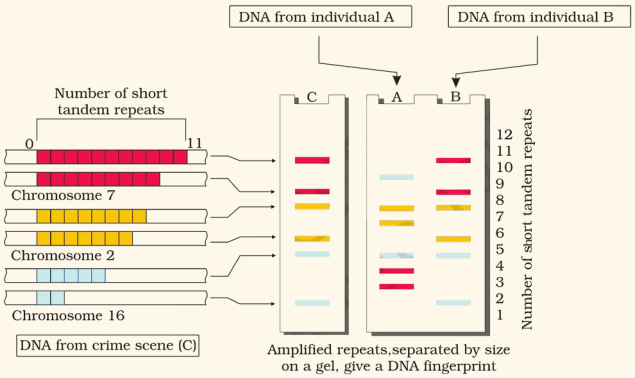


Fig. S.16 Schematic representation of DNA fingerprinting (few representative chromosomes, with VNTRs of different sizes)

→ A VNTR belongs to the class of satellite DNA called **mini-satellite**. A small DNA sequence is arranged **tandemly** in many copy numbers. The copy number varies from chromosome to chromosome in an individual.

→ The numbers of repeats show a very high degree of polymorphism, so the size of VNTR varies from 0.1 to 20 kb. After hybridisation with the VNTR probe, the autoradiogram gives many bands of differing sizes, giving a characteristic pattern.

→ These patterns differ from individual to individual, except in the case of **monozygotic (identical) twins**.

→ The sensitivity of the technique has been increased by the use of polymerase chain reaction (PCR), so DNA from a single cell is enough for analysis.

Applications of DNA fingerprinting

→ **Forensic science**: identifying criminals from blood, hair, semen etc.
→ **Paternity / maternity disputes**.
→ Determining **population and genetic diversities**.
→ **Evolutionary biology**: tracing human origin and migration.

1 minute recap

1. DNA: antiparallel double helix, A=T (2), G≡C (3), pitch 3.4 nm, 10 bp/turn, 0.34 nm; 2.2 m packed by histone octamer into nucleosomes (200 bp).
2. Griffith (transformation), Avery-MacLeod-McCarty (DNase stops it), Hershey-Chase (^{32}P DNA enters): DNA is the genetic material.
3. DNA better for storage, RNA for expression; RNA was the first genetic material.
4. Replication is semi-conservative (Meselson-Stahl, ^{15}N, CsCl); leading continuous, lagging discontinuous, joined by ligase; starts at ori.
5. Transcription unit = promoter + structural gene + terminator; bacteria: one RNA pol with σ and ρ; eukaryotes: pol I, II, III, then splicing, capping, tailing.
6. Genetic code: triplet, 61 sense + 3 stop, degenerate, comma-less, nearly universal, AUG = Met + start.
7. Translation: charging, initiation at AUG, elongation, release factor; 23S rRNA is a ribozyme; UTRs at both ends.
8. Lac operon: i (repressor), z, y, a; lactose = inducer; negative regulation.
9. HGP: 1990-2003, 3164.7 million bp, about 30,000 genes, less than 2% coding; DNA fingerprinting uses VNTR polymorphism.

Exam Alert: is chapter ke sabse repeated PYQs

→ Hershey-Chase (^{32}P/^{35}S), Meselson-Stahl (^{15}N, CsCl) aur lac operon: har saal koi ek 5 marks ka.
→ Transcription unit diagram, tRNA clover-leaf, DNA fingerprinting steps: 3 marks pakke.
→ Coding strand / template strand se mRNA likhna aur Chargaff numerical: 1-2 marks free ke.
→ Genetic code ke features aur RNA polymerase I, II, III ka kaam: MCQ mein bahut aata hai.

Most Important Questions

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM Board fav The bond that links a nitrogenous base to the 1′ carbon of pentose sugar is:
(a) Phosphoester linkage
(b) N-glycosidic linkage
(c) Phosphodiester linkage
(d) Hydrogen bond
Ans: (b) N-glycosidic linkage

Q2. IM Which of the following is a nucleoside?
(a) Adenine
(b) Thymine
(c) Cytidine
(d) Uracil
Ans: (c) Cytidine

Q3. IM The distance between two consecutive base pairs in B-DNA is about:
(a) 0.34 nm
(b) 3.4 nm
(c) 34 nm
(d) 0.034 nm
Ans: (a) 0.34 nm

Q4. IM Board fav If a double-stranded DNA has 30% guanine, the percentage of thymine is:
(a) 30%
(b) 20%
(c) 40%
(d) 70%
Ans: (b) 20%

Q5. IM Board fav Histones are rich in the amino acids:
(a) Glycine and alanine
(b) Aspartate and glutamate
(c) Leucine and valine
(d) Lysine and arginine
Ans: (d) Lysine and arginine

Q6. IM A typical nucleosome contains how many base pairs of DNA?
(a) 20 bp
(b) 100 bp
(c) 200 bp
(d) 2000 bp
Ans: (c) 200 bp

Q7. IM Transcriptionally active, loosely packed and lightly stained chromatin is:
(a) Euchromatin
(b) Heterochromatin
(c) Nucleoid
(d) Centromere
Ans: (a) Euchromatin

Q8. IM In Griffith's experiment, mice died when injected with:
(a) Live R strain
(b) Heat-killed S strain
(c) Heat-killed R strain
(d) Heat-killed S strain + live R strain
Ans: (d) Heat-killed S strain + live R strain

Q9. IM Which enzyme inhibited transformation in the experiment of Avery, MacLeod and McCarty?
(a) Protease
(b) RNase
(c) DNase
(d) Lipase
Ans: (c) DNase

Q10. IM Board fav In the Hershey-Chase experiment, the protein coat of the bacteriophage was labelled with:
(a) ^{32}P
(b) ^{35}S
(c) ^{15}N
(d) ^{14}C
Ans: (b) ^{35}S

Q11. IM Board fav Meselson and Stahl used which heavy isotope to label DNA?
(a) ^{15}N
(b) ^{32}P
(c) ^{35}S
(d) ^{3}H
Ans: (a) ^{15}N

Q12. IM After two generations in ^{14}N medium (Meselson-Stahl), the DNA bands are:
(a) 100% hybrid
(b) 50% heavy and 50% light
(c) 50% hybrid and 50% light
(d) 25% hybrid and 75% light
Ans: (c) 50% hybrid and 50% light

Q13. IM The fragments formed on the lagging strand are joined by:
(a) DNA polymerase
(b) Helicase
(c) RNA polymerase
(d) DNA ligase
Ans: (d) DNA ligase

Q14. IM *E. coli* completes replication of its DNA in about 18 minutes. The average rate of polymerisation is about:
(a) 200 bp per second
(b) 2000 bp per second
(c) 20 bp per second
(d) 2×10^5 bp per second
Ans: (b) 2000 bp per second

Q15. IM In a transcription unit, the promoter is located towards the:
(a) 5′-end of the coding strand (upstream)
(b) 3′-end of the coding strand (downstream)
(c) middle of the structural gene
(d) 5′-end of the template strand
Ans: (a) 5′-end of the coding strand (upstream)

Q16. IM Board fav Which factor helps bacterial RNA polymerase to terminate transcription?
(a) Sigma (σ) factor
(b) Release factor
(c) Rho (ρ) factor
(d) Initiation factor
Ans: (c) Rho (ρ) factor

Q17. IM In eukaryotes, hnRNA is transcribed by:
(a) RNA polymerase I
(b) RNA polymerase II
(c) RNA polymerase III
(d) DNA polymerase
Ans: (b) RNA polymerase II

Q18. IM Board fav Capping of hnRNA involves addition of:
(a) Methyl guanosine triphosphate at the 5′-end
(b) Poly-A tail at the 3′-end
(c) Methyl guanosine at the 3′-end
(d) Adenylate residues at the 5′-end
Ans: (a) Methyl guanosine triphosphate at the 5′-end

Q19. IM Which scientist developed the chemical method of synthesising RNA with defined base combinations?
(a) George Gamow
(b) Marshall Nirenberg
(c) Severo Ochoa
(d) Har Gobind Khorana
Ans: (d) Har Gobind Khorana

Q20. IM How many codons code for amino acids?
(a) 64
(b) 20
(c) 61
(d) 3
Ans: (c) 61

Q21. IM Which is NOT a stop codon?
(a) UAA
(b) AUG
(c) UAG
(d) UGA
Ans: (b) AUG

Q22. IM Board fav Sickle cell anaemia results from the substitution of:
(a) Glutamate by valine at the 6th position of β-globin
(b) Valine by glutamate in α-globin
(c) Lysine by valine in β-globin
(d) Deletion of three bases in β-globin
Ans: (a) Glutamate by valine at the 6th position of β-globin

Q23. IM The actual 3D shape of tRNA is:
(a) Clover-leaf
(b) Double helix
(c) Inverted L
(d) Linear
Ans: (c) Inverted L

Q24. IM Board fav Which RNA acts as a ribozyme in bacteria for peptide bond formation?
(a) 16S rRNA
(b) 5S rRNA
(c) tRNA
(d) 23S rRNA
Ans: (d) 23S rRNA

Q25. IM In the lac operon, the y gene codes for:
(a) β-galactosidase
(b) Permease
(c) Transacetylase
(d) Repressor
Ans: (b) Permease

Q26. IM The inducer of the lac operon is:
(a) Lactose (allolactose)
(b) Glucose
(c) Galactose
(d) Repressor protein
Ans: (a) Lactose (allolactose)

Q27. IM The human chromosome with the fewest genes is:
(a) Chromosome 1
(b) Chromosome 21
(c) X chromosome
(d) Y chromosome (231 genes)
Ans: (d) Y chromosome (231 genes)

Q28. IM Less than what per cent of the human genome codes for proteins?
(a) 0.1 per cent
(b) 2 per cent
(c) 20 per cent
(d) 50 per cent
Ans: (b) 2 per cent

Q29. IM Board fav The probe used by Alec Jeffreys in DNA fingerprinting was:
(a) SNP
(b) BAC
(c) VNTR
(d) EST
Ans: (c) VNTR

Q30. IM The size of VNTRs varies from:
(a) 0.1 to 20 kb
(b) 1 to 200 kb
(c) 20 to 200 bp
(d) 10 to 100 kb
Ans: (a) 0.1 to 20 kb

Q31. IM Which of the following is a pyrimidine found only in RNA?
(a) Thymine
(b) Cytosine
(c) Uracil
(d) Guanine
Ans: (c) Uracil

Q32. IM The pitch of the DNA double helix is:
(a) 0.34 nm
(b) 3.4 nm
(c) 2 nm
(d) 34 nm
Ans: (b) 3.4 nm

Q33. IM The haploid content of human DNA is about:
(a) 4.6×10^6 bp
(b) 48502 bp
(c) 6.6×10^9 bp
(d) 3.3×10^9 bp
Ans: (d) 3.3×10^9 bp

Q34. IM Proteins needed for higher-level packaging of chromatin are:
(a) Non-histone chromosomal (NHC) proteins
(b) Histone octamers
(c) Ribosomal proteins
(d) Repressor proteins
Ans: (a) Non-histone chromosomal (NHC) proteins

Q35. IM Which RNA is the genetic material of Tobacco Mosaic Virus?
(a) DNA
(b) RNA
(c) Protein
(d) Both DNA and RNA
Ans: (b) RNA

Q36. IM Okazaki fragments are formed on:
(a) The leading strand
(b) Both strands
(c) The lagging strand
(d) mRNA
Ans: (c) The lagging strand

Q37. IM In eukaryotes, DNA replication takes place in:
(a) G_1-phase
(b) S-phase
(c) G_2-phase
(d) M-phase
Ans: (b) S-phase

Q38. IM Board fav The strand of DNA that has the same sequence as mRNA (except T for U) is the:
(a) Coding strand
(b) Template strand
(c) Antisense strand
(d) Lagging strand
Ans: (a) Coding strand

Q39. IM The tail added at the 3′-end of hnRNA consists of:
(a) 20–30 guanylate residues
(b) Methyl guanosine triphosphate
(c) 200–300 uridylate residues
(d) 200–300 adenylate residues
Ans: (d) 200–300 adenylate residues

Q40. IM In a cell-free system, the polypeptide formed from poly-U (UUUUUU...) mRNA would be:
(a) Polyphenylalanine
(b) Polyleucine
(c) Polyserine
(d) Polytyrosine
Ans: (a) Polyphenylalanine

Q41. IM Board fav Which of these is the initiator codon?
(a) UAA
(b) UGA
(c) AUG
(d) GUG
Ans: (c) AUG

Q42. IM tRNA was earlier called:
(a) hnRNA
(b) sRNA (soluble RNA)
(c) snRNA
(d) rRNA
Ans: (b) sRNA (soluble RNA)

Q43. IM The binding of a release factor to the stop codon results in:
(a) Initiation of translation
(b) Charging of tRNA
(c) Splicing
(d) Termination of translation
Ans: (d) Termination of translation

Q44. IM The largest known human gene is:
(a) Dystrophin (2.4 million bases)
(b) β-globin
(c) Insulin
(d) Titin
Ans: (a) Dystrophin (2.4 million bases)

Q45. IM The HGP was completed in the year:
(a) 1990
(b) 2000
(c) 2003
(d) 2006
Ans: (c) 2003

Q46. IM In DNA fingerprinting, the separated DNA fragments are transferred to a nitrocellulose or nylon membrane. This step is called:
(a) Hybridisation
(b) Blotting
(c) Electrophoresis
(d) Autoradiography
Ans: (b) Blotting

Q47. IM Board fav Which of the following pairs is correctly matched?
(a) RNA polymerase I: hnRNA
(b) RNA polymerase II: 5S rRNA
(c) RNA polymerase III: tRNA
(d) RNA polymerase I: snRNA
Ans: (c) RNA polymerase III: tRNA

Q48. IM In the Meselson-Stahl experiment, DNA of different densities was separated by:
(a) Gel electrophoresis
(b) CsCl density gradient centrifugation
(c) Paper chromatography
(d) Autoradiography
Ans: (b) CsCl density gradient centrifugation

Q49. IM The anticodon on tRNA for the mRNA codon 5′-AUG-3′ is:
(a) 3′-UAC-5′
(b) 3′-TAC-5′
(c) 5′-AUG-3′
(d) 3′-AUG-5′
Ans: (a) 3′-UAC-5′

Q50. IM Single nucleotide polymorphisms identified in the human genome number about:
(a) 1400
(b) 14,000
(c) 30,000
(d) 1.4 million
Ans: (d) 1.4 million

Q51. IM Board fav **Assertion (A):** RNA is less stable than DNA.
Reason (R): The 2′-OH group present on every nucleotide of RNA is reactive.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.
Ans: (a) Both A and R are true and R is the correct explanation of A.

Q52. IM **Assertion (A):** In bacteria, transcription and translation can take place in the same compartment.
Reason (R): Bacterial mRNA does not require processing and there is no nuclear membrane.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.
Ans: (a) Both A and R are true and R is the correct explanation of A.

Q53. IM Board fav **Assertion (A):** Replication is discontinuous on the template strand with polarity 3′ → 5′.
Reason (R): DNA polymerase catalyses polymerisation only in the 5′ → 3′ direction.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.
Ans: (d) A is false but R is true. (Replication is continuous on the 3′ → 5′ template and discontinuous on the 5′ → 3′ template.)

Q54. IM **Assertion (A):** The genetic code is degenerate.
Reason (R): One codon codes for only one amino acid.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.
Ans: (b) Both A and R are true but R is not the correct explanation of A. (Degeneracy means an amino acid has more than one codon, e.g. Phe has UUU and UUC. R describes a different property: the code is unambiguous and specific, i.e. one codon codes for only one amino acid.)

Q55. 1M **Assertion (A):** Monozygotic twins show the same DNA fingerprint.
Reason (R): VNTR patterns are inherited and identical twins develop from a single zygote.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.
Ans: (a) Both A and R are true and R is the correct explanation of A.

Q56. 1M *Board fav* **Assertion (A):** The lac operon is said to be under negative regulation.
Reason (R): In the absence of lactose, the repressor binds to the operator and prevents transcription.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.
Ans: (a) Both A and R are true and R is the correct explanation of A.

Short answer (2–3 marks)

Q57. 2M Group the following as nitrogenous bases and nucleosides: Adenine, Cytidine, Thymine, Guanosine, Uracil and Cytosine. (NCERT)
Ans: **Nitrogenous bases:** adenine, thymine, uracil, cytosine. **Nucleosides:** cytidine, guanosine (base + pentose sugar).

Q58. 2M *Board fav* If a double-stranded DNA has 20 per cent of cytosine, calculate the per cent of adenine in the DNA. (NCERT)
Ans: By Chargaff's rule, C = G = 20%, so C + G = 40%. A + T = 100 − 40 = 60%. Since A = T, **adenine = 30%.**

Q59. 2M One strand of DNA is 5′-ATGCATGCATGCATGCATGCATGC-3′. Write the complementary strand in 5′ → 3′ direction. (NCERT)
Ans: Complementary strand (written 3′ → 5′): 3′-TACGTACGTACGTACGTACGTACG-5′. Written 5′ → 3′: **5′-GCATGCATGCATGCATGCATGCAT-3′.**

Q60. 2M *Board fav* The coding strand of a transcription unit is 5′-ATGCATGCATGCATGCATGCATGC-3′. Write the sequence of mRNA. (NCERT)
Ans: mRNA has the same sequence as the coding strand, with U in place of T: **5′-AUGCAUGCAUGCAUGCAUGCAUGC-3′.**

Q61. 3M Which property of the DNA double helix led Watson and Crick to hypothesise the semi-conservative mode of replication? Explain. (NCERT)
Ans: **Complementary base pairing** between the two antiparallel strands (A with T, G with C). Since the sequence of one strand determines the other, the strands can separate and each can act as a template for a new complementary strand. Each daughter DNA then has one parental and one new strand.

Q62. 3M Depending on the chemical nature of the template (DNA or RNA) and the nucleic acid synthesised from it, list the types of nucleic acid polymerases. (NCERT)
Ans: (i) **DNA-dependent DNA polymerase**: DNA template, makes DNA (replication). (ii) **DNA-dependent RNA polymerase**: DNA template, makes RNA (transcription). (iii) **RNA-dependent DNA polymerase** (reverse transcriptase): RNA template, makes DNA (retroviruses). (iv) **RNA-dependent RNA polymerase**: RNA template, makes RNA (in some RNA viruses).

Q63. 3M *Board fav* How did Hershey and Chase differentiate between DNA and protein in their experiment? (NCERT)
Ans: They grew one set of bacteriophages on medium with **radioactive phosphorus (^{32}P)**, which labels DNA only (proteins lack P), and another on **radioactive sulfur (^{35}S)**, which labels protein only (DNA lacks S). Bacteria infected with ^{32}P phages became radioactive, showing DNA enters the cell.

Q64. 3M *Board fav* Differentiate between: (a) Repetitive DNA and satellite DNA (b) mRNA and tRNA (c) Template strand and coding strand. (NCERT)
Ans: (a) **Repetitive DNA**: any DNA stretch repeated many times in the genome. **Satellite DNA**: repetitive DNA that separates as small peaks (away from bulk DNA) in density gradient centrifugation; classified as micro- and mini-satellites. (b) **mRNA**: carries codons from DNA, linear, acts as template. **tRNA**: adapter with anticodon and amino acid acceptor end, clover-leaf/inverted L, brings amino acids. (c) **Template strand**: 3′ → 5′, read by RNA polymerase. **Coding strand**: 5′ → 3′, same sequence as RNA (T for U), not read.

Q65. 2M List two essential roles of ribosome during translation. (NCERT)
Ans: (i) It provides the **platform**: the small subunit binds mRNA and the large subunit has two sites where charged tRNAs bind by codon-anticodon pairing. (ii) It acts as a **catalyst** for peptide bond formation (23S rRNA in bacteria is a ribozyme).

Q66. 2M *Board fav* Lactose was added to the medium of *E. coli* and the lac operon was induced. Why does the lac operon shut down some time after addition of lactose? (NCERT)
Ans: β-galactosidase (made by the z gene) hydrolyses lactose into glucose and galactose. As lactose (the inducer) is used up, the repressor is no longer inactivated; it binds the operator again and blocks RNA polymerase, so transcription stops.

Q67. 3M Explain (in one or two lines) the function of: (a) Promoter (b) tRNA (c) Exons. (NCERT)
Ans: (a) **Promoter**: DNA sequence that provides the binding site for RNA polymerase and decides the template strand; it initiates transcription. (b) **tRNA**: adapter that reads the codon by its anticodon and brings the specific amino acid to the ribosome. (c) **Exons**: coding sequences of a split gene that appear in mature mRNA.

Q68. 3M Board fav Why is the Human Genome Project called a mega project? (NCERT)
Ans: (i) It aimed to sequence about 3×10^9 bp; at US $3 per bp the cost was about US $9 billion. (ii) Data from one cell would fill about 3300 books of 1000 pages each. (iii) Storage and analysis needed high-speed computers and gave rise to bioinformatics. (iv) It was a 13-year, multi-nation effort.

Q69. 3M Board fav What is DNA fingerprinting? Mention its applications. (NCERT)
Ans: A quick technique to compare DNA sequences of individuals by identifying differences in **repetitive DNA** (VNTR polymorphism). Applications: forensic science (identifying criminals), paternity disputes, studying population and genetic diversity, evolutionary biology.

Q70. 3M Briefly describe: (a) Transcription (b) Polymorphism (c) Translation (d) Bioinformatics. (NCERT)
Ans: (a) Copying of genetic information from one strand of DNA (template) into RNA by RNA polymerase. (b) Variation at the genetic level arising from mutation; an inheritable variant with frequency above 0.01 in a population is a DNA polymorphism. (c) Polymerisation of amino acids into a polypeptide on ribosomes as per the codons of mRNA. (d) The branch using computers to store, retrieve and analyse biological data such as genome sequences.

Q71. 2M Name the three components of a nucleotide. How is a nucleoside different from a nucleotide?
Ans: Nitrogenous base, pentose sugar, phosphate group. A **nucleoside** = base + sugar (N-glycosidic link); a **nucleotide** = nucleoside + phosphate (phosphoester link at 5′-OH).

Q72. 3M Board fav List any four salient features of the Watson-Crick model of DNA.
Ans: (i) Two antiparallel polynucleotide chains with a sugar-phosphate backbone and bases inside. (ii) A pairs with T by 2 H-bonds; G with C by 3 H-bonds (purine opposite pyrimidine). (iii) Right-handed coil, pitch 3.4 nm, about 10 bp per turn, 0.34 nm between bp. (iv) Base-pair stacking plus H-bonds give stability.

Q73. 2M What is the central dogma? Name the exception and the enzyme responsible.
Ans: Information flows DNA → RNA → Protein (proposed by Crick). In **retroviruses** (e.g. HIV), information flows from RNA to DNA by **reverse transcriptase**.

Q74. 2M Board fav Calculate the length of DNA in a typical mammalian cell. Why must it be packaged?
Ans: 6.6×10^9 bp $\times$ 0.34×10^{-9} m/bp = **about 2.2 m**. A nucleus is only about 10^{-6} m across, so the DNA must be highly condensed (histones, nucleosomes, chromatin fibres).

Q75. 2M Describe the structure of a nucleosome.
Ans: Eight positively charged histone molecules (rich in lysine and arginine) form a **histone octamer**. About **200 bp** of negatively charged DNA wraps around it. Nucleosomes repeat to form chromatin, which looks like beads-on-string under the EM.

Q76. 2M Differentiate between euchromatin and heterochromatin.
Ans: **Euchromatin**: loosely packed, stains light, transcriptionally active. **Heterochromatin**: densely packed, stains dark, transcriptionally inactive.

Q77. 3M Board fav Describe Griffith's experiment and state its conclusion.
Ans: He injected mice with *Streptococcus pneumoniae*: live S (smooth, capsulated) killed mice; live R (rough) did not; heat-killed S did not; heat-killed S + live R killed mice and live S bacteria were recovered. Conclusion: R strain was **transformed** by a "transforming principle" from dead S cells; its biochemical nature was not known.

Q78. 2M How did Avery, MacLeod and McCarty prove that DNA is the transforming principle?
Ans: They purified proteins, DNA and RNA from heat-killed S cells and tested which transformed R cells. Protease and RNase did not affect transformation, but **DNase inhibited it**. So DNA caused transformation.

Q79. 2M Board fav State four criteria a molecule must fulfil to act as genetic material.
Ans: (i) It should be able to generate its replica. (ii) It should be chemically and structurally stable. (iii) It should allow slow changes (mutation) needed for evolution. (iv) It should be able to express itself as Mendelian characters.

Q80. 3M Board fav Give reasons: (a) DNA is a better genetic material than RNA. (b) RNA viruses mutate and evolve faster.
Ans: (a) DNA lacks the reactive 2′-OH, has thymine instead of uracil and is double-stranded with a repair system, so it is chemically more stable. (b) RNA is unstable and reactive, so it mutates at a faster rate; RNA viruses therefore have a shorter life span and evolve faster.

Q81. 2M What is meant by the "RNA world"? Give one evidence.
Ans: The concept that RNA was the first genetic material, acting both as genetic material and catalyst; DNA evolved later from RNA for stability. Evidence: ribozymes, e.g. 23S rRNA catalyses peptide bond formation; splicing and translation evolved around RNA.

Q82. 3M Board fav Why is replication continuous on one strand and discontinuous on the other?
Ans: DNA polymerase adds nucleotides only in the 5′ → 3′ direction. On the template with polarity 3′ → 5′, the new strand grows continuously towards the fork (leading). On the 5′ → 3′ template, it is made in short fragments away from the fork (lagging), later joined by DNA ligase.

Q83. 2M Why are deoxyribonucleoside triphosphates important in replication?
Ans: They serve a **dual purpose**: they act as **substrates** for the new strand and also provide **energy** for polymerisation (by releasing the two terminal phosphates).

Q84. 2M What is an origin of replication? Why is it important in recombinant DNA technology?
Ans: A definite region of DNA where replication begins (DNA polymerase cannot initiate on its own). A foreign DNA piece must be linked to a vector having an **ori** so that it can replicate and propagate in the host.

Q85. 2M Board fav Why is only one strand of DNA transcribed in a given transcription unit?
Ans: (i) If both strands were transcribed, the two RNAs would have different sequences and one segment would code for two different proteins. (ii) The two complementary RNAs would pair to form double-stranded RNA and prevent translation.

Q86. 3M Board fav Draw a schematic transcription unit and label promoter, structural gene, terminator, template and coding strands.
Ans: Diagram as Fig. 5.9: 5′-[Promoter]-[Structural gene]-[Terminator]-3′ on the **coding strand (5′ → 3′)**; the lower **template strand (3′ → 5′)** is read by RNA polymerase. Promoter upstream (5′ side), terminator downstream (3′ side).

Q87. 2M Distinguish between monocistronic and polycistronic structural genes.
Ans: **Monocistronic**: codes for one polypeptide; mostly in eukaryotes; usually split (exons and introns). **Polycistronic**: codes for several polypeptides from one mRNA; mostly in bacteria, e.g. the lac operon (z, y, a).

Q88. 2M What are split genes? What is the significance of introns?
Ans: Eukaryotic structural genes whose coding sequences (**exons**) are interrupted by non-coding sequences (**introns**). Introns are removed by splicing. Their presence is reminiscent of antiquity, and splicing shows the dominance of the RNA world.

Q89. 3M Board fav Describe the three steps of transcription in bacteria.
Ans: **Initiation**: RNA polymerase with σ factor binds the promoter. **Elongation**: it uses NTPs to polymerise RNA complementary to the template, opening the helix as it moves. **Termination**: at the terminator, aided by ρ factor, RNA and polymerase fall off.

Q90. 3M Board fav Name the three RNA polymerases of eukaryotes and the RNAs they transcribe.
Ans: **RNA pol I**: rRNAs 28S, 18S, 5.8S. **RNA pol II**: hnRNA (precursor of mRNA). **RNA pol III**: tRNA, 5S rRNA, snRNAs.

Q91. 3M Board fav Explain the processing of hnRNA in eukaryotes.
Ans: (i) **Splicing**: introns removed and exons joined in a defined order. (ii) **Capping**: methyl guanosine triphosphate added at the 5′-end. (iii) **Tailing**: 200–300 adenylate residues added at the 3′-end, template-independent. The processed molecule (mRNA) moves to the cytoplasm.

Q92. 2M Why did Gamow propose that the genetic code is a triplet?
Ans: There are 4 bases and 20 amino acids. A singlet code gives only 4 and a doublet only 16 (4^2) combinations, too few. A triplet gives $4^3 = 64$ codons, enough for all 20 amino acids.

Q93. 3M Board fav Explain: (a) degenerate code (b) universal code (c) AUG has dual function.
Ans: (a) Some amino acids have more than one codon (e.g. UUU and UUC for Phe). (b) The same codon specifies the same amino acid in almost all organisms (exceptions: mitochondria, some protozoans). (c) AUG codes for methionine and also acts as the initiator codon.

Q94. 2M How does insertion or deletion of one base differ in effect from insertion or deletion of three bases?
Ans: One (or two) bases: the **reading frame shifts** from that point, changing all later codons (frameshift mutation). Three bases (or multiples): one codon (amino acid) is added or lost, and the **reading frame is unaltered** beyond it.

Q95. 3M Board fav Draw a labelled diagram of tRNA (clover-leaf) and state why it is called an adapter molecule.
Ans: Labels: amino acid acceptor end (3′ end with the amino acid), anticodon loop, T loop, D loop (as in Fig. 5.12). It is an **adapter** because it reads the codon on mRNA through its anticodon and also carries the specific amino acid.

Q96. 2M What is charging of tRNA? Why is it needed?
Ans: Activation of amino acids in the presence of **ATP** and their linking to cognate tRNA (aminoacylation). It provides energy so that peptide bond formation between two charged tRNAs becomes energetically favourable.

Q97. 2M What are UTRs? Where are they located and what is their role?
Ans: Untranslated regions of mRNA, present at the **5′-end (before the start codon)** and the **3′-end (after the stop codon)**. They are not translated but are required for efficient translation.

Q98. 3M Board fav What is an operon? Name its components with reference to the lac operon.
Ans: A unit of bacterial DNA in which a polycistronic structural gene is regulated by a common promoter and regulatory genes. Lac operon: regulatory gene **i** (repressor), promoter (p), operator (o) and structural genes **z** (β-galactosidase), **y** (permease), **a** (transacetylase).

Q99. 2M Why is the regulation of the lac operon called negative regulation? Why cannot glucose act as its inducer?
Ans: Because the operon is switched off by the binding of a **repressor** to the operator; the inducer works by removing this block. Glucose is a preferred carbon source and is not the substrate of β-galactosidase, so it cannot inactivate the lac repressor.

Q100. 2M Name the two methodologies used in HGP.
Ans: (i) **Expressed Sequence Tags (ESTs)**: identifying all genes expressed as RNA. (ii) **Sequence annotation**: sequencing the whole genome (coding and non-coding) and then assigning functions to regions.

Q101. 2M What are BAC and YAC? Why were they used in HGP?
Ans: Bacterial artificial chromosome and yeast artificial chromosome: specialised **vectors** used to clone random DNA fragments in bacteria and yeast hosts, so each fragment is amplified for sequencing.

Q102. 2M What are SNPs? How many have been identified in the human genome and why are they useful?
Ans: Single nucleotide polymorphisms: locations where single-base DNA differences occur between humans. About **1.4 million** are identified. They help find chromosomal locations of disease-associated sequences and trace human history.

Q103. 3M What is satellite DNA? Why is it useful in forensic science?
Ans: Repetitive DNA that separates as small peaks from bulk DNA in density gradient centrifugation; it does not code for proteins. It shows high polymorphism, and DNA from every tissue of a person shows the same pattern, so a sample of blood, hair or semen can identify an individual.

Q104. 2M What is a polynucleotide? How is its polarity defined?
Ans: A polymer of nucleotides joined by 3′–5′ phosphodiester bonds. One end has a free phosphate on the 5′ carbon of sugar (5′-end) and the other has a free 3′-OH (3′-end); this gives the chain polarity.

Q105. 2M How many hydrogen bonds are present in a 50 bp DNA segment with 20 adenines?
Ans: A-T pairs = 20 (2 H-bonds each) = 40; G-C pairs = 30 (3 each) = 90. Total = **130 H-bonds**.

Q106. 2M Why is the distance between the two strands of DNA uniform throughout?
Ans: A **purine** (2 rings) always pairs with a **pyrimidine** (1 ring): A with T, G with C. So each base pair has the same width, keeping the strands at a uniform distance.

Q107. 2M Differentiate between the leading and the lagging strand.
Ans: **Leading**: template 3′ → 5′, made continuously towards the fork. **Lagging**: template 5′ → 3′, made discontinuously as fragments away from the fork, joined by DNA ligase.

Q108. 2M Mention the roles of the σ and ρ factors.
Ans: σ (sigma) is the **initiation factor**: it associates with RNA polymerase to recognise the promoter and start transcription. ρ (rho) is the **termination factor**: it helps end transcription at the terminator.

Q109. 2M Write the possible mRNA for the peptide Met-Phe-Phe. Why can there be more than one answer?
Ans: e.g. 5′-AUG UUU UUC-3′ (also AUG UUC UUU, AUG UUU UUU, AUG UUC UUC). Phe has two codons (UUU, UUC), i.e. the code is **degenerate**.

Q110. 2M Why are there no tRNAs for stop codons? How does translation end then?
Ans: Stop codons (UAA, UAG, UGA) do not code for any amino acid, so no tRNA carries an anticodon for them. A **release factor** binds the stop codon and releases the polypeptide.

Q111. 2M Name the levels at which gene expression is regulated in eukaryotes.
Ans: (i) Transcriptional (primary transcript formation), (ii) processing (splicing), (iii) transport of mRNA from nucleus to cytoplasm, (iv) translational level.

Q112. 2M Board fav Mention the role of the i, z, y and a genes of the lac operon.
Ans: **i**: repressor (made constitutively). **z**: β-galactosidase (lactose to glucose + galactose). **y**: permease (entry of lactose). **a**: transacetylase.

Q113. 2M Give two applications of the HGP and name two model organisms sequenced.
Ans: Applications: new ways to diagnose, treat and prevent disorders; understanding all genes and their networks in health and disease. Model organisms: *Caenorhabditis elegans*, *Drosophila*, rice, *Arabidopsis*, yeast, bacteria.

Q114. 2M What is meant by DNA polymorphism? Why is it more common in non-coding DNA?
Ans: An inheritable mutation present at a frequency above 0.01 in a population. It is more common in non-coding DNA because mutations there do not immediately affect the reproductive ability of the individual, so they accumulate.

Q115. 2M Why do monozygotic twins show identical DNA fingerprints while siblings do not?
Ans: Monozygotic twins develop from the same zygote, so their DNA and VNTR copy numbers are identical. Ordinary siblings get different combinations of parental chromosomes, so their VNTR patterns differ (only partly matching each parent).

Q116. 2M Name the components of a transcription unit. What is the polarity of the template strand?
Ans: Promoter, structural gene and terminator. The template strand has polarity **3′ → 5′** (RNA is synthesised 5′ → 3′).

Q117. 2M What is the function of DNA ligase in replication? Why is it called molecular glue?
Ans: It joins the discontinuously synthesised DNA fragments of the lagging strand by forming phosphodiester bonds. Since it seals DNA pieces together, it is called molecular glue (also used in rDNA technology).

Q118. 2M Why is RNA polymerase said to be capable of only elongation in bacteria?
Ans: By itself it can only add nucleotides (elongation). It needs the σ factor to recognise the promoter and initiate, and the ρ factor to terminate transcription.

Q119. 2M Why is hnRNA called heterogeneous nuclear RNA?
Ans: It is found in the nucleus and its molecules vary greatly in size (heterogeneous), since it is the unprocessed primary transcript carrying both exons and introns of different genes.

Q120. 3M Board fav Explain how the lac operon is an example of regulation of enzyme synthesis by its own substrate.
Ans: Lactose is the substrate of β-galactosidase. When present, lactose inactivates the repressor and switches on the genes for β-galactosidase, permease and transacetylase. When lactose is used up, synthesis stops. So the substrate itself controls the synthesis of the enzyme.

Q121. 3M Board fav State the steps involved in whole-genome sequencing in HGP.
Ans: (i) Isolate total DNA and break it into random fragments. (ii) Clone fragments in hosts (bacteria, yeast) using BAC/YAC vectors. (iii) Sequence fragments with automated sequencers (Sanger principle). (iv) Arrange sequences using overlapping regions with computer programs. (v) Annotate and assign sequences to chromosomes.

Q122. 2M Why is DNA fingerprinting used in paternity disputes?
Ans: VNTR polymorphisms are inherited: a child receives half of its bands from the mother and half from the father. Matching the child's band pattern with that of the alleged parents establishes parentage.

Q123. 2M Name the scientists associated with: (a) nuclein (b) lac operon (c) polynucleotide phosphorylase (d) automated DNA sequencing principle.
Ans: (a) Friedrich Meischer, (b) Jacob and Monod, (c) Severo Ochoa, (d) Frederick Sanger.

Long answer (5 marks)

Q124. 5M Board fav Describe the Hershey-Chase experiment. What did it prove? Draw a labelled diagram.
Ans: **Aim:** to find whether DNA or protein is the genetic material. **Procedure:** (i) One batch of T2 bacteriophages was grown with ^{32}P (labels DNA), another with ^{35}S (labels protein coat). (ii) Each batch infected *E. coli*. (iii) After infection, the mixture was agitated in a **blender** to remove viral coats from bacteria. (iv) It was **centrifuged**: bacteria formed a pellet, viral coats stayed in the supernatant. **Result:** with ^{32}P, radioactivity was in the bacteria (pellet); with ^{35}S, radioactivity was in the supernatant, not in bacteria. **Conclusion:** DNA enters the bacterium and directs formation of new phages, so **DNA is the genetic material.** Diagram: Fig. 5.5 (infection, blending, centrifugation for both batches).

Q125. 5M Board fav Describe the Meselson and Stahl experiment. What would happen to the DNA bands after 80 minutes in ^{14}N medium?
Ans: (i) *E. coli* grown for many generations in $^{15}NH_4Cl$, so all DNA became heavy ($^{15}N^{15}N$). (ii) Cells transferred to $^{14}NH_4Cl$; samples taken every generation (20 min). (iii) DNA separated by CsCl density gradient centrifugation. (iv) After 20 min, all DNA was hybrid ($^{15}N^{14}N$); after 40 min, 50% hybrid and 50% light. This proved **semi-conservative replication**. After 80 min (4 generations), 16 DNA molecules are formed from one: 2 hybrid and 14 light, i.e. **12.5% hybrid and 87.5% light**, no heavy DNA.

Q126. 5M Board fav Explain the process of transcription in eukaryotes. How does it differ from that in prokaryotes?
Ans: **Eukaryotes:** (i) Three RNA polymerases: pol I (28S, 18S, 5.8S rRNA), pol II (hnRNA), pol III (tRNA, 5S rRNA, snRNA). (ii) The primary transcript (hnRNA) has exons and introns. (iii) **Splicing** removes introns; **capping** adds methyl guanosine triphosphate at 5′; **tailing** adds 200–300 adenylates at 3′. (iv) Mature mRNA leaves the nucleus for translation. **Differences from prokaryotes:** prokaryotes have a single RNA polymerase (with σ and ρ factors), mostly polycistronic genes without introns, no processing of mRNA, and transcription and translation are coupled in the cytoplasm; in eukaryotes they are separated by the nuclear membrane.

Exam Alert: 5 marks ka answer
Experiment wale answer = **aim, procedure, result, conclusion** + chhota labelled diagram.

Q127. SM Board fav Explain the lac operon with a labelled diagram, in the absence and presence of the inducer.

Ans: **Components:** regulatory gene i (codes repressor, made constitutively), promoter (p), operator (o), structural genes z (β-galactosidase), y (permease), a (transacetylase). **Absence of lactose:** the repressor binds to the operator and prevents RNA polymerase from transcribing z, y, a; operon OFF. **Presence of lactose** (no glucose): a little lactose enters via permease; lactose/allolactose (inducer) binds and inactivates the repressor, which cannot bind the operator. RNA polymerase transcribes z, y, a as one polycistronic mRNA; enzymes metabolise lactose. When lactose is used up, the repressor binds again and the operon shuts off. This is **negative regulation** of enzyme synthesis by its substrate. Diagram: Fig. 5.14.

Q128. SM Board fav List the goals of the Human Genome Project and any six salient features of the human genome.

Ans: **Goals:** identify all about 20,000–25,000 genes; determine the sequence of 3 billion bp; store information in databases; improve data-analysis tools; transfer technologies to other sectors; address ELSI. **Features:** (i) 3164.7 million bases. (ii) Average gene 3000 bases; largest, dystrophin, 2.4 million bases. (iii) About 30,000 genes; 99.9% bases identical in all people. (iv) Functions unknown for over 50% of genes. (v) Less than 2% of the genome codes for proteins; repeated sequences form a large portion. (vi) Chromosome 1 has most genes (2968), Y fewest (231). (vii) About 1.4 million SNPs identified.

Q129. SM Board fav What is DNA fingerprinting? Describe its steps and the principle on which it is based.

Ans: **Definition:** technique to identify differences in repetitive DNA between individuals. **Principle:** polymorphism in satellite DNA, e.g. VNTRs (mini-satellites) whose copy number varies between individuals; VNTR size 0.1–20 kb; patterns are unique except for identical twins and inherited from parents. **Steps** (Alec Jeffreys, Southern blot hybridisation): (i) isolation of DNA; (ii) digestion with restriction endonucleases; (iii) separation of fragments by gel electrophoresis; (iv) blotting onto nitrocellulose or nylon membrane; (v) hybridisation with labelled VNTR probe; (vi) detection by autoradiography. PCR increases sensitivity. **Uses:** forensics, paternity tests, population diversity, evolution.

Exam Alert: lac operon diagram

- → Do panels banao: **inducer absent** (repressor operator pe) aur **inducer present** (repressor + lactose, inactive).
- → Labels: i, p, o, z, y, a, repressor mRNA, RNA polymerase, inducer, β-galactosidase, permease, transacetylase.
- → Repressor **i gene** ka product hai aur **operator** pe baithta hai, promoter pe nahi.

Q130. SM Board fav Describe the process of translation in detail. Mention the role of each RNA.

Ans: (i) **Charging**: amino acids are activated by ATP and linked to their cognate tRNA. (ii) **Initiation**: the small ribosomal subunit binds mRNA; the initiator tRNA (carrying Met) pairs with the start codon AUG; the large subunit joins. (iii) **Elongation**: the next charged tRNA binds the next codon at the second site of the large subunit; a peptide bond forms between the two amino acids, catalysed by 23S rRNA (ribozyme) in bacteria; the ribosome moves one codon forward, and the chain grows. (iv) **Termination**: at a stop codon (UAA/UAG/UGA), a release factor binds and the polypeptide is released; subunits separate. UTRs at both ends help efficient translation. **Roles**: mRNA = template (codons); tRNA = adapter carrying amino acids; rRNA = structural and catalytic.

Q131. SM Board fav Describe DNA replication in *E. coli*, with a labelled diagram of the replicating fork.

Ans: (i) Replication starts at a definite **origin of replication**; the helix opens to form a **replication fork** (opening the whole DNA at once needs too much energy). (ii) The main enzyme is **DNA-dependent DNA polymerase**, using dNTPs as substrate and energy source; it is fast (about 2000 bp/s; whole genome in 18 min) and highly accurate. (iii) It works only 5′ → 3′: on the 3′ → 5′ template the new strand is continuous (leading); on the 5′ → 3′ template it is discontinuous (lagging, Okazaki fragments). (iv) **DNA ligase** joins the fragments. (v) Replication is semi-conservative. Diagram: Fig. 5.8 (template strands, leading and lagging strands, fork, direction 5′ → 3′).

Q132. SM Board fav (a) Explain how DNA is packaged in a eukaryotic nucleus. (b) How is it organised in prokaryotes? (c) Differentiate euchromatin from heterochromatin.

Ans: (a) DNA (about 2.2 m) is negatively charged. Positively charged histones (rich in lysine and arginine) form an octamer; about 200 bp DNA wraps around it to form a **nucleosome**. Nucleosomes repeat as **beads-on-string** chromatin, which coils into **chromatin fibres** and condenses into **chromosomes** at metaphase; higher packaging needs **NHC proteins**. (b) In prokaryotes, there is no defined nucleus; DNA is held with positively charged proteins in large loops in the **nucleoid**. (c) Euchromatin: loose, light-stained, transcriptionally active; heterochromatin: dense, dark-stained, inactive.

Case-based (4 marks)

Q133. 4M Board fav In an experiment, DNase was added to an extract of heat-killed S-strain bacteria, and the extract was then mixed with live R-strain bacteria. (a) Will transformation occur? (b) What would happen if protease were used instead? (c) What does this prove? (d) Name the scientists.

Ans: (a) **No**, DNase destroys DNA, the transforming principle. (b) Transformation **would occur**; protein is not the transforming principle. (c) **DNA** is the genetic material (transforming principle). (d) Avery, MacLeod and McCarty.

Q134. 4M Board fav A student mixed heat-killed S-strain *Streptococcus pneumoniae* with live R strain and injected the mixture into mice. The mice died. (a) What happened to the R strain? (b) Name the scientist who first did this. (c) Which later experiment identified the chemical nature of the transforming principle, and which enzyme stopped transformation? (d) Why did heat not destroy the transforming principle?

Ans: (a) It was **transformed** into virulent S strain (it acquired the ability to make a polysaccharide coat). (b) Frederick Griffith (1928). (c) Avery, MacLeod and McCarty; **DNase** stopped transformation. (d) DNA is chemically and structurally stable; its complementary strands re-join after heating.

Q135. 4M Board fav A DNA segment has the template strand 3′-TACGGATCCATT-5′. (a) Write the coding strand. (b) Write the mRNA. (c) Using the code, give the first codon and its amino acid. (d) Which codon in this mRNA stops translation?

Ans: (a) Coding strand: 5′-ATGCCTAGGTAA-3′. (b) mRNA: 5′-AUGCCUAGGUAA-3′. (c) First codon **AUG** = methionine (start). Codons: AUG (Met), CCU (Pro), AGG (Arg). (d) **UAA** is the stop codon.

Q136. 4M A child was born with anaemia; his red blood cells became sickle-shaped under low oxygen. Analysis showed one base change in the β-globin gene. (a) Name the type of mutation. (b) Which amino acid replaces which? (c) Show the codon change in mRNA. (d) Will insertion of one extra base have a bigger or smaller effect? Why?

Ans: (a) **Point mutation** (base substitution). (b) **Glutamate** (Glu) is replaced by **valine** (Val) at the 6th position of β-globin. (c) GAG → GUG. (d) Usually bigger: insertion of one base causes a **frameshift**, changing every codon after that point.

Q137. 4M Board fav Police found blood stains at a crime scene and collected DNA from three suspects. The scientist digested the DNA, ran a gel, blotted it and used a labelled probe, then exposed an X-ray film. (a) Name the technique and its scientist. (b) Which probe is used and what kind of DNA is it? (c) Name the last step. (d) If only a very small sample is available, how can it be analysed?

Ans: (a) DNA fingerprinting, developed by **Alec Jeffreys**. (b) Labelled **VNTR** probe; a mini-satellite (satellite DNA). (c) **Autoradiography** to detect the hybridised fragments. (d) The DNA is amplified by **PCR**, so even DNA from a single cell can be analysed.

Quick fact

VNTR copy number differs even between homologous chromosomes of one person, so one sample gives **many bands**.

Q138. 4M Board fav *E. coli* cells were grown in a medium containing glucose and lactose. Initially they used glucose; after glucose was exhausted, they began to make β-galactosidase. (a) Name the operon involved. (b) Why was the enzyme not made while glucose was present? (c) What acts as the inducer? (d) What happens to the operon once all lactose is used?

Ans: (a) **Lac operon**. (b) Glucose is the preferred carbon source, so it is used first; the lac operon is expressed only when glucose is absent and lactose is present (glucose is not an inducer). (c) Lactose (allolactose), which inactivates the repressor. (d) The repressor becomes free again, binds the operator and transcription stops.

Q139. 4M An mRNA isolated from the cytoplasm of a eukaryotic cell was much shorter than its gene and had extra residues at both ends that were not coded by the gene. (a) Why is the mRNA shorter than the gene? (b) What is added at the 5′-end? (c) What is added at the 3′-end and how? (d) Which enzyme made its precursor?

Ans: (a) **Introns** were removed by splicing; only exons remain. (b) **Methyl guanosine triphosphate** (cap). (c) A **poly-A tail** of 200–300 adenylate residues, added in a template-independent manner. (d) **RNA polymerase II** (makes hnRNA).

Q140. 4M A researcher found that a gene had 3000 bp, but less than 2 per cent of the whole genome codes for proteins and chromosome 1 has the most genes. (a) Which project gave these facts? (b) When was it completed? (c) Name the largest human gene and its size. (d) Name the vectors used to clone fragments in this project.

Ans: (a) Human Genome Project. (b) 2003 (chromosome 1 sequence completed in May 2006). (c) Dystrophin, 2.4 million bases. (d) BAC (bacterial artificial chromosome) and YAC (yeast artificial chromosome).

Exam Alert: diagrams to practise

- → Fig. 5.2/5.3 DNA double helix (5′/3′, H-bonds, 3.4 nm, 0.34 nm), Fig. 5.4 nucleosome, Fig. 5.9 transcription unit, Fig. 5.12 tRNA aur Fig. 5.14 lac operon: ye paanch sabse zyada aate hain.
- → Replicating fork mein leading/lagging, Okazaki fragments aur ligase label karna mat bhoolna.
- → Polarity (5′, 3′) har sequence wale answer mein likho, warna aadha mark katega.
- → Differentiate wale pairs: template/coding, leading/lagging, euchromatin/heterochromatin, DNA/RNA, ESTs/annotation.

Last-minute check

- ✓ Purines A, G; pyrimidines C, T, U. A=T (2), G≡C (3).
- ✓ ^{32}P = DNA, ^{35}S = protein; ^{15}N + CsCl = Meselson-Stahl.
- ✓ Pol I = rRNA, Pol II = hnRNA, Pol III = tRNA, 5S rRNA, snRNA.
- ✓ UAA, UAG, UGA stop; AUG start (Met). i, z, y, a = repressor, β-gal, permease, transacetylase.

EVOLUTION

Most asked: Miller's experiment, homologous vs analogous organs, industrial melanism, adaptive radiation, Hardy-Weinberg principle, types of natural selection, human evolution

Evolutionary biology is the study of the history of life forms on earth. This chapter follows the story from the **origin of life** to the **evidences and mechanism of evolution** and the **origin of man**.

1. Origin of Life

When we look at stars on a clear night we are, in a way, **looking back in time**: their light started its journey millions of years ago. The origin of life is considered a unique event in the history of the universe.

- → The universe is almost 13.8 billion years old. Huge clusters of galaxies make up the universe; galaxies contain stars and clouds of gas and dust. Compared to the universe, the earth is only a speck.
- → The **Big Bang theory** explains the origin of the universe: a single huge explosion, after which the universe expanded and the temperature came down. Hydrogen and helium formed later, and gases condensed under gravitation to form the galaxies.
- → In the solar system of the Milky Way galaxy, **earth** was formed about 4.5 billion years ago.
- → There was **no atmosphere** on early earth. Water vapour, methane, carbon dioxide and ammonia released from the molten mass covered the surface.
- → UV rays from the sun broke water into hydrogen and oxygen; the lighter H_2 escaped. Oxygen combined with ammonia and methane to form water, CO_2 and others. The **ozone layer** formed. As it cooled, water vapour fell as rain, filling depressions to form **oceans**.
- → Life appeared about 500 million years after the formation of earth, i.e. almost **4 billion years ago**.

A. Theories of origin of life PYQ

Theory	Main idea
Panspermia	early Greek thinkers: units of life called **spores** were transferred to different planets, including earth; still a favourite idea of some astronomers
Spontaneous generation	life came out of decaying and rotting matter like straw, mud, etc.
Biogenesis (Louis Pasteur)	life comes only from pre-existing life; disproved spontaneous generation
Chemical evolution (Oparin and Haldane)	first life came from pre-existing **non-living organic molecules** (e.g. RNA, protein); preceded by formation of organic molecules from inorganic constituents

Pasteur's experiment: in **pre-sterilised flasks**, life did not come from killed yeast, while in another flask **open to air**, new living organisms arose from "killed yeast". The theory of spontaneous generation was dismissed once and for all. But this did not answer how the first life form came on earth.

Oparin-Haldane: conditions on early earth were high temperature, volcanic storms and a reducing atmosphere containing CH_4, NH_3, etc.

B. Miller's experiment (1953) PYQ

- → **S.L. Miller**, an American scientist, created similar conditions on a laboratory scale.
- → He created an electric discharge in a closed flask containing **CH_4, H_2, NH_3 and water vapour at 800°C**.
- → He observed formation of amino acids. Others observed formation of **sugars, nitrogen bases, pigments and fats** in similar experiments.
- → Analysis of **meteorite** content also revealed similar compounds, indicating that similar processes are occurring elsewhere in space.
- → With this limited evidence, the first part of the story, i.e. **chemical evolution**, was more or less accepted.

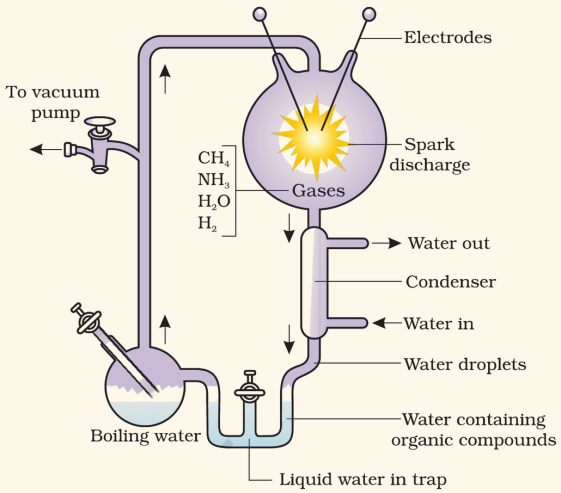


Fig. 6.1 Diagrammatic representation of Miller's experiment

Yaad karo: Miller ki gases

"Mere Hath Nahi Waale" = Methane (CH_4), Hydrogen (H_2), NH_3 (ammonia), Water vapour. 800°C + electric spark = **amino acids**. Oxygen nahi tha, atmosphere **reducing** tha!

- → We have no idea how the first **self-replicating metabolic capsule** of life arose.

→ The first non-cellular forms of life could have originated about **3 billion years** ago. They would have been giant molecules (RNA, protein, polysaccharides, etc.), and these capsules reproduced their molecules perhaps.

→ The first **cellular form of life** did not originate till about 2000 million years ago (mya). These were probably single cells, and all life forms were in a water environment only.

→ This version of abiogenesis (first form of life arose slowly through evolutionary forces from non-living molecules) is accepted by the majority.

Quick fact

Once formed, the first cellular forms of life evolved into the complex biodiversity of today. The story of life on earth is thus a story of chemical evolution followed by biological evolution.

2. Evolution of Life Forms: A Theory

Conventional religious literature tells us about the theory of special creation. It has three connotations:

1. All living organisms (species or types) that we see today were **created as such**.
2. The diversity was **always the same** since creation and will remain the same in future.
3. **Earth is about 4000 years old.**

These ideas were strongly challenged in the **nineteenth century**. Based on observations made during a sea voyage round the world in the sail ship H.M.S. Beagle, **Charles Darwin** concluded:

→ Existing living forms share similarities to varying degrees not only among themselves but also with life forms that existed **millions of years ago**. Many such life forms do not exist any more.

→ There had been **extinctions** of different life forms over the years, just as new forms arose at different periods of the history of earth.

→ There has been **gradual evolution** of life forms. Any population has built-in **variation** in characteristics.

→ Those characteristics which enable some to survive better in natural conditions (climate, food, physical factors, etc.) would outbreed others that are less endowed. This is the idea of **fitness**.

→ According to Darwin, fitness refers ultimately and only to reproductive fitness. Those who are better fit in an environment leave more progeny than others. They will survive more and hence are **selected by nature**: natural selection, a mechanism of evolution.

Alfred Wallace, a naturalist who worked in the **Malay Archipelago**, also came to similar conclusions around the same time. In due course, new types of organisms became recognisable. All existing life forms share similarities and **common ancestors**, but these ancestors were present at different periods of the history of earth (epochs, periods and eras).

Exam Alert

→ "Survival of the fittest" mein fitness = **reproductive fitness** (zyada progeny), sirf taakat nahi.

→ Special creation ke **3 points** yaad rakho; earth 4000 saal purani (galat).

→ Universe **13.8 billion** (naya NCERT), earth **4.5 billion**, life **4 billion** saal pehle.

3. What are the Evidences for Evolution?

A. Paleontological evidence (fossils) PYQ

→ **Fossils** are remains of hard parts of life forms found in rocks.

→ Rocks form **sediments**, and a cross-section of the earth's crust shows sediments arranged one over the other during the long history of earth.

→ Different-aged rock sediments contain fossils of different life forms, which probably died during the formation of that sediment. Some appear similar to modern organisms; others represent extinct organisms (e.g. **dinosaurs**).

→ A study of fossils in different sedimentary layers indicates the **geological period** in which they existed. It shows that life forms varied over time and certain life forms are restricted to certain geological time spans.

→ Ages of fossils are calculated by radioactive dating.

B. Embryological evidence

Proposed by **Ernst Haeckel**, based on features common to all vertebrate embryos but absent in adults. E.g. embryos of all vertebrates, including human, develop a row of vestigial gill slits just behind the head, but these are functional only in fish. This proposal was **disapproved** by **Karl Ernst von Baer**: embryos never pass through the adult stages of other animals. PYQ

C. Comparative anatomy and morphology PYQ

Similarities and differences among organisms of today and those that existed years ago show whether common ancestors were shared.

→ **Homologous organs: same structure (anatomy), different functions.** Whales, bats, cheetah and human (all mammals) share the pattern of forelimb bones: humerus, radius, ulna, carpals, metacarpals and phalanges, although the forelimbs perform different functions (swimming, flying, running, grasping). The same structure developed along different directions due to adaptation to different needs. This is divergent evolution.

→ Other homology examples: **vertebrate hearts or brains**; in plants, the **thorn of *Bougainvillea*** and the **tendril of *Cucurbita***.

→ **Analogous organs: different structure, same function.** Wings of **butterfly and birds** look alike but are not anatomically similar. Analogous structures result from convergent evolution: different structures evolving for the same function.

→ Other analogy examples: **eye of octopus and of mammals**; **flippers of penguins and dolphins**; **sweet potato** (root modification) and **potato** (stem modification): both store food, but one is a root and the other a stem. Similar habitat has selected similar adaptive features in different groups.

Yaad karo: Homo vs Ana

HOMologous = HOMe same (ghar/structure same, kaam alag) = Divergent. **ANAlogous = ANAdar se alag** (structure alag, kaam same) = Convergent. Trick: **"Homo Diwana, Ana Cute"**.

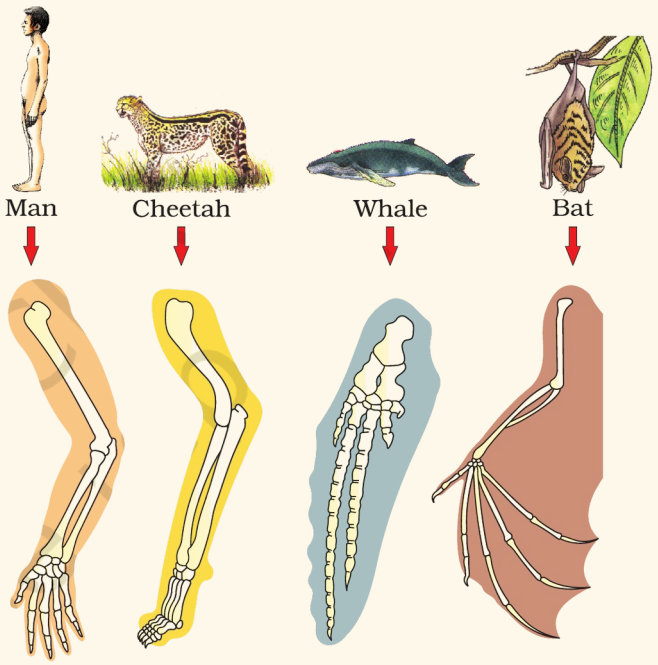


Fig. 6.3 (b) Homologous organs in animals: forelimbs of man, cheetah, whale and bat

Homologous organs	Analogous organs
same origin and basic structure	different origin and structure
different functions	similar functions
show **divergent** evolution	show **convergent** evolution
indicate **common ancestry**	no common ancestry implied
e.g. forelimbs of whale, bat, cheetah, human; thorn of *Bougainvillea* and tendril of *Cucurbita*	e.g. wings of butterfly and bird; eye of octopus and mammal; sweet potato and potato

D. Biochemical evidence

Similarities in **proteins and genes** performing a given function among diverse organisms give clues to common ancestry. These biochemical similarities point to the same shared ancestry as structural similarities.

E. Artificial selection

Man has bred selected plants and animals for agriculture, horticulture, sport or security, and has **domesticated** many wild animals and crops. Intensive breeding has created breeds that differ from other breeds (e.g. **dogs**) but are still of the same group. If man could create new breeds within hundreds of years, nature could have done the same over millions of years.

F. Natural selection in action: industrial melanism PYQ

Before industrialisation (1850s)	After industrialisation (1920)
more **white-winged moths** on trees	more **dark-winged (melanised) moths**
tree trunks covered with almost **white lichens**	tree trunks became **dark** due to smoke and soot; lichens did not grow
dark moths were picked out by predators	white moths were picked out by predators

Fig. 6.4 White-winged moth and dark-winged (melanised) moth on a tree trunk: (a) in unpolluted area (b) in polluted area

→ Explanation: after industrialisation the tree trunks became dark, so the white-winged moths did not survive due to predators, while dark-winged moths survived (camouflage). The **proportion** changed, not the creation of a new kind.

→ In rural areas (no industrialisation), the count of melanic moths stayed low. Lichens are industrial pollution indicators: they will not grow in polluted areas.

→ Remember, **no variant is completely wiped out**.

→ Similarly, excess use of **herbicides, pesticides** etc. has resulted in selection of **resistant varieties** in a much shorter time. The same is true for microbes against which we use antibiotics: resistant organisms appear in months or years, not centuries.

→ These are examples of evolution by anthropogenic action. Evolution is a stochastic process based on **chance events** in nature and **chance mutations** in organisms.

Exam Alert: NCERT exercise (antibiotic resistance)

Bacteria ki population mein pehle se kuch **resistant variants** hote hain. Antibiotic baaki sabko maar deta hai, resistant wale bachte aur multiply karte hain (zyada progeny = **reproductive fitness**). Kuch generations mein poori population resistant: ye **Darwinian natural selection** hai. Antibiotic ne mutation paida nahi kiya, sirf **select** kiya.

Evidences of evolution: one table

Evidence	Example
Palaeontological	fossils in rock layers of different ages show different life forms
Comparative anatomy (homology)	forelimbs of whale, bat, cheetah, human; thorn and tendril
Comparative anatomy (analogy)	wings of butterfly and bird; eyes of octopus and mammals
Biochemical (molecular homology)	similar proteins and genes for a given function
Artificial selection	new breeds of dogs, cows and crops by breeders
Natural selection in action	industrial melanism; antibiotic, herbicide and pesticide resistance

4. What is Adaptive Radiation? PYQ

→ During his journey Darwin visited the **Galapagos Islands**. He observed small black birds, later called Darwin's finches: many varieties on the same island.

→ He conjectured that all varieties evolved **on the island itself**. From the original **seed-eating** features, many forms with altered beaks arose, enabling them to become **insectivorous and vegetarian** finches.

→ The evolution of different species in a given geographical area, starting from a point and literally **radiating to other areas** of geography (habitats), is called adaptive radiation. Darwin's finches are one of the best examples.

→ Another example: **Australian marsupials**. A number of marsupials, each different from the other, evolved from an ancestral stock, all within the Australian island continent.

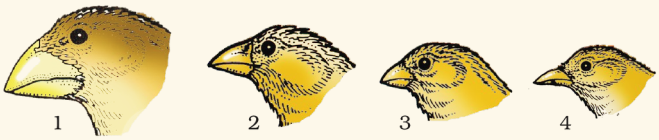


Fig. 6.5 Variety of beaks of finches that Darwin found in the Galapagos Islands

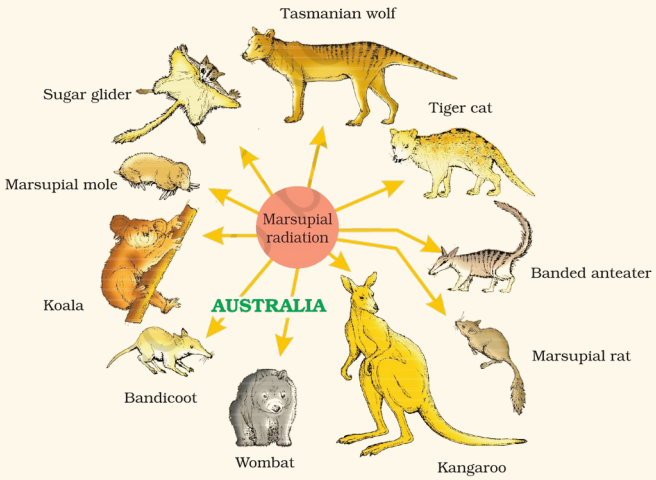


Fig. 6.6 Adaptive radiation of marsupials of Australia

When more than one adaptive radiation appears to have occurred in an isolated geographical area (representing different habitats), one can call this convergent evolution. Placental mammals in Australia also show adaptive radiation into varieties, each appearing "similar" to a corresponding marsupial, e.g. **placental wolf and Tasmanian wolf (marsupial)**.

Paheli-Bujho

Bujho: Paheli, finches ki chonch alag alag kyun thi? Fashion?

Paheli: Nahi re! Khane ke hisaab se. Seed, insect, fruit: har kaam ke liye alag chonch.

Bujho: Matlab ek hi purvaj se poori family alag alag ghar mein shift ho gayi?

Paheli: Haan, yahi adaptive radiation hai. Ek point se chaaron taraf.

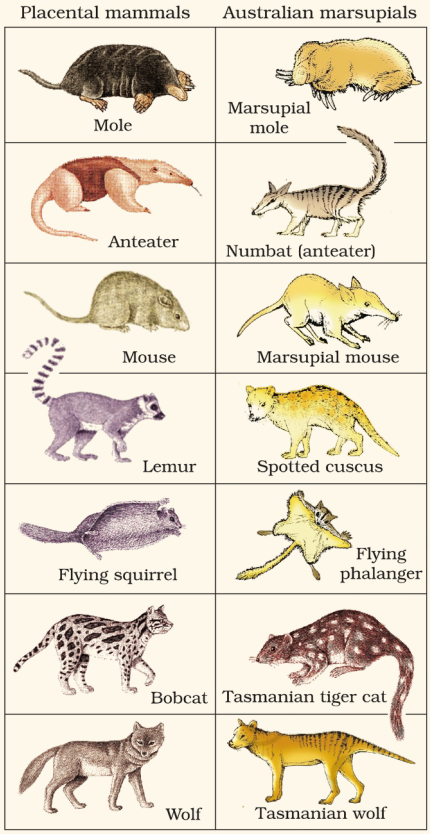


Fig. 6.7 Convergent evolution of Australian marsupials and placental mammals

5. Biological Evolution

→ Evolution by natural selection, in a true sense, would have started when **cellular forms of life** with differences in metabolic capability originated on earth. The essence of Darwinian theory is natural selection.

→ The **rate of appearance of new forms** is linked to the life cycle or life span. Microbes that divide fast can become millions of individuals within hours.

→ A colony of bacteria (say **A**) growing on a given medium has built-in variation in its ability to use a feed component. A change in medium composition brings out only that part of the population (say **B**) that can survive under the new conditions. In due course, this variant population outgrows the others and appears as a new species, **within days**. The same in a fish or fowl would take **millions of years**.

→ We say the fitness of B is better than A under the new conditions: nature selects for fitness. Fitness is based on **inherited** characteristics, so there must be a **genetic basis** for getting selected and to evolve. Adaptive ability is inherited. Fitness is the end result of the ability to adapt and get selected by nature.

→ Branching descent and natural selection are the two key concepts of Darwinian theory of evolution.

A. Lamarck (before Darwin)

French naturalist **Lamarck** said evolution of life forms occurred, driven by use and disuse of organs. Example: **giraffes**, in an attempt to forage leaves on tall trees, elongated their necks and passed this **acquired character** to succeeding generations. Nobody believes this conjecture any more.

B. Basis of natural selection (Darwin) PYQ

Is evolution a process or the result of a process? The world we see is only the success stories of evolution. The work of **Thomas Malthus** on populations possibly influenced Darwin. Natural selection is based on observations which are factual:

1. **Natural resources are limited.**
2. **Populations are stable** in size except for seasonal fluctuation.
3. **Members of a population vary** in characteristics (no two individuals are alike), and most variations are **inherited**.
4. Theoretically population size would grow **exponentially** if everybody reproduced maximally, but in reality it is limited: so there is competition for resources. Only some survive and grow at the cost of others.

Darwin's brilliant insight: variations that are heritable and make resource use better for a few (better adapted to the habitat) enable only those to reproduce and leave more progeny. Over many generations, survivors leave more progeny, the population characteristic changes and new forms appear to arise.

variation (heritable) → overproduction → struggle for existence → survival of the fittest → more progeny → new species

Lamarck vs Darwin: giraffe ki lambi gardan.

Asli baat: Lamarck: use and disuse (acquired traits pass nahi hote). Darwin: variation + natural selection = sahi explanation.

Key terms

- → **Fossil**: remains of hard parts of life forms preserved in rocks.
- → **Gene pool**: total genes and their alleles in a population.
- → **Fitness**: reproductive fitness, i.e. ability to leave more progeny.
- → **Founder effect**: a small drifted population becoming genetically distinct, its members acting as founders.
- → **Saltation**: single-step large mutation causing speciation (de Vries).

6. Mechanism of Evolution

- → What is the origin of variation and how does speciation occur? Even though Mendel had talked of inheritable "factors", Darwin either ignored these or kept silent.
- → In the first decade of the twentieth century, **Hugo de Vries**, based on his work on evening primrose (*Oenothera lamarckiana*), brought forth the idea of **mutations**: large differences arising suddenly in a population.
- → He believed that **mutation causes evolution**, not the minor heritable variations Darwin talked about. He called this mutation-driven speciation saltation (single-step large mutation).
- → Studies in **population genetics** later brought some clarity.

Darwinian variations	*Mutations (de Vries)*
small	large
directional	**random and directionless**
evolution is **gradual**	speciation is sudden: **saltation**

7. Hardy-Weinberg Principle PYQ

- → In a given population, the **frequency of alleles** of a gene or locus can be found. This frequency is supposed to remain fixed, even through generations.
- → The Hardy-Weinberg principle states that allele frequencies in a population are stable and constant from generation to generation. The **gene pool** (total genes and their alleles in a population) remains constant. This is called genetic equilibrium.
- → Sum total of all allelic frequencies is **1**. In a diploid, let **p** = frequency of allele A and **q** = frequency of allele a, so $p + q = 1$.
- → Frequency of AA individuals = p^2 (probability that A appears on both chromosomes), of aa = q^2, of Aa = $2pq$.

The Hardy-Weinberg equation

$p^2 + 2pq + q^2 = 1$, the binomial expansion of $(p + q)^2$. When the measured frequency differs from the expected value, the difference (direction) shows the **extent of evolutionary change**. Disturbance of genetic equilibrium = **evolution**.

Factors affecting Hardy-Weinberg equilibrium PYQ

1. **Gene migration or gene flow**: when a section of a population migrates to another place and population, gene frequencies change in both the original and the new population. New genes/alleles are added to the new population and lost from the old one. If gene migration happens **multiple times**, there is gene flow.
2. **Genetic drift**: if the same change in allele frequency occurs **by chance**, it is genetic drift. Sometimes the change is so different in the new sample of population that they become a different species. The original drifted population becomes the **founders**, and the effect is called the founder effect.
3. **Mutation**: microbial experiments show that pre-existing advantageous mutations, when selected, result in new phenotypes; over a few generations this results in **speciation**.
4. **Genetic recombination**: variation due to recombination during **gametogenesis**.
5. **Natural selection**: heritable variations enabling better survival are enabled to reproduce and leave a greater number of progeny.

Yaad karo: H-W ko disturb karne wale 5

"Mera Gaana Mujhe Rula Naa de" = Migration (gene flow), Genetic drift, Mutation, Recombination, Natural selection.

Exam Alert: H-W numerical

→ Population mein recessive phenotype (aa) = 16% ho, toh $q^2 = 0.16$, **q = 0.4**, p = 0.6.

→ AA = $p^2 = 0.36$ (36%), Aa = 2pq = 0.48 (48%), aa = 0.16 (16%). Total = 1. Hamesha check karo!

Types of natural selection PYQ

Variation due to mutation, recombination during gametogenesis, gene flow or genetic drift changes the frequency of genes and alleles in future generations. Coupled with enhanced reproductive success, natural selection makes it look like a different population. Natural selection can lead to:

Type	Which individuals increase	Result
Stabilising	more individuals acquire the **mean** character value	peak becomes taller and narrower
Directional	more individuals acquire a value **other than the mean** (one extreme)	peak shifts in one direction
Disruptive	more individuals acquire **peripheral** values at **both ends**	two peaks form

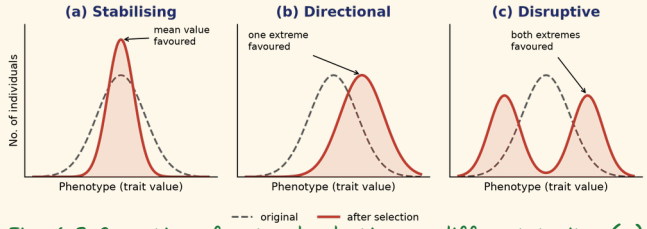


Fig. 6.8 Operation of natural selection on different traits: (a) Stabilising (b) Directional (c) Disruptive

Paheli-Bujho

Bujho: Paheli, founder effect matlab jo pehle aaya wahi boss?

Paheli: Haha, lagbhag! Chhota group nayi jagah gaya, uske genes hi nayi population ka base ban gaye.

Bujho: Aur ye sab chance se hua toh?

Paheli: Toh ye genetic drift hai. Dono Hardy-Weinberg balance bigaadte hain.

8. A Brief Account of Evolution

→ About **2000 mya**, the first cellular forms of life appeared. How non-cellular aggregates of giant macromolecules evolved into cells with a membranous envelope is not known.

→ Some of these cells could release O_2; the reaction could have been similar to the **light reaction of photosynthesis** (water split with captured solar energy).

→ Slowly, single-celled organisms became **multicellular** life forms.

Time (mya)	Event
2000	first cellular forms of life
500	**invertebrates** formed and active
350	**jawless fish** evolved; fish with stout, strong fins (lobefins) moved onto land and back to water
320	sea weeds and a few plants existed
200	some land reptiles went back to water: fish-like reptiles (**Ichthyosaurs**)
65	**dinosaurs** suddenly disappeared
15	**Dryopithecus** and **Ramapithecus** (primates)

→ The **first organisms to invade land were plants**; they were widespread on land when animals invaded land.

→ In **1938**, a fish caught in **South Africa** was a Coelacanth, thought to be extinct. These **lobefins** evolved into the **first amphibians** that lived on both land and water (ancestors of modern frogs and salamanders; no specimens left).

→ Amphibians evolved into **reptiles**, which lay **thick-shelled eggs** that do not dry up in the sun. Modern descendants: turtles, tortoises and crocodiles.

→ In the next **200 million years** or so, reptiles of different shapes and sizes dominated the earth. Giant ferns (pteridophytes) were present but fell to form **coal deposits** slowly.

→ The land reptiles were the **dinosaurs**; the biggest, *Tyrannosaurus rex*, was about **20 feet** tall with huge dagger-like teeth. About **65 mya** they suddenly disappeared. The true reason is not known: some say climatic changes killed them, some say most evolved into birds. Small reptiles of that era still exist.

→ The **first mammals** were like **shrews**. Mammals were viviparous and protected their unborn young inside the mother's body; they were more intelligent in sensing and avoiding danger. When reptiles came down, mammals took over.

→ South America had mammals resembling horse, hippopotamus, bear, rabbit, etc. Due to continental drift, when South America joined North America, these were overridden by North American fauna. Due to the same continental drift, **pouched mammals of Australia survived** because of lack of competition.

→ Some mammals live wholly in water: **whales, dolphins, seals and sea cows**. The most successful story is the evolution of **man**, with language skills and self-consciousness.

Exam Alert: evolution ka sequence

Jawless fish → lobefins (Coelacanth) → **amphibians** → **reptiles** (thick-shelled eggs) → dinosaurs / birds and **mammals** (shrew-like first). Plants pehle land pe aaye, animals baad mein. Mammals **viviparous** the: bachcha maa ke andar safe rehta tha, isliye reptiles ke baad mammals chha gaye.

9. Origin and Evolution of Man PYQ

Stage (time)	Key features
Dryopithecus and Ramapithecus (about 15 mya)	hairy, walked like gorillas and chimpanzees; Ramapithecus more **man-like**, Dryopithecus more **ape-like**
Man-like primates, eastern Africa (3–4 mya)	fossils from **Ethiopia and Tanzania**; not taller than 4 feet but walked **upright**
Australopithecines (2 mya)	East African grasslands; hunted with **stone weapons** but essentially ate fruit

Hominid	Key features
Homo habilis (first human-like being)	brain 650–800 cc; probably did not eat meat
Homo erectus (1.5 mya; fossils in Java, 1891)	brain about 900 cc; probably ate meat

Hominid	Key features
Neanderthal man (1,00,000–40,000 years ago)	brain 1400 cc; near east and central Asia; used hides to protect the body and **buried their dead**
Homo sapiens	arose in Africa, moved across continents and developed into distinct races; modern Homo sapiens arose during the **ice age** (75,000–10,000 years ago)

→ **Prehistoric cave art** developed about 18,000 years ago. Cave paintings by prehistoric humans can be seen at the **Bhimbetka rock shelter** in Raisen district of Madhya Pradesh.

→ **Agriculture** came around 10,000 years ago, and human settlements started. The rest is human history of the growth and decline of civilisations.

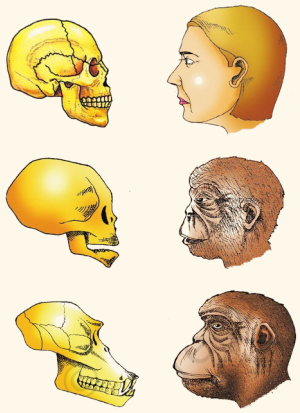


Fig. 6.11 Skulls of adult modern human, baby chimpanzee and adult chimpanzee: the baby chimpanzee skull is more like the adult human skull

Exam Alert: human evolution traps

→ **Ramapithecus** = more man-like, **Dryopithecus** = more ape-like. Ulta mat likhna!

→ Homo habilis probably **did not eat meat**; Homo erectus probably **ate meat**.

→ Neanderthals ne hides use kiye aur apne dead ko **bury** kiya: 1 mark favourite.

Yaad karo: brain size ka ladder

Habilis 650–800 < Erectus 900 < Neanderthal 1400

cc. Trick: **"Habil Ek Neta"** (H, E, N), dimaag badhta gaya, naam ki tarah!

1 minute recap

1. Universe 13.8 billion years, earth 4.5 billion, life about 4 billion years ago; theories: panspermia, spontaneous generation, biogenesis (Pasteur), chemical evolution (Oparin–Haldane).
2. Miller (1953): CH_4, H_2, NH_3, water vapour, 800°C, electric discharge → amino acids.
3. Special creation (3 connotations) challenged by Darwin (H.M.S. Beagle); Wallace (Malay Archipelago) reached the same idea.
4. Evidences: fossils, homology (divergent), analogy (convergent), biochemical similarity, artificial selection, industrial melanism, antibiotic and pesticide resistance.
5. Adaptive radiation: Darwin's finches, Australian marsupials; convergent evolution: placental wolf vs Tasmanian wolf.
6. Darwin: branching descent + natural selection; fitness = reproductive fitness; de Vries: mutation, saltation.
7. Hardy-Weinberg: $p^2 + 2pq + q^2 = 1$; disturbed by gene flow, drift (founder effect), mutation, recombination, natural selection.
8. Selection: stabilising, directional, disruptive.
9. Man: Dryopithecus/Ramapithecus (15 mya) → Australopithecines → H. habilis (650–800 cc) → H. erectus (900 cc) → Neanderthal (1400 cc) → H. sapiens.

MOST IMPORTANT QUESTIONS

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM According to NCERT, the age of the universe is about:
(a) 4.5 billion years (b) 4 billion years
(c) 13.8 billion years (d) 20 million years

Ans: (c) 13.8 billion years

Q2. IM Board fav Which gas was NOT used in Miller's experiment?
(a) Methane (b) Ammonia
(c) Hydrogen (d) Oxygen

Ans: (d) Oxygen

Q3. IM Miller's experiment produced:
(a) Nucleic acids (b) Amino acids
(c) Proteins (d) Living cells

Ans: (b) Amino acids

Q4. IM The theory that life comes only from pre-existing life was demonstrated by:
(a) Louis Pasteur (b) Oparin
(c) S.L. Miller (d) Haldane

Ans: (a) Louis Pasteur

Q5. IM Board fav Which of the following is a pair of homologous organs?
(a) Wings of butterfly and bird
(b) Eye of octopus and mammal
(c) Thorn of *Bougainvillea* and tendril of *Cucurbita*
(d) Sweet potato and potato
Ans: (c) Thorn of *Bougainvillea* and tendril of *Cucurbita*

Q6. IM Analogous organs are a result of:
(a) Divergent evolution (b) Convergent evolution
(c) Adaptive radiation (d) Genetic drift
Ans: (b) Convergent evolution

Q7. IM Sweet potato and potato are an example of analogy because they are respectively modifications of:
(a) Root and stem (b) Stem and root
(c) Both roots (d) Both stems
Ans: (a) Root and stem

Q8. IM The embryological evidence proposed by Haeckel was disapproved by:
(a) Darwin (b) Lamarck
(c) Wallace (d) Karl Ernst von Baer
Ans: (d) Karl Ernst von Baer

Q9. IM Board fav Industrial melanism in moths is an example of:
(a) Mutation (b) Natural selection
(c) Genetic drift (d) Use and disuse
Ans: (b) Natural selection

Q10. IM Lichens are used as indicators of:
(a) Water pollution
(b) Soil fertility
(c) Industrial (air) pollution
(d) Radioactivity
Ans: (c) Industrial (air) pollution

Q11. IM Board fav Darwin's finches are an example of:
(a) Adaptive radiation (b) Convergent evolution
(c) Industrial melanism (d) Founder effect
Ans: (a) Adaptive radiation

Q12. IM Placental wolf and Tasmanian wolf are an example of:
(a) Divergent evolution (b) Homology
(c) Convergent evolution (d) Saltation
Ans: (c) Convergent evolution

Q13. IM According to Darwin, fitness refers to:
(a) Physical strength (b) Long life span
(c) Large body size (d) Reproductive fitness
Ans: (d) Reproductive fitness

Q14. IM Hugo de Vries worked on:
(a) Pea (b) Evening primrose
(c) Snapdragon (d) *Drosophila*
Ans: (b) Evening primrose

Q15. IM The single-step large mutation that causes speciation is called:
(a) Saltation (b) Genetic drift
(c) Gene flow (d) Founder effect
Ans: (a) Saltation

Q16. IM Board fav In a population in Hardy-Weinberg equilibrium, the frequency of heterozygotes is:
(a) p^2 (b) q^2 (c) 2pq (d) p + q
Ans: (c) 2pq

Q17. IM If the frequency of a recessive allele is 0.3, the frequency of homozygous dominant individuals is:
(a) 0.09 (b) 0.49 (c) 0.42 (d) 0.70
Ans: (b) 0.49

Q18. IM Board fav When more individuals acquire peripheral character values at both ends of the distribution curve, the selection is:
(a) Stabilising (b) Directional
(c) Artificial (d) Disruptive
Ans: (d) Disruptive

Q19. IM The fish caught in South Africa in 1938, thought to be extinct, was:
(a) Coelacanth (b) Ichthyosaur
(c) Shark (d) Lungfish
Ans: (a) Coelacanth

Q20. IM The first mammals were like:
(a) Rats (b) Rabbits (c) Shrews (d) Monkeys
Ans: (c) Shrews

Q21. IM Board fav The brain capacity of *Homo erectus* was about:
(a) 650-800 cc (b) 900 cc
(c) 1400 cc (d) 1650 cc
Ans: (b) 900 cc

Q22. IM Which hominid buried their dead and used hides to protect the body?
(a) *Homo habilis* (b) *Homo erectus*
(c) Australopithecines (d) Neanderthal man
Ans: (d) Neanderthal man

Q23. IM Prehistoric cave paintings can be seen at Bhimbetka rock shelter in:
(a) Madhya Pradesh (b) Maharashtra
(c) Rajasthan (d) Karnataka
Ans: (a) Madhya Pradesh

Q24. IM The ship on which Darwin made his voyage round the world was:
(a) H.M.S. Victory (b) H.M.S. Beagle
(c) Titanic (d) Mayflower
Ans: (b) H.M.S. Beagle

Q25. IM Alfred Wallace worked in:
(a) Galapagos Islands (b) Australia
(c) Malay Archipelago (d) South Africa
Ans: (c) Malay Archipelago

Q26. 1M The first non-cellular forms of life could have originated about:
(a) 3 billion years ago (b) 2000 mya
(c) 500 mya (d) 13.8 billion years ago
Ans: (a) 3 billion years ago

Q27. 1M The eye of octopus and the eye of mammals are:
(a) Homologous organs (b) Vestigial organs
(c) Atavistic organs (d) Analogous organs
Ans: (d) Analogous organs

Q28. 1M Board fav Which of the following is NOT a factor affecting Hardy-Weinberg equilibrium?
(a) Gene flow
(b) Genetic drift
(c) Random mating in a large population
(d) Natural selection
Ans: (c) Random mating in a large population

Q29. 1M Fish-like reptiles that evolved about 200 mya were:
(a) Dinosaurs (b) Ichthyosaurs
(c) Coelacanths (d) Pterosaurs
Ans: (b) Ichthyosaurs

Q30. 1M The giant ferns of the reptile era slowly formed:
(a) Coal deposits (b) Petroleum
(c) Limestone (d) Sand dunes
Ans: (a) Coal deposits

Q31. 1M Agriculture started about:
(a) 75,000 years ago (b) 40,000 years ago
(c) 18,000 years ago (d) 10,000 years ago
Ans: (d) 10,000 years ago

Assertion-Reason (1 mark each)

Assertion-Reason key: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q32. 1M Board fav **Assertion (A):** The forelimbs of whale, bat, cheetah and human are homologous organs.
Reason (R): They have the same pattern of bones but perform different functions.
Ans: (a) Both A and R are true and R is the correct explanation of A.

Q33. 1M Board fav **Assertion (A):** After industrialisation in England, dark-winged moths increased in number.
Reason (R): Smoke caused mutations that turned white moths dark.
Ans: (c) A is true but R is false. (Dark moths were already present; they were selected because they were camouflaged on soot-covered trunks.)

Q34. 1M **Assertion (A):** Bacteria can show evolution within days.
Reason (R): The rate of appearance of new forms is linked to the life span of an organism.
Ans: (a) Both A and R are true and R is the correct explanation of A.

Q35. 1M **Assertion (A):** Analogous organs indicate common ancestry.
Reason (R): Analogous organs have different structures but perform similar functions.
Ans: (d) A is false but R is true. (Analogy results from convergent evolution, not common ancestry.)

Short answer (2-3 marks)

Q36. 3M Board fav Explain antibiotic resistance observed in bacteria in light of Darwinian selection theory. (NCERT)
Ans: A bacterial population has natural variation; a few cells are already resistant to an antibiotic. When the antibiotic is used, sensitive bacteria die while resistant ones survive and multiply, leaving more progeny (reproductive fitness). Over generations the population becomes mostly resistant. The antibiotic does not create resistance; it **selects** the pre-existing resistant variants: natural selection.

Q37. 2M Attempt giving a clear definition of the term species. (NCERT)
Ans: A species is a group of organisms with fundamental similarities that can **interbreed freely in nature to produce fertile offspring** and are reproductively isolated from other such groups.

Q38. 3M Board fav Trace the various components of human evolution (brain size and function, skeletal structure, dietary preference). (NCERT)
Ans: **Brain size**: Homo habilis 650-800 cc, H. erectus 900 cc, Neanderthal 1400 cc, larger in H. sapiens, with language and self-consciousness. **Skeleton**: from hairy, gorilla-like walking (Dryopithecus) to upright walking (3-4 mya primates, Australopithecines). **Diet**: Australopithecines mainly fruit, H. habilis probably no meat, H. erectus probably ate meat; agriculture began about 10,000 years ago.

Q39. 2M Board fav Describe one example of adaptive radiation. (NCERT)
Ans: **Darwin's finches** of the Galapagos Islands: from the original seed-eating ancestral finch, many varieties with altered beaks evolved on the same island, becoming insectivorous and vegetarian, each suited to a different food habit. (Or: Australian marsupials evolving from one ancestral stock into many forms.)

Q40. 2M Board fav Can we call human evolution adaptive radiation? (NCERT)
Ans: **No.** Adaptive radiation is the evolution of many different species from a common ancestor, radiating into different habitats in a geographical area at the same time. Human evolution is a sequence of forms (Dryopithecus to Homo sapiens) replacing one another over time, leading to a single surviving species, not a radiation into many habitats.

Q41. 3M Trace the evolutionary stages of horse. (NCERT)
Ans: *Eohippus* (dawn horse, small, four toes on forelimbs) → *Mesohippus* (three toes) → *Merychippus* (three toes, side toes reduced; grazing teeth) → *Pliohippus* (single toe) → *Equus* (modern horse: larger body, single hoofed toe, long limbs, grinding teeth). The trend shows increase in size and reduction of toes for fast running on grasslands.

Q42. 2M What were the conditions on early earth according to Oparin and Haldane? What did they propose?
Ans: High temperature, volcanic storms and a reducing atmosphere with CH_4, NH_3 etc. They proposed that the first life arose from pre-existing non-living organic molecules (RNA, protein), preceded by **chemical evolution** (formation of organic molecules from inorganic constituents).

Q43. 3M Board fav Describe Miller's experiment. What did it prove?
Ans: In a closed flask containing CH_4, H_2, NH_3 and water vapour at 800°C, Miller created an electric discharge (like lightning) and then cooled the products. Amino acids were formed. It proved that organic molecules could form from inorganic ones under early-earth conditions, supporting **chemical evolution**.

Q44. 2M How did Louis Pasteur disprove spontaneous generation?
Ans: He showed that in pre-sterilised flasks, life did not arise from killed yeast, while in a flask open to air, new organisms arose from killed yeast. Hence life comes only from pre-existing life (biogenesis).

Q45. 2M State the three connotations of the theory of special creation.
Ans: (i) All living organisms seen today were created as such. (ii) The diversity was always the same since creation and will remain so. (iii) Earth is about 4000 years old.

Q46. 3M What are fossils? How do they provide evidence for evolution?
Ans: Fossils are remains of hard parts of life forms found in rocks. Rock sediments of different ages contain fossils of different life forms; some resemble modern organisms, others are extinct. This shows that life forms changed over geological time. Their age is found by radioactive dating.

Q47. 3M Board fav Differentiate between homologous and analogous organs with one example each.
Ans: **Homologous**: same basic structure and origin, different functions; divergent evolution; e.g. forelimbs of whale, bat, cheetah and human. **Analogous**: different structure, same function; convergent evolution; e.g. wings of butterfly and bird.

Q48. 3M Explain divergent and convergent evolution with examples.
Ans: **Divergent**: the same ancestral structure develops in different directions for different needs, giving homologous organs (forelimbs of mammals, thorn and tendril).
Convergent: different structures evolve for the same function due to similar habitat, giving analogous organs (wings of bird and butterfly, flippers of penguin and dolphin).

Q49. 2M What was Haeckel's embryological evidence? Why was it rejected?
Ans: Embryos of all vertebrates, including humans, develop a row of vestigial gill slits behind the head, functional only in fish, suggesting common ancestry. Karl Ernst von Baer showed that embryos never pass through the adult stages of other animals, so the proposal was disapproved.

Q50. 3M Board fav Explain industrial melanism. Why are lichens important here?
Ans: In England, before industrialisation (1850s) white-winged moths were more common, as white lichens covered tree trunks and camouflaged them. After industrialisation (1920) smoke darkened the trunks and killed lichens; white moths were picked out by predators and dark moths survived and increased. Lichens indicate pollution as they do not grow in polluted areas.

Q51. 2M What is evolution by anthropogenic action? Give two examples.
Ans: Evolution driven by human activities that act as strong selection pressures. Examples: (i) excess use of herbicides and pesticides selecting resistant varieties; (ii) antibiotics selecting resistant microbes within months or years.

Q52. 3M Board fav What is adaptive radiation? How does convergent evolution arise from it?
Ans: Evolution of different species in a geographical area starting from a point and radiating into different habitats. When more than one adaptive radiation occurs in an isolated area, unrelated groups evolve similar forms for similar habitats, e.g. placental mammals and marsupials in Australia (placental wolf and Tasmanian wolf): convergent evolution.

Q53. 3M Explain the bacterial example of natural selection given in NCERT.
Ans: A bacterial colony A has built-in variation in using a feed component. When the medium changes, only part B that can survive the new conditions grows; over time B outgrows the rest and appears as a new species within days. Fitness of B is better under the new condition: nature selects for fitness.

Q54. 2M What were Lamarck's views? Why are they rejected?
Ans: Lamarck said evolution occurred by use and disuse of organs, and acquired characters are inherited, e.g. giraffes stretched their necks to reach leaves and passed the long neck to offspring. It is rejected because acquired (non-genetic) characters are not inherited.

Q55. 3M Board fav List the factual observations on which Darwin's natural selection is based.
Ans: (i) Natural resources are limited. (ii) Population sizes are stable except seasonal fluctuations. (iii) Members of a population vary, and most variations are inherited. (iv) Populations could grow exponentially but are limited, so there is competition, and only the fitter survive and leave more progeny.

Q56. 2M Differentiate between Darwinian variations and mutations (de Vries).
Ans: Darwinian variations: small, directional, evolution gradual. Mutations: large, sudden, random and directionless; speciation by a single step (saltation).

Q57. 2M Board fav State the Hardy-Weinberg principle. Write its equation.
Ans: Allele frequencies in a population are stable and constant from generation to generation (genetic equilibrium), in the absence of disturbing factors. $p^2 + 2pq + q^2 = 1$, where p and q are frequencies of alleles A and a ($p + q = 1$).

Q58. 2M Board fav Name five factors that affect Hardy-Weinberg equilibrium.
Ans: Gene migration (gene flow), genetic drift, mutation, genetic recombination and natural selection.

Q59. 3M Board fav What is genetic drift? How is it different from gene flow? What is the founder effect?
Ans: **Genetic drift**: change in allele frequency by chance. **Gene flow**: change due to repeated migration of individuals between populations. **Founder effect**: when a small drifted population becomes so different that it forms a new species, the original drifted population acts as founders.

Q60. 3M Board fav In a population of 1000, 90 individuals show a recessive trait (aa). Calculate allele and genotype frequencies.
Ans: $q^2 = 90/1000 = 0.09$, so $q = 0.3$ and $p = 0.7$. AA = p^2 = 0.49 (490), Aa = $2pq$ = 0.42 (420), aa = 0.09 (90).

Q61. 3M Board fav Explain the three types of natural selection with a diagram.
Ans: **Stabilising**: more individuals acquire the mean value (peak narrows). **Directional**: more individuals acquire a value other than the mean (peak shifts to one side). **Disruptive**: more individuals acquire peripheral values at both ends (two peaks). Diagram as Fig. 6.8.

Q62. 2M Why did pouched mammals of Australia survive while South American mammals were overridden?
Ans: Due to continental drift, South America joined North America and its mammals faced competition from North American fauna and were overridden. Australia remained isolated, so its pouched mammals survived because of lack of competition from other mammals.

Q63. 2M Board fav Arrange in order of evolution and give one feature of each: Neanderthal man, Homo habilis, Homo erectus, Australopithecines.
Ans: Australopithecines (stone weapons, ate fruit) → Homo habilis (650-800 cc, first hominid, no meat) → Homo erectus (900 cc, ate meat) → Neanderthal man (1400 cc, buried dead).

Q64. 2M Differentiate between Dryopithecus and Ramapithecus.
Ans: Both existed about 15 mya, were hairy and walked like gorillas and chimpanzees. **Ramapithecus** was more man-like, while **Dryopithecus** was more ape-like.

Q65. 3M List modern-day animals and link each to a corresponding ancient fossil. (NCERT)
Ans: Horse: *Eohippus*; elephant: *Moeritherium*; birds: *Archaeopteryx*; whale: *Pakicetus*; camel: *Protylopus*; frogs/salamanders (amphibians): lobefin fish such as Coelacanth relatives; modern man: Australopithecines; crocodile and birds: dinosaurs (common ancestry).

Q66. 2M Why is the theory of chemical evolution more or less accepted?
Ans: Miller's experiment formed amino acids from CH_4, H_2, NH_3 and water vapour; others obtained sugars, nitrogen bases, pigments and fats; similar compounds were found in meteorites. This limited evidence supports formation of organic molecules from inorganic constituents.

Q67. 2M Give two examples each of homologous and analogous organs in plants.
Ans: Homologous: thorn of *Bougainvillea* and tendril of *Cucurbita* (both modified stems). Analogous: sweet potato (modified root) and potato (modified stem).

Q68. 2M How did continental drift and lack of competition shape mammal evolution?
Ans: When South America joined North America, South American mammals (resembling horse, hippopotamus, bear, rabbit) were overridden by North American fauna; isolated Australia's pouched mammals survived due to lack of competition and underwent adaptive radiation.

Q69. 2M What is meant by "branching descent"?
Ans: All life forms descended from common ancestors, with lineages repeatedly splitting into new forms over time, like branches of a tree; together with natural selection it is a key concept of Darwinian theory.

Q70. 2M Why does evolution appear faster in microbes than in fish or fowl?
Ans: The rate of appearance of new forms is linked to life span. Microbes divide fast and can form millions of individuals within hours, so selected variants become a new population in days; fish or fowl take millions of years as their life spans are in years.

Q71. 2M Board fav What is the founder effect? How is it related to genetic drift?

Ans: When a few individuals of a population migrate and settle a new area, the allele frequencies of the new population may be very different from the original one just by chance. These founders become the ancestors of a new population, which may even become a different species. It is a special case of **genetic drift**, a chance change in allele frequency.

Q72. 2M Give two examples of convergent evolution in plants or animals.

Ans: (i) Wings of butterfly and bird (analogous organs). (ii) Eye of octopus and mammals. (iii) Flippers of penguins and dolphins. (iv) Sweet potato (root) and potato (stem). Any two.

Q73. 2M Why are mammals considered more successful than reptiles in the later history of the earth?

Ans: Mammals were **viviparous** and protected their unborn young inside the mother's body. They were more intelligent in sensing and avoiding danger. When reptiles declined (about 65 mya), mammals took over the land.

Long answer (5 marks)

Q74. 5M Board fav Describe the different evidences of evolution.

Ans: (i) **Paleontological**: fossils in different-aged rock sediments show different life forms; some extinct (dinosaurs); age by radioactive dating. (ii) **Comparative anatomy**: homologous organs (forelimbs of whale, bat, cheetah, human; thorn and tendril) show common ancestry and divergent evolution; analogous organs (wings of butterfly and bird, eye of octopus and mammal) show convergent evolution. (iii) **Embryological** (Haeckel, gill slits), later disapproved by von Baer. (iv) **Biochemical**: similar proteins and genes in diverse organisms. (v) **Artificial selection**: breeds of dogs created in hundreds of years. (vi) **Natural selection in action**: industrial melanism, herbicide, pesticide and antibiotic resistance.

Q75. 5M Board fav Explain the Darwinian theory of evolution. How did de Vries' mutation theory differ from it?

Ans: Darwin (H.M.S. Beagle): (i) populations show heritable variation; (ii) resources are limited and populations stable, so there is competition (struggle for existence); (iii) individuals with useful variations survive and reproduce more: fitness is reproductive fitness; (iv) nature selects them (natural selection); (v) over generations, accumulated variations give new species. Branching descent and natural selection are the two key concepts. **de Vries** (evening primrose): evolution is caused by sudden, large, random, directionless mutations; speciation in a single step (saltation), whereas Darwin's variations are small, directional and evolution gradual.

Q76. 5M Board fav State the Hardy-Weinberg principle. Explain the factors that disturb it. How does this lead to evolution?

Ans: Allele frequencies remain constant over generations; gene pool constant (genetic equilibrium); $p^2 + 2pq + q^2 = 1$. Factors: (i) gene migration/gene flow (alleles added to new and lost from old population); (ii) genetic drift (chance change; founder effect); (iii) mutation (new alleles; advantageous ones selected); (iv) genetic recombination during gametogenesis; (v) natural selection (better adapted leave more progeny). Any change in allele frequency from the expected values indicates evolution; selection can be stabilising, directional or disruptive.

Exam Alert: diagrams and lists to practise

- → Miller ka apparatus (electrodes, CH_4, NH_3, H_2, water vapour, condenser, amino acids), forelimb homology aur 3 types of selection ke graphs.
- → Human evolution table: naam, time, brain size (cc), ek feature. 3 marks pakke.
- → Homology vs analogy ke 2-2 examples (plant + animal) hamesha likho.

Q77. 5M Board fav Describe the origin of life according to the theory of chemical evolution. Explain Miller's experiment with a labelled diagram.

Ans: Early earth (about 4.5 billion years ago) had no free oxygen; water vapour, CH_4, CO_2, NH_3 covered it; UV broke water, ozone formed, rain formed oceans. **Oparin and Haldane**: under high temperature, volcanic storms and a reducing atmosphere, inorganic molecules formed organic molecules (chemical evolution), and the first life arose from these non-living organic molecules (RNA, protein). **Miller (1953)**: closed flask with CH_4, H_2, NH_3 and water vapour at 800°C; electric discharge; condenser; amino acids collected. Others got sugars, nitrogen bases, pigments, fats; meteorites contain similar compounds. First non-cellular life about 3 billion years ago; first cells about 2000 mya. Diagram: Fig. 6.1.

Q78. 5M Board fav Trace the evolution of man from Dryopithecus to Homo sapiens.

Ans: About 15 mya: **Dryopithecus** (ape-like) and **Ramapithecus** (man-like), hairy, walked like gorillas. 3-4 mya: man-like primates in eastern Africa (fossils in Ethiopia, Tanzania), up to 4 feet, walked upright. 2 mya: **Australopithecines**, East African grasslands, stone weapons, ate fruit. **Homo habilis**: first hominid, 650-800 cc, no meat. **Homo erectus** (1.5 mya, Java 1891): 900 cc, ate meat. **Neanderthal** (1,00,000-40,000 years ago): 1400 cc, near east and central Asia, used hides, buried dead. **Homo sapiens**: arose in Africa, spread across continents; modern H. sapiens during the ice age (75,000-10,000 years ago); cave art 18,000 years ago (Bhimbetka, MP); agriculture 10,000 years ago.

Case-based (4 marks)

Q79. 4M Board fav A farmer sprayed the same pesticide every year. In the first year almost all insects died, but after five years the pesticide had little effect. (a) Explain why. (b) Name the process. (c) Is this called evolution by anthropogenic action? Why? (d) Give one similar example from medicine.
Ans: (a) A few insects were already resistant; they survived and reproduced, so the population became mostly resistant. (b) Natural selection. (c) Yes, the selection pressure was created by human activity. (d) Antibiotic-resistant bacteria.

Q80. 4M Board fav In a forest, a bird species has beak sizes ranging from small to large. After a drought, only very hard large seeds and very small soft seeds remained, and birds with medium beaks declined. (a) Name the type of selection. (b) What happens to the distribution curve? (c) Name the other two types. (d) Which type favours the mean value?
Ans: (a) **Disruptive** selection. (b) Two peaks form at both ends; the middle declines. (c) Stabilising and directional. (d) Stabilising selection.

Q81. 4M A student found fossils in successive rock layers: lower layers had only fish, middle layers had amphibians and reptiles, upper layers had mammals. (a) What type of evidence is this? (b) How is the age of fossils found? (c) Which group evolved into the first amphibians? (d) Name the "living fossil" fish caught in 1938.
Ans: (a) Paleontological evidence. (b) Radioactive dating. (c) Lobefins (fish with stout, strong fins). (d) Coelacanth (caught in South Africa).

Q82. 4M Board fav On an isolated island, a few birds of a mainland species arrived after a storm. Over many generations, the island population came to differ greatly from the mainland one in allele frequencies. (a) Name this effect. (b) Is it due to chance or selection? (c) Which principle is disturbed here? (d) Name two other factors that disturb it.
Ans: (a) **Founder effect**. (b) Chance (genetic drift). (c) **Hardy-Weinberg** equilibrium (genetic equilibrium). (d) Gene flow (migration), mutation, recombination or natural selection (any two).

Q83. 4M In Australia, a marsupial "wolf" and a placental wolf look very similar though they are not closely related, while many marsupials (kangaroo, koala, wombat) evolved from one ancestor. (a) Name the process giving many marsupials. (b) Name the process giving the look-alike wolves. (c) Give another example of the first process. (d) Are the organs of the look-alikes homologous or analogous?
Ans: (a) **Adaptive radiation**. (b) **Convergent evolution**. (c) Darwin's finches of the Galapagos. (d) Analogous (similar function and appearance due to similar habitat, not common structure).

Q84. 4M Human blood groups, haemoglobin and cytochrome c show similar sequences among humans, chimpanzees and gorillas, while the forelimbs of whale, bat and cheetah share the same bone pattern. (a) Name the evidence from proteins. (b) Name the evidence from forelimbs. (c) What does similarity in such proteins indicate? (d) Name the kind of evolution shown by the forelimbs.
Ans: (a) **Biochemical (molecular) evidence**. (b) **Comparative anatomy** (homology). (c) Common ancestry: the more similar the sequences, the closer the relationship. (d) **Divergent evolution**.

Q85. 4M Board fav A population of 1000 plants has two alleles, R and r, for flower colour. The frequency of allele r is 0.4. (a) Find the frequency of allele R. (b) Find the expected frequency of heterozygotes. (c) Find the expected number of rr plants. (d) State one condition needed for these values to stay constant.
Ans: (a) $p = 1 - 0.4 = 0.6$. (b) $2pq = 2 \times 0.6 \times 0.4 = 0.48$. (c) $q^2 = 0.16$, so $0.16 \times 1000 = 160$ **plants**. (d) No gene flow, no genetic drift, no mutation, no recombination or no natural selection (any one).

Q86. 4M Board fav Lamarck explained the long neck of the giraffe by the constant stretching of the neck to reach leaves on tall trees, while Darwin explained it by natural selection. (a) What did Lamarck call this idea? (b) Why is it not accepted today? (c) How would Darwin explain the long neck? (d) Who proposed mutation as the cause of evolution, and on which plant?
Ans: (a) Use and disuse of organs (inheritance of acquired characters). (b) Characters acquired during life are not passed on to offspring; only changes in genes are inherited. (c) Giraffes varied in neck length; those with longer necks reached more food, survived and reproduced more, so long-neck variants increased over generations. (d) **Hugo de Vries**, on the **evening primrose**.

Quick fact

Evolution is not a ladder with a goal. Natural selection acts only on variation that is already present, so a trait spreads only if it gives **better fitness in that environment**. Change the environment and the "fittest" can change too, as the peppered moths showed.

Last-minute check

- ✓ Universe 13.8 bn, earth 4.5 bn, life 4 bn years; Miller 1953 = amino acids.
- ✓ Homology = divergent, analogy = convergent; finches = adaptive radiation.
- ✓ $p^2 + 2pq + q^2 = 1$; 5 factors: flow, drift, mutation, recombination, selection.
- ✓ Lamarck = use and disuse; Darwin = natural selection; de Vries = mutation (saltation, evening primrose).
- ✓ Stabilising favours the mean, directional one extreme, disruptive both extremes.
- ✓ Habilis 650–800 cc, erectus 900 cc, Neanderthal 1400 cc; agriculture 10,000 years ago.

Human Health and Disease

Most asked: life cycle of Plasmodium, innate vs acquired immunity, antibody structure, active vs passive immunity, HIV replication, benign vs malignant tumours, drugs and their sources

Health is not just the absence of disease. This chapter covers **common human diseases, immunity, AIDS, cancer** and **drug and alcohol abuse**, the topics boards love for 2, 3 and 5 mark questions.

1. Health and Disease

For a long time health was thought to be a balance of body "**humors**", as asserted by early Greeks like **Hippocrates** and the Indian **Ayurveda** system. People with "black bile" were thought to have a hot personality and fevers. This idea came from pure reflective thought, not experiment.

William Harvey discovered blood circulation by the experimental method, and the thermometer showed normal body temperature in persons with black bile; together these disproved the "good humor" hypothesis. Biology later showed that the **mind influences the immune system** through the neural and endocrine systems, and the immune system maintains our health.

Health is affected by:

1. **Genetic disorders**: deficiencies a child is born with or inherits from parents.
2. **Infections**.
3. **Life style**: food and water, rest and exercise, habits we have or lack.

Key terms

- → **Health**: a state of **complete physical, mental and social well-being**, not merely the absence of disease or physical fitness.
- → **Disease**: when the functioning of one or more organs or systems is adversely affected, with signs and symptoms.
- → **Pathogen**: a disease-causing organism (bacteria, viruses, fungi, protozoans, helminths).

Healthy people are **more efficient at work**, which raises productivity and brings economic prosperity. Health also increases longevity and reduces **infant and maternal mortality**.

Maintaining health: balanced diet, personal hygiene, regular exercise and **yoga**; awareness about diseases, **vaccination**, proper disposal of wastes, control of vectors and hygiene in food and water.

Infectious diseases	Non-infectious diseases
easily transmitted from one person to another	not transmitted from person to person
caused by pathogens	caused by genetic, lifestyle or other factors
e.g. AIDS (fatal), typhoid, common cold, malaria	e.g. **cancer** (major cause of death), diabetes

2. Common Diseases in Humans

Most parasites are pathogens, as they harm the host by living in or on it. Pathogens must adapt to life in the host: e.g. pathogens entering the gut must survive the **low pH of the stomach** and resist digestive enzymes.

A. Typhoid (bacterial) PYQ

- → **Pathogen**: *Salmonella typhi*, a bacterium.
- → **Spread**: enters the **small intestine** through contaminated food and water, then migrates to other organs through blood.
- → **Symptoms**: sustained high fever (39 to 40 °C), weakness, stomach pain, constipation, headache, loss of appetite; **intestinal perforation** and death in severe cases.
- → **Test**: confirmed by the Widal test.

Quick fact: Typhoid Mary

Mary Mallon, a cook, was a typhoid carrier: she had no illness herself but spread typhoid for several years through the food she prepared.

B. Pneumonia (bacterial)

- → **Pathogens**: *Streptococcus pneumoniae* and *Haemophilus influenzae*.
- → **Organ**: infects the **alveoli** of the lungs, which fill with fluid, causing severe problems in respiration.
- → **Symptoms**: fever, chills, cough, headache; in severe cases lips and finger nails turn grey to bluish.
- → **Spread**: inhaling droplets/aerosols from an infected person, or sharing glasses and utensils.

Dysentery, plague and diphtheria are some other bacterial diseases of man.

C. Common cold (viral)

- → **Pathogen**: **Rhino viruses**; one of the most infectious human ailments.
- → **Organ**: infect the nose and respiratory passage, but not the lungs.
- → **Symptoms**: nasal congestion and discharge, sore throat, hoarseness, cough, headache, tiredness; usually last **3–7 days**.
- → **Spread**: droplets from coughs or sneezes, inhaled directly or through contaminated objects (pens, books, cups, doorknobs, keyboard, mouse).

D. Malaria (protozoan) Board fav

Caused by *Plasmodium*, a tiny protozoan. Different species cause different types: *P. vivax*, *P. malaria* and *P. falciparum*. Malignant malaria caused by ***P. falciparum*** is the most serious and can be fatal.

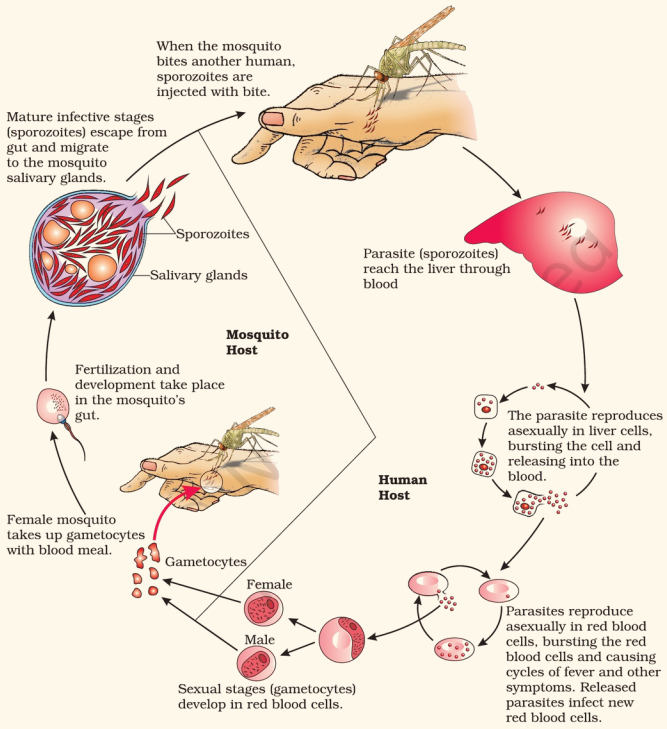


Fig. 7.1 Stages in the life cycle of Plasmodium

1. *Plasmodium* enters the human body as **sporozoites** (infectious form) through the bite of an infected female *Anopheles* mosquito.
2. The parasites first multiply in the **liver cells** (asexual reproduction), then attack the **red blood cells** and rupture them.
3. Rupture of RBCs releases a toxic substance, haemozoin, which causes the **chill and high fever recurring every 3–4 days.**
4. Some parasites form gametocytes in RBCs. When a female *Anopheles* bites an infected person, these enter the mosquito; fertilisation and development take place in the **mosquito gut.**
5. Sporozoites formed in the mosquito are stored in its **salivary glands**, ready to infect the next human.
6. Control therefore targets the vector: mosquito nets, repellents, removal of stagnant water and larvivorous fish such as *Gambusia*.

Yaad karo: malaria ke 2 ghar

Human = intermediate host (asexual stages: liver, phir RBC). **Female Anopheles** = definitive host aur **vector** bhi (sexual stages, gut mein fertilisation). **Haemozoin** = "hai, mujhe thand lag rahi" wala toxin, har 3–4 din fever!

Paheli-Bujho

Bujho: Paheli, malaria wala Plasmodium ek hi jagah kyun nahi rehta?

Paheli: Kyunki life cycle ke liye do host chahiye: insaan aur female Anopheles.

Bujho: Matlab mosquito iska Uber hai?

Paheli: Haan, aur sporozoites salivary glands mein hi wait karte hain!

E. Amoebiasis (protozoan)

- → **Pathogen**: *Entamoeba histolytica*, in the **large intestine**; causes amoebic dysentery.
- → **Symptoms**: constipation, abdominal pain and cramps, stools with excess **mucus and blood clots.**
- → **Spread**: houseflies act as mechanical carriers, carrying the parasite from faeces to food; faecal-contaminated food and water are the main source.

F. Ascariasis (helminth)

- → **Pathogen**: *Ascaris*, the common **round worm**, an intestinal parasite.
- → **Symptoms**: internal bleeding, muscular pain, fever, anaemia and blockage of the intestinal passage.
- → **Spread**: eggs pass out with faeces and contaminate soil, water and plants; a healthy person is infected through contaminated water, vegetables and fruits.

G. Elephantiasis or filariasis (helminth)

- → **Pathogen**: *Wuchereria* (*W. bancrofti* and *W. malayi*), the filarial worms.
- → **Symptoms**: slowly developing **chronic inflammation** of the organs where they live for many years, usually the lymphatic vessels of the lower limbs; genital organs are often affected, causing gross deformities.
- → **Spread**: bite of **female mosquito vectors.**

H. Ringworm (fungal)

- → **Pathogens**: fungi of the genera *Microsporum*, *Trichophyton* and *Epidermophyton*; one of the most common infectious diseases.
- → **Symptoms**: **dry, scaly lesions** on skin, nails and scalp, with intense itching. Heat and moisture help growth, so they thrive in skin folds (groin, between toes).
- → **Spread**: from soil, or by using towels, clothes or even the comb of infected persons.

Exam Alert: disease table (2–3 marks pakke)

Disease	Pathogen	Group	Main organ
Typhoid	*Salmonella typhi*	bacterium	small intestine
Pneumonia	*S. pneumoniae, H. influenzae*	bacteria	alveoli of lungs
Common cold	Rhino viruses	virus	nose, respiratory passage
Malaria	*Plasmodium*	protozoan	liver, RBCs
Amoebiasis	*Entamoeba histolytica*	protozoan	large intestine
Ascariasis	*Ascaris*	helminth	intestine
Filariasis	*Wuchereria*	helminth	lymphatic vessels
Ringworm	*Microsporum, Trichophyton, Epidermophyton*	fungi	skin, nails, scalp

Prevention and control of infectious diseases

→ **Personal hygiene**: keeping the body clean; clean drinking water, food, vegetables and fruits.

→ **Public hygiene**: proper disposal of waste and excreta; periodic cleaning and disinfection of water reservoirs, pools, cesspools and tanks; standard hygiene in public catering. Essential for **food and water-borne** diseases: typhoid, amoebiasis, ascariasis.

→ **Air-borne diseases** (pneumonia, common cold): also avoid close contact with infected persons or their belongings.

→ **Vector-borne diseases** (malaria, filariasis): control or eliminate the vectors and their breeding places.

Vector control measures:

✓ Avoid stagnation of water near homes; clean household coolers regularly.

✓ Use mosquito nets; put wire mesh on doors and windows.

✓ Introduce fishes like *Gambusia* in ponds: they feed on mosquito larvae.

✓ Spray insecticides in ditches, drainage areas and swamps.

These steps matter even more now because of widespread **dengue and chikungunya**, spread by *Aedes* mosquitoes.

Role of biology: vaccines and immunisation programmes have completely eradicated smallpox; polio, diphtheria, pneumonia and tetanus are largely controlled. Biotechnology is making newer, safer vaccines, and antibiotics and other drugs treat infectious diseases effectively.

3. Immunity

We are exposed to many infectious agents every day, but only a few exposures cause disease, because the body defends itself. The overall ability of the host to fight disease-causing organisms, conferred by the immune system, is called immunity. It is of two types: **innate** and **acquired**.

A. Innate immunity PYQ

Non-specific defence present at the time of birth. It works by providing barriers to the entry of foreign agents. It has **four types of barriers**:

Barrier	What it is
1. Physical	**skin** (main barrier); **mucus** coating of the epithelium lining the respiratory, gastrointestinal and urogenital tracts traps microbes
2. Physiological	**acid** in the stomach, **saliva** in the mouth, **tears** from the eyes prevent microbial growth
3. Cellular	leukocytes like **PMNL-neutrophils**, **monocytes** and **natural killer** cells (lymphocytes) in blood, and **macrophages** in tissues phagocytose and destroy microbes
4. Cytokine	virus-infected cells secrete interferons, which protect non-infected cells from further viral infection

Yaad karo: 4 barriers

Papa Pizza Cheese Chahiye = Physical, Physiological, Cellular, Cytokine. Interferon = virus wala "bhai, aage mat jaana" signal!

B. Acquired immunity Board fav

Pathogen-specific and characterised by memory. The first encounter with a pathogen gives a **primary response** of low intensity. A later encounter with the same pathogen gives a highly intensified **secondary or anamnestic response**, because the body remembers the first encounter.

Primary immune response	Secondary (anamnestic) response
on the **first** encounter with a pathogen	on a **later** encounter with the same pathogen
low intensity, slower	**highly intensified**, rapid
memory cells are formed	memory cells recognise the pathogen at once

Both responses are carried out by two special types of lymphocytes in our blood: **B-lymphocytes** and **T-lymphocytes**.

B-lymphocytes	T-lymphocytes
produce an army of proteins called **antibodies** into the blood	do **not** secrete antibodies, but **help B cells** produce them
responsible for **humoral** (antibody-mediated) immune response	responsible for **cell-mediated immunity (CMI)**
act through antibodies in body fluids	responsible for graft rejection

C. Structure of an antibody

Each antibody molecule has four peptide chains: two small **light chains** and two longer **heavy chains**, so an antibody is represented as H_2L_2. Different types are produced in our body: **IgA, IgM, IgE, IgG**, etc.

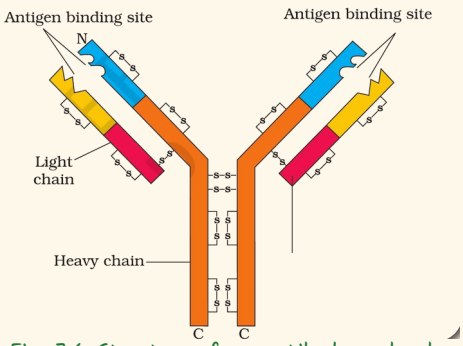
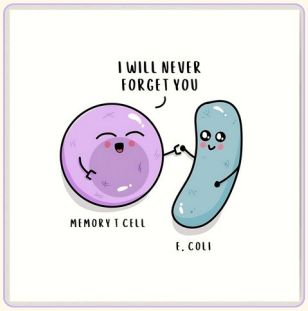


Fig. 7.4 Structure of an antibody molecule

Because antibodies are found in the blood, this response is called the humoral immune response. The second type, the **cell-mediated immune response (CMI)**, is mediated by T-lymphocytes.

Graft rejection: when organs like heart, eye, liver or kidney fail, transplantation may be the only remedy. Grafts from just any source (an animal, another primate or any human) are rejected sooner or later, so **tissue matching and blood group matching** are essential, and the patient takes immunosuppressants lifelong. The body can tell **"self" from "non-self"**, and CMI is responsible for graft rejection.



Memory cell ka promise: ek baar mile ho, ab kabhi nahi bhoolenge.

Asli baat: Memory B and T cells give the fast, strong secondary response.

D. Active and passive immunity PYQ

Active immunity	Passive immunity
host produces its **own antibodies** on exposure to antigens (living or dead microbes or other proteins)	**ready-made antibodies** are given directly
slow; takes time to give its full effective response	**immediate** protection
long-lasting (memory develops)	short-lived (no memory)
e.g. natural infection; deliberate injection of microbes during **immunisation**	e.g. **colostrum** (IgA) to the infant; antibodies from mother to foetus through the **placenta**; antitoxin, anti-venom

Colostrum: the yellowish fluid secreted by the mother in the initial days of lactation has abundant IgA antibodies to protect the infant. This is why mother's milk is essential for the newborn.

E. Vaccination and immunisation

The principle of vaccination is based on the "memory" of the immune system. A preparation of antigenic proteins of a pathogen, or inactivated/weakened pathogen (**vaccine**), is introduced into the body. Antibodies produced against these antigens neutralise the pathogen during actual infection. Vaccines also generate **memory B and T cells** that recognise the pathogen quickly and overwhelm it with a massive production of antibodies.

- → **Passive immunisation**: for deadly microbes needing a quick response, as in **tetanus**, preformed antibodies or antitoxin (a preparation of antibodies to the toxin) are injected. The injection for **snakebite** also contains preformed antibodies against the venom.
- → **Recombinant vaccines**: recombinant DNA technology produces antigenic polypeptides of pathogens in bacteria or yeast, allowing large-scale production, e.g. hepatitis B vaccine from yeast.

Key terms

- → **Antigen**: a foreign substance (microbe or protein) that triggers antibody production.
- → **Vaccine**: antigenic proteins or inactivated/weakened pathogen that build memory without causing disease.
- → **Antitoxin**: a preparation of antibodies against a toxin (e.g. for tetanus).
- → **Anamnestic**: the strong secondary response, based on memory.

Exam Alert

- → Vaccine = **active** immunity (body khud antibody banati hai). Antitoxin / anti-venom = **passive**. Ye ulta likha toh marks gaye.
- → Antibody = H_2L_2; diagram mein antigen binding site, heavy chain, light chain label karo.
- → Colostrum mein **IgA**; allergy mein **IgE**.

F. Allergies PYQ

The exaggerated response of the immune system to certain antigens present in the environment is called **allergy**. The substances causing it are **allergens**, and the antibodies produced are of the IgE type.

- → **Allergens**: mites in dust, pollen, animal dander, etc.
- → **Symptoms**: sneezing, watery eyes, running nose, difficulty in breathing.
- → **Cause**: release of chemicals like **histamine and serotonin** from mast cells.
- → **Diagnosis**: the patient is exposed to or injected with very small doses of possible allergens and the reactions are studied.
- → **Treatment**: **anti-histamine, adrenalin and steroids** quickly reduce symptoms.

Modern life style has lowered immunity and raised sensitivity to allergens: more children in metro cities suffer from

allergies and asthma, possibly because of the protected environment provided early in life.

G. Auto-immunity

Memory-based acquired immunity evolved in higher vertebrates based on the ability to tell foreign organisms from **self-cells**. Sometimes, due to genetic and other unknown reasons, the body attacks its own cells; this damage is called an **auto-immune disease**, e.g. **rheumatoid arthritis**.

H. Immune system in the body

The human immune system consists of **lymphoid organs, tissues, cells** and soluble molecules like antibodies. It recognises foreign antigens, responds to them and remembers them. It also plays a role in allergic reactions, auto-immune diseases and organ transplantation.

Lymphoid organs are organs where origin and/or maturation and proliferation of lymphocytes occur.

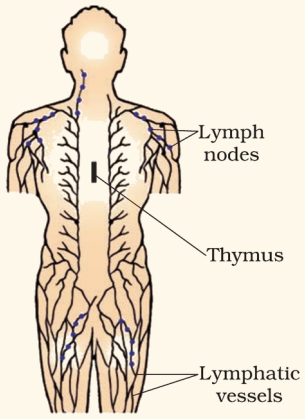


Fig. 7.5 Lymph nodes, lymphatic vessels and thymus in the human body

Primary lymphoid organs	*Secondary lymphoid organs*
bone marrow and **thymus**	**spleen, lymph nodes, tonsils, Peyer's patches** of small intestine and **appendix**
immature lymphocytes differentiate into **antigen-sensitive** lymphocytes	sites for **interaction of lymphocytes with antigen**; lymphocytes then proliferate into effector cells

→ **Bone marrow**: the main lymphoid organ, where all blood cells including lymphocytes are produced.

→ **Thymus**: lobed organ near the heart, beneath the breastbone; quite large at birth but **keeps reducing with age** and becomes very small by puberty. Bone marrow and thymus provide micro-environments for development and maturation of **T-lymphocytes**.

→ **Spleen**: large bean-shaped organ with lymphocytes and phagocytes; acts as a **filter of blood** by trapping blood-borne microbes; has a large reservoir of **erythrocytes**.

→ **Lymph nodes**: small solid structures along the lymphatic system; **trap microbes or antigens** in lymph and tissue fluid, which activate the lymphocytes there.

→ **MALT** (mucosa-associated lymphoid tissue): lymphoid tissue in the lining of the respiratory, digestive and urogenital tracts; about 50 per cent of the lymphoid tissue in the human body.

4. AIDS Board fav

AIDS = Acquired Immuno Deficiency Syndrome: a deficiency of the immune system acquired during an individual's lifetime (so not a congenital disease); "syndrome" means a group of symptoms. AIDS was first reported in **1981** and has killed more than **25 million** persons worldwide.

It is caused by **HIV (Human Immunodeficiency Virus)**, a member of the retroviruses, which have an envelope enclosing an **RNA genome**.

Transmission of HIV generally occurs by:

1. sexual contact with an infected person;
2. transfusion of contaminated blood and blood products;
3. sharing infected needles, as in intravenous drug abusers;
4. from an infected mother to her child through the **placenta**.

High-risk groups: persons with multiple sexual partners, drug addicts who take drugs intravenously, persons needing repeated blood transfusions and children born to HIV-infected mothers. HIV/AIDS is not spread by mere touch or physical contact; it spreads only through body fluids, so infected persons must not be isolated from family and society.

There is always a **time-lag** between infection and the appearance of AIDS symptoms: a few months to many years, usually 5–10 years.

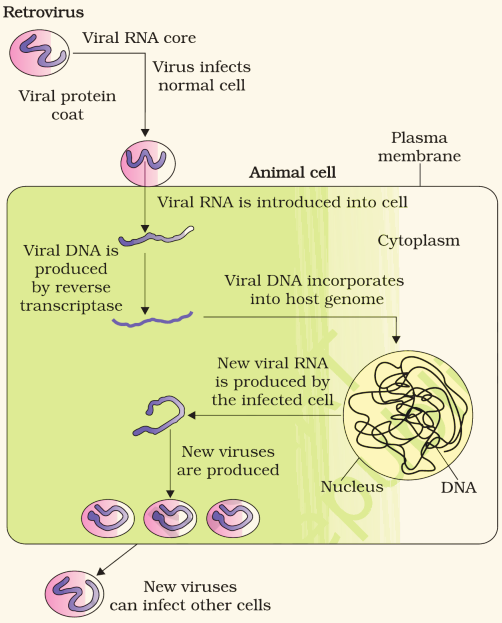
HIV ke 5–10 saal: andar helper T cells khatam, bahar sab "fine".

Asli baat: Time-lag between HIV infection and AIDS symptoms is usually 5–10 years.

Mechanism: how HIV damages immunity

1. HIV enters **macrophages**, where its RNA genome is copied into **viral DNA** by the enzyme reverse transcriptase.
2. Viral DNA is incorporated into the host cell's DNA and directs the cell to make virus particles. Macrophages keep producing virus and act like an **"HIV factory"**.
3. HIV also enters helper T-lymphocytes (T_H), replicates, and the progeny viruses released in blood attack other helper T cells.
4. This repeats, causing a **progressive decrease in helper T cells**. The person suffers bouts of fever, diarrhoea and weight loss.

5. The person becomes so immunodeficient that infections normally overcome (bacteria, especially *Mycobacterium*, viruses, fungi and parasites like *Toxoplasma*) become serious.



NOTE: Infected cell can survive while viruses are being replicated and released

Fig. 7.6 Replication of retrovirus

HIV enters macrophage → reverse transcriptase: RNA to DNA → viral DNA joins host DNA → new viruses → attack helper T cells → T_H count falls → immunodeficiency

→ **Diagnosis**: ELISA (enzyme linked immuno-sorbent assay) is widely used.

→ **Treatment**: **anti-retroviral drugs** are only partially effective; they prolong life but cannot prevent death.

Key terms

- → **Retrovirus**: virus with an RNA genome that is copied into DNA inside the host cell (e.g. HIV).
- → **Reverse transcriptase**: enzyme that makes DNA from an RNA template.
- → **Opportunistic infections**: infections by *Mycobacterium*, *Toxoplasma*, fungi and viruses that a healthy immune system normally overcomes.

Prevention of AIDS

AIDS has no cure, so prevention is the best option. HIV spreads mostly through conscious behaviour patterns, not inadvertently like pneumonia or typhoid; "**don't die of ignorance**". In India the **National AIDS Control Organisation (NACO)** and NGOs educate people, and **WHO** runs programmes to prevent HIV spread:

- ✓ making blood from blood banks **safe from HIV**;
- ✓ using only **disposable needles and syringes** in hospitals and clinics;
- ✓ free distribution of condoms, controlling drug abuse, advocating safe sex;
- ✓ promoting regular check-ups for HIV in susceptible populations.

HIV infection should not be hidden, as it may then spread further. Infected people need help and sympathy, not rejection; society and the medical fraternity must act together.

Paheli-Bujho

Bujho: Paheli, HIV se haath milaya toh mujhe bhi ho jayega?
Paheli: Bilkul nahi! HIV sirf body fluids se failta hai, touch se nahi.
Bujho: Phir log dur kyun bhaagte hain?
Paheli: Ignorance ki wajah se. Patient ko sympathy chahiye, isolation nahi.

5. Cancer

Cancer is one of the most dreaded diseases and a major cause of death worldwide; **more than a million Indians** suffer from it. In our body cell growth and differentiation are highly regulated; in cancer cells these regulatory mechanisms break down (oncogenic transformation).

Contact inhibition: in normal cells, contact with other cells inhibits uncontrolled growth. Cancer cells have lost contact inhibition, so they keep dividing into masses of cells called **tumours**. PYQ

Benign tumour	Malignant tumour
remains confined to its **original location**	cells (neoplastic or tumour cells) grow very rapidly, **invading and damaging** surrounding normal tissues
does not spread to other parts	cells sloughed off reach distant sites through blood and start new tumours (**metastasis**)
causes little damage	starves normal cells by competing for vital nutrients

Metastasis: the property by which cells from a malignant tumour reach distant sites through blood and start a new tumour wherever they lodge. It is the most feared property of malignant tumours. Board fav

Causes of cancer (carcinogens)

Physical, chemical or biological agents that transform normal cells into cancerous neoplastic cells are called carcinogens.

Type	Carcinogen and action
Physical	**ionising radiations** (X-rays, gamma rays) and **non-ionising** radiations (UV) cause DNA damage, leading to neoplastic transformation
Chemical	chemical carcinogens in **tobacco smoke**: a major cause of **lung cancer**
Biological	**oncogenic viruses** have genes called **viral oncogenes**; **cellular oncogenes (c-onc)** or **proto-oncogenes** in normal cells can cause oncogenic transformation when activated under certain conditions

Cancer detection and diagnosis

Early detection allows successful treatment in many cases.

- → **Biopsy and histopathology**: a piece of suspected tissue is cut into thin sections, stained and examined under a microscope by a pathologist.
- → **Blood and bone marrow tests**: for increased cell counts in **leukaemias**.
- → **Radiography** (X-rays), **CT** (computed tomography: X-rays give a 3-D image of internals) and **MRI** (magnetic resonance imaging: strong magnetic fields and non-ionising radiations detect pathological and physiological changes in living tissue).
- → **Antibodies** against cancer-specific antigens detect certain cancers.
- → **Molecular biology** techniques detect genes for inherited susceptibility to certain cancers; such persons may be advised to avoid particular carcinogens (e.g. tobacco smoke for lung cancer).

Quick fact: CT vs MRI

CT (computed tomography)	MRI (magnetic resonance imaging)
uses **X-rays** (ionising)	uses **strong magnetic fields** and non-ionising radiations
generates a 3-D image of internal organs	accurately detects pathological and physiological changes in living tissue

Treatment of cancer

- → **Surgery**: removal of the tumour.
- → **Radiotherapy**: tumour cells are irradiated lethally, taking care of surrounding normal tissue.
- → **Chemotherapy**: drugs kill cancer cells; some are specific for particular tumours. Most have **side effects** like hair loss and anaemia.
- → **Immunotherapy**: tumour cells avoid detection by the immune system, so patients get biological response modifiers such as **α-interferon**, which activate the immune system to destroy the tumour.

Most cancers are treated by a **combination of surgery, radiotherapy and chemotherapy**.

Yaad karo: cancer treatment = SRCI

Sabko Roz Chai Ilachi wali = Surgery, Radiotherapy, Chemotherapy, Immunotherapy (α-interferon). CT = X-ray, MRI = magnet: ye pair confuse mat karna!

6. Drugs and Alcohol Abuse

Use of drugs and alcohol is rising, especially among youth. The commonly abused drugs are opioids, cannabinoids and coca alkaloids; most come from flowering plants, some from fungi.

A. Opioids

Drugs that bind to specific **opioid receptors** in our central nervous system and gastrointestinal tract. **Heroin** (smack) is chemically diacetylmorphine, a white, odourless, bitter crystalline compound obtained by **acetylation of morphine**, which is extracted from the latex of the poppy plant *Papaver somniferum*. Taken by snorting and injection, heroin is a **depressant** that slows down body functions.

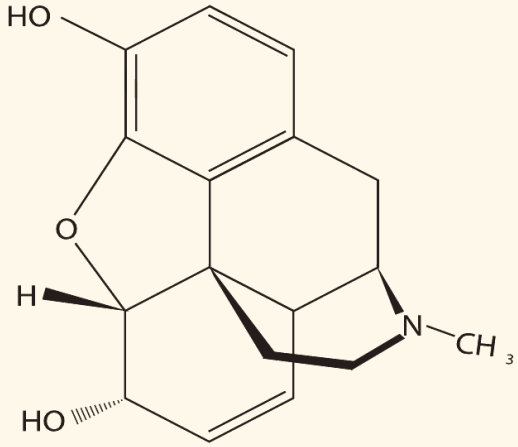


Fig. 7.7 Chemical structure of morphine

B. Cannabinoids

Chemicals that interact with **cannabinoid receptors**, present principally in the brain. Natural cannabinoids come from the **inflorescences** of *Cannabis sativa*. Flower tops, leaves and resin are used to make **marijuana, hashish, charas and ganja**. Taken by inhalation and oral ingestion, they affect the cardiovascular system. Cannabinoids are also abused by some sportspersons.

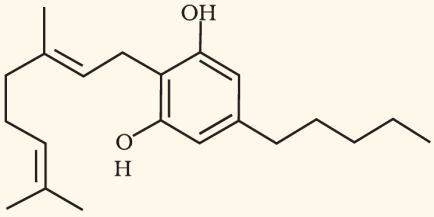


Fig. 7.9 Skeletal structure of a cannabinoid molecule

C. Coca alkaloid (cocaine)

Obtained from the coca plant *Erythroxylum coca*, native to **South America**. It interferes with the transport of the neurotransmitter dopamine. Cocaine (coke or crack) is usually snorted; it is a potent **CNS stimulant**, producing euphoria and increased energy. Excessive doses cause **hallucinations**.

Other plants with hallucinogenic properties: *Atropa belladonna* and *Datura*.

Fig. 7.11 Flowering branch of Datura

Exam Alert: drug table

Drug	Source	Acts on / effect
Heroin (smack)	acetylation of morphine from latex of *Papaver somniferum*	opioid receptors (CNS, GI tract); depressant
Marijuana, hashish, charas, ganja	inflorescences of *Cannabis sativa*	cannabinoid receptors in brain; cardiovascular system
Cocaine (coke, crack)	*Erythroxylum coca*	blocks dopamine transport; CNS stimulant, euphoria, hallucinations

Medicines that are abused: barbiturates, amphetamines, benzodiazepines and similar drugs, normally used to help patients cope with depression and insomnia. **Morphine** is a very effective sedative and painkiller, useful after surgery. Taking such drugs for a purpose other than medicinal use, or in amounts/frequency that impair physical, physiological or psychological functions, is **drug abuse**.

D. Tobacco

Used for more than **400 years**: smoked, chewed or used as snuff. It contains nicotine, an alkaloid, which stimulates the **adrenal gland** to release **adrenaline and nor-adrenaline**, raising blood pressure and heart rate. Smoking also paves the way to hard drugs.

- → Smoking: higher incidence of cancers of **lung, urinary bladder and throat**, bronchitis, emphysema, coronary heart disease, gastric ulcer. Tobacco chewing: cancer of the **oral cavity**.
- → Smoking raises carbon monoxide (CO) in blood and reduces haem-bound oxygen, causing **oxygen deficiency** in the body.

E. Adolescence and drug/alcohol abuse

Adolescence means both "**a period**" and "**a process**" during which a child matures in attitudes and beliefs for effective participation in society. The period of 12–18 years is the adolescence period, a **bridge between childhood and adulthood**, with many biological and behavioural changes; it is a very vulnerable phase.

Causes that push youngsters towards drugs and alcohol:

- → **Curiosity**, need for adventure and excitement, and experimentation; later, use to **escape facing problems**.
- → **Stress** from pressure to excel in academics or examinations.
- → The perception that it is "cool" or progressive, promoted by television, movies, newspapers and the internet.
- → Unstable or unsupportive **family structures** and **peer pressure**.

F. Addiction and dependence PYQ

Addiction	Dependence
a **psychological attachment** to certain effects (euphoria, a temporary feeling of well-being) of drugs and alcohol	the tendency of the body to show a characteristic, unpleasant **withdrawal syndrome** if the regular dose is abruptly stopped
drives people to take them even when not needed or when self-destructive	anxiety, shakiness, nausea and sweating, relieved when use resumes; may be severe, needing medical supervision

With repeated use, the tolerance level of receptors increases, so they respond only to higher doses, leading to greater intake and addiction. Even **a single use can be a fore-runner to addiction**. Dependence makes the person ignore social norms to get funds, causing social adjustment problems.

G. Effects of drug/alcohol abuse

- → **Immediate**: reckless behaviour, vandalism and violence. Excessive doses may cause **coma and death** from respiratory failure, heart failure or cerebral haemorrhage; drug combinations or drugs with alcohol often cause overdosing and death.
- → **Warning signs in youth**: drop in academic performance, unexplained absence from school or college, lack of interest in personal hygiene, withdrawal, isolation, depression, fatigue, aggressive and rebellious behaviour, deteriorating relationships, loss of interest in hobbies, changed sleeping and eating habits, fluctuating weight and appetite.
- → **Far-reaching**: the abuser may turn to **stealing**; causes mental and financial distress to the whole family and friends.
- → **Infections**: intravenous users are much more likely to get AIDS and hepatitis B by sharing needles and syringes; both are chronic and ultimately fatal.
- → **Long-term**: alcohol in adolescence may lead to heavy drinking in adulthood; chronic use damages the **nervous system and liver (cirrhosis)**; use in pregnancy harms the foetus.

Misuse in sports: some sportspersons misuse narcotic analgesics, **anabolic steroids**, diuretics and certain hormones to increase muscle strength and bulk, promote aggressiveness and raise performance.

In females	In males
masculinisation, increased aggressiveness, mood swings, depression	acne, increased aggressiveness, mood swings, depression
abnormal menstrual cycles, excessive hair growth on face and body	reduced size of testicles, decreased sperm production
enlargement of clitoris, deepening of voice	kidney and liver dysfunction, breast enlargement, premature baldness, enlarged prostate

In adolescents of both sexes, steroids cause severe facial and body acne and premature closure of the growth centres of long bones, leading to stunted growth. Effects may be permanent with prolonged use.

H. Prevention and control

"**Prevention is better than cure.**" Such habits start mostly in adolescence, so situations that may push an adolescent towards them should be spotted early; parents and teachers have a special responsibility. Parenting with high nurturance and consistent discipline lowers the risk of substance abuse.

1. **Avoid undue peer pressure**: respect each child's choice and personality; do not push a child beyond his/her threshold in studies, sports or other activities.
2. **Education and counselling**: teach the child to face problems and stress and accept failures as part of life; channelise energy into sports, reading, music, yoga.
3. **Seeking help from parents and peers**: get help immediately, even from close and trusted friends, to vent anxiety and guilt.
4. **Looking for danger signs**: alert parents, teachers and friends should identify the warning signs and inform parents or teachers.
5. **Seeking professional and medical help**: psychologists, psychiatrists, and de-addiction and rehabilitation programmes help the person lead a normal, healthy life.

Paheli-Bujho

Bujho: Paheli, ek baar try karne mein kya jaata hai? Sab "cool" bolenge.

Paheli: Ek baar hi addiction ki shuruaat ban sakti hai. Receptors ki tolerance badhti jaati hai.

Bujho: Matlab dose badhti jaayegi aur chhodna mushkil?

Paheli: Haan! Asli cool wo hai jo seedha "no" bol de.

1 minute recap

1. Health = complete physical, mental and social well-being; Harvey disproved the humor theory.
2. Typhoid (*Salmonella typhi*, Widal), pneumonia (alveoli), common cold (Rhino virus, not lungs), malaria (*Plasmodium*, haemozoin, female *Anopheles*), amoebiasis, ascariasis, filariasis (*Wuchereria*), ringworm (3 fungi).
3. Innate = 4 barriers (physical, physiological, cellular, cytokine/interferon); acquired = specific, memory, B (humoral) and T (CMI) cells.
4. Antibody H_2L_2; IgA in colostrum, IgE in allergy; vaccine = active, antitoxin = passive.
5. Primary lymphoid: bone marrow, thymus; secondary: spleen, lymph nodes, tonsils, Peyer's patches, appendix; MALT about 50%.
6. HIV: retrovirus; reverse transcriptase; macrophage = HIV factory; helper T cells fall; ELISA; NACO.
7. Cancer: loss of contact inhibition, metastasis; carcinogens; biopsy, CT, MRI; surgery, radio, chemo, α-interferon.
8. Opioids (heroin), cannabinoids, cocaine (dopamine); nicotine raises BP; addiction vs dependence; 5 preventive measures.

MOST IMPORTANT QUESTIONS

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM Board fav Typhoid fever is confirmed by:

(a) ELISA (b) Widal test
(c) Biopsy (d) MRI

Ans: (b) Widal test

Q2. IM Which part of the body does the common cold virus NOT infect?

(a) Nose (b) Respiratory passage
(c) Lungs (d) Throat

Ans: (c) Lungs

Q3. IM Board fav The toxic substance responsible for the chill and high fever in malaria is:

(a) Haemozoin (b) Histamine
(c) Serotonin (d) Interferon

Ans: (a) Haemozoin

Q4. IM The infectious stage of *Plasmodium* that enters the human body is:

(a) Gametocyte (b) Merozoite
(c) Trophozoite (d) Sporozoite

Ans: (d) Sporozoite

Q5. IM Where does fertilisation of *Plasmodium* gametes take place?
(a) Human liver
(b) Gut of female *Anopheles*
(c) Human RBCs
(d) Salivary glands of mosquito
Ans: (b) Gut of female *Anopheles*

Q6. IM Which of these act as mechanical carriers of *Entamoeba histolytica*?
(a) Mosquitoes (b) Rats
(c) Houseflies (d) Cockroaches only
Ans: (c) Houseflies

Q7. IM Elephantiasis is caused by:
(a) *Wuchereria* (b) *Ascaris*
(c) *Plasmodium* (d) *Microsporum*
Ans: (a) *Wuchereria*

Q8. IM Which fish is introduced in ponds to control mosquito larvae?
(a) Rohu (b) *Gambusia*
(c) Catla (d) *Labeo*
Ans: (b) *Gambusia*

Q9. IM Board fav Interferons belong to which barrier of innate immunity?
(a) Physical (b) Physiological
(c) Cellular (d) Cytokine
Ans: (d) Cytokine barrier

Q10. IM Board fav An antibody molecule is represented as:
(a) H_4 (b) HL (c) H_2L_2 (d) H_4L_2
Ans: (c) H_2L_2

Q11. IM Board fav Colostrum is rich in:
(a) IgA (b) IgE (c) IgG (d) IgM
Ans: (a) IgA

Q12. IM Antibodies produced in an allergic reaction are of the type:
(a) IgA (b) IgE (c) IgG (d) IgM
Ans: (b) IgE

Q13. IM Graft rejection is mainly due to:
(a) Humoral immunity (b) Innate immunity
(c) Cell-mediated immunity (d) Passive immunity
Ans: (c) Cell-mediated immunity (T cells)

Q14. IM Which of the following is a primary lymphoid organ?
(a) Spleen (b) Lymph node
(c) Tonsil (d) Thymus
Ans: (d) Thymus

Q15. IM MALT constitutes about what part of the lymphoid tissue in the human body?
(a) 25% (b) 50% (c) 75% (d) 10%
Ans: (b) 50 per cent

Q16. IM Rheumatoid arthritis is an example of:
(a) Auto-immune disease (b) Allergy
(c) Viral disease (d) Genetic disorder
Ans: (a) Auto-immune disease

Q17. IM Board fav The enzyme that makes viral DNA from the RNA genome of HIV is:
(a) DNA polymerase (b) RNA polymerase
(c) Reverse transcriptase (d) Ligase
Ans: (c) Reverse transcriptase

Q18. IM HIV mainly destroys:
(a) B-lymphocytes (b) Helper T-lymphocytes
(c) RBCs (d) Platelets
Ans: (b) Helper T-lymphocytes

Q19. IM Board fav The property of normal cells lost by cancer cells is:
(a) Contact inhibition (b) Metastasis
(c) Cell division (d) Differentiation
Ans: (a) Contact inhibition

Q20. IM α-interferon is used in cancer treatment as a:
(a) Chemotherapeutic drug
(b) Carcinogen
(c) Radiotherapy agent
(d) Biological response modifier
Ans: (d) Biological response modifier

Q21. IM Board fav Heroin is chemically:
(a) Morphine (b) Diacetylmorphine
(c) Cannabinoid (d) Nicotine
Ans: (b) Diacetylmorphine

Q22. IM Board fav Cocaine interferes with the transport of:
(a) Adrenaline (b) Serotonin
(c) Dopamine (d) Acetylcholine
Ans: (c) Dopamine

Q23. IM Cannabinoids mainly affect the:
(a) Cardiovascular system (b) Excretory system
(c) Digestive system (d) Skeletal system
Ans: (a) Cardiovascular system

Q24. IM Which of the following diseases is NOT transmitted by mosquitoes?
(a) Malaria (b) Filariasis
(c) Amoebiasis (d) Dengue
Ans: (c) Amoebiasis

Q25. IM Dengue and chikungunya are transmitted by:
(a) *Anopheles* (b) *Aedes*
(c) *Culex* only (d) Houseflies
Ans: (b) *Aedes* mosquitoes

Q26. 1M Which of the following is a cellular barrier of innate immunity?
(a) Natural killer cells (b) Tears
(c) Mucus (d) Interferons
Ans: (a) Natural killer cells

Q27. 1M Which disease has been completely eradicated by vaccination?
(a) Polio (b) Tetanus
(c) Diphtheria (d) Smallpox
Ans: (d) Smallpox

Q28. 1M Board fav Morphine is extracted from:
(a) Latex of *Papaver somniferum*
(b) Leaves of *Cannabis sativa*
(c) *Erythroxylum coca*
(d) *Datura*
Ans: (a) Latex of *Papaver somniferum*

Q29. 1M The age group considered as the adolescence period is:
(a) 10–14 years (b) 12–18 years
(c) 18–24 years (d) 6–12 years
Ans: (b) 12–18 years

Q30. 1M Name the hormones released by the adrenal gland on stimulation by nicotine.
Ans: Adrenaline and nor-adrenaline (they raise blood pressure and heart rate).

Q31. 1M Name the test widely used to diagnose AIDS.
Ans: ELISA (enzyme linked immuno-sorbent assay).

Assertion–Reason (1 mark each)

Assertion–Reason key: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

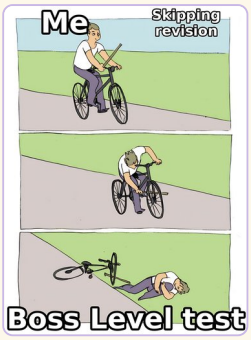
Q32. 1M **A:** Cleaning household coolers helps control malaria. **R:** Mosquito larvae breed in stagnant water.
Ans: (a) Both true and R explains A.

Q33. 1M Board fav **A:** Passive immunity is given to a person bitten by a snake. **R:** A quick response is needed, so preformed antibodies are injected.
Ans: (a) Both true and R explains A.

Q34. 1M **A:** AIDS is a congenital disease. **R:** HIV can pass from an infected mother to her child through the placenta.
Ans: (d) A is false (AIDS is acquired during lifetime), R is true.

Q35. 1M Board fav **A:** Malignant tumours are more dangerous than benign tumours. **R:** Cells of malignant tumours spread to distant sites through blood (metastasis).
Ans: (a) Both true and R explains A.

Q36. 1M Board fav **A:** Drug addicts need higher doses with time. **R:** Repeated use increases the tolerance level of receptors.
Ans: (a) Both true and R explains A.



Notes revise kiye bina seedha MIQ? Boss level test aise hi girata hai.

Short answer (2–3 marks)

Q37. 3M Board fav What are the various public health measures which you would suggest as safeguards against infectious diseases?
Ans: (i) Proper disposal of waste and excreta. (ii) Periodic cleaning and disinfection of water reservoirs, pools, cesspools and tanks. (iii) Standard hygiene practices in public catering. (iv) Control of vectors and their breeding places (no stagnant water, *Gambusia*, insecticide spray, mosquito nets, wire mesh). (v) Immunisation programmes and awareness about diseases.

Q38. 3M In which way has the study of biology helped us to control infectious diseases?
Ans: Biology helped us understand pathogens, their life cycles and modes of transmission, so that preventive measures like hygiene and vector control could be planned. It gave us **vaccines and immunisation** (smallpox eradicated; polio, diphtheria, pneumonia, tetanus controlled), **antibiotics** and other drugs, and biotechnology for newer, safer vaccines (e.g. hepatitis B vaccine from yeast).

Q39. 3M Board fav How does the transmission of each of the following diseases take place? (a) Amoebiasis (b) Malaria (c) Ascariasis (d) Pneumonia
Ans: (a) Food and water contaminated with faecal matter; houseflies act as mechanical carriers. (b) Bite of an infected female *Anopheles* mosquito (vector). (c) Water, vegetables and fruits contaminated with soil carrying *Ascaris* eggs from faeces. (d) Inhaling droplets/aerosols from an infected person, or sharing glasses and utensils.

Q40. 2M What measures would you take to prevent water-borne diseases?
Ans: Drink clean, boiled or filtered water; periodic cleaning and disinfection of water reservoirs and tanks; proper disposal of excreta and sewage so it does not mix with drinking water; keep food covered from houseflies; wash vegetables and fruits well; maintain personal hygiene.

Q41. 2M What does "a suitable gene" mean in the context of DNA vaccines?
Ans: It is a gene of the pathogen that codes for an **antigenic protein**. When introduced (or expressed in bacteria/yeast, as in recombinant vaccines), it produces the antigen, which makes the body produce antibodies and memory cells, without causing disease.

Q42. 2M Board fav Name the primary and secondary lymphoid organs.
Ans: Primary: **bone marrow and thymus**. Secondary: **spleen, lymph nodes, tonsils, Peyer's patches** of small intestine and **appendix**.

Q43. 2M Expand: (a) MALT (b) CMI (c) AIDS (d) NACO (e) HIV
Ans: (a) Mucosa-Associated Lymphoid Tissue. (b) Cell-Mediated Immunity. (c) Acquired Immuno Deficiency Syndrome. (d) National AIDS Control Organisation. (e) Human Immunodeficiency Virus.

Q44. 3M Board fav Differentiate the following and give examples of each: (a) innate and acquired immunity (b) active and passive immunity.
Ans: (a) Innate: non-specific, present at birth, barriers (e.g. skin, stomach acid, neutrophils, interferons). Acquired: pathogen-specific, develops after exposure, has memory (e.g. immunity after measles or a vaccine). (b) Active: own antibodies produced, slow, long-lasting (e.g. vaccination, natural infection). Passive: ready-made antibodies given, immediate, short-lived (e.g. colostrum IgA, antitoxin for tetanus, anti-venom).

Q45. 2M Board fav Draw a well-labelled diagram of an antibody molecule.
Ans: Draw a Y-shaped molecule of 4 polypeptide chains: two long **heavy chains** and two short **light chains** joined by S-S bonds; label **antigen binding sites** at the tips of both arms, heavy chain, light chain (see Fig. 7.4). Represented as H_2L_2.

Q46. 2M Board fav What are the various routes by which transmission of HIV takes place?
Ans: (i) Sexual contact with an infected person. (ii) Transfusion of contaminated blood and blood products. (iii) Sharing infected needles, as in intravenous drug abusers. (iv) From an infected mother to her child through the placenta.

Q47. 3M Board fav What is the mechanism by which the AIDS virus causes deficiency of the immune system of the infected person?
Ans: HIV enters **macrophages**, where reverse transcriptase copies its RNA into viral DNA; this joins the host DNA and the cell makes new viruses (macrophage acts as an HIV factory). HIV also enters **helper T-lymphocytes**, replicates, and progeny viruses attack more helper T cells. Their number falls progressively, so the person cannot fight even ordinary infections (*Mycobacterium*, *Toxoplasma*, fungi, viruses).

Q48. 2M How is a cancerous cell different from a normal cell?
Ans: A normal cell shows **contact inhibition**, regulated growth and differentiation, and stays in place. A cancer cell has lost contact inhibition, divides uncontrollably to form tumours, competes with normal cells for nutrients, and (if malignant) can invade tissues and spread to new sites (metastasis).

Q49. 2M Board fav Explain what is meant by metastasis.
Ans: The property of malignant tumours in which cells sloughed off from the tumour travel through blood to distant sites and start new tumours wherever they lodge. It is the most feared property of malignant tumours.

Q50. 3M List the harmful effects caused by alcohol/drug abuse.
Ans: Reckless behaviour, vandalism and violence; coma and death from respiratory or heart failure or cerebral haemorrhage; AIDS and hepatitis B from shared needles; damage to the nervous system and liver (cirrhosis); harm to the foetus in pregnancy; heavy drinking in adulthood; stealing and mental and financial distress to the family.

Q51. 3M Do you think that friends can influence one to take alcohol/drugs? If yes, how may one protect himself/herself from such an influence?
Ans: Yes, peer pressure is a major cause. Protection: say a firm "no"; choose friends wisely; talk to parents, teachers or trusted friends; channelise energy into sports, music, reading and yoga; remember that even one use can lead to addiction; seek counselling if needed.

Q52. 3M Board fav Why is it that once a person starts taking alcohol or drugs, it is difficult to get rid of this habit?
Ans: Because of their **addictive nature**: repeated use raises the tolerance of receptors, so higher doses are needed. The body also becomes **dependent**: stopping suddenly causes an unpleasant withdrawal syndrome (anxiety, shakiness, nausea, sweating), which pushes the person to continue.

Q53. 3M In your view what motivates youngsters to take to alcohol or drugs and how can this be avoided?
Ans: Motivation: curiosity, adventure, experimentation, academic stress, the idea that it is "cool", media influence, peer pressure, unsupportive family. Avoidance: no undue peer pressure; education and counselling; help from parents and peers; looking for danger signs; professional and medical help (de-addiction and rehabilitation).

Q54. 2M Give the scientific name of the pathogen, the organ affected and one symptom of: (a) typhoid (b) pneumonia.
Ans: (a) *Salmonella typhi*; small intestine (then other organs via blood); sustained high fever of 39–40 °C. (b) *Streptococcus pneumoniae* or *Haemophilus influenzae*; alveoli of lungs; lips and nails turn grey to bluish in severe cases.

Q55. 3M *Board fav* What is allergy? Name the antibody and the chemicals involved, and two drugs used to reduce its symptoms.

Ans: The exaggerated response of the immune system to certain environmental antigens (allergens such as dust mites, pollen, animal dander). Antibodies are of the **IgE** type; **histamine and serotonin** are released from mast cells. Drugs: anti-histamine, adrenalin, steroids (any two).

Q56. 2M Why is the secondary immune response stronger than the primary response?

Ans: During the first encounter the body forms **memory B and T cells**. On a later encounter with the same pathogen, these recognise it quickly and produce a highly intensified (anamnestic) response with a massive production of antibodies.

Q57. 2M Name the disease caused by each and its group: *Microsporum*, *Wuchereria bancrofti*, Rhino virus, *Entamoeba histolytica*.

Ans: Ringworm (fungus); filariasis or elephantiasis (helminth); common cold (virus); amoebiasis or amoebic dysentery (protozoan).

Q58. 2M What is a carcinogen? Give one example each of a physical, a chemical and a biological carcinogen.

Ans: An agent that transforms normal cells into cancerous neoplastic cells. Physical: X-rays, gamma rays or UV. Chemical: carcinogens in tobacco smoke. Biological: oncogenic viruses (with viral oncogenes).

Q59. 2M How does smoking cause oxygen deficiency in the body? Name two diseases associated with smoking.

Ans: Smoking increases **carbon monoxide** in blood, which reduces the concentration of haem-bound oxygen, causing oxygen deficiency. Diseases: cancers of lung, urinary bladder and throat; bronchitis; emphysema; coronary heart disease; gastric ulcer (any two).

Q60. 3M Mention any four side effects of anabolic steroids in males and two in females.

Ans: Males: acne, increased aggressiveness, mood swings, depression, reduced size of testicles, decreased sperm production, kidney and liver dysfunction, breast enlargement, premature baldness, enlarged prostate (any four). Females: masculinisation, abnormal menstrual cycles, excessive facial and body hair, deepening of voice, enlarged clitoris (any two).

Q61. 2M Why are malaria and filariasis called vector-borne diseases? Give the most important measure to control them.

Ans: Their pathogens (*Plasmodium*, *Wuchereria*) are transmitted from person to person by insect vectors: female *Anopheles* for malaria and female mosquitoes for filariasis. The most important measure is to **control or eliminate the vectors and their breeding places**.

Q62. 2M *Board fav* What is the difference between a benign tumour and a malignant tumour? Which one is more dangerous and why?

Ans: Benign: stays at its original site, does not spread, causes little damage. Malignant: neoplastic cells grow rapidly, invade surrounding tissues, starve normal cells, and spread through blood to form new tumours. Malignant is more dangerous because of **metastasis**.

Q63. 3M *Board fav* Describe the four types of barriers of innate immunity with one example each.

Ans: (i) Physical: skin; mucus of respiratory, digestive and urogenital tracts. (ii) Physiological: stomach acid, saliva, tears. (iii) Cellular: neutrophils (PMNL), monocytes, natural killer cells, macrophages. (iv) Cytokine: interferons secreted by virus-infected cells protect non-infected cells.

Q64. 3M Write the functions of (a) thymus (b) spleen (c) lymph nodes.

Ans: (a) Thymus: primary lymphoid organ; provides the micro-environment for maturation of T-lymphocytes; large at birth, shrinks by puberty. (b) Spleen: filters blood by trapping blood-borne microbes; contains lymphocytes and phagocytes; reservoir of erythrocytes. (c) Lymph nodes: trap microbes and antigens in lymph and tissue fluid and activate the lymphocytes there.

Q65. 2M Name the source plant and the receptors on which each acts: (a) heroin (b) marijuana.

Ans: (a) Heroin: from morphine of the latex of *Papaver somniferum* (poppy); acts on opioid receptors in CNS and GI tract. (b) Marijuana: from *Cannabis sativa* (flower tops, leaves, resin); acts on cannabinoid receptors, mainly in the brain.

Q66. 2M How do nicotine and carbon monoxide in tobacco smoke affect the body?

Ans: Nicotine stimulates the adrenal gland to release adrenaline and nor-adrenaline, which raise blood pressure and heart rate. CO increases in blood and reduces haem-bound oxygen, causing oxygen deficiency.

Q67. 2M What is auto-immunity? Give one example.

Ans: When, due to genetic and other unknown reasons, the body's immune system attacks its own (self) cells, causing damage. Example: **rheumatoid arthritis**.

Q68. 2M Why is it said that "health is not just the absence of disease"? List two benefits of good health.

Ans: Health is a state of complete physical, mental and social well-being; a person may have no disease but still lack mental or social well-being. Benefits: more efficiency and productivity at work, economic prosperity, longer life, lower infant and maternal mortality (any two).

Q69. 2M Name two diseases caused by helminths and their pathogens.

Ans: Ascariasis by *Ascaris* (round worm); elephantiasis or filariasis by *Wuchereria bancrofti* / *W. malayi* (filarial worms).

Long answer (5 marks)

Q70. 5M Board fav Describe the life cycle of *Plasmodium*. Why is the female *Anopheles* called the vector? How can malaria be controlled?

Ans: **In humans:** sporozoites (infectious form) enter through the bite of an infected female *Anopheles*; they multiply asexually in liver cells, then attack RBCs and rupture them. Rupture releases **haemozoin**, causing chill and high fever every 3–4 days. Some parasites form gametocytes in RBCs. **In mosquito:** a female *Anopheles* biting an infected person takes up gametocytes; fertilisation and development occur in the mosquito gut; sporozoites formed move to the salivary glands and are injected into the next human. The parasite needs two hosts. **Vector:** the mosquito transmits the parasite from one person to another. **Control:** avoid stagnant water, clean coolers, use nets and wire mesh, *Gambusia* in ponds, insecticide spray in ditches and swamps, timely treatment.

Q71. 5M Board fav What is acquired immunity? Explain humoral and cell-mediated immunity. Describe the structure of an antibody and explain the principle of vaccination.

Ans: Acquired immunity is pathogen-specific and has memory: the primary response is weak, the secondary (anamnestic) response is strong. **Humoral:** B-lymphocytes produce antibodies into the blood. **CMI:** T-lymphocytes mediate it; T cells help B cells and are responsible for graft rejection (self vs non-self). **Antibody:** 4 peptide chains, 2 heavy + 2 light (H_2L_2), Y-shaped, antigen binding sites at the tips; types IgA, IgM, IgE, IgG. **Vaccination:** based on memory; antigenic proteins or weakened/inactivated pathogens are introduced, antibodies and memory B and T cells form, and on actual infection the pathogen is quickly overwhelmed.

Q72. 5M Board fav Explain how HIV is transmitted and how it replicates inside the body. Why do AIDS patients suffer from infections that a healthy person can overcome? List the measures to prevent AIDS.

Ans: **Transmission:** sexual contact, contaminated blood, shared needles, infected mother to child through placenta. **Replication:** HIV (retrovirus, RNA genome) enters macrophages; reverse transcriptase forms viral DNA, which integrates into host DNA; the cell produces new viruses (HIV factory). HIV also enters helper T cells, multiplies, and new viruses attack more helper T cells. **Infections:** helper T cells fall progressively, so immunity collapses; *Mycobacterium*, viruses, fungi and *Toxoplasma* cause disease. Time-lag 5–10 years; ELISA for diagnosis; anti-retroviral drugs only partially effective. **Prevention:** HIV-safe blood, disposable needles and syringes, free condoms, safe sex, control of drug abuse, regular check-ups, education (NACO, NGOs, WHO).

Q73. 5M Board fav What is cancer? Differentiate between benign and malignant tumours. Describe the causes, detection and treatment of cancer.

Ans: Cancer is uncontrolled cell division due to breakdown of the regulatory mechanisms; cancer cells lose contact inhibition and form tumours. **Benign:** confined to its location, little damage. **Malignant:** rapidly growing neoplastic cells invade tissues, starve normal cells and spread through blood (metastasis). **Causes:** carcinogens: physical (X-rays, gamma rays, UV), chemical (tobacco smoke), biological (oncogenic viruses, activated proto-oncogenes/c-onc). **Detection:** biopsy and histopathology, blood and bone marrow tests (leukaemia), radiography, CT, MRI, antibodies against cancer antigens, molecular tests for inherited susceptibility. **Treatment:** surgery, radiotherapy, chemotherapy (side effects: hair loss, anaemia), immunotherapy with biological response modifiers like α-interferon.

Case-based (4 marks)

Q74. 4M Board fav A 17-year-old student shows a sudden drop in marks, skips classes, avoids family, has changed sleeping habits and is often aggressive. His friends say he has started "experimenting" at parties. (a) What do these signs indicate? (b) Name two causes common at this age. (c) Differentiate addiction from dependence. (d) Suggest two steps his parents should take.

Ans: (a) Warning signs of drug/alcohol abuse. (b) Curiosity/experimentation, peer pressure, academic stress, the idea that it is "cool" (any two). (c) Addiction: psychological attachment to effects like euphoria; dependence: the body shows a withdrawal syndrome when the regular dose stops. (d) Talk and counsel with love and consistent discipline; seek professional help (psychologist, de-addiction and rehabilitation programme).

Q75. 4M Board fav A new mother was advised to feed her baby the first yellowish milk. Later the baby got its polio and hepatitis B vaccines. (a) Name the yellowish milk and the antibody in it. (b) Which type of immunity does it give? (c) Which type of immunity do vaccines give and why is it long-lasting? (d) How is the hepatitis B vaccine produced?

Ans: (a) Colostrum; IgA. (b) Passive immunity (ready-made antibodies). (c) Active immunity; the body makes its own antibodies and memory B and T cells. (d) By recombinant DNA technology: the antigenic polypeptide is produced in **yeast**.

Exam Alert: case-based tips

→ Har part (a), (b), (c), (d) alag line pe, ek-ek keyword ke saath.

→ Pathogen ka naam *italics* / underline karke likho.

Q76. 4M A patient had a kidney transplant from an unrelated donor. Doctors matched tissue and blood group, yet prescribed lifelong medicines. (a) Why are the medicines needed? (b) Name these medicines. (c) Which immune response causes rejection, and which cells mediate it? (d) Why can grafts not be taken from any animal or human?

Ans: (a) To prevent rejection of the graft. (b) Immunosuppressants. (c) Cell-mediated immune response, mediated by T-lymphocytes. (d) The body distinguishes "self" from "non-self", so a foreign graft is attacked and rejected sooner or later.

Q77. 4M Board fav During monsoon, many people in a colony got recurring fever with chills every 3-4 days. Water had collected in old tyres and coolers. (a) Name the disease and its pathogen. (b) Name the vector. (c) What causes the recurring fever? (d) Suggest two control measures.

Ans: (a) Malaria; *Plasmodium* (e.g. *P. vivax*; *P. falciparum* causes malignant malaria). (b) Female *Anopheles* mosquito. (c) Rupture of RBCs releases the toxin haemozoin. (d) Remove stagnant water from tyres and coolers; mosquito nets and wire mesh; *Gambusia* in ponds; insecticide spray (any two).

Q78. 4M A doctor removed a small piece of a lump from a patient's breast, stained thin sections and examined them under a microscope. Cells showed no contact inhibition and some had entered blood vessels. (a) Name this diagnostic method. (b) What does the loss of contact inhibition indicate? (c) What is the significance of cells entering blood vessels? (d) Name two imaging techniques to check internal organs.

Ans: (a) Biopsy with histopathological study. (b) The cells are cancerous: they divide uncontrollably. (c) Malignant tumour cells can spread to distant sites and start new tumours (metastasis). (d) CT and MRI (also radiography).

Q79. 4M Board fav Rohan sneezes, has watery eyes and difficulty in breathing whenever he visits his grandmother's dusty storeroom, but is fine elsewhere. (a) Name this condition and the type of antibody involved. (b) Which cells release which chemicals to cause these symptoms? (c) How can the exact cause be found? (d) Name two drugs that give quick relief.

Ans: (a) Allergy (to dust mites); IgE antibodies. (b) Mast cells release histamine and serotonin. (c) He is exposed to or injected with very small doses of possible allergens and the reactions are studied. (d) Anti-histamine, adrenalin or steroids (any two).

Exam Alert: diagrams and lists to practise

- → Fig. 7.1 Plasmodium life cycle (human + mosquito stages), Fig. 7.4 antibody (H_2L_2), Fig. 7.6 HIV replication: ye teen sabse zyada aate hain.
- → Morphine / cannabinoid structure dikha ke "identify" pucha jaata hai: source plant aur receptor bhi likho.
- → Differentiate pairs: innate/acquired, active/passive, humoral/CMI, benign/malignant, addiction/dependence.

Q80. 4M A national athlete was banned after a test showed anabolic steroids. She had developed a deep voice, excessive facial hair and irregular periods. (a) Why do some sportspersons misuse such drugs? (b) Name two other drug groups misused in sports. (c) Name the term for her male-like features. (d) What can steroids do to the bones of adolescents?

Ans: (a) To increase muscle strength and bulk, promote aggressiveness and raise performance. (b) Narcotic analgesics, diuretics, certain hormones (any two). (c) Masculinisation. (d) Premature closure of the growth centres of long bones, causing stunted growth.

Q81. 4M In a hostel, one student has a runny nose, sore throat and hoarseness for 4-5 days, while another has high fever, chills, cough and bluish lips and nails. (a) Name the likely disease of each. (b) Name their pathogens. (c) Which organ is affected in the second case and how? (d) Give one way both spread.

Ans: (a) First: common cold; second: pneumonia. (b) Common cold: Rhino viruses; pneumonia: *Streptococcus pneumoniae* or *Haemophilus influenzae*. (c) Alveoli of the lungs, which fill with fluid and cause severe breathing problems. (d) Droplets from coughs or sneezes of an infected person (also shared utensils or contaminated objects).

Last-minute check

- ✓ Typhoid = Widal; AIDS = ELISA; cancer = biopsy, CT, MRI.
- ✓ Haemozoin = malaria fever; interferon = cytokine barrier and cancer immunotherapy.
- ✓ IgA = colostrum; IgE = allergy; H_2L_2 = antibody.
- ✓ Heroin = depressant (opioid receptors); cocaine = stimulant (dopamine); nicotine = adrenaline.
- ✓ Primary lymphoid: bone marrow, thymus; MALT = about 50% of lymphoid tissue.
- ✓ HIV: macrophage = HIV factory; helper T cells fall; time-lag 5-10 years.
- ✓ Benign = stays; malignant = spreads (metastasis); carcinogens: X-rays, UV, tobacco smoke, oncogenic viruses.
- ✓ Addiction = mind (euphoria); dependence = body (withdrawal syndrome).
- ✓ Smallpox eradicated by vaccination; *Gambusia* eats mosquito larvae.

Yaad karo: pathogen ka group

Bacteria: Typhoid, Pneumonia ("Bhai TP"). Virus: common cold. Protozoa: malaria, amoebiasis ("PMA"). Helminths: ascariasis, filariasis. Fungi: ringworm. Group galat likha toh pura answer galat!

Quick fact

Health is a complete package: a balanced diet, personal hygiene, regular exercise, yoga, and staying away from tobacco, drugs and alcohol protect both the body and the mind, and the immune system that keeps us healthy.

MICROBES IN HUMAN WELFARE

Most asked: microbes and their products (table), primary vs secondary sewage treatment, BOD, biogas plant, Bt and biocontrol, biofertilisers

Not all microbes cause disease. This chapter shows how microbes help us at **home, in industry, in sewage treatment, as biogas, biocontrol agents and biofertilisers.**

1. Microbes: Everywhere and Diverse

Besides macroscopic plants and animals, **microbes** are the major components of biological systems on earth. They are present everywhere: in soil, water, air, inside our bodies and inside other animals and plants.

They live even where no other life-form can exist: deep inside **geysers (thermal vents)** at temperatures as high as 100 °C, deep in the soil, under snow layers several metres thick and in **highly acidic** environments.

Kinds of microbes (microbial diversity):

- → **Cellular**: protozoa, bacteria, fungi (kingdoms **Monera, Protista, Fungi**).
- → **Acellular**: microscopic plant and animal **viruses, viroids** (free RNA) and **prions**, which are proteinaceous infectious agents.

Bacteria and many fungi can be grown on **nutritive media** to form colonies that are visible to the naked eye (Fig. 8.3, colonies in a petri dish). Such cultures are useful in studies on micro-organisms.

Quick fact

Microbes cause many diseases in humans, animals and plants (Chapter 7), but **not all microbes are harmful.** Several are useful to us in diverse ways, and this chapter is about exactly those.

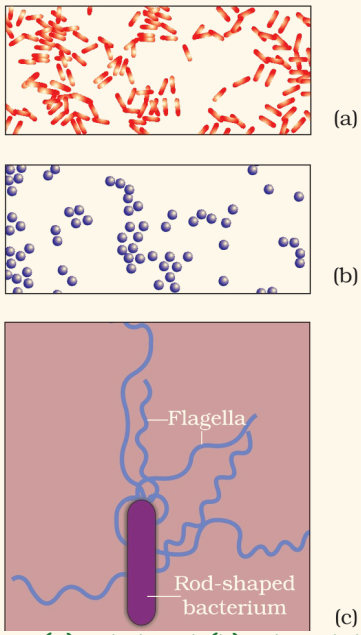


Fig. 8.1 Bacteria: (a) rod-shaped (b) spherical, both 1500X (c) rod-shaped bacterium with flagella, 50,000X

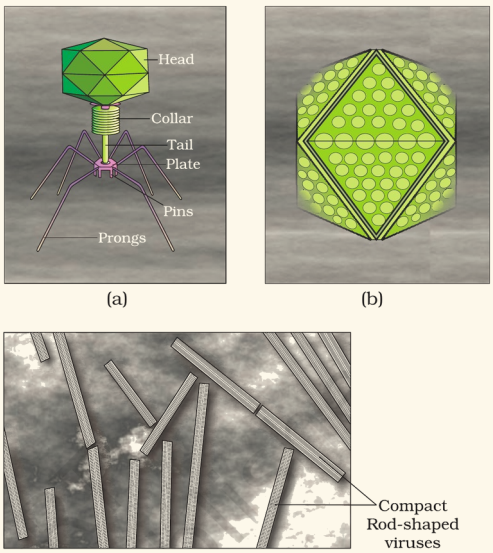


Fig. 8.2 Viruses: (a) bacteriophage (b) adenovirus (respiratory infections) (c) rod-shaped Tobacco Mosaic Virus (TMV)

2. Microbes in Household Products

A. Curd from milk

Lactobacillus and other **lactic acid bacteria (LAB)** grow in milk and convert it into curd. During growth, LAB produce acids that coagulate and partially digest milk proteins.

- → A small amount of curd added to fresh milk acts as the **inoculum or starter**; it contains millions of LAB, which multiply at a suitable temperature.
- → Curd is more nutritious than milk: LAB increase vitamin B_{12}. PYQ
- → In our **stomach** too, LAB check (inhibit) disease-causing microbes.

B. Dough: idli, dosa and bread

The dough for **idli and dosa** is fermented by **bacteria**; its puffed-up look is due to CO_2 gas released during fermentation (glycolysis followed by anaerobic respiration). Bread dough is fermented by **baker's yeast**, *Saccharomyces cerevisiae*.

C. Other fermented foods and cheese

- → **Toddy**: traditional drink of parts of southern India, made by fermenting **sap from palms**.
- → Microbes are used to ferment **fish, soyabean and bamboo-shoots** to make foods.
- → **Cheese** is one of the oldest foods made using microbes; its texture, flavour and taste depend on the **specific microbe** used.

Cheese	Microbe	Effect
Swiss cheese	*Propionibacterium sharmanii* (bacterium)	large holes due to a lot of CO_2
Roquefort cheese	a specific fungus grown on it	ripening and a particular flavour

Yaad karo: Swiss cheese ke chhed

Swiss = Sharmanii. Propionibacterium Phoonk maarta hai (CO_2), isliye cheese mein bade chhed. Roquefort = fungus wala flavour.

Key terms: microbes in the kitchen

Food / drink	Microbe	Gas or product
Curd	*Lactobacillus* (LAB)	lactic acid; vitamin B_{12} rises
Idli, dosa	bacteria	CO_2 puffs the batter
Bread	*Saccharomyces cerevisiae* (baker's yeast)	CO_2 raises the dough
Toddy	microbes fermenting palm sap	alcohol
Swiss cheese	*Propionibacterium sharmanii*	CO_2 makes holes
Roquefort cheese	a fungus	flavour on ripening

Paheli-Bujho

Bujho: Paheli, idli ka batter raat bhar mein phool kaise jaata hai?

Paheli: Bacteria fermentation karte hain aur CO_2 gas banti hai.

Bujho: Matlab batter ke andar bacteria ki party!

Paheli: Haan, aur bread mein yahi kaam yeast karta hai.

Think and answer (NCERT in-text)

→ **Which pathway releases CO_2 in dough?** Glycolysis followed by anaerobic respiration (fermentation): pyruvate is broken down to ethanol and CO_2, or to lactic acid by LAB.

→ **Where do the bacteria for idli-dosa come from?** From the grains themselves, the utensils, the hands and the air: they are naturally present.

3. Microbes in Industrial Products

Industrial production of beverages, antibiotics, acids and enzymes needs microbes grown in very large vessels called fermentors (Fig. 8.4). A whole set-up for this is a fermentation plant (Fig. 8.5).

A. Fermented beverages

Yeast *Saccharomyces cerevisiae*, called **brewer's yeast** here (the same yeast used for bread), ferments **malted cereals and fruit juices** to produce ethanol. The type of drink depends on the raw material and on whether it is distilled.

Malted cereals / fruit juice → Brewer's yeast (anaerobic) → Ethanol + CO_2 → Wine, beer OR distil: whisky, brandy, rum

Without distillation	With distillation of the fermented broth
wine, beer	whisky, brandy, rum
lower alcohol content	higher alcohol content

B. Antibiotics

Antibiotics are chemical substances produced by some microbes that can kill or retard the growth of other (disease-causing) microbes. *Anti* = against, *bio* = life: against the life of pathogens, but **pro-life** for human beings. They are one of the most significant discoveries of the twentieth century.

Key terms: discovery of penicillin

→ **Penicillin** was the first antibiotic discovered, and it was a **chance discovery**. PYQ

→ **Alexander Fleming**, while working on *Staphylococci*, saw a mould growing in an unwashed culture plate; *Staphylococci* could not grow around it. The chemical was named penicillin after the mould *Penicillium notatum*.

→ Its full potential as an effective antibiotic was established by **Ernest Chain and Howard Florey**; it was used to treat American soldiers wounded in World War II.

→ Fleming, Chain and Florey got the **Nobel Prize in 1945**.

Antibiotics greatly improved our capacity to treat deadly diseases like **plague, whooping cough** (kali khansi), **diphtheria** (gal ghotu) and **leprosy** (kusht rog), which used to kill millions all over the globe.

Antibiotic	Source microbe
Penicillin	*Penicillium notatum* (fungus); commercially *P. chrysogenum*
Streptomycin	*Streptomyces griseus* (bacterium)
Tetracycline	*Streptomyces aureofaciens* (bacterium)
Chloramphenicol	*Streptomyces venezuelae* (bacterium)

C. Chemicals, enzymes and other bioactive molecules

Product	Microbe (type)	Use
Citric acid	*Aspergillus niger* (fungus)	organic acid
Acetic acid	*Acetobacter aceti* (bacterium)	vinegar, organic acid
Butyric acid	*Clostridium butylicum* (bacterium)	organic acid
Lactic acid	*Lactobacillus* (bacterium)	organic acid
Ethanol	*Saccharomyces cerevisiae* (yeast)	commercial ethanol

Enzymes and other bioactive molecules:

Product	Microbe (type)	Use
Lipases	microbial enzyme	detergents: remove **oily stains**
Pectinases, proteases	microbial enzymes	**clarify** bottled fruit juices
Streptokinase	*Streptococcus* (bacterium), modified by genetic engineering	**clot buster** after myocardial infarction
Cyclosporin A	*Trichoderma polysporum* (fungus)	**immunosuppressive** agent in organ transplant
Statins	*Monascus purpureus* (yeast)	lower blood **cholesterol**

Statins act by competitively inhibiting the enzyme responsible for the synthesis of cholesterol. Bottled juices are clearer than home-made ones because they are clarified by **pectinases and proteases.** PYQ

Yaad karo: acids aur unke microbes

Citric = Aspergillus (CA: Chief Accountant), Acetic = Acetobacter (A se A), Butyric = Clostridium Butylicum (B se B), Lactic = Lactobacillus (L se L). Sirf citric ka jodi alag hai, wahi fungus hai!

Exam Alert

- Cyclosporin A = **fungus** *Trichoderma polysporum*; Statins = **yeast** *Monascus purpureus*. Inka type (fungus/yeast) MCQ mein poochha jaata hai.
- Streptokinase **bacterium** se banta hai, fungus se nahi.
- *Trichoderma polysporum* (cyclosporin) aur *Trichoderma* (biocontrol) ko mix mat karna.

4. Microbes in Sewage Treatment

Large quantities of waste water are generated every day in cities and towns; its major component is **human excreta**. This municipal waste water is called sewage. It contains large amounts of **organic matter** and microbes, many of them **pathogenic**, so it cannot be discharged directly into rivers.

Sewage is treated in **sewage treatment plants (STPs)** by the heterotrophic microbes naturally present in it, in two stages:

A. Primary treatment (physical)

- Physical removal of particles, large and small, by **filtration and sedimentation.**
- Floating debris is removed by **sequential filtration**; then **grit** (soil and small pebbles) is removed by sedimentation.
- All settled solids form the primary sludge; the supernatant is the **effluent**, taken for secondary treatment.

B. Secondary (biological) treatment

1. Primary effluent goes into large **aeration tanks**; it is constantly **agitated mechanically** and air is pumped in (Fig. 8.6).
2. Useful **aerobic microbes** grow vigorously into flocs: masses of bacteria associated with fungal filaments forming mesh-like structures.
3. The microbes consume the major part of the organic matter, which **significantly reduces the BOD** of the effluent.
4. The effluent then goes to a **settling tank**, where the flocs sediment. This sediment is the activated sludge.
5. A **small part** of the activated sludge is pumped back into the aeration tank as **inoculum**; the major part goes to **anaerobic sludge digesters.**
6. In the digesters, **anaerobic bacteria** digest the bacteria and fungi of the sludge and produce **methane, H_2S and CO_2**: biogas.
7. The treated effluent is released into rivers and streams (Fig. 8.7, aerial view of an STP).

Sewage → Primary treatment → Primary sludge + effluent → Aeration tank (flocs) → Settling tank → Activated sludge → Anaerobic digester → Biogas

Key terms: BOD

- **BOD (biochemical oxygen demand)** = the amount of oxygen that would be consumed if all the organic matter in one litre of water were oxidised by bacteria. PYQ
- The BOD test measures the **rate of uptake of oxygen** by micro-organisms in a water sample, so BOD is an indirect measure of organic matter.
- **Greater BOD = more organic matter = more polluting potential.** Sewage is treated till its BOD is reduced.

Primary treatment	Secondary treatment
physical process	**biological** process
filtration and sedimentation	aeration, flocs, activated sludge, digestion
removes floating debris and grit	removes dissolved **organic matter**
forms primary sludge	forms activated sludge and biogas
BOD is reduced only a little	BOD is **reduced greatly**

Aeration tank	Anaerobic sludge digester
aerobic bacteria and fungi (flocs)	**anaerobic** bacteria
air pumped in, mechanical agitation	no air; closed tank
consume organic matter of effluent	digest bacteria and fungi of the sludge
main result: BOD falls	main result: **biogas** (CH_4, H_2S, CO_2)

Paheli-Bujho

Bujho: BOD zyada hai toh paani zyada saaf hai na? Oxygen zyada!
Paheli: Ulta! Zyada BOD matlab zyada organic matter, yaani zyada ganda paani.
Bujho: Oh, matlab bacteria ki bhookh hi gandagi naapti hai.
Paheli: Bilkul! Isliye untreated sewage ka BOD sabse high hota hai.

No man-made technology has been able to rival microbial treatment of sewage, used for more than a century all over the world. But urbanisation produces far more sewage than our STPs can treat, so untreated sewage often reaches rivers, causing **pollution and water-borne diseases**.

Quick fact: river action plans

The **Ministry of Environment and Forests** initiated the Ganga Action Plan and Yamuna Action Plan to save these rivers from pollution. Under these plans, many STPs are to be built so that only **treated sewage** is discharged into the rivers.

5. Microbes in Production of Biogas

Biogas is a mixture of gases, predominantly methane, produced by microbial activity and used as fuel. The gas a microbe produces depends on the microbe and its substrate: in dough, cheese and beverages the main gas is CO_2.

Certain bacteria growing **anaerobically on cellulosic material** produce large amounts of **methane along with CO_2 and H_2**. They are called methanogens, e.g. *Methanobacterium*. PYQ

- → Found in the **anaerobic sludge** of sewage treatment.
- → Present in the **rumen** (a part of the stomach) of cattle, where they break down **cellulose** and help in cattle nutrition. Humans cannot digest cellulose.
- → So cattle dung (**gobar**) is rich in methanogens, and the biogas from it is called **gobar gas**.

Parts and working of a biogas plant:

- → A **concrete tank, 10–15 feet deep**, in which bio-wastes are collected and a **slurry of dung** is fed.
- → A floating cover over the slurry keeps rising as gas is produced by microbial activity.
- → A gas **outlet** supplies biogas through a pipe to nearby houses for **cooking and lighting**.
- → **Spent slurry** is removed through another outlet and is used as **fertiliser**.

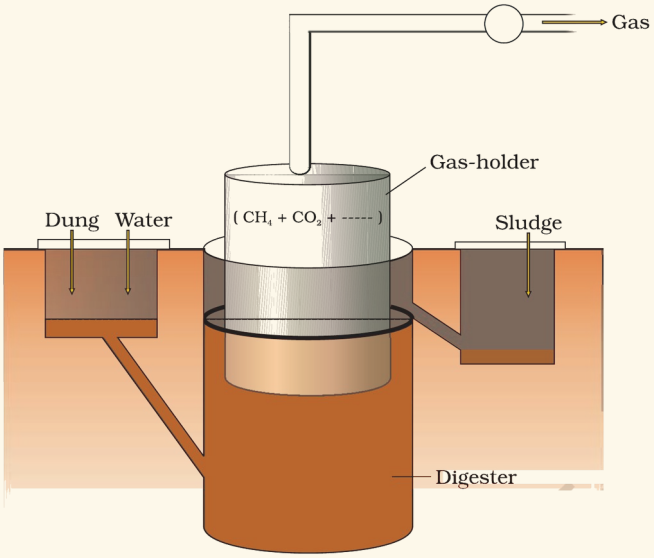


Fig. 8.8 A typical biogas plant

Biogas plants are mostly built in **rural areas**, where cattle dung is available in large quantities. The technology was developed in India mainly by the IARI (Indian Agricultural Research Institute) and KVIC (Khadi and Village Industries Commission). PYQ

Key terms: why biogas is useful

- → **Clean, smokeless fuel** for cooking and lighting in villages; methane burns with a blue flame.
- → **Spent slurry** is a good manure, so nothing is wasted.
- → Better **sanitation**: dung and bio-wastes are used, not left in the open.
- → Saves firewood and cow-dung cakes, so less **deforestation** and less indoor smoke.

Paheli-Bujho

Bujho: Gaay ka gobar gas kaise ban gaya? Magic hai kya?
Paheli: Rumen ke methanogens dung mein aate hain, wahi methane banate hain.
Bujho: Toh gaay chalta-phirta gas plant hai!
Paheli: Haha, aur bacha slurry khet mein khaad ban jaata hai.

6. Microbes as Biocontrol Agents

Biocontrol is the use of biological methods for controlling plant diseases and pests. Chemical insecticides, pesticides and weedicides are toxic to humans and animals and pollute soil, ground water, fruits, vegetables and crops.

The **organic farmer** believes that biodiversity furthers health: pests are not eradicated but kept at **manageable levels** by natural checks and balances. Eradication is undesirable, because beneficial predatory and parasitic insects depend on pests as food or hosts. This **holistic approach** needs knowledge of the field's predators and pests, their life cycles, feeding patterns and habitats.

Biocontrol agent	Controls / how
Ladybird (red-black beetle)	aphids
Dragonflies	mosquitoes
Bacillus thuringiensis (Bt), bacterium	butterfly **caterpillars**; dried spores mixed with water and sprayed on brassicas, fruit trees
Trichoderma (free-living fungus of root ecosystems)	several **plant pathogens**
Baculoviruses (genus *Nucleopolyhedrovirus*)	insects and other arthropods; **species-specific**, narrow spectrum

How Bt works: larvae eat the spores; in the larval gut the toxin is released and the larvae are killed, while other insects remain unharmed. Scientists have introduced *B. thuringiensis* **toxin genes into plants**, making them resistant to insect pests, e.g. Bt-cotton (Chapter 10). PYQ

Baculoviruses have **no negative impact** on plants, mammals, birds, fish or even non-target insects. This is ideal in an integrated pest management (IPM) programme where beneficial insects are conserved, or in an **ecologically sensitive area**.

Chemical pesticides	Biocontrol agents
kill useful and harmful organisms **indiscriminately**	mostly **specific** to the target pest
toxic to humans and animals	safe for humans, animals, non-target insects
pollute soil, ground water, food	no pollution; part of the ecosystem
pests develop resistance	natural checks keep pests at manageable levels

Key terms: IPM

Integrated pest management (IPM) combines biocontrol agents, beneficial predators and only minimal, need-based chemicals, so that pests are kept below damaging levels instead of being wiped out.

Exam Alert

→ Bt = **bacterium**, Trichoderma = **fungus**, Baculovirus = **virus**: teen alag kingdom ke biocontrol agents.

→ Ladybird ke saath **aphids**, dragonfly ke saath **mosquitoes**: ulta mat likhna.

→ Biocontrol se toxic chemicals aur pesticides par dependence kam hoti hai: 2 marks ka point.

7. Microbes as Biofertilisers

Overuse of **chemical fertilisers** has added greatly to pollution, so there is pressure to switch to **organic farming**. Biofertilisers are organisms that **enrich the nutrient quality of the soil**; their main sources are **bacteria, fungi and cyanobacteria**. PYQ

Biofertiliser	Example	Role
Symbiotic N_2-fixing bacteria	*Rhizobium* (root nodules of **legumes**)	fixes atmospheric N_2 into organic forms used by the plant
Free-living N_2-fixing bacteria	*Azospirillum*, *Azotobacter*	enrich soil nitrogen
Mycorrhiza (fungus + root)	many members of genus *Glomus*	absorb **phosphorus** from soil and pass it to the plant
Cyanobacteria (BGA)	*Anabaena*, *Nostoc*, *Oscillatoria*	fix N_2; add organic matter; important in **paddy fields**

Extra benefits to plants with mycorrhiza:

- ✓ resistance to **root-borne pathogens**
- ✓ tolerance to **salinity and drought**
- ✓ overall increase in plant **growth and development**

In return, the **fungus** gets food (carbohydrates made by the plant) and shelter: a classic mutualism, most useful in phosphorus-poor soils. Cyanobacteria are **autotrophic** microbes of aquatic and terrestrial habitats. Many biofertilisers are now sold commercially, and farmers use them to replenish soil nutrients and cut down chemical fertilisers.

Yaad karo: biofertiliser jodi

Rhizobium = Root nodule (legume). Azo wale dono (Azotobacter, Azospirillum) = Akele (free-living). Glomus = Get phosphorus. Paddy = BGA (Anabaena, Nostoc). Mycorrhiza ka fungus = plant ka **phosphorus delivery boy**!

Chemical fertilisers	Biofertilisers
factory-made chemicals (urea, NPK)	living organisms (bacteria, fungi, cyanobacteria)
overuse pollutes soil and water	eco-friendly, no pollution
no organic matter added	add organic matter; improve soil texture
costly, repeated use	cheap, renewable, self-multiplying
act fast but can harm useful soil microbes	act slowly; increase useful soil microbes

Quick fact: more uses of microbes

→ **Single cell protein (SCP)**: microbial biomass such as *Spirulina* (cyanobacterium) grown on waste water, straw or molasses is a protein-rich food and feed.

→ **Soil**: microbes decompose dead matter, recycle nutrients and fix nitrogen, keeping soil fertile.

→ **Medicine**: genetically engineered microbes now make human **insulin** (*E. coli*) and the **hepatitis B vaccine** (yeast); see Chapter 10.

8. Master Table: Microbe and its Use

Microbe	Type	Used for
Lactobacillus	bacterium	curd, lactic acid
Saccharomyces cerevisiae	yeast	bread, beverages, ethanol
Propionibacterium sharmanii	bacterium	Swiss cheese
Penicillium notatum	fungus	penicillin
Aspergillus niger	fungus	citric acid
Acetobacter aceti	bacterium	acetic acid
Clostridium butylicum	bacterium	butyric acid
Streptococcus	bacterium	streptokinase
Trichoderma polysporum	fungus	cyclosporin A
Monascus purpureus	yeast	statins

Microbe	Type	Used for
Methanobacterium	bacterium	biogas (methane)
Bacillus thuringiensis	bacterium	biocontrol of caterpillars
Trichoderma	fungus	biocontrol of plant pathogens
Nucleopolyhedrovirus	virus	biocontrol of insects
Rhizobium, Azotobacter, Azospirillum	bacteria	nitrogen fixation
Glomus	fungus (mycorrhiza)	phosphorus absorption
Anabaena, Nostoc, Oscillatoria	cyanobacteria	N_2 fixation in paddy fields

1 minute recap

1. Microbes are everywhere and diverse: bacteria, fungi, protozoa, viruses, viroids, prions.
2. LAB: milk to curd, more vitamin B_{12}; bacteria ferment idli-dosa dough (CO_2); yeast ferments bread dough; Swiss cheese holes by *Propionibacterium sharmanii*.
3. Brewer's yeast makes ethanol: wine, beer (no distillation); whisky, brandy, rum (distilled).
4. Penicillin (first antibiotic, *Penicillium notatum*); Fleming, Chain, Florey, Nobel 1945.
5. Citric (*Aspergillus niger*), acetic (*Acetobacter aceti*), butyric (*Clostridium butylicum*), lactic (*Lactobacillus*); streptokinase, cyclosporin A, statins.
6. Sewage: primary (physical) then secondary (biological: flocs, BOD falls, activated sludge, anaerobic digester gives biogas).
7. Biogas = mainly methane by methanogens (*Methanobacterium*); gobar gas plant by IARI and KVIC.
8. Biocontrol: ladybird, dragonfly, Bt, *Trichoderma*, baculoviruses (IPM). Biofertilisers: *Rhizobium*, *Azotobacter*, *Azospirillum*, *Glomus*, cyanobacteria.

MOST IMPORTANT QUESTIONS

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM Board fav Lactic acid bacteria improve the nutritional quality of curd by increasing:
(a) Vitamin C
(b) Vitamin B_{12}
(c) Vitamin D
(d) Vitamin A

Ans: (b) Vitamin B_{12}

Q2. IM Board fav The large holes in Swiss cheese are due to CO_2 produced by:
(a) *Lactobacillus*
(b) *Saccharomyces cerevisiae*
(c) *Propionibacterium sharmanii*
(d) *Aspergillus niger*

Ans: (c) *Propionibacterium sharmanii*

Q3. IM Which of the following beverages is produced by distillation of the fermented broth?
(a) Wine
(b) Beer
(c) Toddy
(d) Rum

Ans: (d) Rum

Q4. IM Board fav Cyclosporin A, used as an immunosuppressive agent, is produced by:
(a) *Trichoderma polysporum*
(b) *Monascus purpureus*
(c) *Penicillium notatum*
(d) *Streptococcus*

Ans: (a) *Trichoderma polysporum*

Q5. IM The "clot buster" streptokinase is obtained from:
(a) a fungus
(b) a yeast
(c) a bacterium, *Streptococcus*
(d) a virus

Ans: (c) a bacterium, *Streptococcus*

Q6. IM Bottled fruit juices are clarified using:
(a) Lipases and amylases
(b) Pectinases and proteases
(c) Streptokinase
(d) Cellulases only

Ans: (b) Pectinases and proteases

Q7. IM Activated sludge is the sediment formed in the:
(a) primary settling tank
(b) anaerobic digester
(c) settling tank after secondary treatment
(d) river after discharge

Ans: (c) settling tank after secondary treatment

Q8. 1M Methanogens such as *Methanobacterium* grow:
(a) anaerobically on cellulosic material
(b) aerobically on proteins
(c) in aeration tanks
(d) only in soil
Ans: (a) anaerobically on cellulosic material

Q9. 1M Baculoviruses used as biocontrol agents mostly belong to the genus:
(a) *Bacillus*
(b) *Nucleopolyhedrovirus*
(c) *Trichoderma*
(d) *Glomus*
Ans: (b) *Nucleopolyhedrovirus*

Q10. 1M Board fav Mycorrhizal fungi of the genus *Glomus* mainly help the plant to absorb:
(a) nitrogen
(b) potassium
(c) water only
(d) phosphorus
Ans: (d) phosphorus

Assertion–Reason (1 mark each)

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

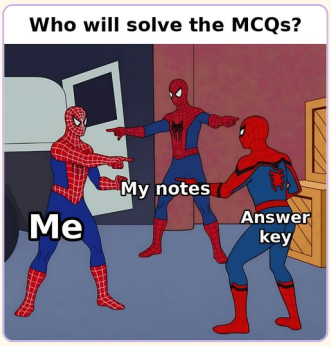
Q11. 1M Board fav **A:** The BOD of sewage falls sharply during secondary treatment. **R:** Aerobic microbes in flocs consume the major part of the organic matter.
Ans: (a) Both true and R explains A.

Q12. 1M **A:** Cattle dung is used to produce biogas. **R:** Dung is rich in methanogens that come from the rumen of cattle.
Ans: (a) Both true and R explains A.

Q13. 1M **A:** Bt toxin kills caterpillars but leaves other insects unharmed. **R:** Bt is a fungus that lives in the soil.
Ans: (c) A true, R false (Bt is a bacterium).

Q14. 1M **A:** Penicillin was a chance discovery. **R:** Fleming saw *Staphylococci* fail to grow around a mould in an unwashed culture plate.
Ans: (a) Both true and R explains A.

Q15. 1M **A:** Whisky and rum contain more alcohol than wine and beer. **R:** Whisky and rum are produced by distillation of the fermented broth.
Ans: (a) Both true and R explains A.



Short answers ki baari: notes aur answer key ek doosre ko point kar rahe hain.
Asli baat: Saare NCERT exercise questions neeche hain: khud likho, phir milao.

Short answer (2–3 marks)

Q16. 2M (NCERT) Bacteria cannot be seen with the naked eye. Which sample would you carry from home to the laboratory to show microbes under a microscope, and why?
Ans: A drop of **curd** (or idli-dosa batter). It contains millions of lactic acid bacteria (*Lactobacillus*), so they are easily seen under a microscope after staining.

Q17. 3M Board fav (NCERT) Give examples to prove that microbes release gases during metabolism.
Ans: (i) Idli-dosa dough and bread dough puff up due to CO_2 from bacteria and yeast. (ii) Large holes in Swiss cheese are made by CO_2 from *Propionibacterium sharmanii*. (iii) Methanogens produce methane, CO_2 and H_2 (biogas) in digesters and in the rumen.

Q18. 3M Board fav (NCERT) In which food would you find lactic acid bacteria? Mention some of their useful applications.
Ans: In **curd**. Uses: convert milk to curd by producing acids that coagulate and partially digest milk proteins; increase vitamin B_{12}; check disease-causing microbes in our stomach; commercial production of lactic acid.

Q19. 2M (NCERT) Name some traditional Indian foods made of wheat, rice and Bengal gram (or their products) which involve microbes.
Ans: Wheat: bread, bhatura, jalebi. Rice (with urad dal): idli, dosa, appam. Bengal gram (besan): dhokla, khaman. All use dough or batter fermented by bacteria or yeast.

Q20. 2M (NCERT) In which way have microbes played a major role in controlling diseases caused by harmful bacteria?
Ans: Microbes produce **antibiotics**, chemicals that kill or retard the growth of disease-causing microbes. E.g. penicillin from *Penicillium notatum*. Antibiotics control plague, whooping cough, diphtheria, leprosy and pneumonia.

Q21. 2M (NCERT) Name any two species of fungus used in the production of antibiotics.
Ans: *Penicillium notatum* (penicillin) and *Penicillium chrysogenum* (commercial penicillin). (Note: *Streptomyces*, which gives streptomycin, is a bacterium, not a fungus.)

Q22. 3M Board fav (NCERT) What is sewage? In which way can sewage be harmful to us?
Ans: Municipal waste water, mainly human excreta, with a lot of organic matter and microbes. It is harmful because it carries **pathogens** (water-borne diseases like cholera, typhoid) and, if discharged untreated, pollutes rivers and raises their BOD, killing aquatic life.

Q23. 2M Board fav (NCERT) What is the key difference between primary and secondary sewage treatment?
Ans: Primary treatment is **physical**: removal of floating debris and grit by filtration and sedimentation. Secondary treatment is **biological**: aerobic microbes (flocs) consume the organic matter, greatly reducing BOD.

Q24. 2M (NCERT) Do you think microbes can also be used as a source of energy? If yes, how?
Ans: Yes. Methanogens like *Methanobacterium* break down cattle dung and other cellulosic wastes anaerobically and produce **biogas** (mainly methane), used for cooking and lighting. Yeast also produces ethanol, used as a fuel.

Q25. 3M Board fav (NCERT) Microbes can be used to decrease the use of chemical fertilisers and pesticides. Explain how.
Ans: Biofertilisers (*Rhizobium*, *Azotobacter*, *Azospirillum*, cyanobacteria fix nitrogen; *Glomus* gives phosphorus) replace chemical fertilisers. Biocontrol agents (Bt, *Trichoderma*, baculoviruses) kill pests and pathogens, replacing pesticides.

Q26. 3M Board fav (NCERT) Water samples A, B and C (river water, untreated sewage, secondary effluent) have BOD 20, 8 and 400 mg/L. Which is most polluted? Label each, assuming river water is relatively clean.
Ans: Most polluted = **C (400 mg/L), untreated sewage**. Lowest BOD, **B (8 mg/L) = river water. A (20 mg/L) = secondary effluent.**

Q27. 2M Board fav (NCERT) Name the microbes from which Cyclosporin A and statins are obtained.
Ans: Cyclosporin A: fungus *Trichoderma polysporum*. Statins: yeast *Monascus purpureus*.

Q28. 3M (NCERT) Find out the role of microbes in (a) single cell protein (SCP) (b) soil.
Ans: (a) Microbes like *Spirulina* and yeasts are grown on cheap wastes to give protein-rich biomass used as food and feed. (b) In soil, microbes decompose organic matter, recycle nutrients, fix nitrogen and form humus, keeping soil fertile.

Q29. 3M (NCERT) Arrange biogas, citric acid, penicillin and curd in decreasing order of importance for human welfare. Give reasons.
Ans: **Penicillin > biogas > curd > citric acid.** Penicillin saves lives from infections; biogas is a clean rural fuel and gives manure; curd is nutritious food with vitamin B_{12}; citric acid is a food preservative and flavour.

Q30. 3M Board fav (NCERT) How do biofertilisers enrich the fertility of the soil?
Ans: *Rhizobium* (root nodules of legumes) and free-living *Azotobacter*, *Azospirillum* fix atmospheric nitrogen; cyanobacteria (*Anabaena*, *Nostoc*) fix nitrogen and add organic matter in paddy fields; mycorrhizal *Glomus* absorbs phosphorus for the plant.

Q31. 2M Board fav What is BOD? Why is the BOD of untreated sewage high?
Ans: BOD is the amount of oxygen consumed if all the organic matter in one litre of water were oxidised by bacteria. Untreated sewage has a lot of organic matter, so microbes need a lot of oxygen to oxidise it.

Q32. 2M Why is a small part of the activated sludge pumped back into the aeration tank?
Ans: It serves as the **inoculum**: it already contains the useful aerobic bacteria and fungi (flocs), so microbial growth and BOD reduction start quickly in the fresh effluent.

Q33. 3M Board fav How does Bt act on caterpillars? How has this been used in Bt-cotton?
Ans: Dried Bt spores are sprayed on plants and eaten by larvae; in the larval gut the toxin is released and kills them, sparing other insects. Bt toxin genes have been introduced into cotton, making Bt-cotton resistant to insect pests.

Q34. 2M Board fav What advantage does the fungus derive from the mycorrhizal association? What does the plant gain?
Ans: The fungus gets **food (sugars) and shelter** from the plant root. The plant gets **phosphorus** absorbed by the fungus, plus resistance to root-borne pathogens and tolerance to salinity and drought.

Q35. 2M Why are biogas plants built more often in rural areas?
Ans: Cattle are kept in large numbers in villages, so **dung** (rich in methanogens) is available in plenty. The biogas meets cooking and lighting needs, and the spent slurry is used as fertiliser in the fields.

Q36. 3M Name the gas produced in (a) idli-dosa dough (b) Swiss cheese (c) anaerobic sludge digester. Which microbes produce them?
Ans: (a) CO_2, by bacteria. (b) CO_2, by *Propionibacterium sharmanii*. (c) Methane with H_2S and CO_2, by anaerobic bacteria (methanogens).

Q37. 2M Board fav Why are baculoviruses considered excellent biocontrol agents?

Ans: They are **species-specific, narrow-spectrum** insecticides with no negative impact on plants, mammals, birds, fish or non-target insects, so they suit IPM programmes and ecologically sensitive areas. Most belong to the genus *Nucleopolyhedrovirus*.

Long answer (5 marks)

Q38. 5M Board fav Describe the steps of sewage treatment in an STP. What is the role of microbes in it? Mention the Ganga and Yamuna Action Plans.

Ans: **Primary:** floating debris removed by sequential filtration, grit by sedimentation; settled solids = primary sludge, supernatant = effluent. **Secondary:** effluent in aeration tanks with mechanical agitation and air; aerobic microbes grow as flocs and eat most of the organic matter, so BOD falls. In the settling tank, flocs sediment as activated sludge; a little is returned as inoculum, the rest goes to anaerobic sludge digesters where anaerobic bacteria digest it and release biogas (methane, H_2S, CO_2). Effluent, now with a far lower BOD, goes to rivers. **Role of microbes:** heterotrophic microbes do all the biological purification. **Action plans:** the Ministry of Environment and Forests started them to build many STPs, so only treated sewage reaches these rivers.

Q39. 5M Board fav (a) What are methanogens? Where are they found? (b) With a labelled diagram, describe a biogas plant. (c) Who developed this technology in India?

Ans: (a) Anaerobic bacteria that grow on cellulosic material and produce large amounts of methane with CO_2 and H_2, e.g. *Methanobacterium*; found in anaerobic sludge and the rumen of cattle (so in dung). (b) Diagram as Fig. 8.8: concrete tank 10–15 feet deep fed with dung slurry; floating gas holder (cover) rises as gas forms; gas outlet pipe to houses for cooking and lighting; spent slurry outlet, used as fertiliser. (c) IARI (Indian Agricultural Research Institute) and KVIC (Khadi and Village Industries Commission).

Q40. 5M Name the microbes and uses of: citric acid, acetic acid, butyric acid, lactic acid, ethanol, lipase, streptokinase, cyclosporin A and statins. How do statins act?

Ans: Citric acid: *Aspergillus niger*; acetic: *Acetobacter aceti*; butyric: *Clostridium butylicum*; lactic: *Lactobacillus*; ethanol: *Saccharomyces cerevisiae*. Lipase: in detergents for oily stains. Streptokinase (*Streptococcus*): clot buster after myocardial infarction. Cyclosporin A (*Trichoderma polysporum*): immunosuppressant in organ transplants. Statins (*Monascus purpureus*): lower cholesterol by **competitively inhibiting** the enzyme of cholesterol synthesis.

Q41. 5M Board fav What are biocontrol agents and biofertilisers? Explain with two examples each how they help organic farming.

Ans: **Biocontrol** = use of biological methods to control pests and diseases. (i) *Bacillus thuringiensis*: spores sprayed on plants; toxin released in the gut of caterpillars kills them. (ii) *Trichoderma*: free-living fungus of root ecosystems that controls several plant pathogens. (Also ladybird for aphids, baculoviruses for insects.) **Biofertilisers** = organisms that enrich soil nutrients. (i) *Rhizobium*: root nodules of legumes, fixes N_2. (ii) *Glomus* (mycorrhiza): absorbs phosphorus for the plant. (Also *Azotobacter*, cyanobacteria in paddy.) **Help:** they reduce toxic pesticides and chemical fertilisers, avoid pollution, keep pests at manageable levels and maintain soil fertility.

Case-based (4 marks)

Q42. 4M Board fav A farmer stops chemical pesticides and fertilisers. He sprays dried spores on his brassicas, releases ladybirds and dragonflies, and adds *Rhizobium* culture for his pulse crop and blue-green algae to his paddy field. (a) Which pest is controlled by ladybirds? (b) How do the sprayed spores kill caterpillars? (c) Why does the farmer not try to eradicate all pests? (d) Name the nitrogen fixers he used.

Ans: (a) Aphids. (b) Bt spores are eaten by larvae; the toxin released in their gut kills them. (c) Beneficial predators and parasites depend on pests as food or hosts; organic farming keeps pests at manageable levels. (d) *Rhizobium* (symbiotic) and cyanobacteria such as *Anabaena*, *Nostoc*.

Q43. 4M In a town's STP, the sewage entering has a BOD of 350 mg/L. After the aeration tank it is 15 mg/L. (a) What does this fall show? (b) What are flocs? (c) What is done with the activated sludge? (d) Which gases are released in the digester?

Ans: (a) Most of the organic matter has been consumed by microbes, so the effluent is much less polluting. (b) Masses of bacteria associated with fungal filaments forming mesh-like structures. (c) A small part returns to the aeration tank as inoculum; the rest goes to anaerobic sludge digesters. (d) Methane, hydrogen sulphide and CO_2 (biogas).

Exam Alert: last-minute pairs

- → Microbe + product table (Section 3C) se har saal 1–2 MCQ: type (bacterium/fungus/yeast) bhi yaad rakho.
- → Primary vs secondary treatment aur BOD wala NCERT numerical (8, 20, 400 mg/L) pakka aata hai.
- → Biogas plant ka diagram: floating cover, slurry inlet, gas outlet, sludge outlet: 4 labels zaroor.
- → Biocontrol (Bt, *Trichoderma*, baculovirus) aur biofertiliser (*Rhizobium*, *Glomus*, BGA) ke 2–2 examples har answer mein likho.
- → Penicillin story: Fleming, *Penicillium notatum*, Chain and Florey, Nobel 1945: ek line mein poora.

Biotechnology: Principles and Processes

Most asked: restriction enzymes and palindromes, pBR322 and insertional inactivation, gel electrophoresis, PCR, competent host, bioreactors and downstream processing

Biotechnology uses living organisms or their enzymes to make useful products. Its two pillars are **genetic engineering** (recombinant DNA technology) and **bioprocess engineering**.

1. What is Biotechnology?

Biotechnology deals with techniques of using **live organisms or enzymes from organisms** to produce products and processes useful to humans. Making curd, bread or wine by microbes is also biotechnology, but in a restricted sense; today the term mostly means processes using **genetically modified organisms** on a large scale.

Key terms: EFB definition

The **European Federation of Biotechnology (EFB)** defines biotechnology as 'the integration of natural science and organisms, cells, parts thereof, and molecular analogues for products and services'. It covers both traditional and modern molecular biotechnology. PYQ

Traditional biotechnology	Modern biotechnology
uses natural microbes as they are	uses genetically modified organisms (GMOs)
curd, bread, wine, cheese, vinegar	in vitro fertilisation (test-tube baby), gene synthesis, DNA vaccines, correcting a defective gene

Quick fact: milestones to remember

Year	Milestone
1963	two enzymes restricting phage growth isolated from *E. coli*: one methylates DNA, the other cuts it (restriction endonuclease)
1972	first recombinant DNA by **Cohen and Boyer**
1980s	PCR developed (Kary Mullis); Taq polymerase made it automatic
today	900+ restriction enzymes from 230+ bacterial strains

Two core techniques of modern biotechnology:

1. **Genetic engineering**: techniques to **alter the chemistry of genetic material** (DNA and RNA), introduce it into host organisms and thus change the phenotype of the host.
2. **Bioprocess engineering**: maintenance of a sterile (microbial contamination-free) ambience in chemical engineering processes, so that only the desired microbe or eukaryotic cell grows in large quantities to make antibiotics, vaccines, enzymes, etc.

Genetic engineering	Bioprocess engineering
changes the **genetic material** of an organism	grows the modified organism on a **large scale**
tools: restriction enzymes, ligase, vectors, host	tools: bioreactors, sterile media, downstream processing
output: recombinant DNA, GMO	output: antibiotics, vaccines, enzymes, hormones

2. Principles of Biotechnology

Sexual reproduction gives **variations** and new combinations of genes, while asexual reproduction preserves the genetic information. Traditional **hybridisation** often brings along undesirable genes with the desired ones.

Genetic engineering avoids this: techniques like **recombinant DNA**, **gene cloning** and **gene transfer** let us isolate and introduce only one or a set of desirable genes, without undesirable ones, into the target organism.

Quick fact: why alien DNA needs an ori

A piece of DNA transferred alone into an alien organism usually **cannot multiply**. It must be part of a chromosome or plasmid with a specific sequence called the origin of replication (ori), where replication starts. A gene linked to an ori can multiply and make many copies: this is **cloning** (making identical copies of a template DNA).

Key terms: first recombinant DNA (1972)

- → Made by **Stanley Cohen and Herbert Boyer** in 1972. PYQ
- → They cut a gene coding for **antibiotic resistance** from a native plasmid of *Salmonella typhimurium* and linked it to a plasmid vector.
- → Cutting was done with **restriction enzymes** ('molecular scissors'); joining with **DNA ligase**.
- → The recombinant plasmid was transferred into *Escherichia coli*, where it replicated using the host's **DNA polymerase** and made multiple copies.

Three basic steps in genetically modifying an organism:

Identify DNA with desirable gene(s) →
Introduce it into the host (via a vector) →
Maintain the DNA in the host and transfer it to its progeny

Key terms: words you must know

Term	Meaning
Recombinant DNA (rDNA)	DNA made by joining DNA from two different sources
Gene cloning	making many identical copies of a gene (or any DNA) in a host
Transformation	introducing a piece of DNA into a host bacterium
GMO / transgenic	organism whose genetic material has been altered by genetic engineering
Host	cell that receives and multiplies the rDNA, e.g. *E. coli*

3. Tools of Recombinant DNA Technology

The key tools are **restriction enzymes, polymerase enzymes, ligases, vectors** and the **host organism.**

Tool	Nickname	Job
Restriction endonuclease	molecular scissors	cuts DNA at a specific site
DNA ligase	molecular glue	joins cut DNA ends
Vector (plasmid, phage)	molecular vehicle	carries DNA into the host
DNA polymerase	DNA copier	makes copies of DNA
Host (*E. coli*)	living factory	multiplies rDNA, makes protein

A. Restriction enzymes (molecular scissors)

In **1963**, two enzymes responsible for restricting the growth of bacteriophage in *E. coli* were isolated. One **added methyl groups** to DNA; the other **cut DNA**. The latter was called restriction endonuclease.

→ The first restriction endonuclease was **Hind II**; it always cuts DNA at a particular point by recognising a specific sequence of six base pairs, its **recognition sequence.** PYQ

→ Today, **more than 900** restriction enzymes are known, isolated from over **230 strains** of bacteria; each recognises a different sequence.

Key terms: naming EcoRI

Letter(s)	Meaning
E	genus: *Escherichia*
c, o	species: *coli*
R	strain: RY 13
I (Roman numeral)	order of isolation from that strain

Restriction enzymes belong to the class **nucleases**:

Exonucleases	Endonucleases
remove nucleotides from the **ends** of DNA	cut at **specific positions within** the DNA
not used to make recombinant DNA	restriction endonucleases are the molecular scissors

Common restriction enzymes and their recognition sites (↓ = cut):

Enzyme	Source bacterium	Site (5′ to 3′)
EcoRI	*Escherichia coli* RY 13	G↓AATTC
BamHI	*Bacillus amyloliquefaciens* H	G↓GATCC
Hind III	*Haemophilus influenzae* Rd	A↓AGCTT
Pst I	*Providencia stuartii*	CTGCA↓G
Sal I	*Streptomyces albus*	G↓TCGAC
Sma I	*Serratia marcescens*	CCC↓GGG (blunt)

Yaad karo: naming ka formula

Genus ka 1 letter + species ke 2 letters + **strain** + **Roman number**. Jaise **Bam**HI = **B**acillus **am**yloliquefaciens, strain **H**, pehla enzyme (**I**). Hind III = **H**aemophilus **in**fluenzae strain **d**, teesra (**III**).

B. Palindromes and sticky ends

Each restriction endonuclease inspects the length of DNA, finds its recognition sequence, binds and cuts each of the two strands at specific points in the **sugar-phosphate backbone.** The recognition sequence is a palindromic nucleotide sequence.

Palindrome in DNA: a base-pair sequence that reads the same on both strands when the orientation of reading is kept the same (5′ to 3′). Example:

5′ – GAATTC – 3′
3′ – CTTAAG – 5′

Restriction enzymes cut a little **away from the centre** of the palindrome, but between the **same two bases** on both strands (EcoRI cuts between G and A). This leaves single-stranded overhangs called sticky ends. They form **hydrogen bonds** with complementary cut counterparts; this stickiness helps **DNA ligase** join them. PYQ

Sticky (cohesive) ends	Blunt ends
single-stranded overhangs left after a **staggered** cut	both strands cut at the **same point**, no overhang
e.g. EcoRI, BamHI, Hind III	e.g. Sma I
join easily by H-bonds + ligase	harder to join; ligation less efficient

Key terms: checking a palindrome

→ Write the complementary strand below the given strand.
→ Read the upper strand 5′ to 3′ (left to right) and the lower strand 5′ to 3′ (right to left).
→ If both read the same, it is a palindrome. **5′-GGATCC-3′**: complement 3′-CCTAGG-5′, read 5′ to 3′ = GGATCC. So it is a palindrome (BamHI site).
→ **5′-GATACC-3′**: complement read 5′ to 3′ = GGTATC, not the same, so **not** a palindrome.

Quick fact: counting fragments

→ A **linear** DNA with **n** sites for an enzyme gives **n + 1 fragments**.

→ A **circular** DNA (plasmid) with **n** sites gives **n fragments**. One site in a plasmid just opens (linearises) it.

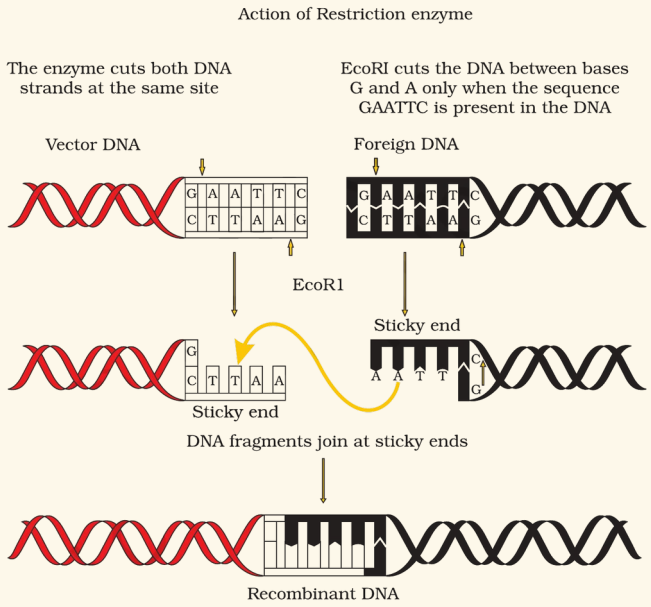


Fig. 9.1 Steps in formation of recombinant DNA by action of restriction endonuclease EcoRI

Exam Alert

→ Same restriction enzyme se **vector aur source DNA dono** kaato: tabhi same sticky ends banenge aur ligase jod payega.

→ Palindrome mein dono strands 5′ se 3′ padho: GAATTC dono par same.

→ EcoRI recognition site 6 bp, cut **G aur A ke beech**.

Paheli-Bujho

Bujho: Paheli, DNA ko kaatne ke liye kaunsi kainchi lagti hai?

Paheli: Restriction endonuclease! Ye palindrome dhoondh ke hi kaatta hai.

Bujho: Aur jodne ke liye? Fevicol?

Paheli: Haha, DNA ligase. Sticky ends ko wahi chipkata hai.

C. Separation of DNA fragments: gel electrophoresis

Restriction digestion gives DNA fragments, which are separated by **gel electrophoresis**:

→ DNA fragments are **negatively charged**, so in an electric field they move towards the **anode (positive electrode)**.

→ The commonly used matrix is **agarose**, a natural polymer extracted from **sea weeds**.

→ Fragments separate by **size**, due to the **sieving effect** of the gel: the **smaller** the fragment, the **farther** it moves. PYQ

→ Separated DNA is stained with **ethidium bromide** and seen under **UV radiation** as bright **orange coloured bands**.

→ The bands are cut out of the gel and extracted from the gel piece. This step is **elution**. Purified fragments are used to make recombinant DNA.

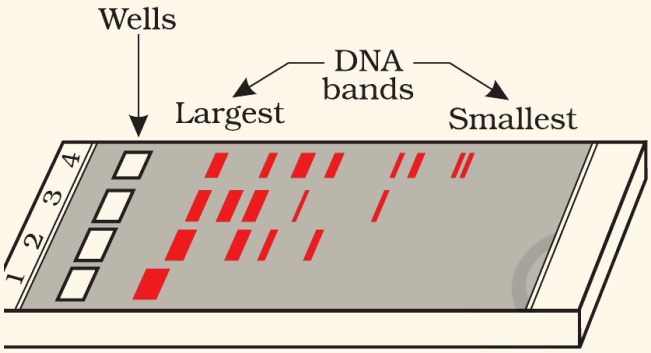


Fig. 9.3 Agarose gel: undigested DNA (lane 1) and digested DNA fragments (lanes 2 to 4)

Part of the set-up	Role
wells at the cathode end	DNA samples are loaded here
agarose gel in buffer	matrix with a sieving effect
cathode (−) and anode (+)	electric field; DNA runs to the anode
ethidium bromide + UV light	stains DNA: orange bands
elution	band cut out and DNA extracted for cloning

Yaad karo: gel ka rule

DNA negative hai, toh positive (anode) ki taraf bhaagta hai. Chhota fragment = halka = sabse aage. Bada fragment = mota = well ke paas atka. EtBr + UV = **orange** bands.

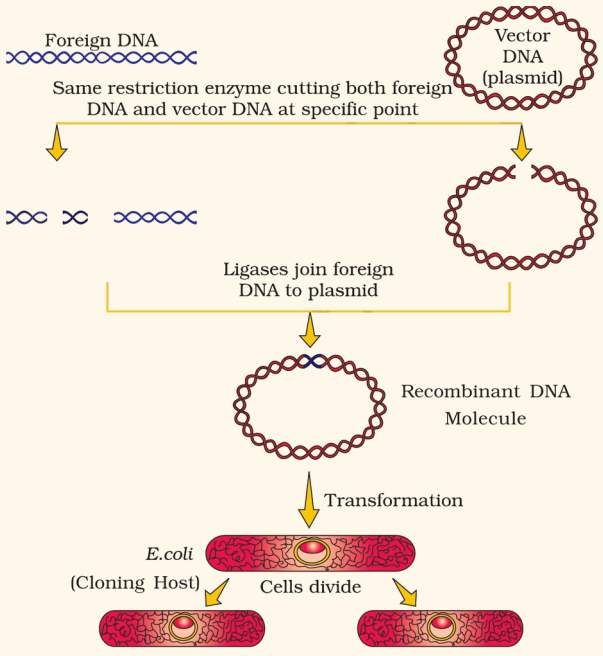


Fig. 9.2 Diagrammatic representation of recombinant DNA technology

Think and answer (NCERT in-text)

→ **At which end of the gel was the sample loaded?** At the **cathode (negative) end**, near the wells; DNA then runs towards the anode.

→ **Why do restriction enzymes not cut the bacterium's own DNA?** The bacterium's DNA is **methylated** by the second (methylase) enzyme, which protects it.

D. Cloning vectors

Plasmids and **bacteriophages** can replicate within bacterial cells independent of chromosomal DNA. Bacteriophages have **very high** copy numbers per cell; some plasmids have only one or two copies, others **15–100**. Linked alien DNA multiplies to the copy number of the vector.

Plasmid vector	Bacteriophage vector
small, circular, extra-chromosomal **dsDNA** of bacteria	virus that infects bacteria
copy number 1–2 to 15–100 per cell	**very high** copy number
e.g. **pBR322**, Ti plasmid	e.g. lambda phage, M13

Key terms: a vector

A **vector** is a DNA molecule (plasmid, phage, disarmed virus) that carries a foreign DNA segment into a host cell and lets it replicate there. The word means 'carrier'.

Key terms: copy number

Copy number = the number of copies of a plasmid (or phage) per cell. Since the ori controls it, a gene linked to a high-copy-number vector is present in many copies and gives more product.

Features required to facilitate cloning into a vector:

1. **Origin of replication (ori)**: sequence where replication starts; any DNA linked to it can replicate. It also controls the copy number, so a high-copy ori is used to get many copies of the target DNA.
2. **Selectable marker**: helps **identify and eliminate non-transformants** and selectively permit the growth of transformants. Genes for resistance to **ampicillin, chloramphenicol, tetracycline or kanamycin** are useful markers for *E. coli* (normal *E. coli* has none of these).
3. **Cloning sites**: the vector needs very few, preferably single, recognition sites for the common restriction enzymes; more than one site makes several fragments and complicates cloning. The foreign DNA is ligated at a site within one of the two antibiotic resistance genes.
4. **Vectors for cloning genes in plants and animals** (below).

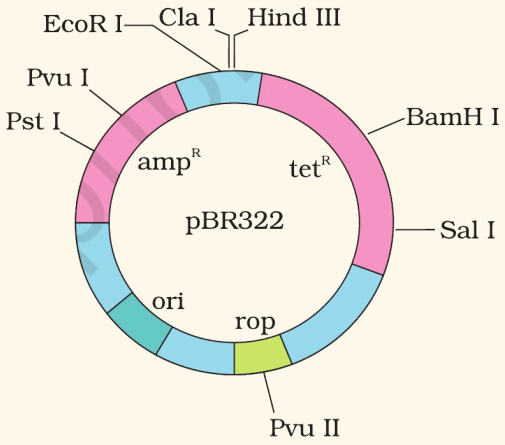


Fig. 9.4 E. coli cloning vector pBR322: restriction sites (Hind III, EcoRI, BamHI, Sal I, Pvu II, Pst I, Cla I), ori, antibiotic resistance genes (amp^R, tet^R) and rop

Part of pBR322	Role
ori	origin of replication
amp^R, tet^R	ampicillin and tetracycline resistance genes (selectable markers)
rop	codes for proteins involved in **replication of the plasmid**
BamHI, Sal I (in tet^R); Pvu I, Pst I (in amp^R)	cloning sites inside resistance genes

Yaad karo: vector ke 3 zaroori features

O-S-C = **O**ri (copy machine), **S**electable marker (pehchaan patra), **C**loning site (ek hi darwaza). pBR322 mein: ori + amp^R + tet^R + rop, aur ek-ek baar aane wale sites.

E. Insertional inactivation

Suppose foreign DNA is ligated at the **BamHI site of the tetracycline resistance gene** in pBR322. The tet^R gene is now inactivated, but amp^R still works:

Cell type	Ampicillin medium	Tetracycline medium
Non-transformant (no plasmid)	dies	dies
Transformant with **non-recombinant** plasmid	grows	grows
Transformant with **recombinant** plasmid	**grows**	**dies**

So recombinants are picked by **replica plating** on the two media. Because this needs simultaneous plating on two antibiotics, it is **cumbersome**; alternative markers were developed.

Key terms: blue-white selection

→ The recombinant DNA is inserted within the coding sequence of the enzyme β-galactosidase, which inactivates the gene: **insertional inactivation.** PYQ

→ Cells are grown on a medium with a **chromogenic substrate.**

→ **Non-recombinant** colonies have an active enzyme and give blue colour; **recombinant** colonies **do not** produce colour (white).

Transformant	Recombinant
cell that has taken up **any** plasmid (with or without insert)	cell whose plasmid carries the **foreign DNA insert**
picked by the selectable marker (e.g. ampicillin)	picked by insertional inactivation (tetracycline, blue-white)
all recombinants are transformants	not all transformants are recombinants

Paheli-Bujho

Bujho: Blue colony aaya, matlab mera gene andar chala gaya?
Paheli: Nahi! Blue matlab enzyme kaam kar raha hai, gene andar nahi gaya.
Bujho: Toh recombinant ka colour?
Paheli: Safed (colourless). Insert ne β-galactosidase gene tod diya.

Selectable marker	Reporter gene
identifies **transformants** (cells with a plasmid)	identifies **recombinants** (plasmid with insert)
usually antibiotic resistance (amp^R, tet^R)	e.g. β-galactosidase (lacZ): colour change
non-transformants die on the antibiotic	recombinants give white (colourless) colonies

F. Vectors for plants and animals

Bacteria and viruses have long known how to deliver genes into plant and animal cells, which transforms the host cells. Scientists use these natural vectors after making them harmless (disarmed).

- → ***Agrobacterium tumefaciens***: a pathogen of several **dicot** plants; it delivers a piece of DNA, **T-DNA**, that transforms normal plant cells into a **tumour** and directs them to make chemicals the pathogen needs.
- → Its tumour inducing (Ti) plasmid has been modified into a cloning vector that is **no longer pathogenic** but still delivers genes of our interest into plants. So *Agrobacterium* is called a **natural genetic engineer** of plants. PYQ
- → **Retroviruses** in animals (and humans) can transform normal cells into **cancerous** cells. Disarmed retroviruses are now used to deliver desirable genes into animal cells.

Exam Alert

- → Ti = **Tumour inducing** plasmid; disarm hone ke baad bhi gene deliver karta hai par tumour nahi banata.
- → Agrobacterium **dicots** ka pathogen hai; T-DNA plant cell ke genome mein judta hai.
- → Retrovirus: animals mein cancer laata hai; disarmed retrovirus = gene therapy ka vector.

G. Competent host (for transformation with recombinant DNA)

DNA is a **hydrophilic** molecule and cannot pass through cell membranes, so bacterial cells must be made competent to take up DNA.

1. Treat cells with a specific concentration of a **divalent cation such as calcium**; this increases the efficiency with which DNA enters through pores in the cell wall.
2. Incubate the cells with recombinant DNA **on ice**.
3. Give a brief heat shock at 42 °C.
4. Put them back **on ice**. This enables the bacteria to take up the recombinant DNA.

Method (other hosts)	How DNA is introduced
Micro-injection	recombinant DNA injected directly into the **nucleus of an animal cell**
Biolistics (gene gun)	plant cells bombarded with high velocity **micro-particles of gold or tungsten** coated with DNA
Disarmed pathogen vectors	allowed to infect the cell; they transfer the recombinant DNA into the host

Yaad karo: competent banane ka formula

Calcium + Thanda + Garam + Thanda: Ca^{2+}, ice, 42 °C heat shock, phir ice. Plant ke liye **gold/tungsten ki goli** (gene gun), animal ke liye **injection** (micro-injection).

Quick fact: why *E. coli* is the favourite host

- → Grows fast (divides about every 20 minutes) on cheap media.
- → Its genetics and plasmids are very well known.
- → Easily made competent and transformed; takes up plasmids like pBR322.

4. Processes of Recombinant DNA Technology

Isolate DNA → Cut with restriction enzyme → Amplify gene (PCR) → Ligate into vector → Transfer into host → Culture host on large scale → Extract product

Step	Tool / chemical used
Isolation of DNA	lysozyme, cellulase, chitinase; RNase, protease; chilled ethanol
Cutting	restriction endonuclease (same for vector and gene)
Checking the cut	agarose gel electrophoresis, EtBr, UV
Amplification	PCR: primers, Taq polymerase, nucleotides
Joining	DNA ligase
Transfer	competent cells (Ca^{2+}, heat shock), micro-injection, gene gun, disarmed vectors
Selection	antibiotic marker, insertional inactivation, blue-white
Large-scale culture	bioreactor
Finishing	downstream processing

A. Isolation of the genetic material (DNA)

DNA is enclosed within membranes and lies with other macromolecules (RNA, proteins, polysaccharides, lipids). To get pure DNA, the cell is broken open with enzymes:

Cell	Main wall material	Enzyme
Bacterium	peptidoglycan	**lysozyme**
Plant cell	cellulose	**cellulase**
Fungus	chitin	**chitinase**

→ RNA is removed by **ribonuclease**; proteins by **protease**; other molecules by suitable treatments.

→ Purified DNA finally precipitates on adding chilled ethanol. It is seen as a collection of **fine threads** in the suspension and can be removed by **spooling** (winding it on a glass rod). PYQ

Think and answer

Why are ribonuclease and protease added? Because RNA and proteins (including histones wrapped around DNA) would contaminate the DNA and interfere with its cutting and ligation; removing them gives pure DNA that can be cut and joined cleanly.

B. Cutting of DNA at specific locations

Purified DNA is incubated with the restriction enzyme at **optimal conditions**. The digestion is checked by **agarose gel electrophoresis** (DNA moves to the anode). The vector DNA is cut with the same enzyme. The cut gene of interest and the cut vector are mixed with **DNA ligase** to make recombinant DNA.

C. Amplification of the gene of interest using PCR

PCR = Polymerase Chain Reaction. Multiple copies of the gene (or DNA) of interest are synthesised **in vitro** using two sets of **primers** (small chemically synthesised oligonucleotides complementary to regions of DNA) and the enzyme **DNA polymerase**. It runs automatically in a machine called a **thermal cycler**, within a few hours. PYQ

Step	What happens
1. **Denaturation**	double-stranded DNA is separated by high temperature
2. **Annealing**	the two primers bind to the complementary regions of the single strands
3. **Extension**	DNA polymerase extends the primers using nucleotides and the genomic DNA as template
Repeat	each cycle doubles the target DNA; usually 25–30 cycles are run

→ The cycle is repeated many times; the DNA segment can be amplified to approximately a billion times (1 billion copies).

→ A **thermostable DNA polymerase**, isolated from the bacterium *Thermus aquaticus* (**Taq polymerase**), is used: it stays active at the high temperature of denaturation.

→ The amplified fragment can then be ligated with a vector for further cloning.

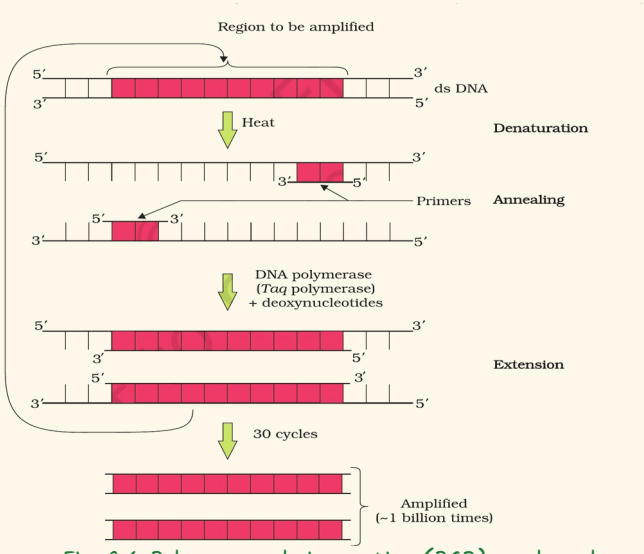


Fig. 9.6 Polymerase chain reaction (PCR): each cycle has three steps: (i) denaturation (ii) primer annealing (iii) extension of primers

Quick fact: uses of PCR

→ Amplifies a **very small amount of DNA** (even from one cell) for cloning.

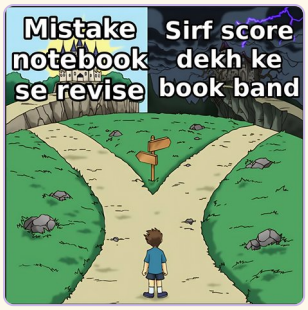
→ **Early diagnosis** of infections like HIV and of genetic mutations, before symptoms appear (Chapter 10).

→ **DNA fingerprinting** and forensic tests (Chapter 5).

→ Detecting **mutations** in genes of suspected cancer patients.

Yaad karo: PCR = DAE

Denaturation (garam karke strands alag) → Annealing (primer chipke) → Extension (Taq ne strand banaya). **"DAE" = Daddy Aaj Exam!** Har cycle mein DNA double: 30 cycles mein lagbhag ek billion copies ($2^{30} \approx 10^9$).



PCR ki 3 steps galat order mein likhi? Mistake notebook mein daalo, warna score dekh ke bhool jaoge.

Asli baat: Denaturation, Annealing, Extension (D-A-E). Taq polymerase garmi mein bhi nahi tootta.

D. Insertion of recombinant DNA into the host cell or organism

Recipient cells are made competent and the recombinant DNA is introduced. If the rDNA carries a gene for **ampicillin resistance**, transformants are selected by spreading cells on **agar plates with ampicillin**: only transformants grow; untransformed cells die. The ampicillin resistance gene is the **selectable marker**.

Gene cloning in a host	*PCR*
in vivo (inside living cells)	**in vitro** (in a tube)
needs vector, host, ligase	needs primers, template, Taq polymerase, nucleotides
slower (days)	fast (hours); about a billion copies

E. Obtaining the foreign gene product

When a cloned gene is expressed under suitable conditions, it makes the protein. A protein encoded by a **recombinant gene** expressed in a heterologous host is a recombinant protein. Cells can be grown in the laboratory, or continuously in a **continuous culture** system, where used medium is drained out from one side and fresh medium added from the other, keeping cells in their **physiologically most active log (exponential) phase**; this gives larger biomass and higher yields.

For large-scale production, bioreactors are used: large volumes (**100–1000 litres**) of culture can be processed. A bioreactor provides optimal conditions for the desired product: temperature, pH, substrate, salts, vitamins, oxygen.

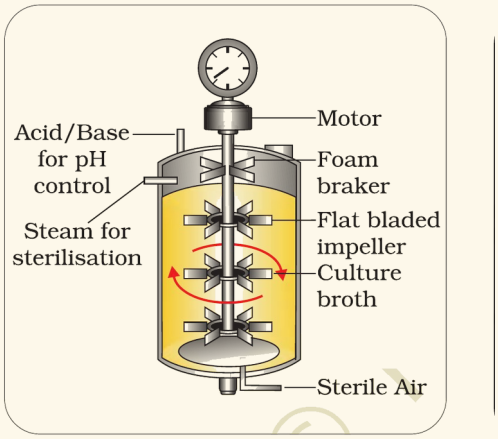


Fig. 9.7 (a) Simple stirred-tank bioreactor

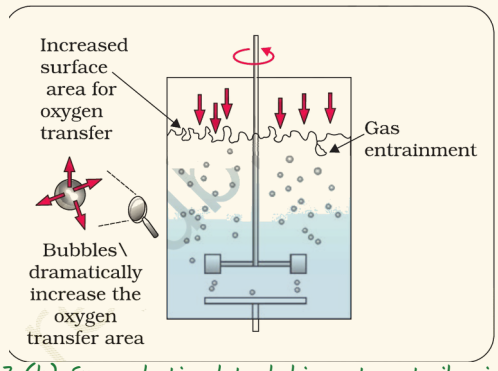


Fig. 9.7 (b) Sparged stirred-tank bioreactor: sterile air bubbles increase the oxygen transfer area

Stirred-tank bioreactor feature	*Function*
cylindrical, or with a **curved base**	helps mixing of the reactor contents
stirrer	even mixing and oxygen availability throughout
sparging (sterile air bubbled in)	increases the oxygen transfer area

Simple stirred-tank	*Sparged stirred-tank*
oxygen mainly through stirring	sterile **air bubbles** sparged through the culture
smaller oxygen transfer area	larger oxygen transfer area, better aeration
flat-bladed impeller mixes the medium	stirrer + sparger give even mixing and oxygen

Shake flask	*Bioreactor*
small volume (100–1000 mL)	large volume (100–1000 L)
poor aeration and mixing	better aeration and mixing
temperature, pH, foam not controlled	temperature, pH, foam, oxygen controlled
no sampling during growth	sampling ports; continuous culture possible

A bioreactor has:

- ✓ an agitator system
- ✓ an oxygen delivery system
- ✓ a foam control system
- ✓ a temperature control system
- ✓ a pH control system
- ✓ sampling ports to withdraw small volumes of culture

F. Downstream processing

After biosynthesis, the product must undergo a series of processes before it is a finished product ready for marketing. These are together called downstream processing: PYQ

→ **Separation and purification** of the product.

→ **Formulation** with suitable **preservatives**.

→ For drugs, **clinical trials** are essential, and strict quality control testing is needed for each product; it varies from product to product.

Downstream step	Why it is needed
Separation	remove cells and medium from the product
Purification	get the product free of other proteins and contaminants
Formulation	add preservatives so the product stays stable
Clinical trials	prove a drug is safe and effective in humans
Quality control	every batch must meet set standards

Exam Alert: bioreactor diagram

→ Labels: motor, flat-bladed impeller (stirrer), foam breaker, culture broth, sparger (air in), steam/sterile air, acid/base for pH, sampling port.

→ Simple vs sparged: sparged mein **air bubbles** se oxygen transfer area badhta hai.

Product in bioreactor → Separation → Purification → Formulation (preservatives) → Clinical trials (drugs) → Quality control → Market

Key terms: enzymes of this chapter

Enzyme	Job
Restriction endonuclease	cuts DNA at palindromic sites
Exonuclease	removes nucleotides from DNA ends
DNA ligase	joins DNA fragments (sticky ends)
DNA polymerase / Taq polymerase	synthesises DNA; extends primers in PCR
Lysozyme / cellulase / chitinase	break bacterial / plant / fungal cell walls
Ribonuclease / protease	remove RNA / proteins during DNA isolation
β-galactosidase	reporter enzyme in blue-white selection

Paheli-Bujho

Bujho: Bioreactor se product nikal gaya, ab seedha dukaan pe bech do!

Paheli: Ruko! Pehle downstream processing: separation, purification, formulation.

Bujho: Aur medicine ho toh?

Paheli: Toh clinical trials aur quality control bhi zaroori.

Key terms: where this chapter leads (Chapter 10)

→ **Human insulin** made in *E. coli* using rDNA (Eli Lilly, 1983).

→ **Bt cotton**: Bt toxin gene put into cotton using these tools.

→ **Gene therapy**: a correct gene delivered by a disarmed virus vector.

→ **Molecular diagnosis** by PCR, and transgenic animals.

NCERT lines that become 1-mark questions

→ Agarose is a natural polymer extracted from **sea weeds**.

→ Bacteriophages have very high copy numbers; plasmids 1-2 to 15-100.

→ DNA is hydrophilic, so it cannot pass through cell membranes.

→ Bioreactors process **100-1000 litres** of culture.

→ PCR can amplify DNA about **a billion** times.

→ The ori also controls the **copy number** of linked DNA.

→ Restriction enzymes cut the **sugar-phosphate backbone** of each strand.

→ Biolistics uses micro-particles of **gold or tungsten**.

→ Bacteria are made competent with the divalent cation **calcium**.

Exam Alert: common mistakes

→ DNA anode ki taraf jaata hai (cathode nahi): DNA negatively charged hai.

→ Recombinant colonies **white** hote hain, blue nahi.

→ Insert BamHI site par = **tetracycline** resistance gaya; ampicillin wala bacha.

→ Hind II pehla enzyme tha; Hind III alag enzyme hai.

→ Taq polymerase bacterium *Thermus aquaticus* se, fungus se nahi.

Chapter at a glance	Key points
Isolation	lysozyme / cellulase / chitinase, RNase, protease, chilled ethanol
Tools	restriction enzymes, ligase, polymerase, vectors, competent host
Separation	agarose gel electrophoresis, EtBr + UV, elution
Vectors	pBR322 (ori, rop, amp^R, tet^R), Ti plasmid, disarmed retroviruses
Selection	antibiotic markers, insertional inactivation, blue-white
Gene transfer	Ca^{2+} + heat shock, micro-injection, biolistics, disarmed vectors
Process	isolate, cut, PCR, ligate, transform, culture, downstream processing
Scale-up	stirred-tank and sparged bioreactors, continuous culture

1 minute recap

1. Biotechnology (EFB definition); two core techniques: genetic engineering and bioprocess engineering.
2. First rDNA: Cohen and Boyer, 1972 (*Salmonella typhimurium* plasmid gene into *E. coli*).
3. Restriction endonucleases (first Hind II) cut palindromes, give sticky ends; ligase joins; EcoRI cuts GAATTC between G and A.
4. Gel electrophoresis: DNA to anode, agarose (sea weed), smaller moves farther; EtBr + UV orange bands; elution.
5. Vector needs ori, selectable marker, single cloning sites; pBR322 (amp^R, tet^R, rop); insertional inactivation; blue-white selection.
6. Ti plasmid of *Agrobacterium*, disarmed retroviruses; competent cells by Ca^{2+}, ice, 42 °C heat shock; micro-injection, biolistics.
7. Process: isolate DNA (lysozyme, cellulase, chitinase; chilled ethanol), cut, PCR (denaturation, annealing, extension; Taq), ligate, transform, culture in bioreactor, downstream processing.
8. Bioreactors (100–1000 L): simple vs sparged stirred-tank; continuous culture keeps cells in the log phase.

Most Important Questions

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM Board fav The first recombinant DNA was constructed by linking an antibiotic resistance gene with a native plasmid of:
(a) *E. coli*
(b) *Salmonella typhimurium*
(c) *Agrobacterium tumefaciens*
(d) *Thermus aquaticus*
Ans: (b) *Salmonella typhimurium* (Cohen and Boyer, 1972)

Q2. IM The first restriction endonuclease discovered was:
(a) EcoRI
(b) BamHI
(c) Hind II
(d) Pst I
Ans: (c) Hind II (recognises a 6 bp sequence)

Q3. IM Board fav In EcoRI, the letter 'R' stands for:
(a) restriction
(b) recombinant
(c) the order of isolation
(d) the strain RY 13
Ans: (d) the strain RY 13 (E = genus, co = species)

Q4. IM Board fav Which of these is a palindromic sequence?
(a) 5′-GAATTC-3′ / 3′-CTTAAG-5′
(b) 5′-GATACC-3′ / 3′-CTATGG-5′
(c) 5′-GACCTT-3′ / 3′-CTGGAA-5′
(d) 5′-GGATCA-3′ / 3′-CCTAGT-5′
Ans: (a) 5′-GAATTC-3′ / 3′-CTTAAG-5′

Q5. IM In agarose gel electrophoresis, DNA fragments move towards the:
(a) cathode, as DNA is positively charged
(b) anode, as DNA is negatively charged
(c) cathode, as DNA is negatively charged
(d) anode, as DNA is positively charged
Ans: (b) anode, as DNA is negatively charged

Q6. IM Separated DNA bands are seen as orange bands after staining with:
(a) iodine
(b) safranin
(c) ethidium bromide and exposure to UV
(d) methylene blue
Ans: (c) ethidium bromide and exposure to UV

Q7. IM In pBR322, the gene 'rop' codes for proteins involved in:
(a) replication of the plasmid
(b) ampicillin resistance
(c) tetracycline resistance
(d) restriction
Ans: (a) replication of the plasmid

Q8. IM Board fav Foreign DNA is inserted at the BamHI site of pBR322. The recombinant cells will:
(a) grow on both media
(b) grow on neither medium
(c) grow on ampicillin but not on tetracycline
(d) grow on tetracycline but not on ampicillin
Ans: (c) grow on ampicillin but not on tetracycline

Q9. IM Bacterial cells are made competent by treating them with:
(a) sodium chloride
(b) calcium (a divalent cation)
(c) chilled ethanol
(d) lysozyme
Ans: (b) calcium (a divalent cation)

Q10. IM Fungal cells are broken open to isolate DNA using:
(a) lysozyme
(b) cellulase
(c) protease
(d) chitinase
Ans: (d) chitinase

Q11. IM Board fav Taq polymerase is obtained from:
(a) *Thermus aquaticus*
(b) *Thermococcus*
(c) *E. coli*
(d) *Bacillus thuringiensis*
Ans: (a) *Thermus aquaticus*

Q12. IM The correct order of steps in one PCR cycle is:
(a) annealing, denaturation, extension
(b) denaturation, annealing, extension
(c) extension, annealing, denaturation
(d) denaturation, extension, annealing
Ans: (b) denaturation, annealing, extension

Q13. IM Which enzyme is called 'molecular glue'?
(a) DNA polymerase
(b) DNA ligase
(c) Restriction endonuclease
(d) Exonuclease
Ans: (b) DNA ligase

Q14. IM The matrix used in gel electrophoresis of DNA is obtained from:
(a) bacteria
(b) fungi
(c) sea weeds
(d) animal bones
Ans: (c) sea weeds

Q15. IM Board fav A circular plasmid has three EcoRI sites. Complete digestion gives:
(a) 3 fragments
(b) 4 fragments
(c) 2 fragments
(d) 6 fragments
Ans: (a) 3 fragments

Q16. IM Plant cells are bombarded with DNA-coated gold or tungsten particles in:
(a) micro-injection
(b) electroporation
(c) heat shock
(d) biolistics
Ans: (d) biolistics

Q17. IM Which of these is NOT a feature needed in a cloning vector?
(a) origin of replication
(b) selectable marker
(c) many recognition sites for the same enzyme
(d) cloning site in a marker gene
Ans: (c) many recognition sites for the same enzyme

Q18. IM In sparged stirred-tank bioreactors, sparging means:
(a) bubbling sterile air through the culture
(b) removing foam
(c) adding fresh medium
(d) controlling pH
Ans: (a) bubbling sterile air through the culture

Q19. IM The *Agrobacterium* plasmid used as a plant vector is:
(a) pBR322
(b) Ti plasmid
(c) 2-micron plasmid of yeast
(d) F plasmid
Ans: (b) Ti plasmid

Q20. IM During DNA isolation, DNA is finally precipitated by adding:
(a) warm water
(b) ethidium bromide
(c) chilled ethanol
(d) calcium chloride
Ans: (c) chilled ethanol

Q21. IM Which of the following is used to cut DNA at specific sites?
(a) Restriction endonuclease
(b) Exonuclease
(c) DNA ligase
(d) DNA polymerase
Ans: (a) Restriction endonuclease

Q22. IM Retroviruses used as vectors in animal cells are first:
(a) mutated to cause cancer
(b) heat killed
(c) disarmed
(d) methylated
Ans: (c) disarmed

Q23. IM Board fav The temperature used for heat shock while making cells competent is:
(a) 4 °C
(b) 42 °C
(c) 72 °C
(d) 94 °C
Ans: (b) 42 °C

Q24. IM The process of removing separated DNA bands from the gel is:
(a) spooling
(b) annealing
(c) ligation
(d) elution
Ans: (d) elution

Q25. 1M The number of copies of a plasmid in a cell is controlled by:
(a) ori
(b) rop
(c) amp^R
(d) tet^R
Ans: (a) ori

Q26. 1M Which enzyme is used to break the bacterial cell wall during DNA isolation?
(a) Cellulase
(b) Lysozyme
(c) Chitinase
(d) Protease
Ans: (b) Lysozyme

Assertion–Reason (1 mark each)

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q27. 1M Board fav **A:** The same restriction enzyme is used to cut both the vector and the source DNA. **R:** It produces the same kind of sticky ends that can be joined by DNA ligase.
Ans: (a) Both true and R explains A.

Q28. 1M **A:** In gel electrophoresis, smaller DNA fragments move farther. **R:** Agarose gel has a sieving effect.
Ans: (a) Both true and R explains A.

Q29. 1M **A:** Bacterial cells must be made competent before transformation. **R:** DNA is a hydrophobic molecule.
Ans: (c) A true, R false (DNA is hydrophilic, so it cannot pass the cell membrane).

Q30. 1M Board fav **A:** Taq polymerase is used in PCR. **R:** It remains active at the high temperature used to denature DNA.
Ans: (a) Both true and R explains A.

Q31. 1M **A:** In blue-white selection, recombinant colonies are blue. **R:** The insert inactivates the β-galactosidase gene.
Ans: (d) A false, R true (recombinants are colourless).

Q32. 1M **A:** Restriction enzymes do not cut the DNA of the bacterium that makes them. **R:** The bacterial DNA is protected by methylation.
Ans: (a) Both true and R explains A.

Q33. 1M **A:** pBR322 is a good cloning vector. **R:** It has two antibiotic resistance genes with unique restriction sites in them.
Ans: (a) Both true and R explains A.

Jab PCR, pBR322 aur gel teeno diagram ke saath likh diye.
Asli baat: Long answers mein labelled diagram + steps = full marks.

Short answer (2–3 marks)

Q34. 3M (NCERT) List some recombinant proteins used in medical practice and their uses.
Ans: Human insulin (diabetes); human growth hormone (dwarfism); interferons (viral infections, some cancers); tissue plasminogen activator and streptokinase (dissolve blood clots); erythropoietin (anaemia); hepatitis B vaccine (immunisation); blood clotting factor VIII (haemophilia); interleukin-2 (cancer therapy); alpha-1 antitrypsin (emphysema); calcitonin (bone disorders).

Q35. 3M Board fav (NCERT) Make a chart showing a restriction enzyme, its substrate DNA, the site where it cuts and the product.
Ans: Enzyme: **EcoRI**. Substrate: DNA with 5′-GAATTC-3′. Cut: between G and A on both strands. Product: two fragments with single-stranded overhangs (sticky ends) 5′-AATT. (Draw as Fig. 9.1.)

Q36. 2M (NCERT) Are enzymes bigger or DNA bigger in molecular size? How do you know?
Ans: DNA is bigger. A restriction enzyme binds to only a short recognition sequence (about 6 bp) of a long DNA molecule and moves along it, inspecting its length, so DNA is far longer than the enzyme.

Q37. 2M (NCERT) What would be the molar concentration of human DNA in a human cell?
Ans: A diploid human cell has about 6.6×10^9 bp. Moles of bp = $6.6 \times 10^9 / 6.022 \times 10^{23} \approx 1.1 \times 10^{-14}$ mol per cell. The molar concentration is this amount divided by the nuclear volume in litres.

Q38. 2M (NCERT) Do eukaryotic cells have restriction endonucleases? Justify.
Ans: No. Restriction endonucleases are found in bacteria, where they protect the cell against bacteriophages by cutting foreign DNA, while the cell's own DNA is protected by methylation. Eukaryotes do not use this defence.

Q39. 3M Board fav (NCERT) Besides better aeration and mixing, what other advantages do stirred-tank bioreactors have over shake flasks?
Ans: Large volumes (100–1000 L) can be processed; temperature, pH, foam and oxygen can be controlled; small samples can be withdrawn through sampling ports; continuous culture is possible, giving higher yields.

Q40. 2M (NCERT) Give five examples of palindromic sequences.
Ans: 5′-GAATTC-3′ (EcoRI); 5′-GGATCC-3′ (BamHI); 5′-AAGCTT-3′ (Hind III); 5′-CTGCAG-3′ (Pst I); 5′-GTCGAC-3′ (Sal I). Each reads the same on the complementary strand from 5′ to 3′. A new one can be made by writing any 3 bases and then their complements in reverse, e.g. ATG + CAT = ATGCAT.

Q41. 2M (NCERT) Recall meiosis: at what stage is a 'recombinant DNA' made naturally?
Ans: At **pachytene of prophase I**, when crossing over between non-sister chromatids of homologous chromosomes produces recombinant chromatids. The enzyme **recombinase** helps, and the chiasmata seen at diplotene mark these sites.

Q42. 3M Board fav (NCERT) How can a reporter enzyme be used to monitor transformation, in addition to a selectable marker?
Ans: A reporter gene such as β-galactosidase (lacZ) is placed so that the insert inactivates it. On a chromogenic substrate, non-recombinant cells give blue colonies and recombinant cells give white colonies, so recombinants are identified visually.

Q43. 3M Board fav (NCERT) Describe briefly: (a) origin of replication (b) bioreactors (c) downstream processing.
Ans: (a) Sequence where replication starts; it also controls the copy number of linked DNA. (b) Large vessels (100–1000 L) giving optimal temperature, pH, oxygen and nutrients for growing cells and making products. (c) Separation, purification and formulation with preservatives of the product, plus clinical trials and quality control, before marketing.

Q44. 3M (NCERT) Explain briefly: (a) PCR (b) restriction enzymes and DNA (c) chitinase.
Ans: (a) In vitro amplification of DNA by repeated cycles of denaturation, annealing of primers and extension by Taq polymerase. (b) Restriction enzymes recognise palindromic sites in DNA and cut both strands, leaving sticky ends. (c) Enzyme that breaks the chitin cell wall of fungi to release DNA during its isolation (like lysozyme for bacteria and cellulase for plants).

Q45. 3M Board fav (NCERT) Distinguish between (a) plasmid DNA and chromosomal DNA (b) RNA and DNA (c) exonuclease and endonuclease.
Ans: (a) Plasmid: small, circular, extra-chromosomal, replicates independently, non-essential genes; chromosomal: large, carries essential genes. (b) RNA: ribose, uracil, usually single-stranded; DNA: deoxyribose, thymine, double-stranded. (c) Exonuclease removes nucleotides from the ends; endonuclease cuts at specific positions within DNA.

Q46. 2M Board fav What are sticky ends? How are they useful?
Ans: Single-stranded overhangs left when a restriction enzyme cuts both strands a little away from the centre of a palindrome. They form hydrogen bonds with complementary cut ends, so DNA ligase can join vector and foreign DNA easily.

Q47. 2M Why should a cloning vector have only one recognition site for a restriction enzyme?
Ans: More than one site would cut the vector into several fragments, which complicates gene cloning. A single site opens the vector at one point so that the foreign DNA is inserted exactly there.

Q48. 3M Board fav What is insertional inactivation? How does it help in selecting recombinants?
Ans: Insertion of foreign DNA within a gene (e.g. tet^R or β-galactosidase) inactivates that gene. Recombinants lose that function (no tetracycline resistance, or no blue colour), so they can be told apart from non-recombinants.

Q49. 3M Board fav How is *Agrobacterium tumefaciens* used as a vector?
Ans: It naturally transfers T-DNA from its Ti plasmid into dicot plant cells, causing tumours. The Ti plasmid is disarmed (made non-pathogenic) and the gene of interest is placed in it, so it delivers the gene into the plant without causing disease.

Q50. 2M Why is chilled ethanol added during DNA isolation?
Ans: Purified DNA precipitates out in chilled ethanol as fine threads, which can be removed by spooling.

Q51. 2M Name the enzymes used to remove RNA and protein during DNA isolation, and those to break bacterial and plant cells.
Ans: RNA: ribonuclease; protein: protease; bacterial cell wall: lysozyme; plant cell wall: cellulase.

Q52. 2M What is continuous culture? Why does it give higher yield?
Ans: Used medium is drained out from one side and fresh medium added from the other. Cells stay in the active log (exponential) phase, giving larger biomass and more product.

Q53. 3M Board fav Name the three steps of PCR and state what happens in each.
Ans: Denaturation: high temperature separates the two DNA strands. Annealing: primers bind to complementary sequences. Extension: Taq polymerase adds nucleotides to the primers, copying the template.

Q54. 2M Identify the palindrome and the enzyme: 5′-AAGCTT-3′. Show the complementary strand.
Ans: Complement 3′-TTCGAA-5′; read 5′ to 3′ it is AAGCTT, the same, so it is a palindrome. It is the recognition site of Hind III (cuts A↓AGCTT).

Q55. 2M Board fav What is the role of 'ori' and 'rop' in pBR322?
Ans: ori is the origin of replication, where replication begins and which controls copy number. rop codes for proteins involved in the replication of the plasmid.

Q56. 2M Why is it necessary to have a selectable marker in a cloning vector?
Ans: Only a few cells take up the vector. The marker (e.g. ampicillin resistance) lets transformants grow on the antibiotic while non-transformants die, so transformants are selected easily.

Q57. 2M Differentiate between micro-injection and biolistics.
Ans: Micro-injection: rDNA injected directly into the nucleus of an animal cell. Biolistics: plant cells bombarded with high-velocity gold or tungsten particles coated with DNA.

Q58. 2M What is elution? Why is it done?
Ans: Cutting out the separated DNA band from the agarose gel and extracting DNA from it. The purified fragment is used for constructing recombinant DNA.

Q59. 2M Mention any four control systems of a bioreactor.
Ans: Agitator system, oxygen delivery system, foam control system, temperature control system, pH control system and sampling ports (any four).

Q60. 2M What are recombinant proteins? Give two examples.
Ans: Proteins coded by a recombinant gene and expressed in a heterologous host, e.g. human insulin made in *E. coli*, hepatitis B vaccine made in yeast.

Q61. 2M Why are disarmed retroviruses used as vectors in animals?
Ans: Retroviruses naturally insert their genetic material into animal cells; once disarmed they cannot cause cancer but still deliver the desired gene into the host cells.

Q62. 2M Board fav A linear DNA molecule has 4 sites for EcoRI. How many fragments are formed? How many if it were circular?
Ans: Linear: 4 + 1 = 5 fragments. Circular: 4 fragments.

Q63. 2M Why is the amp^R gene called a selectable marker? What happens to *E. coli* without the plasmid on an ampicillin plate?
Ans: It allows only cells carrying the plasmid (transformants) to grow on ampicillin, so they are selected. *E. coli* without the plasmid has no resistance and dies.

Q64. 2M Board fav What are primers? Why are two primers needed in PCR?
Ans: Small chemically synthesised oligonucleotides complementary to regions of the template DNA. Two are needed because each strand is copied separately; one primer binds near each end of the target, on opposite strands.

Q65. 2M How is foam controlled in a bioreactor and why is it needed?
Ans: A foam control system breaks foam formed by aeration and agitation; excess foam can block air filters, cause contamination and loss of culture.

Q66. 3M Explain the role of DNA ligase and DNA polymerase in rDNA technology.
Ans: DNA ligase joins the cut ends of vector and foreign DNA (sticky ends) by forming phosphodiester bonds, making rDNA. DNA polymerase (e.g. Taq) synthesises new DNA, amplifying the gene in PCR, and the host's polymerase replicates the rDNA.

Q67. 2M Board fav Which gene of pBR322 is inactivated if foreign DNA is inserted at the Pst I site? How will recombinants behave?
Ans: The ampicillin resistance (amp^R) gene. Recombinants will grow on tetracycline but not on ampicillin.

Q68. 2M What is the significance of the cloning site being inside a selectable marker gene?
Ans: The insert inactivates the marker (insertional inactivation), so recombinants lose that trait and can be distinguished from non-recombinants.

Q69. 2M Why must a vector have a high copy number origin of replication when large amounts of a product are needed?
Ans: The ori controls the number of copies; a high copy number ori gives many copies of the gene per cell, so more recombinant protein is made.

Q70. 3M How is DNA isolated from a plant cell? Name the enzymes in order.
Ans: Cell wall broken by **cellulase**; RNA removed by **ribonuclease**; proteins by **protease**; DNA precipitated by **chilled ethanol** and spooled.

Q71. 2M In the pBR322 diagram, a student inserts a gene at the Sal I site. (a) Which gene is inactivated? (b) On which antibiotic will the recombinants survive?
Ans: (a) tet^R (tetracycline resistance gene), because Sal I lies in it. (b) Ampicillin.

Q72. 2M Board fav Lane A of a gel shows one band near the well, lane B shows three bands farther down. Interpret.
Ans: Lane A: undigested (large, intact) DNA, which moves the least. Lane B: the same DNA cut by a restriction enzyme into three smaller fragments, which move farther towards the anode.

Q73. 2M Why is DNA not seen during gel electrophoresis without staining?
Ans: DNA is colourless. It must be stained with ethidium bromide, which glows orange under UV light, to see the bands.

Q74. 2M Board fav (HOTS) A scientist cut the vector with EcoRI and the foreign DNA with BamHI. Will they join? Why?
Ans: No (not efficiently). EcoRI and BamHI produce different sticky ends (AATT and GATC overhangs), which are not complementary, so they cannot pair and be joined by ligase.

Q75. 2M (HOTS) Why is *E. coli* with a recombinant plasmid grown in a bioreactor rather than an open tank?
Ans: A bioreactor keeps sterile (contamination-free) conditions and controls temperature, pH, oxygen and foam, so only the desired cells grow and give the maximum product.

Q76. 2M What is a recombinant DNA? Why is it called so?
Ans: DNA formed by joining DNA segments from two different sources (e.g. a gene from one organism and a plasmid from a bacterium). It is called recombinant because it combines genetic material in a new way not found in nature.

Q77. 2M Board fav Name the two core techniques of modern biotechnology and state one aim of each.
Ans: Genetic engineering: alter the genetic material to change the phenotype of the host. Bioprocess engineering: keep sterile conditions to grow only the desired microbe or cell in large quantities for products.

Q78. 2M Why can a foreign DNA not multiply on its own when introduced into a host?
Ans: It lacks an origin of replication. Only DNA linked to an ori (in a plasmid or chromosome) can replicate in the host.

Q79. 3M How do sexual and asexual reproduction differ in terms of genetic information? What problem of hybridisation is solved by genetic engineering?
Ans: Sexual reproduction creates variation; asexual preserves the genetic information. Hybridisation also brings undesirable genes; genetic engineering transfers only the desired gene(s).

Q80. 2M What are the two enzymes from *E. coli* that restrict the growth of bacteriophage? What does each do?
Ans: A methylase, which adds methyl groups to DNA (protecting the bacterium's own DNA), and a restriction endonuclease, which cuts (foreign phage) DNA. Together they work as the bacterium's defence against phages.

Q81. 2M State two differences between a simple stirred-tank and a sparged stirred-tank bioreactor.
Ans: In the sparged type, sterile air bubbles are sparged through the culture, giving a larger oxygen transfer area and better aeration; the simple type depends mainly on the stirrer for mixing and oxygen.

Q82. 2M Why is downstream processing necessary before a product is marketed?
Ans: The product made in the bioreactor is mixed with cells, medium and other molecules. It must be separated, purified, formulated with preservatives and, for drugs, tested in clinical trials and quality control before it is safe to sell.

Long answer (5 marks)

Q83. 5M Board fav Describe the steps of recombinant DNA technology from isolation of DNA to obtaining the product.
Ans: (1) **Isolation of DNA**: cells lysed by lysozyme, cellulase or chitinase; RNA and protein removed by ribonuclease and protease; DNA precipitated by chilled ethanol and spooled. (2) **Cutting**: DNA and vector cut by the same restriction enzyme; checked by agarose gel electrophoresis; fragment eluted. (3) **Amplification** of the gene by PCR. (4) **Ligation**: gene and vector joined by DNA ligase to make rDNA. (5) **Transformation**: rDNA put into competent host (Ca^{2+}, ice, 42 °C heat shock, ice); transformants selected on antibiotic medium. (6) **Culturing** the host in a bioreactor to obtain recombinant protein. (7) **Downstream processing**: separation, purification, formulation, clinical trials and quality control.

Q84. 5M Board fav What is PCR? Describe its steps with a labelled diagram. Why is a thermostable polymerase needed?
Ans: PCR (polymerase chain reaction) amplifies a gene segment in vitro. Diagram as Fig. 9.6. (i) **Denaturation**: dsDNA separated by high temperature. (ii) **Annealing**: two primers bind to complementary regions of the template. (iii) **Extension**: DNA polymerase adds nucleotides to the primers. Repeated cycles amplify the DNA about a billion times. A thermostable enzyme (Taq polymerase from *Thermus aquaticus*) stays active through the repeated high-temperature denaturation, so fresh enzyme need not be added each cycle.

Q85. 5M Board fav (a) Draw pBR322 and label ori, rop, amp^R, tet^R and two restriction sites. (b) Explain how recombinants are selected using insertional inactivation, and why blue-white selection is preferred.

Ans: (a) Diagram as Fig. 9.4 (BamHI and Sal I in tet^R; Pst I and Pvu I in amp^R; EcoRI, Hind III, Cla I, Pvu II). (b) Insert at BamHI inactivates tet^R. On ampicillin medium transformants grow; on replica plates with tetracycline, recombinants die while non-recombinants survive; non-transformants die on both. This needs two plates (cumbersome). In blue-white selection the insert inactivates β-galactosidase; on a chromogenic substrate non-recombinants are blue and recombinants colourless, identified on a single plate.

Q86. 5M Board fav (a) What are restriction endonucleases? How are they named? (b) What are palindromes and sticky ends? (c) Explain gel electrophoresis.

Ans: (a) Enzymes that recognise a specific palindromic sequence and cut both DNA strands at specific points; naming: first letter of genus, two of species, strain, Roman numeral for order of isolation (EcoRI: *Escherichia coli* RY 13, first). (b) Palindrome: sequence reading the same on both strands 5′ to 3′, e.g. GAATTC; cutting a little away from the centre leaves single-stranded overhangs (sticky ends) that pair with complementary ends and are joined by ligase. (c) DNA fragments are loaded in wells of agarose gel; in an electric field negatively charged DNA moves to the anode; smaller fragments move farther (sieving); stained with ethidium bromide, seen under UV as orange bands; bands cut and eluted.

Case-based (4 marks)

Q87. 4M Board fav A student digests a DNA sample with EcoRI and runs it on agarose gel with undigested DNA in lane 1. (a) Towards which electrode will the DNA move and why? (b) Which fragments will be nearest the wells? (c) How will the bands be seen? (d) What is the extraction of a band from the gel called?

Ans: (a) Towards the anode, because DNA is negatively charged. (b) The largest fragments (and undigested DNA in lane 1), as the gel's sieving effect slows big fragments. (c) Stain with ethidium bromide and view under UV: orange bands. (d) Elution.

Q88. 3M A plasmid with a single EcoRI site is cut with EcoRI. (a) How many fragments form? (b) What shape does the plasmid take? (c) Which enzyme will close it again with an insert?

Ans: (a) One (the circle simply opens). (b) It becomes linear with two sticky ends. (c) DNA ligase.

Q89. 4M A researcher wants to make a bacterium produce a human protein. She cuts the gene and plasmid with BamHI, joins them, and adds the plasmid to *E. coli* cells kept with $CaCl_2$ on ice, then at 42 °C, then on ice. (a) Why is the same enzyme used for both? (b) Which enzyme joins them? (c) What is the purpose of the Ca^{2+} and heat shock? (d) Why are cells later grown in a bioreactor rather than a flask?

Ans: (a) To get matching sticky ends. (b) DNA ligase. (c) To make the cells competent so that hydrophilic DNA can enter. (d) Bioreactors handle 100–1000 L with controlled temperature, pH, oxygen and foam, giving large-scale yield.

Q90. 4M Plant biotechnologists want to introduce an insect-resistance gene into a dicot crop. (a) Name a natural vector they could use. (b) What is its natural effect on plants? (c) How is it modified? (d) Name a vector-free method for plant cells.

Ans: (a) *Agrobacterium tumefaciens* (Ti plasmid). (b) Its T-DNA transforms plant cells into tumours (crown gall). (c) The Ti plasmid is disarmed, so it is non-pathogenic but still transfers the desired gene. (d) Biolistics (gene gun) using DNA-coated gold or tungsten particles.

Q91. 4M Board fav A gene of interest is inserted into the coding region of β-galactosidase of a vector and plated on a chromogenic substrate. (a) What colour will recombinant colonies show? (b) Why? (c) What are non-recombinant colonies like? (d) Why is this method preferred over replica plating on antibiotics?

Ans: (a) No colour (white). (b) The insert inactivates the β-galactosidase gene (insertional inactivation), so the substrate is not acted upon. (c) Blue, as the enzyme is active. (d) Recombinants are identified on a single plate in one step; no need for simultaneous plating on two antibiotics.

Exam Alert: diagrams and pairs

- → Diagrams: EcoRI action (Fig. 9.1), pBR322 (Fig. 9.4), PCR (Fig. 9.6), stirred-tank bioreactor (Fig. 9.7): har saal koi ek aata hai.
- → Pairs: exo vs endonuclease, plasmid vs chromosomal DNA, simple vs sparged bioreactor, transformant vs recombinant.
- → Recombinant = rDNA wala; transformant = jisme koi bhi plasmid gaya. Dono same nahi!
- → Fragment count: linear mein n + 1, circular mein n.
- → PCR ke 3 steps ka order aur Taq ka source: 1 mark pakka.
- → Gel: chhota fragment sabse door, anode ki taraf.
- → Competent cell: Ca^{2+}, ice, 42 °C heat shock, ice: sequence yaad rakho.
- → Downstream processing: separation, purification, formulation, clinical trials, quality control.
- → Disarmed vectors: Ti plasmid (plants) aur retrovirus (animals).

BIOTECHNOLOGY AND ITS APPLICATIONS

Most asked: Bt cotton and cry genes, RNA interference, genetically engineered insulin, ADA gene therapy, molecular diagnosis (PCR, ELISA), transgenic animals, biopiracy

Biotechnology turns the tools of Chapter 9 into real products: **GM crops, human insulin, gene therapy, early diagnosis and transgenic animals**, along with the **ethical issues** they raise.

1. Introduction

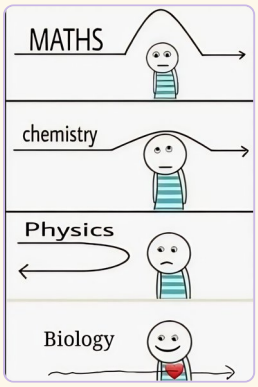
Biotechnology deals with **industrial scale production of biopharmaceuticals and biologicals** using genetically modified microbes, fungi, plants and animals.

Applications of biotechnology include:

- ✓ therapeutics and diagnostics
- ✓ genetically modified crops for agriculture
- ✓ processed food
- ✓ bioremediation
- ✓ waste treatment
- ✓ energy production

Key terms: three critical research areas

1. Providing the **best catalyst** in the form of an improved organism, usually a microbe, or a pure enzyme.
2. Creating **optimal conditions** through engineering for the catalyst to act.
3. **Downstream processing** technologies to purify the protein or organic compound.



Maths upar se, Physics ulta... par Biology seedha dil mein.
Aur biotech toh full pyaar!

Asli baat: Biotech = biology ke tools se insulin, Bt cotton, gene therapy jaise asli products.

2. Biotechnological Applications in Agriculture

Three options for increasing food production:

Agro-chemical based	*Organic agriculture*	*GM crop-based*
fertilisers and pesticides	manure, biofertilisers, biocontrol	crops with useful foreign genes
high yield, but costly and polluting	eco-friendly, but lower yield	high yield, less chemicals; biosafety concerns

The **Green Revolution** succeeded in tripling the food supply, yet it was not enough for the growing population. Yields rose partly due to improved varieties but **mainly** due to better management practices and **agrochemicals** (fertilisers and pesticides). For farmers of the developing world, agrochemicals are often **too expensive**, and conventional breeding with existing varieties cannot raise yields further.

Quick fact: why not just more agrochemicals?

- → They are **too expensive** for many farmers of the developing world.
- → Overuse pollutes soil and water and harms useful organisms.
- → Conventional breeding of existing varieties has reached its limit; hence tissue culture and GM crops.

A. Plant tissue culture

As traditional breeding could not keep pace with demand, tissue culture was developed. In the **1950s** scientists learnt that whole plants could be regenerated from **explants**, i.e., any part of a plant taken out and grown in a test tube under **sterile conditions** in special nutrient media.

Key terms

- → **Explant**: any part of a plant taken out and grown in vitro.
- → **Totipotency**: the capacity to generate a whole plant from any cell or explant. PYQ
- → **Micropropagation**: producing **thousands of plants** in a very short time through tissue culture.
- → **Somaclones**: plants produced by micropropagation; each is genetically identical to the original plant.

The nutrient medium must provide:

- ✓ a **carbon source** such as sucrose
- ✓ **inorganic salts**
- ✓ **vitamins**
- ✓ **amino acids**
- ✓ **growth regulators** like auxins and cytokinins

Many important food plants like **tomato, banana, apple** have been produced on a commercial scale by micropropagation.

B. Virus-free plants by meristem culture

Even if a plant is infected with a virus, its meristem (apical and axillary) is free of virus. So the meristem is removed and grown in vitro to obtain **virus-free plants**. Meristems of **banana, sugarcane and potato** have been cultured successfully. PYQ

Application of tissue culture	Example
Micropropagation (mass multiplication)	tomato, banana, apple
Virus-free plants (meristem culture)	banana, sugarcane, potato
Somatic hybridisation	pomato (tomato + potato)

C. Somatic hybridisation

Single cells are isolated from plants and their **cell walls are digested** to get naked protoplasts (surrounded only by plasma membranes). Protoplasts from two different varieties, each with a desirable character, are **fused** to get hybrid protoplasts, which grow into a new plant.

→ **Somatic hybrids**: the new hybrid plants; the process is **somatic hybridisation**.

→ **Pomato**: a somatic hybrid of tomato and potato; it did **not** have all the desired characters for commercial use.

Plant cells of 2 varieties → Digest cell wall → Naked protoplasts → Fuse → Hybrid protoplast → Somatic hybrid plant

Yaad karo: tissue culture ke 4 kaam

M-V-S-T: Micropropagation (hazaron somaclones), Virus-free (meristem se), Somatic hybrid (pomato), Totipotency (sab ka base). Meristem = virus ka "no entry zone"!

Exam Alert

→ Somaclone = **same** (identical copies); somatic hybrid = **do alag** protoplasts ka fusion.

→ Virus-free ke liye **meristem**, not leaf/root.

→ Medium mein carbon source = **sucrose**; hormones = auxins + cytokinins.

D. Genetically modified organisms (GMO)

Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO). PYQ

Genetic modification of plants has:

1. made crops more tolerant to **abiotic stresses** (cold, drought, salt, heat);
2. reduced reliance on **chemical pesticides** (pest-resistant crops);
3. helped to reduce **post-harvest losses**;
4. increased **efficiency of mineral usage** by plants (prevents early exhaustion of soil fertility);
5. enhanced **nutritional value** of food, e.g. golden rice (vitamin A enriched rice).

GM is also used to create **tailor-made plants** that supply alternative resources to industries: starches, fuels and pharmaceuticals.

Advantages of GM crops	Concerns about GM crops
higher yield, less pesticide	possible allergy or toxicity in some foods
tolerance to drought, salt, cold	gene flow to wild relatives (superweeds)
better nutrition (golden rice)	harm to non-target organisms, loss of biodiversity
less post-harvest loss	pests may evolve resistance; seed cost for farmers

E. Bt crops (pest-resistant plants)

Bt toxin is produced by the bacterium *Bacillus thuringiensis* (Bt). The Bt toxin gene has been cloned and expressed in plants to give resistance to insects without insecticides: in effect a **bio-pesticide**. Examples: **Bt cotton, Bt corn, rice, tomato, potato and soyabean**.

Some strains of *B. thuringiensis* produce proteins that kill certain insects:

Insect group	Examples
Lepidopterans	tobacco budworm, armyworm
Coleopterans	beetles
Dipterans	flies, mosquitoes

Mode of action of Bt toxin:

Bt forms protein crystals (inactive protoxin) → Insect eats it → Alkaline pH of gut solubilises crystals → Active toxin → Binds midgut epithelial cells → Pores, swelling, lysis → Insect dies

Quick fact: why Bt toxin does not kill the bacterium

The toxin exists as an inactive protoxin inside the bacterium. It becomes active only in the **alkaline pH** of the insect gut. So the *Bacillus* itself is safe. PYQ

Specific Bt toxin genes were isolated and put into crops like cotton. The choice of gene depends on the **crop and the target pest**, as most Bt toxins are insect-group specific. The toxin is coded by a gene named ***cry***; its proteins are **Cry proteins**.

cry gene	Pest controlled
cryIAc and *cryIIAb*	cotton bollworms
cryIAb	corn borer

Exam Alert

→ *cryIAc*, *cryIIAb* = cotton bollworm; *cryIAb* = corn borer. Spelling aur capital letters dhyan se.

→ Protoxin inactive hota hai; insect ke gut ki **alkaline** pH (acidic nahi) use activate karti hai.

→ Bt = bacterium; Bt cotton = transgenic plant. Dono alag likho.

Paheli-Bujho

Bujho: Paheli, Bt cotton khud toxin banata hai? Phir hum kyun nahi marte?
Paheli: Toxin sirf alkaline gut wale keedon mein active hota hai.
Bujho: Matlab hamare acidic pet mein bekaar!
Paheli: Haan, aur ye bhi sirf specific insect group pe kaam karta hai.

F. Pest-resistant plants: RNA interference (RNAi)

The nematode *Meloidogyne incognita* infects the **roots of tobacco** plants and greatly reduces yield. A novel strategy based on **RNA interference (RNAi)** was used to prevent this.

- → **RNAi** takes place in all **eukaryotic** organisms as a method of **cellular defence**.
- → It involves silencing of a specific mRNA by a complementary **dsRNA** molecule that binds to and prevents translation of the mRNA.
- → The source of complementary RNA may be an infection by viruses with **RNA genomes** or **mobile genetic elements (transposons)** that replicate via an RNA intermediate.

How the transgenic tobacco was made:

1. Using ***Agrobacterium* vectors**, nematode-specific genes were introduced into the host plant.
2. The DNA was introduced so that it produced both sense and anti-sense RNA in the host cells.
3. Being complementary, the two RNAs formed **dsRNA**, which initiated RNAi and **silenced the specific mRNA of the nematode**.
4. The parasite could not survive in the transgenic host; the plant was **protected**.

Nematode gene via *Agrobacterium* → Sense + anti-sense RNA → dsRNA → RNAi → Nematode mRNA silenced → Nematode dies

Yaad karo: RNAi = spam call block

Nematode ka mRNA = spam call. Plant ne **sense + antisense** RNA bana ke **dsRNA** ka block button daba diya. Call (translation) hi nahi lagi, toh nematode bhookha mar gaya!

Bt cotton	*RNAi tobacco*
protects against **insects** (bollworms)	protects against a **nematode** (*Meloidogyne incognita*)
plant makes a **toxic protein** (Cry)	plant makes **dsRNA**; no toxic protein
gene from *Bacillus thuringiensis*	nematode-specific genes, via *Agrobacterium*
kills insect by gut cell lysis	silences nematode mRNA (no translation)

3. Biotechnological Applications in Medicine

rDNA technology has had an immense impact on healthcare by enabling **mass production of safe and more effective** drugs. Recombinant therapeutics do not induce unwanted immunological responses, unlike similar products from non-human sources. About **30 recombinant therapeutics** are approved worldwide; **12** are marketed in India.

Recombinant therapeutic	*Use*
Human insulin	diabetes
Human growth hormone	dwarfism (growth hormone deficiency)
Interferons	viral infections, some cancers
Tissue plasminogen activator (tPA)	dissolves blood clots (heart attack, stroke)
Erythropoietin	anaemia (stimulates RBC formation)
Hepatitis B vaccine (made in yeast)	prevents hepatitis B
Factor VIII	haemophilia (blood clotting)

A. Genetically engineered insulin

Adult-onset diabetes can be managed by taking insulin at regular intervals. Insulin was earlier extracted from the **pancreas of slaughtered cattle and pigs**; this animal insulin caused allergy or other reactions to the foreign protein in some patients.

- → Insulin has two short polypeptide chains, **chain A and chain B**, linked by disulphide bridges.
- → In mammals it is made as a **pro-hormone (pro-insulin)** with an extra stretch, the **C peptide**, which is removed during maturation.
- → The main challenge was getting insulin **assembled into a mature form**, because bacteria cannot remove the C peptide.
- → Insulin is a peptide hormone of the **beta cells of the islets of Langerhans**; it lowers blood glucose.

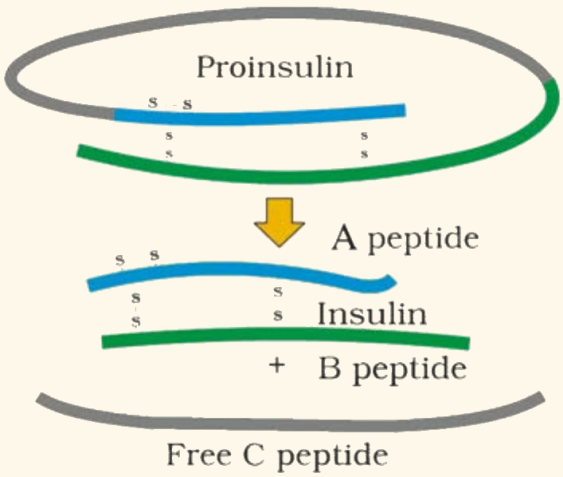


Fig. 10.3 Maturation of pro-insulin into insulin (simplified)

Key terms: Eli Lilly, 1983

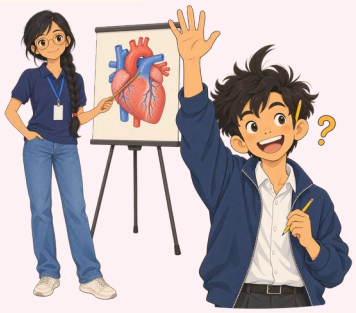
→ In **1983**, the American company **Eli Lilly** prepared two DNA sequences corresponding to chains A and B of human insulin. PYQ
→ They were introduced into **plasmids of *E. coli*** to produce the insulin chains separately.
→ Chains A and B were **extracted and combined by creating disulphide bonds** to form human insulin (humulin).

Think and answer (NCERT in-text)

Can insulin be given orally? No. Insulin is a protein; it would be **digested by proteases** in the stomach and intestine before reaching the blood, so it is injected.

Animal insulin (earlier)	rDNA human insulin
from pancreas of cattle and pigs	made by *E. coli* carrying human genes
slightly different from human insulin	identical to human insulin
allergy, immune reactions in some	no unwanted immune response
limited supply	unlimited, cheaper mass production

Paheli-Bujho



Bujho: Pehle insulin gaay-suar se aata tha? Ab kahan se?
Paheli: Ab *E. coli* banata hai, human genes daal ke. Eli Lilly, 1983.
Bujho: Bacteria ne insulin ki factory khol di!
Paheli: Haan, A aur B chains alag banti hain, phir S–S bond se judti hain.

B. Gene therapy

Gene therapy is a collection of methods that allows correction of a gene defect diagnosed in a child or embryo. Genes are inserted into a person's cells and tissues to treat a disease: a **normal gene** is delivered to take over the function of, and compensate for, the **non-functional gene.** PYQ

Key terms: first gene therapy (1990)

→ Given in **1990** to a 4-year-old girl with adenosine deaminase (ADA) deficiency.
→ ADA is crucial for the **immune system** to function; the disorder is due to **deletion of the gene** for ADA.

Treatment of ADA deficiency	Limitation
Bone marrow transplantation	works only in some children; not completely curative
Enzyme replacement therapy: functional ADA injected	not completely curative; repeated doses
Gene therapy with lymphocytes	lymphocytes are not immortal: periodic infusion needed
ADA gene into **early embryonic cells**	could be a **permanent cure**

Steps of gene therapy for ADA deficiency:

Lymphocytes from patient's blood → Cultured outside the body → Functional ADA cDNA added (retroviral vector) → Cells returned to patient → Repeat periodically

Exam Alert

→ Vector = **retrovirus** (disarmed); gene = functional **ADA cDNA**; cells = **lymphocytes**.
→ Permanent cure tabhi jab gene **early embryonic stage** ke cells (bone marrow ADA gene) mein daala jaaye.
→ Year 1990, age 4 saal: 1-mark ke liye yaad rakho.

C. Molecular diagnosis

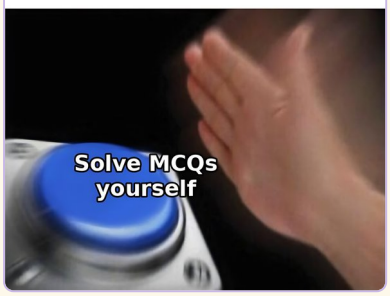
Early diagnosis and understanding pathophysiology are key to treatment. **Conventional methods** (serum and urine analysis) cannot detect disease early. A pathogen is usually suspected only after symptoms appear, when its concentration is already high.

Technique	Principle	Use
PCR	amplifies nucleic acid of even very few pathogen cells	detects **HIV** in suspected AIDS patients; **mutations** in suspected cancer patients; other genetic disorders
rDNA probe + autoradiography	a single-stranded DNA or RNA **tagged with a radioactive molecule (probe)** hybridises to its complementary DNA in a clone of cells	the clone with the **mutated gene does not appear** on the photographic film (no complementarity)
ELISA	antigen-antibody interaction	detects pathogen **antigens** (proteins, glycoproteins) or the **antibodies** made against them

Think and answer (NCERT in-text)

How can PCR detect very low amounts of DNA? PCR **amplifies** the pathogen's DNA (or cDNA from its RNA) about a billion times, so even a few copies present before symptoms become enough to be detected.

Padhai tabhi hogi jab:



Diagnosis ke 3 tools: PCR, probe, ELISA. Aur padhai ka button: khud MCQ solve karo!

Asli baat: PCR = DNA amplify; probe = autoradiography; ELISA = antigen-antibody.

PCR	ELISA
detects the pathogen's **nucleic acid** (DNA/RNA)	detects pathogen **antigens** or **antibodies** against it
works before antibodies are formed	antibody-based test needs time for antibodies to appear
e.g. HIV detection in suspected AIDS patients	e.g. HIV screening tests

4. Transgenic Animals

Animals whose DNA has been manipulated to possess and express an extra (foreign) gene are **transgenic animals**. Transgenic rats, rabbits, pigs, sheep, cows and fish have been produced, but **over 95 per cent** of all existing transgenic animals are mice. PYQ

Reason	What is done / example
Normal physiology and development	study how genes are regulated and affect body functions, e.g. complex growth factors like **insulin-like growth factor**
Study of disease	models for human diseases: **cancer, cystic fibrosis, rheumatoid arthritis, Alzheimer's**; test new treatments
Biological products	human protein **α-1-antitrypsin** to treat **emphysema**; attempts for phenylketonuria (PKU) and cystic fibrosis
Vaccine safety	transgenic mice test the safety of **polio vaccine**; could replace monkeys
Chemical safety (toxicity) testing	animals carry genes making them **more sensitive** to toxic substances; results in less time

Key terms: Rosie, the first transgenic cow (1997)

- → Produced human protein-enriched milk (2.4 grams per litre). PYQ
- → The milk contained human **alpha-lactalbumin**.
- → It was nutritionally a **more balanced product for human babies** than natural cow milk.

Yaad karo: transgenic animals ke 5 reasons

"Pyaari Didi Bina Vaccine Chal" = Physiology, Disease study, Biological products, Vaccine safety, Chemical safety. Rosie gaay = 2.4 g/L human α-lactalbumin.

Foreign gene isolated → Injected into fertilised egg (zygote) → Embryo implanted in surrogate mother → Transgenic offspring → Gene expressed (e.g. in milk)

Quick fact

A transgenic animal carries the foreign gene in **every cell**, including germ cells, so it passes the gene to its offspring. Products like human proteins are often made in the **milk**, which is easy to collect.

5. Ethical Issues

Manipulation of living organisms cannot go on without regulation; **ethical standards** are needed to judge the morality of activities that may help or harm living organisms. GM organisms may have unpredictable results when introduced into ecosystems.

Key terms: GEAC

The Indian Government has set up organisations such as the GEAC (Genetic Engineering Approval Committee), which decides on the **validity of GM research** and the **safety of introducing GM organisms** for public services. PYQ

Ethical / biological concern	Example
unpredictable results in ecosystems	GM genes spreading to wild plants
animal welfare	suffering of transgenic animals used in research
safety of GM food	allergy or toxicity tests needed
patents on bio-resources	Basmati, turmeric, neem

A. Patents and the Basmati case

There is growing public anger that companies get **patents** for products and technologies based on genetic materials, plants and biological resources long identified, developed and used by **farmers and indigenous people** of a region.

- → India has an estimated **200,000 varieties of rice**; its rice diversity is one of the richest in the world.
- → **27 documented varieties of Basmati** are grown in India, known for unique aroma and flavour; Basmati is mentioned in ancient texts, folklore and poetry.
- → In 1997, an **American company** got patent rights on Basmati rice through the **US Patent and Trademark Office**, allowing it to sell a 'new' variety in the US and abroad.
- → This variety was actually derived from Indian farmers' varieties: Indian Basmati was **crossed with semi-dwarf varieties** and claimed as an invention. The patent extends to **functional equivalents**, so others selling Basmati could be restricted.

→ Attempts have also been made to patent uses and processes based on Indian traditional herbal medicines, e.g. **turmeric and neem.**

B. Biopiracy

Biopiracy is the use of bio-resources by multinational companies and other organisations without proper authorisation from the countries and people concerned, and **without compensatory payment.** PYQ

Industrialised nations	Developing / underdeveloped nations
financially rich	financially poorer
poor in biodiversity and traditional knowledge	**rich** in biodiversity and traditional knowledge
exploit traditional knowledge to save time, effort and cost	need laws for benefit sharing

Realising this injustice and poor benefit sharing, nations are making laws against unauthorised exploitation. The Indian Parliament has cleared the second amendment of the Indian Patents Bill, which considers such issues, including patent terms, emergency provisions and research and development initiatives.

Key terms: words to define

Term	Meaning
GMO	organism whose genes are altered by manipulation
Transgenic	organism carrying and expressing a foreign gene
Bio-pesticide	a pest-killing product of biological origin, e.g. Bt toxin
Patent	right granted by a government to an inventor to stop others from making or selling the invention
Biopiracy	unauthorised, uncompensated use of bio-resources

Paheli-Bujho

Bujho: Basmati toh apna hai, phir videshi company ka patent kaise?

Paheli: 1997 mein US company ne patent le liya, Indian Basmati cross karke.

Bujho: Ye toh seedhi chori hai!

Paheli: Isi ko biopiracy kehte hain. Isliye Patents Bill amend hua.

Field	Product / technique	Key point
Agriculture	tissue culture, Bt crops, RNAi plants, golden rice	more yield, less chemicals, better nutrition
Medicine	human insulin, gene therapy	safe, no immune reaction; corrects a defective gene
Diagnosis	PCR, rDNA probe, ELISA	detects disease before symptoms
Research and industry	transgenic animals	disease models, human proteins in milk, safety tests
Society	GEAC, Patents Bill	biosafety, protection from biopiracy

Exam Alert: numbers to remember

→ 1950s: plant regeneration from explants learnt.
→ ~30 recombinant therapeutics worldwide; 12 marketed in India.
→ 1983 Eli Lilly insulin; 1990 first gene therapy (4-year-old girl); 1997 Rosie (2.4 g/L) and Basmati patent.
→ 200,000 rice varieties in India; 27 documented Basmati varieties; over 95% transgenic animals are mice.
→ *cryIAc*, *cryIIAb*: bollworm; *cryIAb*: corn borer.
→ Gene therapy vector: retrovirus; cells: lymphocytes.

1 minute recap

1. Three critical areas: best catalyst, optimal conditions, downstream processing.
2. Tissue culture: explant, totipotency, micropropagation (somaclones), virus-free meristem culture, somatic hybridisation (pomato).
3. GMO benefits: stress tolerance, less pesticide, less post-harvest loss, mineral efficiency, golden rice (vitamin A).
4. Bt toxin: inactive protoxin, activated by alkaline gut pH; *cryIAc*, *cryIIAb* (bollworm), *cryIAb* (corn borer).
5. RNAi: sense + anti-sense RNA form dsRNA, silence *Meloidogyne incognita* mRNA in tobacco.
6. Insulin: chains A and B made separately in *E. coli* (Eli Lilly 1983), joined by disulphide bonds; C peptide removed in pro-insulin.
7. Gene therapy: ADA deficiency, 1990, lymphocytes + retroviral vector; molecular diagnosis: PCR, probe, ELISA.
8. Transgenic animals (95% mice); Rosie 1997 (2.4 g/L alpha-lactalbumin); GEAC; Basmati patent 1997; biopiracy.
9. Ethics: GM organisms can have unpredictable effects in ecosystems, so research and release are regulated.
10. Recombinant therapeutics: ~30 approved worldwide, 12 marketed in India.

Most Important Questions

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM Board fav The capacity of a plant cell to generate a whole plant is called:
(a) pluripotency
(b) differentiation
(c) totipotency
(d) dedifferentiation
Ans: (c) totipotency

Q2. IM Plants obtained by micropropagation are called:
(a) somatic hybrids
(b) somaclones
(c) transgenics
(d) protoplasts
Ans: (b) somaclones

Q3. IM Board fav Virus-free plants are obtained by culturing the:
(a) apical and axillary meristem
(b) leaf mesophyll
(c) root cortex
(d) seed coat
Ans: (a) apical and axillary meristem

Q4. IM Pomato is an example of:
(a) a transgenic plant
(b) a somaclone
(c) a mutant
(d) a somatic hybrid
Ans: (d) a somatic hybrid

Q5. IM Golden rice is rich in:
(a) iron
(b) vitamin A
(c) vitamin C
(d) lysine
Ans: (b) vitamin A

Q6. IM Board fav Bt toxin is activated in the insect gut because of:
(a) acidic pH
(b) neutral pH
(c) alkaline pH
(d) high temperature
Ans: (c) alkaline pH

Q7. IM Board fav The proteins encoded by *cryIAc* and *cryIIAb* control:
(a) cotton bollworms
(b) corn borer
(c) nematodes
(d) aphids
Ans: (a) cotton bollworms

Q8. IM *Meloidogyne incognita* infects the roots of:
(a) cotton
(b) rice
(c) maize
(d) tobacco
Ans: (d) tobacco

Q9. IM Board fav RNA interference involves:
(a) silencing of DNA by methylation
(b) silencing of a specific mRNA by complementary dsRNA
(c) degradation of proteins
(d) mutation of the gene
Ans: (b) silencing of a specific mRNA by complementary dsRNA

Q10. IM The two polypeptide chains of insulin are linked by:
(a) hydrogen bonds
(b) peptide bonds
(c) disulphide bridges
(d) phosphodiester bonds
Ans: (c) disulphide bridges

Q11. IM Eli Lilly produced human insulin by rDNA technology in:
(a) 1983
(b) 1990
(c) 1997
(d) 1972
Ans: (a) 1983

Q12. IM Board fav The first clinical gene therapy was given for:
(a) haemophilia
(b) ADA deficiency
(c) cystic fibrosis
(d) sickle-cell anaemia
Ans: (b) ADA deficiency

Q13. IM ELISA is based on the principle of:
(a) DNA hybridisation
(b) PCR amplification
(c) autoradiography
(d) antigen-antibody interaction
Ans: (d) antigen-antibody interaction

Q14. IM More than 95 per cent of existing transgenic animals are:
(a) cows
(b) sheep
(c) mice
(d) fish
Ans: (c) mice

Q15. 1M Board fav The first transgenic cow, Rosie, produced milk containing human:
(a) alpha-lactalbumin
(b) alpha-1-antitrypsin
(c) insulin
(d) casein
Ans: (a) alpha-lactalbumin

Q16. 1M α-1-antitrypsin produced by transgenic animals is used to treat:
(a) diabetes
(b) emphysema
(c) PKU
(d) AIDS
Ans: (b) emphysema

Q17. 1M In India, the body that decides on the safety of GM organisms is:
(a) ICMR
(b) CSIR
(c) GEAC
(d) WHO
Ans: (c) GEAC

Q18. 1M Board fav (NCERT) Crystals of Bt toxin do not kill the bacteria themselves because:
(a) bacteria are resistant to the toxin
(b) toxin is immature
(c) toxin is inactive
(d) bacteria enclose toxin in a special sac
Ans: (c) the toxin is inactive

Q19. 1M Somatic hybridisation requires removal of the cell wall to obtain:
(a) explants
(b) naked protoplasts
(c) somaclones
(d) callus
Ans: (b) naked protoplasts

Q20. 1M Which organism is the source of the vector used to make RNAi tobacco?
(a) *Bacillus thuringiensis*
(b) *E. coli*
(c) *Agrobacterium tumefaciens*
(d) retrovirus
Ans: (c) *Agrobacterium tumefaciens*

Q21. 1M ADA deficiency is caused by:
(a) deletion of the gene for adenosine deaminase
(b) excess of ADA
(c) a viral infection
(d) a chromosomal non-disjunction
Ans: (a) deletion of the gene for adenosine deaminase

Q22. 1M In 1997, the US patent on Basmati rice was granted to:
(a) the Indian government
(b) IARI
(c) a European university
(d) an American company
Ans: (d) an American company

Assertion-Reason (1 mark each)

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q23. 1M Board fav **A:** Meristem culture gives virus-free plants. **R:** The apical and axillary meristems are free of virus even in infected plants.
Ans: (a) Both true and R explains A.

Q24. 1M **A:** Bt toxin does not harm the bacterium that makes it. **R:** The toxin exists as an inactive protoxin in the bacterium.
Ans: (a) Both true and R explains A.

Q25. 1M Board fav **A:** Insulin cannot be given orally. **R:** Insulin is a protein that is digested in the alimentary canal.
Ans: (a) Both true and R explains A.

Q26. 1M **A:** Gene therapy using lymphocytes gives a permanent cure for ADA deficiency. **R:** Lymphocytes are not immortal.
Ans: (d) A false, R true (periodic infusion is needed).

Q27. 1M **A:** Somaclones are genetically identical to the parent plant. **R:** They are produced by meiosis.
Ans: (c) A true, R false (they arise by mitosis from explants).

Short answer (2-3 marks)

Q28. 2M Board fav (NCERT) Which part of the plant is best suited for making virus-free plants and why?
Ans: The apical and axillary **meristem**. Even in an infected plant, the meristem is free of virus (the virus does not reach the actively dividing tissue fast enough), so plants grown from it in vitro are virus-free.

Q29. 2M Board fav (NCERT) What is the major advantage of producing plants by micropropagation?
Ans: Thousands of genetically identical plants (somaclones) are produced in a very short time and small space, all season, keeping the desirable traits of the parent; disease-free plants can also be raised.

Q30. 2M (NCERT) What are the components of the medium used for propagation of an explant in vitro?
Ans: A carbon source (sucrose), inorganic salts, vitamins, amino acids and growth regulators such as auxins and cytokinins; agar is added to solidify it.

Q31. 3M Board fav (NCERT) What are transgenic bacteria? Illustrate with one example.
Ans: Bacteria carrying and expressing a foreign gene. E.g. *E. coli* carrying the DNA sequences for human insulin chains A and B (Eli Lilly, 1983) produce insulin chains, which are combined by disulphide bonds to give human insulin.

Q32. 3M (NCERT) Compare the advantages and disadvantages of producing GM crops.
Ans: Advantages: stress tolerance, less pesticide use, less post-harvest loss, efficient mineral use, better nutrition (golden rice), industrial products. Disadvantages: possible allergy or toxicity, gene flow to wild relatives, harm to non-target organisms and biodiversity, pest resistance, costly seeds and patent issues.

Q33. 3M Board fav (NCERT) What are Cry proteins? Name an organism that produces them. How has man exploited them?
Ans: Insecticidal (Bt toxin) proteins coded by *cry* genes, produced by *Bacillus thuringiensis*. The *cry* genes have been put into crops (Bt cotton, Bt corn) so the plants make their own bio-pesticide and resist bollworms and corn borers.

Q34. 2M (NCERT) Suggest a method to remove oil (hydrocarbon) from seeds based on rDNA technology.
Ans: Identify the gene coding for the key enzyme of oil (fatty acid / triglyceride) synthesis in the seed and silence it (by RNAi or by inactivating the gene), so the seed stores little or no oil.

Q35. 2M (NCERT) What is golden rice?
Ans: A genetically modified rice variety enriched with vitamin A (beta-carotene, its precursor), giving the grain a golden colour; meant to reduce vitamin A deficiency.

Q36. 2M (NCERT) Does our blood have proteases and nucleases?
Ans: Blood plasma has some proteases (e.g. clotting factors such as thrombin, and plasmin) and nucleases in small amounts; they are mostly present in an inactive form or are tightly regulated, so blood proteins and nucleic acids are not freely digested.

Q37. 3M (NCERT) How can orally active protein pharmaceuticals be made? What is the main problem?
Ans: Proteins can be packed in protective coats (encapsulation, nanoparticles), modified to resist enzymes, or made in edible transgenic plants. The main problem is digestion by proteases and acids in the gut and poor absorption into blood.

Q38. 2M Differentiate between somaclones and somatic hybrids.
Ans: Somaclones: genetically identical plants from micropropagation of one plant. Somatic hybrids: plants from the fusion of protoplasts of two different varieties or species (e.g. pomato), carrying characters of both.

Q39. 3M Board fav How does Bt toxin kill insects?
Ans: Inactive protoxin crystals are eaten by the insect; the alkaline pH of the gut solubilises and activates them; the active toxin binds to midgut epithelial cells and creates pores, causing swelling, lysis and death of the insect.

Q40. 2M What is the source of complementary RNA in RNAi?
Ans: Infection by viruses with RNA genomes, or mobile genetic elements (transposons) that replicate via an RNA intermediate.

Q41. 2M Why was animal insulin replaced by rDNA insulin?
Ans: Animal insulin (from cattle and pig pancreas) caused allergy and other reactions to the foreign protein in some patients and its supply was limited; rDNA insulin is identical to human insulin and can be mass produced.

Q42. 3M Board fav What is the C peptide? Why was it a challenge in making insulin by rDNA technology?
Ans: An extra stretch present in pro-insulin, removed during maturation. Bacteria cannot process pro-insulin into mature insulin, so Eli Lilly made chains A and B separately and joined them by disulphide bonds.

Q43. 3M Board fav Why is gene therapy using lymphocytes not a permanent cure for ADA deficiency? What could be a permanent cure?
Ans: Genetically engineered lymphocytes are not immortal, so periodic infusion is needed. Introducing the ADA gene (isolated from marrow cells) into cells at early embryonic stages could be a permanent cure.

Q44. 2M Why can conventional methods not detect diseases early? Name three molecular methods.
Ans: Serum and urine tests detect a pathogen only after it has multiplied and caused symptoms. Molecular methods: PCR, recombinant DNA probes (with autoradiography) and ELISA.

Q45. 3M Board fav How does a radioactive probe identify a mutated gene?
Ans: A single-stranded DNA/RNA probe tagged with a radioactive molecule hybridises with complementary DNA in a clone of cells and is detected by autoradiography. A clone with the mutated gene does not appear on the film, because the probe cannot pair with it.

Q46. 2M What are transgenic animals? Why are most of them mice?
Ans: Animals whose DNA is manipulated to possess and express a foreign gene. Mice are small, breed fast, are cheap to keep and their genetics is well known, so over 95% of transgenic animals are mice.

Q47. 3M How are transgenic animals useful in vaccine safety and chemical safety testing?
Ans: Transgenic mice test the safety of vaccines such as polio vaccine before use in humans (could replace monkeys). Animals made more sensitive to toxic substances are exposed to chemicals so toxicity results come in less time.

Q48. 2M Board fav What is biopiracy? Give an example.
Ans: Use of bio-resources by multinational companies or organisations without authorisation from, or compensation to, the country and people concerned. E.g. the 1997 US patent on Basmati rice derived from Indian varieties; attempts to patent turmeric and neem.

Q49. 2M What is the role of GEAC?
Ans: The Genetic Engineering Approval Committee decides on the validity of GM research and the safety of introducing GM organisms for public services in India.

Q50. 2M What is meant by 'bio-pesticide' in the context of Bt crops?
Ans: The plant itself makes the insecticidal Bt (Cry) protein from a cloned bacterial gene, so insects are killed without spraying chemical insecticides.

Q51. 2M Name the three insect groups killed by Bt toxins with one example each.
Ans: Lepidopterans (tobacco budworm, armyworm), coleopterans (beetles), dipterans (flies, mosquitoes).

Q52. 2M Why are recombinant therapeutics preferred over products from non-human sources?
Ans: They are identical to human proteins, so they do not induce unwanted immunological responses; they are safer, more effective and can be mass produced.

Q53. 2M What is the role of the retroviral vector in ADA gene therapy?
Ans: The disarmed retrovirus carries the functional ADA cDNA into the cultured lymphocytes and inserts it into their genome, so the cells start making ADA.

Q54. 2M Give two examples of human diseases for which transgenic models exist and one human protein produced by transgenic animals.
Ans: Cancer, cystic fibrosis, rheumatoid arthritis, Alzheimer's (any two); α-1-antitrypsin (for emphysema) or alpha-lactalbumin in Rosie's milk.

Q55. 2M Board fav Why was Rosie's milk considered better for human babies?
Ans: It contained human alpha-lactalbumin (2.4 g per litre), making it nutritionally more balanced for human babies than natural cow milk. Rosie (1997) was the first transgenic cow.

Q56. 2M State any two ways in which GM crops increase food security.
Ans: They tolerate abiotic stresses (drought, salt, cold, heat), reduce pest damage and post-harvest losses, and use minerals efficiently, so yields stay high and soil fertility is not exhausted early (any two).

Q57. 2M What are the industrial uses of tailor-made GM plants?
Ans: They supply alternative resources to industries in the form of starches, fuels and pharmaceuticals, made directly by the crop plants. This reduces dependence on chemical synthesis.

Long answer (5 marks)

Q58. 5M Board fav (NCERT) What is gene therapy? Illustrate using the example of ADA deficiency.
Ans: Gene therapy corrects a diagnosed gene defect by delivering a normal gene into the person or embryo to take over the function of the non-functional gene. **ADA deficiency:** deletion of the adenosine deaminase gene; ADA is crucial for the immune system. First gene therapy: 1990, a 4-year-old girl. Other treatments: bone marrow transplant, enzyme replacement therapy (ADA injection); neither is completely curative. **Steps:** lymphocytes from the patient's blood are cultured outside the body; functional ADA cDNA is introduced using a retroviral vector; cells are returned to the patient. As lymphocytes are not immortal, infusions are repeated periodically. Introducing the ADA gene from marrow cells into early embryonic cells could be a permanent cure, as all cells formed later would carry the normal gene.

Q59. 5M Board fav (NCERT) Diagrammatically represent the steps in cloning and expressing a human gene (e.g. growth hormone gene) in *E. coli*.
Ans: Flow: isolate human DNA (or mRNA, make cDNA) → cut the growth hormone gene and a plasmid (e.g. pBR322) with the same restriction enzyme → ligate with DNA ligase to form rDNA → make *E. coli* competent (Ca^{2+}, heat shock) and transform → select transformants on antibiotic and recombinants by insertional inactivation → grow recombinant *E. coli* in a bioreactor → extract and purify human growth hormone (downstream processing).

Q60. 5M Board fav Explain how (a) Bt cotton and (b) RNAi tobacco are made resistant to pests. How do they differ?

Ans: (a) *cry* genes (*cryIAc, cryIIAb*) from *Bacillus thuringiensis* are put into cotton; the plant makes inactive protoxin; in the alkaline gut of bollworms it becomes active, binds midgut cells, makes pores and kills the insect. (b) Nematode-specific genes are introduced through *Agrobacterium* so that both sense and anti-sense RNA are made; they form dsRNA, which triggers RNAi and silences the specific mRNA of *Meloidogyne incognita*; the nematode cannot survive. Difference: Bt kills by a toxic protein; RNAi blocks translation of a parasite mRNA without any toxin. Both reduce the need for chemical pesticides.

Q61. 5M What are transgenic animals? Explain five reasons for producing them, with examples.

Ans: Animals with a manipulated DNA expressing a foreign gene (95% mice). (1) Normal physiology and development: e.g. role of insulin-like growth factor. (2) Study of disease: models for cancer, cystic fibrosis, rheumatoid arthritis, Alzheimer's. (3) Biological products: α-1-antitrypsin for emphysema; Rosie (1997) gave milk with human alpha-lactalbumin, 2.4 g/L. (4) Vaccine safety: transgenic mice for polio vaccine testing. (5) Chemical safety testing: animals carrying genes that make them more sensitive to toxic substances are exposed to chemicals, giving quicker results.

Case-based (4 marks)

Q62. 4M Board fav A farmer's banana plantation is infected with a virus. A scientist takes the shoot tip of an infected plant and grows thousands of healthy plants in a laboratory. (a) Which part did he culture and why? (b) Name the technique. (c) What will the new plants be called? (d) Name two nutrients in the medium.

Ans: (a) Apical meristem, as it is free of virus. (b) Meristem culture / micropropagation (tissue culture). (c) Somaclones (virus-free). (d) Sucrose, inorganic salts, vitamins, amino acids, auxins and cytokinins (any two). All the plants will be genetically identical to the parent.

Q63. 4M Board fav A diabetic patient develops an allergy to insulin from pig pancreas. The doctor shifts him to 'human insulin made by bacteria'. (a) Why was he allergic? (b) How is human insulin made in bacteria? (c) Why must it be injected? (d) What is removed from pro-insulin in our body?

Ans: (a) Pig insulin is a foreign protein, slightly different from human insulin. (b) DNA sequences for chains A and B are put into *E. coli* plasmids; the chains are made separately, extracted and joined by disulphide bonds. (c) It is a protein and would be digested in the gut. (d) The C peptide.

Q64. 4M An Indian scientist finds that a foreign company has patented a rice variety derived from Indian Basmati and a process using turmeric. (a) What is this act called? (b) How many Basmati varieties are documented in India? (c) Which Indian law was amended to address this? (d) Why are developing nations targets?

Ans: (a) Biopiracy. (b) 27. (c) The Indian Patents Bill (second amendment). (d) They are rich in biodiversity and traditional knowledge, while industrialised nations are rich financially but poor in these.

Q65. 4M Board fav A child has frequent severe infections. Tests show he lacks the enzyme adenosine deaminase. (a) Why does this cause infections? (b) Name two conventional treatments. (c) Outline the gene therapy given. (d) Why must it be repeated?

Ans: (a) ADA is crucial for the immune system; without it immunity fails. (b) Bone marrow transplantation; enzyme replacement therapy (ADA injection). (c) His lymphocytes are cultured outside the body, functional ADA cDNA is introduced with a retroviral vector, and the cells are returned. (d) Lymphocytes are not immortal, so periodic infusion is needed.

Q66. 4M A research team makes transgenic mice that are very sensitive to a cancer-causing chemical, and another team plans to test a new batch of polio vaccine on transgenic mice. (a) Name the two uses. (b) What advantage do sensitive mice give? (c) Which animal could the vaccine mice replace? (d) What percentage of transgenic animals are mice?

Ans: (a) Chemical safety (toxicity) testing and vaccine safety testing. (b) Results are obtained in less time. (c) Monkeys. (d) Over 95 per cent.

Q67. 3M A plant breeder fuses protoplasts of tomato and potato. (a) What is removed first from the cells? (b) What is the fused product called? (c) Name the resulting plant and state why it failed commercially.

Ans: (a) The cell wall (by enzymes), giving naked protoplasts. (b) A hybrid protoplast (it grows into a somatic hybrid). (c) Pomato; it did not have all the desired combination of characters for commercial use.

Exam Alert: most asked

- → Bt toxin ka mechanism (protoxin, alkaline pH, midgut pores) aur *cry* genes: har saal.
- → RNAi ke steps (sense + antisense = dsRNA) aur nematode ka naam.
- → Insulin: A, B chains, disulphide bond, C peptide, Eli Lilly 1983.
- → ADA gene therapy steps, PCR / ELISA diagnosis, Rosie aur biopiracy (Basmati 1997).
- → Tissue culture: totipotency, somaclones, meristem (virus-free), pomato: 1-2 marks pakke.
- → GEAC ka full form aur kaam bhi yaad rakho.

Organisms and Populations

Most asked: population attributes and age pyramids, exponential and logistic growth (dN/dt), BIDE, life history variation, predation, competition (Gause), parasitism, commensalism, mutualism

Ecology at the level of the **population**: how populations are described, how they **grow**, and how **species interact** with each other.

1. Introduction

Ecology is the subject that deals with the interactions among organisms and between the organism and its physical (abiotic) environment. It gives a holistic perspective to biology. PYQ

Ecology is basically concerned with **four levels of biological organisation**:

Organisms → Populations → Communities → Biomes

The living world is understood by studying processes at every level: macromolecules, cells, tissues, organs, individual organisms, populations, communities, ecosystems and biomes. In this chapter we study ecology at the **population level**.

2. Populations

In nature we rarely find isolated, single individuals of any species; most live in groups in a well-defined geographical area, **share or compete** for similar resources and potentially **interbreed**, thus constituting a population. PYQ

Key terms

- → **Population**: a group of individuals of the **same species** living in a defined geographical area, sharing or competing for resources and potentially interbreeding.
- → **Community**: an assemblage of populations of **different species** living together and interacting in an area.
- → **Interbreeding**: implies sexual reproduction, but a group of individuals produced by **asexual reproduction** is also generally considered a population for ecological studies.

Examples of populations:

- ✓ all the **cormorants** in a wetland
- ✓ **rats** in an abandoned dwelling
- ✓ **teakwood trees** in a forest tract
- ✓ **bacteria** in a culture plate
- ✓ **lotus plants** in a pond

Population ecology is important because it links ecology to **population genetics and evolution**. Although an individual organism has to cope with a changed environment, it is at the population level that natural selection operates to evolve the desired traits.

3. Population Attributes

A population has certain attributes that an **individual organism does not have**. An individual may have births and deaths, but a population has birth rates and death rates. PYQ

Attribute	Individual	Population
Birth / death	born once, dies once	birth rate, death rate
Sex	male or female	sex ratio
Age	one age	age distribution (pyramid)
Number	one	population density (N)

A. Birth rate and death rate

In a population these rates are expressed as **per capita** births or deaths, i.e., rate of increase or decrease **with respect to members** of the population.

Worked example (NCERT)

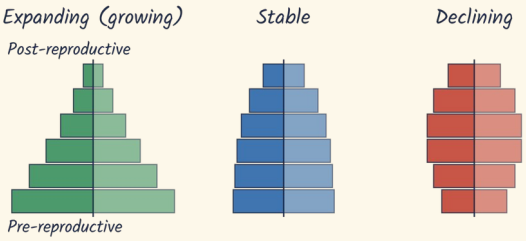
- → A pond had **20 lotus plants** last year; through reproduction **8 new plants** are added. Birth rate = 8/20 = 0.4 offspring per lotus per year.
- → In a laboratory population of **40 fruit flies**, 4 died in a week. Death rate = 4/40 = 0.1 individuals per fruit fly per week.

B. Sex ratio

An individual is either a male or a female, but a population has a sex ratio, e.g. 60 per cent of the population are females and 40 per cent males.

C. Age distribution and age pyramids

At any given time a population is composed of individuals of different ages. If the age distribution (per cent individuals of a given age or age group) is plotted for the population, the resulting structure is called an age pyramid. For human populations, the age pyramids generally show the age distribution of **males and females** in a combined diagram.



Bars = age groups (left: male, right: female)

Fig. 11.1 Representation of age pyramids for human population

Pyramid shape	Base (pre-reproductive)	Status
Triangular, broad base	largest group	**expanding** (growing)
Bell-shaped	about equal to reproductive	**stable**
Urn-shaped, narrow base	smaller than reproductive	**declining**

Exam Alert

- → Age pyramid ke 3 age groups: **pre-reproductive, reproductive, post-reproductive.**
- → Broad base = growing; bell = stable; urn (base patla) = declining. Diagram aur shape ka naam dono likho.
- → The shape of the pyramid reflects the **growth status** of the population.

D. Population density (N)

The size of the population tells us a lot about its status in the habitat. Ecological processes we study (outcome of competition with another species, impact of a predator, effect of a pesticide) are always evaluated in terms of a **change in population size.**

- → Size, technically called population density (designated as N), need not be measured in numbers only.
- → It can be as low as **<10** (Siberian cranes at Bharatpur wetlands in any year) or go into **millions** (*Chlamydomonas* in a pond).
- → Total number is generally the most appropriate measure, but in some cases it is **meaningless or difficult** to determine.
- → In an area with **200 carrot grass (*Parthenium hysterophorus*)** plants and only **a single huge banyan tree** with a large canopy, the low density of banyan underestimates its enormous role; here percent cover or biomass is a more meaningful measure.
- → Total number is also difficult to count when the population is huge, e.g. bacteria in a Petri dish.

Method	How density is measured	Example
Total count	count every individual	trees in a small area
Biomass / percent cover	weight or area covered	a banyan tree; plants in a field
Relative density	number caught per unit effort	fish caught per trap
Indirect estimate	signs without seeing or counting the organism	**tigers** by **pug marks and fecal pellets**

For certain ecological investigations there is no need to know absolute densities; **relative densities** serve the purpose equally well, e.g. the number of fish caught per trap is a good enough measure of total population density in a lake.

Yaad karo: population ke 4 gun

"BSAD" = Birth/death rate, Sex ratio, Age pyramid, Density. Ye sab population ke hote hain, akele individual ke nahi. Banyan ek hi ho, par percent cover mein boss hai!

4. Population Growth

The size of a population for any species is **not a static parameter**. It keeps changing with time, depending on food availability, predation pressure and adverse weather. These changes tell us whether the population is **flourishing or declining.**

Four basic processes change population density:

Process	Effect on N	Meaning
Natality (B)	increase	number of births during a given period in the population
Immigration (I)	increase	individuals of the same species that come into the habitat from elsewhere
Mortality (D)	decrease	number of deaths in the population during a given period
Emigration (E)	decrease	individuals of the population who left the habitat and gone elsewhere

N_t at time t → + Births (B) + Immigration (I) → − Deaths (D) − Emigration (E) → N_{t+1}

$$N_{t+1} = N_t + [(B + I) - (D + E)]$$ population density at time t + 1

- → Population density will **increase** if B + I is more than D + E; otherwise it will decrease. PYQ
- → Under normal conditions, **births and deaths** are the most important factors.
- → Immigration and emigration assume importance only under special conditions, e.g. if a **new habitat is just being colonised**, immigration may contribute more than birth rates.

Paheli-Bujho

Bujho: Paheli, ek talab mein naye mendhak aa gaye, kuch chale gaye. Population ka kya hua?

Paheli: BIDE lagao: B + I zyada toh badhegi, D + E zyada toh ghategi.

Bujho: Matlab aane wale plus, jaane wale minus!

Paheli: Bilkul. Aur normal mein births aur deaths sabse important.

A. Growth models

Does the growth of a population with time show any specific and predictable pattern? Two models explain it.

(i) Exponential growth

Resource (food and space) availability is essential for unimpeded growth. When resources in the habitat are unlimited, each species has the ability to realise fully its innate potential to grow in number, as Darwin observed while developing his theory of natural selection. Then the population grows in an **exponential or geometric** fashion.

If in a population of size N the birth rates (not total number but per capita births) are represented as **b** and death rates (again per capita) as **d**, the increase or decrease in N during a unit time period t (dN/dt) is:

$$dN/dt = (b - d) \times N$$ let $(b - d) = r$; then $dN/dt = rN$

- → **r** is the intrinsic rate of natural increase, a very important parameter for assessing the impact of any biotic or abiotic factor on population growth. PYQ
- → The equation describes the exponential or geometric growth pattern and results in a J-shaped curve when N is plotted against time.

Population	Value of r
Norway rat	0.015
Flour beetle	0.12
Human population of India (1981)	0.0205

The integral form of the exponential growth equation is:

$$N_t = N_0 e^{rt}$$ $e = 2.71828$

Here N_t = population density after time t, N_0 = population density at time zero, r = intrinsic rate of natural increase and e = the base of natural logarithms.

Quick fact: how fast is exponential?

- → Darwin showed that even a slow-growing animal like the **elephant** could reach enormous numbers in the absence of checks.
- → The chessboard story: a king put 1 grain of rice on the first square, 2 on the second, 4 on the third, doubling each time; the treasury could not fill all **64 squares**.
- → A single *Paramecium* dividing once a day would reach a mind-boggling number in 64 days, if food and space stayed unlimited.

(ii) Logistic growth

No population in nature has **unlimited resources** to permit exponential growth. This leads to **competition** between individuals for limited resources; eventually the 'fittest' individual will survive and reproduce. The governments of many countries have also realised this and introduced various restraints to limit human population growth.

- → In nature, a given habitat has enough resources to support a maximum possible number, beyond which no further growth is possible. This limit is the carrying capacity (K) of that species in that habitat. PYQ
- → A population growing in a habitat with limited resources shows initially a **lag phase**, followed by phases of **acceleration and deceleration**, and finally an **asymptote**, when the population density reaches the carrying capacity.
- → A plot of N in relation to time t results in a sigmoid (S-shaped) curve. This type of growth is called **Verhulst-Pearl logistic growth**.

$$dN/dt = rN[(K - N)/K]$$ K = carrying capacity

Here N = population density at time t and r = intrinsic rate of natural increase.

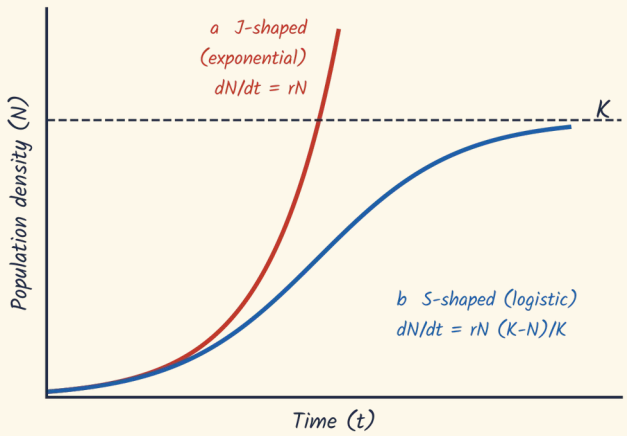


Fig. 11.2 Population growth curves: (a) when resources are not limiting, the plot is exponential; (b) when resources are limiting, the plot is logistic; K is carrying capacity

Since resources for growth of most animal populations are finite and become limiting sooner or later, the logistic growth model is considered more realistic. PYQ

Exponential growth	Logistic growth
resources unlimited	resources limited
J-shaped curve	S-shaped (sigmoid) curve
dN/dt = rN	dN/dt = rN (K – N)/K
no carrying capacity	population levels off at K
no competition	competition for resources
rarely seen; short time only	realistic, seen in nature

Worked example (NCERT exercise)

A population growing exponentially doubles in 3 years. Find r. $N_t = 2N_0$, t = 3. So $2 = e^{3r}$, ln 2 = 3r, r = 0.693/3 = 0.231 per year (about 23 per cent per year).

Yaad karo: J vs S

J = "Jackpot" growth, resources unlimited, rukne ka naam nahi. S = "Samajhdaar" growth, K pe aake ruk jaati hai. Phases: **Lag, Acceleration, Deceleration, Asymptote** = "LADA".

Exam Alert

→ Logistic curve ka diagram (K line ke saath) 3–5 marks mein almost har saal.

→ Formula mein r = b – d; K = carrying capacity. Units dhyan se likho.

→ Jab N = K, toh (K – N)/K = 0, isliye dN/dt = 0: growth ruk jaati hai.

B. Life history variation

Populations evolve to maximise their reproductive fitness, also called **Darwinian fitness** (high r value), in the habitat in which they live. Under a particular set of selection pressures, organisms evolve towards the most efficient reproductive strategy.

Strategy	Examples
Breed only **once** in a lifetime	**Pacific salmon** fish, **bamboo**
Breed **many times** in a lifetime	most **birds and mammals**
Large number of small-sized offspring	**oysters**, pelagic fishes
Small number of large-sized offspring	birds, mammals

Ecologists suggest that life history traits of organisms have evolved in relation to the **constraints imposed by the abiotic and biotic components** of the habitat in which they live. Evolution of life history traits in different species is an important area of ecological research.

5. Population Interactions

In nature, animals, plants and microbes do not and cannot live in isolation; they interact in various ways to form a **biological community**. Even in minimal communities, many interactive linkages exist, though all may not be readily apparent.

Interspecific interactions arise from the interaction of populations of **two different species**. They could be beneficial (+), detrimental (–) or neutral (0) to one of the species or both.

Species A	Species B	Name of interaction
+	+	Mutualism
–	–	Competition
+	–	Predation
+	–	Parasitism
+	0	Commensalism
–	0	Amensalism

Both parties benefit in **mutualism** and both lose in **competition**. In **parasitism and predation** only one species benefits (parasite and predator) and the interaction is detrimental to the other (host and prey). **Commensalism**: one benefits, the other is neither benefited nor harmed. **Amensalism**: one is harmed, the other is unaffected. PYQ

Jab paper mein pucha jaaye: interaction ka sign batao. Spidey sense on!

Asli baat: Predation aur parasitism dono (+/–), commensalism (+/0), amensalism (–/0).

A. Predation

Predation is nature's way of **transferring to higher trophic levels the energy fixed by plants**. When we think of predator and prey, a tiger and a deer readily come to mind, but a **sparrow eating any seed** is no less a predator. Herbivores are not very different from predators, though animals eating plants are categorised separately as **herbivores**.

Important roles of predators:

1. They act as **conduits for energy transfer** across trophic levels.
2. They **keep prey populations under control**. Without predators, prey species could reach very high densities and cause ecosystem instability.
3. They help in **maintaining species diversity** in a community, by reducing the intensity of competition among competing prey species.

Quick fact: famous predation examples

→ When certain exotic species are introduced into a geographical area, they become invasive and spread fast because the invaded land has **no natural predators**. The prickly pear cactus introduced into **Australia** in the early 1920s spread rapidly into millions of hectares of rangeland; it was brought under control only after a **cactus-feeding predator (a moth)** from its natural habitat was introduced.

→ **Biological control** methods in agricultural pest control are based on the ability of the predator to regulate the prey population. PYQ

→ A common example: the fish *Gambusia* is introduced into ponds to feed on mosquito larvae.

Quick fact: predators and diversity

→ In the rocky intertidal communities of the American Pacific Coast, the starfish *Pisaster* is an important predator. When all starfish were removed from an enclosed area, **more than 10 species of invertebrates became extinct within a year** due to interspecific competition.

→ If a predator is too efficient and overexploits its prey, the prey might become extinct and, following it, the predator too will become extinct for lack of food. This is why predators in nature are **'prudent'**.

Defences of prey against predation:

Defence	Example
Camouflage (cryptically coloured, merge with the background)	some insects and frogs
Poisonous / distasteful	**Monarch butterfly**: distasteful due to a special chemical acquired by feeding on a poisonous weed during its caterpillar stage
Morphological defence in plants	**thorns** of *Acacia* and cactus
Chemical defence in plants	***Calotropis*** makes highly poisonous **cardiac glycosides**; nicotine, caffeine, quinine, strychnine, opium are also defences against grazers and browsers

For plants, herbivores are the predators. Nearly **25 per cent of all insects** are known to be **phytophagous** (feeding on plant sap and other parts of plants). The problem is acute for plants because, unlike animals, they **cannot run away**. Many plants produce chemicals that make the herbivore sick when eaten, inhibit feeding or digestion, disrupt its reproduction or even kill it. That is why you never see cattle or goats browsing on *Calotropis*.

Yaad karo: plant ki chemical army

"Nikki Kafi Queen Stray Opera" = Nicotine, Caffeine, Quinine, Strychnine, Opium. Aur *Calotropis* = cardiac glycosides, isliye bakri bhi door bhaagti hai!

B. Competition

When Darwin spoke of the struggle for existence and survival of the fittest, he was convinced that **interspecific competition** is a potent force in organic evolution. It is generally believed that competition occurs when closely related species compete for the same resources that are limited, but this is not entirely true.

→ **Totally unrelated species** could also compete for the same resource: in some shallow South American lakes, visiting **flamingoes and resident fishes** compete for their common food, the **zooplankton**.

→ Resources need not be limiting for competition to occur. In interference competition, the feeding efficiency of one species might be reduced due to the interfering and inhibitory presence of the other species, even if resources (food and space) are abundant.

→ Competition is best defined as a process in which the **fitness of one species** (measured by its r, the intrinsic rate of increase) is significantly lower in the presence of another species. PYQ

Quick fact: evidence for competition

→ The **Abingdon tortoise** in the Galapagos Islands became extinct **within a decade after goats were introduced** on the island, apparently due to the greater browsing efficiency of the goats.

→ Competitive release: a species whose distribution is restricted to a small area due to a competitively superior species expands its range dramatically when the competing species is experimentally removed.

→ **Connell's** field experiments on the rocky sea coasts of Scotland: the larger and competitively superior barnacle ***Balanus*** dominates the intertidal area and excludes the smaller barnacle ***Chthamalus*** from that zone.

→ Herbivores and plants appear to be more adversely affected by competition than carnivores.

Key terms: Gause's competitive exclusion principle

It states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually. This may be true if resources are limiting, but not otherwise. PYQ

Species facing competition might evolve mechanisms that promote co-existence rather than exclusion. One such mechanism is resource partitioning. If two species compete for the same resource, they could avoid competition by choosing, for instance, **different times for feeding or different foraging patterns**.

Quick fact: MacArthur's warblers

MacArthur showed that **five closely related species of warblers** living on the same tree were able to avoid competition and co-exist due to **behavioural differences in their foraging activities** (each fed in a different part of the tree).

Intraspecific competition	Interspecific competition
between members of the **same** species	between members of **different** species
needs are identical, so very intense	needs overlap partly
regulates population size	may lead to exclusion or resource partitioning

Paheli-Bujho

Bujho: Paheli, ek hi ped pe paanch warbler?
Ladai nahi hoti?

Paheli: Nahi, sab alag hisse mein khaate hain. Isko resource partitioning kehte hain.

Bujho: Jaise hostel mein sab ka alag shelf!

Paheli: Exactly. Warna Gause ke hisaab se ek out ho jaata.

C. Parasitism

Considering that the parasitic mode of life ensures **free lodging and meals**, it is not surprising that parasitism has evolved in so many taxonomic groups, from plants to higher vertebrates. Many parasites have evolved to be host-specific (they can parasitise only a single species of host) in such a way that both host and parasite tend to **co-evolve**: if the host evolves special mechanisms to reject or resist the parasite, the parasite has to evolve mechanisms to counteract and neutralise them.

Adaptations of parasites:

- ✓ loss of unnecessary **sense organs**
- ✓ presence of **adhesive organs or suckers** to cling on to the host
- ✓ loss of **digestive system**
- ✓ **high reproductive capacity**

→ Life cycles are often complex, involving one or two intermediate hosts or vectors. The **human liver fluke** (a trematode) depends on two intermediate hosts (a snail and a fish); the **malarial parasite** needs a vector (mosquito) to spread to other hosts. PYQ

→ Most parasites **harm the host**: they may reduce its survival, growth and reproduction and reduce its population density. They might render the host more **vulnerable to predation** by making it physically weak.

Ectoparasites	Endoparasites
feed on the external surface of the host	live inside the host body at different sites (liver, kidney, lungs, RBCs)
lice on humans, ticks on dogs, many marine fish infested with ectoparasitic copepods	liver fluke, malarial parasite, tapeworm
Cuscuta, a parasitic plant on hedge plants, has lost its chlorophyll and leaves; it derives nutrition from the host	life cycles more complex; morphological and anatomical features greatly simplified; emphasis on reproductive potential

Key terms: brood parasitism

In birds, the parasitic bird lays its eggs in the nest of its host and lets the host incubate them. The cuckoo (koel) lays its eggs in the nest of a crow. During evolution, the eggs of the parasitic bird have evolved to **resemble the host's eggs in size and colour** to reduce the chances of the host detecting and ejecting them. PYQ

Exam Alert

→ Female mosquito parasite nahi maana jaata, chahe blood pe depend karti hai (reproduction ke liye).

→ *Cuscuta* = ectoparasite plant (no chlorophyll, no leaves).

→ Koel + crow = brood parasitism; eggs mimic size aur colour.

D. Commensalism

This is an interaction in which one species benefits and the other is neither harmed nor benefited. PYQ

Commensal (benefits, +)	Host (unaffected, 0)	How
Orchid (epiphyte)	**mango** branch	grows on the branch for support and light
Barnacles	back of a **whale**	get transport and new feeding places
Cattle egret	grazing **cattle**	eats insects stirred up from the vegetation by cattle; otherwise hard to find
Clown fish	**sea anemone**	protected from predators by its stinging tentacles

E. Mutualism

This interaction confers benefits on both the interacting species.

→ **Lichens**: an intimate mutualistic relationship between a **fungus** and photosynthesising **algae or cyanobacteria**.

→ **Mycorrhizae**: associations between **fungi and the roots of higher plants**. The fungi help the plant in the **absorption of essential nutrients** from the soil, while the plant provides the fungi with **energy-yielding carbohydrates**.

The most spectacular and evolutionarily fascinating examples of mutualism are found in **plant-animal relationships**. Plants need the help of animals for **pollinating** their flowers and **dispersing** their seeds. In return, plants offer rewards or fees: **pollen and nectar** for pollinators and **juicy, nutritious fruits** for seed dispersers.

Key terms: fig and wasp

Many species of **fig trees** have a tight one-to-one relationship with a pollinator species of wasp: a given fig species can be pollinated only by its partner wasp. The **female wasp** uses the fruit not only as an **oviposition (egg-laying) site** but uses the developing seeds within the fruit for **nourishing its larvae**. The wasp pollinates the fig inflorescence while searching for suitable egg-laying sites; in return the fig offers the wasp some of its developing seeds as food for the larvae.

Key terms: sexual deceit in Ophrys

- → Orchids show a bewildering diversity of floral patterns, many evolved to attract the right pollinator insect and ensure pollination.
- → The Mediterranean orchid *Ophrys* employs '**sexual deceit**': one petal of its flower bears an uncanny resemblance to the **female of the bee** in size, colour and markings.
- → The male bee is attracted to what it perceives as a female, '**pseudocopulates**' with the flower, and is dusted with pollen; it transfers the pollen to another flower.
- → If the female bee's colour patterns change even slightly during evolution, pollination success will be reduced unless the orchid flower **co-evolves** to maintain the resemblance.

F. Amensalism

An interaction in which one species is harmed and the other is unaffected (−/0). For example, the mould *Penicillium* secretes penicillin, which kills bacteria while the mould itself is unaffected; similarly, a large tree shading the ground can suppress small plants below it.

Interaction	Sign	Example
Mutualism	+/+	lichens, mycorrhizae, fig-wasp
Competition	−/−	flamingoes and fishes; *Balanus* and *Chthamalus*
Predation	+/−	tiger-deer, sparrow-seed, starfish *Pisaster*
Parasitism	+/−	*Cuscuta*, liver fluke, cuckoo (brood)
Commensalism	+/0	orchid on mango, cattle egret, clown fish
Amensalism	−/0	*Penicillium* and bacteria

Yaad karo: interactions ka sign

Mutualism = dost (+/+), **Competition = dushman** (−/−), **Predator / Parasite = khaane wala** (+/−), **Commensal = muft ki sawari** (+/0), **Amensal = bina wajah nuksaan** (−/0).

Exam Alert

- → Orchid on mango = **commensalism**, parasitism nahi: orchid apna khana khud banata hai.
- → Lichen aur mycorrhiza = mutualism; *Ophrys* = sexual deceit (pseudocopulation).
- → Clown fish + sea anemone = commensalism (NCERT ke hisaab se).

1 minute recap

1. Population: same species, defined area, share resources, interbreed; natural selection acts at this level.
2. Attributes: birth rate, death rate (per capita), sex ratio, age pyramid (expanding, stable, declining), density N (number, biomass, percent cover, relative, pug marks).
3. $N_{t+1} = N_t + [(B + I) - (D + E)]$.
4. Exponential: $dN/dt = rN$, $N_t = N_0e^{rt}$, J-curve. Logistic: $dN/dt = rN(K - N)/K$, S-curve, K = carrying capacity; more realistic.
5. Life history: salmon and bamboo breed once; oysters many small offspring; birds and mammals few large.
6. Predation: prickly pear and moth, *Pisaster*, Monarch butterfly, *Calotropis*. Competition: flamingoes, Abingdon tortoise, *Balanus*, Gause, warblers.
7. Parasitism: host-specific, liver fluke (snail, fish), *Cuscuta*, cuckoo. Commensalism: orchid, barnacles, egret, clown fish. Mutualism: lichen, mycorrhiza, fig-wasp, *Ophrys*.

MOST IMPORTANT QUESTIONS

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM Board fav Which of the following is an attribute of a population but not of an individual?

(a) birth
(b) death
(c) sex ratio
(d) age

Ans: (c) sex ratio

Q2. IM In a pond with 20 lotus plants, 8 new plants are added in a year. The birth rate is:

(a) 0.8
(b) 0.4 per lotus per year
(c) 2.5
(d) 0.25

Ans: (b) 0.4 per lotus per year

Q3. IM Board fav An urn-shaped age pyramid indicates a population that is:

(a) expanding
(b) stable
(c) growing rapidly
(d) declining

Ans: (d) declining

Q4. IM Tiger census in national parks is based on:
(a) pug marks and fecal pellets
(b) total count by sight
(c) number caught per trap
(d) percent cover
Ans: (a) pug marks and fecal pellets

Q5. IM For a single huge banyan tree in an area with 200 carrot grass plants, the best measure of population size is:
(a) number
(b) relative density
(c) percent cover or biomass
(d) age pyramid
Ans: (c) percent cover or biomass

Q6. IM Board fav Population density increases when:
(a) (B + I) > (D + E)
(b) (B + I) < (D + E)
(c) (B + D) > (I + E)
(d) B = D
Ans: (a) (B + I) > (D + E)

Q7. IM Board fav In dN/dt = rN, r stands for:
(a) rate of respiration
(b) intrinsic rate of natural increase
(c) carrying capacity
(d) resistance of environment
Ans: (b) intrinsic rate of natural increase

Q8. IM The value of r for the human population of India in 1981 was:
(a) 0.015
(b) 0.12
(c) 0.0205
(d) 0.205
Ans: (c) 0.0205

Q9. IM A J-shaped growth curve is obtained when:
(a) resources are unlimited
(b) resources are limited
(c) the population reaches K
(d) emigration is high
Ans: (a) resources are unlimited

Q10. IM The logistic growth curve is also called:
(a) geometric growth
(b) J-curve
(c) Darwinian growth
(d) Verhulst-Pearl logistic growth
Ans: (d) Verhulst-Pearl logistic growth

Q11. IM Which organism breeds only once in its lifetime?
(a) sparrow
(b) Pacific salmon
(c) dog
(d) oyster
Ans: (b) Pacific salmon

Q12. IM Board fav Cattle egret and grazing cattle show:
(a) mutualism
(b) predation
(c) commensalism
(d) parasitism
Ans: (c) commensalism

Q13. IM The interaction between a fungus and roots of higher plants is:
(a) mycorrhiza (mutualism)
(b) lichen
(c) parasitism
(d) amensalism
Ans: (a) mycorrhiza (mutualism)

Q14. IM Board fav *Ophrys* uses which strategy for pollination?
(a) nectar reward
(b) sexual deceit
(c) brood parasitism
(d) camouflage
Ans: (b) sexual deceit

Q15. IM The prickly pear cactus in Australia was controlled by introducing:
(a) goats
(b) a fungus
(c) a virus
(d) a cactus-feeding moth
Ans: (d) a cactus-feeding moth

Q16. IM The Abingdon tortoise became extinct after the introduction of:
(a) goats
(b) rats
(c) cats
(d) dogs
Ans: (a) goats

Q17. IM Board fav *Calotropis* protects itself from grazers by producing:
(a) nicotine
(b) quinine
(c) cardiac glycosides
(d) caffeine
Ans: (c) cardiac glycosides

Q18. 1M Board fav (NCERT) Select the statement which best explains parasitism:
(a) one organism is benefited
(b) both organisms are benefited
(c) one organism is benefited, other is not affected
(d) one organism is benefited, other is affected
Ans: (d) one organism is benefited, the other is affected

Assertion-Reason (1 mark each)

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q19. 1M Board fav **A:** The logistic growth model is more realistic. **R:** Resources for most populations are finite and become limiting.
Ans: (a) Both true and R explains A.

Q20. 1M **A:** Monarch butterfly is avoided by its predators. **R:** It acquires a poisonous chemical by feeding on a poisonous weed as a caterpillar.
Ans: (a) Both true and R explains A.

Q21. 1M Board fav **A:** Five species of warblers co-exist on one tree. **R:** They forage in different parts of the tree (resource partitioning).
Ans: (a) Both true and R explains A.

Q22. 1M **A:** Female mosquito is a parasite. **R:** It needs blood for reproduction.
Ans: (d) A false, R true: it is not considered a parasite.

Q23. 1M **A:** Competition occurs only between closely related species. **R:** Flamingoes and fishes compete for zooplankton.
Ans: (d) A false, R true: unrelated species can also compete.

Short answer (2-3 marks)

Q24. 2M Board fav (NCERT) List the attributes that populations possess but not individuals.
Ans: Birth rate, death rate, sex ratio, age distribution (age pyramid) and population density.

Q25. 2M Board fav (NCERT) If a population growing exponentially doubles in size in 3 years, what is r?
Ans: $N_t = N_0 e^{rt}$; $2N_0 = N_0 e^{3r}$; $\ln 2 = 3r$; $r = 0.693/3 = 0.231$ per year.

Q26. 3M Board fav (NCERT) Name important defence mechanisms in plants against herbivory.
Ans: Morphological: thorns (*Acacia*, cactus). Chemical: poisonous or distasteful chemicals such as cardiac glycosides (*Calotropis*), nicotine, caffeine, quinine, strychnine and opium that make herbivores sick, inhibit feeding or digestion, disrupt reproduction or kill them.

Q27. 2M Board fav (NCERT) An orchid grows on the branch of a mango tree. How do you describe this interaction?
Ans: Commensalism. The orchid (epiphyte) gets support and better light, while the mango tree is neither harmed nor benefited.

Q28. 2M (NCERT) What is the ecological principle behind biological control of pest insects?
Ans: Predation: a predator (or parasite) regulates the prey population. Natural enemies of the pest are introduced to keep the pest population under control without chemicals.

Q29. 2M (NCERT) Define population and community.
Ans: Population: a group of individuals of the same species in a defined area, sharing or competing for resources and potentially interbreeding. Community: all the populations of different species living and interacting in an area.

Q30. 3M Board fav (NCERT) Define with one example each: (a) commensalism (b) parasitism (c) camouflage (d) mutualism (e) interspecific competition.
Ans: (a) One benefits, other unaffected: orchid on mango. (b) One benefits at the cost of the other: *Cuscuta* on hedge plants. (c) Cryptic colouration to merge with surroundings: some insects and frogs. (d) Both benefit: lichens. (e) Fitness of one species lowered in the presence of another: flamingoes and fishes for zooplankton.

Q31. 3M (NCERT) List any three important characteristics of a population and explain.
Ans: (1) Birth rate: per capita births per unit time, e.g. 0.4 per lotus per year. (2) Death rate: per capita deaths, e.g. 0.1 per fruit fly per week. (3) Age distribution: proportion in pre-reproductive, reproductive and post-reproductive groups, shown by age pyramids (expanding, stable, declining). (Sex ratio and density also accepted.)

Q32. 3M Why is population density not always measured as a number? Give examples.
Ans: Number may be meaningless or hard to count. A single huge banyan tree plays a bigger role than 200 *Parthenium* plants, so percent cover or biomass is used; bacteria in a Petri dish are too many to count; tigers are estimated from pug marks and fecal pellets.

Q33. 2M Board fav Write the equation for population density at time t + 1 and explain each term.
Ans: $N_{t+1} = N_t + [(B + I) - (D + E)]$; N_t = density at time t, B = births (natality), I = immigration, D = deaths (mortality), E = emigration.

Q34. 2M When does immigration become more important than birth rate?
Ans: When a new habitat is just being colonised; individuals arriving from outside then contribute more to population growth than births.

Q35. 3M Board fav Differentiate between exponential and logistic growth.
Ans: Exponential: unlimited resources, $dN/dt = rN$, J-shaped curve, no carrying capacity. Logistic: limited resources, $dN/dt = rN(K - N)/K$, S-shaped curve with lag, acceleration, deceleration and asymptote at K.

Q36. 2M What is carrying capacity? What happens to growth when $N = K$?
Ans: The maximum number of individuals of a species that a habitat can support with its resources. When $N = K$, $(K - N)/K = 0$, so $dN/dt = 0$ and growth stops.

Q37. 2M What is meant by Darwinian fitness? Give two life history strategies.
Ans: Reproductive fitness (high r value) that populations evolve to maximise in their habitat. Breeding once (Pacific salmon, bamboo) or many times (birds, mammals); many small offspring (oysters) or few large ones (birds, mammals).

Q38. 3M Board fav Explain three roles of predators in nature.
Ans: (1) Transfer energy fixed by plants to higher trophic levels. (2) Keep prey populations under control, preventing instability. (3) Maintain species diversity by reducing competition among prey, e.g. removal of *Pisaster* made more than 10 invertebrate species extinct within a year.

Q39. 2M Why are predators said to be 'prudent'?
Ans: If a predator overexploits its prey, the prey may become extinct and then the predator too dies out for lack of food. So in nature predators avoid overexploiting their prey.

Q40. 2M What is interference competition?
Ans: Competition in which the feeding efficiency of one species is reduced by the interfering and inhibitory presence of another species, even when food and space are abundant.

Q41. 3M Board fav State Gause's competitive exclusion principle. How do species escape it?
Ans: Two closely related species competing for the same limiting resources cannot co-exist indefinitely; the inferior one is eliminated. Species escape it by resource partitioning, e.g. feeding at different times or in different places, as in MacArthur's warblers.

Q42. 3M What is competitive release? Give Connell's example.
Ans: A species restricted by a superior competitor expands its range when the competitor is removed. On the Scottish coast, the larger barnacle *Balanus* excludes the smaller *Chthamalus* from the intertidal zone; when *Balanus* is removed, *Chthamalus* spreads.

Q43. 2M Board fav List four adaptations of parasites.
Ans: Loss of unnecessary sense organs, adhesive organs or suckers to cling to the host, loss of digestive system, and high reproductive capacity.

Q44. 2M Board fav What is brood parasitism? Give an example.
Ans: A parasitic bird lays eggs in the host's nest and lets the host incubate them, e.g. the cuckoo (koel) lays eggs in a crow's nest; its eggs resemble the host's in size and colour.

Q45. 2M Distinguish between ectoparasites and endoparasites with examples.
Ans: Ectoparasites feed on the external surface of the host (lice on humans, ticks on dogs, copepods on fish, *Cuscuta*). Endoparasites live inside the host body (liver fluke, malarial parasite); their life cycles are more complex and body features simplified.

Q46. 3M Board fav Explain the mutualism between fig and wasp.
Ans: Each fig species is pollinated only by its partner wasp. The female wasp lays eggs in the fig fruit and its larvae feed on some developing seeds; while searching for egg-laying sites, the wasp pollinates the fig inflorescence.

Q47. 3M What are lichens and mycorrhizae? Why are they mutualistic?
Ans: Lichen: fungus + photosynthetic alga or cyanobacterium; the alga makes food, the fungus gives shelter and absorbs water. Mycorrhiza: fungus + roots of higher plants; the fungus helps absorb nutrients, the plant gives carbohydrates. Both partners benefit.

Q48. 2M Give two examples of commensalism other than the orchid on mango.
Ans: Barnacles growing on the back of a whale; cattle egret feeding near grazing cattle; clown fish living among the stinging tentacles of a sea anemone (any two).

Long answer (5 marks)

Q49. SM Board fav (NCERT) With a suitable diagram describe the logistic population growth curve.

Ans: Draw the S-shaped curve of N against t with a dashed line K. In a habitat with limited resources, growth shows a lag phase, then acceleration, deceleration, and finally an asymptote when N reaches the carrying capacity K. Equation: $dN/dt = rN[(K - N)/K]$, where r is the intrinsic rate of natural increase. As N approaches K, $(K - N)/K$ approaches 0 and growth slows to zero. Called Verhulst-Pearl logistic growth; more realistic than exponential growth because resources become limiting.

Q50. SM Describe the exponential growth model of a population. Derive its equation and give its curve.

Ans: When resources are unlimited, each species realises its full potential and grows geometrically. If b and d are per capita birth and death rates, $dN/dt = (b - d)N$; putting $r = b - d$, $dN/dt = rN$. The integral form is $N_t = N_0e^{rt}$ ($e = 2.71828$). The plot of N against t is J-shaped. r values: Norway rat 0.015, flour beetle 0.12, India (1981) 0.0205. Example: Darwin's elephants, chessboard story.

Q51. SM Board fav Describe the various defences of prey against predators with examples.

Ans: Animals: camouflage (cryptic colouration of insects and frogs); being poisonous or distasteful (Monarch butterfly, which gets a chemical from a poisonous weed as a caterpillar). Plants cannot run away, so: morphological defences such as thorns (*Acacia*, cactus); chemical defences such as cardiac glycosides in *Calotropis* and nicotine, caffeine, quinine, strychnine and opium, which make herbivores sick, inhibit feeding or digestion, disrupt reproduction or kill them. Nearly 25% of insects are phytophagous.

Q52. SM Explain parasitism with its types, adaptations and examples. Why do host and parasite co-evolve?

Ans: Parasitism (+/–): the parasite gets free lodging and food and harms the host. Many parasites are host-specific, so host and parasite co-evolve: when the host evolves resistance, the parasite evolves counter-measures. Adaptations: loss of sense organs, suckers or adhesive organs, loss of digestive system, high reproductive capacity; complex life cycles (liver fluke: snail and fish hosts; malarial parasite: mosquito vector). Ectoparasites: lice, ticks, copepods, *Cuscuta*. Endoparasites: liver fluke, malarial parasite. Brood parasitism: koel lays eggs in a crow's nest.

Q53. SM Board fav Explain competition with examples. Include Gause's principle and resource partitioning.

Ans: Competition (–/–) lowers the fitness (r) of one species in the presence of another. It may occur between unrelated species (flamingoes and fishes for zooplankton) and even without limiting resources (interference competition). Evidence: Abingdon tortoise extinct after goats were introduced; competitive release, e.g. *Balanus* excluding *Chthamalus* (Connell). Gause: two closely related species competing for the same limiting resources cannot co-exist indefinitely. Resource partitioning allows co-existence: MacArthur's five warbler species foraging in different parts of a tree.

Case-based (4 marks)

Q54. 4M Board fav A student counts a population of 500 deer in a forest. In one year, 60 deer are born, 30 die, 10 enter from a nearby forest and 20 leave. (a) Find N_{t+1}. (b) Find the per capita birth rate. (c) Is the population growing? (d) Name the four processes.

Ans: (a) $500 + [(60 + 10) - (30 + 20)] = 520$. (b) $60/500 = 0.12$ per deer per year. (c) Yes, since B + I (70) > D + E (50). (d) Natality, immigration, mortality, emigration.

Q55. 4M Starfish were removed from a rocky coast. Within a year more than 10 species of invertebrates disappeared. (a) Name the starfish. (b) What role did it play? (c) Why did the invertebrates disappear? (d) What is this role of predators called?

Ans: (a) *Pisaster*. (b) It was a key predator controlling prey populations. (c) Without the predator, competition among prey species increased and the weaker ones were eliminated. (d) Maintaining species diversity by reducing competition among prey.

Q56. 4M Board fav A crow notices nothing odd as it incubates eggs in its nest, but one chick hatches as a koel. (a) Name the interaction. (b) Which bird is the parasite? (c) How did the eggs escape detection? (d) Under which broad interaction does it fall?

Ans: (a) Brood parasitism. (b) The koel (cuckoo). (c) Its eggs resemble the crow's eggs in size and colour. (d) Parasitism (+/–).

Exam Alert: most asked

- → Logistic curve diagram + equation, aur exponential vs logistic difference: har saal.
- → Interaction pehchano: orchid-mango, egret-cattle, fig-wasp, koel-crow, *Ophrys*.
- → Age pyramid ke 3 shapes; r ka numerical (doubling time).
- → Gause ka principle, resource partitioning (warblers), aur predator ke 3 roles.

ECOSYSTEM

Most asked: GPP vs NPP, steps of decomposition, 10 per cent law, GFC vs DFC, inverted pyramids, limitations of ecological pyramids

An **ecosystem** is the functional unit of nature. This chapter covers its **structure, productivity, decomposition, energy flow and ecological pyramids.**

1. Ecosystem: Structure and Function

An **ecosystem** is a functional unit of nature, where living organisms interact among themselves and also with the surrounding **physical environment.**

→ Size varies greatly: from a **small pond** to a **large forest or a sea.**
→ Many ecologists regard the entire biosphere as a global ecosystem, a composite of all local ecosystems on Earth.
→ Being too big and complex to study at one time, it is divided into two basic categories:

Terrestrial ecosystems	*Aquatic ecosystems*
forest, grassland, desert	pond, lake, wetland, river, estuary
on land; temperature and soil moisture are key factors	in water; light and nutrients are key factors

Man-made ecosystems: crop fields and an **aquarium** are ecosystems created and maintained by humans.

Natural ecosystem	*Man-made (artificial) ecosystem*
develops on its own, without human help	created and maintained by humans
self-regulating and self-sustaining	needs constant human inputs (water, fertiliser, feed)
high species diversity; long food chains	low diversity (often one crop); short food chains
e.g. forest, pond, ocean	e.g. crop field, aquarium, garden

A. Structure of an ecosystem

Interaction of **biotic** (living) and **abiotic** (non-living) components results in a physical structure that is characteristic for each type of ecosystem.

→ **Species composition**: identification and enumeration of the plant and animal species of an ecosystem.
→ **Stratification**: vertical distribution of different species occupying different levels. In a forest, **trees** occupy the top vertical stratum, **shrubs** the second, and **herbs and grasses** the bottom layers. PYQ

Quick fact

Species composition and stratification are the **two main structural features** of an ecosystem. Stratification lets many species share one area by using light at different heights.

B. Functional aspects

The components of an ecosystem function as a **unit** when we consider four aspects:

1. **Productivity**
2. **Decomposition**
3. **Energy flow**
4. **Nutrient cycling**

Abiotic components	*Biotic components*
inorganic substances: water, air, soil minerals, CO_2, O_2	**producers**: green plants, phytoplankton (autotrophs)
organic compounds of dead matter (proteins, carbohydrates, humus)	**consumers**: herbivores, carnivores, omnivores (heterotrophs)
climatic factors: light, temperature, rainfall, humidity	**decomposers**: bacteria, fungi (saprotrophs)

C. Pond: a model ecosystem

A **pond** is a shallow water body and a fairly self-sustainable unit. It shows all four basic components of an ecosystem clearly, so it explains even the complex interactions of an aquatic ecosystem.

Component	*In a pond*
Abiotic	water with dissolved inorganic and organic substances; rich soil deposit at the bottom
Climatic regulators	solar input, cycle of temperature, day-length and other climatic conditions regulate the rate of function of the whole pond
Autotrophs (producers)	phytoplankton, some algae, and floating, submerged and marginal plants at the edges
Consumers	zooplankton, free-swimming and bottom-dwelling forms
Decomposers	fungi, bacteria and flagellates, especially abundant at the **bottom**

The pond performs all functions of any ecosystem and of the biosphere:

1. Conversion of **inorganic into organic** material by autotrophs with the radiant energy of the sun.
2. Consumption of autotrophs by **heterotrophs.**
3. **Decomposition and mineralisation** of dead matter to release nutrients back for reuse by autotrophs; these events repeat over and over.

There is a unidirectional movement of energy towards the higher trophic levels, and its **dissipation and loss as heat** to the environment. PYQ

Paheli-Bujho

Bujho: Paheli, pond toh bas paani hai na? Isme ecosystem kahan hai?

Paheli: Paani, mitti, dhoop abiotic hain; phytoplankton producer, zooplankton consumer.

Bujho: Aur neeche wala keechad?

Paheli: Wahi decomposers ka ghar hai: fungi, bacteria. Poora set ek pond mein!

2. Productivity

A **constant input of solar energy** is the basic requirement for any ecosystem to function and sustain.

- → **Primary production** = the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis. Expressed as **weight (g m^{-2})** or **energy (kcal m^{-2})**.
- → **Productivity** = the rate of biomass production; expressed in **g m^{-2} yr^{-1}** or **(kcal m^{-2}) yr^{-1}**, so that different ecosystems can be compared.

Gross primary productivity (GPP)	Net primary productivity (NPP)
rate of production of organic matter during photosynthesis	GPP minus respiration losses (R)
total energy fixed by producers	energy left after producers use some in respiration
a considerable amount is used by plants in **respiration**	the **available biomass** for consumption by heterotrophs (herbivores and decomposers)

GPP − R = NPP (Board fav)

Secondary productivity is the rate of formation of new organic matter by consumers. PYQ

Factors affecting primary productivity

- → the **plant species** inhabiting the area
- → environmental factors (light, temperature, water)
- → availability of **nutrients**
- → **photosynthetic capacity** of plants

Hence productivity varies in different types of ecosystems.

Quick fact: numbers to remember

- → Annual NPP of the whole biosphere is about **170 billion tons** (dry weight) of organic matter.
- → Oceans cover about **70%** of the surface, yet their productivity is only **55 billion tons**; the rest is on land.

Exam Alert: why is ocean productivity low?

- → Light penetrates only the upper layers of water, so photosynthesis is limited to a thin zone.
- → Nutrients (especially nitrogen and phosphorus) are scarce in open surface water; they settle in deep water.
- → In aquatic ecosystems, **light** is the limiting factor for productivity (NCERT fill-in-the-blank).

Yaad karo: GPP aur NPP

GPP = poori salary. **R** = ghar ka kharcha (respiration). **NPP** = jo bachaa, wahi dusron (herbivores, decomposers) ke kaam aata hai. Salary − kharcha = savings!

3. Decomposition

The earthworm is called the farmer's **friend** because it helps in the breakdown of complex organic matter and in **loosening of the soil.**

Decomposition: decomposers break down complex organic matter into inorganic substances like carbon dioxide, water and nutrients.

Detritus: dead plant remains such as **leaves, bark, flowers** and dead remains of animals, including **fecal matter**. Detritus is the raw material for decomposition. PYQ

Steps of decomposition

Step	What happens
1. Fragmentation	**detritivores** (e.g. earthworm) break down detritus into smaller particles
2. Leaching	water-soluble inorganic nutrients go down into the soil horizon and get precipitated as **unavailable salts**
3. Catabolism	**bacterial and fungal enzymes** degrade detritus into simpler inorganic substances
4. Humification	accumulation of a dark coloured amorphous substance, **humus**
5. Mineralisation	humus is further degraded by some microbes, releasing **inorganic nutrients**

All the steps of decomposition operate simultaneously on the detritus. Humification and mineralisation occur during decomposition in the **soil.**

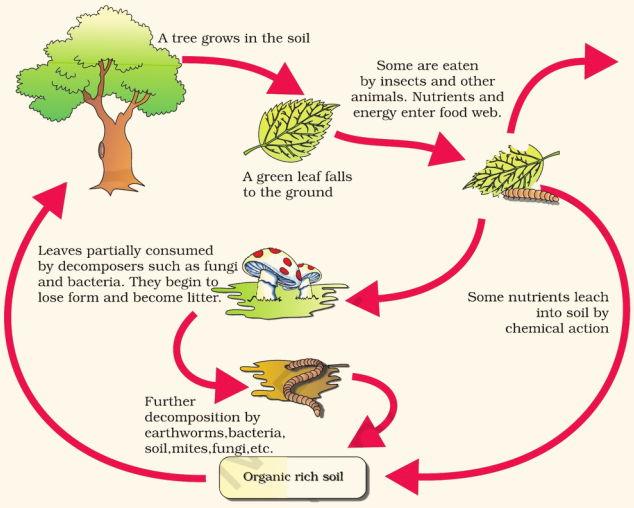


Fig. 12.1 Decomposition cycle in a terrestrial ecosystem

Humus

→ dark coloured, **amorphous** substance

→ highly resistant to microbial action; decomposes at an extremely slow rate

→ **colloidal** in nature, so it serves as a **reservoir of nutrients**

Factors affecting the rate of decomposition

Decomposition is largely an oxygen-requiring process. Its rate is controlled by the **chemical composition of detritus** and **climatic factors**.

Decomposition is faster	Decomposition is slower
detritus rich in **nitrogen** and water-soluble substances like **sugars**	detritus rich in **lignin** and **chitin**
warm and moist environment	**low temperature** and **anaerobiosis** (lack of oxygen)
active soil microbes	inhibited microbes, so organic matter **builds up**

Temperature and soil moisture are the most important climatic factors; they regulate decomposition through their effect on the activities of soil microbes. PYQ

Yaad karo: steps of decomposition

Fir **L**ate **C**lass **H**ui **M**ummy = **F**ragmentation, **L**eaching, **C**atabolism, **H**umification, **M**ineralisation.

Exam Alert

→ Fragmentation is done by **detritivores** (physical); catabolism by **enzymes of bacteria and fungi** (chemical). Dono ko mix mat karna.

→ Leaching mein nutrients **unavailable salts** ban jaate hain: ye loss hai, gain nahi.

→ NCERT summary: decomposition involves three processes, fragmentation, leaching and catabolism; humification and mineralisation happen in the soil.

4. Energy Flow

→ Except for the **deep sea hydro-thermal ecosystem**, the sun is the only source of energy for all ecosystems on Earth.

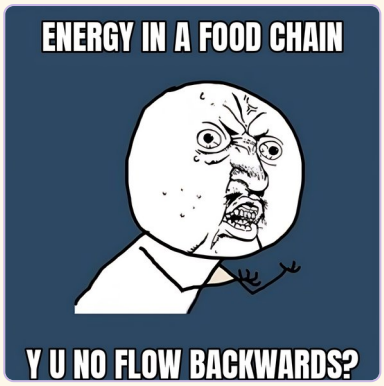
→ Of the incident solar radiation, **less than 50%** is photosynthetically active radiation (PAR).

→ Plants capture only **2–10% of the PAR**, and this small amount of energy sustains the entire living world. PYQ

→ All organisms depend on producers for food, directly or indirectly; so there is a unidirectional flow of energy from the sun to producers and then to consumers.

Quick fact: thermodynamics

Unidirectional flow obeys the **first law** (energy is converted, not created). Ecosystems are also not exempt from the **second law**: they need a constant supply of energy to synthesise the molecules they require, to counteract the universal tendency towards increasing **disorderliness**.



Bujho ka sapna: energy wapas producer tak laut aaye.
Asli baat: Energy flow is ***unidirectional***; nutrients cycle, energy does not.

A. Producers and consumers

→ **Producers**: green plants. Terrestrial: **herbaceous and woody plants**. Aquatic: **phytoplankton, algae and higher plants**.

→ **Consumers (heterotrophs)**: all animals depend on plants, directly or indirectly.

Consumer	Feeds on	Examples
Primary consumer (herbivore)	producers (plants)	insects, birds, mammals on land; **molluscs** in water
Secondary consumer (primary carnivore)	herbivores	frog, small birds, fox
Tertiary consumer (secondary carnivore)	primary carnivores	snake, hawk, wolf, man

No energy that is trapped into an organism remains in it for ever: the energy trapped by the producer is either **passed on to a consumer** or the organism dies. Death of an organism is the beginning of the detritus food chain.

B. Food chains

Grazing food chain (GFC): starts from green plants.

Grass (producer) → Goat (primary consumer) → Man (secondary consumer)

Detritus food chain (DFC): begins with dead organic matter. It is made up of **decomposers**, heterotrophic organisms, mainly **fungi and bacteria**, also called **saprotrophs** (sapro: to decompose). They secrete digestive enzymes that break down dead and waste materials into simple inorganic materials, which they then absorb. PYQ

Grazing food chain	Detritus food chain
starts with living green plants (producers)	starts with dead organic matter (detritus)
energy comes directly from the sun	energy comes from detritus
major conduit of energy flow in **aquatic** ecosystems	a **much larger fraction** of energy flows through it in **terrestrial** ecosystems
herbivores, carnivores	decomposers: fungi, bacteria (saprotrophs)

C. Food web

The DFC may be connected with the GFC at some levels: some organisms of the DFC are **prey to GFC animals**, and some animals like **cockroaches and crows** are omnivores. These natural interconnections of food chains make a **food web**.

Food chain	Food web
a single, straight sequence of who eats whom	many interconnected food chains
each organism has one food source at one level	an organism may have many food sources
removal of one link disturbs the whole chain	alternatives exist, so it gives **stability**

Grass ne mehnat se khaana banaya, deer ne seedha energy le li.

Asli baat: Producers fix solar energy; herbivores (2nd trophic level) take it from them.

D. Trophic levels

Based on the source of their nutrition, organisms occupy a specific place in the food chain, their **trophic level**.

Trophic level	Organisms	Examples (land / water)
First	producers (plants)	grass, trees / phytoplankton
Second	primary consumers (herbivores)	cow, grasshopper / zooplankton
Third	secondary consumers (carnivores)	birds, fishes, wolf
Fourth	tertiary consumers (top carnivores)	man, lion, hawk

→ The amount of energy decreases at successive trophic levels.

→ When an organism dies it becomes detritus (dead biomass), an energy source for decomposers.

→ Organisms at each trophic level depend on those at the **lower** trophic level for energy.

E. Standing crop

Each trophic level has a certain mass of living material at a particular time, called the **standing crop**. It is measured as the **mass of living organisms (biomass)** or the **number** in a unit area. Biomass is expressed as fresh or dry weight; dry weight is more accurate because water content varies with season and time and has no energy value. PYQ

F. The 10 per cent law

Only 10 per cent of the energy is transferred to each trophic level from the lower trophic level (the rest is used in respiration and lost as heat). So the number of trophic levels in a **grazing food chain is restricted** (producer, herbivore, primary carnivore, secondary carnivore). Board fav

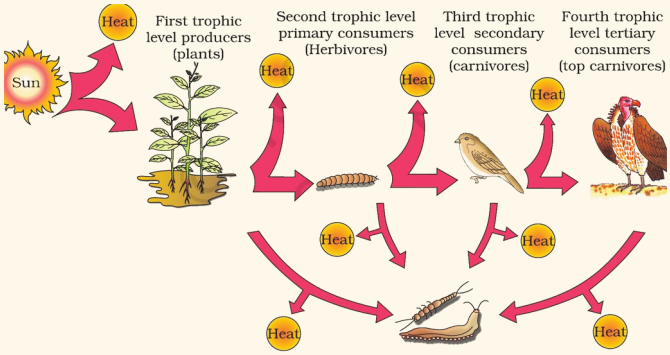


Fig. 12.3 Energy flow through different trophic levels

Level	Energy (with 1,000,000 J of sunlight)
Sunlight	1,000,000 J
Producers (1% converted to NPP)	10,000 J
Primary consumers	1,000 J
Secondary consumers	100 J
Tertiary consumers	10 J

Exam Alert: 10% law numericals

→ If producers have 10,000 J, the **third** trophic level gets 10,000 x 10% x 10% = **100 J**.

→ Reverse: a top carnivore with 10 J at the 4th level needs 10,000 J at the producer level.

→ A detritus food chain is not limited this way in the same manner, since decomposers feed on dead matter from all levels.

Paheli-Bujho

Bujho: Lion itna bada hai, phir bhi jungle mein kam kyun hote hain?

Paheli: 10 percent law! Har level pe sirf 10% energy aage jaati hai.

Bujho: Matlab lion ke hisse mein bas crumbs?

Paheli: Haan, isliye food chain 4–5 levels se lambi nahi hoti.

5. Ecological Pyramids

The **base** of a pyramid is broad and it narrows towards the **apex**. A similar shape is obtained when the food or energy relationship between organisms at different trophic levels is expressed in terms of **number, biomass or energy**.

→ Base = **producers** (first trophic level); apex = **tertiary or top level consumer**.

→ Three types: (a) pyramid of number, (b) pyramid of biomass, (c) pyramid of energy.

Fig. 12.4 (a) Pyramid of numbers in a grassland: only 3 top carnivores on nearly 6 million plants. (b) Upright pyramid of biomass: sharp decrease at higher levels

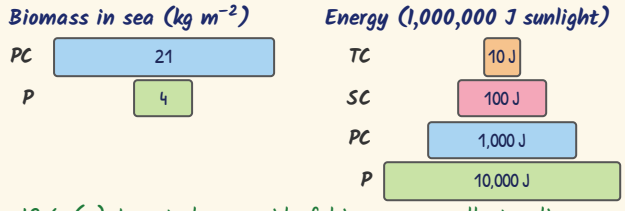


Fig. 12.4 (c) Inverted pyramid of biomass: small standing crop of phytoplankton supports a large standing crop of zooplankton. (d) Ideal pyramid of energy: producers convert only 1% of sunlight into NPP

Important points

→ Any calculation of energy content, biomass or numbers has to include all organisms at that trophic level; generalisations from a few individuals are not true.

→ A given organism may occupy **more than one trophic level** simultaneously. The trophic level represents a functional level, not a species. A **sparrow** is a primary consumer when it eats seeds, fruits and peas, and a secondary consumer when it eats insects and worms. PYQ

Pyramid	Usually	Exception
Number	upright (grassland)	**inverted**: one big tree supports many insects, which support small birds and fewer large birds (tree ecosystem: spindle-shaped/inverted)
Biomass	upright (land)	**inverted in sea**: biomass of fishes (and zooplankton) far exceeds that of phytoplankton
Energy	**always upright**	**never inverted**: some energy is always lost as heat at each step

Each bar in the energy pyramid indicates the amount of energy present at each trophic level in a given time or annually per unit area.

In most ecosystems all three pyramids are **upright**: producers are more in number and biomass than herbivores, and herbivores more than carnivores. Energy at a lower trophic level is always more than at a higher level.

Quick fact: the sea paradox

The sea's inverted biomass pyramid is not a paradox: phytoplankton have a very **small standing crop** but a very **high turnover rate**; they multiply fast and are eaten fast, so a small mass at any moment feeds a larger mass of consumers over time.

Limitations of ecological pyramids

1. They do not take into account the **same species belonging to two or more trophic levels.**
2. They assume a **simple food chain**, something that almost never exists in nature; they do not accommodate a **food web.**
3. **Saprophytes** are not given any place, even though they play a vital role in the ecosystem.

Yaad karo: kaun ulta ho sakta hai?

Number: ped pe keede (ulta). **Biomass**: samundar (ulta). **Energy**: kabhi ulta nahi, kyunki har step pe heat ka "tax" katta hai!

Paheli-Bujho

Bujho: Samundar mein phytoplankton kam aur machhliyan zyada? Ye kaise chalta hai?

Paheli: Phytoplankton bahut tez badhte hain, aur tez khaye bhi jaate hain.

Bujho: Toh pyramid ulta, par energy wala?

Paheli: Energy pyramid hamesha seedha. Heat loss kabhi ulta nahi hota.

Feature	Number	Biomass	Energy
measures	individuals per unit area	dry weight per unit area (standing crop)	energy per unit area per time (J or kcal m^{-2} yr^{-1})
can be inverted?	yes (tree)	yes (sea)	**never**
most accurate picture	least	better	**best** (includes time, turnover)

6. Nutrient Cycling and Ecosystem Services

Nutrient cycling: the storage and movement of nutrient elements through the various components of the ecosystem; nutrients are **used repeatedly** through this process (also called biogeochemical cycles).

Gaseous cycle	Sedimentary cycle
reservoir: **atmosphere or hydrosphere**	reservoir: **Earth's crust**
e.g. **carbon** (nitrogen too)	e.g. **phosphorus** (sulphur too)
faster; element exists as a gas (CO_2)	slower; released by weathering of rocks

The **oceans** are the major reservoir of carbon on Earth (NCERT fill-in-the-blank).

Ecosystem services: products of ecosystem processes, e.g. purification of air and water by forests. Healthy ecosystems also mitigate droughts and floods, cycle nutrients, generate fertile soils, give wildlife habitat and pollinate flowering plants.

NCERT in-text questions, answered

→ **Why is ocean productivity low?** Light is limited to surface layers and nutrients are scarce there.
→ **Unidirectional flow and the first law?** Yes: energy is only converted (solar to chemical to heat), never created or destroyed.
→ **Why is dry weight more accurate?** Water content varies and carries no energy.
→ **Is a DFC limited like a GFC?** Less so: decomposers draw on dead matter from every trophic level, so the DFC is not restricted to a fixed number of levels in the same way.
→ **How would you classify human beings?** Omnivores working at the 2nd, 3rd and even 4th trophic levels.
→ **Sea paradox?** Phytoplankton have a small standing crop but a very fast turnover.

Key terms

→ **Stratification**: vertical distribution of species at different levels.
→ **Detritus**: dead plant and animal remains, incl. fecal matter.
→ **Humus**: dark, amorphous, colloidal, slow-decomposing reservoir of nutrients.
→ **PAR**: photosynthetically active radiation (less than 50% of sunlight).
→ **Standing crop**: mass or number of living organisms at a trophic level at a given time.
→ **Saprotrophs**: decomposers (fungi, bacteria) living on dead matter.
→ **Trophic level**: functional level of an organism in a food chain.

1 minute recap

1. Ecosystem = functional unit; structure: species composition + stratification.
2. Four functions: productivity, decomposition, energy flow, nutrient cycling.
3. GPP – R = NPP; biosphere NPP 170 billion t, oceans only 55 billion t.
4. Decomposition: fragmentation, leaching, catabolism, humification, mineralisation; fast if warm, moist, N-rich; slow if lignin, chitin, cold, anaerobic.
5. Sun is the only energy source (except deep-sea hydrothermal); PAR below 50%; plants capture 2–10% of PAR.
6. GFC dominates water, DFC dominates land; 10% law limits trophic levels.
7. Pyramids: energy always upright; number (tree) and biomass (sea) can be inverted; 3 limitations.

Exam Alert: most asked in boards

→ GPP – R = NPP with units; why ocean productivity is low.
→ 5 steps of decomposition + factors (lignin/chitin slow, warm-moist fast).
→ Why energy pyramid is never inverted; sea biomass pyramid inverted; tree number pyramid inverted.

Most Important Questions

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM Board fav Which one of the following has the largest population in a food chain?
(a) Producers
(b) Primary consumers
(c) Secondary consumers
(d) Decomposers
Ans: (d) Decomposers

Q2. IM Board fav The second trophic level in a lake is:
(a) Phytoplankton
(b) Zooplankton
(c) Benthos
(d) Fishes
Ans: (b) Zooplankton

Q3. IM Secondary producers are:
(a) Herbivores
(b) Producers
(c) Carnivores
(d) None of the above
Ans: (d) None of the above (there is no such category; producers are always primary)

Q4. IM Board fav The percentage of photosynthetically active radiation (PAR) in the incident solar radiation is about:
(a) 100%
(b) 50%
(c) 1–5%
(d) 2–10%
Ans: (b) 50% (less than 50 per cent)

Q5. IM Plants capture what percentage of PAR?
(a) 50%
(b) 1%
(c) 2–10%
(d) 20%
Ans: (c) 2–10%

Q6. IM Board fav Which pyramid is always upright?
(a) Pyramid of number
(b) Pyramid of biomass
(c) Pyramid of energy
(d) All of these
Ans: (c) Pyramid of energy

Q7. IM Decomposition is slow when detritus is rich in:
(a) Sugars
(b) Lignin and chitin
(c) Nitrogen
(d) Water-soluble substances
Ans: (b) Lignin and chitin

Q8. IM The step of decomposition in which water-soluble nutrients go down into the soil and are precipitated as unavailable salts is:
(a) Leaching
(b) Catabolism
(c) Humification
(d) Mineralisation
Ans: (a) Leaching

Q9. IM Board fav Fill in the blanks: (a) Plants are called ___ because they fix CO_2. (b) In an ecosystem dominated by trees, the pyramid of numbers is ___ type. (c) In aquatic ecosystems the limiting factor for productivity is ___. (d) Common detritivores are ___. (e) The major reservoir of carbon on earth is ___.
Ans: (a) producers (autotrophs); (b) inverted; (c) light; (d) earthworms; (e) oceans.

Q10. IM Annual net primary productivity of the whole biosphere is approximately:
(a) 55 billion tons
(b) 170 billion tons
(c) 115 billion tons
(d) 700 billion tons
Ans: (b) 170 billion tons (dry weight)

Q11. IM Which is the only ecosystem that does not depend on the sun for energy?
Ans: The deep sea hydro-thermal ecosystem.

Q12. IM Name the dark, amorphous, colloidal substance formed during decomposition.
Ans: Humus (formed by humification).

Q13. IM Board fav If 20,000 J of energy is trapped by producers, how much energy is available to the secondary consumers?
(a) 200 J
(b) 2,000 J
(c) 20 J
(d) 2 J
Ans: (a) 200 J (20,000 to 2,000 to 200 by the 10 per cent law)

Q14. IM A sparrow eating seeds and also insects shows that:
(a) it is a decomposer
(b) it is a producer
(c) one species can occupy more than one trophic level
(d) pyramids are always inverted
Ans: (c) one species can occupy more than one trophic level

Q15. IM The rate of formation of new organic matter by consumers is called:
(a) GPP
(b) NPP
(c) Secondary productivity
(d) Standing crop
Ans: (c) Secondary productivity

Q16. IM Which of these is the reservoir of the sedimentary cycle of phosphorus?
(a) Atmosphere
(b) Earth's crust
(c) Hydrosphere
(d) Biosphere
Ans: (b) Earth's crust

Q17. IM Board fav Vertical distribution of different species occupying different levels in an ecosystem is:
(a) Stratification
(b) Species composition
(c) Succession
(d) Zonation
Ans: (a) Stratification

Q18. IM Which of the following organisms are called saprotrophs?
(a) Herbivores
(b) Phytoplankton
(c) Omnivores
(d) Fungi and bacteria that decompose dead matter
Ans: (d) Fungi and bacteria that decompose dead matter

Q19. IM Which of the following is an example of an ecosystem service?
(a) Deforestation
(b) Purification of air and water by forests
(c) Biomagnification
(d) Mining
Ans: (b) Purification of air and water by forests

Q20. IM Board fav The unit of productivity is:
(a) g m^{-2}
(b) kcal m^{-2}
(c) g m^{-2} yr^{-1}
(d) J
Ans: (c) g m^{-2} yr^{-1}

Assertion-Reason (1 mark each)

Key: (a) both A and R true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

Q21. IM Board fav **A:** The pyramid of energy is always upright. **R:** Some energy is lost as heat at every trophic level.
Ans: (a) Both true; R correctly explains A.

Q22. IM **A:** The pyramid of biomass in the sea is generally inverted. **R:** The biomass of fishes far exceeds that of phytoplankton.
Ans: (a) Both true; R correctly explains A.

Q23. IM **A:** In a terrestrial ecosystem, a much larger fraction of energy flows through the detritus food chain. **R:** In an aquatic ecosystem, the grazing food chain is the major conduit of energy flow.
Ans: (b) Both true, but R does not explain A.

Q24. IM **A:** Decomposition is faster in cold and dry regions. **R:** Low temperature and anaerobiosis inhibit decomposition.
Ans: (d) A is false (warm and moist conditions favour decomposition); R is true.

Short answer (2-3 marks)

Q25. 3M Board fav Distinguish between grazing food chain and detritus food chain.

Ans: GFC starts with living green plants and gets energy directly from the sun; it is the major conduit of energy flow in aquatic ecosystems. DFC starts with dead organic matter (detritus), is made of decomposers (fungi, bacteria) and carries a much larger fraction of energy in terrestrial ecosystems. Example GFC: grass, goat, man.

Q26. 2M Distinguish between production and decomposition.

Ans: Production: synthesis of organic matter (biomass) from inorganic raw materials by producers using solar energy during photosynthesis; energy is stored. Decomposition: breakdown of complex organic matter (detritus) into inorganic substances like CO_2, water and nutrients by decomposers; nutrients are released.

Q27. 2M Board fav Distinguish between upright and inverted pyramids with one example each.

Ans: Upright: the base (producers) is largest and values decrease towards the apex, e.g. pyramid of numbers in a grassland, pyramid of biomass on land, every pyramid of energy. Inverted: values increase towards the apex, e.g. pyramid of numbers in a tree ecosystem, pyramid of biomass in the sea.

Q28. 2M Distinguish between food chain and food web.

Ans: Food chain: a single linear sequence of organisms through which energy passes (grass, goat, man). Food web: a network of interconnected food chains; organisms have alternative food sources, so the ecosystem is more stable.

Q29. 2M Distinguish between litter and detritus.

Ans: Litter: dead plant material lying on the soil surface (leaves, twigs, bark). Detritus: all dead plant remains (leaves, bark, flowers) and dead remains of animals including fecal matter; it is the raw material for decomposition. Litter is a part of detritus.

Q30. 2M Board fav Distinguish between primary and secondary productivity.

Ans: Primary productivity: rate of biomass production by producers during photosynthesis (g m^{-2} yr^{-1}), divided into GPP and NPP. Secondary productivity: rate of formation of new organic matter by consumers.

Q31. 3M Board fav What is primary productivity? Give a brief description of the factors that affect it.

Ans: The rate of biomass or organic matter production per unit area over a time period by plants during photosynthesis. It depends on the plant species of the area, environmental factors (light, temperature, water), availability of nutrients and the photosynthetic capacity of plants, so it varies between ecosystems.

Q32. 2M Board fav Define GPP and NPP. How are they related?

Ans: GPP: rate of production of organic matter during photosynthesis. NPP: GPP minus respiration losses; the biomass available to heterotrophs. GPP − R = NPP.

Q33. 3M Why is the productivity of oceans low even though they cover about 70% of the Earth's surface?

Ans: Light penetrates only the upper layers, so photosynthesis is limited to a thin zone; nutrients are scarce in surface waters. Light is the limiting factor in aquatic ecosystems. Hence oceans produce only about 55 of the 170 billion tons of annual NPP.

Q34. 2M Board fav What is stratification? Explain with an example.

Ans: Vertical distribution of different species occupying different levels. In a forest, trees occupy the top stratum, shrubs the second and herbs and grasses the bottom layers.

Q35. 2M Name the four functional aspects of an ecosystem.

Ans: Productivity, decomposition, energy flow and nutrient cycling.

Q36. 2M What is humification? Why is humus important?

Ans: Accumulation of a dark coloured amorphous substance, humus, during decomposition in the soil. Humus is highly resistant to microbial action and decomposes very slowly; being colloidal, it acts as a reservoir of nutrients.

Q37. 2M Differentiate between fragmentation and catabolism.

Ans: Fragmentation: physical breakdown of detritus into smaller particles by detritivores such as earthworms. Catabolism: chemical breakdown of detritus into simpler inorganic substances by enzymes of bacteria and fungi.

Q38. 2M Board fav How do temperature and soil moisture influence decomposition?

Ans: They regulate the activity of soil microbes. Warm and moist conditions favour decomposition; low temperature and anaerobiosis inhibit it, so organic matter builds up.

Q39. 2M Why is decomposition slower in detritus rich in lignin and chitin?

Ans: Lignin and chitin are complex, tough compounds that few microbes can break down easily, while nitrogen-rich detritus with water-soluble sugars is decomposed quickly.

Q40. 2M Why is the earthworm called the farmer's friend?

Ans: It helps in the breakdown (fragmentation) of complex organic matter and in loosening the soil, improving aeration and fertility.

Q41. 3M Board fav What is the 10 per cent law? How does it limit the length of a food chain?

Ans: Only about 10% of the energy at one trophic level is transferred to the next; the rest is used in respiration and lost as heat. Energy falls so fast that after 4-5 levels too little is left to support another level, so grazing food chains are short.

Q42. 2M Board fav What is standing crop? Why is dry weight a more accurate measure of biomass?
Ans: Standing crop is the mass of living material (biomass) or number of organisms at a trophic level at a particular time, per unit area. Dry weight is more accurate because water content varies with season and organism and carries no energy value.

Q43. 2M "The trophic level represents a functional level, not a species." Explain with an example.
Ans: A species can feed at different levels at the same time. A sparrow eating seeds, fruits and peas is a primary consumer, but eating insects and worms it is a secondary consumer.

Q44. 3M Board fav Write any three limitations of ecological pyramids.
Ans: (i) They do not consider the same species at two or more trophic levels. (ii) They assume a simple food chain and do not accommodate a food web. (iii) Saprophytes are given no place though they play a vital role.

Q45. 2M Board fav Why is the pyramid of energy never inverted?
Ans: When energy flows from one trophic level to the next, some energy is always lost as heat (respiration). Each higher level therefore always has less energy than the level below.

Q46. 3M Draw the pyramid of numbers for a tree ecosystem with insects, small birds and large birds. What is its shape?
Ans: One tree at the base supports many insects, which support fewer small birds and still fewer large birds. The pyramid is inverted (spindle shaped) at the base, since one producer supports many herbivores.

Q47. 2M How would you classify human beings in a food chain?
Ans: Humans are omnivores: primary consumers when eating plant food and secondary or tertiary consumers when eating meat or fish. So they function at more than one trophic level.

Q48. 3M Explain how the detritus food chain is linked to the grazing food chain.
Ans: Some organisms of the DFC (earthworms, insects) are prey to GFC animals (birds), and omnivores such as cockroaches and crows feed in both. Death of any GFC organism adds detritus to the DFC. These links form a food web.

Q49. 2M What is meant by unidirectional flow of energy? Is it in keeping with the first law of thermodynamics?
Ans: Energy moves from the sun to producers and then to consumers and never back; at each step part is lost as heat. Yes: energy is only converted from one form to another (solar to chemical to heat), not created or destroyed.

Q50. 2M Board fav What are gaseous and sedimentary nutrient cycles? Give one example of each with its reservoir.
Ans: Nutrient cycling is the storage and movement of nutrients through ecosystem components. Gaseous cycle: reservoir is the atmosphere or hydrosphere, e.g. carbon. Sedimentary cycle: reservoir is the Earth's crust, e.g. phosphorus.

Q51. 2M What are ecosystem services? Give two examples.
Ans: Products of ecosystem processes that benefit humans, e.g. purification of air and water by forests, pollination of crops, mitigation of floods and droughts, formation of fertile soil.

Q52. 3M Explain why a terrestrial ecosystem has more energy flowing through the detritus food chain.
Ans: On land a large part of plant biomass (leaves, wood, litter) is not eaten by herbivores; it dies and becomes detritus that decomposers use. Hence a much larger fraction of energy flows through the DFC on land, while in water the GFC dominates.

Q53. 2M Board fav In a food chain, a frog has 100 J of energy. How much energy was present in the grass (producer) if the chain is grass, grasshopper, frog? State the law used.
Ans: 10 per cent law: grasshopper had 1,000 J and grass had 10,000 J.

Q54. 3M Board fav What would happen to an ecosystem if all its decomposers were removed?
Ans: Dead organic matter would pile up, nutrients would stay locked in detritus and would not return to the soil (no mineralisation); producers would run short of nutrients, productivity would fall and the ecosystem would collapse.

Q55. 2M Differentiate between detritivores and decomposers.
Ans: Detritivores (e.g. earthworm) feed on detritus and break it into smaller particles (fragmentation, physical). Decomposers (bacteria, fungi) secrete enzymes and chemically break detritus into inorganic substances (catabolism) which they absorb.

Q56. 2M Why are the producers always placed at the base of an ecological pyramid?
Ans: Producers form the first trophic level; they fix solar energy and all other organisms depend on them directly or indirectly, so they hold the largest energy (and usually number and biomass).

Q57. 2M Explain the role of the second law of thermodynamics in ecosystems.
Ans: Ecosystems need a constant supply of energy to synthesise the molecules they require and to counteract the universal tendency towards increasing disorderliness; at each transfer energy is lost as heat, so energy must keep flowing in from the sun.

Q58. 2M Why do the energy contents of successive trophic levels decrease? Name two ways in which energy is lost at each level.
Ans: Organisms use most of the energy they take in for their own life processes. Energy is lost (i) as heat during respiration and (ii) in undigested or uneaten parts and dead matter that pass to decomposers. Only about 10% is stored as biomass for the next level.

Q59. 2M Name the decomposers of a pond and state where they are most abundant. What is their role?
Ans: Fungi, bacteria and flagellates; most abundant at the bottom of the pond. They decompose and mineralise dead matter, releasing nutrients for reuse by the autotrophs.

Q60. 2M What are omnivores? Give two examples and state their role in linking food chains.
Ans: Animals that feed on both plants and animals, e.g. cockroaches, crows (and humans). Feeding at several trophic levels, they connect different food chains, including the grazing and detritus chains, into a food web.

Q61. 2M Which climatic factors regulate the rate of function of a pond ecosystem?
Ans: Solar input (light), the cycle of temperature, day-length and other climatic conditions.

Q62. 2M Give one example each of a natural and a man-made ecosystem. Why does a crop field need constant human input?
Ans: Natural: forest or pond. Man-made: crop field or aquarium. A crop field has low diversity and is not self-sustaining; it needs water, fertilisers, pest control and harvesting by humans.

Q63. 2M Explain why a pond is called a self-sustainable unit.
Ans: It has all four functional aspects: producers fix solar energy, consumers feed on them, decomposers mineralise dead matter and return nutrients to producers, and energy flows in from the sun continuously; so it maintains itself without outside inputs other than sunlight.

Q64. 2M What is detritus? Name its main components and the food chain it starts.
Ans: Dead plant remains (leaves, bark, flowers) and dead remains of animals including fecal matter. It is the raw material for decomposition and starts the detritus food chain.

Q65. 2M Board fav What is mineralisation? How is it different from humification?
Ans: Mineralisation: release of inorganic nutrients when humus is further degraded by some microbes. Humification: formation of humus, a dark, amorphous, resistant, colloidal substance. Humification stores nutrients; mineralisation releases them.

Long answer (5 marks)

Q66. 5M Board fav Describe the components of an ecosystem, using a pond as an example.
Ans: Abiotic: water with dissolved inorganic and organic substances, soil at the bottom; solar input, temperature and day-length regulate it. Biotic: producers (phytoplankton, algae, floating, submerged and marginal plants), consumers (zooplankton, free-swimming and bottom-dwelling forms), decomposers (fungi, bacteria, flagellates at the bottom). Functions: autotrophs convert inorganic to organic matter using sunlight; heterotrophs consume autotrophs; decomposition and mineralisation return nutrients. Energy flows one way and is lost as heat. Structure: species composition and stratification.

Q67. 5M Board fav Define decomposition. Describe the processes and products of decomposition.
Ans: Decomposition: breakdown of complex organic matter (detritus) into inorganic substances by decomposers. Processes: fragmentation by detritivores; leaching of water-soluble nutrients into the soil as unavailable salts; catabolism by bacterial and fungal enzymes; humification forming humus; mineralisation releasing inorganic nutrients. Products: CO_2, water, inorganic nutrients and humus. Rate: fast in warm, moist conditions with N-rich, sugary detritus; slow with lignin, chitin, low temperature, anaerobiosis.

Q68. 5M Board fav Give an account of energy flow in an ecosystem.
Ans: Sun is the only source (except deep-sea hydrothermal); less than 50% of sunlight is PAR and plants capture 2–10% of it. Producers fix it as food; it passes to primary, secondary and tertiary consumers through grazing food chains, or to decomposers through detritus food chains (GFC dominates water, DFC dominates land). Flow is unidirectional; at each level energy is used in respiration and lost as heat, so only about 10% passes on (10 per cent law): e.g. 1,000,000 J sunlight, 10,000 J producers, 1,000 J, 100 J, 10 J. This limits food chains to 4–5 levels and makes the energy pyramid upright.

Q69. 5M Board fav Define ecological pyramids. Describe with examples the pyramids of number and biomass. Why can the pyramid of energy never be inverted?
Ans: Graphic representation of number, biomass or energy at successive trophic levels; producers form the base, top consumers the apex. Number: upright in a grassland (5,842,000 plants, 708,000 PC, 354,000 SC, 3 TC); inverted in a tree ecosystem. Biomass: upright on land (809, 37, 11, 1.5 kg m^{-2}); inverted in sea (phytoplankton 4, zooplankton 21 kg m^{-2}). Energy: always upright because energy is lost as heat at each transfer. Limitations: species at many levels, food webs, saprophytes ignored.

Q70. 5M Board fav (a) Define productivity and distinguish between GPP, NPP and secondary productivity. (b) Why does productivity vary between ecosystems? (c) Compare the productivity of oceans and land.

Ans: (a) Productivity is the rate of biomass production per unit area over time (g m^{-2} yr^{-1} or kcal m^{-2} yr^{-1}). GPP: rate of production of organic matter in photosynthesis. NPP = GPP − R, the biomass available to herbivores and decomposers. Secondary productivity: rate of formation of new organic matter by consumers. (b) It depends on the plant species, environmental factors (light, temperature, water), nutrient availability and photosynthetic capacity. (c) Annual NPP of the biosphere is about 170 billion tons (dry weight); oceans, though covering 70% of the surface, give only 55 billion tons because light and nutrients limit them; the rest is from land.

Ecosystem ke numericals galat hue? Mistake notebook mein likho.

Asli baat: 10% law ke sawal 2 baar solve karo: board mein pakke aate hain.

Case-based (4 marks)

Q71. 4M Board fav In a forest, a student counted 1 large banyan tree, 4,000 insects on it, 40 small birds feeding on the insects and 3 hawks. (a) Draw the pyramid of numbers. (b) Is it upright or inverted? (c) Would the pyramid of energy for the same system be inverted? Why? (d) Which pyramid limitation does the bird that also eats fruit show?

Ans: (a) Narrow base (1 tree), wide second bar (4,000 insects), then 40, then 3. (b) Inverted at the base (spindle shaped). (c) No; energy is always upright, as heat is lost at each level and the tree holds the most energy. (d) The same species at more than one trophic level.

Q72. 4M In a grassland, producers fix 20,000 kJ m^{-2} yr^{-1} (GPP) and use 8,000 kJ in respiration. (a) Calculate NPP. (b) Using the 10 per cent law, how much energy reaches the tertiary consumers? (c) Where does the rest of the energy go? (d) Name the food chain that carries most energy in this land ecosystem.

Ans: (a) NPP = GPP − R = 20,000 − 8,000 = 12,000 kJ m^{-2} yr^{-1}. (b) 12,000 to 1,200 to 120 to 12 kJ at the tertiary level. (c) Used in respiration and lost as heat, and passed to decomposers as dead matter. (d) The detritus food chain.

Q73. 4M Board fav Two bags of leaves were buried: bag A in a warm, humid tropical forest and bag B in a cold, waterlogged bog. After a year, A had almost disappeared while B was largely intact. (a) Explain the difference. (b) Name the two climatic factors involved. (c) Which step converts the remains into a nutrient reservoir? (d) Why would pine needles rich in lignin decay slowly even in A?

Ans: (a) Warm, moist, aerobic conditions favour microbial activity in A; cold and anaerobic (waterlogged) conditions inhibit it in B, so organic matter builds up. (b) Temperature and soil moisture. (c) Humification (humus, colloidal reservoir of nutrients). (d) Lignin is resistant; detritus rich in lignin and chitin decomposes slowly.

Q74. 4M Board fav A pond was studied: phytoplankton 4 kg m^{-2}, zooplankton 21 kg m^{-2}. (a) Draw the pyramid of biomass. (b) Name its type. (c) How can a small producer biomass support a larger consumer biomass? (d) Draw the pyramid of energy for the same pond.

Ans: (a) Small base (4) and wide top (21). (b) Inverted pyramid of biomass. (c) Phytoplankton have a small standing crop but a very high turnover rate: they reproduce and are eaten quickly. (d) Upright: producers have the most energy, which decreases at each level.

Q75. 4M Board fav Read the food web of a grassland: grass is eaten by grasshoppers, rabbits and mice; frogs eat grasshoppers; snakes eat frogs and mice; hawks eat snakes, rabbits and mice. (a) Write the longest food chain. (b) At which trophic levels does the hawk feed? (c) Why is a food web more stable than a single food chain? (d) Name the organisms that would start the detritus food chain when the hawk dies.

Ans: (a) Grass, grasshopper, frog, snake, hawk. (b) Third level (eating rabbits and mice), fourth and fifth (eating snakes in the two chains). (c) Each organism has alternative food sources; loss of one species does not break all energy paths. (d) Decomposers: bacteria and fungi (with detritivores such as earthworms and insects).

Q76. 4M Scientists measured the energy of a lake: producers 8,000 kJ, primary consumers 800 kJ, secondary consumers 80 kJ. (a) Which law is shown? (b) Predict the energy of tertiary consumers. (c) Can this pyramid ever be inverted? (d) Name the main food chain that carries energy in this lake.

Ans: (a) The 10 per cent law of energy transfer. (b) 8 kJ. (c) No; the pyramid of energy is always upright because energy is lost as heat at each transfer. (d) The grazing food chain (the major conduit in aquatic ecosystems).

Exam Alert: diagrams and pairs

- → Must draw: decomposition cycle (Fig. 12.1), energy flow (Fig. 12.3), all 4 pyramids with numbers.
- → Differences ke 6 NCERT pairs (GFC/DFC, production/decomposition, upright/inverted, chain/web, litter/detritus, primary/secondary productivity): har saal 2–3 marks.
- → Numericals: GPP − R = NPP aur 10% law; units likhna mat bhoolna.

Biodiversity and Conservation

Most asked: levels of biodiversity, latitudinal gradient (3 hypotheses), species-area relationship (Z value), Evil Quartet, rivet popper hypothesis, in situ vs ex situ, hotspots, sacred groves

Biodiversity is the variety of life at **genetic, species and ecosystem levels**. This chapter covers its **patterns, importance, loss and conservation**.

1. Biodiversity

The diversity of life on Earth is enormous: there are more than **20,000 species of ants**, **3,00,000 species of beetles**, **28,000 species of fishes** and nearly **20,000 species of orchids**.

Biodiversity is the term popularised by the sociobiologist Edward Wilson to describe the combined diversity at all levels of biological organisation. Ecosystems are not only diverse in species but also show diversity at all levels, from macromolecules within cells to biomes.

A. Three levels of biodiversity Board fav

Level	Meaning	NCERT example
(i) Genetic diversity	high diversity at the genetic level within a single species over its distributional range	*Rauwolfia vomitoria* in different Himalayan ranges varies in potency and concentration of **reserpine**; India has more than **50,000** strains of rice and **1,000** varieties of mango
(ii) Species diversity	diversity at the species level	the **Western Ghats** have greater amphibian species diversity than the **Eastern Ghats**
(iii) Ecological diversity	diversity at the ecosystem level	India (deserts, rain forests, mangroves, coral reefs, wetlands, estuaries, alpine meadows) has greater ecosystem diversity than a Scandinavian country like **Norway**

Yaad karo: 3 levels

Good Students Excel = Genetic, Species, Ecological. Rice ke 50,000 strains = genetic; Western Ghats ke frogs = species; desert se coral reef tak = ecological.

Quick fact

It has taken **millions of years** of evolution to accumulate this rich diversity, but we could lose all of it in less than two centuries if the present rates of species losses continue.

B. How many species are there on Earth and in India?

→ According to the **IUCN (2004)**, the total number of plant and animal species **described** so far is slightly more than **1.5 million**.

→ For many groups (especially tropical ones) inventories are incomplete. Ecologists estimate the total by a **statistical comparison** of the temperate-tropical species richness of an exhaustively studied group (insects) with other groups, and extrapolate this ratio to estimate the total number of species. PYQ

→ Extreme estimates range from **20 to 50 million**, but a more conservative and scientifically sound estimate by Robert May places global species diversity at about **7 million**.

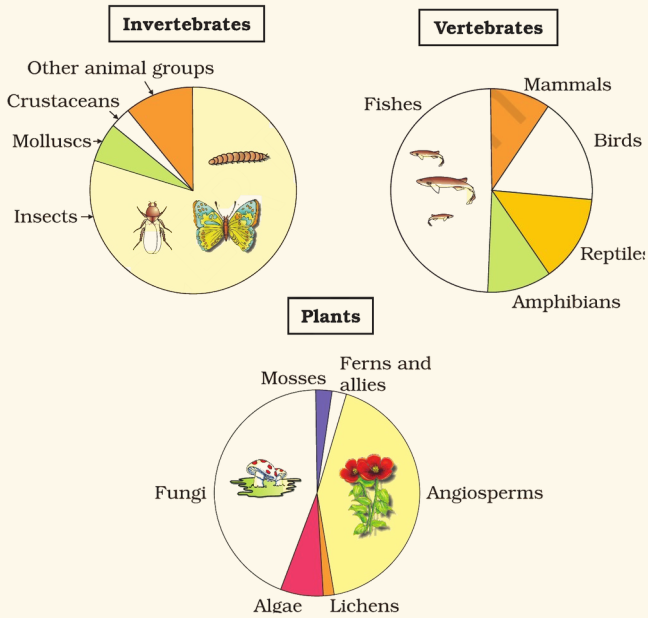


Fig. 13.1 Global biodiversity: proportionate number of species of major taxa of plants, invertebrates and vertebrates

→ More than **70%** of all species recorded are **animals**; plants (including algae, fungi, bryophytes, gymnosperms and angiosperms) comprise no more than **22%**.

→ Among animals, insects are the most species-rich group, making up more than **70%** of the total: out of every 10 animals on the planet, **7 are insects**.

→ The number of **fungi** species in the world is more than the combined total of the species of fishes, amphibians, reptiles and mammals.

Quick fact: prokaryotes are missing

Biodiversity estimates do not give figures for **prokaryotes**. Conventional taxonomic methods are not suitable for identifying microbial species and many species are **not culturable** under laboratory conditions. Using biochemical or molecular criteria, their diversity alone might run into **millions**.

C. India as a mega diversity country

Fact	Value
India's share of world land area	only **2.4%**
India's share of global species diversity	**8.1%**
Rank	one of the **12 mega diversity countries**
Species recorded	nearly **45,000** plants and **twice as many** animals
Yet to be discovered (by May's 22%)	more than **1,00,000** plant and **3,00,000** animal species

Exam Alert

→ 7 million = Robert May; 1.5 million = described so far (IUCN 2004). Dono ko confuse mat karna.

→ Biodiversity term = **Edward Wilson**; rivet popper = **Paul Ehrlich**; species-area = **Humboldt**; outdoor plots = **David Tilman**.

→ India: 2.4% land, 8.1% species: ye pair 1 mark mein seedha aata hai.

Paheli-Bujho

Bujho: Paheli, duniya mein kitne species hain? Gin ke batao!

Paheli: Describe hue sirf 1.5 million. Robert May ke hisaab se lagbhag 7 million.

Bujho: Toh baaki sab chhupe kahan hain?

Paheli: Tropics ke jungle mein aur microbes mein, jo lab mein ugte hi nahi!

2. Patterns of Biodiversity

A. Latitudinal gradients

Diversity of plants and animals is not uniform throughout the world. In general, species diversity decreases as we move away from the equator towards the poles. With very few exceptions, the **tropics (23.5° N to 23.5° S)** harbour more species than temperate or polar areas. PYQ

Place (latitude)	Bird species
Colombia (near equator)	nearly **1,400**
India (mostly tropical)	more than **1,200**
New York (41° N)	**105**
Greenland (71° N)	only **56**

→ A tropical forest like **Ecuador** has up to **10 times** as many species of vascular plants as a forest of equal area in a temperate region like the Midwest of the USA.

→ The largely tropical Amazonian rain forest (South America) has the **greatest biodiversity on Earth**: more than 40,000 species of plants, 3,000 of fishes, 1,300 of birds, 427 of mammals, 427 of amphibians, 378 of reptiles and more than 1,25,000 invertebrates. There may be at least **two million insect species** there still waiting to be discovered and named.

Why do the tropics have more species? (3 hypotheses)

Board fav

1. **Time:** speciation is generally a function of time. Temperate regions were subjected to **frequent glaciations** in the past, while tropical latitudes remained relatively undisturbed for millions of years and thus had a long evolutionary time for species diversification.
2. **Constant environment:** tropical environments are less seasonal, relatively more **constant and predictable**. Such constant environments promote niche specialisation and lead to greater species diversity.
3. **More solar energy:** more solar energy is available in the tropics, which contributes to **higher productivity**; this in turn might contribute indirectly to greater diversity.

Yaad karo: tropics jeet gaye

T S E = Time (glaciation nahi hua), Stable mausam (niche specialisation), Energy (zyada dhoop, zyada productivity). Teeno likhoge to 3 marks pakke!

B. Species-area relationships

During his pioneering explorations of South American jungles, the German naturalist and geographer Alexander von Humboldt observed that within a region, species richness **increased with increasing explored area, but only up to a limit.**

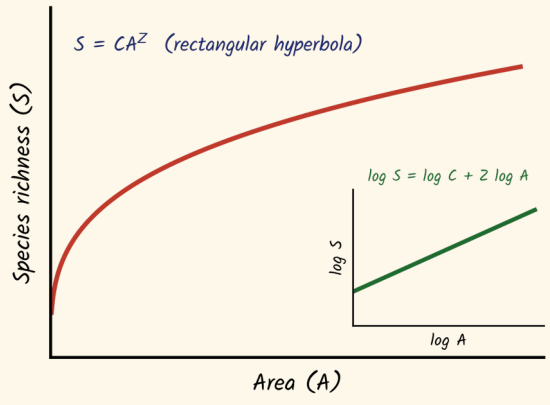


Fig. 13.2 Species-area relationship (on a log scale the relationship becomes linear)

For a wide variety of taxa (angiosperm plants, birds, bats, freshwater fishes), the relation between species richness and area turns out to be a **rectangular hyperbola**. On a logarithmic scale it is a **straight line**, described by:

$$\log S = \log C + Z \log A$$ (Board fav)

Symbol	Meaning
S	species richness
A	area
Z	slope of the line (**regression coefficient**)
C	Y-intercept

→ Regardless of the taxonomic group or the region (plants in Britain, birds in California or molluscs in New York State), the slopes are amazingly similar: Z lies in the range 0.1 to 0.2.

→ For very large areas like **entire continents**, the slope is much steeper: Z = 0.6 to 1.2. For frugivorous (fruit-eating) birds and mammals in tropical forests of different continents, the slope is **1.15**.

→ A **steeper slope** means species richness rises faster with area, i.e. larger areas add many more new species. PYQ

Exam Alert: species-area numericals

→ Small area (region): Z = 0.1–0.2. Continent: Z = 0.6–1.2 (frugivores 1.15).

→ Graph mein curve = rectangular hyperbola; log-log = straight line. Axes label zaroor karna: S on Y, A on X.

→ Z = slope = regression coefficient: ye naam board mein poochha jaata hai.

C. Importance of species diversity to the ecosystem

Ecologists believe that communities with more species tend to be more stable than those with fewer species. A stable community:

→ should not show too much **variation in productivity** from year to year;

→ must be either **resistant or resilient** to occasional disturbance (natural or man-made);

→ must be resistant to **invasions by alien species**.

David Tilman's long-term ecosystem experiments using **outdoor plots**: plots with more species showed less year-to-year variation in total biomass; he also showed that increased diversity contributed to **higher productivity**. PYQ

Rivet popper hypothesis Board fav

An analogy used by the Stanford ecologist Paul Ehrlich to show why species richness matters:

Airplane analogy	Ecosystem meaning
airplane	ecosystem
thousands of rivets joining its parts	species
every passenger pops a rivet to take home	species become extinct one by one
initially flight safety is not affected; over time the plane becomes dangerously weak	loss of species may not affect the ecosystem at first, but over time functioning weakens
loss of rivets on the **wings** is more serious than on seats or windows	loss of **key species** that drive major ecosystem functions is a more serious threat

Paheli-Bujho

Bujho: Ek-do species chali gayi toh kya farak padega?
Paheli: Plane ke rivets socho. Ek-do nikle toh plane udta rahega.
Bujho: Aur agar wings ke rivets nikle?
Paheli: Toh crash! Key species gayi toh ecosystem bhi gir jaata hai.

3. Loss of Biodiversity

→ Colonisation of tropical Pacific Islands by humans is said to have led to the extinction of more than **2,000 species of native birds**.

→ The IUCN Red List (2004) documents the extinction of **784 species** (338 vertebrates, 359 invertebrates and 87 plants) in the last **500 years**. PYQ

→ Recent extinctions: **dodo** (Mauritius), **quagga** (Africa), **thylacine** (Australia), **Steller's sea cow** (Russia) and three subspecies (**Bali, Javan, Caspian**) of tiger. The last twenty years alone saw the disappearance of **27 species**.

→ Extinctions across taxa are not random: **amphibians** appear to be more vulnerable.

→ More than **15,500 species** worldwide face the threat of extinction.

Group	Share facing threat of extinction
Birds	12%
Mammals	23%
Amphibians	32%
Gymnosperms	31%

The sixth mass extinction

Five episodes of mass extinction happened before humans appeared. The current '**Sixth Extinction**' is different: it is 100 to 1,000 times faster than in pre-human times, and **human activities** are responsible. If present trends continue, nearly **half of all species** on Earth might be wiped out within the next 100 years.

Type of extinction	Feature
Natural (background)	slow replacement of species by better adapted ones as conditions change; always going on
Mass extinction	a large number of species lost in a short geological time due to catastrophes; five such episodes before humans
Anthropogenic	caused by human activities (habitat loss, over-exploitation, invasions); the present **sixth extinction**

Effects of loss of biodiversity in a region

1. decline in **plant production**
2. lowered resistance to environmental perturbations such as **drought**
3. increased variability in certain ecosystem processes such as plant productivity, water use, and **pest and disease cycles**

Causes of biodiversity losses: the Evil Quartet Board fav

1. **Habitat loss and fragmentation**: the most important cause. Tropical rain forests once covered more than **14%** of Earth's land surface; now no more than **6%**. The **Amazon rain forest** (the 'lungs of the planet', harbouring probably millions of species) is being cut and cleared for cultivating **soya beans** or for conversion to grasslands for raising **beef cattle**. Degradation by **pollution** also threatens many species. When large habitats are broken into fragments, mammals and birds requiring **large territories** and animals with **migratory habits** are badly affected.
2. **Over-exploitation**: when biological need turns into 'greed'. Many species extinctions in the last 500 years (**Steller's sea cow, passenger pigeon**) were due to over-exploitation by humans; many marine fish populations around the world are over-harvested, endangering some commercially important species.
3. **Alien species invasions**: some introduced species turn invasive and cause decline or extinction of indigenous species. The **Nile perch** introduced into **Lake Victoria** (east Africa) led to the extinction of more than **200 species of cichlid fish**. Invasive weeds: **carrot grass (*Parthenium*), *Lantana* and water hyacinth (*Eichhornia*)**. The illegal introduction of the African catfish *Clarias gariepinus* for aquaculture threatens indigenous catfishes in our rivers.
4. **Co-extinctions**: when a species becomes extinct, the plant and animal species associated with it in an **obligatory** way also become extinct. When a host fish becomes extinct, its unique assemblage of **parasites** also meets the same fate; in coevolved plant-pollinator **mutualism**, extinction of one leads to extinction of the other.

Yaad karo: Evil Quartet

Habitat Over Alien Co = "HOAC": ghar toota (habitat), lalach (over-exploitation), baahar wala ghus aaya (alien), dost ke saath gaya (co-extinction).

Quick fact

By the time you finish reading this chapter, about **1,000 more hectares** of rain forest would have been lost (NCERT).

4. Biodiversity Conservation

A. Why should we conserve biodiversity?

1. **Narrowly utilitarian**: humans derive countless direct economic benefits from nature: food (cereals, pulses, fruits), firewood, fibre, construction material, industrial products (tannins, lubricants, dyes, resins, perfumes) and medicines. More than 25% of drugs sold worldwide are derived from plants, and **25,000 species** of plants contribute to traditional medicines of native peoples. **Bioprospecting**: exploring molecular, genetic and species-level diversity for products of economic importance; nations with rich biodiversity can expect enormous benefits.
2. **Broadly utilitarian**: biodiversity plays a major role in many **ecosystem services**. The fast-dwindling Amazon forest produces, through photosynthesis, about 20% of the total oxygen in Earth's atmosphere. **Pollination** (without which plants cannot give fruits or seeds) is done by bees, bumblebees, birds and bats. Nature also gives aesthetic pleasures: walking through thick woods, watching spring flowers in full bloom, or waking up to a bulbul's song.
3. **Ethical**: we share this planet with millions of species; every species has an intrinsic value, even if it is not of current or any economic value to us. We have a **moral duty** to care for their well-being and pass on our biological legacy in good order to future generations.

Paheli-Bujho

Bujho: Paheli, ped-paudhe bachaane se mujhe kya milega?
Paheli: Dawaiyaan, khaana, oxygen. Amazon akela 20% oxygen deta hai!
Bujho: Aur agar kisi kaam ka na ho?
Paheli: Phir bhi bachao. Har species ki apni value hai: ethical argument.

B. How do we conserve biodiversity?

When we conserve and protect the **whole ecosystem**, its biodiversity at all levels is protected (to save the tiger, we save the whole forest). This is **in situ (on site) conservation**.

When an organism is endangered and needs urgent measures, **ex situ (off site) conservation** is the desirable approach.

In situ conservation

Biodiversity hotspots: regions with very high levels of species richness and a **high degree of endemism** (species confined to that region and not found anywhere else).

- → Initially **25** hotspots were identified; **9** more were added, so the total is **34 hotspots**. They are also regions of accelerated habitat loss.
- → Three hotspots cover India's high biodiversity: **Western Ghats and Sri Lanka, Indo-Burma, and Himalaya.** PYQ
- → Hotspots together cover **less than 2%** of Earth's land area, yet harbour an extremely high number of species; strict protection could reduce the ongoing mass extinctions by almost **30%**.

Protected area (India)	Number
Biosphere reserves	14
National parks	90
Wildlife sanctuaries	448

National park	Wildlife sanctuary	Biosphere reserve
area reserved for wildlife where the whole habitat is protected	area where mainly the animals are protected	large multipurpose area protecting wildlife, traditional life of tribals and genetic resources
no human activity (grazing, forestry, cultivation) allowed	limited activities (collection of forest products, grazing) may be allowed	divided into **core** (no human activity), **buffer** (limited activity, research) and **transition** zones (human settlements)
e.g. Jim Corbett (Uttarakhand), Kaziranga (Assam)	e.g. Ranganathittu Bird Sanctuary (Karnataka)	e.g. Nilgiri, Nanda Devi, Sundarbans

Quick fact: famous protected areas

- → **Jim Corbett** (Uttarakhand): India's first national park (1936).
- → **Kaziranga** (Assam): one-horned rhinoceros. **Gir** (Gujarat): Asiatic lion. **Sundarbans** (West Bengal): Royal Bengal tiger in mangroves.
- → **Nilgiri**: India's first biosphere reserve (1986). **Project Tiger** was launched in 1973.

Sacred groves: tracts of forest set aside by religious and cultural traditions, where all trees and wildlife are venerated and given **total protection**.

Sacred groves are found in	State
Khasi and Jaintia Hills	Meghalaya
Aravalli Hills	Rajasthan
Western Ghat regions	Karnataka and Maharashtra
Sarguja, Chanda and Bastar areas	Madhya Pradesh

In Meghalaya, the sacred groves are the last refuges for a large number of rare and threatened plants. Board fav

Quick fact: Red Data Book and Ramsar sites

- → The **IUCN Red List (Red Data Book)** records the conservation status of species: extinct, extinct in the wild, critically endangered, endangered, vulnerable, near threatened and least concern.
- → **Ramsar sites** are wetlands of international importance protected under the Ramsar Convention (1971); wetlands support many endemic and migratory species.

Ex situ conservation

Threatened animals and plants are taken out of their natural habitat and placed in special settings where they can be protected and given special care.

- → **Zoological parks, botanical gardens and wildlife safari parks**; many animals extinct in the wild continue to be maintained in zoos.
- → **Cryopreservation**: gametes of threatened species can be preserved in viable and fertile condition for long periods.
- → Eggs can be fertilised **in vitro**, and plants can be propagated by **tissue culture**.
- → **Seed banks**: seeds of different genetic strains of commercially important plants are kept for long periods.

In situ conservation	Ex situ conservation
on site: protection in the natural habitat	off site: protection outside the natural habitat
whole ecosystem is protected	individual threatened species are protected
hotspots, biosphere reserves, national parks, sanctuaries, sacred groves	zoos, botanical gardens, safari parks, seed banks, cryopreservation
cheaper; species keep evolving naturally	for species needing urgent help

C. International efforts

- → **Earth Summit, Rio de Janeiro (1992)**: all nations were called upon to take appropriate measures for conservation of biodiversity and sustainable utilisation of its benefits (this led to the **Convention on Biological Diversity**).
- → **World Summit on Sustainable Development, Johannesburg, South Africa (2002)**: **190 countries** pledged to achieve, by 2010, a significant reduction in the current rate of biodiversity loss at global, regional and local levels.

NCERT in-text questions, answered

- **Does it matter if we have 20,000 or 15,000 species of ants?** The rivet popper analogy says yes: each lost species weakens the ecosystem a little, and losing key species can make it collapse.
- **What do steeper slopes (z = 0.6–1.2) mean?** Over very large areas, species richness increases much faster with area.
- **Can we put an economic value on oxygen from forests?** Compare the cost of a cylinder of oxygen in a hospital with the free oxygen a forest supplies continuously.
- **Why only 22% recorded?** Most tropical species and microbes remain undiscovered and undescribed.

Endemic species	Exotic (alien) species
native and confined to one region	introduced from another region
high endemism marks a hotspot	some turn invasive and wipe out natives
e.g. many amphibians of the Western Ghats	e.g. Nile perch, *Parthenium*, *Lantana*, African catfish

Key terms

- **Biodiversity**: variety of life at genetic, species and ecological levels.
- **Endemism**: species confined to one region only.
- **Hotspot**: region of high species richness and high endemism.
- **Bioprospecting**: exploring biodiversity for economically important products.
- **Co-extinction**: extinction of obligately linked species together.
- **Cryopreservation**: preserving gametes or tissues at very low temperature.
- **Sacred grove**: forest patch protected by religious tradition.

1 minute recap

1. Biodiversity (Edward Wilson): genetic, species, ecological.
2. 1.5 million described (IUCN 2004); about 7 million estimated (Robert May); animals over 70%, plants 22%; insects over 70% of animals.
3. India: 2.4% land, 8.1% species, one of 12 megadiverse countries.
4. Diversity falls from equator to poles: time, stable environment, more energy.
5. Humboldt: $\log S = \log C + z \log A$; z = 0.1–0.2 (regions), 0.6–1.2 (continents).
6. Tilman plots and Ehrlich's rivet popper: more species, more stability.
7. 784 extinctions in 500 years; sixth extinction 100–1,000 times faster; Evil Quartet.
8. Conserve: utilitarian (narrow, broad) and ethical; in situ (34 hotspots, 14 BR, 90 NP, 448 WLS, sacred groves) and ex situ.

Most Important Questions

Objective, assertion-reason, short, long and case-based, with answers

Objective (1 mark each)

Q1. IM Board fav The term biodiversity was popularised by:
(a) Paul Ehrlich
(b) Edward Wilson
(c) Robert May
(d) David Tilman
Ans: (b) Edward Wilson

Q2. IM The global species diversity estimated by Robert May is about:
(a) 1.5 million
(b) 20 million
(c) 7 million
(d) 50 million
Ans: (c) 7 million

Q3. IM Board fav Which of the following shows genetic diversity?
(a) Variation in reserpine content of *Rauwolfia vomitoria* in different Himalayan ranges
(b) Amphibians of Western Ghats vs Eastern Ghats
(c) Deserts and mangroves of India
(d) Coral reefs vs alpine meadows
Ans: (a) Variation in reserpine content of *Rauwolfia vomitoria* in different Himalayan ranges

Q4. IM Board fav In the species-area relationship for regions, the value of z lies in the range:
(a) 0.1 to 0.2
(b) 0.6 to 1.2
(c) 1.5 to 2.0
(d) 2 to 5
Ans: (a) 0.1 to 0.2

Q5. IM The rivet popper hypothesis was proposed by:
(a) Alexander von Humboldt
(b) David Tilman
(c) Paul Ehrlich
(d) Edward Wilson
Ans: (c) Paul Ehrlich

Q6. IM Board fav The most important cause of biodiversity loss is:
(a) Over-exploitation
(b) Alien species invasion
(c) Co-extinction
(d) Habitat loss and fragmentation
Ans: (d) Habitat loss and fragmentation

Q7. 1M The Nile perch introduced into Lake Victoria caused the extinction of more than 200 species of:
(a) Catfish
(b) Cichlid fish
(c) Frogs
(d) Water birds
Ans: (b) Cichlid fish

Q8. 1M Board fav Which of the following is an example of ex situ conservation?
(a) National park
(b) Sacred grove
(c) Seed bank
(d) Biosphere reserve
Ans: (c) Seed bank

Q9. 1M How many biodiversity hotspots are there in the world, and how many cover India?
Ans: 34 hotspots in all; 3 cover India (Western Ghats and Sri Lanka, Indo-Burma, Himalaya).

Q10. 1M Board fav Name the three important components (levels) of biodiversity. (NCERT)
Ans: Genetic diversity, species diversity and ecological diversity.

Q11. 1M Which group appears to be the most vulnerable to extinction?
(a) Amphibians
(b) Birds
(c) Mammals
(d) Insects
Ans: (a) Amphibians (32% of species threatened)

Q12. 1M India has ___ % of the world's land area and ___ % of global species diversity.
Ans: 2.4% of land area and 8.1% of species diversity.

Q13. 1M Name the place with the greatest biodiversity on Earth.
Ans: The Amazonian rain forest of South America.

Q14. 1M Sacred groves of the Khasi and Jaintia Hills are found in:
(a) Rajasthan
(b) Meghalaya
(c) Madhya Pradesh
(d) Karnataka
Ans: (b) Meghalaya

Q15. 1M The number of species of fungi in the world is:
(a) less than that of mammals
(b) equal to that of birds
(c) more than the combined species of fishes, amphibians, reptiles and mammals
(d) about 70% of all species
Ans: (c) more than the combined species of fishes, amphibians, reptiles and mammals

Q16. 1M Which of these is NOT a part of the Evil Quartet?
(a) Habitat loss
(b) Over-exploitation
(c) Mutualism
(d) Co-extinction
Ans: (c) Mutualism

Q17. 1M Board fav India has how many biosphere reserves, national parks and wildlife sanctuaries respectively (NCERT)?
(a) 14, 90 and 448
(b) 90, 14 and 448
(c) 448, 90 and 14
(d) 34, 90 and 448
Ans: (a) 14, 90 and 448

Assertion-Reason (1 mark each)

Key: (a) both A and R true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

Q18. 1M Board fav **A:** Tropics have greater species diversity than temperate regions. **R:** Tropical latitudes remained relatively undisturbed for millions of years, giving a long time for speciation.
Ans: (a) Both true; R correctly explains A.

Q19. 1M **A:** Hotspots cover less than 2% of the Earth's land area. **R:** Strict protection of hotspots could reduce the ongoing mass extinctions by almost 30%.
Ans: (b) Both true, but R does not explain A.

Q20. 1M **A:** Co-extinction occurs when a host species becomes extinct. **R:** Parasites that are obligately dependent on a host cannot survive without it.
Ans: (a) Both true; R correctly explains A.

Q21. 1M **A:** Biodiversity estimates include all prokaryotes. **R:** Many microbes cannot be cultured under laboratory conditions.
Ans: (d) A is false (prokaryotes are not included in the estimates); R is true.

Short answer (2-3 marks)

Q22. 3M Board fav How do ecologists estimate the total number of species present in the world? (NCERT)
Ans: By statistical comparison of temperate-tropical species richness of an exhaustively studied group of organisms (such as insects) with other groups, and extrapolating this ratio to estimate the total number of species; Robert May's estimate is about 7 million.

Q23. 3M Board fav Give three hypotheses for explaining why tropics show the greatest levels of species richness. (NCERT)
Ans: (i) Speciation is a function of time: temperate regions faced frequent glaciations, tropics remained undisturbed for millions of years. (ii) Tropical environments are less seasonal, more constant and predictable, promoting niche specialisation. (iii) More solar energy in the tropics gives higher productivity, which may indirectly support greater diversity.

Q24. 3M Board fav What is the significance of the slope of regression in a species-area relationship? (NCERT)
Ans: The slope (Z, the regression coefficient) shows how fast species richness increases with area. It is 0.1–0.2 for regions irrespective of taxa; a steeper slope (0.6–1.2, e.g. 1.15 for frugivorous birds and mammals of tropical forests of different continents) for very large areas means richness rises rapidly with area.

Q25. 3M Board fav What are the major causes of species losses in a geographical region? (NCERT)
Ans: The Evil Quartet: (i) habitat loss and fragmentation (rain forests from 14% to 6%), (ii) over-exploitation (Steller's sea cow, passenger pigeon), (iii) alien species invasions (Nile perch in Lake Victoria, *Parthenium*, *Lantana*, water hyacinth), (iv) co-extinctions (host and parasite).

Q26. 3M Board fav How is biodiversity important for ecosystem functioning? (NCERT)
Ans: Communities with more species are more stable: less year-to-year variation in productivity, resistance or resilience to disturbance and resistance to alien invasions. Tilman's plots showed less variation in biomass and higher productivity with more species. The rivet popper hypothesis shows that loss of key species can threaten ecosystem functions.

Q27. 3M Board fav What are sacred groves? What is their role in conservation? (NCERT)
Ans: Forest tracts protected by religious and cultural traditions, where all trees and wildlife are venerated and fully protected (Khasi and Jaintia Hills of Meghalaya, Aravalli Hills of Rajasthan, Western Ghats of Karnataka and Maharashtra, Sarguja, Chanda and Bastar of MP). They are the last refuges for many rare and threatened plants (in situ conservation).

Q28. 2M Among the ecosystem services are control of floods and soil erosion. How is this achieved by the biotic components of the ecosystem? (NCERT)
Ans: Plant roots bind the soil particles and prevent erosion; vegetation cover and litter reduce the impact of rain and slow surface run-off, letting water soak into the soil; this reduces floods and keeps the topsoil in place.

Q29. 3M The species diversity of plants (22%) is much less than that of animals (72%). What could explain the greater diversification of animals? (NCERT)
Ans: Animals are mobile and have nervous and sensory systems, so they could adapt to and exploit many more habitats and food niches; diverse feeding habits, behaviour and small body size (insects) with short life cycles allowed rapid speciation.

Q30. 2M Can you think of a situation where we deliberately want to make a species extinct? How would you justify it? (NCERT)
Ans: Disease-causing organisms such as the smallpox virus, polio virus or the malarial parasite and its mosquito vector. Their elimination saves human lives and reduces suffering; such pathogens have no known beneficial role for humans.

Q31. 3M Board fav Differentiate between narrowly utilitarian and broadly utilitarian arguments for conservation.
Ans: Narrowly utilitarian: direct economic benefits such as food, fibre, firewood, industrial products and medicines (over 25% of drugs from plants). Broadly utilitarian: indirect benefits through ecosystem services such as oxygen production (Amazon about 20%), pollination, flood control and aesthetic pleasure.

Q32. 2M What is bioprospecting? Why is it important for a country like India?
Ans: Exploring molecular, genetic and species-level diversity for products of economic importance. Nations with rich biodiversity, like India, can expect to reap enormous benefits from it.

Q33. 2M Board fav What are biodiversity hotspots? Name the hotspots that cover India.
Ans: Regions with very high species richness and a high degree of endemism. India: Western Ghats and Sri Lanka, Indo-Burma, and Himalaya.

Q34. 3M Board fav Differentiate between in situ and ex situ conservation with two examples each.
Ans: In situ: protection of species in their natural habitat by protecting the whole ecosystem, e.g. national parks, biosphere reserves, sacred groves. Ex situ: protection outside the natural habitat, e.g. zoos, botanical gardens, seed banks, cryopreservation.

Q35. 2M Explain co-extinction with two examples.
Ans: When a species becomes extinct, the species associated with it obligately also become extinct: (i) a host fish and its parasites, (ii) a plant and its specific pollinator in coevolved mutualism.

Q36. 2M How does habitat fragmentation affect animals?
Ans: Large habitats broken into small pieces cannot support mammals and birds that need large territories, or animals with migratory habits; their populations decline.

Q37. 2M Give two examples each of extinction due to over-exploitation and due to alien species.
Ans: Over-exploitation: Steller's sea cow, passenger pigeon. Alien species: cichlid fishes of Lake Victoria due to the Nile perch; indigenous catfishes threatened by the African catfish *Clarias gariepinus*.

Q38. 2M What did David Tilman's experiments show?
Ans: In long-term outdoor plots, those with more species showed less year-to-year variation in total biomass, and increased diversity contributed to higher productivity.

Q39. 2M Board fav Why is the present extinction called the sixth extinction, and how is it different?
Ans: Five mass extinctions occurred before humans. The present one is the sixth; it is 100 to 1,000 times faster than in pre-human times and is caused by human activities.

Q40. 2M Name the methods by which threatened species can be preserved ex situ using modern techniques.
Ans: Cryopreservation of gametes, in vitro fertilisation of eggs, plant tissue culture, and seed banks for genetic strains of commercially important plants.

Q41. 2M What are the effects of loss of biodiversity on a region?
Ans: Decline in plant production, lowered resistance to perturbations like drought, and increased variability in ecosystem processes such as plant productivity, water use, and pest and disease cycles.

Q42. 3M Board fav Differentiate between a national park and a biosphere reserve.
Ans: National park: area reserved for wildlife where no human activity like grazing, forestry or cultivation is allowed, e.g. Jim Corbett. Biosphere reserve: large multipurpose protected area with a core zone (no human activity), buffer zone (limited activity, research) and transition zone (human settlements); it protects wildlife, tribal life and genetic resources, e.g. Nilgiri.

Q43. 3M What is endemism? Why are hotspots given priority for conservation?
Ans: Endemism: species confined to a particular region and not found anywhere else. Hotspots have very high species richness and endemism, and face accelerated habitat loss; though they cover less than 2% of land, protecting them could cut ongoing mass extinctions by almost 30%.

Q44. 2M Board fav "Amazon forest is called the lungs of the planet." Justify. Why is it being destroyed?
Ans: It is estimated to produce about 20% of the total oxygen of Earth's atmosphere through photosynthesis. It is being cut and cleared for cultivating soya beans and for conversion to grasslands for raising beef cattle.

Q45. 2M Give the IUCN Red List (2004) figures for extinctions in the last 500 years.
Ans: 784 species: 338 vertebrates, 359 invertebrates and 87 plants.

Q46. 2M Name four animals that became extinct in recent times and their countries.
Ans: Dodo (Mauritius), quagga (Africa), thylacine (Australia), Steller's sea cow (Russia); also Bali, Javan and Caspian tigers.

Q47. 2M What percentage of birds, mammals, amphibians and gymnosperms face the threat of extinction?
Ans: Birds 12%, mammals 23%, amphibians 32% and gymnosperms 31%.

Q48. 2M Board fav Why are prokaryotes not included in biodiversity estimates?
Ans: Conventional taxonomic methods are not suitable for identifying microbial species, and many species are not culturable in the laboratory. With biochemical or molecular criteria their diversity might run into millions.

Q49. 2M What is the ethical argument for conserving biodiversity?
Ans: Every species has an intrinsic value, even if it has no current economic value to us. We share the planet with millions of species and have a moral duty to care for their well-being and pass on our biological heritage to future generations.

Q50. 2M Why are amphibians more vulnerable to extinction?
Ans: Amphibians live partly in water and partly on land and have a thin, moist, permeable skin; habitat loss, pollution and changes in water bodies affect them strongly. NCERT records that extinctions are not random and that 32% of amphibian species are threatened.

Q51. 2M What is the role of seed banks and cryopreservation in conservation?
Ans: Seed banks store seeds of different genetic strains of commercially important plants for long periods; cryopreservation keeps gametes of threatened species viable and fertile for long periods. Both preserve genetic diversity ex situ for future use and breeding.

Q52. 3M Board fav Name two international summits for biodiversity conservation and state what was decided.
Ans: Earth Summit, Rio de Janeiro (1992): all nations to take measures for conservation and sustainable use of biodiversity. World Summit on Sustainable Development, Johannesburg (2002): 190 countries pledged to significantly reduce the rate of biodiversity loss by 2010.

Q53. 2M Why do we need ex situ conservation when in situ conservation protects whole ecosystems?
Ans: Some species are so endangered that they need urgent measures; they cannot survive in their degraded natural habitats. Ex situ methods (zoos, botanical gardens, seed banks, cryopreservation, tissue culture) keep them alive and allow breeding and later reintroduction.

Q54. 2M What is species richness? How does it change with area?
Ans: Species richness is the number of species in an area. According to Humboldt, it increases with the explored area, but only up to a limit (species-area relationship).

Q55. 2M Explain how the introduction of the African catfish threatens Indian rivers.
Ans: *Clarias gariepinus* was illegally introduced for aquaculture; being an alien predator, it competes with and preys on native fishes, posing a threat to the indigenous catfishes of our rivers.

Q56. 2M What is meant by "need turning into greed" in the context of biodiversity loss?
Ans: Humans have always depended on nature for food and shelter (need), but over-exploitation for profit beyond need (greed) has driven species like Steller's sea cow and passenger pigeon to extinction and over-harvested marine fish.

Q57. 2M Board fav How can loss of a pollinator species lead to loss of a plant species?
Ans: In coevolved plant-pollinator mutualism, the plant depends on one specific pollinator; if the pollinator becomes extinct, the plant cannot reproduce and also becomes extinct (co-extinction).

Q58. 2M Explain why it is said that the tropics had more evolutionary time for diversification.
Ans: Temperate regions were subjected to frequent glaciations in the past, which wiped out many species and reset diversification, whereas tropical latitudes remained relatively undisturbed for millions of years. Since speciation is a function of time, the tropics accumulated more species.

Q59. 3M How does a stable community differ from an unstable one? Give three features.
Ans: A stable community (i) shows little variation in productivity from year to year, (ii) is resistant or resilient to occasional natural or man-made disturbances, and (iii) resists invasion by alien species. Communities with more species tend to be more stable.

Q60. 2M Board fav Why is the loss of a key species more serious than the loss of other species?
Ans: Key species drive major ecosystem functions (like rivets on the wings of a plane in the rivet popper analogy); their loss weakens or collapses the whole ecosystem, while loss of other species (rivets on seats) may not have an immediate effect.

Long answer (5 marks)

Q61. 5M Board fav (a) Describe the three levels of biodiversity with examples. (b) Why is India called a mega diversity country?
Ans: (a) Genetic: diversity within a species (*Rauwolfia vomitoria* reserpine potency, 50,000 rice strains, 1,000 mango varieties). Species: number of species in a region (Western Ghats have more amphibian species than Eastern Ghats). Ecological: diversity of ecosystems (India with deserts, rain forests, mangroves, coral reefs, wetlands, estuaries, alpine meadows vs Norway). (b) With only 2.4% of world land area, India has 8.1% of global species diversity (about 45,000 plant and twice as many animal species recorded) and is one of the 12 mega diversity countries.

Q62. 5M Board fav (a) What is the species-area relationship? Draw the graph and write its equation. (b) Explain the values of Z.
Ans: (a) Humboldt observed that species richness increases with explored area, but only up to a limit; the curve is a rectangular hyperbola (Fig. 13.2) and becomes a straight line on a log scale: log S = log C + Z log A (S = species richness, A = area, C = Y-intercept, Z = slope or regression coefficient). (b) For regions, Z = 0.1–0.2 irrespective of taxa or region (plants in Britain, birds in California, molluscs in New York State). For very large areas such as continents, Z = 0.6–1.2 (1.15 for frugivorous birds and mammals of tropical forests), meaning richness rises much faster with area.

Q63. 5M Board fav Explain the Evil Quartet with examples. Which cause is the most important?
Ans: Habitat loss and fragmentation (most important): rain forests reduced from 14% to 6% of land; Amazon cleared for soya beans and cattle; fragmentation harms wide-ranging and migratory animals. Over-exploitation: Steller's sea cow, passenger pigeon, over-harvested marine fish. Alien species invasions: Nile perch caused extinction of 200+ cichlids in Lake Victoria; *Parthenium*, *Lantana*, water hyacinth; African catfish. Co-extinctions: host and parasite, plant and pollinator.

Q64. 5M Board fav Why should we conserve biodiversity? Describe in situ and ex situ methods of conservation.

Ans: Reasons: narrowly utilitarian (food, fibre, medicines; bioprospecting), broadly utilitarian (ecosystem services: Amazon gives 20% of oxygen; pollination; aesthetics), ethical (intrinsic value, moral duty). In situ: protecting whole ecosystems: 34 hotspots (3 in India), 14 biosphere reserves, 90 national parks, 448 sanctuaries, sacred groves. Ex situ: zoos, botanical gardens, safari parks, cryopreservation of gametes, in vitro fertilisation, tissue culture, seed banks. International: Earth Summit 1992, Johannesburg 2002.

Q65. 5M Board fav (a) What is meant by a biodiversity hotspot? Name the hotspots covering India. (b) Explain the role of sacred groves and protected areas in in situ conservation.

Ans: (a) A region with very high species richness and a high degree of endemism; there are 34 in the world (25 initially, 9 added), together covering less than 2% of land but harbouring a very high number of species; strict protection could reduce ongoing mass extinctions by almost 30%. India: Western Ghats and Sri Lanka, Indo-Burma, Himalaya. (b) Protected areas: 14 biosphere reserves, 90 national parks and 448 wildlife sanctuaries protect whole habitats. Sacred groves (Khasi and Jaintia Hills of Meghalaya, Aravalli Hills of Rajasthan, Western Ghats of Karnataka and Maharashtra, Sarguja, Chanda and Bastar of MP) are fully protected by tradition and are the last refuges for many rare and threatened plants.

Case-based (4 marks)

Q66. 4M Board fav A student compared bird species: Colombia (near the equator) about 1,400, New York (41° N) 105, Greenland (71° N) 56. (a) Name the pattern. (b) Give any two reasons for it. (c) Which forest on Earth has the greatest biodiversity? (d) Why does India have more than 1,200 bird species?

Ans: (a) Latitudinal gradient: diversity decreases from the equator towards the poles. (b) Tropics had a long undisturbed evolutionary time (no frequent glaciation); constant, predictable environments promote niche specialisation; more solar energy gives higher productivity (any two). (c) The Amazonian rain forest. (d) Much of India lies in tropical latitudes.

Q67. 4M In the species-area study of two regions: region X gave $z = 0.15$ and a continent-level study gave $z = 1.15$ for frugivorous birds and mammals. (a) Write the equation used. (b) What does z represent? (c) What does the higher z value indicate? (d) Who first described this relationship?

Ans: (a) $\log S = \log C + z \log A$. (b) The slope of the line (regression coefficient). (c) A steeper slope: species richness increases much faster with area over very large areas such as continents. (d) Alexander von Humboldt.

Q68. 4M Board fav A lake had many native fish species. After a large predatory fish was introduced for fishing, many native species vanished, and some parasites specific to those fishes also disappeared. (a) Name the two causes of loss shown. (b) Give the NCERT example of the first cause. (c) Name two invasive weeds in India. (d) Suggest one way to prevent such losses.

Ans: (a) Alien species invasion and co-extinction. (b) Nile perch in Lake Victoria led to the extinction of more than 200 species of cichlid fish. (c) *Parthenium* (carrot grass), *Lantana* or water hyacinth (*Eichhornia*). (d) Strict checks on the introduction of exotic species (quarantine), and in situ protection of the native ecosystem.

Q69. 4M A district has a forest with a core area where no human activity is allowed, a surrounding zone used for research and education, and an outer zone with villages and farms. (a) Identify this protected area. (b) Name the three zones. (c) Give one Indian example. (d) Is this in situ or ex situ conservation?

Ans: (a) A biosphere reserve. (b) Core zone, buffer zone and transition zone. (c) Nilgiri (or Nanda Devi, Sundarbans). (d) In situ conservation.

Q70. 4M Data: Species richness of a group was recorded in a set of islands of increasing area. A log-log plot gave a straight line with slope 0.18. (a) What is this relationship called? (b) What does the slope represent? (c) Is 0.18 typical for such regions? (d) What happens to the slope if the study covers whole continents?

Ans: (a) Species-area relationship ($\log S = \log C + z \log A$). (b) z, the regression coefficient. (c) Yes, z normally lies between 0.1 and 0.2 for regions. (d) It becomes much steeper, 0.6 to 1.2.

Q71. 4M Board fav A medicinal plant growing in different Himalayan ranges showed different concentrations of its active chemical. (a) Name the plant and the chemical. (b) Which level of biodiversity does this show? (c) Give one more example of this level from India. (d) Why is this level important for crops?

Ans: (a) *Rauwolfia vomitoria*; reserpine. (b) Genetic diversity. (c) More than 50,000 strains of rice or 1,000 varieties of mango in India. (d) Genetic variation is the raw material for breeding new varieties with high yield, disease resistance and stress tolerance.

Exam Alert: final checklist

- → Graph: species-area curve with log-log inset; labels S, A.
- → Names: Wilson, May, Humboldt, Tilman, Ehrlich: MCQ ke favourites.
- → Numbers: 1.5 million, 7 million, 2.4%/8.1%, 784 (338+359+87), 34 hotspots, 14/90/448, 12/23/32/31%.
- → In situ vs ex situ table aur Evil Quartet examples: 3-5 marks har saal.

Last-Night Revision

Most asked: chapter-wise one-liners, key differences, must-draw diagrams, scientists, numbers, full forms, most expected questions

Exam se pehle ki raat: **har chapter ke key facts, differences aur diagrams** ek jagah. Read it twice, then sleep well.

1. Sexual Reproduction in Flowering Plants

→ Anther: bilobed, dithecous, **tetrasporangiate**; wall layers: epidermis, endothecium, middle layers, **tapetum** (nourishes pollen; dense cytoplasm, often multinucleate).

→ PMC (2n) undergoes meiosis to form a microspore tetrad; pollen grain: exine of **sporopollenin** (most resistant organic material), germ pores, intine of cellulose and pectin.

→ Mature pollen: vegetative cell (big, forms tube) + generative cell (forms 2 male gametes); shed at **2-celled stage in over 60%** of angiosperms.

→ Pollen viability: rice and wheat about 30 minutes; members of Rosaceae, Leguminosae and Solanaceae for months. Pollen stored in liquid nitrogen (−196 °C) in pollen banks.

→ Ovule: funicle, hilum, integuments, micropyle, chalaza, nucellus; MMC gives 4 megaspores, 1 functional (**monosporic**); 3 free-nuclear divisions give the **7-celled, 8-nucleate** embryo sac.

→ Egg apparatus = 2 synergids (with **filiform apparatus**) + egg; 3 antipodals at chalazal end; central cell with 2 polar nuclei.

→ Autogamy, geitonogamy (genetically = autogamy), xenogamy. *Viola, Oxalis, Commelina*: chasmogamous + cleistogamous flowers (cleistogamy = assured seed-set).

→ Wind: light non-sticky pollen, well-exposed stamens, feathery stigma, one ovule per ovary (grasses, maize tassels). Water: *Vallisneria, Hydrilla, Zostera* (pollen with mucilage).

→ Outbreeding devices: pollen release and stigma receptivity not synchronised, different positions of anther and stigma, **self-incompatibility**, unisexual flowers (castor, maize monoecious; papaya dioecious).

→ Artificial hybridisation: **emasculation** (remove anthers) + **bagging**, dust desired pollen, rebag.

→ Double fertilisation: syngamy (2n zygote) + triple fusion (3n PEN). Endosperm develops before embryo; coconut water = free-nuclear, kernel = cellular endosperm.

→ Albuminous seeds (wheat, maize, castor) vs non-albuminous (pea, groundnut); perisperm in black pepper and beet.

→ False fruits: apple, strawberry, cashew (thalamus). Parthenocarpy: banana. Apomixis: grasses, Asteraceae; polyembryony: Citrus, mango.

Pericarp	Perisperm
wall of fruit (from ovary wall)	residual persistent nucellus in seed
in all fruits	black pepper, beet

Yaad karo

Embryo sac = **7 cell, 8 nucleus** (central cell mein 2 nuclei). PEN = **3n**, zygote = **2n**, nucellus = 2n, endosperm = 3n.

2. Human Reproduction

→ Testes in scrotum (2–2.5 °C below body temperature); each testis about **250 lobules** with 1–3 seminiferous tubules; spermatogonia + **Sertoli cells** (nourish); **Leydig cells** make androgens.

→ Duct path: seminiferous tubules, rete testis, vasa efferentia, epididymis, vas deferens, ejaculatory duct, urethra. Glands: seminal vesicles, prostate, bulbourethral.

→ Female: ovaries, oviduct (infundibulum with fimbriae, ampulla, isthmus), uterus (perimetrium, myometrium, endometrium), cervix, vagina. Mammary gland: 15–20 lobes with alveoli.

→ GnRH from hypothalamus; LH acts on Leydig cells (androgens), FSH on Sertoli cells (spermiogenesis).

→ Sperm: head (acrosome + nucleus), neck, middle piece (mitochondria), tail; 200–300 million per ejaculation; at least 60% normal shape and 40% vigorous motility for fertility.

→ Oogenesis starts in the foetus; primary oocytes arrested in prophase I; 60,000–80,000 follicles left at puberty; secondary oocyte released at ovulation; meiosis II completes only after sperm entry.

→ Menstrual cycle (28/29 days): menstrual phase 3–5 days, follicular phase, ovulation on about day 14 (**LH surge**), luteal phase (corpus luteum secretes progesterone). Menarche at puberty; menopause about 50 years.

→ Fertilisation in the **ampullary region**; sperm X or Y decides the sex of the baby.

→ Cleavage: morula (8–16 cells), blastocyst (trophoblast + inner cell mass); implantation in endometrium; inner cell mass has **stem cells**.

→ Placenta hormones: hCG, hPL, estrogens, progestogens; relaxin from ovary in later pregnancy.

→ Development: heart after 1 month; limbs and digits by end of 2nd month; major organ systems by end of 1st trimester; first movements and hair on head in 5th month; by end of 24 weeks body covered with fine hair, eyelids separate.

→ Parturition: foetal ejection reflex, oxytocin causes strong uterine contractions; gestation about 9 months. Colostrum has **IgA** antibodies.

Spermatogenesis	Oogenesis
starts at puberty, in testes	starts in foetus, in ovary
1 primary spermatocyte gives 4 sperms	1 primary oocyte gives 1 ovum + polar bodies
continuous	cyclic, stops at menopause

Yaad karo

Sperm ka raasta: **S**eminiferous, **R**ete, **V**asa efferentia, **E**pididymis, **V**as deferens, **E**jaculatory duct, **U**rethra.

3. Reproductive Health

→ India was among the first countries to start national family planning programmes (**1951**); now called RCH (Reproductive and Child Health Care) programmes.

→ Population: about 350 million at independence, close to 1 billion by 2000, crossed 1.2 billion in May 2011.

→ Natural methods: periodic abstinence (days 10–17), withdrawal (coitus interruptus), lactational amenorrhea (up to 6 months after parturition).

→ Barriers: condoms (Nirodh), diaphragms, cervical caps, vaults. IUDs: non-medicated (Lippes loop), copper releasing (CuT, Cu7, Multiload 375), hormone releasing (Progestasert, LNG-20).

→ Cu ions suppress sperm motility and fertilising capacity; IUDs increase phagocytosis of sperms.

→ Oral pills (progestogens or progestogen-estrogen, 21 days); **Saheli** (once a week, non-steroidal) from **CDRI Lucknow**; injections and implants; emergency contraceptives within **72 hours**.

→ Surgical: vasectomy (vas deferens), tubectomy (fallopian tube); highly effective but poor reversibility.

→ MTP legalised in India in 1971; MTP (Amendment) Act 2017: within 12 weeks on the opinion of one registered medical practitioner; 12–24 weeks needs two.

→ STIs: gonorrhoea, syphilis, genital herpes, chlamydiasis, genital warts, trichomoniasis, hepatitis-B, HIV; hepatitis-B, genital herpes and HIV are not completely curable; highest incidence in 15–24 years.

→ ART: IVF-ET (test tube baby); **ZIFT** (zygote or embryo up to 8 blastomeres into fallopian tube), **IUT** (more than 8 blastomeres into uterus), **GIFT** (ovum into fallopian tube), **ICSI**, **AI/IUI**.

ZIFT	GIFT
zygote or early embryo (up to 8 blastomeres) transferred	ovum (gamete) transferred
into fallopian tube	into fallopian tube of a female who cannot produce ova but can provide the environment

Yaad karo

IUD types: **N**on-medicated (Lippes), **C**opper (CuT), **H**ormone (LNG-20) = "**NCH**". 8 blastomere tak tube (ZIFT), zyada ho to uterus (IUT).

4. Principles of Inheritance and Variation

→ Mendel (1856–1863), garden pea, **7 pairs** of contrasting traits; laws: dominance, segregation (purity of gametes), independent assortment.

→ Monohybrid F2: phenotype 3:1, genotype 1:2:1; dihybrid F2: 9:3:3:1; test cross (with recessive parent): 1:1 or 1:1:1:1.

→ Incomplete dominance: snapdragon (*Antirrhinum*) flower colour, F2 1:2:1 for both. Codominance: ABO blood group (I^A, I^B, i: 6 genotypes, 4 phenotypes); also multiple alleles.

→ Pleiotropy: one gene, many effects (phenylketonuria). Polygenic: human skin colour.

→ Chromosomal theory: **Sutton and Boveri**; linkage and recombination: **Morgan** (*Drosophila*); gene mapping: Sturtevant.

→ Sex determination: XO (grasshopper), XY (humans, *Drosophila*), ZW (birds: female heterogametic); honey bee haplodiploid (male haploid by parthenogenesis).

→ Sickle-cell anaemia: GAG to GUG, Glu replaced by Val at 6th position of beta globin chain; autosomal recessive.

→ Haemophilia and colour blindness: X-linked recessive (colour blindness about 8% males, 0.4% females). Phenylketonuria, thalassemia: autosomal recessive.

→ Chromosomal disorders: Down's syndrome (trisomy 21), Klinefelter's (47, XXY), Turner's (45, XO).

Incomplete dominance	Codominance
F1 is intermediate (pink snapdragon)	both alleles fully expressed in F1 (AB blood group)
neither allele dominant	both alleles dominant

Klinefelter's syndrome	Turner's syndrome
47, XXY (male)	45, XO (female)
gynaecomastia, sterile	sterile, ovaries rudimentary, lack secondary sexual characters

Yaad karo

Ratio set: **3:1, 1:2:1, 9:3:3:1, 1:1**. Down = 21 ki teen copy; Klinefelter = XXY (ladka); Turner = XO (ladki).

5. Molecular Basis of Inheritance

→ DNA double helix: **Watson and Crick (1953)** using X-ray data of Wilkins and Rosalind Franklin; Chargaff's rule A = T, G = C; pitch 3.4 nm, 10 bp per turn, 0.34 nm between bp.

→ Human haploid genome 3.3 x 10^9 bp; E. coli 4.6 x 10^6 bp; lambda phage 48,502 bp; phi X 174 has 5,386 nucleotides.

→ Nucleosome: about 200 bp of DNA wrapped on a histone octamer; euchromatin (loose, active) vs heterochromatin (dense, inactive).

→ Griffith (1928): transformation with *Streptococcus pneumoniae* S and R strains; Avery, MacLeod, McCarty: DNA is the transforming principle; Hershey and Chase (1952): ^{32}P (DNA) and ^{35}S (protein) proved DNA is genetic material.

→ RNA was the first genetic material; DNA is more stable (2'-OH absent, thymine instead of uracil).

→ Semiconservative replication: **Meselson and Stahl (1958)** with ^{15}N in E. coli and CsCl density gradient; Taylor with *Vicia faba*.

→ DNA polymerase works 5' to 3' only: continuous leading strand, discontinuous lagging strand (Okazaki fragments joined by DNA ligase).

→ Transcription unit: promoter, structural gene, terminator; bacteria: one RNA polymerase with sigma (initiation) and rho (termination) factors.

→ Eukaryotes: RNA pol I (rRNAs 28S, 18S, 5.8S), II (hnRNA), III (tRNA, 5S rRNA, snRNAs); hnRNA processing: splicing, capping (methyl guanosine triphosphate at 5'), tailing (poly-A at 3').

→ Genetic code: triplet, 64 codons, 61 for amino acids, stop UAA, UAG, UGA; AUG = start + methionine; degenerate, unambiguous, nearly universal, comma-less.

→ tRNA = adapter (clover leaf, inverted L in 3D); in bacteria 23S rRNA acts as ribozyme (peptidyl transferase).

→ Lac operon (Jacob and Monod): i (repressor), z (beta-galactosidase), y (permease), a (transacetylase); lactose is the inducer; negative regulation.

→ HGP (1990–2003): 3164.7 million bp; about 30,000 genes (estimate); dystrophin largest gene (2.4 million bases); less than 2% codes for proteins; 99.9% bases same in all people; chromosome 1 most genes (2968), Y fewest (231); 1.4 million SNPs.

→ DNA fingerprinting (Alec Jeffreys): VNTR probe; steps: isolation, digestion by restriction enzymes, electrophoresis, Southern blotting, hybridisation, autoradiography.

Template strand	Coding strand
3' to 5' polarity	5' to 3' polarity
used to make RNA	same sequence as RNA (T instead of U); not transcribed

Yaad karo

Stop codons: **U Are Away, U Are Gone, U Go Away** = UAA, UAG, UGA. Lac operon genes: **i z y a** = "**izy a**".

6. Evolution

→ Earth about 4.5 billion years old; life appeared about 500 million years later, almost 4 billion years back.

→ Oparin and Haldane: chemical evolution; **S.L. Miller (1953)**: CH_4, H_2, NH_3, water vapour at 800 °C with electric discharge gave amino acids. Pasteur disproved spontaneous generation.

→ Homologous organs (divergent evolution): forelimbs of whale, bat, cheetah, human; thorn of *Bougainvillea* and tendril of *Cucurbita*.

→ Analogous organs (convergent evolution): wings of butterfly and bird, eyes of octopus and mammals, flippers of penguin and dolphin, sweet potato (root) and potato (stem).

→ Industrial melanism: *Biston betularia* in England after industrialisation; resistance to antibiotics, herbicides and pesticides: evolution by anthropogenic action.

→ Adaptive radiation: Darwin's finches (Galapagos), Australian marsupials; convergent evolution: placental mammals vs Australian marsupials.

→ Darwin: branching descent and natural selection; Alfred Wallace (Malay Archipelago) reached similar conclusions; fitness = reproductive fitness.

→ Hugo de Vries (evening primrose, *Oenothera*): mutations, saltation (single step large mutation).

→ Hardy-Weinberg: $p^2 + 2pq + q^2 = 1$; disturbed by gene migration (gene flow), genetic drift (founder effect), mutation, recombination, natural selection.

→ Selection types: stabilising (more of mean), directional (shift to one extreme), disruptive (both extremes).

→ Coelacanth caught in South Africa (1938): lobefin thought extinct; dinosaurs disappeared about 65 mya.

→ Human evolution: *Dryopithecus*, *Ramapithecus* (about 15 mya); Australopithecines (about 2 mya); *Homo habilis* (650–800 cc, probably no meat); *Homo erectus* (Java 1891, about 1.5 mya, 900 cc, ate meat); Neanderthal (1400 cc, 1,00,000–40,000 years back, buried dead); *Homo sapiens* arose in Africa.

→ Cave art about 18,000 years ago (Bhimbetka, Raisen district, MP); agriculture about 10,000 years back.

Homologous organs	Analogous organs
same structure and origin, different functions	different structure and origin, same function
divergent evolution	convergent evolution

Yaad karo

Brain size: habilis **650–800**, erectus **900**, Neanderthal **1400** cc. HW ke 5 factors: **G**ene flow, **D**rift, **M**utation, **R**ecombination, **N**atural selection.

7. Human Health and Disease

→ Typhoid: *Salmonella typhi*, Widal test (Mary Mallon = Typhoid Mary). Pneumonia: *Streptococcus pneumoniae*, *Haemophilus influenzae*; lips and nails grey to bluish.

→ Common cold: rhinoviruses (nose and respiratory passage, not lungs). Malaria: *Plasmodium vivax, malariae, falciparum* (falciparum malignant); female *Anopheles*; haemozoin causes chill and fever.

→ Amoebiasis: *Entamoeba histolytica* (houseflies carriers); ascariasis: *Ascaris*; filariasis (elephantiasis): *Wuchereria bancrofti, W. malayi*; ringworm: *Microsporum, Trichophyton, Epidermophyton*.

→ Innate immunity barriers: physical (skin, mucus), physiological (stomach acid, saliva, tears), cellular (PMNL-neutrophils, monocytes, NK cells, macrophages), cytokine (interferons).

→ Acquired immunity: B cells (antibodies, humoral) and T cells (cell-mediated, graft rejection); primary and secondary (anamnestic) responses. Antibody = H_2L_2; IgA, IgM, IgE, IgG.

→ Active (own antibodies, slow, long-lasting) vs passive (ready-made antibodies: colostrum IgA, placenta IgG, anti-tetanus serum).

→ Allergy: IgE, histamine and serotonin from mast cells; treated with antihistamines, adrenalin, steroids. Autoimmunity: rheumatoid arthritis.

→ Primary lymphoid organs: bone marrow, thymus; secondary: spleen, lymph nodes, tonsils, Peyer's patches, appendix; MALT about 50% of lymphoid tissue.

→ AIDS: HIV (retrovirus), macrophage = HIV factory, attacks helper T cells; ELISA test; antiretroviral drugs; NACO.

→ Cancer: loss of contact inhibition; benign vs malignant (metastasis); carcinogens: X-rays, gamma rays, UV, tobacco smoke, oncogenic viruses (proto-oncogenes); detection: biopsy, CT, MRI, monoclonal antibodies; treatment: surgery, radiotherapy, chemotherapy, alpha-interferon.

→ Drugs: heroin (smack, diacetylmorphine) from morphine of *Papaver somniferum*, depressant; cannabinoids (*Cannabis sativa*) affect cardiovascular system; cocaine (*Erythroxylum coca*) interferes with dopamine transport; *Atropa belladonna, Datura* hallucinogenic. Nicotine raises adrenaline and nor-adrenaline.

Active immunity	Passive immunity
body makes its own antibodies	ready-made antibodies given
slow, long-lasting	quick, short-lived
vaccination, infection	colostrum, anti-tetanus serum

Benign tumour	Malignant tumour
remains confined	spreads by metastasis
little damage	invades and damages tissues

Yaad karo

Filaria ka "W" = **W**uchereria; ringworm ke 3 = **M T E**.
Colostrum = **IgA**, allergy = **IgE**, placenta = **IgG**.

8. Microbes in Human Welfare

→ Curd: *Lactobacillus* (LAB) also increases vitamin B_{12}; bread: *Saccharomyces cerevisiae* (baker's yeast); toddy from palm sap.

→ Swiss cheese holes: CO_2 by *Propionibacterium sharmanii*; Roquefort cheese ripened by fungi.

→ Wine and beer without distillation; whisky, brandy, rum by distillation (brewer's yeast).

→ Penicillin: Alexander Fleming (*Penicillium notatum*, on *Staphylococci*); potential established by Ernst Chain and Howard Florey.

→ Acids: citric (*Aspergillus niger*), acetic (*Acetobacter aceti*), butyric (*Clostridium butylicum*), lactic (*Lactobacillus*).

→ Lipases in detergents; pectinases and proteases clarify fruit juices; streptokinase (*Streptococcus*) clot buster; cyclosporin A (*Trichoderma polysporum*) immunosuppressant; statins (*Monascus purpureus*) lower cholesterol.

→ Sewage: primary treatment (physical: filtration, sedimentation); secondary (biological: aeration, flocs reduce BOD; activated sludge; anaerobic digesters give biogas). Ganga and Yamuna Action Plans.

→ Biogas: methanogens (*Methanobacterium*) in cattle rumen; methane + CO_2 + H_2S; gobar gas plants (IARI and KVIC).

→ Biocontrol: ladybird (aphids), dragonflies (mosquitoes), *Bacillus thuringiensis* (caterpillars), *Trichoderma* (plant pathogens), baculoviruses (*Nucleopolyhedrovirus*) in IPM.

→ Biofertilisers: *Rhizobium* (root nodules), *Azospirillum*, *Azotobacter* (free-living), mycorrhiza (*Glomus*: phosphorus), cyanobacteria (*Anabaena, Nostoc, Oscillatoria*) in paddy.

Primary sewage treatment	Secondary sewage treatment
physical: filtration and sedimentation	biological: microbes in aeration tanks
removes floating debris, grit	reduces BOD; forms activated sludge

Yaad karo

Acid set: Citric = Aspergillus, Acetic = Acetobacter, Butyric = Clostridium, Lactic = Lactobacillus.

9. Biotechnology: Principles and Processes

→ Two core techniques: genetic engineering and bioprocess engineering (sterile conditions).

→ First rDNA (1972): **Stanley Cohen and Herbert Boyer**: antibiotic resistance gene linked with a native plasmid of *Salmonella typhimurium*.

→ Restriction enzymes (first in 1963): EcoRI = *Escherichia coli* RY13, first enzyme; recognise **palindromic** sequences, cut to give sticky ends; ligase joins.

→ Gel electrophoresis: agarose (from sea weeds); DNA (negative) moves to the anode; small fragments move farther; stained with ethidium bromide, orange bands under UV; elution.

→ Vector features: ori, selectable marker (ampicillin, tetracycline, chloramphenicol, kanamycin resistance), cloning sites; pBR322: BamHI in tet^R, PstI and PvuI in amp^R.

→ Insertional inactivation: insert in beta-galactosidase gene gives colourless colonies (recombinant) vs blue (non-recombinant) with chromogenic substrate.

→ Natural vectors: *Agrobacterium tumefaciens* (Ti plasmid), retroviruses (disarmed).

→ Competent host: $CaCl_2$ + heat shock (ice, 42 °C, ice); microinjection (animal cells); biolistics/gene gun (gold or tungsten, plants).

→ DNA isolation: lysozyme (bacteria), cellulase (plant), chitinase (fungi); RNase, protease; chilled ethanol precipitates DNA (spooling).

→ PCR (Kary Mullis): denaturation, annealing (primers), extension (**Taq polymerase** from *Thermus aquaticus*).

→ Bioreactors (100–1000 litres): stirred tank, sparged; downstream processing: separation, purification, formulation, clinical trials, quality control.

Exonuclease	Endonuclease
removes nucleotides from the ends	cuts at specific positions within DNA
no recognition sequence	restriction endonucleases recognise palindromes

Yaad karo

PCR = **D A E**: Denature (94 °C), Anneal, Extend (Taq). EcoRI: Escherichia + coli + RY13 + I (first).

10. Biotechnology and its Applications

→ Bt toxin from *Bacillus thuringiensis*: inactive protoxin activated in the **alkaline gut** of insect; cryIAc and cryIIAb (cotton bollworms), cryIAb (corn borer).

→ RNAi: dsRNA silences mRNA of nematode *Meloidogyne incognita* in tobacco roots; introduced through *Agrobacterium* vectors.

→ GM crops: tolerant to abiotic stress, less pesticide use, less post-harvest loss, better mineral use, enhanced nutritional value (e.g. vitamin A enriched rice).

→ Human insulin: **Eli Lilly (1983)** made A and B chains separately in E. coli and joined them by disulphide bonds; proinsulin has an extra C-peptide.

→ Gene therapy: first in 1990, a 4-year-old girl with ADA (adenosine deaminase) deficiency; lymphocytes with functional ADA cDNA; permanent cure by introducing the gene at early embryonic stage.

→ Molecular diagnosis: rDNA technology, PCR, ELISA; probe + autoradiography detects mutated genes.

→ Transgenic animals: over 95% are mice; Rosie (1997) cow: human protein-enriched milk (2.4 g/litre, alpha-lactalbumin); alpha-1-antitrypsin for emphysema; vaccine safety (polio) and chemical safety testing.

→ GEAC (Genetic Engineering Appraisal Committee) checks validity of GM research and safety of GM organisms.

→ Biopiracy: patent on Basmati rice by an American company (1997); India has 27 documented Basmati varieties; turmeric and neem also involved; Indian Patents Bill amended.

Bt cotton	RNAi tobacco
makes Bt toxin protein	makes dsRNA that silences pest mRNA
against bollworms (insects)	against nematode *Meloidogyne incognita*

Yaad karo

Cotton = **cryIAc, cryIIAb**; corn = **cryIAb**. Insulin 1983 Eli Lilly; gene therapy 1990 ADA; Rosie 1997.

11. Organisms and Populations

→ Population attributes: birth rate, death rate (per capita), sex ratio, age distribution (age pyramids: expanding, stable, declining), population density.

→ Density: numbers, percent cover, biomass, or indirect estimates (pug marks, faecal pellets for tigers).

→ Population growth: natality and immigration increase it; mortality and emigration decrease it: $N_{t+1} = N_t + [(B + I) - (D + E)]$.

→ Exponential growth: $dN/dt = rN$; $N_t = N_0e^{rt}$; J-shaped curve. r for Norway rat 0.015, flour beetle 0.12, India (1981) 0.0205.

→ Logistic growth (Verhulst-Pearl): $dN/dt = rN[(K - N)/K]$; S-shaped (sigmoid) curve; K = carrying capacity.

→ Life history: some breed once (Pacific salmon, bamboo), others many times (most birds, mammals); some produce many small offspring (oysters), others few large (birds, mammals).

→ Interactions: mutualism (+,+), competition (-,-), predation and parasitism (+,-), commensalism (+,0), amensalism (-,0).

→ Predation: prickly pear cactus in Australia controlled by a cactus-feeding moth; starfish *Pisaster* removal led to extinction of more than 10 invertebrate species.

→ Competition: Gause's competitive exclusion; Abingdon tortoise extinct after goats came to Galapagos; Connell: *Balanus* excludes *Chathamalus* in Scotland; resource partitioning (MacArthur's warblers).

→ Parasitism: *Cuscuta* (no chlorophyll), brood parasitism (cuckoo lays eggs in crow's nest), ectoparasites (lice, ticks), endoparasites.

→ Commensalism: orchid on mango branch, barnacles on whale, cattle egret and cattle, clown fish and sea anemone.

→ Mutualism: lichens, mycorrhizae, fig and wasp, *Ophrys* orchid sexual deceit with bees.

Exponential growth	*Logistic growth*
resources unlimited	resources limited
J-shaped curve	S-shaped curve with carrying capacity K
$dN/dt = rN$	$dN/dt = rN(K - N)/K$

Yaad karo

Interactions ke signs: **++** mutualism, **--** competition, **+-** predation/parasitism, **+0** commensalism, **-0** amensalism.

12. Ecosystem

→ Ecosystem = functional unit of nature; structure: species composition + stratification (trees, shrubs, herbs).

→ Four functions: productivity, decomposition, energy flow, nutrient cycling.

→ GPP - R = NPP; secondary productivity = rate of new organic matter by consumers; biosphere NPP 170 billion tons, oceans 55 billion tons.

→ Decomposition: fragmentation (detritivores), leaching, catabolism (enzymes), humification (humus), mineralisation; fast if warm, moist, N-rich; slow with lignin, chitin, cold, anaerobiosis.

→ Sun only energy source (except deep-sea hydrothermal); PAR less than 50%; plants capture 2-10% of PAR.

→ GFC dominates aquatic ecosystems; DFC carries more energy on land; 10 per cent law limits trophic levels.

→ Pyramid of energy always upright; number inverted in tree ecosystem; biomass inverted in sea (phytoplankton 4, zooplankton 21 kg m^{-2}).

→ Limitations of pyramids: species at many levels ignored, no food webs, saprophytes ignored.

GFC	*DFC*
starts with green plants	starts with detritus
major in water	major on land

13. Biodiversity and Conservation

→ Levels: genetic (*Rauwolfia* reserpine; 50,000 rice strains), species (Western Ghats amphibians), ecological (India vs Norway).

→ 1.5 million described (IUCN 2004); about 7 million (Robert May); animals over 70%, plants 22%; insects over 70% of animals.

→ India: 2.4% land, 8.1% species diversity; one of 12 mega diversity countries.

→ Latitudinal gradient: Colombia 1,400 bird species, New York 105, Greenland 56; Amazon has the greatest biodiversity.

→ Species-area (Humboldt): $\log S = \log C + Z \log A$; $Z = 0.1$-0.2 (regions), 0.6-1.2 (continents).

→ Tilman plots and Ehrlich's rivet popper hypothesis: more species, more stability.

→ IUCN 2004: 784 extinctions in 500 years (338 vertebrates, 359 invertebrates, 87 plants); sixth extinction 100-1,000 times faster.

→ Evil Quartet: habitat loss (most important), over-exploitation, alien invasions (Nile perch), co-extinctions.

→ In situ: 34 hotspots (3 in India), 14 biosphere reserves, 90 national parks, 448 sanctuaries, sacred groves; ex situ: zoos, seed banks, cryopreservation.

→ Earth Summit Rio (1992); Johannesburg (2002): 190 countries to cut biodiversity loss by 2010.

14. Scientists and Their Contributions

Scientist	*Contribution*
Gregor Mendel	laws of inheritance (garden pea, 1856-1863)
Sutton and Boveri	chromosomal theory of inheritance
T.H. Morgan	linkage and recombination in *Drosophila*
Alfred Sturtevant	gene mapping using recombination frequency

Henking	discovered the X body (1891)
Hugo de Vries	mutation theory; saltation (*Oenothera*)
Frederick Griffith	transformation in *Streptococcus pneumoniae* (1928)
Avery, MacLeod, McCarty	DNA is the transforming principle

Hershey and Chase	DNA is the genetic material (1952)
Watson and Crick	double helix model of DNA (1953)
Erwin Chargaff	A = T, G = C
Meselson and Stahl	semiconservative replication (1958)

Jacob and Monod	lac operon
George Gamow	triplet code proposed
Har Gobind Khorana	synthesised RNA with defined bases
Marshall Nirenberg	cell-free system for protein synthesis; decoded codons

Severo Ochoa	polynucleotide phosphorylase (enzyme to make RNA)
Alec Jeffreys	DNA fingerprinting
Oparin and Haldane	chemical origin of life
S.L. Miller	experiment on origin of life (1953)

Louis Pasteur	disproved spontaneous generation
Charles Darwin	natural selection, branching descent
Alfred Wallace	natural selection (Malay Archipelago)
Lamarck	use and disuse of organs

Hardy and Weinberg	genetic equilibrium
Alexander Fleming	penicillin
Ernst Chain, Howard Florey	established penicillin as an antibiotic
Cohen and Boyer	first recombinant DNA (1972)

Kary Mullis	PCR
Edward Wilson	popularised the term biodiversity
Robert May	estimated 7 million species
Alexander von Humboldt	species-area relationship

David Tilman	diversity and ecosystem stability (outdoor plots)
Paul Ehrlich	rivet popper hypothesis
G.F. Gause	competitive exclusion principle
Verhulst and Pearl	logistic growth

Joseph Connell	competition experiments with barnacles

Exam Alert

→ Naam + year + organism: teeno saath yaad karo (Meselson-Stahl, 1958, E. coli, ^{15}N).

→ MCQ mein scientist ka match-the-following almost har saal aata hai.

15. Important Numbers at a Glance

Fact	Number
Embryo sac	7 cells, 8 nuclei
Pollen shed at 2-celled stage	over 60% of angiosperms
Testicular lobules per testis	about 250
Sperms per ejaculate	200–300 million

Primary follicles at puberty	60,000–80,000
Menstrual cycle; ovulation	28/29 days; about day 14
Mendel's contrasting traits	7 pairs
ABO genotypes / phenotypes	6 / 4

Colour blindness	about 8% males, 0.4% females
Human haploid genome	3.3 x 10^9 bp (HGP: 3164.7 million bp)
Distance between bp; pitch	0.34 nm; 3.4 nm
Codons: total / sense / stop	64 / 61 / 3

Human genes (estimate)	about 30,000
DNA same in all humans	99.9%
SNPs identified	1.4 million
Homo habilis / erectus / Neanderthal brain	650–800 / 900 / 1400 cc

Earth's age	about 4.5 billion years
MALT share of lymphoid tissue	about 50%
Restriction enzymes known	more than 900 (from over 230 strains)
Bioreactor volume	100–1000 litres

Transgenic animals that are mice	over 95%
Rosie milk protein	2.4 g per litre
Basmati varieties in India	27 documented
PAR in sunlight; captured by plants	less than 50%; 2–10% of PAR

Biosphere NPP; oceans	170 billion t; 55 billion t
Species described; estimated	1.5 million; about 7 million
India land; species share	2.4%; 8.1%
Extinctions (IUCN 2004)	784 in 500 years

Hotspots; in India	34; 3
India: BR / NP / WLS	14 / 90 / 448

16. Important Full Forms

Short	Full form
PMC / MMC	pollen mother cell / megaspore mother cell
PEN	primary endosperm nucleus
hCG / hPL	human chorionic gonadotropin / human placental lactogen
IUD	intra uterine device
MTP	medical termination of pregnancy
STI	sexually transmitted infection
ART	assisted reproductive technologies
IVF / ET	in vitro fertilisation / embryo transfer
ZIFT / GIFT / IUT	zygote intra fallopian transfer / gamete intra fallopian transfer / intra uterine transfer
ICSI	intra cytoplasmic sperm injection
AI / IUI	artificial insemination / intra-uterine insemination
RCH	Reproductive and Child Health Care
VNTR	variable number of tandem repeats
SNP	single nucleotide polymorphism
HGP	Human Genome Project
BAC / YAC	bacterial / yeast artificial chromosome
EST	expressed sequence tags
hnRNA	heterogeneous nuclear RNA
HIV / AIDS	human immunodeficiency virus / acquired immuno deficiency syndrome
ELISA	enzyme linked immuno-sorbent assay
MALT	mucosa associated lymphoid tissue
PMNL	polymorpho-nuclear leukocytes
NACO	National AIDS Control Organisation
LSD	lysergic acid diethylamide
LAB	lactic acid bacteria
BOD	biochemical oxygen demand
IPM	integrated pest management
KVIC / IARI	Khadi and Village Industries Commission / Indian Agricultural Research Institute
PCR	polymerase chain reaction
rDNA	recombinant DNA
ADA	adenosine deaminase
GEAC	Genetic Engineering Appraisal Committee
GM / GMO	genetically modified (organism)
RNAi	RNA interference
GPP / NPP	gross / net primary productivity
PAR	photosynthetically active radiation
GFC / DFC	grazing / detritus food chain
IUCN	International Union for Conservation of Nature and Natural Resources
CDRI	Central Drug Research Institute

Paheli-Bujho

Bujho: Paheli, kal paper hai aur sab kuch yaad karna hai!

Paheli: Ghabrao mat. Ye revision do baar padho, diagrams ek baar bana lo.

Bujho: Aur raat bhar jaagu?

Paheli: Bilkul nahi! 7 ghante neend, tabhi dimaag fresh rahega.

17. Organisms and Their Uses

Organism	Use / role
Lactobacillus	curd; lactic acid; vitamin B_{12}
Saccharomyces cerevisiae	bread, beverages, ethanol
Propionibacterium sharmanii	Swiss cheese (holes)
Penicillium notatum	penicillin
Aspergillus niger	citric acid
Acetobacter aceti	acetic acid
Clostridium butylicum	butyric acid
Trichoderma polysporum	cyclosporin A (immunosuppressant)
Monascus purpureus	statins (lower cholesterol)
Streptococcus	streptokinase (clot buster)
Methanobacterium	biogas (methane)
Bacillus thuringiensis	Bt toxin, biocontrol of caterpillars
Trichoderma	biocontrol of plant pathogens
Nucleopolyhedrovirus	species-specific insecticide (baculovirus)
Rhizobium	N_2 fixation in legume root nodules
Azotobacter, Azospirillum	free-living N_2 fixers
Glomus (mycorrhiza)	phosphorus absorption
Anabaena, Nostoc, Oscillatoria	biofertilisers in paddy fields
Thermus aquaticus	Taq polymerase for PCR
Agrobacterium tumefaciens	Ti plasmid vector for plants
Meloidogyne incognita	nematode controlled by RNAi in tobacco
Salmonella typhimurium	plasmid used in first rDNA
Biston betularia	industrial melanism
Rauwolfia vomitoria	reserpine; genetic diversity

18. Key Differences: Quick Table

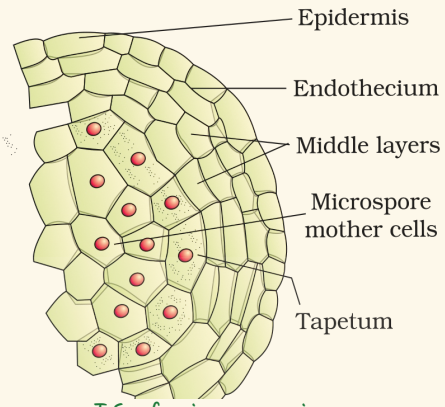
Pair	First	Second
Autogamy / Geitonogamy	pollen to stigma of same flower	pollen to another flower of the same plant
Chasmogamous / Cleistogamous	open flowers, anthers and stigma exposed	closed flowers, assured seed-set
Epicotyl / Hypocotyl	embryonal axis above cotyledons, ends in plumule	axis below cotyledons, ends in radicle
Coleoptile / Coleorrhiza	sheath over plumule in grass	sheath over radicle in grass
Morula / Blastocyst	solid ball of 8–16 cells	hollow, with trophoblast and inner cell mass
Vasectomy / Tubectomy	vas deferens cut or tied	fallopian tube cut or tied
Euchromatin / Heterochromatin	loosely packed, light, active	densely packed, dark, inactive
Leading / Lagging strand	continuous synthesis	discontinuous (Okazaki fragments)
Exons / Introns	coding sequences, appear in mRNA	intervening, removed by splicing
Divergent / Convergent evolution	same structure, different functions (homologous)	different structures, same function (analogous)
Innate / Acquired immunity	present at birth, non-specific	developed later, specific, with memory
Humoral / Cell-mediated	B cells, antibodies	T cells, graft rejection
Primary / Secondary lymphoid organs	bone marrow, thymus (lymphocytes mature)	spleen, lymph nodes, tonsils, Peyer's patches
Biofertiliser / Biocontrol agent	enriches soil nutrients (*Rhizobium*)	controls pests (Bt, ladybird)
Plasmid / Chromosomal DNA	small, circular, extra-chromosomal, used as vector	main genome of the cell
In situ / Ex situ	in natural habitat	outside natural habitat

19. Must-Draw Diagrams (with labels)

Exam Alert

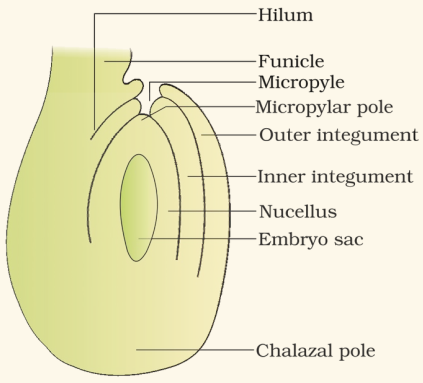
→ Har diagram mein **heading + minimum 5 labels**; labels right side, seedhi lines se.

→ Pencil se banao, shading avoid karo; neat diagram = full marks.



T.S. of microsporangium

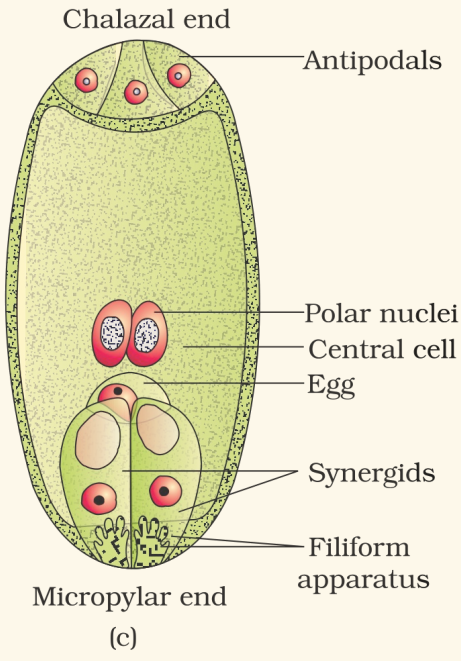
Labels: epidermis, endothecium, middle layers, tapetum, sporogenous tissue



(d)

Anatropous ovule

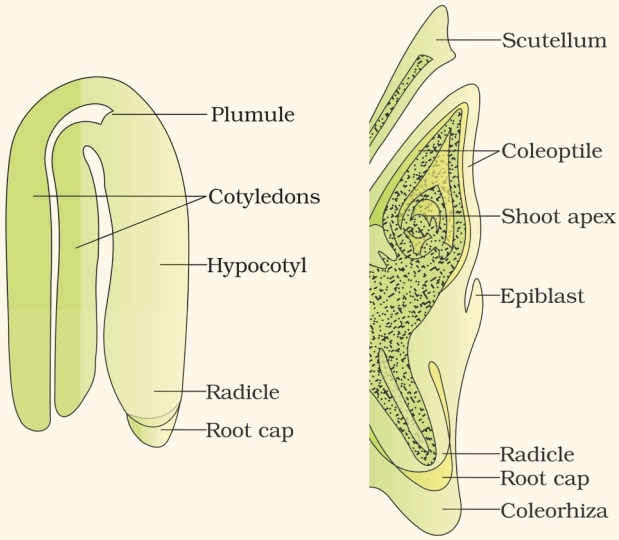
Labels: funicle, hilum, micropyle, outer and inner integuments, nucellus, embryo sac, chalaza



(c)

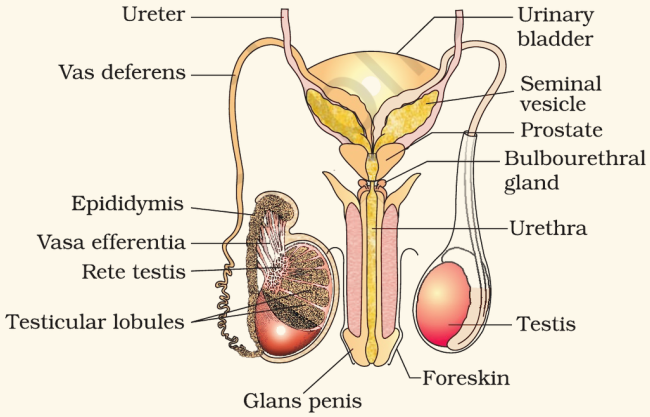
Mature embryo sac

Labels: synergids, filiform apparatus, egg, central cell, polar nuclei, antipodals



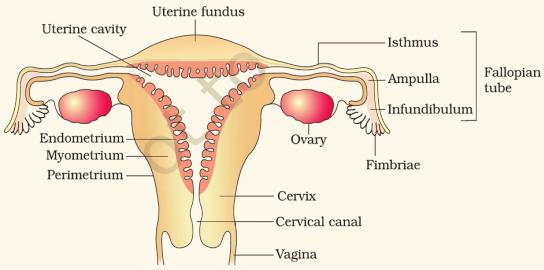
Dicot embryo and L.S. of grass embryo

Labels: plumule, cotyledons, hypocotyl, radicle, root cap; scutellum, coleoptile, coleorrhiza



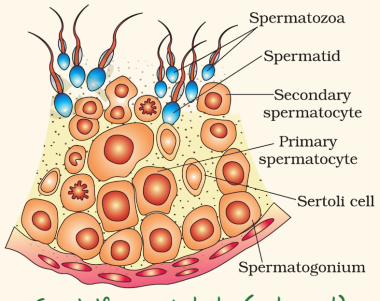
Male reproductive system

Labels: testis, epididymis, vas deferens, seminal vesicle, prostate, bulbourethral gland, urethra



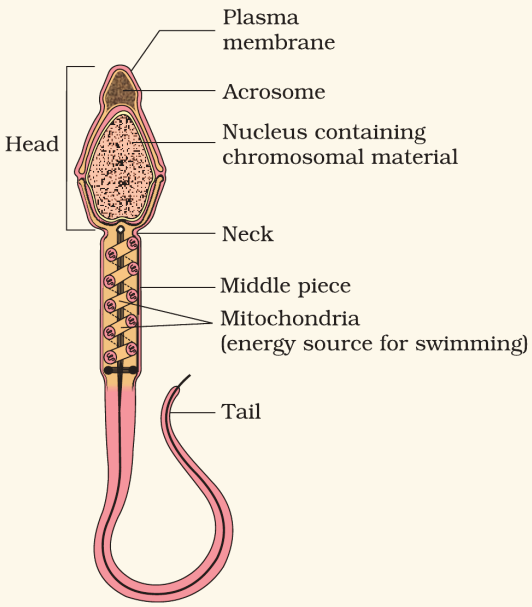
Female reproductive system

Labels: ovary, fimbriae, infundibulum, ampulla, isthmus, uterus, endometrium, myometrium, cervix, vagina



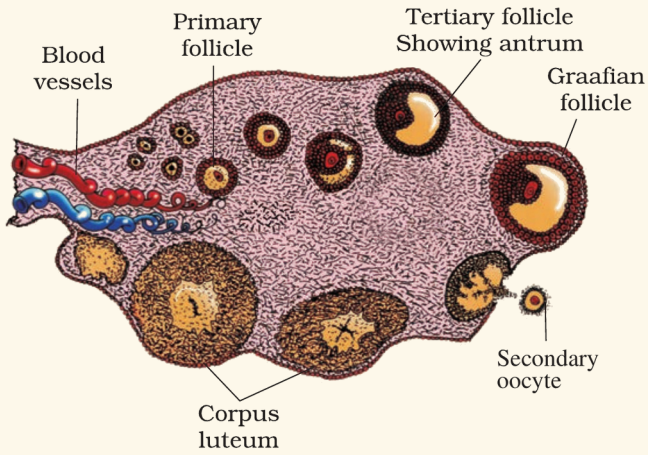
Seminiferous tubule (enlarged)

Labels: spermatogonia, spermatocytes, spermatids, sperms, Sertoli cells, Leydig cells



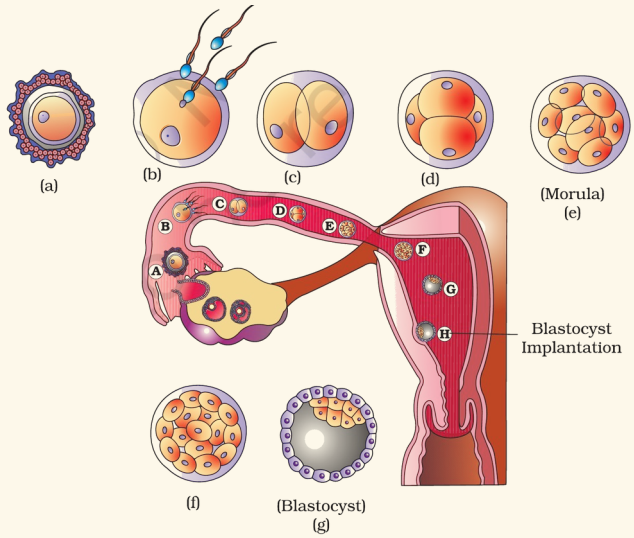
Structure of a sperm

Labels: acrosome, nucleus, neck, middle piece with mitochondria, tail



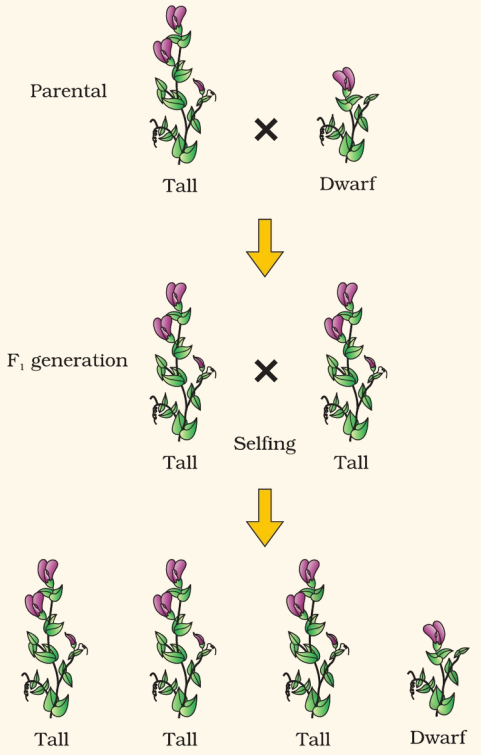
Section of ovary

Labels: primary, secondary, tertiary and Graafian follicles, ovum, corpus luteum



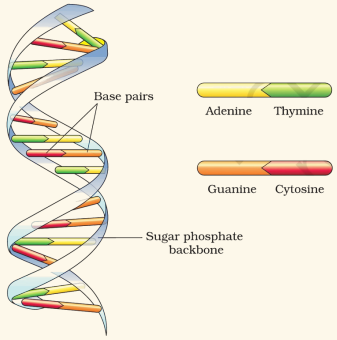
Transport of ovum and implantation

Labels: ovulation, fertilisation in ampulla, morula, blastocyst, implantation



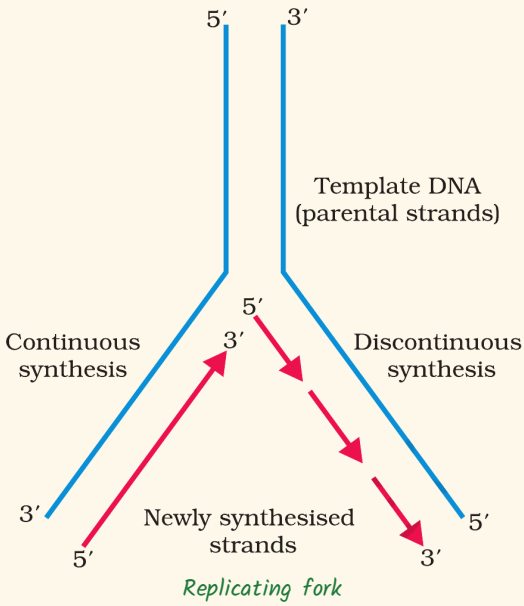
Monohybrid cross

Labels: P, F1, F2, gametes, 3:1 phenotype, 1:2:1 genotype



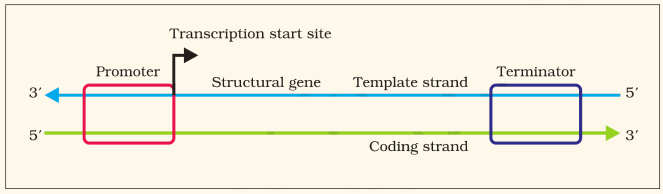
DNA double helix

Labels: sugar-phosphate backbone, base pairs, 3.4 nm pitch, 0.34 nm, 2 nm width



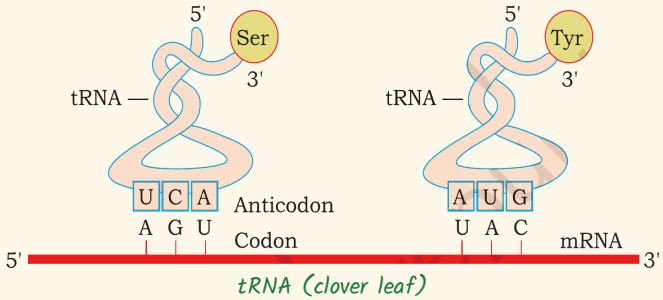
Replicating fork

Labels: template strands, leading strand, lagging strand, Okazaki fragments, 5' and 3' ends



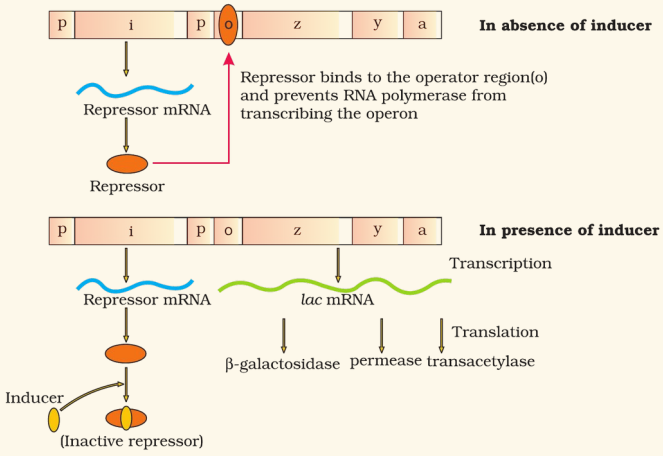
Transcription unit

Labels: promoter, structural gene, terminator, template and coding strands



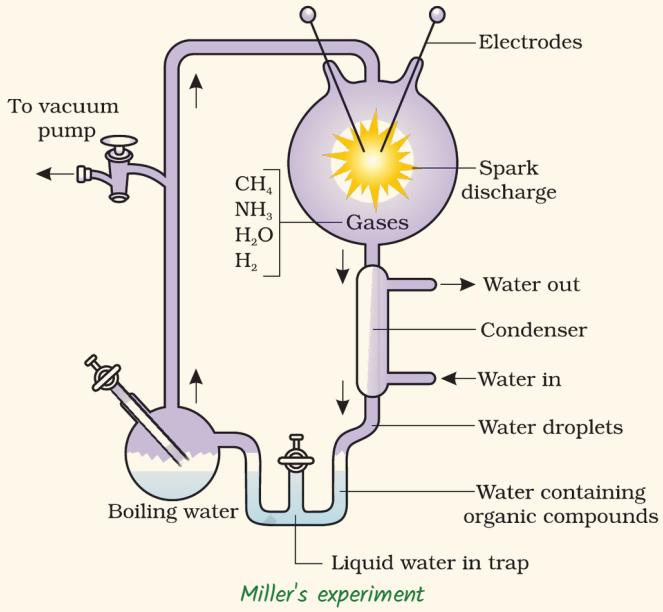
tRNA (clover leaf)

Labels: anticodon loop, amino acid acceptor end, T loop, D loop



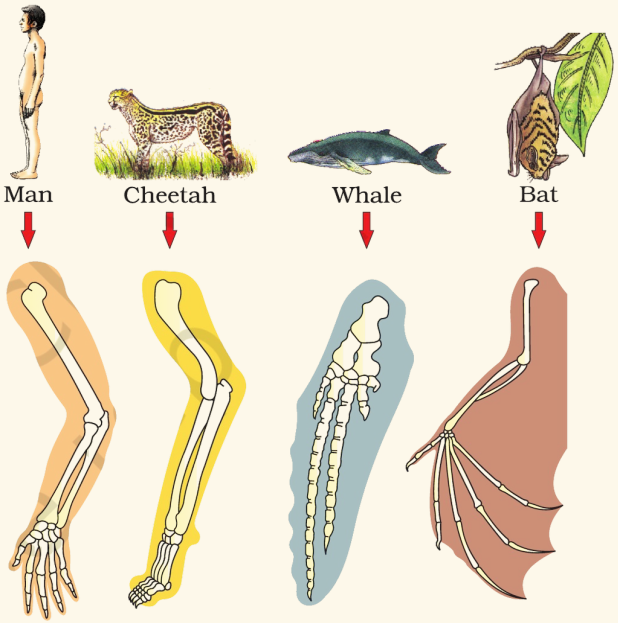
Lac operon

Labels: i gene, repressor, operator, promoter, z, y, a; inducer (lactose)



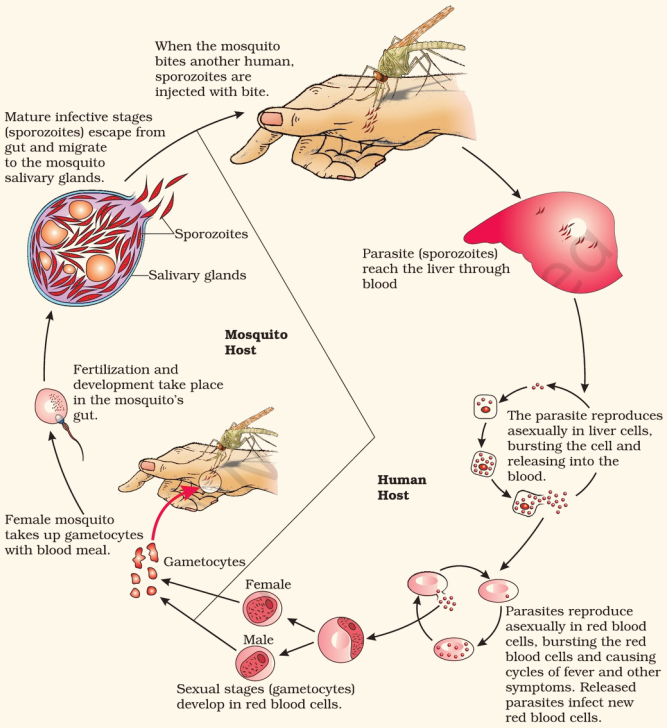
Miller's experiment

Labels: electrodes, CH_4, NH_3, H_2, H_2O vapour, condenser, boiling water, trap with amino acids



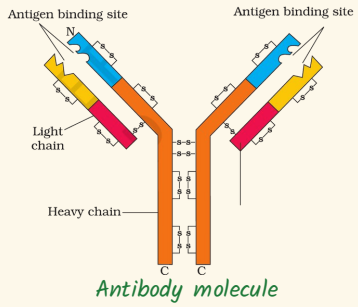
Homologous forelimbs

Labels: humerus, radius, ulna, carpals, metacarpals, phalanges in man, cheetah, whale, bat



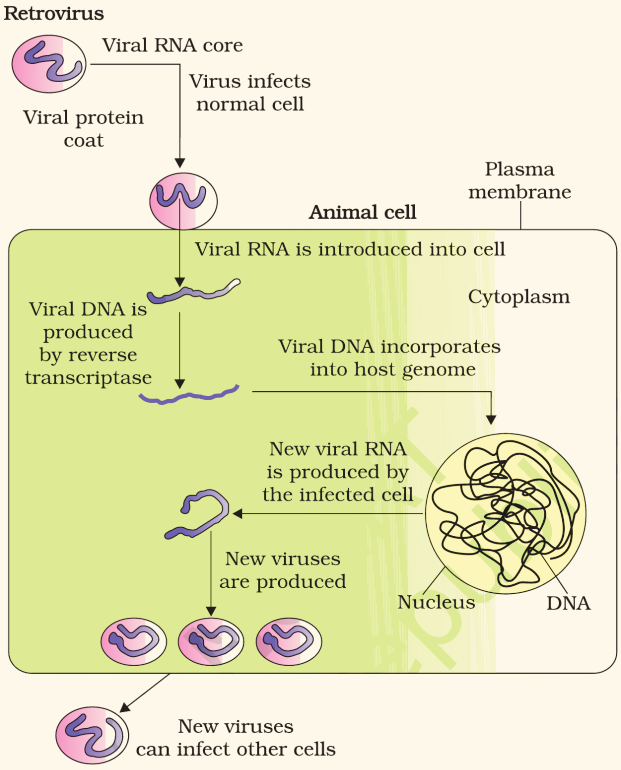
Life cycle of Plasmodium

Labels: sporozoites, liver cells, RBCs, haemozoin, gametocytes, female *Anopheles*, salivary glands



Antibody molecule

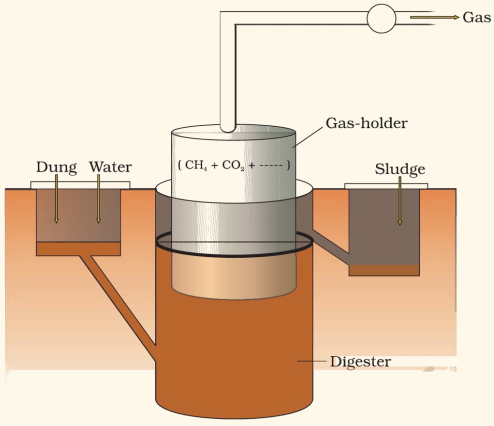
Labels: heavy chains, light chains, antigen binding sites, disulphide bonds (H_2L_2)



NOTE: Infected cell can survive while viruses are being replicated and released

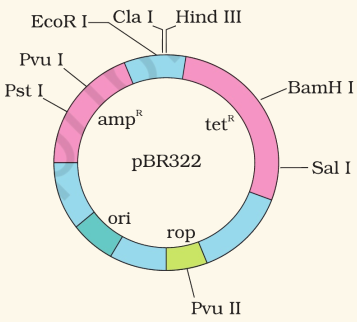
Replication of retrovirus (HIV)

Labels: viral RNA, reverse transcriptase, viral DNA integrated in host DNA, new viruses



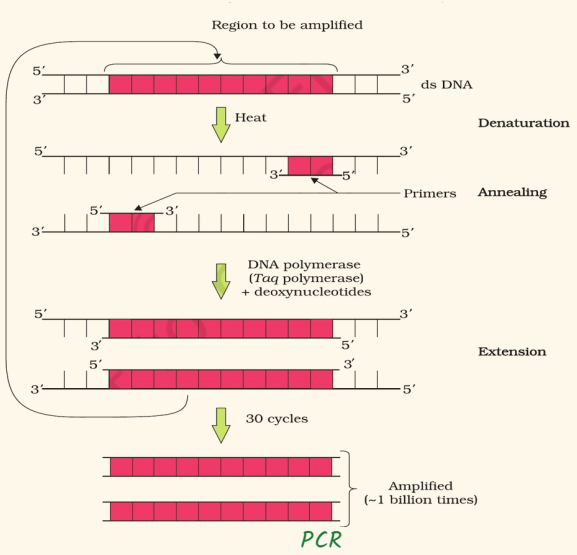
Biogas plant

Labels: dung + water, digester, gas holder, sludge outlet, gas outlet



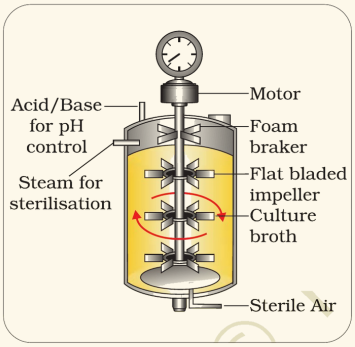
pBR322 vector

Labels: ori, rop, amp^R, tet^R, BamHI, EcoRI, PstI, PvuI, HindIII, SalI



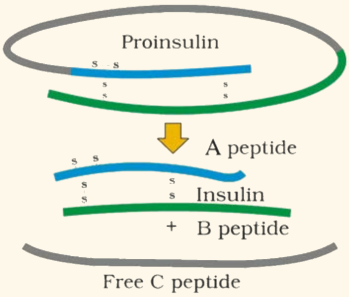
PCR

Labels: denaturation, annealing (primers), extension (Taq polymerase)



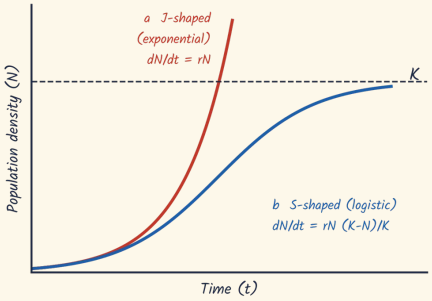
Stirred-tank bioreactor

Labels: motor, stirrer, foam breaker, steam, acid/base, culture broth



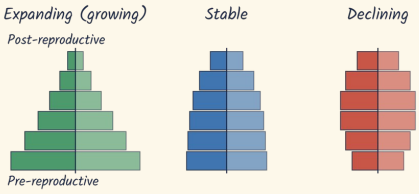
Maturation of insulin

Labels: pro-insulin, A chain, B chain, C peptide, disulphide bonds



Population growth curves

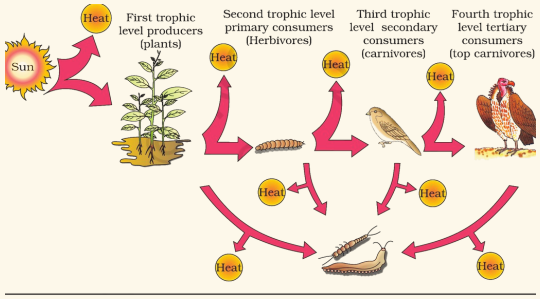
Labels: J-shaped (exponential) and S-shaped (logistic) curves, carrying capacity K



Bars = age groups (left: male, right: female)

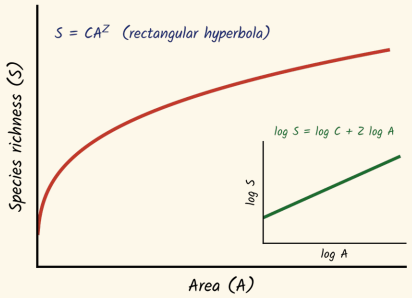
Age pyramids

Labels: pre-reproductive, reproductive, post-reproductive; expanding, stable, declining



Energy flow

Labels: sun, producers, herbivores, carnivores, heat loss at each level



Species-area relationship

Labels: species richness vs area; log scale straight line with slope Z

MOST IMPORTANT QUESTIONS

Chapter-wise top board questions with crisp answers

Ch 1 Sexual Reproduction in Flowering Plants

Q1. 3M Board fav Draw a labelled diagram of the T.S. of a mature anther / microsporangium.

Ans: Label: epidermis, endothecium, middle layers, tapetum, sporogenous tissue / pollen grains, connective. Tapetum nourishes pollen; endothecium helps dehiscence.

Q2. 3M Board fav Describe the structure of a mature embryo sac.

Ans: 7-celled, 8-nucleate: egg apparatus (2 synergids with filiform apparatus + 1 egg) at micropylar end, 3 antipodals at chalazal end, large central cell with 2 polar nuclei.

Q3. 2M Board fav What is double fertilisation? Why is it called so?

Ans: Two fusions in one embryo sac: syngamy (male gamete + egg = 2n zygote) and triple fusion (male gamete + 2 polar nuclei = 3n PEN).

Q4. 2M Board fav List three outbreeding devices.
Ans: Non-synchronised pollen release and stigma receptivity; anther and stigma at different positions; self-incompatibility; unisexual flowers (monoecious or dioecious).

Q5. 3M Board fav What is apomixis? Why is it important for hybrid seed industry?
Ans: Formation of seeds without fertilisation (asexual reproduction that mimics sexual reproduction), e.g. Asteraceae, grasses. Hybrid characters do not segregate, so farmers can reuse the same hybrid seed every year.

Ch 2 Human Reproduction

Q6. 2M Board fav Draw a labelled diagram of a human sperm.
Ans: Head (acrosome with enzymes, nucleus), neck, middle piece (mitochondria for energy), tail (motility); plasma membrane envelops the whole body.

Q7. 5M Board fav Explain the hormonal control of the menstrual cycle.
Ans: Follicular phase: FSH and LH rise, follicles grow, estrogen regenerates endometrium. LH surge (mid-cycle) causes ovulation (about day 14). Luteal phase: corpus luteum secretes progesterone to maintain endometrium. No fertilisation: corpus luteum degenerates, menstruation begins.

Q8. 3M Board fav Differentiate between spermatogenesis and oogenesis (any three).
Ans: Puberty vs foetal start; 4 sperms vs 1 ovum + polar bodies per meiocyte; continuous vs cyclic (stops at menopause); in testes vs ovary.

Q9. 2M Board fav What is the role of the placenta?
Ans: Supplies oxygen and nutrients to the embryo, removes CO_2 and wastes, and acts as an endocrine tissue producing hCG, hPL, estrogens and progestogens.

Q10. 2M Board fav Describe the structure of a blastocyst and its implantation.
Ans: Outer trophoblast and inner cell mass; trophoblast attaches to endometrium and inner cell mass differentiates into embryo; uterine cells divide and cover the blastocyst (implantation).

Ch 3 Reproductive Health

Q11. 3M Board fav Explain the mode of action of copper-releasing IUDs and oral pills.
Ans: Cu ions suppress sperm motility and fertilising capacity; IUDs increase phagocytosis of sperms. Pills inhibit ovulation and implantation and alter cervical mucus to retard entry of sperms.

Q12. 2M Board fav What are ZIFT and IUT? Differentiate them.
Ans: IVF embryo with up to 8 blastomeres transferred into fallopian tube (ZIFT); more than 8 blastomeres into uterus (IUT).

Q13. 2M Board fav Write the principles to remain free of STIs.
Ans: Avoid sex with unknown or multiple partners; always use condoms; consult a qualified doctor early for diagnosis and complete treatment in case of doubt.

Q14. 2M Board fav Why is MTP legalised with conditions in India?
Ans: To get rid of unwanted pregnancies (failure of contraception, rape) or when continuation is harmful to mother or foetus, while preventing misuse such as female foeticide.

Ch 4 Principles of Inheritance and Variation

Q15. 3M Board fav Explain codominance with ABO blood groups.
Ans: Gene I has alleles I^A, I^B, i; I^A and I^B are codominant (AB group shows both), both dominant over i; 6 genotypes and 4 phenotypes (A, B, AB, O).

Q16. 2M Board fav A colour-blind man marries a normal (non-carrier) woman. What is the probability of a colour-blind son?
Ans: Zero: sons get their X from the mother (normal); all daughters will be carriers.

Q17. 2M Board fav Explain sex determination in honey bee.
Ans: Fertilised eggs (2n = 32) develop into females (queen or worker); unfertilised eggs develop into males (drones, n = 16) by parthenogenesis: haplodiploid system.

Q18. 3M Board fav Explain the cause of sickle-cell anaemia.
Ans: Autosomal recessive; substitution of glutamic acid by valine at the 6th position of the beta globin chain due to GAG to GUG; RBCs become sickle shaped under low oxygen.

Q19. 5M Board fav Work out a dihybrid cross and state the law it explains.
Ans: RRYY x rryy; F1 RrYy; F2 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green; law of independent assortment.

Ch 5 Molecular Basis of Inheritance

Q20. 5M Board fav Describe the Meselson and Stahl experiment.
Ans: E. coli grown in $^{15}NH_4Cl$, then shifted to ^{14}N; DNA separated by CsCl density gradient: after 1 generation all hybrid (intermediate), after 2 generations hybrid and light in equal amounts: semiconservative replication.

Q21. 5M Board fav Explain the lac operon.
Ans: Regulator i makes repressor that binds operator and blocks transcription of z, y, a. Lactose (inducer) binds repressor and inactivates it; RNA polymerase transcribes z (beta-galactosidase), y (permease), a (transacetylase). Negative regulation.

Q22. 3M Board fav What is the Hershey-Chase experiment?
Ans: Bacteriophages grown with ^{32}P (DNA labelled) or ^{35}S (protein labelled) infected E. coli; after blending and centrifuging, radioactivity of ^{32}P was found in bacteria: DNA enters, so DNA is the genetic material.

Q23. 3M Board fav Explain the processing of hnRNA in eukaryotes.
Ans: Splicing (introns removed, exons joined), capping (methyl guanosine triphosphate added at 5' end), tailing (adenylate residues added at 3' end) give mature mRNA.

Q24. 3M Board fav List the steps of DNA fingerprinting.
Ans: Isolation of DNA, digestion with restriction enzymes, electrophoresis, Southern blotting to membrane, hybridisation with labelled VNTR probe, autoradiography.

Ch 6 Evolution

Q25. 2M Board fav Give two examples each of homologous and analogous organs.
Ans: Homologous: forelimbs of whale, bat, cheetah and human; thorn of *Bougainvillea* and tendril of *Cucurbita*. Analogous: wings of butterfly and bird; eye of octopus and mammals.

Q26. 3M Board fav Explain industrial melanism as evidence for natural selection.
Ans: Before industrialisation white-winged moths were more (lichen-covered trees); after it, trunks darkened by soot, dark moths escaped predators and increased: selection of the better adapted form.

Q27. 3M Board fav State the Hardy-Weinberg principle and five factors affecting it.
Ans: Allele frequencies stay constant: $p^2 + 2pq + q^2 = 1$. Disturbed by gene migration, genetic drift, mutation, genetic recombination, natural selection.

Q28. 2M Board fav Explain adaptive radiation with an example.
Ans: Evolution of different species from one ancestor radiating into different habitats: Darwin's finches with varied beaks on Galapagos; Australian marsupials.

Ch 7 Human Health and Disease

Q29. 5M Board fav Describe the life cycle of *Plasmodium*.
Ans: Sporozoites enter human through female *Anopheles* bite, multiply in liver cells, then attack RBCs and release haemozoin (chill and fever). Gametocytes taken up by mosquito; fertilisation and development in stomach wall; sporozoites stored in salivary glands.

Q30. 2M Board fav Draw the structure of an antibody molecule.
Ans: Y-shaped, two heavy and two light chains (H_2L_2) joined by disulphide bonds; antigen binding sites at tips.

Q31. 3M Board fav How does HIV multiply in the body?
Ans: HIV enters macrophages; RNA is copied into viral DNA by reverse transcriptase and integrates into host DNA; infected cells make virus particles (macrophage = HIV factory); virus attacks helper T cells, reducing their number and causing immunodeficiency.

Q32. 2M Board fav Differentiate between benign and malignant tumours.
Ans: Benign: confined to original location, little damage. Malignant: proliferating mass of neoplastic cells that invade and spread to distant sites by metastasis.

Ch 8 Microbes in Human Welfare

Q33. 5M Board fav Explain primary and secondary treatment of sewage.
Ans: Primary: physical removal by filtration and sedimentation (primary sludge, effluent). Secondary: effluent aerated, aerobic microbes form flocs that consume organic matter and reduce BOD; settled activated sludge partly re-used, rest digested anaerobically to give biogas.

Q34. 2M Board fav Name the microbes producing citric acid, statins, cyclosporin A and streptokinase.
Ans: *Aspergillus niger*; *Monascus purpureus*; *Trichoderma polysporum*; *Streptococcus*.

Q35. 2M Board fav How are biofertilisers useful? Give examples.
Ans: They enrich soil nutrients naturally: *Rhizobium* in legume nodules, free-living *Azotobacter* and *Azospirillum*, mycorrhiza (*Glomus*) for phosphorus, cyanobacteria (*Anabaena, Nostoc*) in paddy.

Ch 9 Biotechnology: Principles and Processes

Q36. 3M Board fav What is insertional inactivation? How is it used to select recombinants?
Ans: Inserting the foreign DNA within a marker gene (e.g. beta-galactosidase) inactivates it; recombinant colonies give no colour with chromogenic substrate, non-recombinants give blue colonies.

Q37. 3M Board fav Explain the three steps of PCR.
Ans: Denaturation (high temperature separates strands), annealing (primers bind), extension (Taq polymerase extends primers); repeated cycles amplify DNA up to billion times.

Q38. 3M Board fav Draw pBR322 and mark its important sites.
Ans: ori, rop, amp^R (with PvuI, PstI sites), tet^R (with BamHI, SalI sites), EcoRI, ClaI, HindIII, PvuII sites.

Q39. 2M Board fav How is a bacterial cell made competent to take up DNA?
Ans: Treat with divalent cation like calcium; incubate cells with rDNA on ice, give a brief heat shock at 42 °C, then back on ice.

Ch 10 Biotechnology and its Applications

Q40. 3M Board fav How does Bt toxin kill insects but not the bacterium?
Ans: Bt toxin exists as inactive protoxin in bacteria; in the alkaline pH of insect gut it is activated, binds midgut epithelial cells, creates pores, causing swelling, lysis and death.

Q41. 3M Board fav Explain how human insulin was produced by rDNA technology.
Ans: Eli Lilly (1983) prepared two DNA sequences for A and B chains, introduced them into plasmids of E. coli to make chains separately; chains combined by disulphide bonds to form human insulin.

Q42. 3M Board fav What is RNA interference? How is it used to make pest-resistant plants?
Ans: Silencing of specific mRNA by complementary dsRNA; nematode-specific genes introduced into tobacco via *Agrobacterium* produce sense and antisense RNA forming dsRNA, which silences the nematode mRNA so the parasite cannot survive.

Q43. 2M Board fav What is biopiracy? Give an example.
Ans: Use of bio-resources and traditional knowledge of a country by multinational companies without authorisation or compensation, e.g. patent on Basmati rice by an American company (1997).

Ch 11 Organisms and Populations

Q44. 3M Board fav Write the logistic growth equation and explain the terms.
Ans: $dN/dt = rN[(K - N)/K]$; N = population density, r = intrinsic rate of natural increase, K = carrying capacity; gives a sigmoid curve.

Q45. 2M Board fav Give examples of commensalism and mutualism.
Ans: Commensalism: orchid on mango branch, barnacles on whale, cattle egret and cattle, clown fish and sea anemone. Mutualism: lichens, mycorrhizae, fig and wasp.

Q46. 2M Board fav What is brood parasitism? Give an example.
Ans: A parasitic bird lays eggs in the nest of its host and the host incubates them: the cuckoo (koel) lays eggs in the crow's nest; its eggs resemble the host's eggs.

Ch 12 and 13 Ecology

Q47. 3M Board fav Explain the steps of decomposition.
Ans: Fragmentation by detritivores, leaching of soluble nutrients, catabolism by microbial enzymes, humification (humus) and mineralisation (release of inorganic nutrients).

Q48. 2M Board fav Why is the pyramid of energy always upright?
Ans: Energy is lost as heat at every trophic level, so each level has less energy than the one below.

Q49. 3M Board fav Explain the Evil Quartet.
Ans: (i) Habitat loss and fragmentation (most important; e.g. Amazon rain forest cleared for soya beans and cattle). (ii) Over-exploitation (Steller's sea cow, passenger pigeon). (iii) Alien species invasions (Nile perch in Lake Victoria wiped out over 200 cichlid species; *Parthenium*, *Lantana*, water hyacinth; African catfish). (iv) Co-extinctions (a host fish and its specific parasites; plant and its pollinator).

Q50. 3M Board fav Explain the species-area relationship.
Ans: $\log S = \log C + Z \log A$; $Z = 0.1$–0.2 for regions, 0.6–1.2 for continents (Humboldt).

Q51. 2M Board fav Differentiate between in situ and ex situ conservation.
Ans: In situ: in natural habitat (hotspots, national parks, sacred groves). Ex situ: outside it (zoos, seed banks, cryopreservation).

Assertion-Reason Practice (1 mark each)

How to answer
(a) Both A and R true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

Q52. 1M A: Tapetum is the innermost wall layer of the microsporangium. R: Tapetum nourishes the developing pollen grains.
Ans: (b) Both true; R does not explain why it is innermost.

Q53. 1M A: Geitonogamy is genetically similar to autogamy. R: In geitonogamy the pollen comes from the same plant.
Ans: (a)

Q54. 1M A: Apomictic seeds of a hybrid can be used year after year. R: In apomixis there is no segregation of characters.
Ans: (a)

Q55. 1M A: The sex of a baby is decided by the father. R: Sperms carry either an X or a Y chromosome.
Ans: (a)

Q56. 1M A: Corpus luteum is essential for maintaining the endometrium. R: Corpus luteum secretes large amounts of progesterone.
Ans: (a)

Q57. 1M A: Saheli is a non-steroidal oral contraceptive. R: Saheli has to be taken daily.
Ans: (c) It is taken once a week.

Q58. 1M A: Haemophilia is more common in males. R: Males have only one X chromosome.
Ans: (a)

Q59. 1M A: In birds the female decides the sex of the chick. R: Females are heterogametic (ZW) in birds.
Ans: (a)

Q60. 1M A: DNA is a better genetic material than RNA. R: The 2'-OH group makes RNA reactive and less stable.
Ans: (a)

Q61. 1M A: The lagging strand is synthesised discontinuously. R: DNA polymerase can polymerise only in the 5' to 3' direction.
Ans: (a)

Q62. 1M A: The genetic code is degenerate. R: Some amino acids are coded by more than one codon.
Ans: (a)

Q63. 1M A: Wings of a butterfly and a bird are homologous organs. R: They perform the same function.
Ans: (d) They are analogous organs.

Q64. 1M A: Colostrum gives passive immunity to the newborn. R: Colostrum contains IgA antibodies.
Ans: (a)

Q65. 1M A: Cocaine produces euphoria. R: It interferes with the transport of dopamine.
Ans: (a)

Q66. 1M A: Large holes in Swiss cheese are made by bacteria. R: *Propionibacterium sharmanii* produces a large amount of CO_2.
Ans: (a)

Q67. 1M A: Restriction enzymes cut DNA at specific palindromic sites. R: Palindromes read the same on both strands in the same direction (5' to 3').
Ans: (a)

Q68. 1M A: Bt toxin does not kill *Bacillus thuringiensis*. R: The toxin exists in the bacterium as an inactive protoxin.
Ans: (a)

Q69. 1M A: The pyramid of energy is always upright. R: Energy is lost as heat at each trophic level.
Ans: (a)

Q70. 1M A: Tropics have greater biodiversity. R: Tropical environments are less seasonal and receive more solar energy.
Ans: (a)

Q71. 1M A: Habitat loss is the most important cause of extinction. R: Tropical rain forests once covered 14% of the land but now cover no more than 6%.
Ans: (a)

Case-Based Practice (4 marks each)

Q72. 4M Board fav A couple is unable to conceive even after two years of unprotected cohabitation. Tests show the male has a very low sperm count, while the female is normal. (a) Name the technique in which sperm from the husband or a donor is introduced into the vagina or uterus. (b) Name the technique in which a sperm is directly injected into the ovum. (c) If the embryo formed in vitro has 6 blastomeres, where should it be transferred and what is the technique called?
Ans: (a) Artificial insemination (AI) / intra-uterine insemination (IUI). (b) ICSI (intra cytoplasmic sperm injection). (c) Into the fallopian tube: ZIFT.

Q73. 4M Board fav In a family, a normal couple has a son who is haemophilic. (a) Write the genotypes of the parents. (b) What is the probability that their next son will be haemophilic? (c) Why is the disease rare in females?
Ans: (a) Mother X^HX^h (carrier), father X^HY. (b) 50% (1 in 2 sons). (c) A female must have the allele on both X chromosomes, which needs a carrier mother and a haemophilic father, and such females rarely survive.

Q74. 4M Board fav E. coli was grown for many generations in $^{15}NH_4Cl$ and then transferred to $^{14}NH_4Cl$. (a) What will be the density of DNA after one generation? (b) What proportion of DNA will be light after two generations? (c) What does the experiment prove?
Ans: (a) All hybrid (intermediate density). (b) 50% light and 50% hybrid. (c) DNA replicates semiconservatively.

Q75. 4M Board fav A student finds that a sewage sample has a very high BOD. (a) What does high BOD indicate? (b) In which stage of sewage treatment is BOD reduced, and how? (c) What happens to the excess activated sludge?
Ans: (a) A large amount of organic matter (high pollution potential). (b) Secondary treatment: aerobic microbes form flocs and consume organic matter. (c) It is pumped into anaerobic sludge digesters, where bacteria digest it and produce biogas.

Q76. 4M Board fav A scientist inserts a gene into the BamHI site of pBR322 and transforms E. coli. (a) Which antibiotic resistance will be lost in recombinants? (b) How will you select recombinants? (c) Name an alternative selection method that uses colour.

Ans: (a) Tetracycline resistance (BamHI lies in tet^R). (b) Recombinants grow on ampicillin but not on tetracycline (replica plating). (c) Insertional inactivation of beta-galactosidase: recombinants give colourless colonies, non-recombinants blue.

Q77. 4M Board fav In a pond, phytoplankton biomass is 4 kg m^{-2} and zooplankton biomass is 21 kg m^{-2}. (a) Draw the shape of the pyramid of biomass. (b) Why is it so? (c) Which pyramid can never be inverted?

Ans: (a) Inverted. (b) Phytoplankton have a small standing crop but reproduce very fast, supporting a larger biomass of zooplankton. (c) The pyramid of energy.

Practice MCQs (Section A style)

Q78. 1M The number of nuclei in a mature embryo sac is
(a) 7
(b) 8
(c) 6
(d) 3

Ans: (b) 8

Q79. 1M Which of the following is a non-albuminous seed?
(a) wheat
(b) maize
(c) pea
(d) castor

Ans: (c) pea

Q80. 1M Leydig cells secrete
(a) estrogen
(b) androgens
(c) FSH
(d) inhibin only

Ans: (b) androgens

Q81. 1M hCG, hPL and relaxin are produced in women
(a) at puberty
(b) only during pregnancy
(c) at menopause
(d) throughout life

Ans: (b) only during pregnancy

Q82. 1M Multiload 375 is a
(a) hormone-releasing IUD
(b) copper-releasing IUD
(c) non-medicated IUD
(d) oral pill

Ans: (b) copper-releasing IUD

Q83. 1M A cross between Tt and tt gives tall and dwarf plants in the ratio
(a) 3:1
(b) 1:2:1
(c) 1:1
(d) 9:3:3:1

Ans: (c) 1:1

Q84. 1M Which of the following is an example of pleiotropy?
(a) ABO blood group
(b) phenylketonuria
(c) skin colour
(d) snapdragon flower colour

Ans: (b) phenylketonuria

Q85. 1M The enzyme that makes hnRNA in eukaryotes is
(a) RNA polymerase I
(b) RNA polymerase II
(c) RNA polymerase III
(d) DNA polymerase

Ans: (b) RNA polymerase II

Q86. 1M In the lac operon, lactose acts as
(a) repressor
(b) inducer
(c) operator
(d) promoter

Ans: (b) inducer

Q87. 1M Which brain capacity belongs to *Homo erectus*?
(a) 650–800 cc
(b) 900 cc
(c) 1400 cc
(d) 450 cc

Ans: (b) 900 cc

Q88. 1M Which of the following is a primary lymphoid organ?
(a) spleen
(b) lymph node
(c) thymus
(d) tonsil

Ans: (c) thymus

Q89. 1M Cyclosporin A is obtained from
(a) *Monascus purpureus*
(b) *Trichoderma polysporum*
(c) *Streptococcus*
(d) *Aspergillus niger*

Ans: (b) *Trichoderma polysporum*

Q90. IM DNA fragments in an agarose gel are visualised by staining with
(a) safranin
(b) ethidium bromide
(c) methylene blue
(d) iodine
Ans: (b) ethidium bromide

Q91. IM The first transgenic cow was named
(a) Dolly
(b) Rosie
(c) Molly
(d) Polly
Ans: (b) Rosie

Q92. IM Cuckoo laying eggs in a crow's nest is an example of
(a) commensalism
(b) brood parasitism
(c) mutualism
(d) amensalism
Ans: (b) brood parasitism

Q93. IM The value of Ƶ for very large areas like continents is
(a) 0.1–0.2
(b) 0.6–1.2
(c) 2–3
(d) 0.01
Ans: (b) 0.6–1.2

20. Tricks and Mnemonics

Topic	Trick
Anther wall (outside to inside)	"Every Exam Makes Tension" = Epidermis, Endothecium, Middle layers, Tapetum
Embryo sac	"7–8 ka pahada": 7 cells, 8 nuclei; 3 + 3 + 1 (central cell with 2 nuclei)
Oviduct parts (ovary to uterus)	"I Am Indian" = Infundibulum, Ampulla, Isthmus
Uterine wall (outside to inside)	"Papa Mummy Endless" = Perimetrium, Myometrium, Endometrium
Menstrual phases	"MFOL" = Menstrual, Follicular, Ovulatory, Luteal
Barrier contraceptives	"CDCV" = Condoms, Diaphragms, Cervical caps, Vaults
STIs not fully curable	"HHH" = Hepatitis-B, Herpes (genital), HIV
Chromosomal disorders	"21 ki teen copy = Down"; "XXY = extra X wala ladka (Klinefelter)"; "X0 = ek X wali ladki (Turner)"
Purines	"All Good" = Adenine, Guanine (pure as gold); pyrimidines C, T, U
RNA polymerases (eukaryote)	"RHT" = I rRNA, II hnRNA, III tRNA
Human evolution order	"DRAHHNS" = Dryopithecus, Ramapithecus, Australopithecus, habilis, erectus, Neanderthal, sapiens
Innate immunity barriers	"PPCC" = Physical, Physiological, Cellular, Cytokine
Antibody types	"GAME" = IgG, IgA, IgM, IgE
Biofertilisers	"RAAG + cyanobacteria" = Rhizobium, Azotobacter, Azospirillum, Glomus
Tools of rDNA	"RLVH" = Restriction enzymes, Ligase, Vector, Host
Decomposition steps	"FLCHM" = Fragmentation, Leaching, Catabolism, Humification, Mineralisation
Population interactions	"++ --": mutualism ++, competition --, predation/parasitism +-, commensalism +0, amensalism -0
Evil Quartet	"HOAC" = Habitat loss, Over-exploitation, Alien species, Co-extinction

21. Common Mistakes That Cost Marks

→ Writing "pollen grain is the male gamete": the pollen grain is the male gametophyte; it forms two male gametes.
→ Calling the endosperm of angiosperms haploid: it is triploid (3n) because of triple fusion.
→ Saying ovulation releases an ovum: in humans the secondary oocyte is released; meiosis II completes after sperm entry.
→ Writing that the placenta makes oxytocin: oxytocin comes from the maternal pituitary.
→ Confusing ZIFT and GIFT: ZIFT transfers a zygote or early embryo; GIFT transfers an ovum.
→ Writing the genotype ratio for phenotype: monohybrid phenotype is 3:1, genotype is 1:2:1.
→ Saying the coding strand is transcribed: the template strand (3′ to 5′) is transcribed.
→ Writing "Okazaki fragments on the leading strand": they form on the lagging strand.
→ Calling HIV a DNA virus: HIV is a retrovirus with RNA genome and reverse transcriptase.
→ Saying antibiotics kill viruses: antibiotics act on bacteria, not viruses.
→ Writing EcoRI produces blunt ends: it produces sticky ends (GAATTC, cut between G and A).

→ Saying DNA moves towards the cathode in electrophoresis: it moves towards the anode (positive electrode).

→ Drawing the pyramid of biomass in sea as upright: it is inverted.

→ Not underlining scientific names in handwritten answers.

One-Mark Rapid Fire (answers in brackets)

1. Name the nutritive layer of the anther wall. (tapetum)
2. Which cell of the embryo sac has the filiform apparatus? (synergid)
3. Name the hormone that triggers ovulation. (LH)
4. Where does fertilisation take place in humans? (ampullary region of the fallopian tube)
5. Name a hormone-releasing IUD. (LNG-20 / Progestasert)
6. What is the genotype of a person with blood group O? (ii)
7. Which disorder is caused by trisomy of chromosome 21? (Down's syndrome)
8. Name the enzyme that joins Okazaki fragments. (DNA ligase)
9. Which codon is the initiation codon? (AUG)
10. Who proposed the mutation theory? (Hugo de Vries)
11. Name the test used to confirm typhoid. (Widal test)
12. Which cells are called the HIV factory? (macrophages)
13. Name the source of statins. (*Monascus purpureus*)
14. Which enzyme is used in PCR? (Taq polymerase)
15. Which cry gene controls corn borer? (cryIAb)
16. Write the formula of logistic growth. ($dN/dt = rN(K - N)/K$)
17. Which pyramid is always upright? (pyramid of energy)
18. Name the Indian biodiversity hotspots. (Western Ghats and Sri Lanka, Indo-Burma, Himalaya)

23. Must-Do Topics, Chapter-wise

Ch	Topics that repeat in board papers
1	embryo sac, double fertilisation, outbreeding devices, apomixis and polyembryony, pollen-pistil interaction
2	gametogenesis, menstrual cycle, fertilisation and implantation, placenta hormones
3	contraceptive methods with mode of action, ART (ZIFT, GIFT, ICSI), STIs, MTP
4	dihybrid cross, test cross, codominance (ABO), sex determination, pedigree, genetic disorders
5	Meselson-Stahl, Hershey-Chase, lac operon, transcription in eukaryotes, genetic code, DNA fingerprinting
6	homologous vs analogous organs, Hardy-Weinberg, natural selection types, human evolution
7	*Plasmodium* life cycle, immunity types, antibody structure, HIV replication, drugs
8	sewage treatment, biogas, biofertilisers, biocontrol, microbes and their products
9	restriction enzymes, pBR322, insertional inactivation, PCR, competent host, bioreactors
10	Bt cotton, RNAi, insulin, gene therapy (ADA), transgenic animals, biopiracy
11	growth equations and curves, age pyramids, population interactions with examples
12	productivity, decomposition, energy flow and 10 per cent law, ecological pyramids
13	species-area relationship, Evil Quartet, rivet popper, in situ and ex situ conservation

22. Exam Hall Time Plan (3 hours)

Time	What to do
First 15 min (reading)	read the full paper, tick choices, plan long answers
0–25 min	Section A: 16 MCQ and Assertion-Reason
25–55 min	Section B: 5 very short answers (2 marks)
55–110 min	Section C: 7 short answers (3 marks)
110–130 min	Section D: 2 case-based questions
130–170 min	Section E: 3 long answers with diagrams
Last 10 min	check labels, units, question numbers

Revision tracker: tick karo

- ✓ Ch 1–3 one-liners + diagrams (Reproduction)
- ✓ Ch 4–6 crosses, codons, evolution dates
- ✓ Ch 7–8 diseases table + microbes list
- ✓ Ch 9–10 tools, PCR, pBR322, applications
- ✓ Ch 11–13 equations, pyramids, Evil Quartet
- ✓ Scientists, numbers, full forms tables
- ✓ Assertion-Reason + case-based + MCQs solved
- ✓ Common mistakes page read twice

Paheli-Bujho

Bujho: Paheli, sab tick ho gaya!
Paheli: Shabaash! Ab ek timed sample paper aur phir aaram.
Bujho: Kal exam hall mein milte hain, full confidence ke saath!

ALL THE BEST FOR YOUR BOARD EXAM!

Most asked: revise twice, draw every diagram once, solve one timed paper, sleep well

Yeh notebook yahin khatam hoti hai, par tumhari **tayyari** ab final round mein hai.

1. Paheli aur Bujho ki taraf se

Paheli-Bujho

Bujho: Paheli, poori book khatam! Ab kya?
Paheli: Ab revision, diagrams aur ek timed sample paper.
Bujho: Aur agar exam mein kuch bhool gaye toh?
Paheli: Keywords likho, diagram banao, step by step. Marks milenge!
Bujho: All the best, sabko!

2. Exam Day Checklist

- → Admit card, two blue pens, pencil, eraser, scale, sharpener: raat ko hi bag mein.
- → Reach the centre 30 minutes early; water bottle (transparent) saath rakho.
- → Reading time: choice questions tick karo, long answers ka order plan karo.
- → Har answer ke saath diagram ya flow chart jahan possible ho.
- → Last 10 minutes: question numbers, labels and units check karo.

Exam Alert

- → Scientific names italic nahi likh sakte, toh **underline** karo (genus capital, species small).
- → Ratio, number aur year galat mat likho: 3:1, 9:3:3:1, 1953, 1958.
- → Diagram bina labels ke = aadhe marks. Labels hamesha seedhi line se.

3. 10 Lines to Read in the Exam Hall Corridor

1. Embryo sac: 7 cells, 8 nuclei; PEN is 3n.
2. Sperm path: rete testis, vasa efferentia, epididymis, vas deferens.
3. Ovulation about day 14 by LH surge; corpus luteum makes progesterone.
4. ZIFT up to 8 blastomeres, IUT more than 8.
5. Monohybrid 3:1, dihybrid 9:3:3:1, test cross 1:1.
6. Stop codons UAA, UAG, UGA; start AUG.
7. Meselson-Stahl 1958: semiconservative replication.
8. Pyramid of energy is always upright.
9. PCR: denaturation, annealing, extension (Taq).
10. Evil Quartet: habitat loss is the biggest cause of extinction.

4. Must-Draw Diagrams: Tick Karo

Diagram checklist

Chapter	Diagrams
1	T.S. anther, anatropous ovule, embryo sac, dicot embryo, monocot seed
2	male and female reproductive systems, T.S. testis, sperm, T.S. ovary, blastocyst
3	IUDs (CuT, LNG-20)
4	monohybrid and dihybrid crosses, pedigree symbols
5	nucleosome, replication fork, transcription unit, tRNA, lac operon
6	Miller's apparatus, homologous forelimbs
7	antibody, life cycle of *Plasmodium*, HIV replication
8	sewage treatment plant, biogas plant
9	pBR322, gel electrophoresis, PCR, bioreactors
10	maturation of insulin
11	exponential and logistic growth curves, age pyramids
12	pyramids, energy flow, decomposition cycle
13	species-area curve, pie charts of species

Nitesh Bhaiya ki baat

Tumne poori Class 12 Biology ek notebook mein padh li, yeh khud mein badi baat hai. Ab bas revise karo, likh ke practice karo, aur khud pe bharosa rakho. Board exam tumhara hai!

Nitesh Bhaiya | Eduspark

Not in board exam (2026-27)

Reproduction in Organisms, Strategies for Enhancement in Food Production, Environmental Issues, and the "Organism and its Environment" part of Organisms and Populations (removed in the rationalised NCERT). Is book mein sirf current syllabus hai.

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This book is free for every student. Share it with friends who need it, and keep studying with **Eduspark**: notes, one shots and Most Important Questions for every subject.

Thank you

Thank you for studying with this notebook. Best wishes for your board exam and for everything that comes after it.